Encore® and Eclipse®

Start-up/Service Manual

MDE-3804Q Part 1 of 2

Computer Programs and Documentation

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Approvals

Gilbarco is an ISO 9001:2008 registered company.

Underwriters Laboratories (UL):

UL File#	Products listed with UL
MH1941	All Gilbarco pumps and dispensers that bear the UL listing mark.
MH8467	Transac System 1000 and PAM 1000
E105106	Dell DHM Minitower
E165027	G-SITE and Passport Systems

California Air Resources Board (CARB):

Executive Order #	Product
G-70-52-AM	Balance Vapor Recovery
G-70-150-AE	VaporVac

National Conference of Weights and Measures (NCWM) - Certificate of Conformance (CoC):

Gilbarco pumps and dispensers are evaluated by NCWM under the National Type Evaluation Program (NTEP). NCWM has issued the following CoC:

CoC#	Product	Model #	CoC#	Product	Model #
02-019	Encore	Nxx	02-036	Legacy	Jxxx
02-020	Eclipse	Exx	02-037	G-SITE Printer (Epson)	PA0307
02-025	Meter - C Series	PA024NC10		G-SITE Distribution Box	PA0306
	Meter - C Series	PA024TC10		G-SITE Keyboard	PA0304
02-029	CRIND	—		G-SITE Mini Tower	PA0301
02-030	TS-1000 Console	—		G-SITE Monitor	PA0303
	TS-1000 Controller	PA0241		G-SITE Printer (Citizen)	PA0308
	Distribution Box	PA0242	02-038	C+ Meter	T19976
	Meter - EC Series	PA024EC10	02-039	Passport	PA0324
	VaporVac Kits	CV	02-040	Ecometer	T20453
			05-001	Titan	KXXY Series

Trademarks

Non-registered trademarks

Applause™ Media System	G-CAT™	MPD™	Surge Management System™
CIM™	Gilbert™	Optimum Series™	Tank Monitor™
C-PAM™	G-SITE® Link™	PAM 1000™	TCR™
Dimension™	G-SITE® Lite™	PAM 5000™	Titan™
Ecometer™	G-SITE™	PAM™	The Advantage™ Series
ECR™	Highline™	Passport™	Trimline
EMC™	Horizon™	SMART CRIND™	Ultra-Hi™
Eclipse™	InfoScreen™	SMART Meter™	ValueLine™
e-CRIND™	MultiLine™	SmartPad™	
FlexPay™	Making Things Better™	Super-Hi™	

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Legacy®
Performer®
Transac®
TRIND®
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Additional U.S. and foreign trademarks pending.
All product names, logos, and brands are the property of their respective owners and are for identification purposes only.
Use of these names, logos, and brands does not imply endorsement.

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1 – Read This First

Introduction

Purpose

Sections of this manual combine to provide start-up and service information for Encore® and Eclipse® series units. This manual is a general service guide and not a replacement for Gilbarco®-certified training. Certified training includes instructions regarding safety procedures, use of test equipment and common tools, wiring requirements, and electrical service procedures. This section contains a general introduction section, service protocol, safety, and model number information.

The following chapters are covered in this manual:

Topics in This Section

Related Documents

Document Number	Title	GOLD℠ Library
FE-341	Field Wiring Diagram Eclipse Series Dispensers	• Encore and Eclipse • Encore and Eclipse Installers • Engineering Diagrams • Field Wiring
MDE-2531	Gilbarco Pump and Dispenser Start-up and Service Manual	• Pump and Dispenser Start-up and Service Manual • Service Manual
MDE-2540	The Advantage® Series, Legacy® and MPD® Series Owner's Manual	• Advantage and Legacy Models • Domestic Warranty and Owner's Manuals • Export Warranty and Owners Manuals
MDE-2562	MPD-3 and The Advantage Series Single-Line CRIND, CRIND with Monochrome LCD, Infoscreen®, External CRIND, C-PAM™ CRIND Service Manual	• CRIND and TRIND • Service Manual
MDE-2755	STP Control and Dispenser Isolation Relay Box (PA0287)	• Advantage and Legacy Models • Encore and Eclipse • Encore and Eclipse Installers
MDE-3022	Field Performance Test Kit K94123 for VaporVac®	Advantage and Legacy Models
MDE-3026	Standardized Service Coding/Terminology	Gilbarco Forms
MDE-3802	Encore and Eclipse Site Preparation Manual	• Encore and Eclipse • Encore and Eclipse Installers • Footprint and Elevations • Site Prep
MDE-3860	Programming Quick Reference Guide	• Encore and Eclipse • Encore and Eclipse Installers
MDE-3893	Encore and Eclipse Series Owner's Manual	• Encore and Eclipse • Encore and Eclipse Installers • Domestic Warranty and Owner's Manuals • Export Warranty and Owners Manuals
MDE-3985	Encore Installation Manual	• Encore and Eclipse • Encore and Eclipse Installers • Footprint and Elevations
MDE-3986	Eclipse Installation Manual	• Encore and Eclipse • Encore and Eclipse Installers
MDE-4084	Encore Junction Box Retrofit Kit M01483K00X Installation Instructions	• Encore and Eclipse • Encore and Eclipse Installers
MDE-4096	Blackmer® Global Dispenser Pump Operation, Maintenance and Kit Installation Manual	• Encore and Eclipse • Encore and Eclipse Installers • Gasboy Q, A and E Series Pumps/Dispensers • Service Manual • Site Prep
MDE-4133	Power Supply Kit M00016K003 Installation Manual	• Encore and Eclipse • Encore and Eclipse Installers
MDE-4226	Encore/Eclipse Installation Checklist (Form A)	• Encore and Eclipse • Encore and Eclipse Installers • Gilbarco Forms
MDE-4227	Encore/Eclipse Start-up Checklist (Form B)	• Encore and Eclipse • Encore and Eclipse Installers • Gilbarco Forms
MDE-4228	Encore Commissioning Checklist (Form C)	• Encore and Eclipse • Encore and Eclipse Installers • Gilbarco Forms

Document Number	Title	GOLD℠ Library
MDE-4281	Calibration Quick Reference Card Encore and Eclipse Units	• Encore and Eclipse • Encore and Eclipse Installers • Service Manual
MDE-4447	Gilbarco Global Pumping Unit Operation and Service Manual	• Pump and Dispenser Start-up • Service Manual
MDE-4897	Installation Checklist for Diesel Exhaust Fluid (DEF) Units (Form A)	• Encore and Eclipse • Encore and Eclipse Installers • Gilbarco Forms
MDE-4898	Start-up Checklist for Diesel Exhaust Fluid (DEF) Units (Form B)	• Encore and Eclipse • Encore and Eclipse Installers • Gilbarco Forms
MDE-4899	Commissioning Checklist for Diesel Exhaust Fluid (DEF) Units (Form C)	• Encore and Eclipse • Encore and Eclipse Installers • Gilbarco Forms
MDE-4950	Encrypted Pulser Retrofit Kits (M12514K00X) Installation Instructions for Encore 500/S/700 S/S E-CIM™	Encore and Eclipse
MDE-5354	Ultra-Hi Encrypted Encoder Retrofit Kits (M15177K00X, M15178K00X, M15179K00X, M15180K00X) Installation Instructions for Encore 500/S/700 S/S E-CIM	Encore and Eclipse
PT-1936	Encore Series Pumps and Dispensers Illustrated Parts Manual	• Encore and Eclipse • Encore and Eclipse Installers • Parts Manual
PT-1937	Encore 300, Encore 500/500 S, Encore 550, Encore 700 S, Eclipse Recommended Spare Parts Manual	• Encore and Eclipse • Parts Manual
PT-1938	Eclipse Illustrated Parts Manual	• Encore and Eclipse • Encore and Eclipse Installers • Parts Manual
RP100-90	Petroleum Equipment Institute (PEI) Recommended Practices for Underground Liquid Storage Systems	N/A

Note: In addition to these documents, Gilbarco Product Service Bulletins (PSBs) and monthly newsletter “ASC Update” are excellent sources of service information.

Abbreviations and Acronyms

Term	Description
AIM	Automatic Identification Manufacturers
ASC	Authorized Service Contractor
ATC	Automatic Temperature Compensation
BIOS	Basic Input/Output System
CC	Command Code
CCN	CRIND® Control Node
CIM	Customer Interface Module
CNG	Compressed Natural Gas
CoC	Certificate of Compliance
CRIND	Card Reader in Dispenser
CSR	Customer Service Report
D-Box	Distribution Box
DEF	Diesel Exhaust Fluid
DLT	Displaying Last Transaction
EC	Error Code
ESD	Electrostatic Discharge

Term	Description
FC	Function Code
FFD	Film and Foil Dispenser
GDP	Global Dispenser Pump (Blackmer unit)
GGE	Gasoline Gallon Equivalent
GLE	Gasoline Liter Equivalent
GOLD	Gilbarco Online Documentation
GPM	Gallons per Minute
IC	Integrated Circuit
IFSF	International Forecourt Standards Forum
I.S.	Intrinsic Safety
ISD	In-station Diagnostic
J-box	Junction Box
LAN	Local Area Network
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LNG	Liquefied Natural Gas
LON	Local Operating Network
LPG	Liquefied Petroleum Gas
LPM	Liters per Minute
LSD	Least Significant Digit
LTD	Large Touch Display
MIP	Microprocessor Interface Program
MOC	Major Oil Company
MPD	Multi Product Dispenser
MPP	Multi Product Pump
MSD	Most Significant Digit
NFPA	National Fire Protection Association
OIML	International Organization of Legal Metrology
OSHA	Occupational Safety and Health Administration
PCA	Printed Circuit Assembly
PCB	Printed Circuit Board
PCN	Pump Control Node
PCV	Pressure Control Valve/Proportional Control Valve
PDN	Pump Door Node
PEI	Petroleum Equipment Institute
PFCV	Proportional Flow Control Valve
POS	Point of Sale
PPP	Programmable Pump Preset
PPU	Price per Unit
PSI	Pounds (of Pressure) per Square Inch
RF	Radio Frequency
SAE	Society of Automotive Engineers
STP	Submersible Turbine Pump
TLS	Tank Level Sensor

Term	Description
TRIND®	Transmitter/Receiver in Dispenser
TRP	Technician Resources Page
UHF	Ultra High Frequency
UL®	Underwriters Laboratories
USB	Universal Serial Bus
UST	Underground Storage Tank
VAC	Voltage Alternate Current
VCF	Volume Correction Factor
VDC	Voltage Direct Current
VUT	Volume Unit Type
W&M	Weights and Measures

Intended Users

This manual is intended for Gilbarco Authorized Service Contractors (ASCs) who have been trained and certified (on completion of Gilbarco-certified training courses). These ASCs must be aware of basic troubleshooting techniques, such as reading a volt/ohm meter, reading pressure, and so on. ASCs must also understand the differences in Gilbarco products, options, and functionality.

If you have not been through Gilbarco training, contact the Gilbarco training department.

Service Protocol

Requesting Further Technical Assistance

A service technician requesting technical assistance must do the following:

1. Be at the site.
2. Have the following available when placing the call:
   - Unit model and serial number
   - Site name and telephone number
   - Your technician number
   - Problem description and history
   - All required recommended spare parts
   - Manuals available for reference
3. Call the Technical Support Department at 1-800-743-7501. If assigned previously, refer to the Customer Service Report (CSR) number.

Using Replacement Parts

Use only genuine Gilbarco replacement parts. Use of other parts will void warranty and can affect unit conformance to various national, local, or state codes.

Whom to Contact at Gilbarco

Information	Contact
Schedule Training	Gilbarco Training Department at 1-336-547-5743
Technical Assistance	Gilbarco Technical Support at 1-800-743-7501
Warranty Service and Information	Gilbarco Tele-support at 1-800-800-7498
Explanation of Gilbarco’s Warranty Policy	For assistance, contact your local Gilbarco Distributor.
Technical Literature, Parts Manuals, and other Documents	Gilbarco Literature Department at 1-336-547-5661

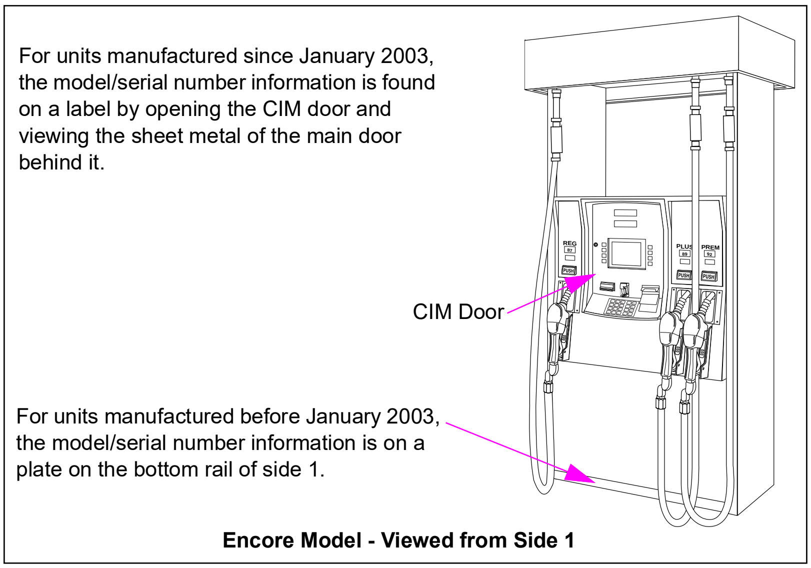
Model Information

Where to Find Model Information on Unit

Depending on the date of manufacture, Encore and Eclipse series units will have the model/serial number information located in different locations of the unit. The following information describes where to find the model/serial numbers, examples of the format, and coding pertaining to model number and date.

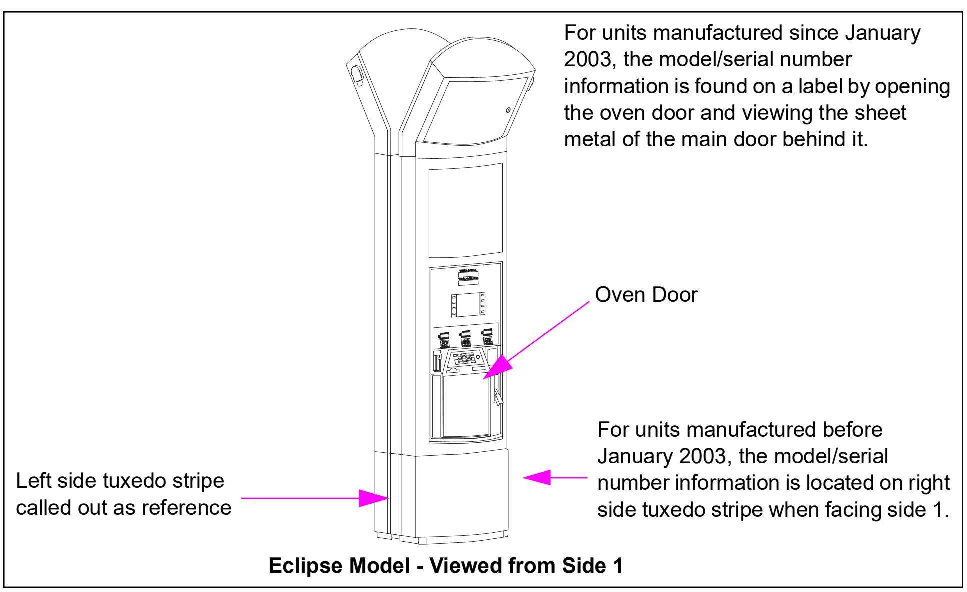
Encore

Figure 1-1: Encore Model/Serial Number Information Locations



Eclipse

Figure 1-2: Eclipse Model/Serial Number Information Locations



Model/Serial Number Tag Formats

Figure 1-3: Model/Serial Number Example for Units Manufactured Before January 2003

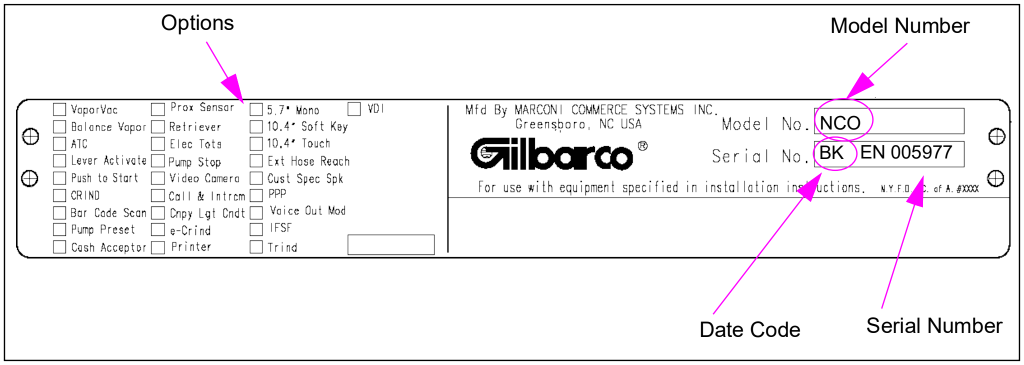
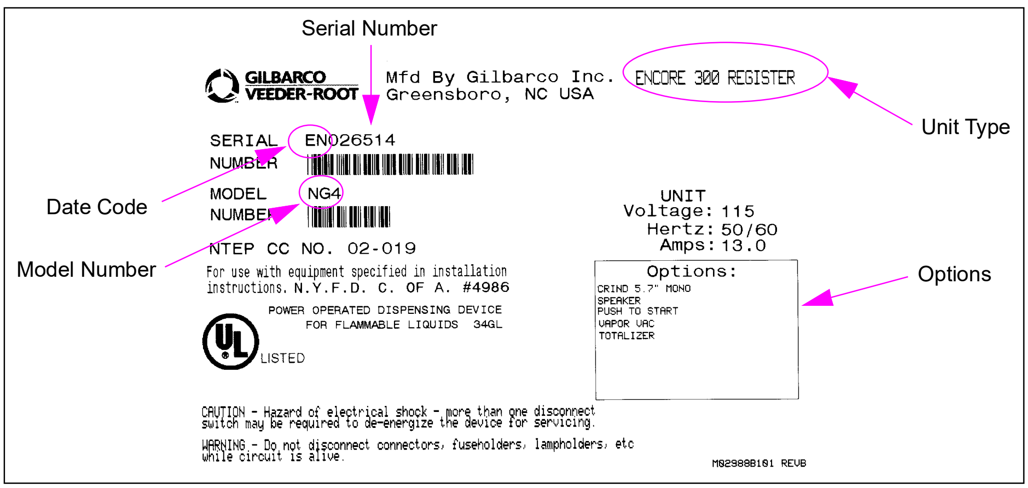


Figure 1-4: Model/Serial Number Example for Units Manufactured Since January 2003



Definitions of Fields

Term	Definition
Model Number	Contains unit configuration information.
Options	A list of customer-driven options per unit.
Serial Number	Identifies the unit by customer.
Unit Type	Provides the type of unit. For example, Encore 500 or Eclipse.

Model Codes

Encore

The model number provides information on the configuration and options of the unit. In Figure 1-3 on page 1-7, the model number NCO corresponds to an Encore unit, pump type, and multi-hose, single grade unit, as described in the following table:

N = represents an Encore unit (first column of data)
C = multi-hose pump (second column of data)
O = represents 1 grade (third column of data)

To determine the model code on a Gilbarco pump or dispenser, refer to the following table:

N	X	X
Encore	**A** = Multi-Hose Dispenser	**0** = 1 Grade **1** = 2 Grade **2** = 3 Grade **3** = 4 Grade
	C = Multi-Hose Pump	**0** = 1 Grade **1** = 2 Grade **2** = 3 Grade **3** = 4 Grade
	F = Multi-Hose Universal Blender	**0** = 5 Grade 3 - 2 (2 hose) Multi-Hose Universal Blender **1** = 5 Grade 3 - 1 - 1 (3 hose) Multi-Hose Universal Blender **2** = 5 Grade 3 - 1 + 1 (3 hose)* Multi-Hose Universal Blender **4** = 5 Grade 3 - 1 + 1 (3 hose)** Multi-Hose Universal Blender **6** = 6 Grade 3 - 2 - 1 + 1 (4 hose) Multi-Hose Universal Blender **7** = 6 Grade 4 - 1 + 1 (3 hose) Multi-Hose Universal Blender **8** = 5 Grade 3 - 1 - 1 (3 hose) Multi-Hose Universal Blender
	G = Single-hose	**0** = 3 Grade Dispenser **1** = 3 Grade + 1 Dispenser **2** = 3 Grade Pump **3** = 3 Grade Pump + 1 **4** = 2 Grade Single-hose Multi Product Dispenser (MPD) **5** = 2 Grade Single-hose Pump
	J = Multi-hose Blender	**0** = 3 Grade Blender Dispenser **1** = 3 Grade Blender Pump **2** = 4 Grade Blender + 1 Dispenser **3** = 4 Grade Blender + 1 Pump **4** = 3 + 2 Multi-hose Hybrid Blender **5** = 5 + 0 Multi-hose Hybrid Blender **6** = 4 + 1 Multi-hose Hybrid Blender
	L = X+1 Blender	**0** = 2 + 1 Grade Blender Dispenser **1** = 3 + 1 Grade Blender Dispenser **2** = 4 + 1 Grade Blender Dispenser **4** = 2 + 1 Grade Blender Pump **5** = 3 + 1 Grade Blender Pump **6** = 4 + 1 Grade Blender Pump **8** = 5+1 Grade Blender Pump
	N = X+0 Blender	**1** = 3 + 0 Grade Blender Dispenser **2** = 4 + 0 Grade Blender Dispenser **3** = 5 + 0 Grade Blender Dispenser **5** = 3 + 0 Grade Blender Pump **6** = 4 + 0 Grade Blender Pump **7** = 5 + 0 Grade Blender Pump
	P = High Flow (1 Grade)	**3** = Ultra-Hi™ Master **4** = Ultra-Hi Combo **5** = Ultra-Hi Satellite **6** = Ultra-Hi Master - Single-sided Dual Product **8** = Ultra-Hi Satellite - Single-sided Dual Product

N	X	X
Encore	**R** = Special	**0** = No Hydraulics **1** = Robot **2** = Simulator **3** = Pumpless Pump **4** = No Electronics

*Center hose right of CIM
**Center hose left of CIM

Encore Generic

Family	Code	Description
MPDs	NA0	MPD Dispenser 1-Grade
	NA1	MPD Dispenser 2-Grade
	NA2	MPD Dispenser 3-Grade
	NA3	MPD Dispenser 4-Grade (500 only)
	NA4	MPD Dispenser 1-Grade, DEF Only
	NC0	MPD Pump 1-Grade
	NC1	MPD Pump 2-Grade
	NC2	MPD Pump 3-Grade
	NC3	MPD Pump 4-Grade (500 Only)
Multi-Hose Universal Blender	NF0	Multi-Hose Universal Blender 5 Grade 3 - 2 (2 hose)
	NF1	Multi-Hose Universal Blender 5 Grade 3 - 1 - 1 (3 hose)
	NF2	Multi-Hose Universal Blender 5 Grade 3 - 1 + 1 (3 hose)*
	NF4	Multi-Hose Universal Blender 5 Grade 3 - 1 + 1 (3 hose)**
	NF6	Multi-Hose Universal Blender 6 Grade 3 - 2 - 1 + 1 (4 hose)
	NF7	Multi-Hose Universal Blender 6 Grade 4 - 1 + 1 (3 hose)
	NF8	Multi-Hose Universal Blender 5 Grade 3 - 1 - 1 (3 hose), which reserves “+1” for diesel.
Single-hose MPD	NG0	Single-hose MPD Dispenser, 3-Grade
	NG1	Single-hose + 1 MPD Dispenser, 4-Grade (500 only)
	NG2	Single-hose Multi-product Pump (MPP) Pump, 3-Grade
	NG3	Single-hose + 1 MPP Pump, 4-Grade (500 only)
	NG4	Single-hose MPD Dispenser, 2-Grade
	NG5	Single-hose MPP Pump, 2-Grade
X + 0 Blender	NN0	Blender Dispenser 2+0
	NN1	Blender Dispenser 3+0
	NN2	Blender Dispenser 4+0
	NN3	Blender Dispenser 5+0
	NN5	Blender Pump 3+0
	NN6	Blender Pump 4+0
	NN7	Blender Pump 5+0
X + 1 Blender	NL0	Blender Dispenser 2+1
	NL1	Blender Dispenser 3+1
	NL2	Blender Dispenser 4+1
	NL3	Blender Dispenser 3+1+1 Film and Foil Dispenser (FFD)
	NL4	Blender Pump 2+1

Family	Code	Description
Multi-hose Blender	NL5	Blender Pump 3+1
	NL6	Blender Pump 4+1
	NJ0	Multi-hose Blender Dispenser
	NJ2	Multi-hose +1 Blender Dispenser
	NJ1	Multi-hose Blender Pump
	NJ3	Multi-hose +1 Blender Pump
	NJ4	Multi-hose Hybrid Blender 3+2 (FFD)
	NJ5	Multi-hose Hybrid Blender 5+0 (Standard and Alternate)
	NJ6	Multi-hose Hybrid Blender 4+1 (Standard and Alternate)
	NL8	Blender Pump 5+1
Ultra-Hi	NP0	Super-Hi Dispenser
	NP1	Super-Hi Master
	NP2	Super-Hi Combo
	NP3	Ultra-Hi Master Dispenser
	NP4	Ultra-Hi Combo Dispenser
	NP5	Ultra-Hi Satellite Dispenser
	NP6	Ultra-Hi Master Dispenser - Single-sided Dual Product Unit
	NP8	Ultra-Hi Satellite Dispenser - Single-sided Dual Product Unit
	NPA	Ultra-Hi Master Diesel and DEF
	NPB	Ultra-Hi Master/Satellite Combo Diesel and DEF
	NPC	Ultra-Hi Master Dispenser - Single-sided Dual Product Unit with DEF
	NPD	Ultra-Hi - TBD
Specials	NR0	No Hydraulics, Dual, 1 Nozzle Boot
	NR1	Robot
	NR2	Simulator
	NR3	Pumpless Pump
	NR4	No Electronics

*Center hose right of CIM
**Center hose left of CIM

DEF Only Units - NA4

N	A	4
Encore	Multi-hose Dispenser	DEF only

DEF+1 Units

Model #	Ultra-Hi Dispenser Model Description	Encore Inlets
NPA	Ultra-Hi Master Dispenser + DEF 1-side	X from Tank
NPB	Ultra-Hi Combo Dispenser + DEF 1-side	X from Tank Y from Tank

Eclipse

The model number provides information on the configuration and options of the unit. For units with totalizers installed in tuxedo stripe (for stripe location, see Figure 1-2 on page 1-6), totalizer will always be on left tuxedo stripe when facing side 1.

Note: If a unit does not have a totalizer, do not assume side location by unused knockouts in tuxedo stripe. All tuxedo stripes have totalizers knockouts, whether a totalizer can be installed or not.

Model #	Unit Type	Grade/Hoses
EG0	Dispenser Single-hose	3 Grade, 2 Hose
EL0	Blender Dispenser X+1	2+1 Grade, 4 Hose
EL1		3+1 Grade, 4 Hose
EL2		4+1 Grade, 4 Hose
EL3		5+1 Grade, 4 Hose
EN0	Blender Dispenser X+0	2+0 Grade, 2 Hose
EN1		3+0 Grade, 2 Hose
EN2		4+0 Grade, 2 Hose
EN3		5+0 Grade, 2 Hose

Date Codes

The date code provides the month and year of manufacture. In Figure 1-3 on page 1-7, an Encore unit stamped BK was built in February 2001. To determine date codes, refer to the following chart:

Month Codes			
A = January	D = April	G = July	K = October
B = February	E = May	H = August	L = November
C = March	F = June	J = September	M = December
Year Codes			
A = 1992	L = 2002	A = 2012	L = 2022
B = 1993	M = 2003	B = 2013	M = 2023
C = 1994	N = 2004	C = 2014	N = 2024
D = 1995	P = 2005	D = 2015	P = 2025
E = 1996	R = 2006	E = 2016	R = 2026
F = 1997	S = 2007	F = 2017	S = 2027
G = 1998	T = 2008	G = 2018	T = 2028
H = 1999	U = 2009	H = 2019	U = 2029
J = 2000	W = 2010	J = 2020	W = 2030
K = 2001	X = 2011	K = 2021	X = 2031

Recommended Tool

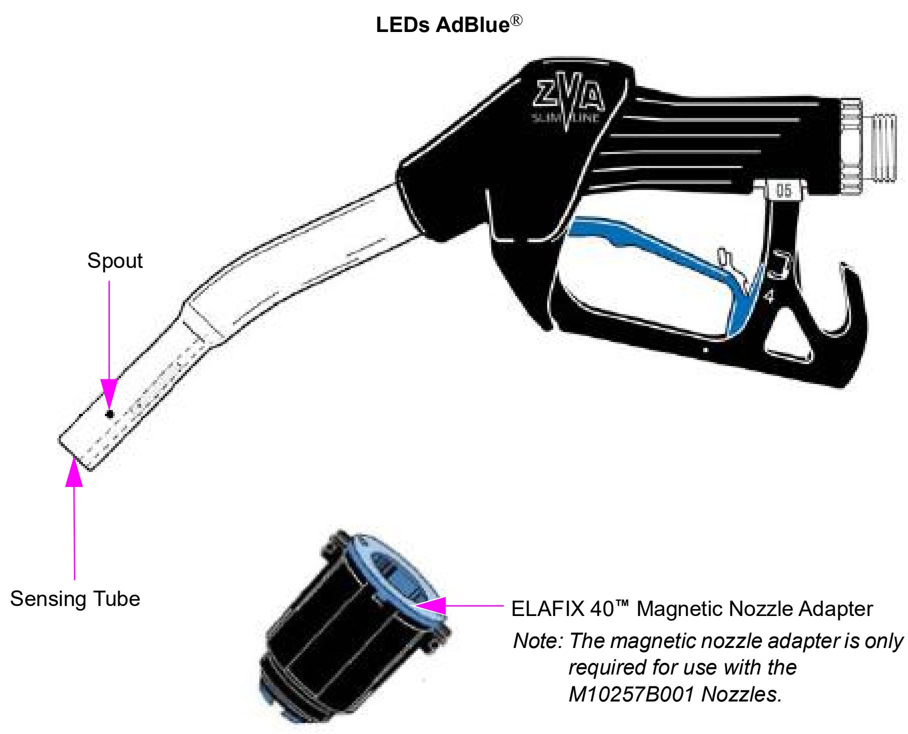
A Magnetic Nozzle Adapter (M10656B001) is required for each technician servicing the DEF units.

Note: The magnetic nozzle adapter is required only for M10257B001 Nozzles. M10257B002 Nozzles do not require this adapter.

Parts

DEF Nozzle (M10257B001 and M10257B002)

Figure 1-5: DEF Dispensers with LEDs DEF Nozzle



The DEF nozzle is designed to dispense DEF. No other nozzle type is currently approved. The DEF nozzle is manufactured by Elaflex®/OPW® (Elaflex Slimline LEDs).

- The DEF nozzle is a non-vapor recovery nozzle. Conventional performance and troubleshooting procedures for non-vapor recovery nozzles are also applicable to the DEF nozzle.
- In addition to fluid compatibility, some nozzles require a magnetic nozzle adapter to be slipped over the spout to dispense DEF. Normally, this adapter is part of the construction of the vehicles DEF tanks. The magnetic nozzle adapter helps to prevent DEF from being dispensed into the diesel tank that may result in severe damage to the engine.
  The technician must have a magnetic nozzle adapter or its equivalent to dispense DEF through the nozzle during tests. Test magnetic nozzle adapter is available from Gilbarco.
- The nozzle handle also contains the activation magnet that is part of the pump handle system.
- The nozzle can freeze if the cabinet heater fails, power is lost, or the nozzle door does not drop down completely during cold weather. The nozzle must be checked for leaks and operation after thawing.

Check Valve Assembly (M10170B001)

The Check Valve (M10170B001) is compatible with DEF. Conventional troubleshooting procedures for other check valves are also applicable to the DEF check valve.

Dispenser DEF Strainer (M10820B001)

Filtration is required to protect nozzles and valves. The earlier versions of the dispenser relied completely on a filter tank system. The later versions of the dispensers have a strainer. The 100-micron strainer and assembly is compatible with DEF. The strainer must be cleaned after the first air purging and calibration to ensure good flow rates. Refer to "Purging Air from the System" on page 7-7.

Hose Breakaway/Swivel (M10258B001)

The Swivel (M10258B001) is compatible with DEF. If the swivel freezes in very cold weather with no heat in the cabinet, the breakaway can pop. It can be reset as per the manufacturer's instructions. Check for leaks and proper retention after repair. Follow any manufacturer's recommendations on repair procedure and the maximum number of times the breakaway can be reset.

Special Fuels/Fluids

The DEF standard gallonage dispenser was designed to provide quick and reliable dispensing of DEF to 2010 and later model trucks, agriculture equipment, and other potential vehicles requiring DEF. DEF is a non-flammable fluid, that can freeze at low temperatures and is used in conjunction with the vehicles catalytic converter to reduce emissions. DEF when injected into exhaust from diesel engines before a catalytic converter reduces NO_x (Nitrogen Oxide) emissions immensely. In addition to emission reduction, the technology allows greater diesel fuel economy for the vehicle by allowing the vehicle to be optimized for fuel economy with the new DEF technology.

⚠ CAUTION

Applicable during installation and operation of the dispenser: DEF freezes at approximately -11.5 °C (11 °F). Power to the dispenser and heater must always remain ON in cold weather. If power is lost and the temperature drops below this point, the system must be checked for freeze damage before restart. For sites that experience occasional power losses or for sites that are located in very cold climates, it is recommended that a backup power generator be used to maintain constant power to the dispenser. Do not use any additives to lower the freezing point of DEF. Additives of any type must not be used in DEF. Freezing can cause damaged or inoperative hose breakaways, fluid lines or components, valves, nozzles, and meters.
Prolonged storage at temperatures above 25 °C (77 °F) can impair the quality of DEF and reduce its shelf life.

⚠ CAUTION

DEF is mildly corrosive. It can corrode components that are made from incompatible materials and reduce their integrity. The use of incompatible materials may lead to leaks and spills, and can contaminate and degrade the DEF. When dispensing DEF, verify with the manufacturer if the material of all plumbing components are compatible with the DEF being dispensed.

⚠ CAUTION

Do not use prover cans meant for engine fuel with DEF or vice versa. Use stainless steel prover cans for DEF. DEF and engine fuel must not be mixed with each other or contaminated. Otherwise, damage to a vehicle's engine or pollution control devices can occur. DEF crystallizes as its water base evaporates. Pouring out liquid will not guarantee that no corrosive DEF remains in the prover can. DEF must not be contaminated with diesel fuel, contaminants, or other fluids or materials. Such contamination can cause serious damage to vehicles catalytic converters.

- Conventional fluid handling precautions are also applicable to DEF.
- DEF is non-flammable and not explosive.
- When exposed to air, the water in DEF will evaporate and result in development of urea crystals. Crystals may be fine and sharp. They will dissolve in water.
- Clean any DEF spill with water and dry the area with clean rugs, especially areas that contain metallic parts. Ensure that spilled DEF does not reenter the tank. Spilled DEF can be slippery and will corrode metallic parts. Wear eye protection and rubber gloves during any cleanup activity.
- DEF can be returned to the storage tank only if it is not contaminated. Contaminated DEF must be disposed of in an environmentally safe manner. Do not dump DEF in storm sewers or any location where the fluid or its constituents may enter a water source.
- DEF is much heavier than fuels such as gasoline. Be careful to avoid injury when lifting heavier prover cans. Use proper safe lifting techniques.

⚠ CAUTION

When stored at temperatures above 37 °C (100 °F), DEF may break down into ammonia. Ensure that you avoid inhaling any toxic ammonia vapors when opening cabinets, tank vaults, or other areas where DEF may have leaked or spilled.

⚠ WARNING

DEF can cause serious eye injury if sprayed into the eyes or may affect those with sensitive skin. Wear protective gloves and eye protection, as required. Flush eyes immediately with water, if sprayed.

Ensure that eyewash stations and safety showers are close to the work location.

If DEF Freezes

If DEF freezes, proceed as follows:

1. Remove power to the pump and dispenser.

2. Check for any damage to breakaways, nozzles, hoses/hanging hardware, and meter. Perform required repairs. Valve damage can often be detected by looking for seals slightly protruding from the valve body (compared to the new valve). Hose breakaways may only partially separate.

3. Correct the issue that is causing the loss of heat.

4. Thaw components of the dispenser. As long as electronic boards or connections are not present in the area, dispenser units can be thawed using directed exhaust from a vehicle. They can also be thawed using a non-flame heating device.
   Notes: 1) Electronic boards or connections must not be present in the area when you thaw the components of the dispenser, as any resulting condensation may damage the electronic boards or connections.
   2) Electrical cords or heating device must not be used within the designated hazardous areas around any fueling pump or dispenser. To identify the designated hazardous area, refer to MDE-3985 Encore Installation Manual.

5. Check for leaks. Some leaks can be significant. You may have to turn off the unit immediately.

6. Check if there is any damage to the aboveground tank system or heated aboveground plumbing. For assistance, consult the tank manufacturer.

   For more details on the significance and impact of DEF freeze, refer to "DEF Heating System" on page 8-29.

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2 – Important Safety Information

Notes: 1) Save this Important Safety Information section in a readily accessible location.

2) Although DEF is non-flammable, Diesel is flammable. Therefore, for DEF cabinets that are attached to Diesel dispensers, follow all the notes in this section that pertain to flammable fuels.

This section introduces the hazards and safety precautions associated with installing, inspecting, maintaining, or servicing this product. Before performing any task on this product, read this safety information and the applicable sections in this manual, where additional hazards and safety precautions for your task will be found. Fire, explosion, electrical shock, or pressure release could occur and cause death or serious injury, if these safe service procedures are not followed.

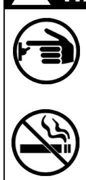
Preliminary Precautions

You are working in a potentially dangerous environment of flammable fuels, vapors, and high voltage or pressures. Only trained or authorized individuals knowledgeable in the related procedures should install, inspect, maintain, or service this equipment.

Emergency Total Electrical Shut-Off

The first and most important information you must know is how to stop all fuel flow to the pump/dispenser and island. Locate the switch or circuit breakers that shut off all power to all fueling equipment, dispensing devices, and Submerged Turbine Pumps (STPs).

WARNING



The EMERGENCY STOP, ALL STOP, and PUMP STOP buttons at the cashier's station WILL NOT shut off electrical power to the pump/dispenser. This means that even if you activate these stops, fuel may continue to flow uncontrolled.

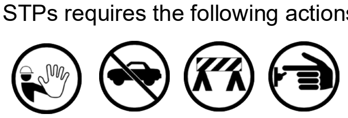
You must use the TOTAL ELECTRICAL SHUT-OFF in the case of an emergency and not the console's ALL STOP and PUMP STOP or similar keys.

Total Electrical Shut-Off Before Access

Any procedure that requires access to electrical components or the electronics of the dispenser requires total electrical shut off of that unit. Understand the function and location of this switch or circuit breaker before inspecting, installing, maintaining, or servicing Gilbarco equipment.

Evacuating, Barricading, and Shutting Off

Any procedure that requires access to the pump/dispenser or STPs requires the following actions:



- An evacuation of all unauthorized persons and vehicles from the work area
- Use of safety tape, cones, or barricades at the affected unit(s)
- A total electrical shut-off of the affected unit(s)

Read the Manual

Read, understand, and follow this manual and any other labels or related materials supplied with this equipment. If you do not understand a procedure, call the Gilbarco Technical Assistance Center (TAC) at 1-800-743-7501. It is imperative to your safety and the safety of others to understand the procedures before beginning work.

Follow the Regulations

Applicable information is available in National Fire Protection Association (NFPA) 30A; *Code for Motor Fuel Dispensing Facilities and Repair Garages*, NFPA 70; *National Electrical Code (NEC)*, Occupational Safety and Health Administration (OSHA) regulations and federal, state, and local codes. All these regulations must be followed. Failure to install, inspect, maintain, or service this equipment in accordance with these codes, regulations, and standards may lead to legal citations with penalties or affect the safe use and operation of the equipment.

Replacement Parts

Use only genuine Gilbarco replacement parts and retrofit kits on your pump/dispenser. Using parts other than genuine Gilbarco replacement parts could create a safety hazard and violate local regulations.

Federal Communications Commission (FCC) Warning

This equipment has been tested and found to comply with the limits for a Class A digital device pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy, and if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense. Changes or modifications not expressly approved by the manufacturer could void the user's authority to operate this equipment.

Safety Symbols and Warning Words

This section provides important information about warning symbols and boxes.

Alert Symbol

This safety alert symbol is used in this manual and on warning labels to alert you to a precaution which must be followed to prevent potential personal safety hazards. Obey safety directives that follow this symbol to avoid possible injury or death.

Signal Words

These signal words used in this manual and on warning labels tell you the seriousness of particular safety hazards. The precautions below must be followed to prevent death, injury, or damage to the equipment:

DANGER: Alerts you to a hazard or unsafe practice which will result in death or serious injury.

WARNING: Alerts you to a hazard or unsafe practice that could result in death or serious injury.

CAUTION with Alert symbol: Designates a hazard or unsafe practice which may result in minor injury.

CAUTION without Alert symbol: Designates a hazard or unsafe practice which may result in property or equipment damage.

Working With Fuels and Electrical Energy

Prevent Explosions and Fires

Fuels and their vapors will explode or burn, if ignited. Spilled or leaking fuels cause vapors. Even filling customer tanks will cause potentially dangerous vapors in the vicinity of the dispenser or island.

DEF is non-flammable. Therefore, explosion and fire safety warnings do not apply to DEF lines.

No Open Fire

Open flames from matches, lighters, welding torches or other sources can ignite fuels and their vapors.

No Sparks - No Smoking

Sparks from starting vehicles, starting or using power tools, burning cigarettes, cigars or pipes can also ignite fuels and their vapors. Static electricity, including an electrostatic charge on your body, can cause a spark sufficient to ignite fuel vapors. Every time you get out of a vehicle, touch the metal of your vehicle, to discharge any electrostatic charge before you approach the dispenser island.

Working Alone

It is highly recommended that someone who is capable of rendering first aid be present during servicing. Familiarize yourself with Cardiopulmonary Resuscitation (CPR) methods, if you work with or around high voltages. This information is available from the American Red Cross. Always advise the station personnel about where you will be working, and caution them not to activate power while you are working on the equipment. Use the OSHA Lockout/Tagout procedures. If you are not familiar with this requirement, refer to this information in the service manual and OSHA documentation.

Working With Electricity Safely

Ensure that you use safe and established practices in working with electrical devices. Poorly wired devices may cause a fire, explosion or electrical shock. Ensure that grounding connections are properly made. Take care that sealing devices and compounds are in place. Ensure that you do not pinch wires when replacing covers. Follow OSHA Lockout/Tagout requirements. Station employees and service contractors need to understand and comply with this program completely to ensure safety while the equipment is down.

Hazardous Materials

Some materials present inside electronic enclosures may present a health hazard if not handled correctly. Ensure that you clean hands after handling equipment. Do not place any equipment in the mouth

WARNING

In the event of inclement weather, including snow, ice, or flooding that makes driving conditions dangerous, please avoid servicing units. Always use available door stops to secure upper doors against unwanted/unexpected movement, especially during high winds. If necessary, reschedule service to avoid damage to the equipment. Weather may change unexpectedly; be aware of local weather conditions. During service, if conditions develop making service unsafe, close the unit(s) and proceed to a safe location.

WARNING

The pump/dispenser contains a chemical known to the State of California to cause cancer.

WARNING

The pump/dispenser contains a chemical known to the State of California to cause birth defects or other reproductive harm.

Gilbarco Veeder-Root encourages the recycling of our products. Some products contain electronics, batteries, or other materials that may require special management practices depending on your location. Please refer to your local, state, or country regulations for these requirements.

In an Emergency

Inform Emergency Personnel

Compile the following information and inform emergency personnel:

- Location of accident (for example, address, front/back of building, and so on)
- Nature of accident (for example, possible heart attack, run over by car, burns, and so on)
- Age of victim (for example, baby, teenager, middle-age, elderly)
- Whether or not victim has received first aid (for example, stopped bleeding by pressure, and so on)
- Whether or not a victim has vomited (for example, if swallowed or inhaled something, and so on)

WARNING

Gasoline/DEF ingested may cause unconsciousness and burns to internal organs. Do not induce vomiting. Keep airway open.
Oxygen may be needed at scene. Seek medical advice immediately.

WARNING

DEF generates ammonia gas at higher temperatures. When opening enclosed panels, allow the unit to air out to avoid breathing vapors. If respiratory difficulties develop, move victim away from source of exposure and into fresh air. If symptoms persist, seek medical attention.

WARNING

Gasoline inhaled may cause unconsciousness and burns to lips, mouth and lungs.
Keep airway open.
Seek medical advice immediately.

WARNING

Gasoline/DEF spilled in eyes may cause burns to eye tissue.
Irrigate eyes with water for approximately 15 minutes.Seek medical advice immediately.

WARNING

Gasoline/DEF spilled on skin may cause burns.
Wash area thoroughly with clear water.
Seek medical advice immediately.

WARNING

DEF is mildly corrosive. Avoid contact with eyes, skin, and clothing. Ensure that eyewash stations and safety showers are close to the work location. Seek medical advice/recommended treatment if DEF spills into eyes.

IMPORTANT: Oxygen may be needed at scene if gasoline has been ingested or inhaled. Seek medical advice immediately.

Lockout/Tagout

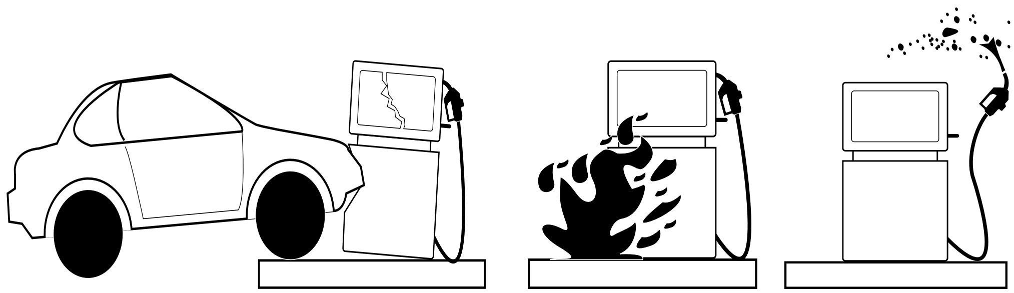
Lockout/Tagout covers servicing and maintenance of machines and equipment in which the unexpected energization or start-up of the machine(s) or equipment or release of stored energy could cause injury to employees or personnel. Lockout/Tagout applies to all mechanical, hydraulic, chemical, or other energy, but does not cover electrical hazards. Subpart S of 29 CFR Part 1910 - Electrical Hazards, 29 CFR Part 1910.333 contains specific Lockout/Tagout provision for electrical hazards.

Hazards and Actions

WARNING

Spilled fuels, accidents involving pumps/dispensers, or uncontrolled fuel flow create a serious hazard.
Fire or explosion may result, causing serious injury or death.
Follow established emergency procedures.
DEF is non-flammable. However, it can create a slip hazard. Clean up spills promptly.

The following actions are recommended regarding these hazards:



Collision of a Vehicle with Unit | Fire at Island | Fuel Spill

- Do not go near a fuel spill or allow anyone else in the area.
- Use station EMERGENCY CUTOFF immediately. Turn off all system circuit breakers to the island(s).
- Do not use console E-STOP, ALL STOP, and PUMP STOP to shut off power. These keys do not remove AC power and do not always stop product flow.
- Take precautions to avoid igniting fuel. Do not allow starting of vehicles in the area. Do not allow open flames, smoking or power tools in the area.
- Do not expose yourself to hazardous conditions such as fire, spilled fuel or exposed wiring.
- Call emergency numbers.

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3 – Start-up and Commissioning

About This Section

Gilbarco requires start-up and commissioning of its pumps and dispensers installed in the United States. This procedure ensures proper installation and warranty registration of the equipment. It is also recommended that you follow this procedure outside the United States.

Topics in This Section

Certificate of Compliance (CoC) Numbers

CoC numbers are required during installation and start-up of new sites. You may be asked for these numbers by any Weights and Measures (W&M) official. Refer to the following table:

Unit Name	Gilbarco CoC Number
Encore	02-019
Eclipse	02-020

Commissioning Instructions

Most of the *MDE-4228 Encore Commissioning Checklist (Form C)* is self-explanatory. Ensure that you read it completely before commissioning the site, paying particular attention to instructions and notes. Additional information is supplied to further define some areas and to explain how the *MDE-4226 Encore/Eclipse Installation Checklist (Form A)* (Figure 3-1 on page 3-3), the *MDE-4227 Encore/Eclipse Start-up Checklist (Form B)* (Figure 3-2 on page 3-4) and the *MDE-4228 Encore Commissioning Checklist (Form C)* (Figure 3-3 on page 3-5) are related.

1 The commissioning ASC must have previously read and understood the latest versions of *MDE-3804 Encore and Eclipse Start-up/Service Manual*, *MDE-3985 Encore Installation Manual*, *MDE-3986 Eclipse Installation Manual*, and *MDE-3802 Encore and Eclipse Site Preparation Manual*, before performing any commissioning.

2 The commissioning ASC must read and be very familiar with the required checks and instructions as outlined under *MDE-4226 Encore/Eclipse Installation Checklist (Form A)*. It is important that the commissioner clearly explains to the station manager or representative about the nature of problems uncovered in the checklists and the implications of not correcting them.

3 In *MDE-4228 Encore Commissioning Checklist (Form C)*, under “Model/Serial Numbers” there are check boxes for the commissioner to record whether he/she has obtained Form A and Form B for each unit being commissioned. The checklists must be left in the electronics cabinet of each individual dispenser by the installing contractor and start-up ASC.
The commissioner must obtain these forms and keep a copy of each as outlined on the individual forms with *MDE-4228 Encore Commissioning Checklist (Form C)* for the site. The signed forms must be retained by the commissioning ASC for a minimum of three years.

4 The commissioning ASC is not responsible for completing Form A and Form B for each commissioned unit, nor does the commissioning fee paid by Gilbarco Veeder-Root® cover such effort.

5 In case Form A or Form B are not complete or are missing for each unit, the commissioner must notify the station manager or representative about the requirement to have the checklists completed and sent to the commissioner. This ensures the warranty coverage with the proper and safe operation of the equipment.
Note: Unit problems created by improper installation or start-up are not covered within warranty.

After receiving the forms, the commissioner will retain the completed form with his/her commissioning records for the site. The ASC must carry Form A and Form B, in case if it is not present in the dispensers.

6 Gilbarco may request a copy of Form A, Form B, or Form C during commissioning “call in”, if a unit is reported as not passing inspection or not having been inspected per Form A or Form B.

7 Under “Required Signatures”, the station manager or representative must read and check either box A or B, and then sign Form C. When box B is appropriate, the commissioner must ensure that all details as covered under box B have been explained to the station manager or representative, before requesting his/her signature.

Sample Install, Start-up, and Commissioning Forms

Figure 3-1: Sample Installation Checklist (MDE-4226)

MDE-4226E Encore®/Eclipse® Installation Checklist
(Form A) · July 2013

GILBARCO VEEDER-ROOT

Site and Installer Information

SITE INFORMATION	INSTALLER INFORMATION
Unit Serial Number:	Company Name:
Site Name/Number:	Company Address:
Street:	Street:
City/State/Zip:	City/State/Zip:

INSTRUCTIONS: Complete this checklist for each pump/dispenser. Information regarding "Station Electrical" must be completed on only one pump/dispenser form. **After completing the form, leave the original in the pump/dispenser cabinet, away from the wiring or circuit boards.** Keep the copy for your records.

IMPORTANT NOTE: This checklist does not include all the requirements for the safe or proper installation of a pump/dispenser. However, this checklist includes certain checks for some of the most important items and items that are occasionally ignored during installation. The installer must read and follow the installation manual completely, for equipment installation.

Individual Pump/Dispenser Hydraulic/Mechanical-related

Item	Procedure	Check if OK/Complete
1	Where required, shear valves are installed as per the manufacturer instructions.	
2	Flexible pipe is not used within the dispenser. Flexible pipe that meets local and state codes can be used below the dispenser.[1]	
3	Pumps used only with above-ground tanks: Vacuum-actuated pressure regulating valve is installed at the inlet for each product.[1]	
4	Pumps with only in-ground tanks: Check valve is installed for each product inlet.[1]	
5	Hose breakaways are installed as per code and manufacturer recommendations.	
6	After the dispenser has dispensed Diesel Exhaust Fluid (DEF), power to the dispenser must be left on [when the outside temperature drops below 20 °F (-6.67 °C) and the unit is not being used].[2]	
7	All hanging hardware is Underwriters Laboratories (UL ®) or equivalently listed, CARB-certified[1] (where required) and checked for continuity.[1]	
8	Unit is anchored to the island or skid tank as per installation instructions.	
9	Vapor recovery piping has no traps or sags in dispenser sump.[1]	
10	Correct nozzles, piping, and brand panels are used for each grade.	
11	All code, regulatory agency or customer-specified safety warning signs, labels, or decals have been installed.	
12	Hose lengths beyond standard lengths are not being used without special retrievers. No more than 6 inches of the hose touches the ground with the nozzle in the nozzle boot for vacuum-assisted vapor recovery hoses.[1] Vapor balance hoses do not touch the ground.[1]	
13	Air has been purged from the lines using shear valve[1] for dispensers and nozzle for pumps. For DEF dispensers, purge at the nozzle.[2]	
14	Piping has been secured to the lower piping brace as per the installation instructions (Encore® only).[1]	
15	The unit and plumbing are checked for leaks. There are no leaks at the hanging hardware.	
16	Correct filters [M008007B010, M10316B010, and M08006B010 (gasoline) or M08007B030 (diesel)] manufactured by PetroClear® have been used for the Ecometer.[1]	
17	If converting directly from gasoline to ethanol blends or from diesel to gasoline, tanks and lines have been cleaned in accordance with "Important Considerations When Changing Fuel Types" found in *MDE-3985 Encore Installation Manual*.[1]	

[1] *Not applicable to DEF units.*
[2] *Applicable only to DEF units.*

Individual Pump/Dispenser Electrical-related

1	Units without factory-installed or factory kit junction boxes (J-boxes) only: Conduit into the electronic cabinet is potted as per the installation instructions and no unfilled openings exist between the electronics cabinet and the lower piping area.	
2	Conduit at the base of the unit is potted as per local, state and national codes, and as per *MDE-3985 Encore Installation Manual*.	
3	Conduit(s) to the electronic cabinet for non J-box equipped units is secured to the lower brace and potted as per *MDE-3985 Encore Installation Manual*.	
4	Field-installed J-box is secured to the lower frame.	
5	Twisted-pair (not shielded) wire is used for two-wire communication for newly wired systems.	
6	All wiring to the dispenser is stranded copper of gauge and insulation type specified in the installation instructions.	
7	Ground wires are properly connected at the pump/dispenser.	
8	Contractor supplied conduit and J-box are approved for hazardous locations involved and properly sized for wiring and number of connections as per the codes.	
9	All J-box covers are replaced, all bolts used are tightened, and all openings are plugged.	
10	All speaker, intercom, and other class 2 wiring are located in separate conduits and J-boxes from the power wiring.	

Station Electrical (Check this section only once per site)

1	Conduits to pumps/dispensers are potted as per the code requirement at the station building.	
2	Emergency power cutoff are installed as per code to remove all power from the fueling equipment in case of an emergency.	
3	Dispensers only: Submersible Turbine Pump (STP) isolation relays are used for all dispensers.	
4	Grounds to the dispensers are properly wired as per the codes and installation instructions.	
5	Properly sized circuit breakers for the units and unit options are involved.	
6	All pump/dispenser wiring is properly spaced and isolated from the wiring to electrically noisy devices as per *MDE-3985 Encore Installation Manual*.	
7	Power wiring to the pumps/dispensers are not shared by other devices.	
8	All circuit breakers to the pumps/dispensers are clearly labeled and readily accessible.	
9	All Distribution Boxes (D-Boxes) are clearly labeled showing the pump/dispenser connections.	
10	All pumps/dispensers are wired to the same phase of electrical power.	
11	New site wiring to pumps/dispensers was Megger®-tested. Old site wiring was continuity-tested and short-tested with a digital meter.	

Required Signature

Installer Name (Print):	Technician Number (if applicable):
Signature:	Date:

NOTE: This is a two-part form. Place the top copy (original) in the pump/dispenser cabinet, away from the wiring and circuit boards. Keep the bottom copy for your records.

Figure 3-2: Sample Start-up Checklist (MDE-4227)

MDE-4227C Encore®/Eclipse® Start-up Checklist
(Form B) · July 2013

GILBARCO VEEDER-ROOT

Site and Start-up Technician Information

SITE INFORMATION	START-UP TECHNICIAN INFORMATION
Unit Serial Number:	Company Name:
Site Name/Number:	Company Address:
Street:	Street:
City/State/Zip:	City/State/Zip:

INSTRUCTIONS: Before powering the unit, the start-up technician/Authorized Service Contractor (ASC) should verify that the unit has been inspected as per *MDE-4226 Encore®/Eclipse® Installation Checklist (Form A)*. Keep the copy of this form for your records and place the original in the unit. The commissioning technician will use this form and *MDE-4226 Encore/Eclipse Installation Checklist (Form A)* to complete commissioning. Depending upon pre-agreement with the installer, calibration, or VaporVac® performance testing may be done by the installer.

IMPORTANT NOTE: This checklist does not include all the requirements for a complete start-up of a pump/dispenser. However, included are certain checks for some of the most important items and items occasionally missed during start-up. The start-up technician must read and completely follow *MDE-3985 Encore Installation Manual* for the equipment and have a full knowledge of the requirements of the station and the units involved. Incomplete forms may delay commissioning.

Start-up Checklist

Item	Procedure	Check if OK/Complete
1	The installation checklist has been completed. Refer to Installation Checklist *[MDE-4226 Encore/Eclipse Installation Checklist (Form A)]*.	
2	Unit is programmed to station requirements.	
3	Backlights, brand panel lights, etc., are all functional.	
4	Proper communication with the Point of Sale (POS) is established.	
5	Push-to-Start and Push-to-Stop buttons are functional.	
6	Pump/dispenser handles and grade select buttons activate the correct Submersible Turbine Pump (STP).	
7	Price Per Units (PPUs) set by the POS match brand panels. PPU for the grade in operation is not blanked when the handle is raised.	
8	Blender only: Blend ratios set as per instructions from site representative.	
9	Flow rates do not exceed the regulations or codes.	
10	Verify meter calibration. Recalibrate as necessary (refer to INSTRUCTIONS).	
11	Card Reader IN Dispenser (CRIND®) only: CRIND displays, contactless card readers, cash acceptors, scanners, and printers pass diagnostic card testing.	
12	Cash Acceptor only: Cash acceptor door switches activate alarms as installed.	
13	VaporVac units only: VaporVac passes A/L tests (refer to INSTRUCTIONS).	
14	Automatic Temperature Compensation (ATC) equipped units only: Code 303 verifies that each meter temperature is being measured.	
15	Transmitter/Receiver IN Dispenser (TRIND®) only: Proper response obtained with hand-held and vehicular-mount transponder testers depending upon the unit equipment.	
16	Pulser shipping tie-wraps have been removed.	

Required Signature

Start-up Technician Name (Print):	Technician Number (if applicable):
Signature:	Date:

Note: *This is a two-part form. Place the top (original) copy in the pump/dispenser away from the wiring and circuit boards and keep the bottom copy for your records. An incomplete or missing form may delay commissioning of the unit/site and may void warranty.*

Figure 3-3: Sample Commissioning Checklist (MDE-4228)

MDE-4228L Encore®/Eclipse® Commissioning Checklist
(Form C) · July 2013

GILBARCO VEEDER-ROOT

Site Information

Commissioning SR #

SITE INFORMATION	
Site Name/Number:	Street:
Phone Number:	City/State/Zip:
Passport™ Only	Software and Service Pack Version:

INSTRUCTIONS: This form is to be completed by the commissioner. The commissioner is defined as a person with the appropriate Gilbarco® training, usually an Authorized Service Contractor (ASC), necessary to complete the commissioning process and train station personnel. *MDE-4226 Encore®/Eclipse® Installation Checklist (Form A)* and *MDE-4227 Encore/Eclipse Start-up Checklist (Form B)* are prerequisites to this form. The commissioner should retrieve the original and a copy of Form A and Form B from the dispensers/pumps. Give the copy to the station representative and save the original for your records. Keep the original of this commissioning form, Form A, and Form B in your records for three years. The commissioner will discuss results of all the checklists with the station manager/supervisor.

IMPORTANT NOTE: The commissioning ASC should verify that the units have been inspected per the Installation Checklist (Form A) and the Start-up Checklist (Form B). Failure to complete all forms may delay commissioning completion and may affect warranty. In case all forms are not complete or available, the commissioner must explain the options for completing commissioning to the station owner.

COMMISSIONING: The commissioner should have all of the required forms and signatures ready, and call **1-888-800-7498** to report commissioning. Log the SR # at the top right of form. Do not mail commissioning form. Information is taken over the phone.

POS Type and Name: ______________________________

Model/Serial Numbers

Item	Check if Received		Model Number	Serial Number	Baud Rate	Biodiesel or E85? (Y/N)	Software Versions		
	Form A	Form B					Pump/Dispenser	CRIND®	Door Node*
1									
2									
3									
4									
5									
6									
7									
8									

**Door Node information is applicable only for Encore 500 and Eclipse units.*

Blend Ratio Programming

***ONLY REQUIRED IF SPECIFIED BY THE CUSTOMER*. Complete for blenders only. Customer representative must sign.**

Grade Name	Low Product Name	High Product Name	% of Low Product

Customer representative's signature indicates blend ratios listed are same as those provided to the ASC.

Customer Representative	

Individual Unit Inspections

Item	Procedure	Check if OK/Complete
1	Installation and start-up checklists have been completed and results reviewed with station manager/supervisor.	
2	All piping is secured to lower piping brace for Encore dispensers/pumps.	
3	Units without factory installed or factory kit junction boxes (J-box) only: Conduits into electronic cabinet are potted and no unfilled openings exist between the electronics cabinet and the lower piping area.	
4	Minimum wire gauges meet Gilbarco field wiring diagram specifications at pump/dispenser entry.	
5	All wiring is stranded, gas and oil resistant, Ethernet®, and Digital Subscriber Line (DSL) cables meet requirements per *MDE-3985 Encore Installation Manual*.	
6	Twisted-pair wiring used for two-wire data lines for stations with new wiring.	
7	No leaks in hydraulic assembly, hose outlets, and hanging hardware.	
8	Units with J-box have all J-box covers and bolts installed.	
9	(Encore 500/500 S/Eclipse only) Check software versions and record versions above. If not latest production released versions, install (download from laptop/PC) and generate a separate SR # by calling Gilbarco Call Center at 1-888-800-7498.	

Station Requirements (including training for station manager/supervisor)

Item	Procedure	Check if OK/Complete
1	All dispenser/pump wiring properly spaced and isolated from wiring to electrically noisy devices per *MDE-3802 Encore and Eclipse Site Preparation Manual*.	
2	Breakers marked with dispenser ID numbers.	
3	A copy of the unit warranty statement and the policy has been reviewed with and given to the site owner/manager.	
4	Dispenser emergency shutdown procedures discussed per *MDE-3893 Encore and Eclipse Series Owner's Manual*.	
5	Reviewed by owner/manager for retrieving and reconciling pump totals using manager keypad.	
6	Reviewed by owner/manager for price setting procedures using Point of Sale (POS) and manager keypad.	
7	Reviewed by owner/manager for when and how to switch dispensers to standalone operation.	
8	Reviewed by owner/manager for Encore 500/Eclipse V1.8.00 and later software units, security procedures for station approval of programming access by service technicians.	
9	Advised owner/manager that nondefault PIN (access) codes are to be installed as part of commissioning to enhance security after explaining security benefits and that lost-code additional-service labor will not be covered by warranty. If station refuses to allow nondefault PIN codes to be used, do not modify codes and do not check box to the right (will not invalidate warranty).	
10	Card Reader IN Dispenser (CRIND) device only: Provided Universal Serial Bus (USB) Printer Maintenance Cards to manager.	
11	CRIND device only: Reviewed by owner/manager/cashiers and applicable maintenance personnel for paper/jam removal procedures, specific paper requirements, and how to order and replace CRIND paper.	
12	Reviewed by owner/manager and applicable maintenance personnel for periodic maintenance inspections (leaks, rust, hanging hardware, belts, etc.) and recommended frequency (refer to *MDE-3893 Encore and Eclipse Series Owner's Manual*).	
13	Station owner/manager was given an Owner's Manual (*MDE-3893 Encore and Eclipse Series Owner's Manual*) and received explanation of information in the manual.	
14	Station owner/manager was given an Owner's Programming Manual (*MDE-4732 Encore and Eclipse Series Owner's Programming Manual*), received explanation of information in the manual, and advised on keeping programming information secure.	
15	Station owner/manager received information about noncompliance that can affect warranty or station safety when all forms were not completed or problems were discovered.	

Required Signatures

Station Manager Information (station manager must check one of the following and sign)		
A ☐ I have been advised that all dispensers have been inspected per the checklists and no problems were found needing correction.	**B** ☐ I have been advised that either some equipment inspections have not been completed or problems requiring correction were found during inspections. Should the equipment not have been properly installed, it may be subject to one or more of the following problems: functions improperly, fails prematurely, creates a safety hazard, and not covered by warranty. I have further been given information regarding inspections or corrections required. I have also been advised on how to correct these situations and provide results to the commissioning ASC to ensure warranty coverage.	
Station Manager Name (print):	Signature:	Date:
Commissioning ASC Name:		
ASC Technician Number:	Signature:	Date:

***Note**: This is a two-part form. Top (original) - Keep for your records. Bottom sheet - Give to station manager.*

Installation, Start-up, and Commissioning Checklists

Refer to the following manuals:
- *MDE-4897 Installation Checklist for Diesel Exhaust Fluid (DEF) Units (Form A)*
- *MDE-4898 Start-up Checklist for Diesel Exhaust Fluid (DEF) Units (Form B)*
- *MDE-4899 Commissioning Checklist for Diesel Exhaust Fluid (DEF) Units (Form C)*

Start-up and Commissioning

Installation Start-up Information - DEF

Dispensers can be supplied individually or as part of a complete aboveground tank system premounted to the tank platform. If shipped as a complete system with tank, for any issues regarding the tank system, platform, mounting of the dispenser on the platform, and wiring or plumbing to the dispenser, installers must contact the tank manufacturer. Gilbarco Warranty and Technical Services only support the dispensers for complete systems shipped with tank and attached tank and pump system. The installer/start-up technician must be aware of the following information about installations:
- Calibration procedures are identical for Encore S units with electronic calibration.
- To produce flow through the nozzle, a test nozzle Ring Magnet (M10656B001) must be used.
- Install a standard Encore S pulser to the Pump Control Node (PCN) when purging air to the unit for the first time as the meter is sensitive to aeration of the fluid. For additional information, refer to "Purging Air from the System" on page 7-7.
- Some tank systems have no air separation system with the pump. Air mixed with fluid will cause the meter to be erratic during calibration testing/setting. Before suspecting issues with the meter, investigate air leaks to the inlet of the pump.
- During cold weather, power to the dispenser and tank system must always be maintained after DEF is delivered and dispensed through the dispenser. This is done to avoid severe damage to the dispenser and potentially the tank system, regardless of whether the dispenser is in use or not. The dispenser has provisions to lock the nozzle.
- DEF is non-flammable or explosive. In general, other dispensers operate in hazardous locations with hazardous fuels. Also, consider the proximity of the DEF dispenser to those pumps/dispensers with hazardous fuel, if certain normal dispenser installation requirements are not followed. For any relaxation of installation requirements, such as potting, contact local authorities/regulators. When in doubt, install as per normal dispenser guidelines for hazardous locations and fuels. If a DEF dispenser is moved or a fueling pump/dispenser is installed such that the DEF dispenser is now in a hazardous area, it may be required to upgrade the installation.
- If a unit is frozen, a Gilbarco ASC must investigate for potential damage to the dispenser and make appropriate repairs. Freeze damage caused by not maintaining power to the dispenser during cold weather is not covered within warranty. For additional information, refer to "Error Codes" on page 8-20.

4 – Preventive Maintenance and Inspection

About This Section

This section provides information on preventive maintenance and associated inspections.

Topics in This Section

Section Overview

Preventive maintenance is essential to prevent unexpected downtime of units. The "Suggested Maintenance Chart" on page 4-2 provides maintenance recommendations. The "Sample Maintenance Checklist" on page 4-4 shows a sample of site survey/preventive maintenance checklist, a valuable tool for recording site configuration information and tracking preventive maintenance action taken at the site.

How to Use This Section

Discuss the suggested maintenance items listed in "Suggested Maintenance Chart" on page 4-2 with the station owner. To familiarize yourself with preventive maintenance tasks that you may have to perform at the site, review "Suggested Maintenance Chart" on page 4-2.

Preventive Maintenance Checklists

Suggested Maintenance Chart

For detailed information, refer to the specific item in the component section. Discuss this information with the station owner. Remind the owner about the importance of preventive maintenance and inspections.

IMPORTANT INFORMATION

Check maintenance instructions for third-party component manufacturers (hoses, nozzles, shear valves, and so on) for their maintenance recommendations. Comply to whatever recommendation is more stringent relative to inspection frequency and ensure that inspection routines include their requirements as well.

WARNING

In the event of inclement weather, including snow, ice, or flooding that makes driving conditions dangerous, please avoid servicing units. Always use available door stops to secure upper doors against unwanted/unexpected movement, especially during high winds. If necessary, reschedule service to avoid damage to the equipment. Weather may change unexpectedly, be aware of local weather conditions. During service, if conditions develop making service unsafe, close the unit(s) and proceed to a safe location.

Item	Inspection/Procedure	Frequency
General inspection	Check all areas of pump/dispenser for damage or sharp edges. Replace any missing warning labels.	Once a week or on receiving a complaint.
Hoses	Check for leaks and damage. Check hose covering for weakness, splits, cuts, bulges, flattened spots, wet spots, soft spots, and reinforcement showing. Look for holes and cuts in vapor recovery hoses. Refer to "Hoses and Flow Restrictors" on page 7-26.	
Hose retrievers	Check for frayed or broken cables, and cables wrapped around hoses. Refer to "Testing Hose and Hanging Hardware Continuity" on page 7-35. Check if retractors are operating properly.	
Leaks, external	Check all hoses, nozzles, swivels, couplings, casting, and so on. Repair immediately if leak is found.	
Nozzles and boot areas	Check for leaks, missing parts, and damage. Check vapor recovery boots (bellows) for proper seal and damage. Check VaporVac nozzle spouts for blockage, loose spouts, and damage (that is, dents). Refer to "Adjusting Nozzle Hook for Specific Nozzles" on page 7-37.	
Breakaways	Check for secure connection to hose, and for any leaks or wet spots. Refer to "Breakaways" on page 7-28. *Note: Some manufacturer's breakaways are resettable and others require replacement after a drive-off. For information, refer to manufacturer's documentation.*	Once a week or after drive-offs.
Leaks, internal	Check all hydraulic connections and seals including meters, valves, and so on. Repair immediately if wetness or dripping fuel is found. Some staining of parts around seals is normal. Monitor these places more closely.	Once a month, or on receiving a complaint, or after drive-offs.

Item	Inspection/Procedure	Frequency
Filter change and strainer cleaning	For detailed instructions, refer to "Servicing Filters" on page 7-20. Follow all safety precautions. Frequency of filter changes will depend on fuel and installation cleanliness. Corroded tanks, poor quality fuel, and initial conversion to alcohol-enhanced fuels may require more frequent filter changes and strainer cleanings.	Once a month or 50,000 gallons after new installs. Every 300,000 gallons or six months thereafter.
Wash unit	For detailed instructions, refer to *MDE-3893 Encore and Eclipse Series Owner's Manual.*	Once a month or as required.
Wax units	For detailed instructions, refer to *MDE-3893 Encore and Eclipse Series Owner's Manual*.	Once a month (in harsh environments) or every six months.
Stainless steel sheathing and lower doors	For stains on stainless steel, use a cleaner specifically formulated for cleaning stainless steel. Gilbarco recommends Bar Keepers Friend®. Ensure that you thoroughly rinse off the cleaner. Do not spray water/cleaner directly at or into card readers, cash acceptors, or printer chutes.	Once every three months (once a month in harsh environments).
Door locks Meters Nozzles, hooks/shafts Pumps, pulleys, belts, and belt tension Shear valves	Lubricate with a good grade of lock oil. Check calibration and calibrate, if required. Refer to "About Calibrating Meters" on page 7-18. Check operation. Clean and lubricate with silicone grease, if required. Check belts for fraying/cracks. Check pulleys for excessive wear in grooves and excessive bearing play. There must not be more than 1-inch of play on either side of the belt. Refer to "Belts and Pulleys" on page 7-66. Check valve operation and lubricate. Refer to "Shear Valves" on page 7-75.	Every six months.
CRIND Card Reader	Clean the card reader using Cleaner Card (Q11482).	Once a week or on receiving a complaint.
CRIND Printer	Clean print head using Printer Cleaning Card (Q13400).	Every three months or on receiving a complaint.
Display Backlight bulbs and Eclipse ad panel bulbs	Check for bad bulbs and replace, as required.	Once a month.
Displays	Check and repair, as required.	Once a week or on receiving a complaint.
Underground piping for sites using VaporVac system	Perform a static pressure decay test. Refer to "Common Vapor Recovery Testing Procedures" on page 7-81.	Once a year.
DEF spills	DEF is slippery and mildly corrosive. Clean with water in case of DEF spills.	On occurrence.

Sample Maintenance Checklist

Figure 4-1: Sample Maintenance Checklist

GILBARCO VEEDER-ROOT

SITE INSPECTION/PREVENTIVE MAINTENANCE CHECKLIST

Customer's Name ______
Manager's Name ______
Street Address ______
City ______
Phone# (   )______
Type of Vapor Recovery System Installed ______

Store # ______
Date of Service ______
✓ = OK or Service Performed   "O" = Broken, Missing, Etc.
Explain All Items Marked "O" In Space Provided
Type of POS System Installed: Passport™   G-SITE®   TS-1000™
Other POS System ______

POS PREVENTIVE MAINTENANCE

Workstation #	1	2	3	4	5	6	7	8
1. Clean Printer(s)								
2. Remove Printer Paper Dust								
3. Check Printer Operation								
4. Check for Dedicated Isolated Circuit								
5. Lubricate Cash Drawer Assembly (per AGP spec)								
6. Check Condition of Cables and Power Cords (cuts/frays/bends)								
7. Check Condition of Plugs								
8. Clean Keyboard Case (G-SITE & TS-1000)								
9. Clean & Dust Console & Surrounding Area								
10. Clean Card Reader with Cleaning Card Q12534-170								
11. Check Hard Drive & Note Bad Sectors (G-SITE only)								
12. Clean Floppy Drive with Q13476-101 (see instructions)								
13. Check Presence & Condition of R20391-01 FD Dust Cover (486 & Pentium® G-SITE only)								
14. Vacuum (not blow) Power Supply Air Inlet & Front Bezel								
15. Degauss Monitor (Passport only)								
16. Check UPS Battery (Passport only)								

Workstation #1 Model # ______ Serial # ______
Workstation #2 Model # ______ Serial # ______
Workstation #3 Model # ______ Serial # ______
Workstation #4 Model # ______ Serial # ______
Workstation #5 Model # ______ Serial # ______
Workstation #6 Model # ______ Serial # ______
Workstation #7 Model # ______ Serial # ______
Workstation #8 Model # ______ Serial # ______

PUMP, DISPENSER & CRIND® PREVENTIVE MAINTENANCE

Dispenser #	1	2	3	4	5	6	7	8	9	10	11	12
1. Check Hoses*												
2. Check Nozzles (including Connection to Pumps/Dispensers), Retaining Springs & Swivels*												
3. Check Breakaway Couplings*												
4. Check Condition of Hose Retrievers (if used)												
5. Check Lamps & Lenses*												
6. Check for Exterior Damage/Rust*												
7. Check Operation/Clean Displays												
8. Lubricate All Locks												
9. Ensure Unit Is Bolted To Island**												
10. Check Bezel & PPU Gaskets												
11. Check Shear Valve Operation, Mounting & Lube												
12. Check Plumbing for Weeps/Leaks												
13. Check for Install Filter Instruction Tag, and Install if Tag is Not Present (see MDE-2215)												
14. Check & Lubricate All Linkages												
15. Ensure Calibration Seals Are Intact												
16. Check Junction Box Bolts/Plugs												
17. Check Pump Handle Switch												
18. Check Vapor Recovery Plumbing**												
19. Check Vapor Recovery Bellows/Hoses for Holes/Cuts/Cracks												
20. Verify that Vapor Recovery System Back Pressure and/or Efficiency has been Tested per Regulatory Agencies												
21. Check Belts On Pumps												
22. Ensure Electrical Panel Circuits Are Clearly Marked												
23. Ensure Free Access To Emergency Cut-off Switch												
24. Check CRIND for Correct Installation & Type of Paper												
25. Check Operation/Clean Card Reader (use Q11482 Cleaning Card, see MDE-3562)												
26. Check Operation/Clean Keypad												
27. Perform CRIND Diagnostics (see MDE-3562)												
28. Verify that Safety Information is Intact/Legible												

Pump #1 Type ______ Model # ______ Serial # ______
Pump #2 Type ______ Model # ______ Serial # ______
Pump #3 Type ______ Model # ______ Serial # ______
Pump #4 Type ______ Model # ______ Serial # ______
Pump #5 Type ______ Model # ______ Serial # ______
Pump #6 Type ______ Model # ______ Serial # ______
Pump #7 Type ______ Model # ______ Serial # ______
Pump #8 Type ______ Model # ______ Serial # ______
Pump #9 Type ______ Model # ______ Serial # ______
Pump #10 Type ______ Model # ______ Serial # ______
Pump #11 Type ______ Model # ______ Serial # ______
Pump #12 Type ______ Model # ______ Serial # ______

* Notify Dealer
** Only perform this task if a new pump or dispenser has been installed, or if an existing pump or dispenser was removed and replaced

Note any problems w/equipment ______

I certify that all inspections and maintenance items have been performed and problems noted and explained to the Station Manager:

INCIDENT# ______ Date ______ Station Manager ______ Date ______

ASC# ______ ASC Name ______ Tech. Name ______ Tech # ______

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■ White — Contractor Copy ■ Yellow — Customer Copy — Mail to Maintenance Supervisor Weekly ■ Pink — Site Copy ■ Golden Rod — Mail to Gilbarco Inc. F-53

5 – Pump Programming

About This Section

This section provides instructions for programming Encore and Eclipse series dispenser pump functions from the manager keypad using only the manager keypad and main transaction along with the grade Price per Unit (PPU) Liquid Crystal Display (LCDs) for displaying programming information.

Topics in This Section

General Programming Information

Terms Used in This Section

Term	Definition
6 X 6 X 4 Display	Standard display unit for the U.S. Market. It is a six-digit money, six-digit volume, and a four-digit PPU display.
Allocation	Maximum amount of fuel that can be dispensed from a nozzle for a given transaction.
ASC	Authorized Service Contractor
ATC	Automatic Temperature Compensation
Coldstart	Clears pump and configuration data
Cold Coldstart	Encore 300 procedure that performs a coldstart plus clears certain information (WatchDog timer state and so on) not cleared by a normal coldstart. *Note: Not commonly performed.*
Command Code (CC)	Listing under Command Level for setting software parameters.
Command Level	A level of programming accessed by PIN(s)
Dispenser	In this document, “dispenser” is used interchangeably to describe a dispensing unit.
Display Blanking	Function turns all displays “ON” or “OFF” on side of the dispenser. If the displays are blank, then a transaction is not allowed.
DLT	Display Last Transaction
Error Code (EC)	A number that identifies an error condition reported by the software.

Term	Definition
Five Button Preset (5-button Preset)	This is a keypad that contains five buttons using which the customer can enter a predetermined value and/or increment or decrement the selected sale or volume value before fueling.
Firmware	Basic unit/component operational instructions contained and sold on a hardware electronic chip.
Function Code (FC)	One or more procedures within a CC.
Grade	Indicates the fuel grade available at the nozzle. In case of blender dispensers, a grade is created by blending products.
LCD	Liquid Crystal Display
LSD	Least Significant Digit
Manager Keypad	A device where programming and access of other unit information such as totals can be accessed.
Money Display	A six-digit display that during normal operation shows the total value of the fuel dispensed.
MSD	Most Significant Digit
Option Code	A choice that is available after entering a FC.
PIN	Personal Identification Number is a Security PIN Code used to access the programming.
Programmable Pump Preset (PPP)	This option is used to set up specific volume of fuel or dollar amounts to be dispensed using specific keypad buttons.
PPU Display	Display that contains the PPU of the associated grade.
Product	Indicates the base fuel grade available in the storage tank. Not to be confused with “Grade”. However, in a non-blending environment, a grade is same as the product.
Prover Can	A calibrated can used by W&M officials and calibrating technicians to accurately check or calibrate meters in a fuel dispenser.
Software	Basic unit/component operational instructions downloaded from a laptop or other source, and stored within the dispenser.
Side 1	Indicates the electrical access side of the dispenser. The manager keypad is accessed from side 1 of the dispenser.
Volume Display	A six-digit display that shows the total volume of product dispensed during normal operation.
VUT	Volume Unit Type (liters or US gallons)
Warmstart	Accomplished by powering down the dispenser, repowering, and exiting programming modes. Resets the unit processors.

Programming Mode

Dispenser programming mode can only be entered when both sides of dispenser are in an idle mode. Nozzles must be in nozzle boots and no grade select buttons can be pressed. Dispenser programming results in unit being placed offline with regard to external communications. To start programming the dispenser, press **F1** on the manager keypad.

The following sub-sections describe general dispenser operation while in the programming mode.

Programming Levels

There are three basic programming levels associated with Encore 300, and four programming levels with Encore 500/S/700 S and Eclipse units.

For Encore 500/S/700 S, entry into a security level allows use of all the programming features within that security level and below. For Encore 300, entry into a security level allows use of all the programming features within that security level and not below. Thus, if the configuration Level 4 Security PIN Code is entered, all configuration, Level 3, Level 2, and Level 1 CCs are allowed.

⚠ WARNING

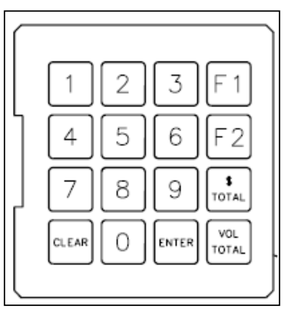
Surfaces of Eclipse series units are non-metallic. Magnetic-backed keypads will not adhere to Eclipse unit surfaces as they will to others. In any situation where power is on, and gasoline and its vapors are present or potentially present (for example, calibration), do not attempt to use the manager keypad with electronics cabinet open.

Fuel and its vapors may ignite, leading to serious injury or death.

For Eclipse units, run the keypad cable over door and close door, allowing keypad to be suspended on unit and accessed from outside, for all procedures performed with power applied or where there is a possibility of fuel or fuel vapor being present.

Manager Keypad Layout

Figure 5-1: Manager Keypad Layout



Manager Keypad Key Definitions and Use

Key(s)	Definition/Use
0-9	Numeric values
F1	Function 1 – Used to start dispenser programming and sequence among programming and FCs. In general, pressing **F1** will take the user back to a previous programming function selection. Pressing **F1** from the normal dispenser state will always initiate programming mode. *Note: Displays are always activated during programming mode.*
F2	Function 2 – Used to exit programming mode and return to normal mode. A unit warmstart also occurs after pressing **F2** while in the programming mode.
$Totals	Money Totals – Used to display money totals by side and grade. This key does not require a Security PIN Code. CLEAR key is used to exit money totals mode.
Vol. Total	Volume Totals – Used to display volume totals by side and grade. This key does not require a Security PIN Code. CLEAR key is used to exit volume totals mode.

Key(s)	Definition/Use
ENTER	Value entry keys – sends the entered value to pump. *Note: If you do not press ENTER during programming, that particular programming step will not be saved.*
CLEAR	CLEAR key – Used to clear last keypad entry, exit money, and volume total mode.

Display Conventions

Programming Convention

The programming digit positions for main money and volume displays are shown in Figure 5-2. This applies even to cases where more than six display digits are available for display. Information is displayed on all grade PPUs when required and restricted to digits “4” through “1”, unless otherwise noted.

Figure 5-2: Programming Digit Positions for Main Money and Volume Displays

Main Display $	6	5	4	3	2	1
Main Display V	6	5	4	3	2	1
Grade 1 PPU			4	3	2	1

6 Digit Money/Volume and 4 Digit PPU Layout

Main Display $	6	5	4	3	2	1
Main Display V	6	5	4	3	2	1
Grade 1 PPU		5	4	3	2	1

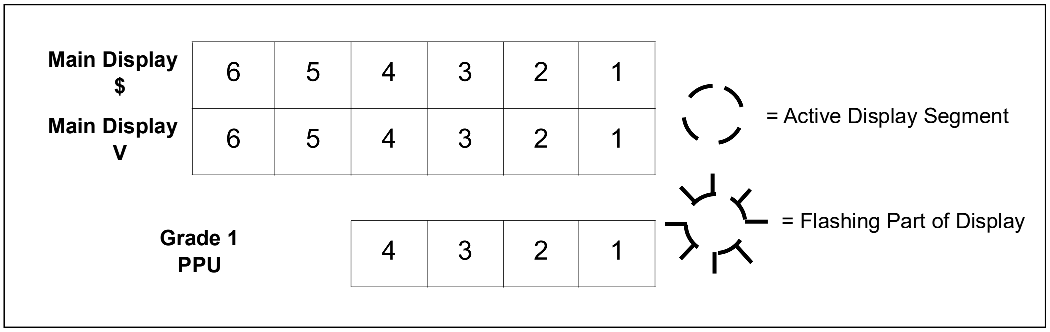
6 Digit Money/Volume and 5 Digit PPU Layout

Main Display $	7	6	5	4	3	2	1
Main Display V	7	6	5	4	3	2	1
Grade 1 PPU			5	4	3	2	1

7 Digit Money/Volume and 5 Digit PPU Layout

Note: Note that 7th digit programming guides demonstrate 6 digit money/volume and 4 digit PPU. The 7th digit of money/volume is always unused and remains blank (‘ ’) while the 5th PPU digit is only used for manual price entry (See Encore 500 Programming).

Figure 5-3: Programming Digit Positions on Displays



During programming, CC is always shown left justified in the main money display, starting at digit position “6”. Other display information is dependent on specific programming CCs and FCs within the CC. The flashing field on the LCD display is shown in large bold italic font (for example, **2**).

Programming parameters are shown in LCD displays as soon as parameter selection key is pressed. Selected parameters continue flashing until you press either **ENTER** or **F2**. Parameters are entered into the memory only after you press **ENTER**.

Locating Specific Programming Information

Specific programming information related to product series begins on the following pages:

For Information on	Go to	Page
Encore 300 Series	Encore 300 Programming Pump/Dispenser	5-7
Encore 500/S/700 S and Eclipse Series	Programming Encore 500/S/700 S and Eclipse Series Units	5-39

Firmware/Software Versions

Encore 300 Firmware Versions

	Version	
Unit Type	**With Digital Valves**	**With Proportional Valves**
MPD/Dual/Quad/Ultra High Frequency (UHF)	20.X.X	10.X.X
MPD/Dual/Quad Special Export Version	21.X.XX	11.X.XX
Single-hose MPD	22.X.X	12.X.X
Single Blender	25.X.X	15.X.X
Multi (6) Blender	27.X.X	17.X.X
ATC Option (ATC Controller)	30.X.X	30.X.X

Programming Errors

If data entered (CC, FC, and Parameter) is invalid, data field will go blank for two seconds and a double beep will sound.

General Programming Operation

On pressing **F1**, dispenser will go offline and all main display LCDs will blank out and money display will show flashing 8888. To begin programming, you have to enter a four-digit Security PIN Code for the required security level. As each digit of Security PIN Code is entered, a dash (-) will be displayed on the main money display for that digit. After all security digits are entered, press and release **ENTER**. If Security PIN Code is accepted, main money display will change to flashing 0000. Otherwise, money display will show flashing 8888 to indicate an invalid Security PIN Code.

Note: To begin programming, you must enter the four-digit Security PIN Code.

If a valid Security PIN Code is entered, user may then enter CC and proceed with dispenser programming as described in the sections that follow. If an invalid Security PIN Code is entered, user must simply re-enter the code and press **ENTER** again. Otherwise, user may press **F2** to exit programming mode.

After a valid Security PIN Code is entered, display will show 0000 until first digit of a CC is entered. After entering the first digit, display will blank out showing the entered digit as flashing and left justified to position 6. As each new digit is entered, digits flash and follow in positions 5, 4, and so on. When a complete CC is entered, it will flash until you press **ENTER**.

After a programming CC is activated, the default or first selectable entry for that mode will be shown as flashing. Defaults and further keypad entries will show as flashing digits and will update display as they are keyed in. This action continues until you press **ENTER**. After you press **ENTER,** the next programming field will start flashing indicating an operator action is required. This keypad/display functionality will continue until you exit the programming mode.

Notes: 1) It is important to note that each programming level utilizes a Security PIN Code as shown in the following table. FCs within a CC begins at 1.

2) Entering the highest level Security PIN Code for Encore 500/S/700 S and Eclipse series units will grant you access to all the lower level Security PIN Codes.

Entry into a CC will present data as the default value or last programmed values for that CC. Only one CC and FC/Parameter may be programmed/changed at a time.

The following table shows currently allocated Security and CCs that have been assigned for use in Encore and Eclipse. These PIN Codes can be programmer modified as described later.

Note: Gilbarco recommends that Security PIN Codes be changed after installation of pumps and dispensers.

Programming Level	Default Security PIN Code	Encore 300 CC Range	Encore 500/S/700 S CC Range
Level 1	2222	1-9	20-39
Level 2	1503	10-12	40-69
Level 3	1309	13-17	70-89
Level 4*	0128	N/A	84-99

**Only on Encore 500/S/700 S and Eclipse units.*

The following sub-sections describe programming commands for each programming level.

Important Information About Using V1.7.58 and Later for Encore 500/S/700 S and Eclipse Dispensing Pump Software

As the calibration mode is automatically entered when the W&M switch is toggled for these versions of software to program Level 3 and 4 commands (70 and higher), a different approach must be used. Earlier, it was possible to toggle the W&M switch before entering the programming mode. Now, you must be in Level 3 or 4 before toggling the switch. Follow the instructions provided at the beginning of Level 3 and Level 4 programming for this version of software.

The calibration procedure assumes that either a 5 gallon or 20 liter prover can will be used. If another size can is used, you must change the default can size in CC82 as the new can size will be stored in memory for future calibrations. If the special can size is not likely to be used for the next calibration, it is very important to reprogram the can size back to default after completing the calibration. It is also wise to verify the can size on the money display during calibration.

EC26 will not clear automatically after a unit has been first calibrated. A warmstart is required now. This will be primarily noted after installing a new service board and calibrating or performing a true coldstart (drop back to V1.3.74 pump node software and then reload to the current version), where recalibration is required.

When performing combined verification test and recalibration, electronic and electromechanical totals will increase as during a normal sale.

Encore 300 Programming Pump/Dispenser

Contents in This Sub-section

Encore 300 Programming Quick Reference

CCs and FCs Quick Reference Chart

The following chart shows CCs and FCs for different levels. For more details, refer to the following pages:

Level One CCs (refer to page 5-9)

- 1 Programming PPU Mode
- 2 Setting Mode of Operation
- 3 Programming Volume Allocation
- 4 Blanking Display Manually and Selecting Money/Volume Preset
  - 1 Manual Blank Displays
  - 2 Five Button Preset
- 6 Clearing Memory for EC31 and 35 only
- 7 Setting Totals Input Mode
- 8 Displaying Software Version Number

Level Two CCs (refer to page 5-17)

- 10 Configuring The System
  - 1 Unit Type
  - 2 Money/PPU Decimal Total
  - 3 Volume/Calculated PPU Decimal Point
  - 4 Displayed PPU Decimal Point/Conversion Factor
  - 5 STP Preset
  - 6 Beeper (ON/OFF)
  - 7 Not used in this version
  - 8 Two-wire Display Blank Option
  - 9 Manual Display Blank Option/5 or 6 Digit Cash
  - 10 Decimal/Comma
  - 11 Preselect Timeout
  - 12 Pump Timeout
  - 13 Slowdown (Ultra-High Only)
  - 14 Push to Start (Blenders Only)
  - 15 Zero PPU Option
  - 16 Beeper Timeout
  - 17 Stop Control
  - 18 Zero Previous Transaction
  - 19 Lamp Test
  - 20 Programmable Pump Preset Value
  - 21 PPU Blinking Option
  - 22 Volume Decimal Point, MPD, Single Hose only
  - 23 Program STP Control
  - 24 Hose Pressurization Options
- 11 Not used
- 12 Programming Pumps
  - 1 Two-wire ID
  - 2 Cash/Credit at Pump
  - 3 Side Exists
  - 4 PPU Options

Level Three CCs (refer to page 5-26)

- 13 Changing PIN
- 14 Not used
- 15 Various Timeouts
- 16 Programming Blend Ratios
- 17 Entering Conversion Factors/Performing Master Resets
  - 1 Conversion Factor
  - 2 Master Reset

Encore 300 Programming and Data Access Instructions

This section contains information and steps to program the Encore 300 pump/dispenser. Certain programming levels will also allow you to view pump/dispenser data.

Note: For a simplified listing of functions and settings, refer to MDE-4039 Encore 300 Programming Quick Reference Guide.

⚠ CAUTION

When any work activity needs to be performed outside the scope of “normal” operation (SW download, terminal activation, etc), you must disconnect all communication cables (CAT5/6) to the POS to prevent possible interference and related issues when performing the required task(s).

Level One CCs

To access Level One CCs, proceed as follows:

1. Press **F1** on the manager keypad.
2. Enter your PIN (default 2222).

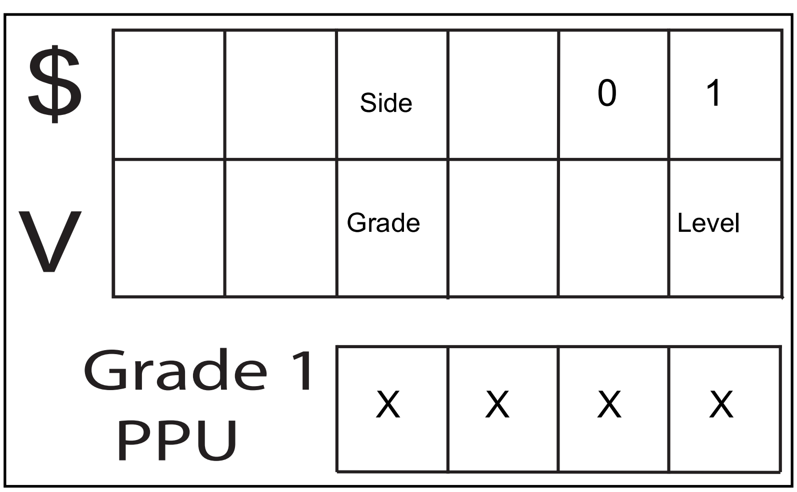
To access a different Level One CC (from any CC), proceed as follows:

1. Press **F1** on the manager keypad.
2. Enter the CC number to access.
3. Press **ENTER**.

CC1: Programming PPU Mode

Program the PPU for each side, grade, and price level (Cash/Credit).

Figure 5-4: Programming PPU Mode



1 Press **1** and then press **ENTER** on the manager keypad to access this CC.

2 Select side and press **ENTER**.

To Select Side	Press Configuration
Side 1 [side with Junction Box (J-box)]	1
Side 2	2
Both sides	3

3 Select Grade and press **ENTER**. Price level number flashes.
Note: The number of hoses/grades vary for each model.

Dispenser Type	Grade Numbers
MPD	1, 2, 3
MPD 4	Not available
Quad	1, 2
Dual 1: 6 Blender	1, 2, 3
3+0 Blender	1, 2, 3
5+0 Blender	1, 2, 3, 4, 5
3+1 Blender	1, 2, 3, 6

4 Select Price Level and press **ENTER**. The PPU flashes.

To Select	Press Configuration
Cash price levels	1
Credit price levels	2

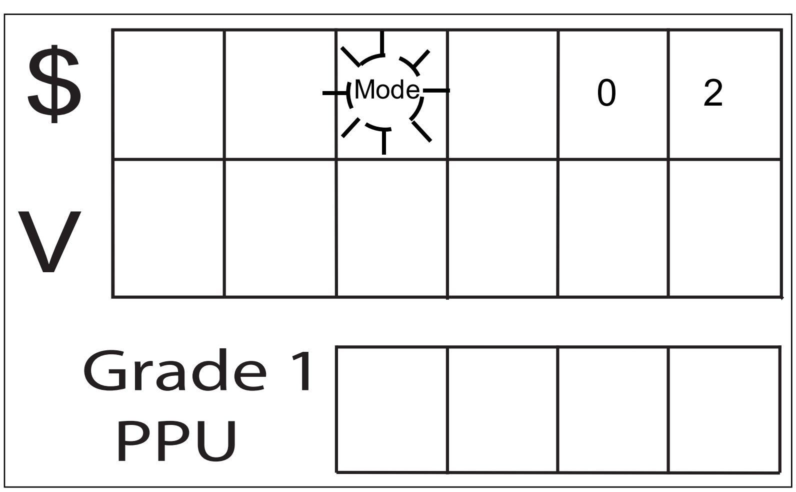
5 Enter new PPU. Start with Most Significant Digit [MSD (for example, 1.259; press 1, 2, 5, 9)]. Press **ENTER**.

6 Repeat steps 2 through 6 to program all sides, grades, and price levels.

CC2: Program Two-wire/Standalone

Choose the mode of operation: Two-wire for console control or standalone for non-console control for your pump/dispenser.

Figure 5-5: Program Two-wire/Standalone



1 Press **2** and then press **ENTER** on the manager keypad to access this CC.

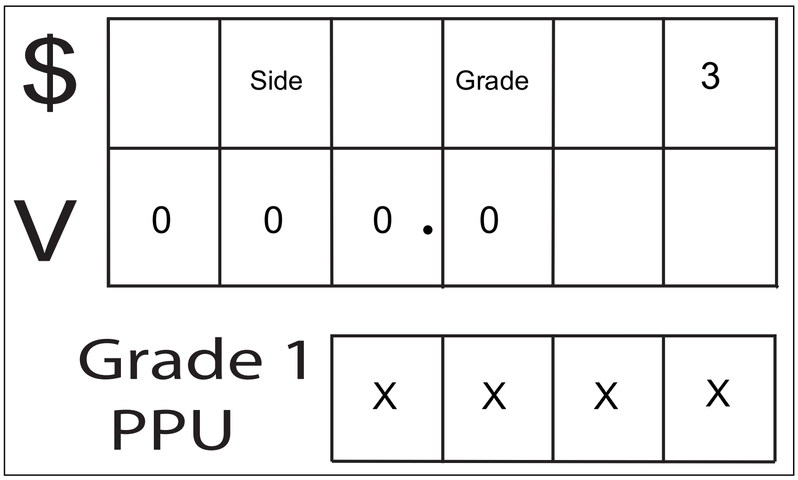
2 Select Mode and press **ENTER**.

To Select Mode	Press Configuration
Standalone (without console control)	0
Two-wire operation (with console control) *Note: This is the default.*	1

CC3: Programming Volume Allocation

Set a maximum limit on fuel volume pumped per customer.

Figure 5-6: Programming Volume Allocation



1 Press **3** and then press **ENTER** on the manager keypad to access this CC.

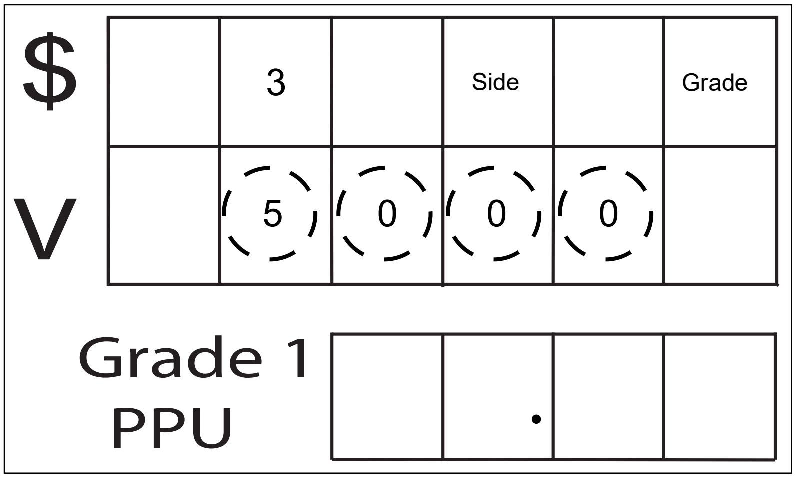
2 Select side and press **ENTER**.

To Select	Press Configuration
Side 1 (side with J-box)	1
Side 2	2
Both sides	3

3 Select Grade and then press **ENTER**.
Note: The number of grades vary for each model.

4 Enter volume allocation amount. Start with MSD. For example, to enter 500 gallons, press **5** and then press **0**. The default is 000.0 (no allocation limit). You cannot enter the hundredth digit.

Figure 5-7: Volume Allocation Display



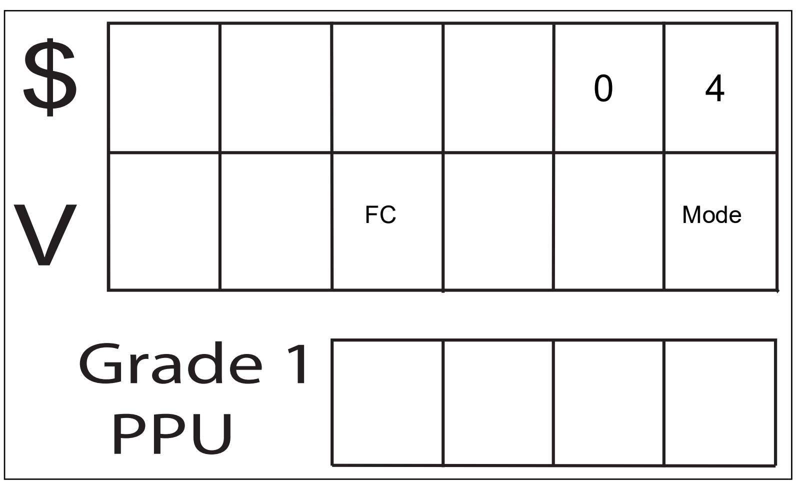
5 Press **ENTER**. Side number flashes.

6 Repeat steps 1 on page 5-11 through 5 to program all sides, grades, and price levels.

CC4: Blanking Display Manually and Selecting Five Button Money (Cash/Volume) Preset

This manual blank function is typically used for standalone units to blank out the displays when the station is closed but the power is not removed from the pump/dispenser.

Figure 5-8: Blanking Display



1 Press **4** and then press **ENTER** on the manager keypad to access this CC.

2 Select FC1 or 2, and press **ENTER**.

FC1: Manual Blank Displays

To Select This Option	Press Configuration
Displays OFF	0
Displays on (Default)	1

FC2: Five Button Preset

To Select This Option	Press Configuration
No Five Button Preset* (Default)	0
Money Preset	1
Volume Preset	2
Incremental Presets	3

**If PPP is present, unit cannot be programmed to Five Button Operation.*

Note: CC5 and FC8 of CC10 on *page 5-20* *can also be used to blank out displays from the console.*

CC5: Programming Customer Presets

This customer preset function is only used if units have the special 5-button or customer programmable preset option installed.

Use CC5 to program the preset buttons in 5-button and cumulative presets so that the customer can select a preset amount of fuel to dispense (for example, $5 of fuel).

Note: Programming is required only when this option is available. Units can also be preset using most consoles.

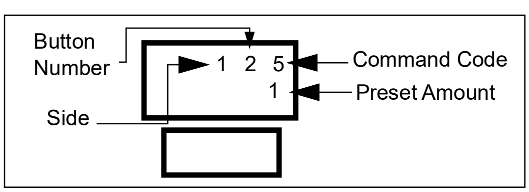
Two types of customer presets are available:

- 5-button customer preset
- Customer programmable preset

Note: Changing the customer presets may require a graphic change, which must be performed by an ASC.

Diagram: Command Code 5

The following diagram shows the settings for Command Code 5.



Fields Defined

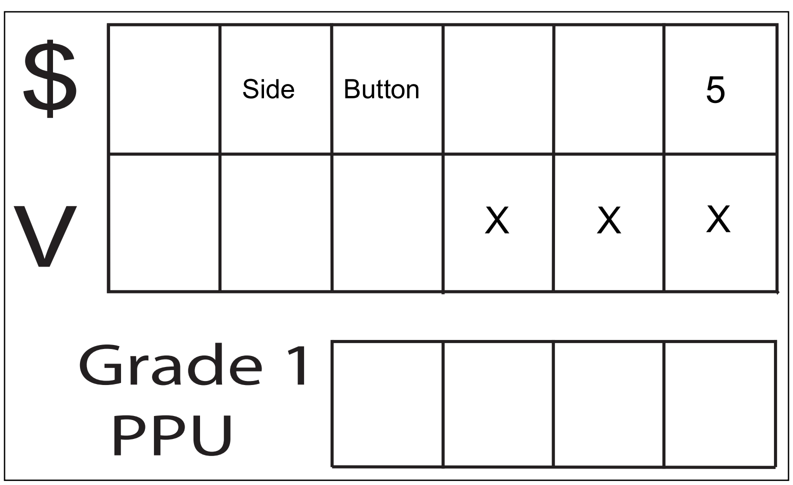
The following table provides descriptions of each field on the screen for Command Code 5.

Field	Description	Options
Command Code	Displays the Command Code for setting PPU values	N/A
Button Number	Preset button number	• 1—Leftmost or top • 2—Second from left or top • 3—Third from left or top • 4—Fourth from left or top • 5—Bottom or right
Preset Amount	Amount assigned to a particular button	1 through 999. Defaults are: • 1—$1 • 2—$5 • 3—$10 • 4—$15 • 5—Automatically set for FILL
Side of unit	Displays the side of the unit to program	• 1—Side 1 (A) • 2—Side 2 (B) • 3—Both sides

Procedure: Programming Customer Presets

After placing the unit in Program Mode (refer to “Programming Mode” on page 5-2), follow these steps to program the manual blank display. For the description and option information for each field, refer to the table in the section “Fields Defined”.

Figure 5-9: Programming Customer Presets



1 Press **5** and then press **ENTER** on the manager keypad to access this CC.

2 Select button number **2** and press **ENTER**.

Button: Preset Button Number Displays

To Select This Option	Press Configuration
Leftmost or Top	1
Second from Left or Top	2
Third from Left or Top	3
Fourth from Left or Top	4
Bottom or Right	5

3 Select the new preset amount in whole numbers (for example, 5 = 5.00).

4 Press **ENTER**. The CC flashes.

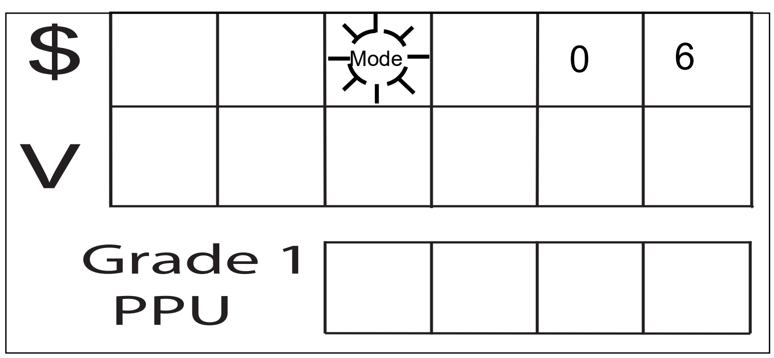
5 Repeat this procedure for each preset button you would like to program and do one of the following:
- To enter another CC, press **F1**.
- To exit the program mode, press **F2**.

CC6: Clearing Memory

Perform a Totals Memory Clear. Record all totals before reset. After performing a totals memory clear, a master reset CC17, FC2, must also be completed.

Note: A totals memory clear works only if EC31 (Totals Data Error) or EC35 (Configuration Data Error) occurs.

Figure 5-10: Clearing Memory



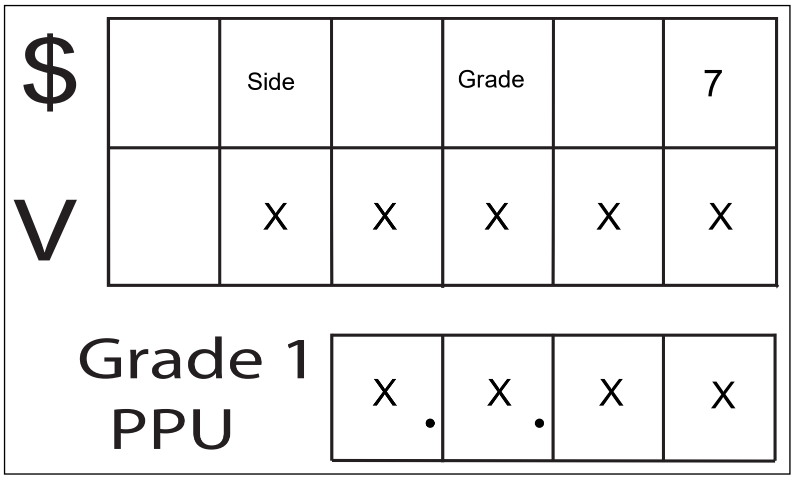
1 Press **6** and then press **ENTER** to access this CC.

2 Press **1** and then press **ENTER**. System performs power-up reset and continues in normal operation. You must reprogram the unit.

CC7: Setting Totals Input Mode

Money total is displayed when Totals Input Mode is entered. Press **VOL TOTAL** to enter volume totals. Press **$TOTAL** to change back to money total.

Figure 5-11: Setting Totals Input Mode



CC7 is active only after clearing memory (refer to “CC6: Clearing Memory” on page 5-15), performing a master reset, or during an install.

1 Press **7** and then press **ENTER** on the manager keypad to access this CC.

2 Select side and press **ENTER**.

To Select	Press Configuration
Side A (side with J-box)	1
Side B	2

3 Select Grade and press **ENTER**.
Note: For Blenders, grade 7 = high product and grade 8 = low product.

4 Enter total starting with MSD. Press **ENTER**.

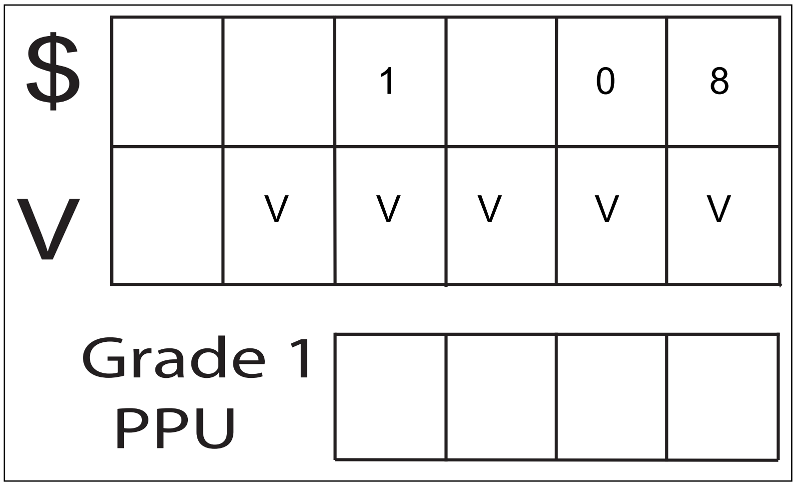
5 Repeat steps 1 through 4 for each side and grade.

6 Press **VOL TOTAL** on the manager keypad and repeat steps 3 to 5 for volume totals.

CC8: Displaying Software Version

Display software version installed in dispenser. Version number is displayed on the volume display.

Figure 5-12: Displaying Software Version



To access CC8, press **8** and then press **ENTER**.

Level Two CCs

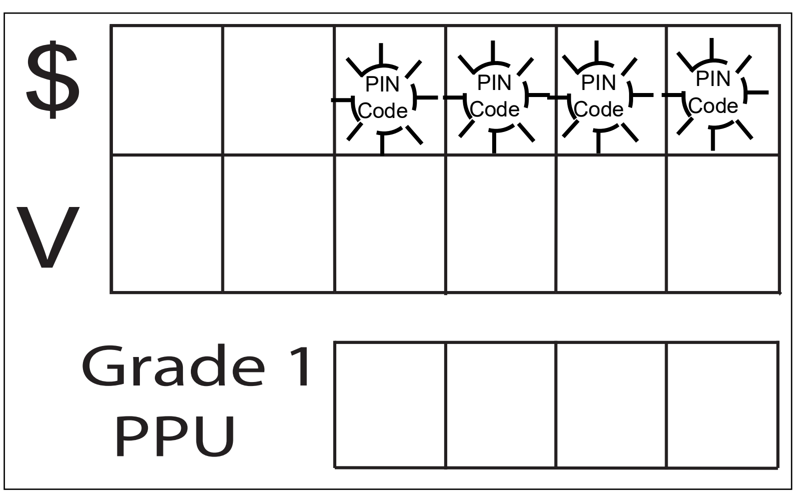
Do not allow station owners to program these Service CCs. Before programming, make a note of your unit's model number and software version. Refer to "CC8: Displaying Software Version".

Note: ASCs are required to use this programming information during the start-up procedure.

To Access Level Two CCs:

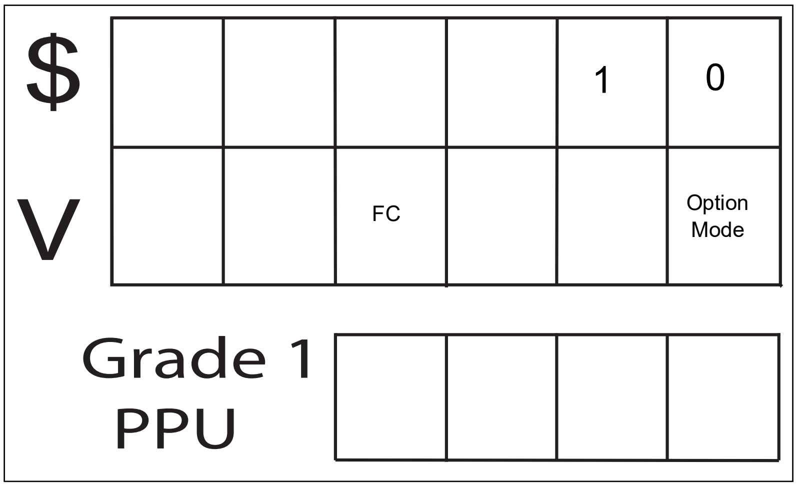
1 Press **F1** on the manager keypad, enter the Level 1 PIN Code, and then press **ENTER**.

Figure 5-13: Level 1 PIN Code on the Manager Keypad



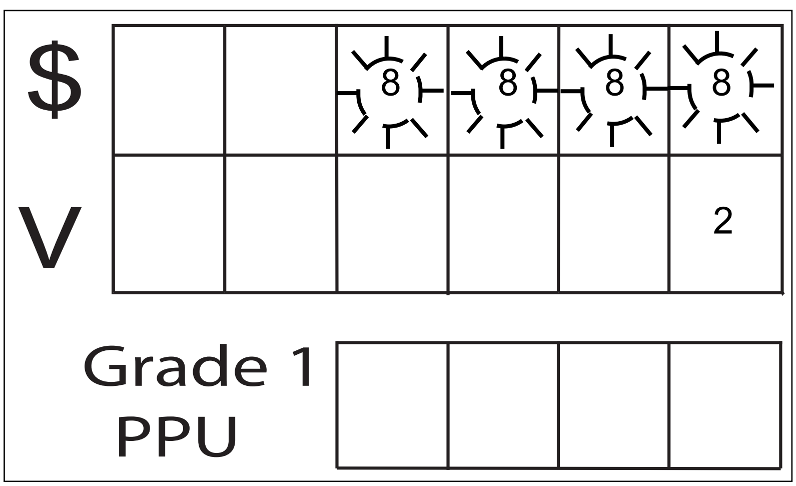
2 Enter the Level 2 CC (10-12) and press **ENTER**. A decimal point is displayed on the money display and 2 is displayed on the volume display.

Figure 5-14: Money Display



3 Enter your Level 2 PIN Code (Default 1503) and press **ENTER**.

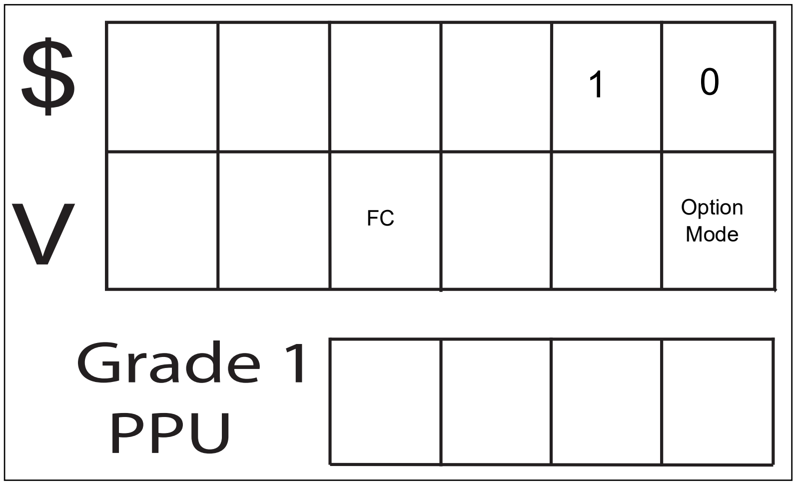
Figure 5-15: Level 2 PIN Code



CC10: Configuring the System

Note: Settings affect both fueling positions.

Figure 5-16: Configuring the System



1 Press **1**, **0**, and then press **ENTER** on the manager keypad to access this CC.

2 Enter a FC number (1-24) and press **ENTER**.

3 Enter a Configuration Code and press **ENTER**.

4 Choose another FC number or press **F2** to exit.

FC1: Unit Type

It is recommended to leave these settings at the system defaults.

For This Unit Type	Press Configuration
1 Product	1
2 Products	2
3 Products	3

FC2: Money Decimal Point

For This Money Decimal Point	Press Configuration
XXXXX.	1
XXXX.X	2
XXX.XX (Default)	3
XXX.XXX	4

FC3: Calculation PPU Decimal Point

U.S. W&M requires Configuration Code 4.

For This Calculation PPU Decimal Point	Press Configuration
XXXXX.	1
XXX.X	2
XX.XX	3
X.XXX (Default)	4

FC4: Displayed PPU Decimal Point

Use this FC when you calculate the PPU in dollars but display the decimal point in cents (for example, for Canada). You must program FC3 before FC4, if the displayed decimal point is different from the calculation decimal point.

For This Displayed PPU Decimal Point	Press Configuration
N/A	0
XXXX	1
XXX.X	2
XX.XX	3
X.XXX (Default)	4

FC5: Submersible Turbine Pump (STP) Prestart

Use this function to program the STP to turn on after pump handle activation. Turn this function "ON" for mechanical line leak detectors, or "OFF" for most electrical line leak detectors.

To Program the STP	Press Configuration
Off	0
On (Default) with pump handle activation	1

FC6: Beeper and VaporVac Alarm Option

Use this function to program the beeper to sound when you lift pump handles, authorize the dispenser, or press keypad switches. The VaporVac alarm beeps until warmstart occurs.

To Program the Beeper	And the VaporVac Alarm	Press Configuration
Off (Default)	Off	0
On	Off	1
Off	On*	2*
On	On*	3*

**VaporVac alarm must be set to "ON" for units in Mexico.*

FC7: Vapor Sense Option

Automatically set to "ON" for units with VaporVac option.

To Program the Vapor Sense Option	Press Configuration
Off (Default)	0
On	1

FC8: Two-wire Display Blank Option

Use this function to blank the displays and disable the dispenser using the console. You must perform a $980.00 preset post pay at the console to blank the displays and a $981.00 preset post pay to turn them back on. Also, for details on how to manually blank out displays at the pump/dispenser, refer to CC4, FC1 on page 5-13.

Note: Dispenser backlights remain on even after digits are blanked out. Pumps and dispensers cannot be used when displays are blanked since handles and grade select button are disabled.

To Program the Two-wire Blank Option	Press Configuration
Off (Default)	0
On	1

FC9: 5 or 6 Digit Cash

Use this function to program five or six digits of cash. You must program control console/cash register to same mode for proper operation.

To Program the Digits of Cash	Press Configuration
5 Digits of cash (Default)	0
6 Digits of cash	1

FC10: Decimal/Comma

Use this function to set commas instead of decimals for all displays.

To Program the Decimal/Comma	Press Configuration
Decimal (Default)	0
Comma	1

FC12: Pump Timeout

Pump Timeout sets the maximum continuous amount of time after starting the fueling that a customer can halt fuel delivery before the sale is terminated. This function can be used as a security feature to monitor tampering with the unit. Setting too small a time leads to customer inconvenience.

For This Timeout	Press Configuration
None (Default)	0
5 seconds	1
15 seconds	2
30 seconds	3
45 seconds	4
60 seconds	5
75 seconds	6
90 seconds	7
120 seconds	8

FC13: Slowdown (Ultra-Hi Units Only)

Models NP0, NP3, NP4, and NP5.

Note: This programming mode or CC13, FC4 can be used by Ultra-Hi models. The last FC programmed will be used by the unit regardless of the setting of the other code.

Program the point at which the dispenser goes into slowdown on all non-blenders. Not all software version offers all of the selections noted here. Normally, stations not offering prepays can be set at minimum times. Higher values can be selected if preset targets are not hit consistently. Select three preset sales minimums at the setting to allow the dispenser to “learn” the new value before drawing conclusions regarding a setting’s ability to correct the preset problem.

For This Slowdown	Press Configuration
0.070 gallons (0.265 liters)	0
0.120 gallons [0.454 liters (Default)]	1
0.170 gallons (0.643 liters)	2
0.220 gallons (0.833 liters)	3
0.270 gallons (1.02 liters)	4
0.320 gallons (1.21 liters)	5
0.370 gallons (1.40 liters)	6

For This Slowdown	Press Configuration
0.420 gallons (1.59 liters)	7
0.470 gallons (1.78 liters)	8
0.520 gallons (1.97 liters)	9

FC14: Push-to-Start for G6 (+1 of 3+1 Blender Only)

When Grade 6 (3+1 blender) programming has been enabled, the G6 (+1) grade selection button must be pressed before fuel can be delivered for G6.

To Program Zero PPU Pricing	Press Configuration
Disabled/not installed	0
Enabled	1

FC15: Zero PPU Pricing

Allows fuel dispensing with a PPU of zero (fleet operations only)

To Program Zero PPU Pricing	Press Configuration
Disabled (Default)	0
Enabled	1

FC16: Beeper Timeout

Use with FC12 to give an audible signal after pump timeout. After pump timeout, the beeper beeps for the programmed seconds and deactivates the dispensing unit with EC29. EC29 is dismissed by returning the nozzle to the nozzle boot.

Beeper Timeout	Press Configuration
None (Default)	0
5 seconds	1
15 seconds	2
30 seconds	3
45 seconds	4
60 seconds	5
75 seconds	6
90 seconds	7
120 seconds	8

FC17: Stop Control

Deactivates one or both sides of a dispenser when optional Push-to-Stop Button is pressed.

To Program Stop Control	Press Configuration
For one side (Default)	0
For two sides	1

FC18: Zero Previous Transaction

Programs the dispenser to reset to zero the transaction displayed after:

- The pump handle is activated (raised) and unit is authorized
- No fuel is dispensed
- The handle is deactivated (lowered).

Normally, the previous sale will be displayed. You must use Configuration Code 1 for New Jersey installs.

To Zero Previous Transactions	Press Configuration
After lamp test (Default)	0
Before lamp test	1

FC19: Lamp Test (Leak Detector Test Setup)

Sets the amount of time in seconds, the display shows all “8”s during the reset cycle at the beginning of a transaction. This time is typically set to allow the STP mechanical leak detectors to run a complete test. For electronic line leak detectors, normally this is set to “OFF”. However, follow the recommendations of line leak detector manufacturer to set time (not an actual test of lamps):

To Perform	Press Configuration
Not used	0
One second lamp test	1
Two second lamp test	2
Three second lamp test (Default)	3
Four second lamp test	4
Five second lamp test	5
Six second lamp test	6
Seven second lamp test	7
Eight second lamp test	8
Nine second lamp test	9

FC20: Programmable Pump Preset Value

Normally set to “0”. Program to “1” for use with six-digit cash programming.

To Set Programmable Pump Preset Value	Press Configuration
X1 (Default)	0
X10	1

FC21: PPU Blinking Option

Program the PPU displays in the price bar to blink under certain conditions.

To Set PPU Blinking	Press Configuration
Off	0
On (Default)	1

FC22: Volume Display Decimal Point

Use this function to program the decimal point out of the main volume display. When programmed to “1”, the decimal point is not shown on the display. FC22 is not approved by U.S. W&M.

To Set Volume Display	Press Configuration
XXX.XXX (Default)	0
XXXX.XX (Export Only)	1

Note: If set to “1”, decimal point will not show. The decimal point shown in the table is for reference only.

FC23: Program STP Control

Use this function to activate all STPs on all transactions (regardless of which grade is selected). Use this Code for Encore MPD single-hose units only (NG0, NG2, NG4, or NG5).

To Set STP Control	Press Configuration
Off (Default)	0
On	1

FC24: Hose Pressurization Option

Use this function to pressurize the hose during lamp tests of the first transaction after a power-up or the first transaction after a period of inactivity.

Note: This function must not be used to correct situations with leaky check valves, nozzles, or vapor recovery hoses. This programming feature can be useful to correct minor leaks from the meter back toward the tank which cause calibration issues, fuel delivery indication when the fueling position is activated with the nozzle is closed, and so on. Recent U.S. W&M requirements states that the configuration #1 and later firmware versions must not be programmed differently.

To Set Hose Pressurization Option	Press Configuration
Off (Default)	0
On, up to 10 minutes	1
On, up to 30 minutes	2

CC12: Programming Pump/Dispenser ID and Certain Configurations

Program the pump as follows. Each fueling position side must be programmed.

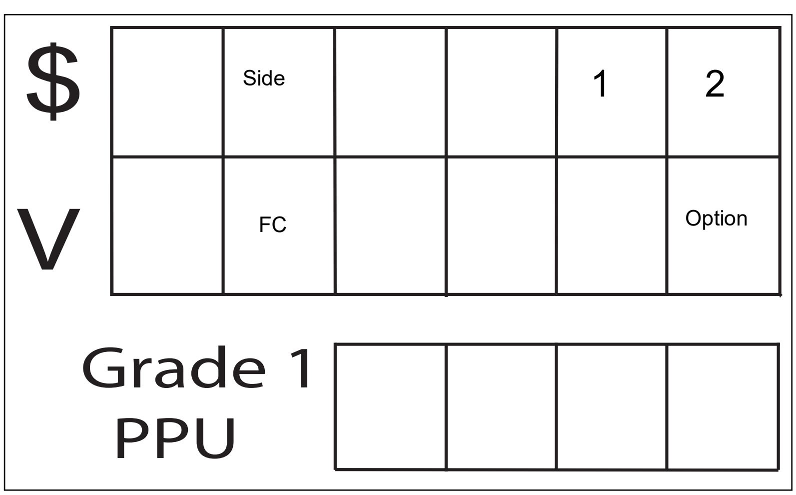
1 Press **1**, **2**, and then press **ENTER** to access this CC.

2 Select side and press **ENTER**.

To Select	Press Configuration
Side 1 [A (side with J-box)]	1
Side 2 (B)	2

3 Enter a FC number (1-5) and press **ENTER**.

Figure 5-17: Programming Pump/Dispenser ID



4 Enter a Configuration Code Option and press **ENTER**.

5 Choose another FC number or press **F2** to exit.

FC1: Two-wire ID
Do not use the same ID (address) number for units (or unit sides) on the same data loop. Side “1” always defaults to 7 and Side “2” always defaults to “11”.

To assign a number 1-16, press that number on the keypad and press **ENTER**.

FC2: Cash/Credit at Pump
When programmed to “Yes,” the customer must select Cash or Credit at the unit. This FC must always be set at “0” for Encore 300.

FC3: Side Exists
Use this option to disable unused side of single-sided units only:

To Program	Press Configuration
Does not exist	0
Exists (Default)	1

FC4: PPU Options
This option is used mainly with single level CRIND units and split island operation.

- If the unit has single-level PPU displays with switches (M02652A003) or without switches (M02652A004), the PPU will display the price regardless of the level sent from the console or CRIND.
- If the unit has dual-level PPU displays with switches (M02652A003) or without switches (M02265A008), the Level 2 (bottom) display is used for single level price posting when top display is covered.

.

To Program	Press Configuration
Level 1 price bar displays level sent by console or CRIND card reader	0
Normal (Default)	1
Level 2 price bar, Level 1 prices, Level 1, Level 2 prices	2
Level 2 price bar displays level sent by console or CRIND card reader (V70.6 and later)	3

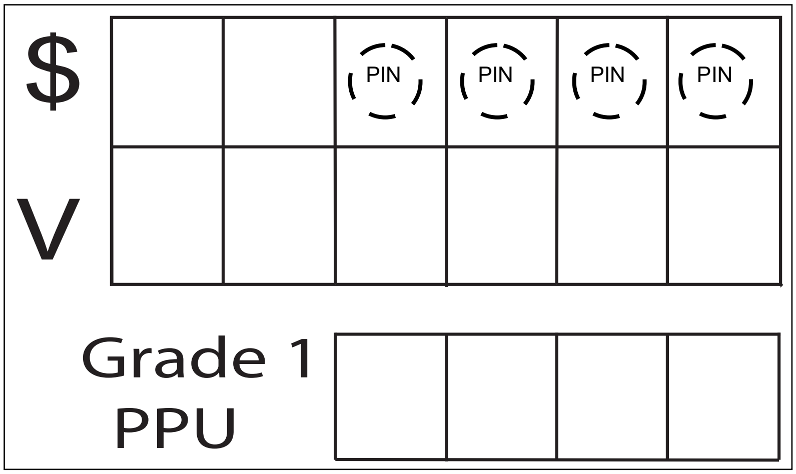
FC5:
Not available in this pump software version.

Level Three CCs

To Access Level Three CCs:

1 Press **F1** on the manager keypad, enter the Level 3 PIN Code, and then press **ENTER**.

Figure 5-18: Accessing Level Three CCs on the Manager Keypad



CC13: Changing PIN Code

Use the following procedure to change PIN Code for added security. Current PIN Code must be accessed before it can be changed. Revised PIN Codes enhance unit security. They must be securely stored by the station and ASC.
Note: Do not allow station owners to program these Service CCs.

To access Level 3 CCs, proceed as follows:

1 Press **F1** on the manager keypad and then enter the Level 1 PIN.

2 Enter the Level 2 CC (10-12) and press **ENTER**.

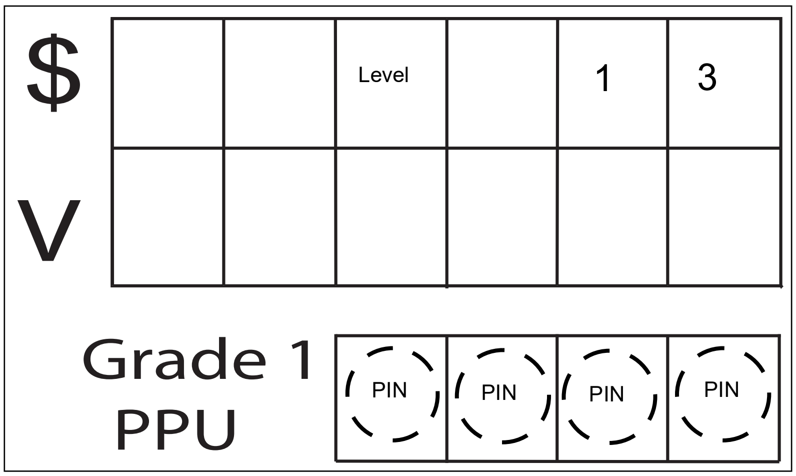
3 Enter the Level 2 PIN Code (Default 1503) and press **ENTER**.

4 Enter the Level 3 CC (13 or higher) and press **ENTER**.

5 Enter the Level 3 PIN Code (Default 1309) and press **ENTER**.

> **⚠ CAUTION**
>
> Carefully monitor keystroke activity for these steps. If you enter an incorrect PIN number (different than intended), a coldstart, reprogramming, and recalibration of the unit will be required.

Figure 5-19: Changing PIN Code



CC15: Set Timeouts

CC15 is entered in Level 3 programming to set dispenser timeout options and for extra slow down value, proceed as follows:

All entered values can be in the range of 0-999 seconds and 0-0.999 gallons or liters for the slow down value.

The following table lists the function codes and their options:

FC	Option
1	Pre-select Timeout (0-999 seconds)
2	Dispenser Timeout (0-999 seconds)
3	Beeper Timeout (0-999 seconds)
4	Slow Down Value (0-0.999 gallons or liters)
5	Unused
6	Initial Dispenser Timeout (0-999 seconds)
7	Initial Beeper Timeout (0-999 seconds)

For programming method, refer to "CC53: Set Timeouts" on page 5-77.

FC Descriptions for CC15

FC1: Set Pre-select Timeout
Pre-select timeout will set pre-select timeout timer period allowed for grade select buttons to be pressed, before handle is raised for a non-nozzle activated unit. Timer starts when the dispenser is authorized and times out until the customer lifts the pump handle. After pre-select timeout has occurred, grade selected is lost and must be reselected by customer. Pump does NOT change state when this timeout occurs. A zero value means there is no timeout. Pump preset is also under the control of pre-select timeout. Allowed range for this timer is from 0 to 999 seconds. Default value for this function is 15 seconds.

FC2: Set Dispenser Timeout
Dispenser timeout is measured from when pulses STOP flowing, until handle is lowered, to remind driver or attendant to remove the nozzle from tank. If Beeper Timeout is enabled and set to a value greater than 0, beeper must beep once a second (until Beeper Timeout timer runs down) after the dispenser timeout time has occurred. Default value for this function is 0 seconds (no timeout). If fueling is resumed, beeper will stop sounding and timer is reset.

FC3: Set Beeper Timeout
Beeper timeout value determines the amount of time beeper sounds after dispenser timeout has occurred before the transaction is ended. Allowed range for this timeout is 0 to 999 seconds. Default value is 0 or no beeper. Beeper must sound once a second for the number of seconds programmed.

FC4: Slowdown Volume
Slowdown volume is a setting used to determine when the dispenser goes into slowdown mode before reaching transaction preset value. Allowed range is 0.000 to 0.999 gallons or liters. It is recommended to never program less than 0.200 gallons or 0.600 liters. When set, this value overrides the slowdown setting in CC10, FC13. Setting to 0 disables this slowdown setting.

FC6: Set Initial Dispenser Timeout
Initial Dispenser timeout is measured from when transactions starts if no pulses are ever received to remind driver or attendant to remove the nozzle from tank or depress the nozzle handle. If Initial Beeper Timeout is enabled and set to a value greater than 0, beeper must beep once a second (until Initial Beeper Timeout timer runs down) after the Initial Dispenser timeout time has occurred. Default value for this function is 0 seconds (no timeout). If fueling starts, beeper will stop sounding and timer is reset.

FC7: Set Initial Beeper Timeout
Initial Beeper timeout value determines the amount of time beeper sounds after dispenser timeout has occurred before the transaction is ended. Allowed range for this timeout is 0 to 999 seconds. Default value is 0 or no beeper. Beeper must sound once a second for the number of seconds programmed.

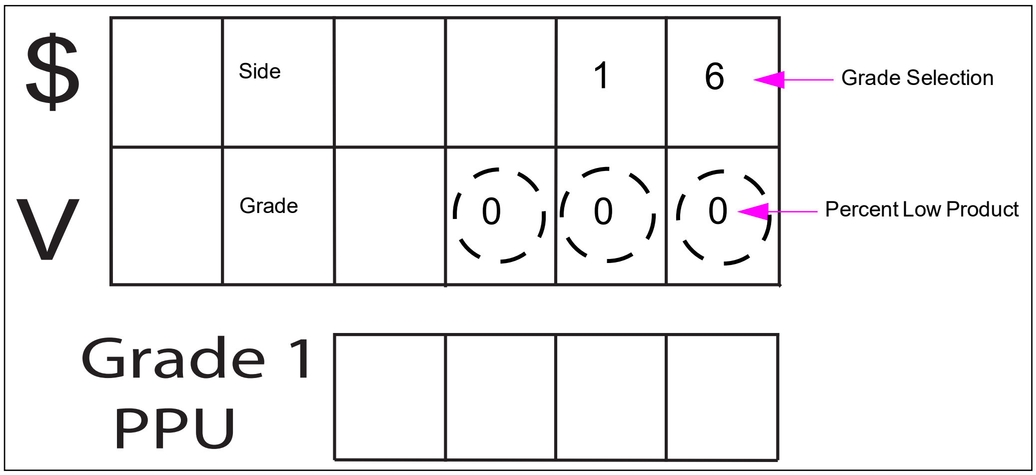
CC16: Programming Blend Ratios (Blenders Only)

1 For blenders only, press **1**, **6**, and then press **ENTER** to access this CC.

2 Select side (1-3) and press **ENTER**.

To Program	Press Configuration
Side 1 (side with J-box)	1
Side 2	2
Both sides	3

Figure 5-20: Programming Blend Ratios



Use the following table when programming grades:

For These Models	Program These Grades
Dual	Grade 1
3 + 0	Grades 1, 2, and 3
5 + 0	Grades 1, 2, 3, 4, and 5
3 + 1	Grades 1, 2, 3, and 6
Six-Hose (Fixed) Blenders	Grade 1, 2, and 3

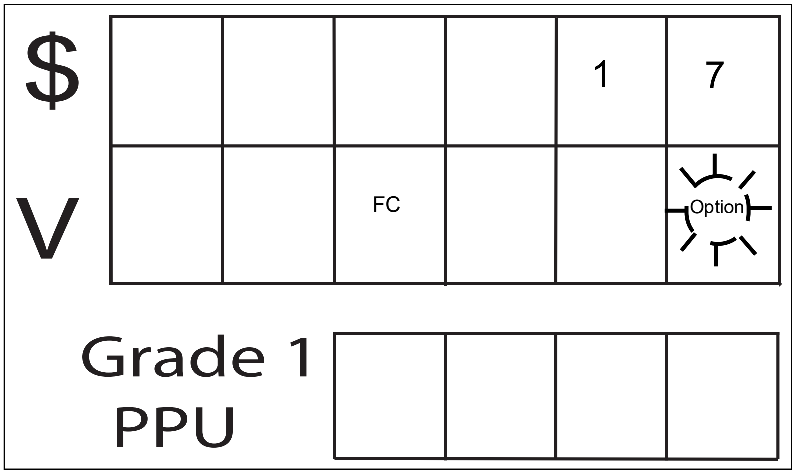
Note: For three grades, program as Grade 1, 3, 5, and 6 for an unblended hose (certain modifications may allow Grades 1, 2, 3, and 6).

3 Enter low product blend percentage and press **ENTER**. The high product blend percentage calculates automatically.

4 Repeat the sequence for remaining grades and side, if required.

CC17: Programming Conversion Factors and Performing Master Resets

Figure 5-21: Programming Conversion Factors and Performing Master Resets



To program conversion factors and perform a master reset, proceed as follows:

1 Press **1**, **7**, and then press **ENTER** on the manager keypad to access this CC.

2 Press **1** and then press **ENTER** to program the conversion factors.

FC1: Conversion Factor

Note: Units will not operate without a conversion factor programmed.

To Program	Press Configuration
No conversion factor (Default)	0
U.S. gallons	1
Imperial gallons	2
Liters	3
1,012 pulses per gallon	4

3 Select the conversion factor (0-4) and then press **ENTER**.

4 Press **2** and then press **ENTER** to perform a master reset.

5 Press **1** and then press **ENTER**. The unit will perform a master reset.

FC2: Master Reset

To Program	Press Configuration
N/A	0
Master reset	1

CC18

Do not use.

CC90: Encore 300 Electronic Calibration Procedure

Ensure that you follow the listed steps in the given order. Failure to follow the steps in the given order will result in calibration not being properly completed properly. Electronic calibration must always be manually verified before returning the dispenser to service. Additional calibration information can be found in "Pump Programming" on page 5-1.

IMPORTANT INFORMATION

Never enter into CC90 directly from any other CC if programming has just been done. If this is done, the calibration steps will open to function and display properly, but actually will not. Always start with pressing **F1** and use appropriate PIN Codes to go directly into the calibration mode.

IMPORTANT INFORMATION

If ATC is installed, product must be dispensed to exactly 20 liters in a 20-liter test can with no offset entered. Failure to do this will result in significant measurement errors in the net volume in the normal operating mode (200 and 300 modes).

1 Turn on (to right) the W&M switch (see Figure 5-74 on page 5-100).

2 Press **F1** on the manager keypad.

3 Enter CC90 and press **ENTER**.

4 You will be prompted to enter the prover can size in the measuring units (for example, gallons, liters, and so on) that is programmed in CC17. After entering prover can size, press **ENTER**.

5 Remove the nozzle, and then select grade or lift lever.
Note: Only pure product can be calibrated.

6 For units without ATC, dispense the product into the prover can as close to zero volume as possible and hang up nozzle. For units with ATC, dispense exactly 20 liters into the calibration can. The volume display will show the number of uncalibrated pulses. Any reactivation of the handle switch before completion of the calibration procedure will result in lost calibration.
Note: If a blender is being calibrated where either a pure low or pure high grade does not exist, then you must change a blend ratio in CC16 temporarily to create the missing pure high or pure low grade. For the missing pure grade to calibrate, revert the temporary grade to actual one.

7 Lower the pump handle or return nozzle to boot (nozzle activated lever).

8 Enter the offset (number of cubic inches or centiliters away from zero) and press **ENTER**.

9 Enter 0 if the offset is above 0 (zero) or 9 if the offset is below 0 (zero). Press **ENTER**.

10 The volume display will flash the adjusted volume (for example, 5.008 gallons is slightly less than +2 cubic inches on the prover scale). Now, you have the option of pressing CLEAR to return to step 9 to re-enter the offset data.
Note: One cubic inch is 0.0043 gallon.

11 Press **ENTER** to return to the main display to enter new prover can size, side, or grade. The volume display will display the adjusted volume amount (for example, 5.008 must equal +2 cubic inches on the prover can scale).

12 When calibration is complete, turn off the W&M switch to exit the programming mode.

Note: Programming will be lost if the W&M switch is turned off before the completion of calibration programming.

13 Press **F2** to return the unit to service.

Additional Manager Functions for Encore 300

The following manager functions are available when the unit is in normal operation.

Verifying Blend Ratios, Changes to Blend Ratios, and Configuration Changes

To verify blend ratios, changes to blend ratios, and configuration changes for blenders, proceed as follows:

1 Press **ENTER** on the manager keypad.

The PPUs will display the programmed blend ratios.

Note: The volume display will list the number of times the unit of volume configuration changes have been made. The sale display will show the number of times blend ratios have been changed.

2 Press **CLEAR** to exit.

DLT

The last transaction displays for 15 minutes after AC power is lost.

Unit Totals Retrieval Convention

Totals Examples

For both $ TOTAL and VOL TOTAL keys, press **ENTER** to toggle between grade and side selection.

Non-resettable Money and Volume Totals

The dispenser maintains a set of non-resettable totals for each dispensed fuel grade. These totals reflect total money and volume since dispenser installation time. The following are keypad and display examples for viewing dispenser non-resettable money and volume totals.

Money and Volume Totals

Use the $ TOTAL key to retrieve money totals for each fuel grade. The $ TOTAL and VOL TOTAL keys do not require a Security PIN Code. Use the CLEAR key to exit money totals mode. The “1” shown in the leftmost $ display indicates that this total is a non-resettable money total. The VOL TOTAL key displays the volume for the grade selected. Volume for both sides of the unit can be viewed. Use the CLEAR key to exit the volume total mode. This convention comes from the Advantage series product line and is maintained here for ASC and site manager familiarity.

Displaying Pump Money and Volume Totals

To view side 1 totals, proceed as follows:

1. Press **$TOTAL** on the manager keypad. Combined Cash and Credit total will be displayed for grade 1, side 1.
2. Select Grade(s) using keypad. Read $ totals for each grade selected. The number of hose(s)/grades(s) vary for each model.
   Note: For Blenders, grade 7 = high product, and grade 8 = low product.
3. Press **ENTER** to view side 2 money totals.
4. Press **VOL TOTAL**. Volume total is displayed for the grade selected.
5. Select Grade(s) using keypad. Read volume totals for each grade selected. Press **ENTER** to view side 2 totals.
6. Select side 2. Repeat steps 1 through 5 for side 2.
7. Press **CLEAR** to exit.

Audit Trail Information

Figure 5-22: E300 Dispenser Audit Trail Display - Newer, No Main PPU Display

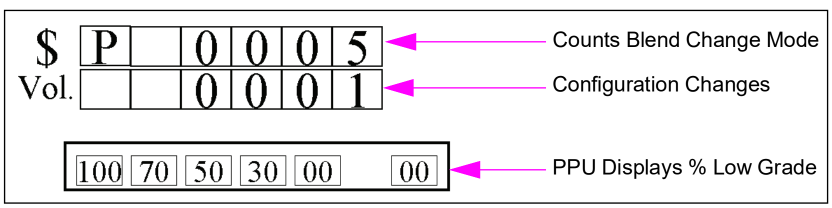
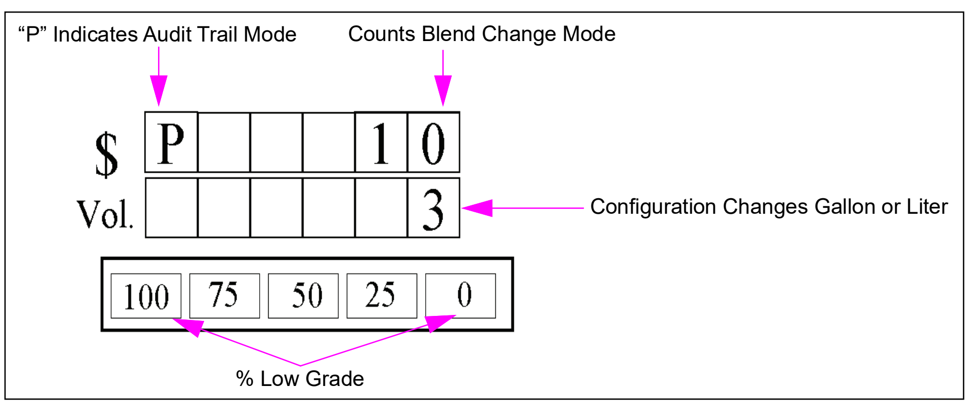


Figure 5-23: Encore/Eclipse Audit Trail Display - Press Enter First Time



Programming the ATC Option

To program the ATC option, proceed as follows:

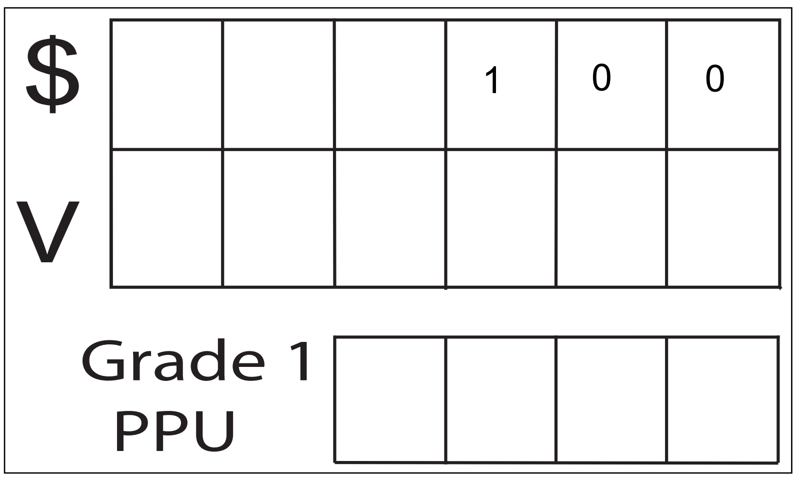
1 Place the programming switch located on the ATC controller board in the "ON" position.

2 At power-up:
- Programmed models will flash "104" and then normal information will flash.
- Non-programmed models will flash "100" in PPU and require the models to be programmed.

How to Program Each Fueling Position

1 Enter 100.

2 Press **ENTER**.



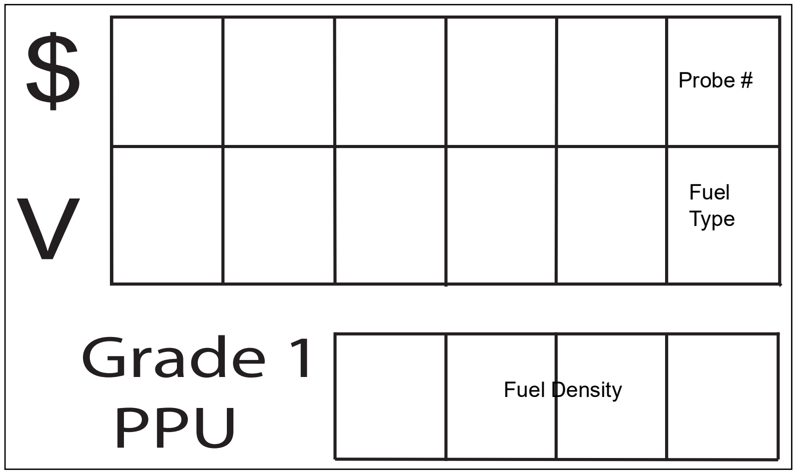
Note: If the program switch is "OFF", it will not work.

The money position will display "1". The money position shows which probe is selected. (Probe: 1 through 8)

The volume position will display “1”. The volume position shows which fuel type has been selected (Fuel Type: 1-gasoline, 2-diesel).

The PPU position will display “730”. The PPU position shows which fuel density has been selected for the corresponding probe (Fuel Density: gasoline-730, diesel-840).

Note: The default value for Fuel Density is 730.



The software will sequence through each meter position for you to assign the required fuel type.

3 Press **ENTER** after selecting the fuel type.

4 Sequence through each meter position and verify the required fuel type by pressing **ENTER**.

5 Place the programming switch on the ATC controller board in the “OFF” position.

6 Press **F2** to exit ATC programming mode.

7 W&M auditor will put a seal on the programming switch.

8 Close the electronics door as described in *MDE-2531 Pump and Dispenser Start-up and Service Manual*.

How to Access ATC Mode Information

View Last Transaction

1 Enter 300.

3 0 0

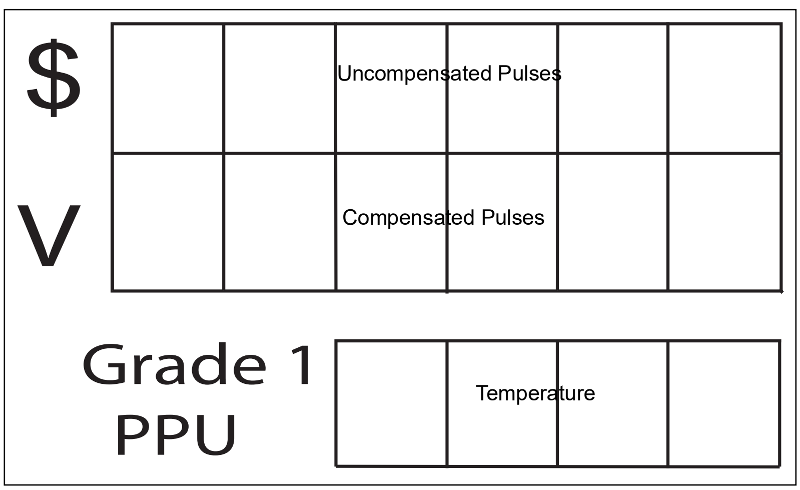
2 Press **ENTER**.

ENTER

3 Press **1** for Meter Position One information.

1

4 Press **ENTER**.



5 Record the information.

6 Continue to press each position number for corresponding position information.

7 Press **ENTER**.

ENTER

8 Press **F2** to exit ATC Mode (wait for one minute timeout if “F2” is non-functional).

F2

Definition/Function of ATC Codes

100 Programming Code

For instructions on programming ATC option, refer to "Programming the ATC Option" on page 5-34.

200 Inspection Mode

Access 200 Inspection mode to obtain a precise reading of ATC system operation during an actual transaction. This is the most accurate means of checking ATC system accuracy because temperature readings are displayed during transaction.

Pure product only: Program a blender for 100%. Otherwise, the PPU display shows EC105-product error.

- Money display - shows uncompensated volume.
- Volume display - shows compensated volume.
- PPU display - shows real-time temperature (in celsius).

300 View Last Transaction

Access 300 View Last Transaction mode after a transaction. This mode can be used as a basic check to ensure that the ATC system is operating properly. Temperature displayed in this mode is an average during the last transaction. Do not use this mode to verify the accuracy of the ATC system.

There are no blender restrictions.

- Money display - shows uncompensated volume.
- Volume display - shows compensated volume.
- PPU display - shows average temperature (in celsius).

301 Display Volume Correction Factor (VCF)

- Money display - shows product number.
- PPU display - shows volume correction factor.

302 Display Density

PPU display - shows selected meter's programmed density information.

303 Display Temperature

PPU display - shows selected meter's current product temperature.

304 Display Gross Volume Totals

Volume and PPU displays - shows a ten digit number representing the selected meter's cumulative uncompensated volume total for a pure product.

500 Software Version

Main display - shows software version number.

600 Troubleshoot Probe

Access 600 Troubleshoot Probe mode to troubleshoot probes. There must be a small change in readings for every second if T-meter and controller board are working properly. Each probe can be checked by entering meter number.

- Money display - pulse width in terms of counts.
- Volume display - probe resistance in ohms.
- PPU display - shows the temperature of probe (in celsius).

601 Probe History

Access 601 Probe History mode to display probe history since ATC was installed. This is useful for troubleshooting intermittent probes or probe connections. Access each probe by entering meter number.

- Money display - shows probe number entered.
- Volume display - shows total number of probe range errors logged.
- PPU display - shows total number of missing probe pulse errors.

ECs

Error	EC	Description
Programming Error	100	All fuel types not programmed
Rom Checksum	101	Controller ROM checksum error
Ram Error	102	Controller RAM write/read error
T Meter Dead	103	Not receiving any T-meter pulses
T Meter Sync	104	Receiving T-meter pulses, but with errors
Product Error	105	Multiple product selected
Ramp Range Error	106	Gilbarco use only
Range Error For Probe	107	Probe out of range
Missing Error	108	Probe information missing
Pulser Error	109	Pulser fail
Reset JJ Error	110	Reset jumper is installed
Ram Corrupted	111	RAM data corrupted
No Input Timer	112	No timer A interrupts occurring
No Output Timer	113	No timer B interrupts occurring
Power Fail Error	114	Power fail signal stuck
Recursion Error	115	Read T-meter interrupt recursion

Note: For additional troubleshooting information, refer to MDE-2531 Pump and Dispenser Start-up and Service Manual.

Programming Encore 500/S/700 S and Eclipse Series Units

⚠ CAUTION

When any work activity needs to be performed outside the scope of “normal” operation (SW download, terminal activation, etc), you must disconnect all communication cables (CAT5/6) to the POS to prevent possible interference and related issues when performing the required task(s).

Contents in this Sub-section

Software Versions

Software versions can be obtained by using the Laptop Tool for a complete listing, W&M Review/Audit Trail (only on dispensers with software V02.9.06 or later) or at power up for a partial listing (door node and pump software version only).

PIN Code Entry

Programming Level	Default PIN Code	CC Range
Level 1	2222	20-39
Level 2	1503	40-69
Level 3	1309	70-89
Level 4	0128	84-99

- *Press* ***F1*** *to perform additional programming.*
- *Press* ***ENTER*** *after making selection in all levels.*
- *Press* ***F2*** *to exit programming.*

Note: Use of a higher level Encore 500/S/700 S and Eclipse PIN Code will allow access to lower level programming codes as well (for example, keying the Level 3 PIN Code will allow access to Level 1, 2, and 3 programming but not Level 4).

Encore 500/S/700 S and Eclipse Programming Quick Reference

⚠ CAUTION

When any work activity needs to be performed outside the scope of “normal” operation (SW download, terminal activation, etc), you must disconnect all communication cables (CAT5/6) to the POS to prevent possible interference and related issues when performing the required task(s).

CCs and FCs Quick Reference Chart

The following chart shows level CCs and FCs, wherever applicable. For more details, refer to the following pages:

Level One CCs
(Starts on page 5-42)

- 20 Set PPU Values
  Note: Not allowed when in two-wire operation mode in software V02.9.06 and later.
- 21 Not applicable
- 22 Set Preset Types and Values
  - 1 Set Preset Type
  - 2 Set Preset Amounts
  - 3 Set PPU ON/OFF
  - 4 Volume Rounding Digit
  - 5 Money Rounding Digit
- 24 Set Operations Mode
  - 1 Two-wire*
  - 2 Standalone
- 25 Set Allocation Volume
- 27 Review ATC
  - 2 Display Volume Correction Factor
  - 3 Display Fuel Density
  - 4 Display Temperature
  - 5 Gross Volume Totals
  - 6 Software Version
  - 7 Real Time Transaction
    Note: This is no longer available in current software. Refer to "Real-time Data Review - ATC" on page 7-89
  - 8 Engineering Audit
- 28 Change First Level PIN
- 31 Set Allocation Money

Level Two CCs
(Starts on page 5-51)

- 40 Set Two-wire ID (pump number)
- 41 Set PPU Options
- 42 Set Decimal Point Options
  - 1 Main Display Money
  - 2 Main Display Volume
  - 3 PPU for Calculations
  - 4 PPU for Display
- 43 PPU Blinking Options
- 45 Set Lamp Test Time
- 46 Set Comma/Decimal Point
- 47 Two-wire Options
  Note: Not allowed when in two-wire operation mode in software V02.9.06 and later.
  - 1 Set Reported Money Size
  - 2 Set Reported Volume Format
  - 3 Set VRC Stated in Transaction Data
  - 4 Status Response Format
  - 5 Baud Rate
  - 6 Lost Communication Function
  - 7 Real-Time Data Format
  - 8 Preset Method Without Pump Stop
  - 9 Remote Preset Display
- 48 Zero Previous Transaction
- 49 STP Controls
  - 1 Set STP Pre-start
  - 2 Set STP Control
  - 3 Set STP Side Mapping
- 51 Set Main Beeper
- 52 Set VaporVac Alarms
- 53 Set Timeouts
  - 1 Set Beeper
  - 2 Set Pre-select
  - 3 Set Dispenser
  - 4 Initial Dispenser Timeout
  - 5 Initial Beeper Timeout
  - 6 Two-wire No Communication Timeout
  - 7 Display Backlight Timeout
  - 8 Maximum Authorize Time
  - 9 Maximum Dispense Time
- 54 Set Slowdown Value
- 55 Set Hose Pressurization
  Note: This is no longer allowed in US gallons.
- 56 Set Stop Button Control
- 60 Change Date/Time
  - 1 Set Time
  - 2 Set Date
  - 3 Set Daylight Savings Time
- 62 Two-wire Remapping Grade

Level Three CCs
(Starts on page 5-88)

- 71 Set Volume Units
- 72 Set Blend Ratios
- 73 Set/Display Money Totals
  Note: This is no longer allowed to be programmed after dispensing commissioning.
- 74 Set/Display Volume Totals
  Note: This is no longer allowed to be programmed after dispensing commissioning.
- 75 Set Fuel Density
- 76 Set Money Rounding Method
- 77 Change PIN Code
  - 1 Level One
  - 2 Level Two
  - 3 Level Three
  - 4 Level Four
- 78 Electronic Calibration
  Note: For historical use only.
- 80 Set Maximum Flow Rate
- 82 Set Calibration Can Size
- 83 Set Alternate Fuel Mode

Level Four CCs
(Starts on page 5-117)

- 84 Set Pulser Type
  - 0 Standard Pulser
  - 1 Encrypted Pulser
- 90 Set Unit Type
- 91 Select Installed Options
  - 1 Preset Type
  - 2 Push to Start
  - 3 Set VaporVac
  - 5 Set Push to Stop
  - 6 Set ATC
  - 7 Grade Select Function
  - 8 Misc. Protection Method
  - 11 Set Totalizer function
  - 12 Set Nozzle activated Pump Handle
  - 13 Set Meter Type
    Note: For historical use only.
  - 26 Set Operation Mode Protection
  - 30 Set Pulser Error Reporting
  - 31 Set Unauthorized Flow Reporting
  - 32 Set South Korea Roundup Method
  - 33 Set Price Change Method
  - 47 PC Serial Communication Protocol
- 92 Set Side Exits
- 94 Set Zero PPU Option
- 96 Feature Set
- 98 Inlet Assignment

Encore 500/S/700 S and Eclipse Programming Instructions

Level One CCs

Level 1 commands are those commands most commonly performed on site by station owner/operator.

Programming Level	Default PIN Code	CC Range
Level 1	2222	20-39

- *Press **F1** to perform additional programming.*
- *Press **ENTER** after making selection in all levels.*
- *Press **F2** to exit programming.*

Note: Entry of Level 2, 3, or 4 PIN Codes will also allow entry into Level 1 programming.

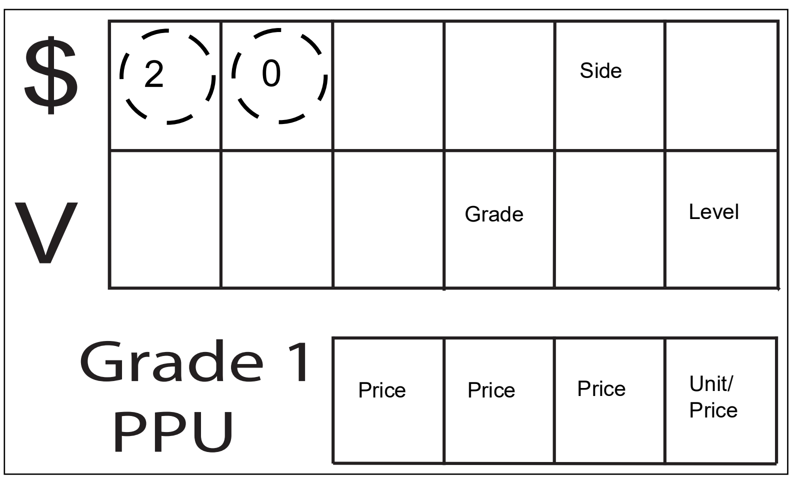
The following CCs described show the layout and digit position meaning for programming features:

CC20: Set PPU Values

Layout and digit position meaning for this programming feature are shown in Figure 5-24.

With pump software V02.9.06P and later, Set PPU Values is not available unless the operation mode is set to Standalone. Attempting to enter CC20 will double beep.

Figure 5-24: Setting the PPU Values



1 Press **2**, **0**, and then press **ENTER** to access this CC.

2 Select side and press **ENTER**.

To Program	Press Configuration
Side 1 (side with J-box or side with W&M switch)	1
Side 2	2
Both Sides	3

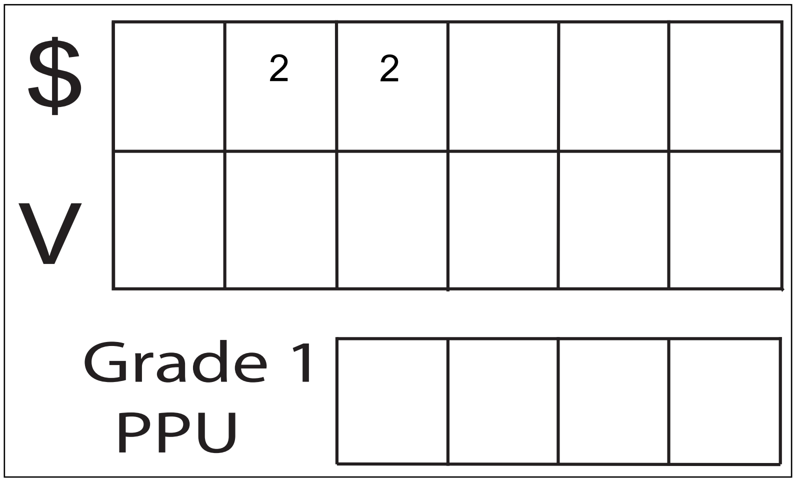
3 Select Price Level (Cash/Credit).

4 Enter the new PPU number.
Note: Repeat for other side, grade, and price level.

CC22: Set Preset Types and Values

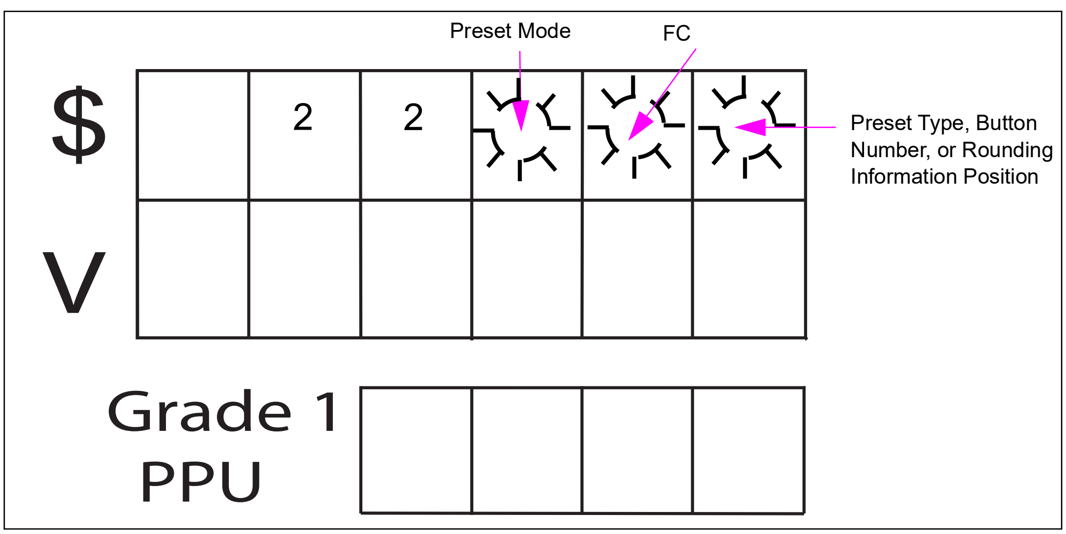
CC22 allows programming the preset keypad setup information when the preset keypad is configured in “CC91: Select Installed Options” on page 5-120. This section can be used to program specific setup information for buttons, extra keys, and special operations.

Figure 5-25: Setting the Preset Types and Values



Press **2**, **2**, and then press **ENTER** to access this CC. For more information on position of FC information and data, see If “NO KEYPAD PRESET” was set in “CC91: Select Installed Options” on page 5-120, then a programming error display will result as described in the dispenser programming section.

Figure 5-26: Setting Money and Volume Units



FC1: Set Preset Type

Set Preset Type is only used with 5-button or incremental preset keypads and is unused for PPP. This function is used to dictate whether the keypad is for Money or Volume preset entry. Set 1 for Money or 2 for Volume.

FC2: Set Preset Amounts

Set Preset Amounts is only used with the 5-button or incremental preset keypads and is unused for PPP. This FC expects more information for Button Number and will allow entry of up to three digits to assign each button of the Incremental or 5-button keypads. Values can be 0 to 999 for buttons 1, 2 and 3.

FC3: Set PPP On/Off

Set PPP On/Off is only used with PPP and is unused for 5-button and incremental keypads. This FC is used to allow easy access to temporarily turn PPP keypads off. Values here are 0 for off, 1 for On and 2 for and On mode that disables the use of the P1 and P2 keys.

FC4: Configure Volume Rounding Digit

The Configure Volume Rounding Digit configures a special preset change where the user can press the volume key (or volume roundup key in 5-button or incremental keypads) to dispense to the next configured volume digit. This can be used in conjunction with the Money Rounding Digit.

Roundup Configuration	Roundup Digit on Display (Round to the Next X Digit on Volume Display)
0	No Volume Roundup Allowed
1	xxxxXx
2	xxxXxx
3	xxXxxx
4	xXxxxx
5	Xxxxxx

FC5: Configure Money Rounding Digit

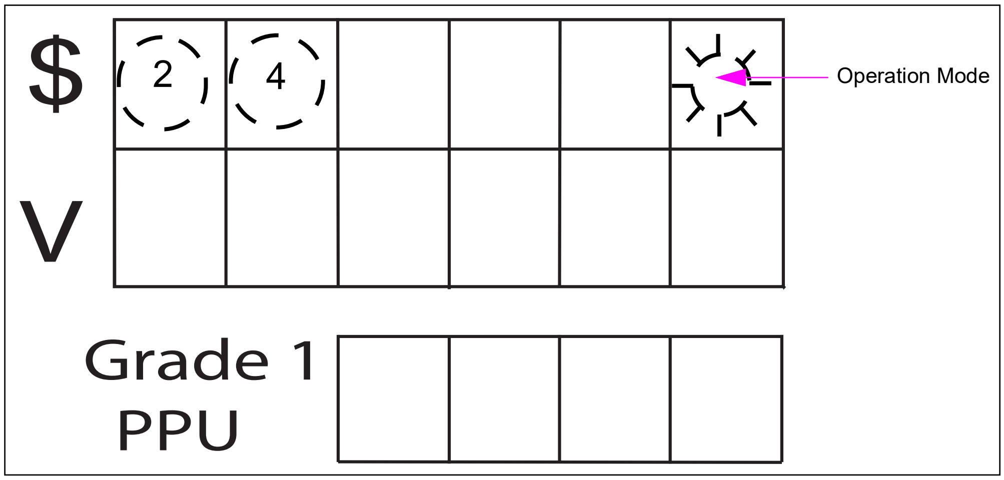
The Configure Money Rounding Digit configures a special preset change where the user can press the money key (or money roundup key in 5-button or incremental keypads) to dispense to the next configured money digit.

Roundup Configuration	Roundup Digit on Display (Round to the Next X Digit on Money Display)
0	No Money Roundup Allowed
1	xxxxXx
2	xxxXxx
3	xxXxxx
4	xXxxxx
5	Xxxxxx

CC24: Set Operations Mode

CC24 is entered in Level 1 programming to set the dispenser operating mode. Operation mode setting determines how the dispenser is controlled remotely or if it operates without external control. Available choices for operation mode setting are shown in Figure 5-27.

Figure 5-27: Setting Operations Mode



1 Press **2**, **4**, and then press **ENTER** to access this CC.

2 Select Mode of Operation option using the following table:

Option	Operation Mode
1 (Default)	Two-wire
2	Standalone

3 Press **ENTER** to accept mode of operation.

CC25: Set Volume Allocation

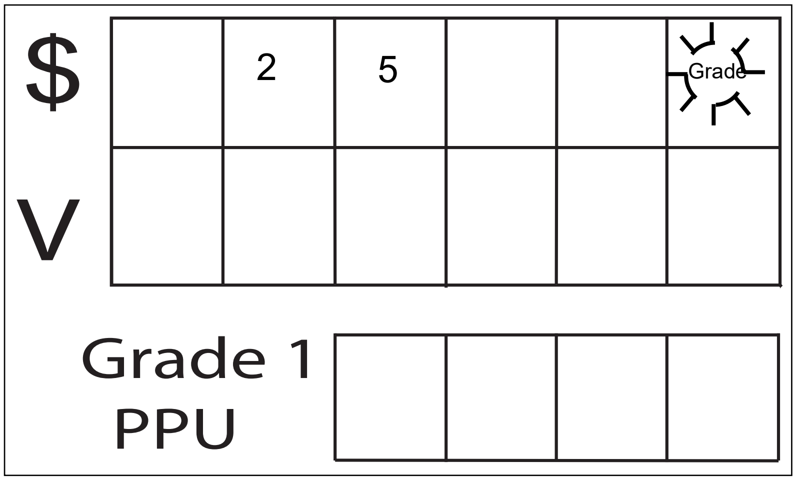
CC25 is entered in Level 1 programming to set side and grade allocation volumes.

Note: Decimal point position may change in main volume display depending on its location from display decimal point programming. Default allocation setting is none or all zeros.

1 Press **2**, **5**, and then press **ENTER** to access this CC.

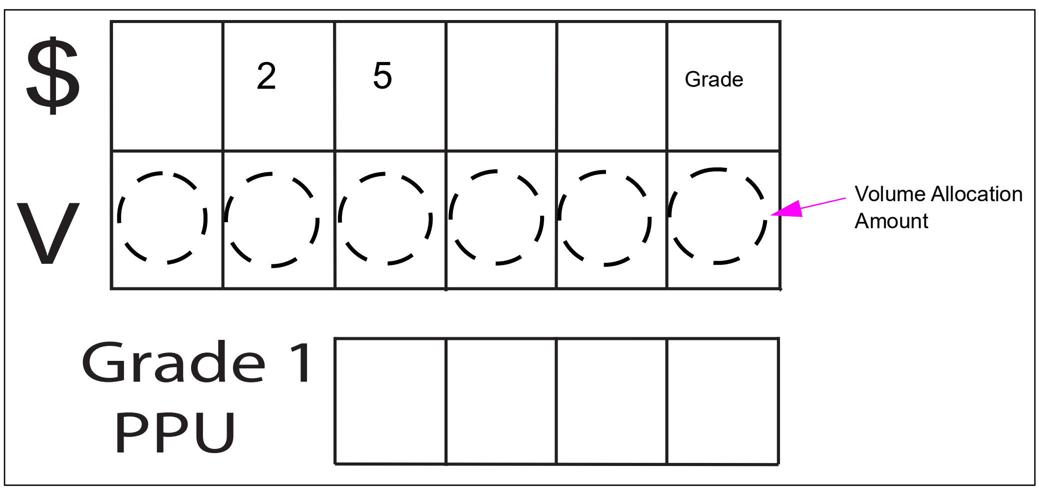
Note: Layout and meaning of the digit position for this programming feature are shown in *Figure 5-28*.

Figure 5-28: Grade Selection



2 Select Grade and press **ENTER**.

Figure 5-29: Setting Volume Allocation Amount



3 Set Volume Allocation Amount in the volume display and press **ENTER**.

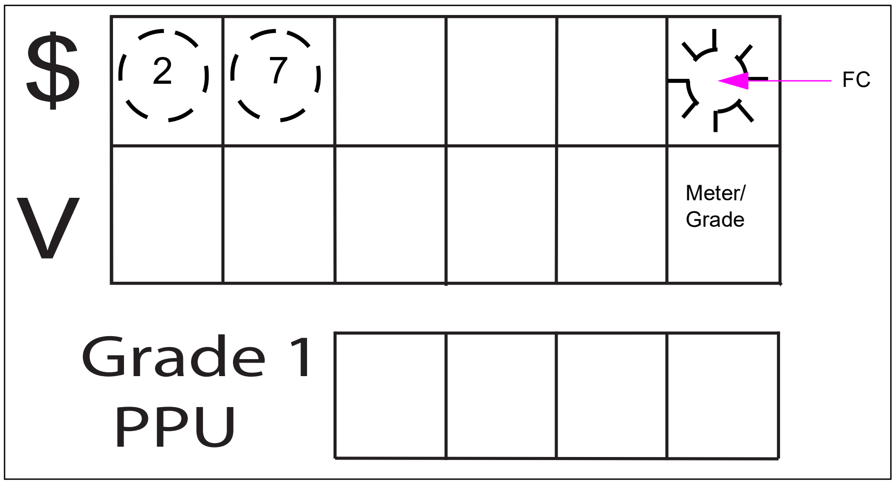
CC27: Review ATC Data

CC27 is entered in Level 2 programming to access ATC data. FCs in this programming level provide read only access to ATC operating parameters. No changes can be made to operating parameters at this level. Default FC (1) provides audit of last transaction data.

Notes: 1) Attempting to use this CC when ATC is not installed will result in a display error as described in *"Pump Programming" on page 5-1**.*

2) Complete ATC data will be shown in displays depending on specific FCs as described in sub-sections that follow.

Figure 5-30: Reviewing ATC Data



1 Press **2**, **7**, and then press **ENTER** to access this CC.

2 Select FC number (1-8) using the following table and press **ENTER**.

The following table shows available FCs and keypad programming for CC27:

FC	ATC Functions and Description	FC Keypad Programming in CC27
1 (Default)	Audit last transaction: Provides display of uncompensated and compensated product volumes (single meter) in main money and volume displays respectively. Average transaction temperature for the selected inlet is shown in PPU display for selected meter. ATC data is reported after inlet selection has been entered in main volume display. *Note: On blended product, one must view both pure products and manually add them to create blended totals (volume and sale).*	• Press **ENTER** (accepts default FC). • Select meter. • Press **ENTER**. • Select next meter. • Press **ENTER**.
2	Display VCF: Provides display of VCF for a selected product (meter). VCF is based on currently sensed meter temperature. ATC data is reported after meter selection has been entered in main volume display.	• Select FC 2. • Press **ENTER**. • Select meter. • Press **ENTER**. • Select Grade. • Press **ENTER**.
3	Display fuel density: Provides display of density for a selected meter. Displayed density for a particular grade reflects density assignment made in Level 4 programming. ATC data is reported after grade selection has been entered in main volume display.	• Select FC 3. • Press **ENTER**. • Select meter. • Press **ENTER**. • Select next meter. • Press **ENTER**.
4	Display temperature: Provides display of the temperature in Celsius for a selected meter.	• Select FC 4. • Press **ENTER**. • Select meter. • Press **ENTER** (refer to Note 1). • Select next meter. • Press **ENTER** (refer to Note 1).
5	Display gross volume totals: Provides display of gross volume totals for a selected meter.	• Select FC5. • Press **ENTER**. • Select meter. • Press **ENTER**. • Select next meter. • Press **ENTER**.
6	Display software version: Provides display of ATC neuron software version number.	• Select FC 6. • Press **ENTER**.
7	Real time transaction: *Note: This is only valid for pump software before V01.8.30. For Real-Time ATC review, refer to “Real-time Data Review - ATC” on page 7-89.* Provides means to monitor ATC operation during a transaction. Monitored data includes uncompensated volume, compensated volume, and product temperature for product selected. Product selection will be accomplished by sensing the coincidence of grade authorization and pulser activity. After product selection, both money and volume displays are zeroed and product temperature will be displayed in PPU display. During real-time transaction mode, money display on each side will show uncompensated volume for selected product. Volume display on each side will display compensated volume for selected product. Finally, PPU display on each side will show current temperature in Celsius. End of a transaction is determined when all grade authorization signals become inactive. End of a transaction results in both money and volume displays being blanked with 8s shown in each PPU digit.	• Select FC 7. • Press **ENTER** (refer to Notes 1 and 2). • Press **ENTER** (avoid timeout).

FC	ATC Functions and Description	FC Keypad Programming in CC27
8	Engineering audit: Engineering audit mode is intended for use by Gilbarco Engineering personnel to judge performance of ATC system. Mode provides information on T-meter operation and temperature data conversion. Typically, customer or ASC is not required to access this mode. Money display shows probe width and volume display shows temperature in Celsius. Values are updated continuously.	• Select FC 8. • Press **ENTER**. • Select Grade. • Press **ENTER** (refer to Note 3). • Select next grade. • Press **ENTER** (refer to Note 3).

Notes:

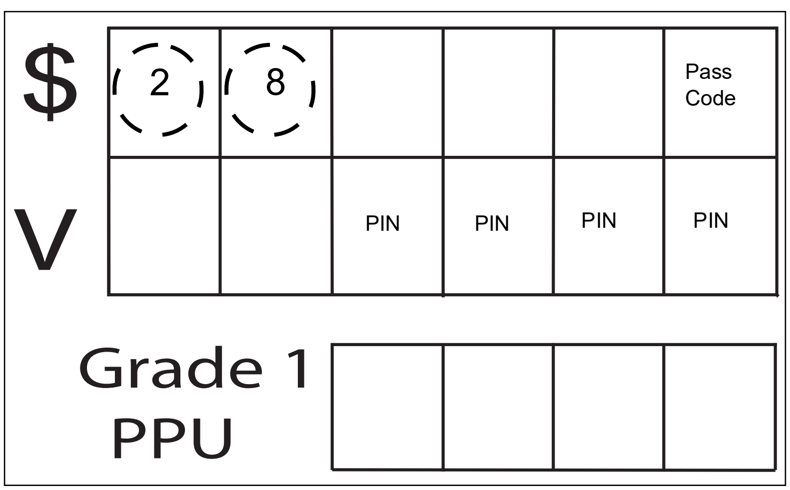
1. *Display indicates '0' for a negative temperature value and is blank for a positive value (Current PPU display cannot display a "-" character).*
2. *When you press **ENTER**, display will blank money and volume displays of both sides and show 8s in the PPUs. When fuel flow is detected, display will show real-time uncompensated volume in money display, compensated volume in volume display, and temperature in PPU displays. When the transaction is complete, display will retain transaction data for one minute and then revert to blank money and volume displays with 8s shown in PPUs. One minute timeout can be avoided by pressing **ENTER**.*
3. *"-" Indicates a negative temperature value and is blank for a positive value. Money display shows probe width and volume display shows temperature in Celsius. Values are updated continuously.*

CC28: Change First Level PIN

Allows site manager to change site management level (Level 1) PIN Code. Entering new digits for new code, changes the display as digits are entered. Display does not show actual PIN digit. It only shows "0" to signify that a number was entered. Same PIN Code must be entered twice to be accepted.

First pass at entering PIN Code is signaled by a "1" appearing in Pass Code display. Second pass is signaled by a "2" appearing in Pass Code display. Acceptance of a PIN Code is signaled by a "3" appearing in Pass Code display. A "0" in Pass Code display indicates that the PIN Code was not accepted. Layout and digit position are shown in Figure 5-31.

Figure 5-31: Changing the First Level PIN



1. Press **2**, **8**, and then press **ENTER** to access this CC.
   Note: Pass Code displays 1 for first pass.
2. Enter PIN Code: XXXX and press **ENTER**.
   Note: Pass Code displays 2, prompting a second pass.
3. Enter PIN Code again and press **ENTER**.

CC31: Set Money Allocation

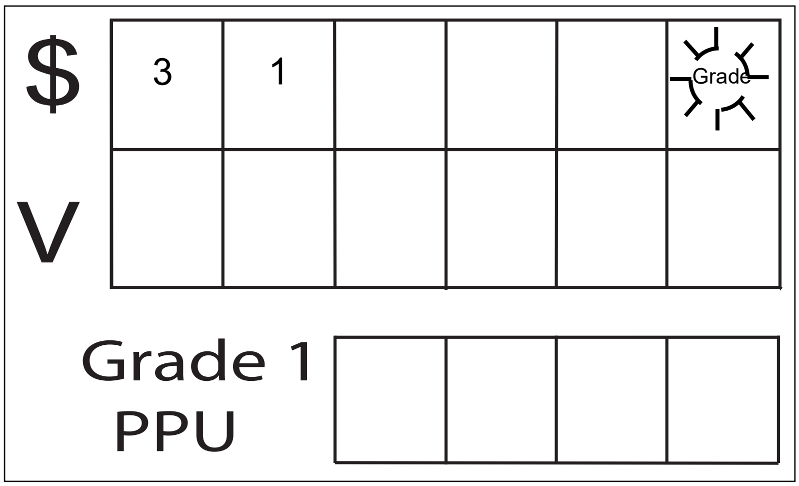
CC31 is entered in Level 1 programming to set side and grade money allocation.

Note: Decimal point position may change in money display depending on its location from display decimal point programming. Default allocation setting is none or all zeros.

1 Press **3**, **1**, and then press **ENTER** to access this CC.

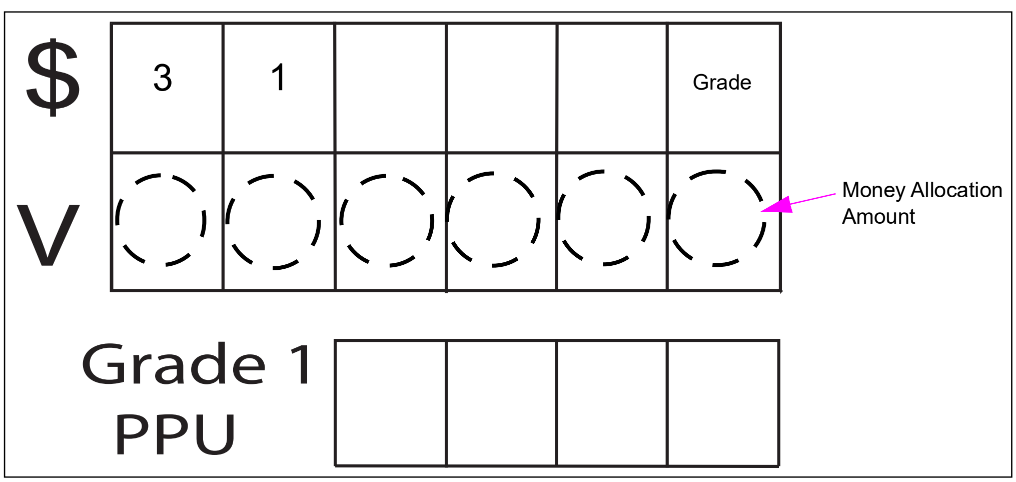
Note: Layout and meaning of the digit position for this programming feature are shown in Figure 5-32.

Figure 5-32: Grade Selection



2 Select Grade and press **ENTER**.

Figure 5-33: Setting Money Allocation Amount



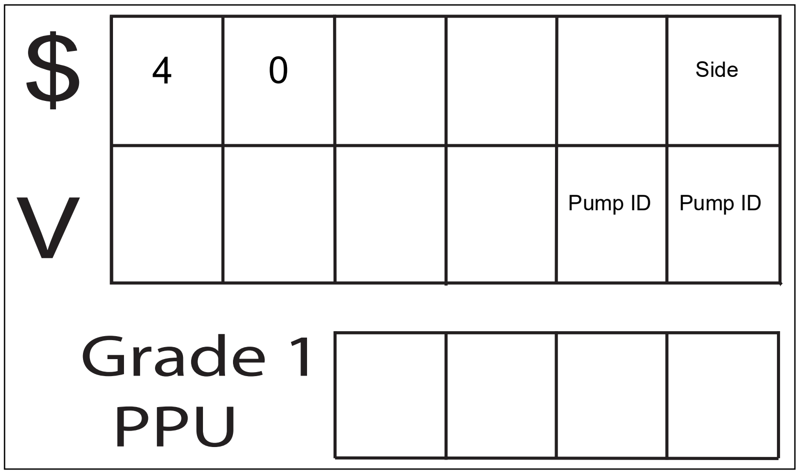
3 Set Money Allocation Amount in the Money display and press **ENTER**.

Level Two CCs

CC40: Set Two-wire ID (Pump Number)

CC40 is entered in Level 2 programming to set the pump ID number. Address range is 1-16. Pump side ID default values are side 1 = 7 and side 2 = 11. Layout and digit position meaning for this programming feature are shown in Figure 5-34.

Figure 5-34: Setting Pump ID Number

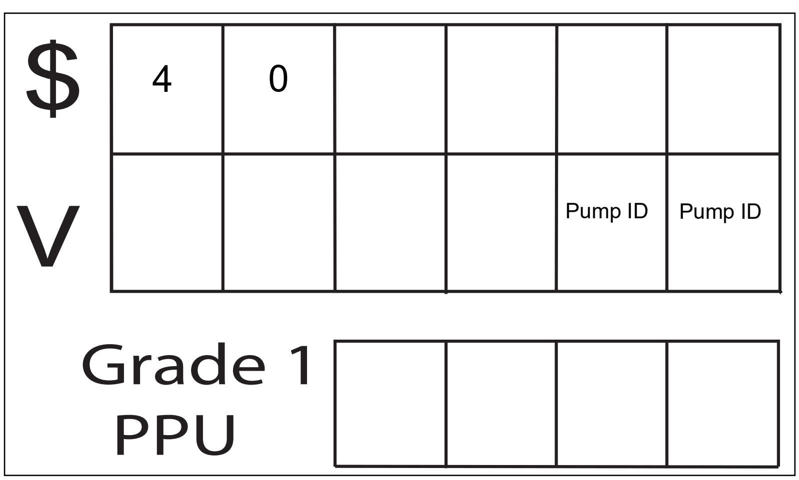


1 Press **4**, **0**, and then press **ENTER** to access this CC.

2 Select side and press **ENTER**.

To Program	Press Configuration
Side 1 (side with J-box or side with W&M switch)	1
Side 2	2

Figure 5-35: Pump ID Number

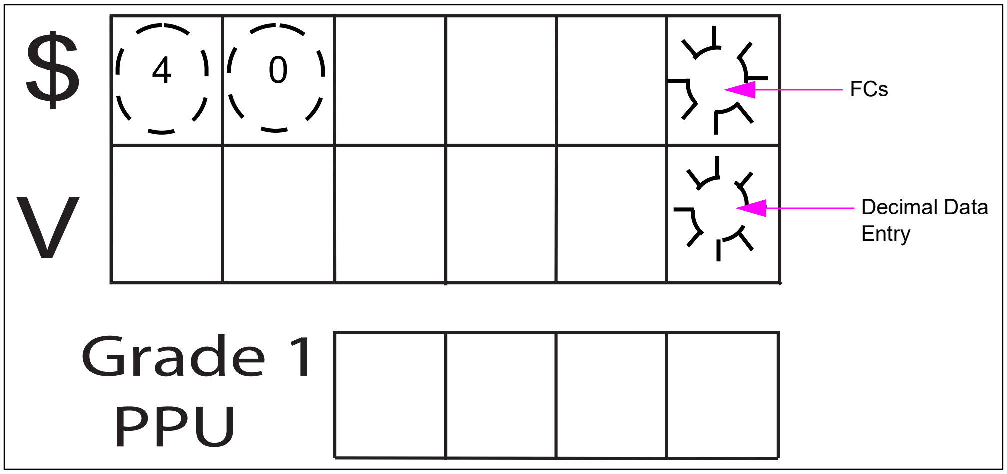


3 Enter address (1-16) and press **ENTER**.

CC42: Set Decimal Point Options

CC42 is entered in Level 2 programming to assign decimal point locations. FCs 1 through 4 are selected to set the location of money and volume decimal points for display and calculation purposes. Entry into CC42 will select FC1 by default. Layout and meaning of each digit position for this programming feature are shown in Figure 5-36.

Figure 5-36: Set Decimal Point Options



1 Press **4**, **2**, and then press **ENTER** to access this CC.

2 Select FC (1-4) using the following table and press **ENTER**.

FC1: Set Money Decimal Point Position for Display

The following table indicates locations where main money display decimal point can be positioned through CC42, FC1:

Position Parameter Options	Decimal Point Positions
1	X
2	X.X
3 (Default)	X.XX
4	X.XXX

1 Select FC1 and then press **ENTER**.
Note: Accepts default FC - Main Money Display.

2 Select Position Parameter Option (1-4) and press **ENTER**.

FC2: Set Volume Decimal Point Position for Display

The following table indicates the locations where main money display decimal point can be positioned through CC42, FC2:

Position Option	Decimal Point Positions
3	XXXX.XX
4 (Default)	XXX.XXX

1 Select FC2 and then press **ENTER**.

2 Select Position Parameter Option (1-4).

3 Press **ENTER** to save the setting.

FC3: Set PPU Decimal Point Position for Calculations (Canada Only)

The following table indicates locations where PPU calculation decimal point can be located. Default PPU decimal point for calculations is position 4.
Note: PPU grade mapping is not allowed in pump software version later than V01.7.85.

1 Select FC3 and press **ENTER** to access this CC.

2 Select Position Parameter Option (1-4) and press **ENTER**.

Position Option	Decimal Point Positions
1	X
2	X.X
3	X.XX
4 (Default)	X.XXX

FC4: Set PPU Decimal Point Position for Display

The following table indicates where PPU display decimal point can be positioned. Default PPU decimal point for display is position 4.

1 Select FC4 and then press **ENTER** to access this CC.

2 Select Position Parameter Option (1-4) using the following table and press **ENTER**.

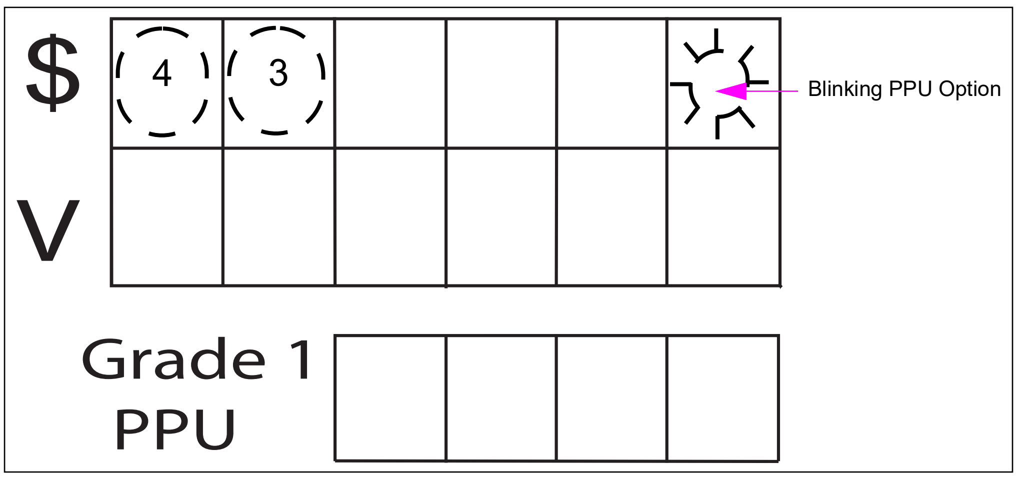
Position Option	Decimal Point Positions
1	X
2	X.X
3	X.XX
4 (Default)	X.XXX

CC43: PPU Blinking Options

CC43 is entered in Level 2 programming to set PPU blinking functionality. This allows programmer to set PPU displays for blinking on and off under certain conditions. On a single-hose blender, MPD, or Master/Satellite, setting this option to “ON” will cause all PPU displays to blink after pump handle is raised (or in this case if nozzle is removed from an auto-on pump handle). Blinking stops when grade select button is pressed. Default for this option is selection 1. This programming option affects all sides of dispenser – all sides are programmed the same.

Note: CC43 is used to specify PPU blinking options for dispensers without grade indicator lights installed with the PPU displays. Because of this, this option is typically not used with Encore S/700 S units. When CC96 is set to 1, regardless of dispenser model type, this CC will function as described.

Figure 5-37: PPU Blinking Options



1. Press **4**, **3**, and then press **ENTER** to access this CC.
2. Select “ON”/“OFF” Option using the following table.

   The following table shows the available options for this CC:

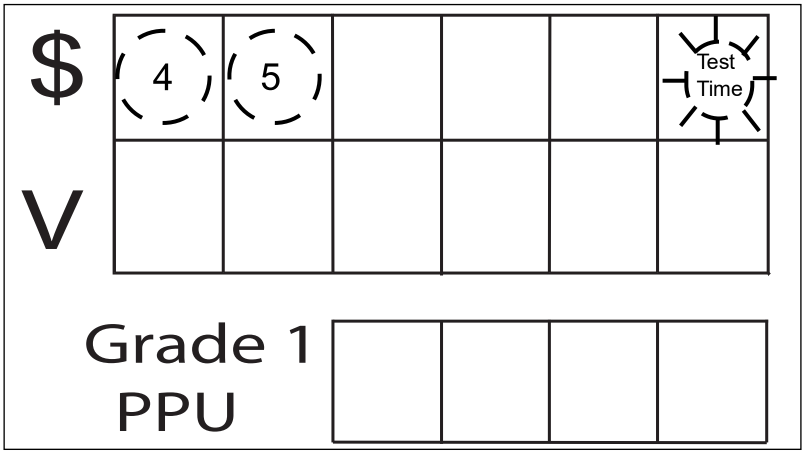
Blinking Code Option	State
1 (Default)	On
2	Off

3. Press **ENTER**.

CC45: Set Lamp Test Time

CC45 is entered in Level 2 programming to set the time duration of lamp test. Lamp test time determines the amount of time allotted for the system to run a line leak test before opening the dispenser valves. Maximum allowed lamp test time is 9 seconds, minimum is 0 seconds, and default test time is 3 seconds. Basic test sequence is all 8s, blanks, 0s, transaction PPU.

Figure 5-38: Setting Lamp Test Time



1 Press **4**, **5**, and then press **ENTER** to access this CC.

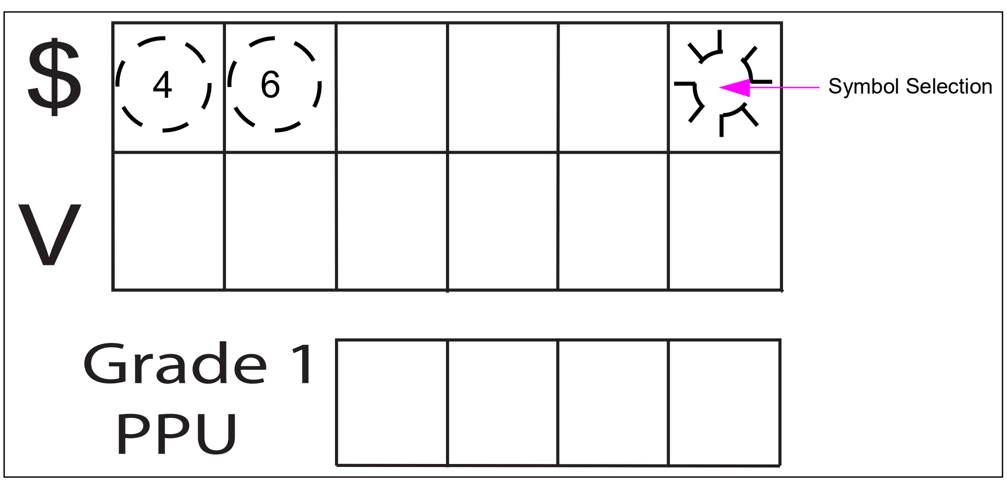
2 Select Lamp Test time (**0-9** seconds).

3 Press **ENTER**.

CC46: Set Comma/Decimal Point

CC46 is entered in Level 2 programming to select comma or decimal point display at main transaction and PPU displays.

Figure 5-39: Setting Comma/Decimal Point



1 Press **4**, **5**, and then press **ENTER** to access this CC.

2 Select the symbol option (1-2) using the following table:

Symbol Options	Symbol
1 (Default)	Decimal Point
2	Comma

3 Press **ENTER**.

CC47: Set Reported Money Size

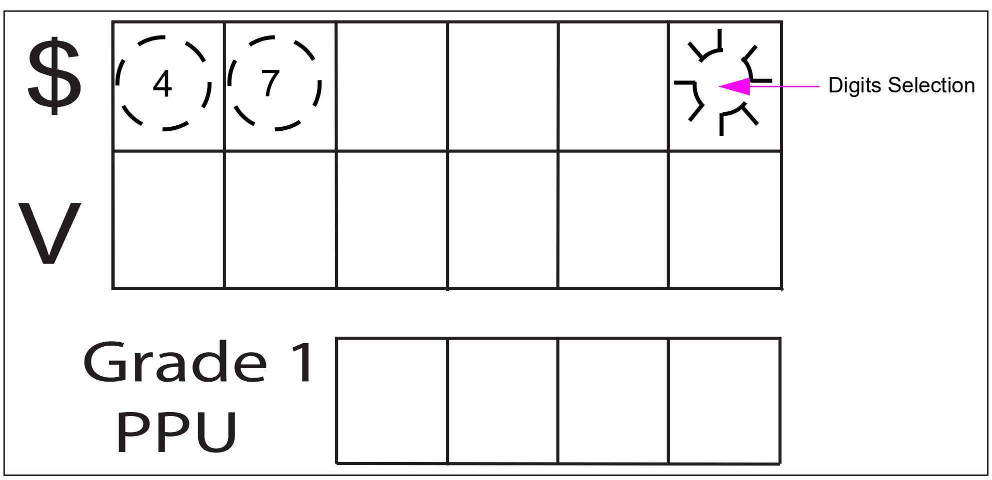
Note: This section is for pump software version earlier than V02.9.06 only. For information on pump software version later than V02.9.06, proceed to “CC47: Set Two-wire Options” on page 5-57.

CC47 is entered in Level 2 programming to set the method by which transaction money amount is returned to console over two-wire communication loop. This CC is for third-party international consoles supporting six digits. Selection 1 is default and returns money in the following format with an implied decimal point as:

XXX.XXY

For programming option 1, least significant digit (shown as Y in the above example) is not used and must be discarded. All known consoles in the U.S. use this as default. Option 2 means that all six digits of money display are returned with an implied decimal point as selected for money display. In this case, all six digits of money reported over the two-wire communications loop are meaningful.

Figure 5-40: Setting the Reported Money Size



1 Press **4**, **7**, and then press **ENTER** to access this CC.

2 Select the Digit Selection Option (1-2) using the following table:

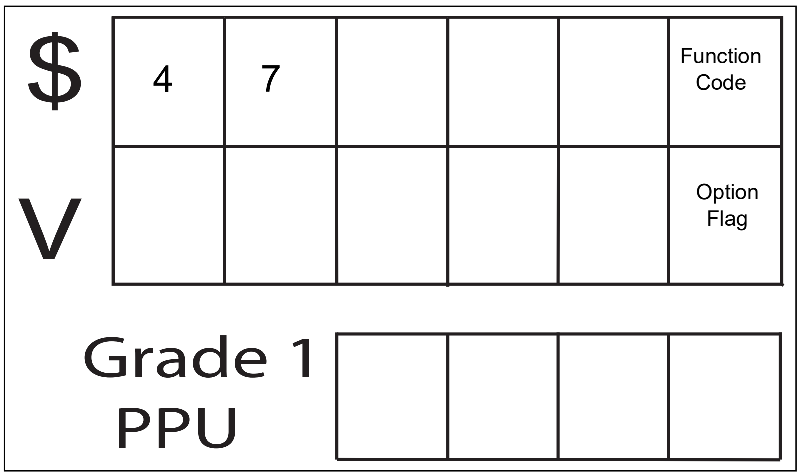
Digit Selection Option	Return Digits
1 (Default)	Return 5 Digits
2	Return All 6 Digits

3 Press **ENTER**.

CC47: Set Two-wire Options

This command allows the user, technician, or factory associate to set up the two-wire communications parameters. Historically, CC47 was solely used to set the 5 or 6 digit two-wire money mode. However, now this CC is divided into multiple FCs. However, if, over ZModem, a configuration file is received that sets CC47 with no FC, this will be interpreted as setting only the 5 or 6 digit money mode. This CC is protected by the security switch. Without the switch being open, the unit will double beep and prevent entering a new value. All changes will be logged into the Regulatory Log.

Figure 5-41: Setting the Reported Money Size



The following table describes the relationship between FCs and installable options:

FC	Dispenser Option Description
1 (Default)	Set Reported Money Size
2	Set Reported Volume Format
3	Set VRC States in Transaction Data
4	Set Status Response Format
5	Set Baud Rate
6	Set Communications Lost Method
7	Set Real-time Data Format
8	Set Preset With Pump Stop Method
9	Set Remote Preset Display

FC1: Set Reported Money Size

For programming option 1, the least significant digit (shown as Y in the example above) is not used and should be discarded. All known consoles or Point of Sale (POS) in the USA use this default.

Option 2 means that all six digits of the money display are returned with an implied decimal point as selected for the money display. In this case all 6 digits of the money reported over the 2-wire communications loop are meaningful.

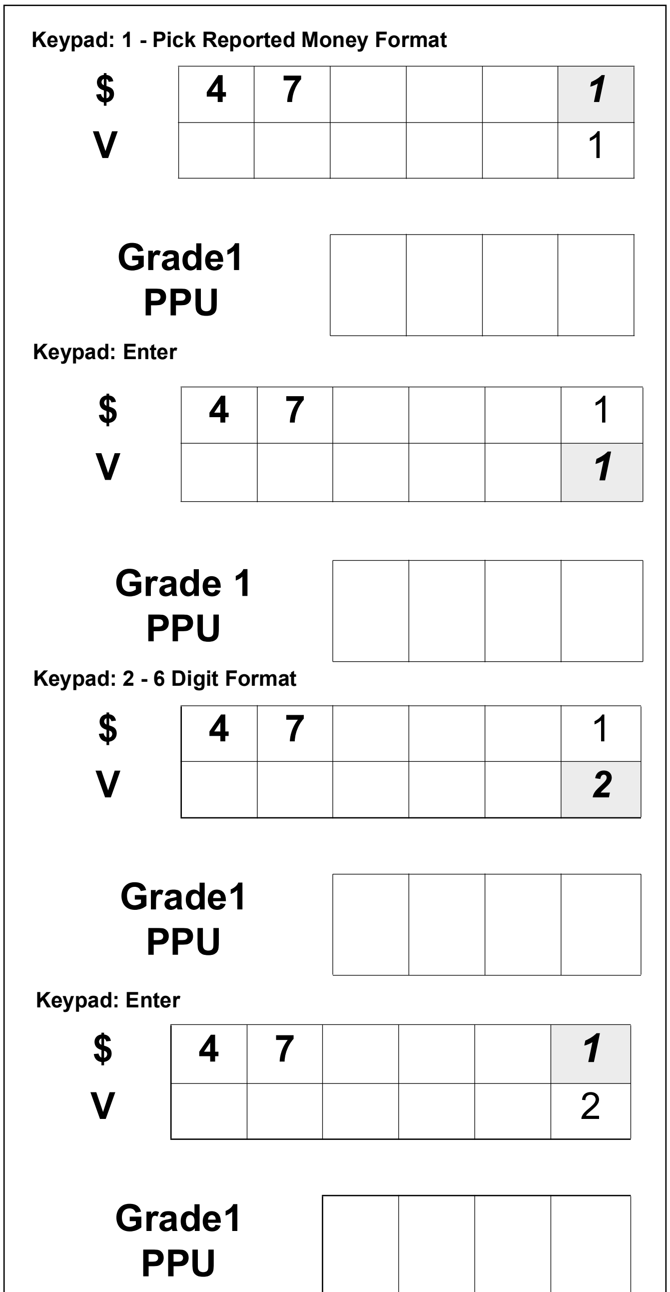
Option 3 and 4 set to extended two-wire modes.

The following table shows the available options for this FC:

Selection	Option
1 (Default)	5 Digit Money
2	6 Digit Money
3	7 Digit Money (Extended Two-wire)
4	8 Digit Money (Extended Two-wire)

Figure 5-42 shows an example on how to set report money size as 6 digit.

Figure 5-42: Setting the Reported Money Size as 6 Digit - Example



Note: The F1 key will allow additional programming or the F2 key may be used to exit programming.

FC2: Set Reported Volume Format

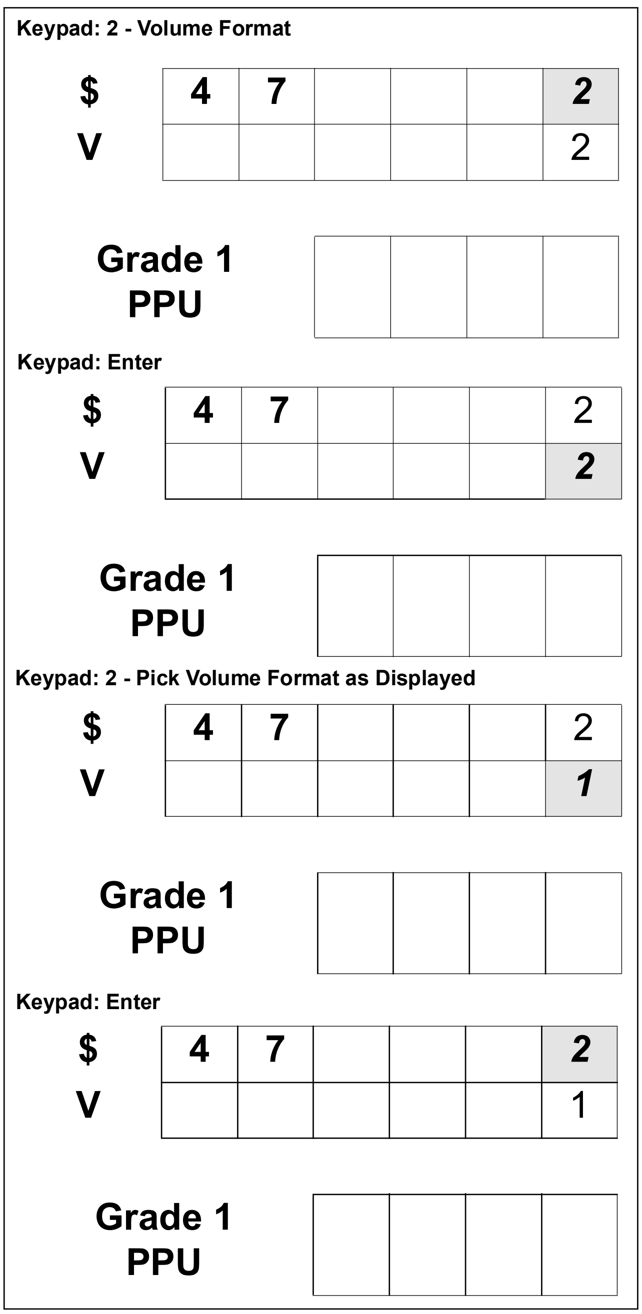
For standard two-wire modes (CC47 FC1 set to 1 or 2), this function can be used to report volume in either XXX.XXX format by default or report volume as it is displayed. This is the volume as reported in Real-Time volume or Transaction Data only.

The following table shows the available options for this FC:

Selection	Option
1	Volume As Displayed
2 (Default)	Volume In XXX.XXX

Figure 5-43 shows an example on how to set the volume in “Volume as Displayed” format.

Figure 5-43: Setting the Volume Format to “Volume as Displayed” - Example



FC3: Set VRC States in Transaction Data

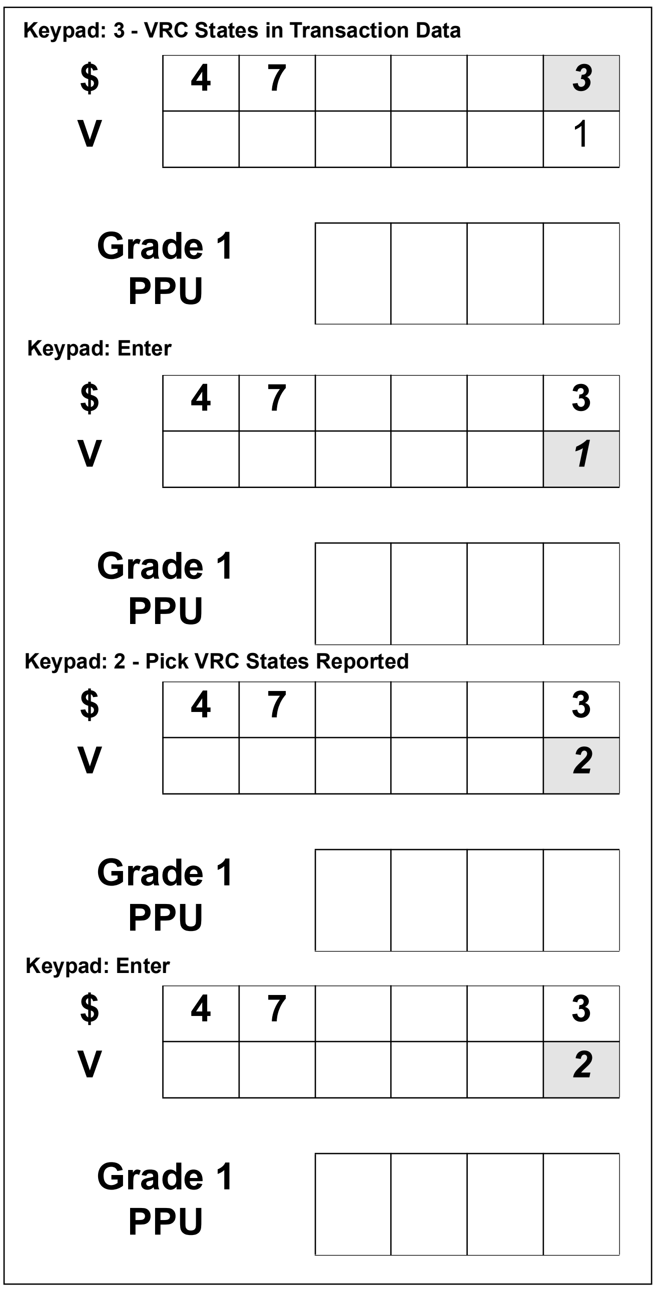
This option will allow setting VRC states for Vaporix Vapor Recovery in the Error Bits for the transaction data response messages.

The following table shows the available options for this FC:

Selection	Option
1 (Default)	Disabled (Standard Transaction Errors)
2	Vaporix Errors in Transaction Data

Figure 5-44 shows an example on how to set the option to Vaporix Error Reporting.

Figure 5-44: Setting the Option to Vaporix Error Reporting - Example



FC4: Set Status Response Format

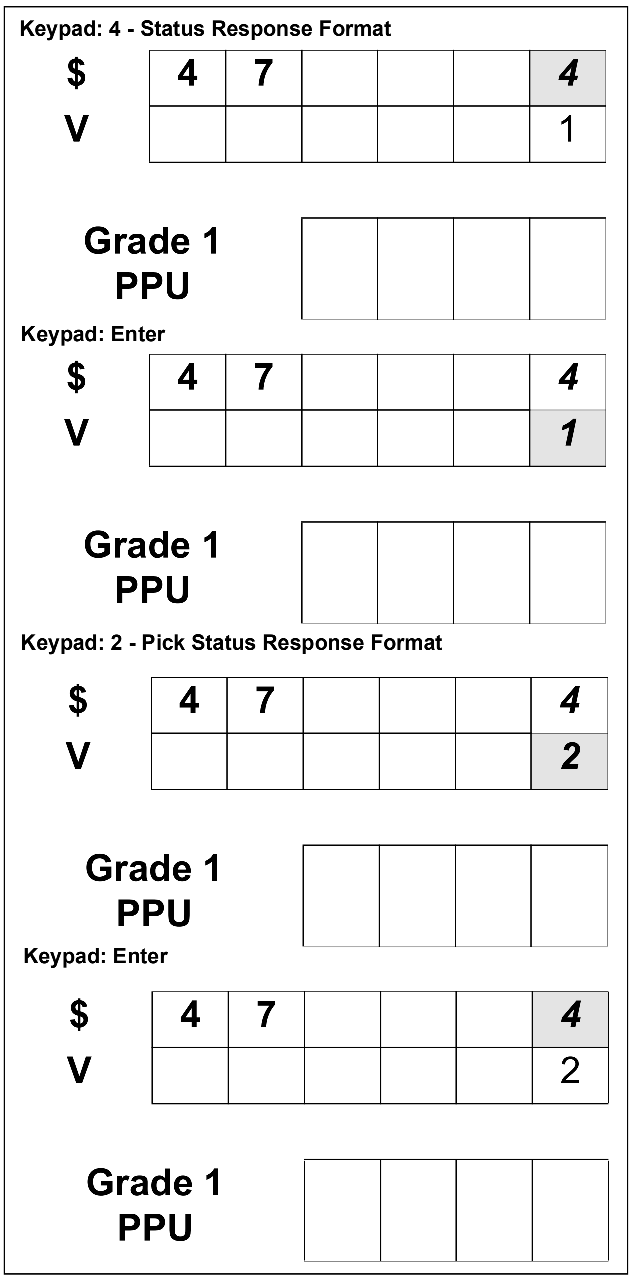
For standard or extended two-wire modes, an alternate status response message can be configured for specialty equipment to gather extra information during BUSY, STOP, and FEOT/PEOT pump states.

The following table shows the available options for this FC:

Option	Definition
1 (Default)	Standard Response
2	Alternate Response

Figure 5-45 shows an example on how to set alternate status poll responses.

Figure 5-45: Setting the Alternate Status Poll Responses - Example



FC5: Set Baud Rate

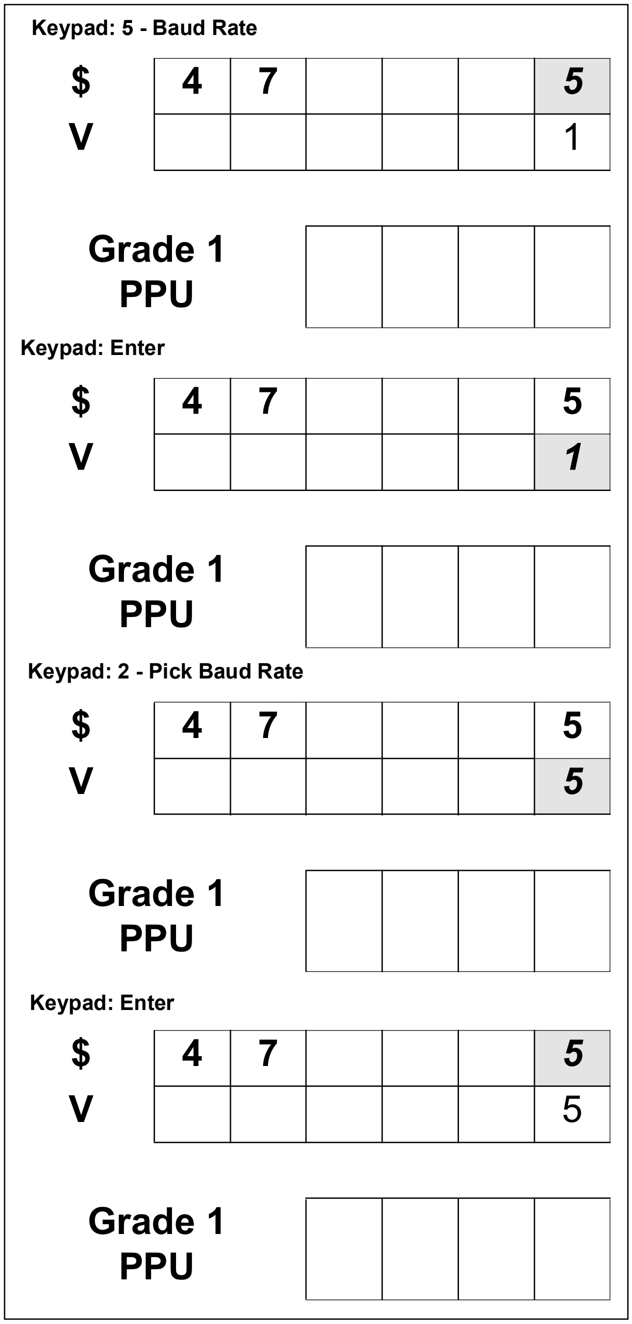
Many markets require different two-wire baud rates. The Encore 500/500 S can handle any baud rates specified here, depending on hardware used external to the pump for managing communications.

The following table shows the available options for this FC:

Option	Definition
1 (Default)	5787 (Standard)
2	1200
3	2400
4	4800
5	9600
6	19200

Figure 5-46 shows an example on how to set a baud rate of 9600.

Figure 5-46: Setting the Baud Rate of 9600 - Example



FC6: Lost Communication Function

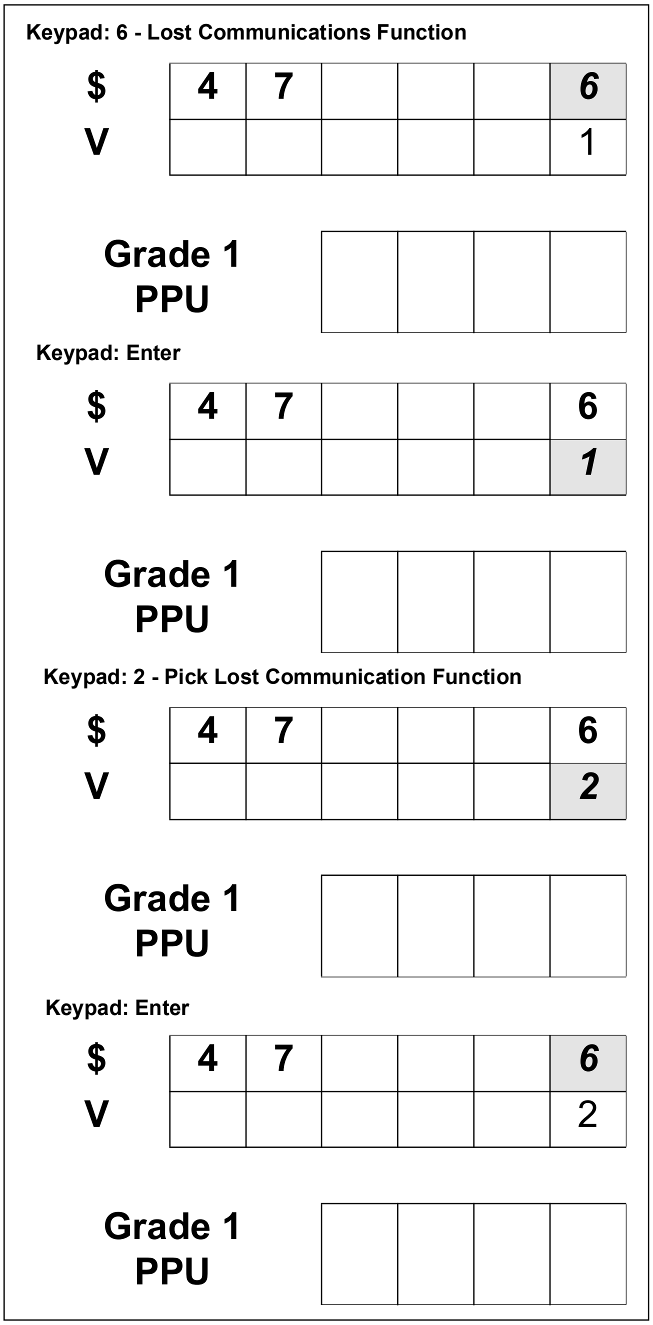
The setting of this function determines the method for how the pump operates when communicating to a controlling system over two-wire cease. The default method of the pump, for both safety and theft reasons, is to suspend the transaction based upon the timeout setting in CC53 FC6. However, a special option can be set to allow the current transaction to continue dispensing fuel, despite communications being lost.

The following table shows the available options for this FC:

Selection	Option
1 (Default)	Suspend (Based on Timeout)
2	Continue

Figure 5-47 is an example showing setting to continue on communications lost.

Figure 5-47: Setting to Continue on Communications Lost - Example



FC7: Real-Time Data Format

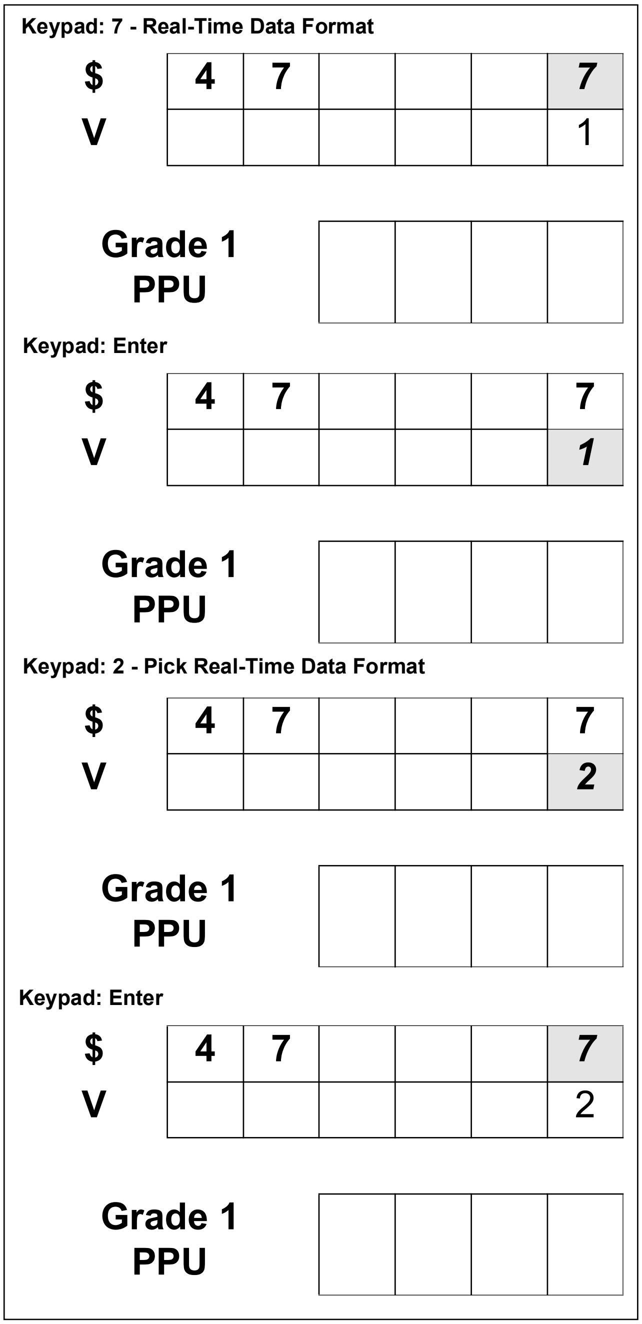
The default setting for two-wire in standard or extended mode is Real-Time Money. An alternate Real-Time Volume is configurable to be sent over the same Real-Time Money message where the Volume is representative of the Transaction Volume format. Two new options exist for Real-Time All Data where sends Money, Volume, and Grade; and an Alternate Real-Time Volume (command 7) that can augment Real-Time Money. This setting must be compatible with the controller that the pump communicates to.

The following table shows the available options for this FC:

Selection	Option
1 (Default)	Real-Time Money
2	Real-Time Volume
3	Real-Time All Data
4	Real-Time Money with Alternate Real-Time Volume

Figure 5-48 shows an example on how to set Real-Time Volume.

Figure 5-48: Setting Real-Time Volume - Example



FC8: Preset Mode Without Pump Stop

The setting of this function determines the method for how the pump handles two-wire Preset Commands. By the Gilbarco two-wire specification, a pump can only accept one preset message and all subsequent presets will not be honored unless a pump stop is sent to cancel the previous preset. Setting this option to 2 will allow all preset messages to be handled, in essence, overriding the previous preset as long as a pump is in OFF, AUTH, or CALL state.

The following table shows the available options for this FC:

Selection	Option
1 (Default)	Allow One Preset
2	Allow Any Preset Until Fueling Starts

Figure 5-49 shows an example on how to preset mode without pump stop.

Figure 5-49: Setting Preset Mode Without Pump Stop - Example

Keypad: 8 - Preset Mode Without Pump Stop

$	**4**	**7**				***8***
V						1

Grade 1 PPU				

Keypad: Enter

$	**4**	**7**				8
V						***1***

Grade 1 PPU				

Keypad: 2 - Pick Preset Mode

$	**4**	**7**				8
V						***2***

Grade 1 PPU				

Keypad: Enter

$	**4**	**7**				***8***
V						2

Grade 1 PPU				

FC9: Remote Preset Display

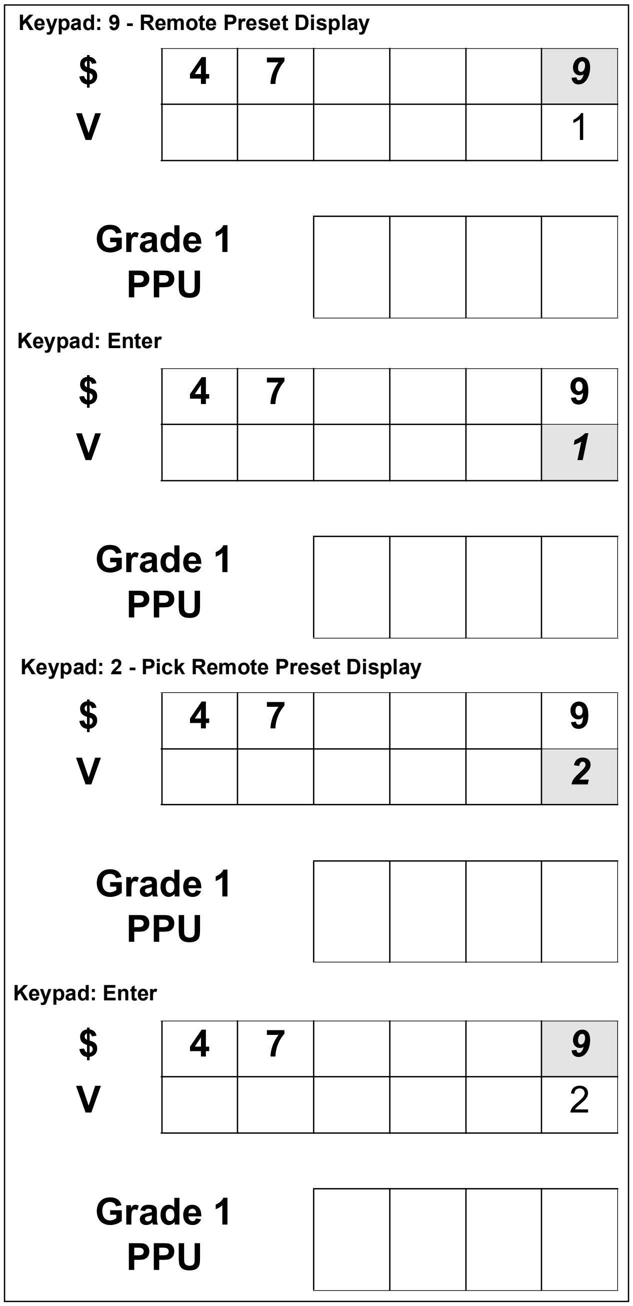
The setting of this function determines whether the remote preset display functionality is enabled or not. Setting this value to 1 prevents remote preset display from occurring. Setting this value to do will show a remote preset (received from a controller on Gilbarco two-wire). A money preset will be shown in the money display and subsequently a volume preset will be shown in the volume display. The method of remote preset display will be dependent on the function of CC48 (Zero Previous Transaction) on whether the opposite display is blank or not and whether the display will flash or not. The duration of the remote preset display is dependent on the setting of the preselect timeout in CC53, FC2. If no timeout is set, the remote preset display will be persistent. If a timeout is set, the remote preset display will be shown for the number of seconds configured. When the display is reverted, the previous transaction data is refreshed and displayed.

The following table shows the available options for this FC:

Selection	Option
1 (Default)	No Remote Preset Display
2	Remote Preset Display Configured

Figure 5-50 shows an example on how to set remote preset display.

Figure 5-50: Setting Remote Preset Display - Example



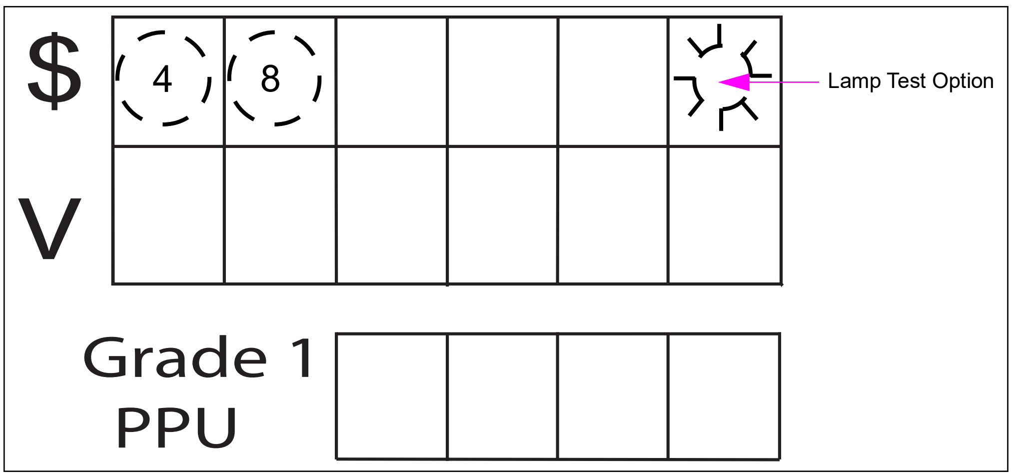
CC48: Zero Previous Transaction

CC48 is entered in Level 2 programming to determine when dispenser displays will be zeroed out and previous transaction cleared internally. Default is option 1 after lamp test. Option 2, before lamp test, results in the dispenser being reset to zero, the transaction displayed after:

- The pump handle is activated (raised) and unit is authorized
- No fuel is dispensed
- The handle is deactivated (lowered)

Note: Option 2 is required in New Jersey and is generally known as the New Jersey Display Option.

Figure 5-51: Zero Previous Transaction



1 Press **4**, **8**, and then press **ENTER** to access this CC.

2 Select (1-2) for the Lamp Test Option using the following table:

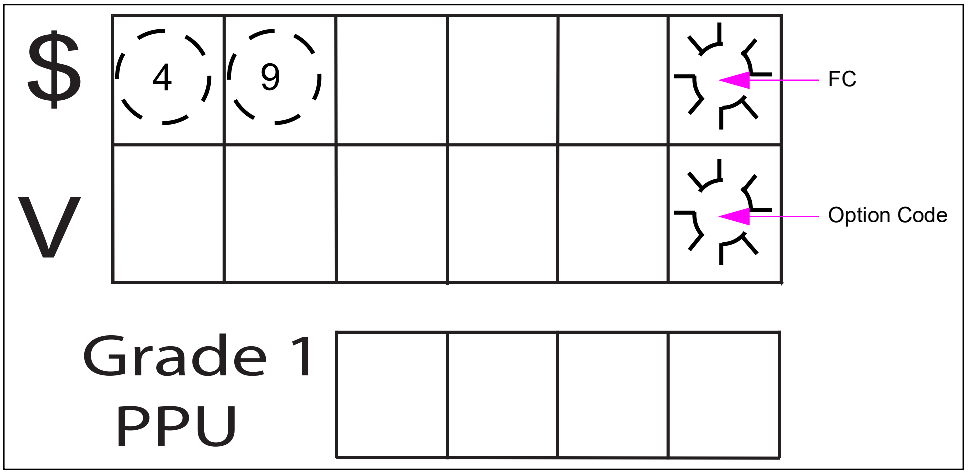
Option	Setting
1 (Default)	After Lamp Test
2	Before Lamp Test
3	Before Lamp Test or After Preset Keypad Use (Chile)

3 Press **ENTER**.

CC49: STP Controls

CC49 programming is used to set when the STPs will turn on, which STPs will be activated, and where the STP relay signal originates. Sub-sections that follow describe each FC programming option. Layout and meaning of each digit position for this programming feature are shown in Figure 5-52.

Figure 5-52: STP Controls

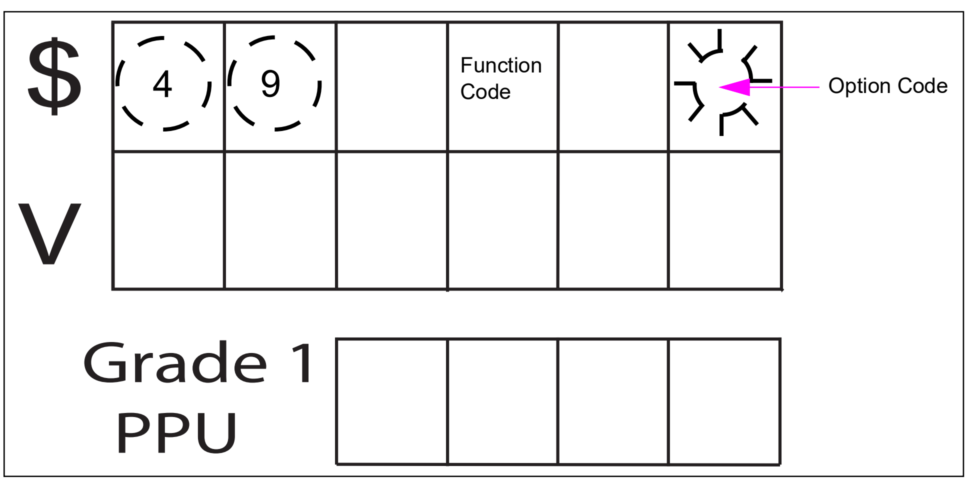


1 Press **4**, **9**, and then press **ENTER** to access this CC.

2 Select the FC (1-3) using the following listing.

3 Select the appropriate Option Code.

Figure 5-53: Selecting the Option Code



FC1: Set STP Pre-start

STP pre-start must be programmed depending on the type of leak detection implemented in the dispenser. Pre-start must be set to "ON" for mechanical line leak detectors and "OFF" for electrical line leak detectors. Default for this option is selection 1, "PRESTART OFF". "PRESTART ON" turns the STPs on when the handle switch is activated.

1 Select option (1-2) using the following table:

Option	Function
1 (Default)	Pre-start OFF
2	Pre-start ON

2 Press **ENTER**.

FC2: Set STP Control

STP control determines how STPs will be controlled during a fueling operation. Option 1 is default and results in only the STP(s) required for fueling operation to be turned on. Option 2 turns on all STPs anytime there is a fueling operation.

1 Select option (1-2) using the following table:

2 Press **ENTER**.

Option	STP Option
1 (Default)	Turn on ONLY selected STP (Default)
2	Turn on ALL STPs

FC9: Set STP Side Mapping

STP/side mapping is used in the event that a dispenser (only applicable to Unit Type 1) needs to provide different products (hence different STPs) for the same grade selection on opposite sides. The DEFAULT for this option is selection 1, 'ONLY STP 1 USED'. This configuration will have no change on any pump other than Unit Type 1. For more information, refer to "CC90: Set Unit Type" on page 5-119.

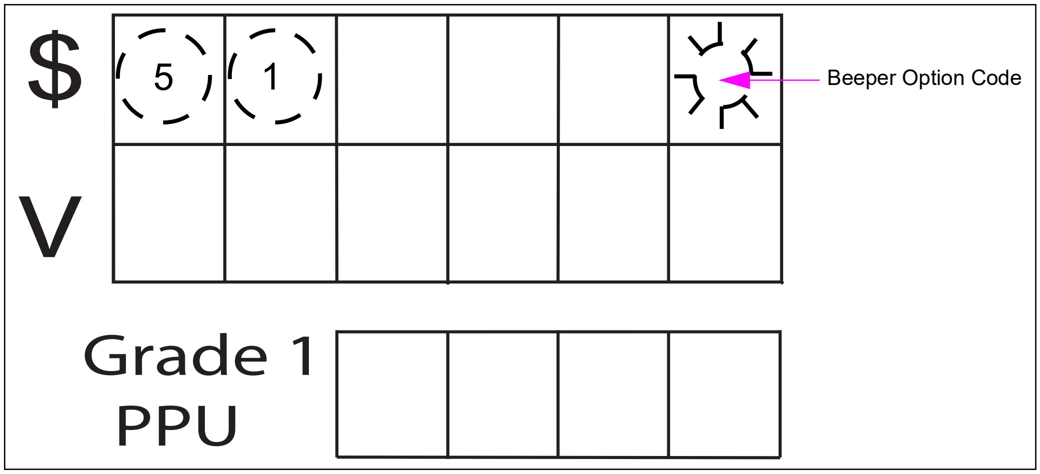
Option	STP Side Mapping
1 (Default)	Only STP 1
2	Side 1 is STP 1, Side 2 is STP 2

IMPORTANT INFORMATION
CC50 is only available in pump software V01.7.85L or earlier.

CC51: Set Main Beeper

CC51 is entered in Level 2 programming to set beeper functionality. The following table indicates beeper programming options. If beeper option is set to “ON”, the beeper will sound when pump handles are lifted, when the dispenser is authorized, and when any pump switches/keypads are activated. Default for this option is selection 1 - beeper “ON”.

Figure 5-54: Setting the Main Beeper



1 Press **5**, **1**, and then press **ENTER** to access this CC.

2 Select option (1-2) using the following table:

Option Code	Beeper
1 (Default)	ON
2	OFF

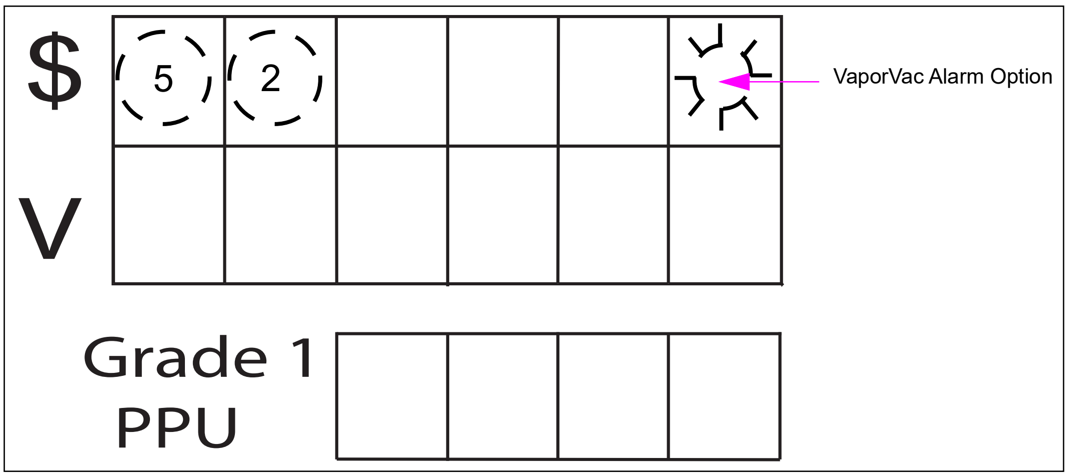
3 Press **ENTER**.

CC52: Set VaporVac Alarm

CC52 is entered in Level 2 programming to set VaporVac alarm functionality. The following table indicates alarm programming options. If VaporVac alarm is set to “ON”, the beeper will turn on at a VaporVac error and will not turn off until the dispenser has been warmstarted. *Note: In Mexico, the VaporVac alarm must be set to “ON”.*

Default for this option is selection 1 - VaporVac alarm “ON”.

Figure 5-55: Setting VaporVac Alarm



1 Press **5**, **2**, and then press **ENTER** to access this CC.

2 Select option using the following table:

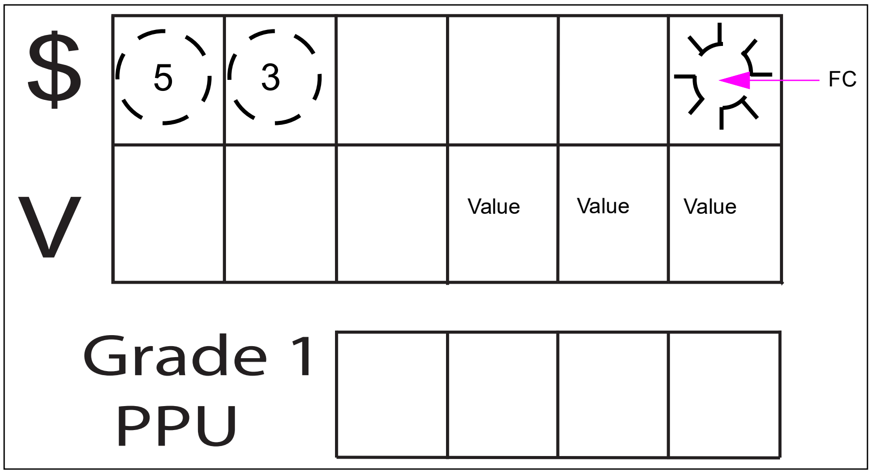
Option Code	VaporVac Alarm
1 (Default)	ON
2	OFF

3 Press **ENTER**.

CC53: Set Timeouts

CC53 is entered in Level 2 programming to set the dispenser timeout options. FCs 1 through 3 are selected to set various dispenser timeouts. Sub-sections that follow describe each timeout programming option. Entry into CC53 will select FC1 by default. Layout and digit position meaning for this programming feature are shown in Figure 5-56.

Figure 5-56: Setting Timeouts

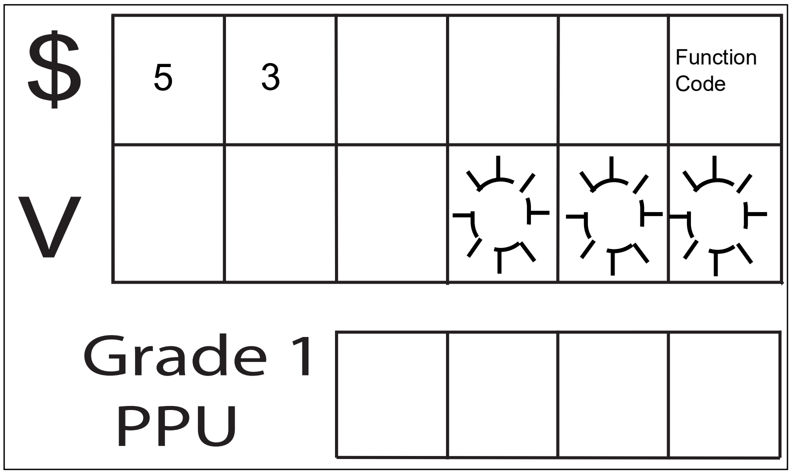


1 Press **5**, **3**, and then press **ENTER** to access this CC.

2 Select option (1-3) using the following table:
Note: For more information, refer to "FC Descriptions for CC53" on page 5-78.

FC Option	Timeout Option
1	Beeper Timeout
2	Pre-select Timeout
3	Dispenser Timeout
4	Initial Dispenser Timeout
5	Initial Beeper Timeout
6	Two-wire No Communications Timeout
7	Display Backlight Timeout
8	Maximum Authorize Time
9	Maximum Dispensing Time
10	Disable Dispenser Timeout
11	Stop Door Security Alarm Timeout

Figure 5-57: FC Settings



3 Press **ENTER**.

FC Descriptions for CC53

FC1: Set Beeper Timeout

This programming option determines the amount of time the beeper sounds after the dispenser timeout has expired. This timer starts as soon as the dispenser timeout expires and continues to run (beeping once per second) until it has expired, the nozzle is replaced, or pulses are received. Upon expiration, the fueling point will deauthorize with EC29. The allowed range for this timeout is 0 to 999 seconds. The DEFAULT value is 0 or no beeper timeout. If this option is set to 0, the fueling point will deauthorize immediately following the dispenser timeout.

FC2: Pre-select Timeout

Pre-select timeout will set pre-select timeout timer period allowed for grade select buttons to be pressed, before handle is raised for a non-Nozzle activated unit. Timer starts when the dispenser is authorized and times out until the customer lifts the pump handle. After pre-select timeout has occurred, grade selected is lost and must be re-selected by customer. Pump does NOT change state when this timeout occurs. A zero value means there is no timeout. Pump Preset is also under the control of pre-select timeout. If pre-select timeout occurs, preset keypad entry must re-entered. Allowed range for this timer is from 0 to 999 seconds. Default value for this function is 15 seconds.

FC3: Dispenser Timeout

This programming option will set the time allowed for a pump to remain authorized without receiving pulses before starting the beeper timeout. This timer starts as soon as pulses stop being received and continues to run until it has expired, the nozzle is replaced, or pulses are received. Upon expiration, the fueling point will start the Beeper Timeout. The DEFAULT value for dispenser timeout is 0 seconds.

If fueling is resumed, beeper will stop sounding and timer is reset.

Note: If this is a nozzle activated (Option 91 Function 12) unit, this value is ignored and treated as if value was set to "0". ***NOZZLE ACTIVATED PUMPS (AUTO-ON) MUST NOT ALLOW PRE-SELECTION OF GRADE or PUSH-TO START****. Pump Preset can still be preselected.*

FC4: Initial Dispenser Timeout
This programming option will set the time allowed for a pump to remain authorized without receiving pulses before starting the Initial Beeper Timeout. This timer starts as soon as lamp test finishes and continues to run until it has expired, the nozzle is replaced, or pulses are received. Upon expiration, the fueling point will start the Initial Beeper Timeout. The allowed range for this timeout is 0 to 999 seconds. The DEFAULT value for Initial Dispenser Timeout is 0 seconds. This option is only available in pump software 01.8.30 or 02.8.30 and greater.

FC5: Initial Beeper Timeout
This programming option determines the amount of time the initial beeper sounds after the initial dispenser timeout has expired. This timer starts as soon as the initial dispenser timeout expires and continues to run (beeping once per second) until it has expired, the nozzle is replaced, or pulses are received. Upon expiration, the fueling point will deauthorize with EC29. The allowed range for this timeout is 0 to 999 seconds. The DEFAULT value is 0 or no beeper timeout. If this option is set to 0, the fueling point will deauthorize immediately following the initial dispenser timeout. This option is only available in pump software 01.8.30 or 02.8.30 and greater.

FC6: Two-wire No Communication Timeout
This programming option will set the amount of time allowed that the dispenser can receive no communications while fueling until the sale is suspended. The sale can be restarted with an authorization. The DEFAULT value for the two-wire No Communication Timeout is 30 seconds. This option will be ignored if CC47 FC6, Lost Communications Function, is set to “Continue Transaction”. The allowed range for this timeout is 0 to 999 seconds. This option is only available in pump software 02.8.86 or greater.

FC7: Display Backlight Timeout
This programming option will set the amount of time allowed for an idle pump that will dim the display backlights for Door Nodes that support this feature. Presently, Door Node 5 and Door Node 5 7 Digit are the only two model Door Nodes that support this. And Idle pump is any pump not in Keypad Dispenser Programming, W&M Review, Totals Review, Authorized, with Handle out or Grade selected. In any of these scenarios, the display backlight will be normal. Any time any of these features stops, then timer will start for the duration of this configured time in seconds. Once the timer expires, the backlight will dim. The DEFAULT value for the display backlight timeout is 0 seconds. This option is only available in pump software 02.9.42 or later.

FC8: Maximum Authorize Time
This programming option will set the amount of time allowed to remain in two-wire authorize state before a transaction starts. When configured for 0, Authorize state is allowed indefinitely. Programming to a non-zero number is allowed in increments of 10. Programming for less than 10 will result in a 0 value being used. Programming to 11 means that 10 is stored. The reason for this is that the internal International Forecourt Standards Forum (IFSF) data only stores this value as 0 to 250 values stored as increments of 10. Maximum value entered can be 2500.

When this timeout is configured, this is the maximum time Authorize state is allowed before putting the pump back to the OFF (Idle) state.

The DEFAULT value for the Maximum Authorize Time is 0 seconds (off).

This option is only available in 03.0.01 or later.

FC9: Maximum Dispense Time
This programming option will set the amount of time allowed to perform a transaction once a transaction starts and motors/valves are actuated. When configured for 0, dispensing time can be for as long as the customer allows the transaction to be or until a limit is reached. Programming to a non-zero number is allowed in increments of 10. Programming for less than 10 will result in a 0 value being used. Programming to 11 means that 10 is stored. The reason for this is that the internal IFSF data only stores this value as 0 to 250 values stored as increments of 10. Maximum value entered can be 2500.

When this timeout is configured, this is the maximum time allowed for dispensing. In two-wire mode, when the limit is reached the pump will go to the STOP state and can be reauthorized. When in standalone mode, the pump goes to an OFF state and cannot continue.

The DEFAULT value for the Maximum Fill Time is 0 seconds (off).

This option is only available in 03.0.01 or later.

FC10: Disable Dispenser Timeout
FC10 is used to set the disable timeout feature. If CC91 FC37 or FC42 is set to 3 or 5, this value sets the durations (in minutes) for which the side or dispenser will be disabled after the upper door or hydraulics panel is closed. For example, if the value is 10 and the door or panel is closed, the side or dispenser is automatically enabled after 10 minutes. If the alarm option is enabled (CC91 FC38), the alarm will shut off when the side or dispenser has been enabled.

FC11: Stop Door Security Alarm Timeout
FC11 is used to set the alarm timeout feature. If CC91 FC38 is set to 2, this value sets the duration (in seconds) for which the alarm will sound, provided it is not silenced by clearing the condition that initiated it. For example, if it is set to 600 (10 minutes) and the upper door or hydraulics panel is left open, the alarm will turn off after 10 minutes even though the door or panel remains open. If the condition that initiated the alarm is cleared (door or panel is closed and the unit is enabled) before the timeout is reached, the alarm will turn off.

FC12: End of Transaction Timeout
FC12 is used to set the two-Wire End of Transaction timeout feature. The timeout values are entered in minutes. The default value is zero, which means no timeout. This FC is introduced at one of our customer requests to resolve the fuel sale completion issue at their bank. PCN will send an EOT to the FCC/POS after the timeout even if the nozzle is not returned to the nozzle boot.

CC54: Set Slowdown Valve

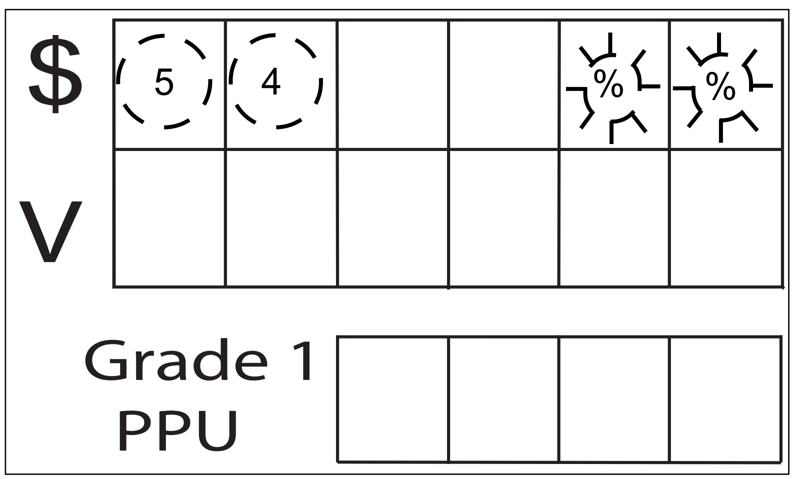
Note: This option is not allowed to be programmed when the Volume Unit Type is set to US gallons.

For Software V01.7.85 and Earlier Only

CC54 is entered in Level 2 programming to determine when the dispenser goes into slowdown mode before reaching transaction preset value. Programmer enters slowdown value using digit keys. Slowdown value is in terms of a percentage of pulses per unit volume with a default value of 20% (pulses/unit). Range of values is 0 to 99 percent. A value of 0 means no slowdown. Layout and meaning of each digit position for this programming feature are shown in Figure 5-58.

This parameter is set to reduce “preset” problems with missed targets. Experimentation is typically required to find a suitable value. It is typically used when multiple units are experiencing preset problems.

Figure 5-58: Setting Slowdown Valve - 1



1 Press **5**, **4**, and then press **ENTER** to access this CC.

2 Select percentage (**0-99**).

3 Press **ENTER**.

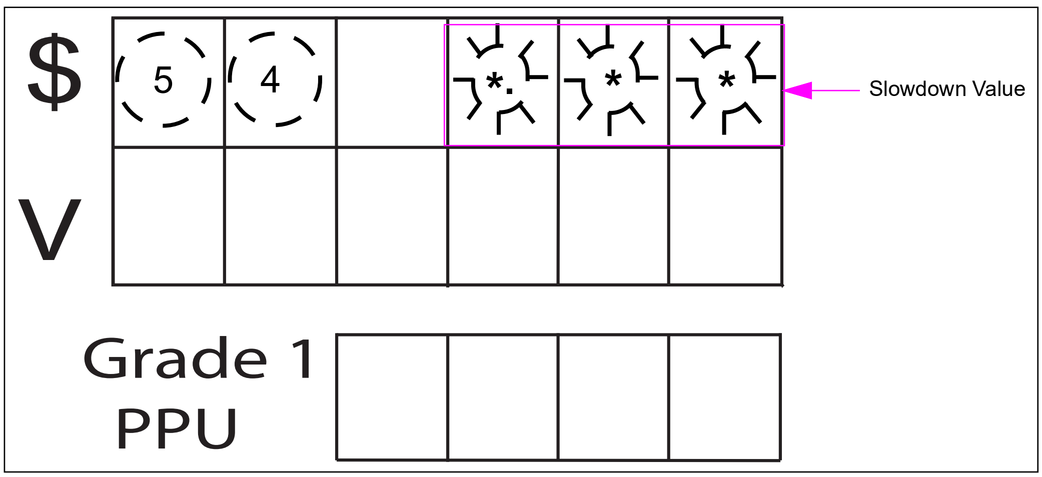
CC54

For Software V01.7.85 and Later

CC54 is entered in level two programming (or higher) to determine when the dispenser goes into slowdown mode prior to reaching the transaction preset value. The programmer enters the slowdown value using the digit keys. The slowdown value is in terms of volume units with a DEFAULT value of 0.20. The range of values is 0.20 to 9.99 for US gallons and 0.30 to 9.99 for liters. All changes will be logged into the Regulatory Log.

Minimums have been implemented as of pump software 02.9.42P and later. Slowdown cannot be set less than 0.6 liters or 0.2 gallons.

Figure 5-59: Setting Slowdown Valve - 2



1 Press **5**, **4**, and then press **ENTER** to access this CC.

2 Enter the amount of volume before preset to enter slow down mode.

3 Press **Enter**.

CC55: Set Hose Pressurization

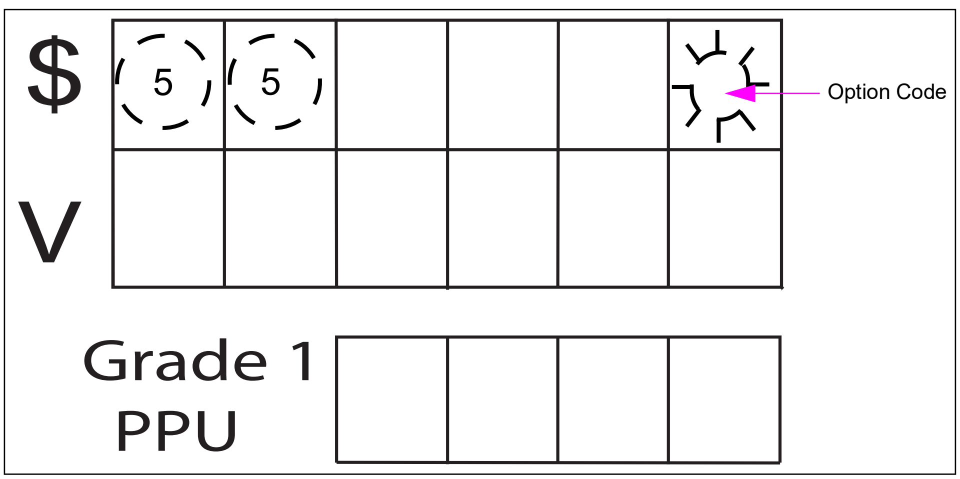
Note: This option is not allowed to be programmed when the Volume Unit Type is set to US gallons.

CC55 is entered in Level 2 programming to pressurize hose before dispensing, by not registering flow up to 20 pulses or 1 second, whichever comes first.
Note: This flow will open on the customers display.

This option is used in cold climates, where fuel in hose may have contracted after a period of inactivity. This can cause what is commonly referred to as ‘meter jump’ at the beginning of a transaction. Default value is selection 1, “OPTION OFF”. This function will pressurize hose during lamp test of first transaction after power-up or during lamp test of first transaction after a period of inactivity defined by programming option code.
Note: Fuel passing through meter during hose pressurization will not be registered on displays nor counted toward the sale before start of delivery.

IMPORTANT INFORMATION
CC55 is not intended as a fix for defective equipment such as a leaky meter, meter check valve, nozzle or vapor recovery hoses. Always fix the problems quickly.

Figure 5-60: Setting Hose Pressurization



1 Press **5**, **5**, and then press **ENTER** to access this CC.

2 Select option (1-3) using the following table:

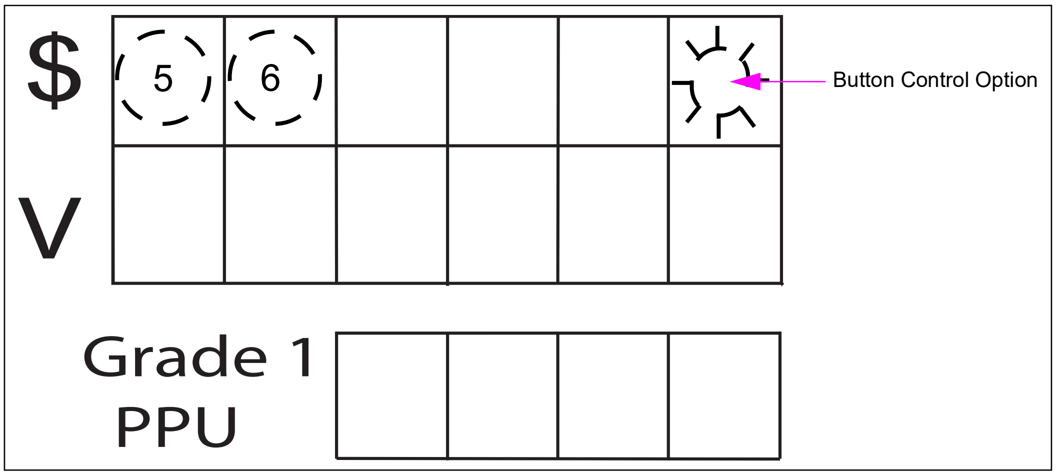
Option	Hose Option
1 (Default)	Option OFF
2	Unit has sat inactive for 10 minutes
3	Unit has sat inactive for 30 minutes.

3 Press **ENTER**.

CC56: Set Stop Button Control

CC56 is entered in Level 2 programming to set stop button control. Programming is allowed but will only be effective if a “STOP” button has been selected as INSTALLED in the Level 4 programming mode. Option 1 means that each STOP button will only stop its own side. Selection two means that STOP button will stop both sides of the dispenser. Default is selection 1 for one side only.

Figure 5-61: Setting Button Control Options



1 Press **5**, **6**, and then press **ENTER** to access this CC.

2 Select option (1-2) using the following table:

Option	Button Control
1 (Default)	Stop one side only
2	Stop both sides

3 Press **ENTER**.

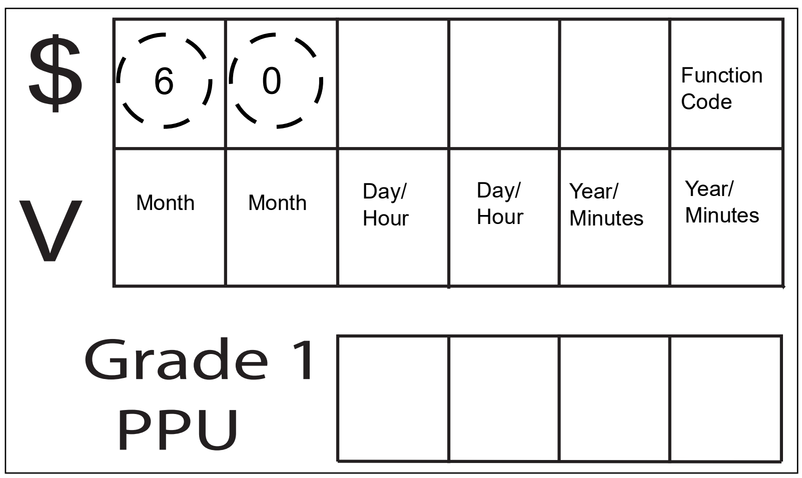
CC60: Set Clock and Calendar

Settings for time and date programming may be completed from the manager keypad or sent from console/controller. As a result, the dispenser always uses last settings made by either programmer or console/controller. Not setting the correct values may affect reports, closing shifts, and dispenser event logs. Properly time stamped event logs are valuable in service situations.

There are two FCs applicable to CC60, FC1 for setting time and FC2 for setting date. Default selection after entering into CC60 is FC1 for setting time.

Layout and digit positions are shown in Figure 5-62.

Figure 5-62: Setting Clock and Calendar



1 Press **6**, **0**, and then press **ENTER** to access this CC.

FC1: Set Dispenser Time
Dispenser time is set in 24 hour format. Programmer enters the time in terms of hours and minutes. If time programming is not done before, then time will default to 12:00. The following table shows the display assignments for setting the time:

Main Volume Display
HHMM

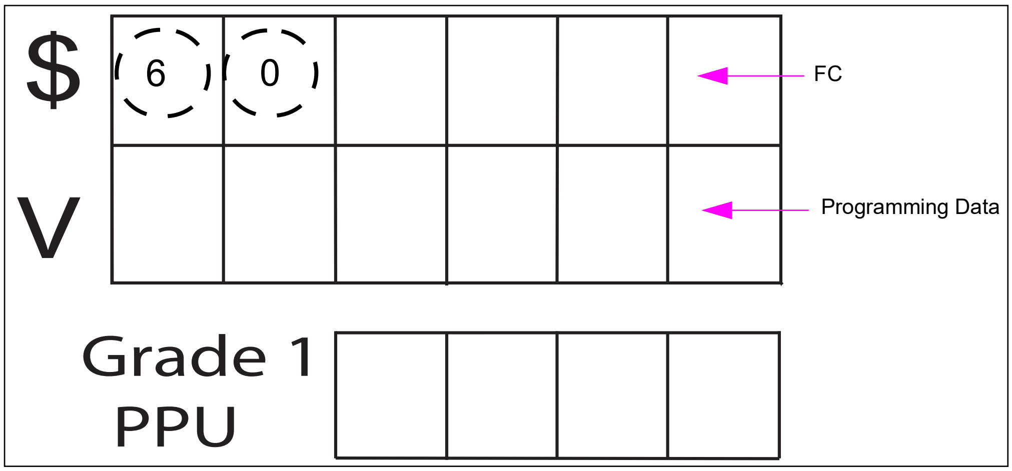
FC2: Set Dispenser Date
Date is programmed by entering month, day, and year in that order. If the date is not programmed before, then default date will be shown as 010100 (January 1, 2000). The following table shows the display assignments for setting the time:

Main Volume Display
MMDDYY

FC3: Set Dispenser Daylight Savings Time Mode

Daylight Savings Time is used in many areas of the USA, Canada, and Mexico based on rules dictated by regional and national governments. This FC will set the use of automatic daylight savings time adjustment based on rules as of January 2014 for USA, Canada, or Mexico. The default setting for this option is disabled.

Figure 5-63: Setting Daylight Saving Time Mode



The following table shows the values for the Daylight Savings Time programming:

Option	Daylight Savings Time
1 (Default)	Disabled
2	USA
3	Canada
4	Mexico

CC62: Two-wire Remapping Grade

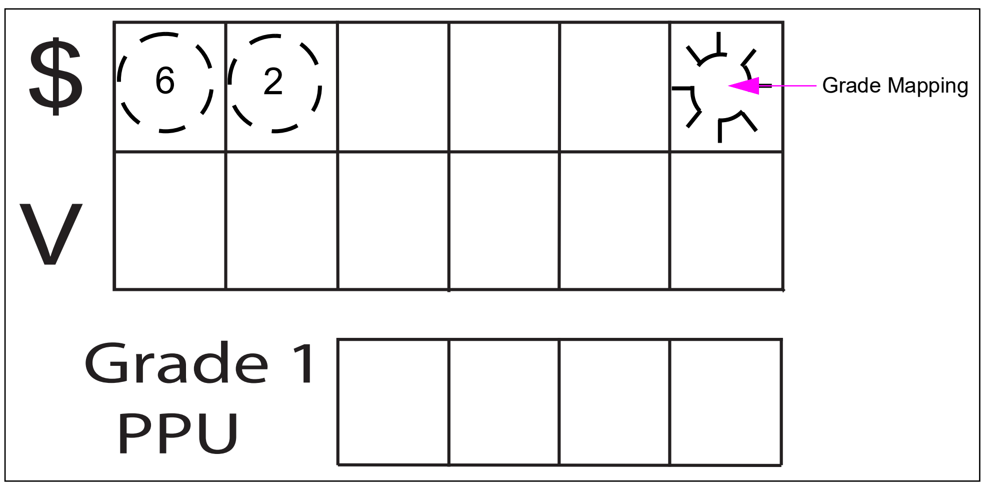
CC62 is used to remap grades of a blender 3+0 or 3+1 configuration onto two-wire protocol. Configuration switches grades 2 and 3 of a 3+0 or 3+1 blender to 3 and 5 respectively. Feature is set when the dispenser is connected to POS system requiring different grade configuration. Also, this feature can be used to remap the grade of a 3+2 blender.

The following table lists more information:

Selection	3+0 (+1) Blender Grade Mapping	2+2 Blender Grade Mapping	3+2 Blender Grade Mapping
1 (Default)	Two-wire [1-2-3 (-6)]	1-2-3-6	1-2-3-4-6
2	Other POS Systems [(1-3-5 (-6)]	1-2-3-4	1-2-3-4-5

Note: Pump and dispenser types 2+2 and 3+2 blenders are only available in pump software V02.9.74P or later.

Figure 5-64: Two-wire Remapping Grade



1 Enter **6**, **2**, and press **ENTER** to access this CC.

2 Select grade mapping options using the following table:

Selection	3+0 Blender Grade Mapping
1 (Default)	Two-wire (1-2-3)
2	Other POS System (1-3-5)

3 Press **ENTER**.

Level Three CCs

If using pump software V1.7.58 or later
In this version of software, the calibration mode is automatically entered when the W&M switch is toggled. Earlier versions of the software allowed you to toggle the programming switch before entering the programming mode. Now, you must be in the programming mode before toggling the W&M switch. For additional information, refer to "Important Information About Using V1.7.58 and Later for Encore 500/S/700 S and Eclipse Dispensing Pump Software" on page 5-7.

To enter Level 3 programming levels with V1.7.58 or later software for programming purposes (except those for calibration), proceed as follows:

1 Press **F1** on the manager keypad.

2 Toggle the W&M switch to enter the higher level programming.

3 Enter PIN Code for higher level programming, when required.

4 When finished programming, turn the W&M switch off.

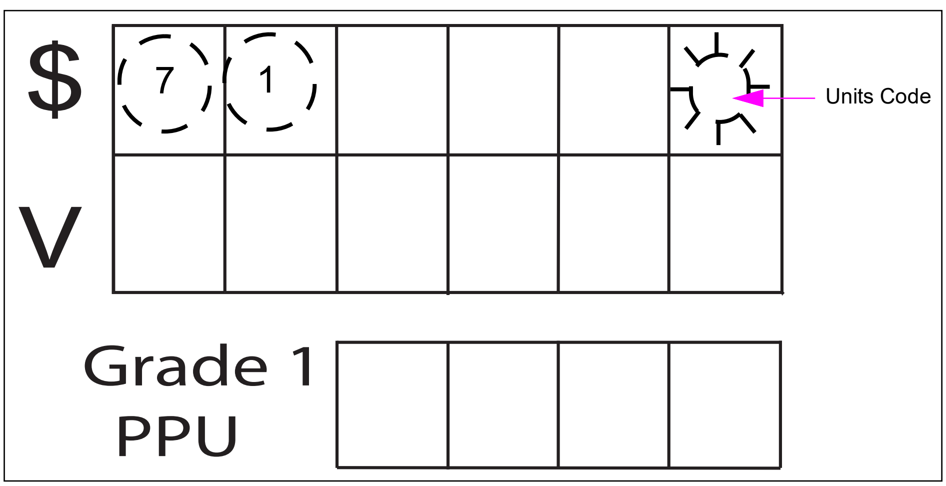
5 Press **F2** to exit programming.

If the pump uses software before V1.7.58, toggle the W&M switch before entering the programming mode.

CC71: Set Volume Units

Layout and meaning of each digit position for this programming feature are shown in Figure 5-65.

Figure 5-65: Setting Volume Units



1 Press **7**, **1**, and then press **ENTER** to access this CC.

2 Select volume unit code (1-3) using the following table:

Volume Units Code	Unit Type
1 (Default)	U.S. gallons
2	Liters
3	Imperial gallons

3 Press **ENTER**.

CC72: Set Blend Ratios

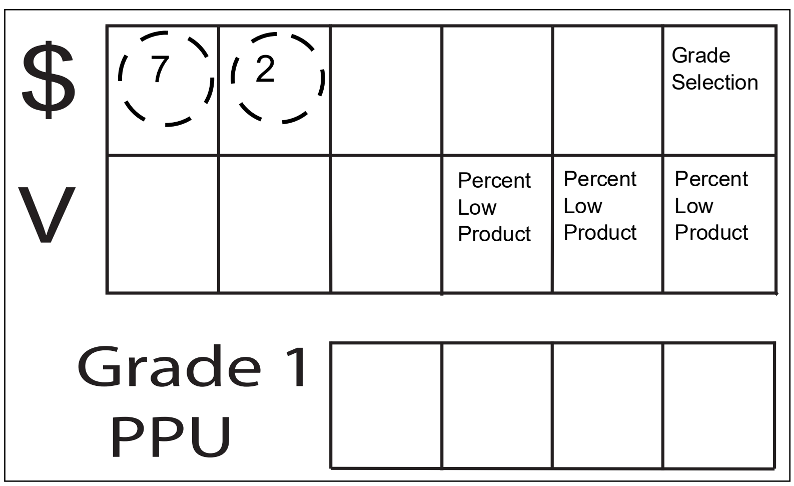
Set blend ratios allows a technician to program all blend ratios for all grades that can be blended according to dispenser type. Blend programming is allowed only for grades that can be blended. Thus, non-blended grades are treated as pure, 100% products. For conventional blenders, valid entries for percent figures are from 0 to 100. Entries above 100 will result in a programming error condition as described in the Pump Programming section. Default value is 120.

*Note: The default value is not a usable blend ratio and as a result, blend ratios **must be programmed before the dispenser will operate**. Unprogrammed blend ratios will open as flashing 888 on the Volume display for each product.*

> **⚠ CAUTION**
>
> The blend programming process for the NF Series Blender has CHANGED, and should be programmed only by a Gilbarco Certified Technician.
> If programmed incorrectly, wrong fuel blends could result in equipment damage.
> For more information, refer to *MDE-5413 Multi-Hose Universal Blender (NF Series) Quick Reference*.

Layout and digit positions are shown in Figure 5-66.

Figure 5-66: Setting Blend Ratios (Conventional Blenders)



1. Press **7**, **2**, and then press **ENTER** to access this CC.
2. Select Grade (1-6) using the following table for conventional blenders:

Grade Selection	Low Product Blend Percentage
1-6	0 to 100

3. Select percent low product (0-100%).
   Note: For NF Series Blender, the following blend values are not allowed: 1, 2, 3, 4 and 96, 97, 98, 99. Valid values include 0, 5-95, 100.
4. Press **ENTER**.
   *Note: No Default - **MUST** be programmed.*

Sample Blend Programming NJ5 Standard (CC90 = 60)

Button Grade	Delivered Product	NJ5 Example Inlet Product Percentages W (Low Grade)	X (High Grade)	Y (E85)	Entry to Program for the Target Delivered Product
1	87 (Grade 1)	94%	0%	6%	94
2	91 (Grade 2)	0%	100%	0%	100
3	93 (Grade 3)	0%	91%	9%	91
4	E15 (Grade 4)	88%	0%	12%	88
5	E85 (Grade 5)	0%	0%	100%	0

Sample Blend Programming NJ6 Standard (CC90 = 61)

Button Grade	Delivered Product	NJ6 Example Inlet Product Percentages W (Low Grade)	X (High Grade)	Y (E85)	Z (Diesel)	Entry to Program for the Target Delivered Product
1	87 (Grade 1)	94%	0%	6%	0%	94
2	91 (Grade 2)	0%	100%	0%	0%	100
3	93 (Grade 3)	0%	91%	9%	0%	91
4	E15 (Grade 4)	88%	0%	12%	0%	88
5	Diesel (Grade 5)	0%	0%	0%	100%	100

Sample Blend Programming NJ5 Alternate (CC90 = 62)

Button Grade	Delivered Product	NJ5 Example Inlet Product Percentages W (Low Grade)	X (High Grade)	Y (E85)	Entry to Program for the Target Delivered Product
1	87 (Grade 1)	94%	0%	6%	94
2	91 (Grade 2)	0%	100%	0%	100
3	93 (Grade 3)	0%	91%	9%	91
4	E15 (Grade 4)	88%	0%	12%	88
5	E85 (Grade 5)	0%	0%	100%	0

Sample Blend Programming NJ6 Alternate (CC90 = 63)

Button Grade	Delivered Product	NJ6 Example Inlet Product Percentages W (Low Grade)	X (High Grade)	Y (E85)	Z (Diesel)	Entry to Program for the Target Delivered Product
1	87 (Grade 1)	94%	0%	6%	0%	94
2	91 (Grade 2)	0%	100%	0%	0%	100
3	93 (Grade 3)	0%	91%	9%	0%	91
4	E15 (Grade 4)	88%	0%	12%	0%	88
5	Diesel (Grade 5)	0%	0%	0%	100%	100

CC73: Set/Display Money Totals

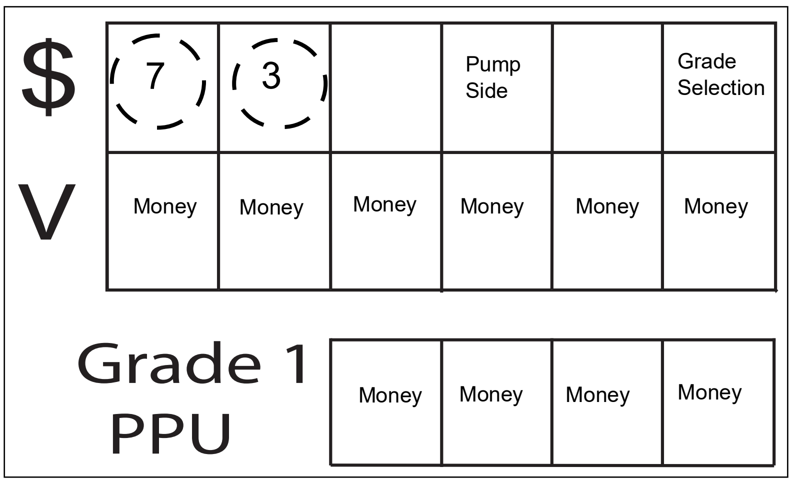
Allows programmer to read and set money totals. Initially, displays will show zeros in all locations.

Note: This function can only be performed at first initialization of dispenser and before total flow exceeds 250 volume units.

After totals of a grade reach 250 volume units that cannot be reset, only grades used in dispenser will accept money totals. Note that the position of decimal point in totals values will depend on earlier programmed location of decimal point in the dispenser display. Layout and meaning of each digit position for this programming feature are shown Figure 5-67.

Layout and digit positions are shown in Figure 5-67.

Figure 5-67: Money Totals Display



1. Press **7**, **3**, and then press **ENTER** to access this CC.
2. Select side (1 or 2) and press **ENTER**.
3. Select Grade (1-6) and press **ENTER**.
4. Set or Display Money Total.
5. Press **ENTER**.

CC74: Set/Display Volume Totals

Allows the programmer to read and set volume totals. Initially, for a new pump controller board, displays will show zeros in all locations.

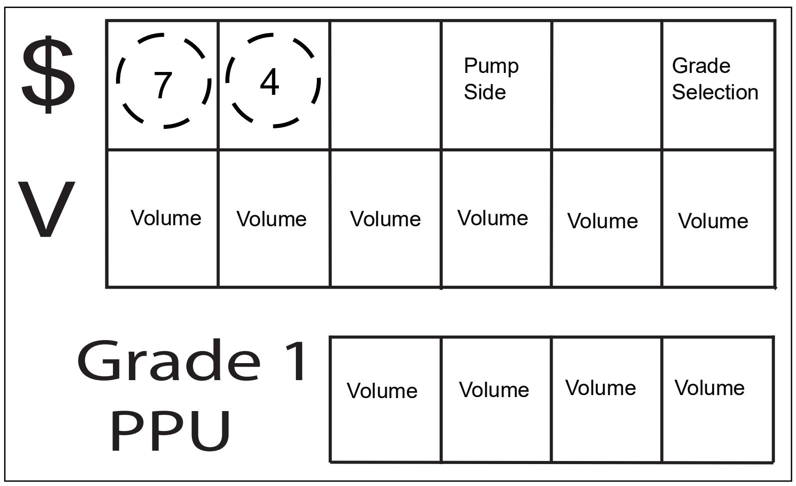
Note: Function can only be performed at first initialization of the dispenser and before the total flow exceeds 250 volume units.

After totals of a grade reach 250 volume units, they cannot be reset. Only grades used in the dispenser will accept volume totals.

Note: Position of decimal point in totals values will depend on earlier programmed location of decimal point in the dispenser display.

Layout and meaning of each digit position for this programming feature are shown in Figure 5-68.

Figure 5-68: Volume Totals Display



1 Press **7**, **4**, and then press **ENTER** to access this CC.

2 Select side (1-2) and press **ENTER**.

3 Select Grade (1-6: up to 8 for blenders) and press **ENTER**.

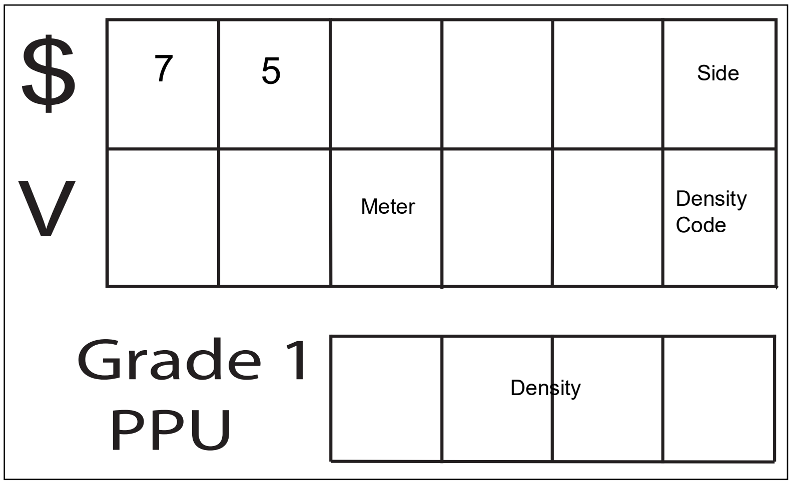
4 Set or Display Total.

5 Press **ENTER**.

CC75: Set Fuel Density (ATC and VaporVac Only)

Allows programmer to set density of fuel to be dispensed under ATC. Values can be set to 730 for gasoline and 840 for diesel fuel. Only the number of meters that exist (sensed by dispenser controller software) are shown. In this mode, meters can be cycled repeatedly pressing **ENTER**. Layout and meaning of each digit position for this programming feature are shown in Figure 5-69.

Figure 5-69: Setting the Fuel Density



1 Press **7**, **5**, and then press **ENTER** to access this CC.

2 Select side (1-2) and press **ENTER** to accept side.

3 Select Meter and press **ENTER** to accept meter.

4 Select Density (1 or 2).

The following tables shows the allowed fuel density:

Density Codes	Fuel Density
0 - NOT PROGRAMMED (DEFAULT)	N/A
1 - GASOLINE	730
2 - DIESEL	840
3 - DEF	1087
4 - ETHANOL	785

5 Press **ENTER** to accept density.

Deactivating VaporVac Pumps for Diesel Pumping in CC75

VaporVac motors are not required to run while dispensing diesel products. Motors can be set not to run when diesel is pumped using this CC. The default setting for Encore and Eclipse units with VaporVac is all products vapor motors turned on.

To dispense a product without turning on VaporVac motor (for example, diesel), proceed as follows while in CC75:

1 Select Meter.
Note: Selecting a meter automatically sets the density for the corresponding meter on the opposite side for that product.

2 Press **ENTER**.

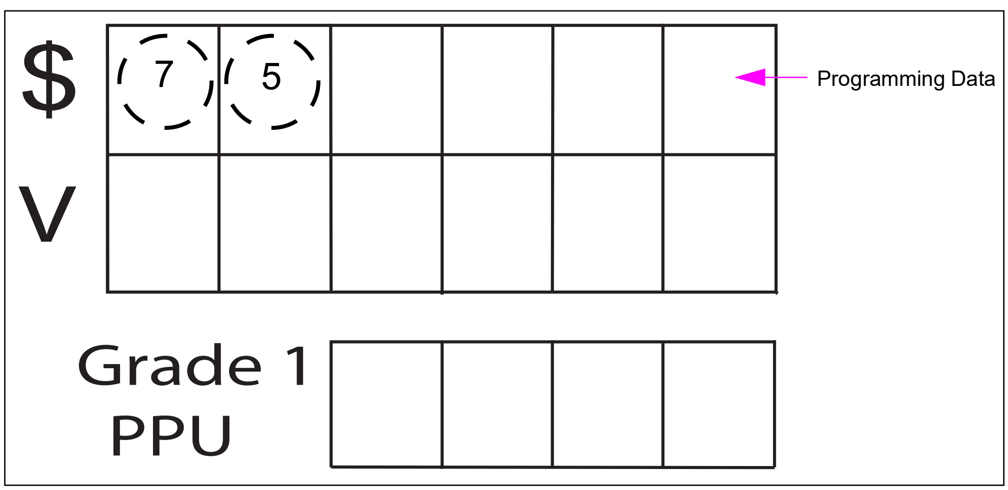
3 Select Density.
Note: Selecting 2 (diesel) will deactivate VaporVac pump for that meter.

CC76 - Money Rounding Display Method

This programming function allows the programmer to select the rounding method for the dispenser. The DEFAULT is selection 0, 'Nearest.001'. This CC is protected by the security switch. Without the switch being open, the unit will double beep and prevent entering a new value. All changes will be logged into the Regulatory Log.

The layout and digit position meaning for this programming feature is as shown in Figure 5-70.

Figure 5-70: Selecting the Rounding Method



1 Press **7**, **6**, and then press **ENTER** to access this CC.

2 Enter the programming information based on table:

Volume Units Code	Unit Type
0 (DEFAULT)	Nearest 0.001
1	Nearest 0.005 (Kuwait)

Rounding table for Option 2 (Only applies if CC42, FC1 is set to 4 - xxx.xxx format):

Calculated Amount	Display/Reported Amount
xxx.x00	xxx.x00
xxx.x01	xxx.x05
xxx.x02	xxx.x05
xxx.x03	xxx.x05
xxx.x04	xxx.x05
xxx.x05	xxx.x05
xxx.x06	xxx.x10
xxx.x07	xxx.x10
xxx.x08	xxx.x10
xxx.x09	xxx.x10
xxx.x10	xxx.x10

Note: This feature is only available in pump software 01.8.00L or later.

CC77: Change PIN Code

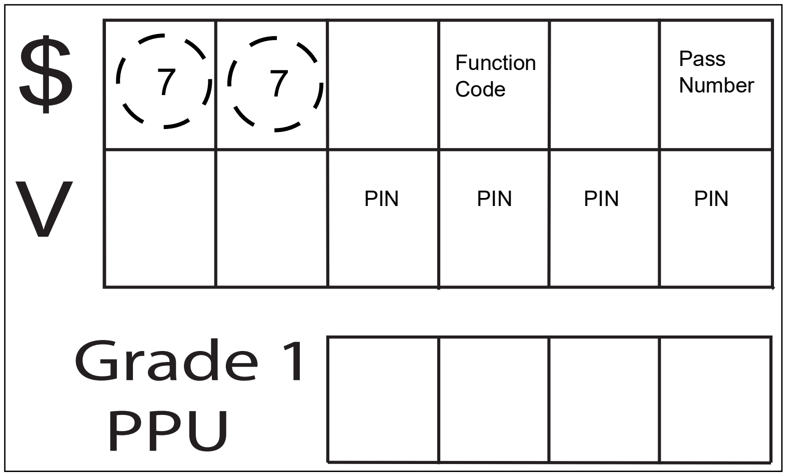
PIN Codes are a security feature. Changing PIN Codes from default can enhance station security. Revised PIN Codes must be safely stored by the station and ASC such that they can be accessed, as required. They must not be misplaced or lost because a “cold coldstart” is required to reset the code. As an alternate option, you can revert to V1.3.74 or earlier software, reload the latest version software, and then reprogram.
*Note: PIN Code **must** be entered twice.*

This CC allows the programmer to set new PIN Codes for entry into the four different programming levels. When accessed, all codes are shown as “- - - -”. Entering new digits for new codes changes the display, as they are entered. Display does not show actual PIN digit, but shows “0” to denote that a number was entered. Same PIN Code must be entered twice for it to be allowed.

First pass at entering a PIN Code is signaled by “1” appearing in the Pass Code display. Second pass is signaled by “2” appearing in the Pass Code display. Acceptance of a PIN Code is signaled by “3” appearing in the Pass Code display. A “0” in the Pass Code display indicates that the PIN Code was not accepted.

Four FCs are provided for changing each programming level PIN Code. All FC/PIN Code change operations have identical procedures. Default selection after entering into CC77 is for FC1 – change level one PIN Code. Layout and meaning of each digit position for this programming feature are shown in Figure 5-71.

Figure 5-71: Changing the PIN Code



The following table shows the FCs available for this CC:

FC	Option
1 (Default)	Change PIN Code - Level 1
2	Change PIN Code - Level 2
3	Change PIN Code - Level 3
4	Change PIN Code - Level 4

1 Press **7**, **7**, and then press **ENTER** to access this CC.

2 Program/assign new PIN Code.

3 Press **ENTER** to accept new PIN Code.

Change PIN Codes – FCs 1- 4

PIN Code levels correspond to FCs in CC77 (for example, FC1 for Level 1, 2 for Level 2, and so on). Note that all PIN Code assignments work the same.

Keypad Programming: New PIN Code

- Select Level (1-4) for code change.

Notes: 1) If Level 1, no selection is required as Level 1 is the default.

- *ENTER – accept level PIN Code for change.*
- *Enter PIN Code.*
- *ENTER – complete first pass, pass number now shows 2.*
- *Re-enter PIN code.*
- *ENTER – complete second pass.*

*2) If pass number display shows value of "0", it indicates that first and second PIN Code entries did not agree. Press **F1** to start PIN Code entry sequence from the beginning. If the new code is entered correctly twice, pass number display will display a value of 3 to indicate the acceptance of new PIN Code.*

Electronic Calibration

WARNING

You are working in a potentially dangerous environment of flammable fuels/vapors and high voltage.

Fuel and its vapors may ignite, leading to serious injury or death. Fire, explosion, or electrical shock can result in severe injury or death if you do not follow safety procedures.

Surfaces of Eclipse Series units are non-metallic. Magnetic-backed keypads will not adhere to Eclipse unit surfaces as they will to others. In any situation where power is on and gasoline and its vapors are present or potentially present (for example, calibration), do not attempt to use the manager keypad with electronics cabinet open.

When calibrating Eclipse units, run keypad cable over door and close door, allowing keypad to be suspended on unit and accessed from outside for all procedures done with power applied or where there is a possibility of fuel or fuel vapor presence.

All meters must be calibrated before the dispenser functions. If the unit is a blender where the percentage for low grade product is not 0% or 100% for high grade product, you must temporarily set the blend to 100% for the low grade product meter or 0% for high grade product meter. If all meters are not calibrated, dispenser will display "Not Calibrated" EC when the dispenser is placed in normal mode.

CAUTION

Before you proceed, read and understand "Purging Air from the System" on page 7-7. Operating the unit with air in the system can cause damage to meter from overspeed.

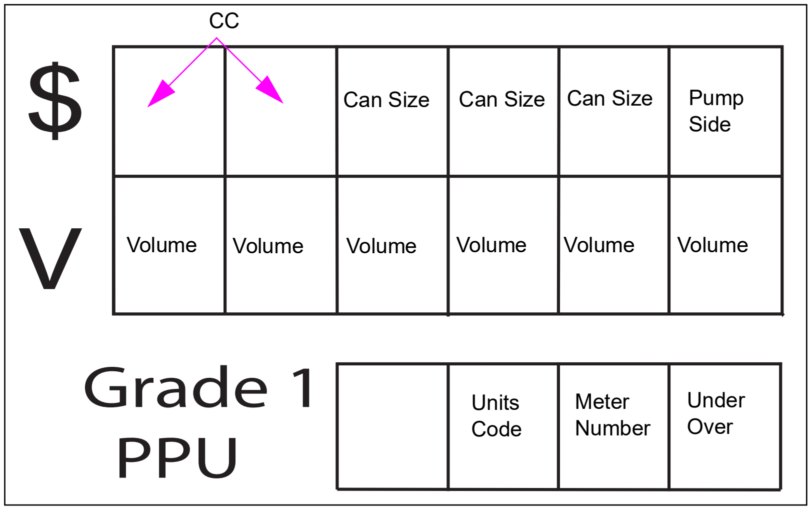
The unit must be properly configured to CC71 before calibration. After completing calibration, the W&M switch must be sealed per local authority. All replacement meters must be calibrated after installation and testing.

Figure 5-72: Manager Keypad

Calibration procedure requires that each parameter be entered before advancing to next procedural step. If a parameter is not entered, dispenser/display will wait until the parameter is entered. Entries are not checked for errors, and are thus accepted and used as entered by programmer. Entry into meter calibration will default to side 1 with no units programmed by default.

Layout and digit position are shown in Figure 5-73.

Figure 5-73: Layout and Digit Position Display



For information on diagnosing calibration problems, refer to "Error Codes" on page 8-20.

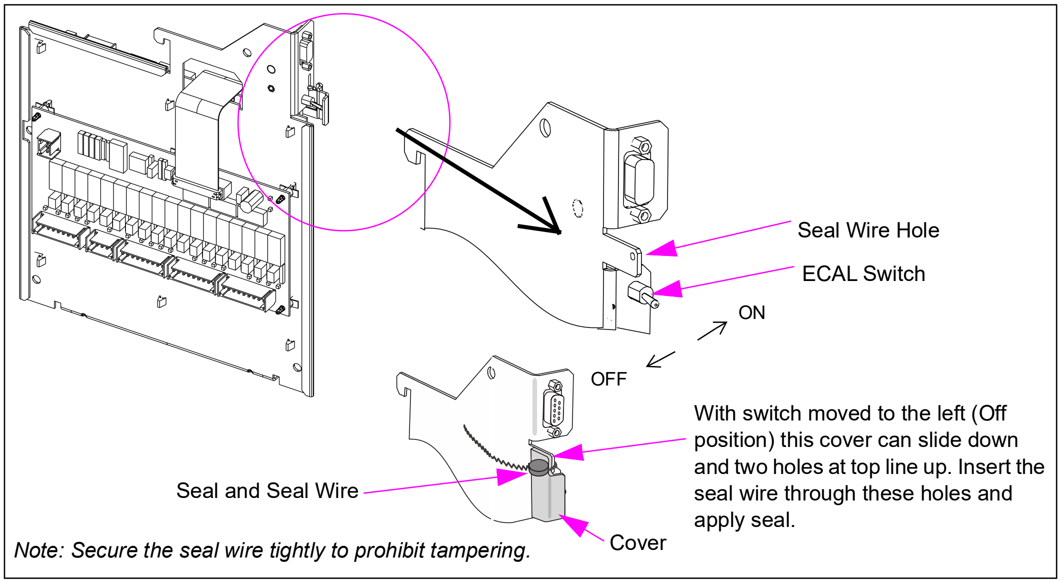
Pump software V1.7.58 and later for Encore 500/S/700 S and Eclipse contains enhancements to simplify the meter calibration verification and calibration process. It is no longer required to run a separate verification process to determine if meter is properly calibrated and then run a separate calibration setting if so required.

With this Version Pump Software	Refer to
V1.7.58 or later	"Enhanced Meter Calibration and Audit" on page 5-101
Before V1.7.58	"Standard Meter Calibration Procedure (Pre-V1.7.58 Software)" on page 5-106

After completing calibration, the W&M switch must be sealed per local authority. All replacement meters must be calibrated after installation and testing.

Install the updated software to take advantage of the enhancements.

Figure 5-74: W&M Switch



Enhanced Meter Calibration and Audit

Pump software V1.7.58 for Encore 500/S/700 S and Eclipse with C+ meters/V meters contains enhancements to simplify C+ meter/V meter calibration and provides the capability for W&M audit trail. It is no longer required to run a separate verification process to determine if meter was previously calibrated and then run a calibration process, if required.

IMPORTANT INFORMATION
To ensure accuracy for pump software V1.7.58 and later, it is **Very Important** that the filling of the prover be accurate and within +/- 1 cubic inch of the zero reading. If the calibrator fails to stop flow within this limit, then the calibration process must be repeated.

For more information, refer to "First Time Meter Calibration Procedure" on page 5-102 and "Combined Verification and Calibration Procedure" on page 5-104.

Setup

Calibration can be performed in standalone mode or through console control operation.

If the unit volume type is programmed to gallons, the prover can size will default to 5 gallons and if programmed to liters, the default is 20 liters. If a different size prover can is being used, it can be programmed in CC82. This can size will be retained in memory.

If the Volume Unit Type (VUT) or blends must be programmed, the programming mode is entered in normal manner except that the calibration switch must be turned "ON" after entering the programming mode and turned "OFF" before exiting the programming mode to save the changes.

Special considerations for V1.7.58 software:

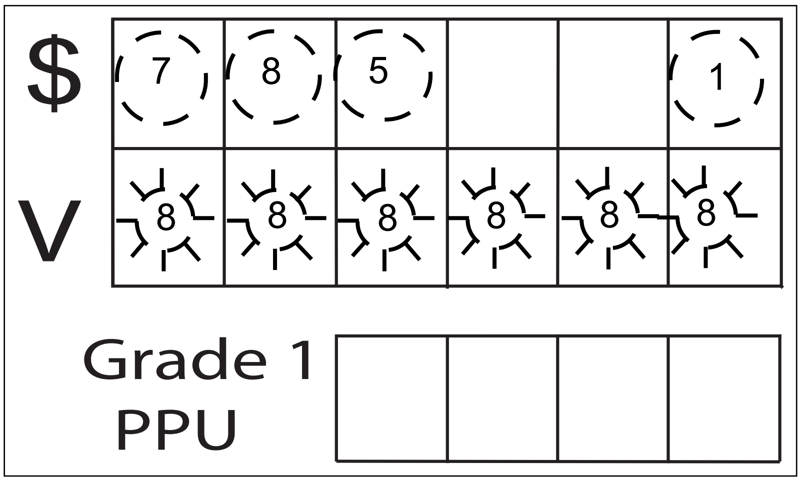
- EC26 will not clear automatically after calibrating the meter. A warmstart is now required.
- When performing the combined verification and calibration, the electronic totals will be increased as during a normal sale.
- When performing the first time calibration procedure, the totals are not updated.

First Time Meter Calibration Procedure

This calibration process is most commonly used for units that are not already calibrated. Factory precalibration must be verified.

1 Move the cover to expose the W&M switch (see Figure 5-74 on page 5-100). Flip the switch to "ON" to turn on the main display on both sides of the unit. The PPU shows the number of days that have lapsed since the last calibration. If this meter has not been verified or calibrated before (factory precalibration is not counted), the PPU displays flashing 8s.

Figure 5-75: PPU Display



2 Setting the prover can size and VUT:
- The three MSDs of the money display contains the prover can size which is 5 gallons in this case.
- The Least Significant Digit (LSD) of the money display shows the VUT: 1= U.S. gallons and 2 = liters. Option 1 is shown in this case.

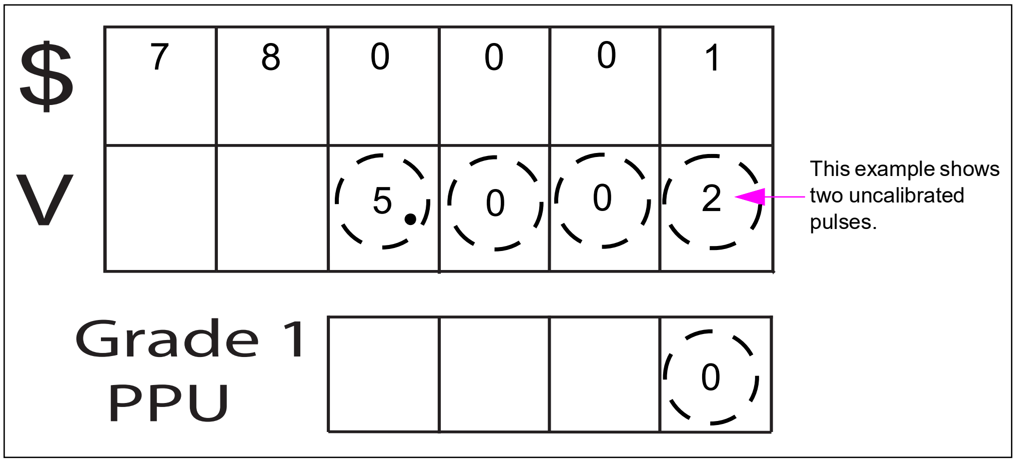
If the VUT or blends have not been programmed, the PPUs will flash 26 for "No VUT" and/or 4307 for "No Blends".

If the prover can size is not shown, refer to CC82 instructions. You must turn off the W&M calibration switch before changing CC82 can sizing.

3 Remove nozzle and select pure product (grade) by raising the pump handle or remove the nozzle and push the grade select button of the meter to be calibrated.
Note: All PPUs except for the one associated with the meter being calibrated will be blank.

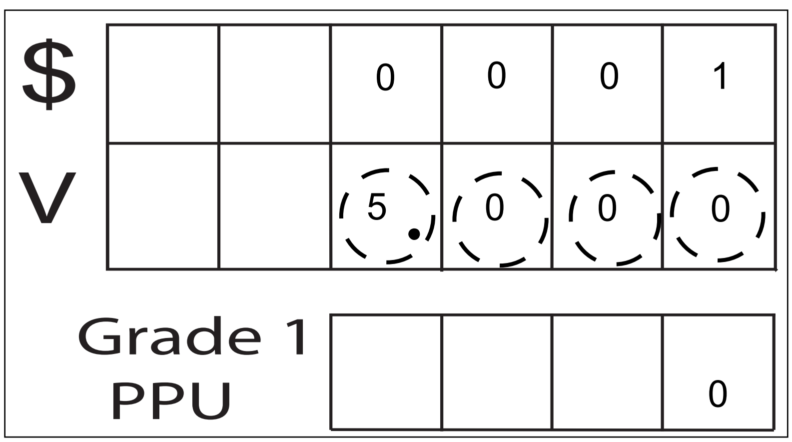
4 Dispense the fuel into the appropriate prover can and ensure that you stop within +/- 1 cubic inch. If the calibrator fails to stop the flow within this limit, you must repeat the calibration process.

Figure 5-76: Two Uncalibrated Pulses



5 Lowering the handle or replacing the nozzle will make the volume display value to change to the programmed prover can value. This meter is now calibrated and the volume display shows the amount corresponding with the prover can size.

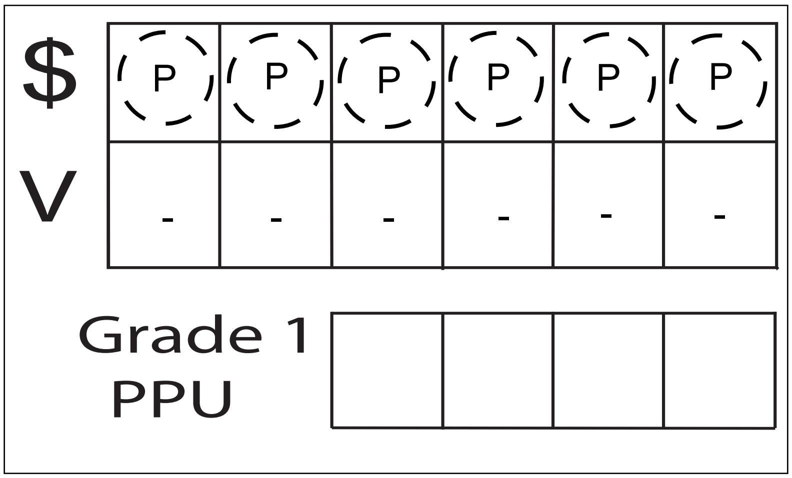
Figure 5-77: Meter Calibration



6 If other meters must be recalibrated, repeat steps 2 (on page 5-102) through 5 for each meter. It is not required to re-enter prover can size and VUT information after each meter calibration. This allows calibration of both sides of the unit without returning to the manager keypad.

7 After all meters are calibrated, flip the W&M switch to “OFF” position. When the switch is placed in “OFF” position, the new calibration values are first saved to redundant data storage. The Money display shows “PPPPPP” and the volume display shows “------”, and then returns the unit to normal operation.

Figure 5-78: Flipping the Switch



8 Install seal wire and apply seal as determined by the governing authority.

Combined Verification and Calibration Procedure

This alternate approach is for all units previously calibrated at some point in time. Before this enhancement, the technician had to run a verification test to determine if recalibration was required and run through the calibration process if the meter required recalibration.

If the technician, or W&M official is checking for calibration, then 5 gallons (20 liters) are dispensed into the prover can, stopping at the zero mark on the can. Compare the prover can size along with the displayed amount and use the appropriate set of instructions.

Main Display and Prover Can Match

If the pump displays the amount in the can +/-1 cubic inch, then that meter is within tolerance and you can move to the next meter.

Main Display and Prover Can Do Not Acceptably Match

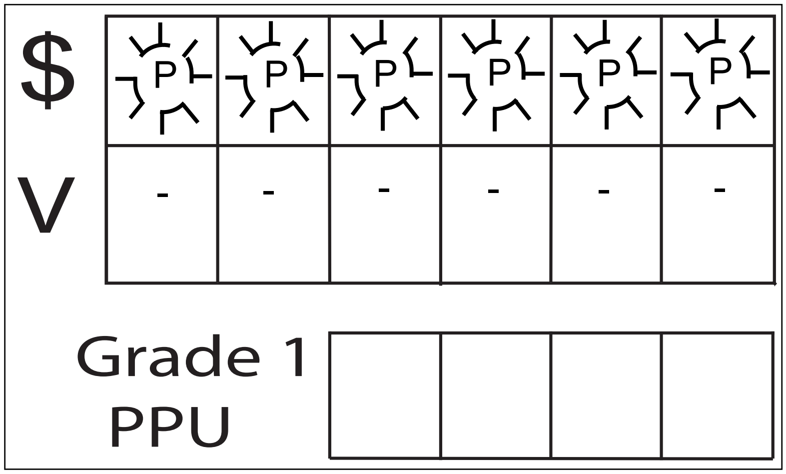
1 **STOP** - Do not lower the pump handle or replace the nozzle at this time, but continue to the following procedures.

2 Cut the seal wire on the calibration switch and turn the switch to “ON” position, wait for confirmation (two beeps), switch back to “OFF” position, and again wait for confirmation (two beeps).

3 After lowering the pump handle or replacing the nozzle, the money display will flash "PPPPPP" and volume display will show the volume in the prover can for two seconds, before saving the new calibration value in storage.

Note: The display then returns to the transaction value that was displayed before the calibration procedure was performed.

Figure 5-79: Meter Calibration and Verification



4 No other transaction can be active on other side of the unit during this procedure. The unit will not go into calibration mode if more than one pump handle is activated.

Note: The CRIND may reboot during this procedure.

5 This meter is now calibrated.

Note: Electronic and electromechanical totals are updated when this method is used.

Audit Trail Feature

While in calibration mode of operation, pressing **ENTER** on the manager keypad the specified number of times (as listed in the following steps) will display the data described in "Audit Table Descriptions" on page 5-106.

1 To access the audit feature, place the unit in normal mode of operation and press **ENTER** on the manager keypad as shown below:

- Four times for a blender
- Three times for a non-blender

The displays now show the information conforming to the audit trail for the associated meter. Refer to "Audit Table Descriptions" on page 5-106.

2 Press **CLEAR** on the manager keypad to return to normal operation.

Audit Table Descriptions

	Press ENTER	Unit Type 1-7	Unit Type 8-9	Unit Type 12-19
X1	Money Display	N/A	BC	BC
	Volume Display	VUTC	VUTC	VUTC
	PPU Displays	DLC	Blend Ratio	Blend Ratio
X2	Money Display	N/A	BC	BC
	Volume Display	VUTC	VUTC	VUTC
	PPU Displays	CT	DLC	DLC
X3	Money Display	N/A	BC	BC
	Volume Display	VUTC	VUTC	VUTC
	PPU Displays	PPVUT	CT	CT
X4	Money Display	No Change	BC	BC
	Volume Display	No Change	VUTC	VUTC
	PPU Displays	No Change	PPVUT	PPVUT
Legend	BC = Total times blends have been changed			
	CT = Total times associated meter calibrated			
	DLC = Days since last associated meter calibration			
	PPVUT = Pulses per VUT for associated meter			
	VUTC = Total time VUT changed			
	VUT = Volume Unit Type (liters or US gallons)			

Standard Meter Calibration Procedure (Pre-V1.7.58 Software)

The calibration procedure assumes that either 5 gallon or 20 liter prover can will be used. If another can size is used, you must change the default can size in CC82. As the new can size will be stored in memory for future calibrations, if the special can size is not likely to be used for the next calibration, it is very important to reprogram the can size back to default after completing calibration. You may also verify the can size on the money display during calibration.

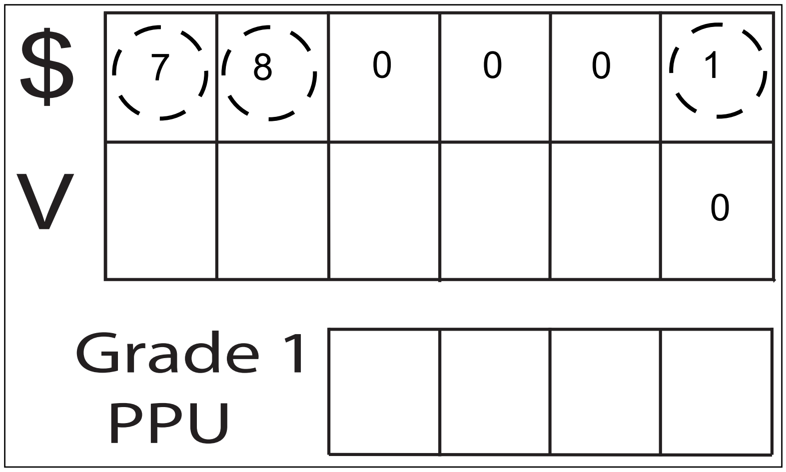
EC26 will not clear automatically after a unit has been first calibrated. A warmstart is now required. This will be primarily noted after installing a new service board and calibrating, or performing a true coldstart (drop back to V1.3.74 pump node software and then reload to the current version) where recalibration is required.

When performing the combined verification test and recalibration, electronic and electromechanical totals will increase as normally as during a normal sale.
Note: Use this calibration procedure for units with pump software before version V1.7.58.

1. Before entering CC78, the W&M switch must be turned "ON" (see Figure 5-74 on page 5-100).
2. Press **1** on the manager keypad (see Figure 5-72 on page 5-98).
3. Press **0128** and then press **ENTER**.
4. Press **78** and then press **ENTER**.

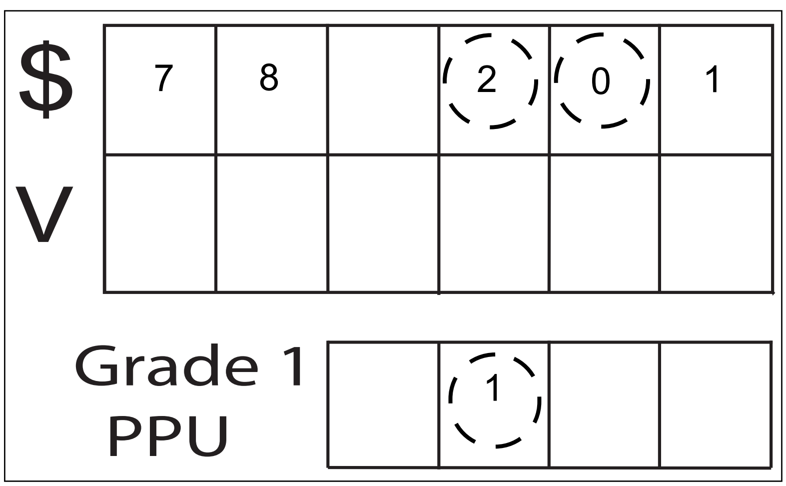
The following example shows CC78 and unit side 1:

Figure 5-80: CC78 and Unit Side 1



5 Enter the prover can size. The default value is “0”. You must enter a volume from 1 to 999 to proceed to the next step. In this example, the prover can size is 20 U.S. gallons. Current volume unit is selected in CC71 and reported in the Unit Code field (shown as 1 in the following example):

Figure 5-81: Current Volume Unit Reported in the Unit Code Field

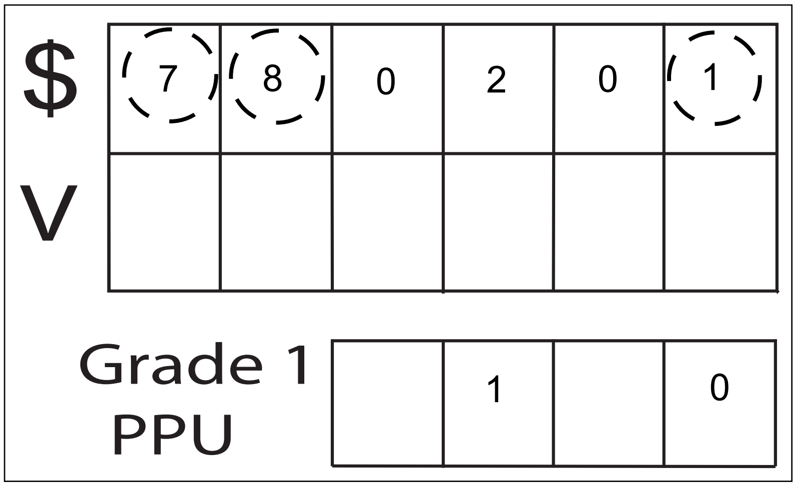


The following table shows the volume units and their codes:

Prover Can Volume Units	Unit Codes
No Units Programmed (Default)	0
U.S. gallons	1
Liters	2
Imperial gallons	3

6 Press **ENTER** to accept the default or press **2** to select side 2. Then, insert the calibration side. Default value is 1 as shown in Figure 5-82.

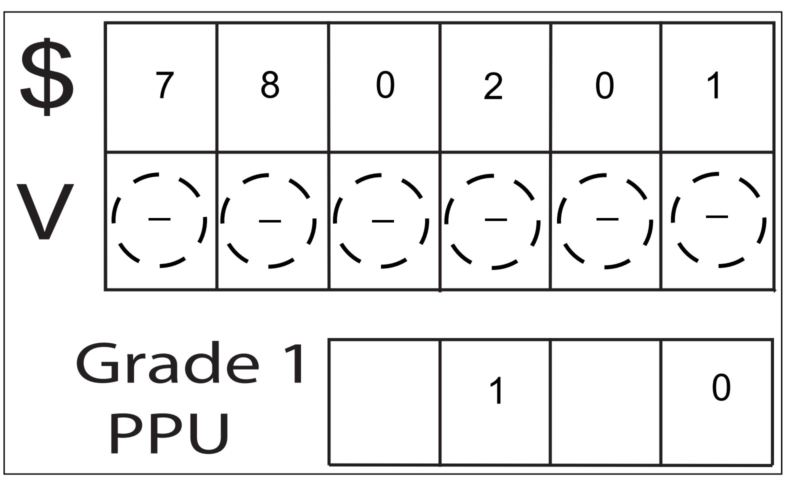
Figure 5-82: Inserting the Calibration Side



To Select This Side	Complete This Action
1 (Side A)	Press **ENTER** (default setting).
2 (Side B)	Press **2** on the manager keypad and then press **ENTER**.

7 Activate the fueling position. The volume display shows all dashes to indicate that the pump is ready to start dispensing fuel.

Figure 5-83: Volume Display Showing Dashes

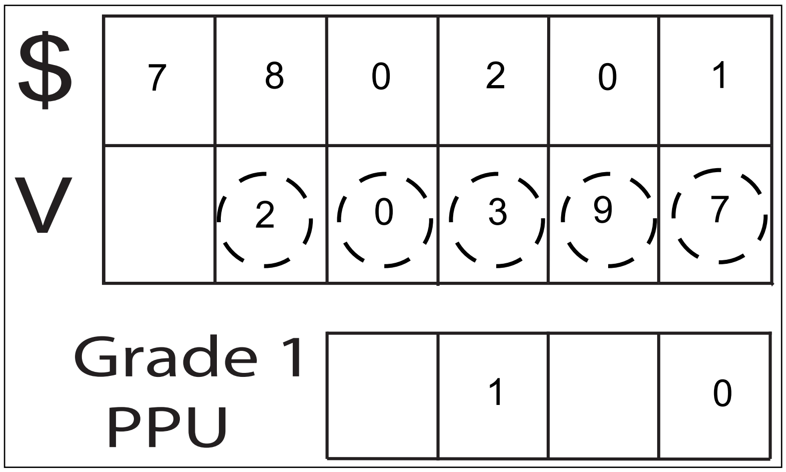


8 Fill the prover can to within +/- 1 cubic inch of the 0 mark. Return the pump handle to “OFF” position when done.

Note: For blenders, calibration can be performed only on pure product grades (ratio 100 or 0 percent). A double beep sound will signal an incorrect grade selection.

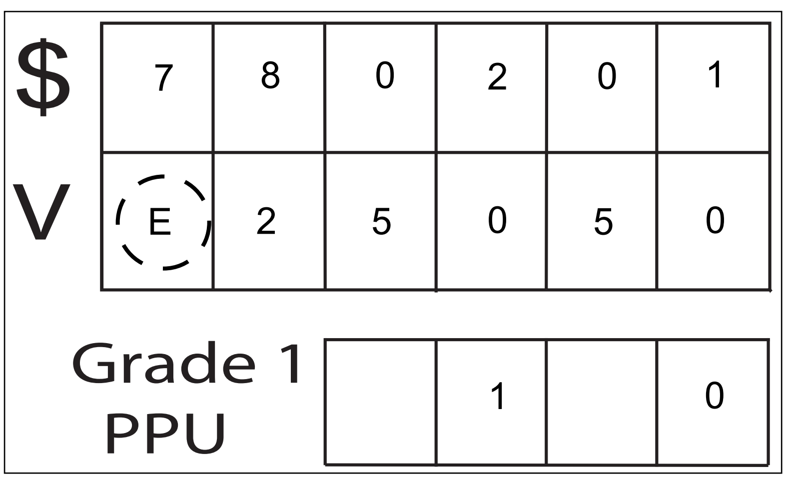
During the fuel dispensing process, real-time raw pulses are shown in the Volume display area (see Figure 5-84).

Figure 5-84: Real-time Raw Pulses Shown in the Volume Display



Final raw pulses are shown until **ENTER** is pressed. Pressing any other key will signal error by sounding a double beep. If an error occurs during this phase, it will be accompanied by “E” on the volume display.

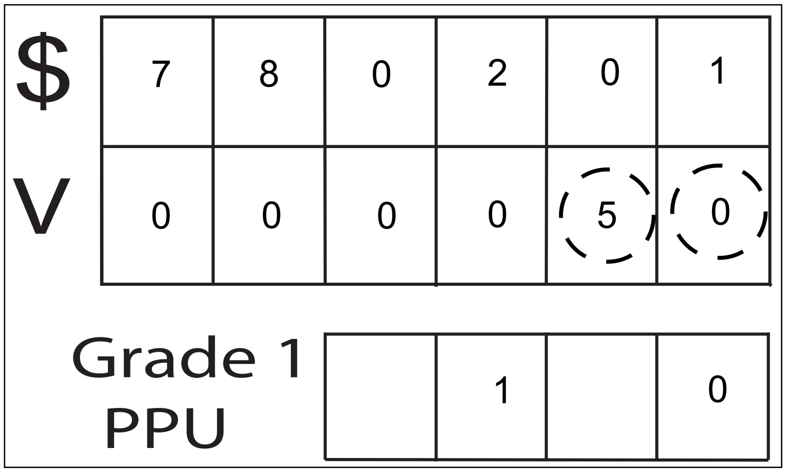
Figure 5-85: Error Shown on Volume Display



9 The display field is the Volume field. Enter the overflow or underflow value from reading the scale on the prover can. Unit volume is expressed difference as shown in the following table:

Prover Can Volume Units	Difference Volume Units
Liters	Millimeters
U.S. gallons	Cubic Inches
Imperial gallons	

Figure 5-86: The Display Field is the Volume Field

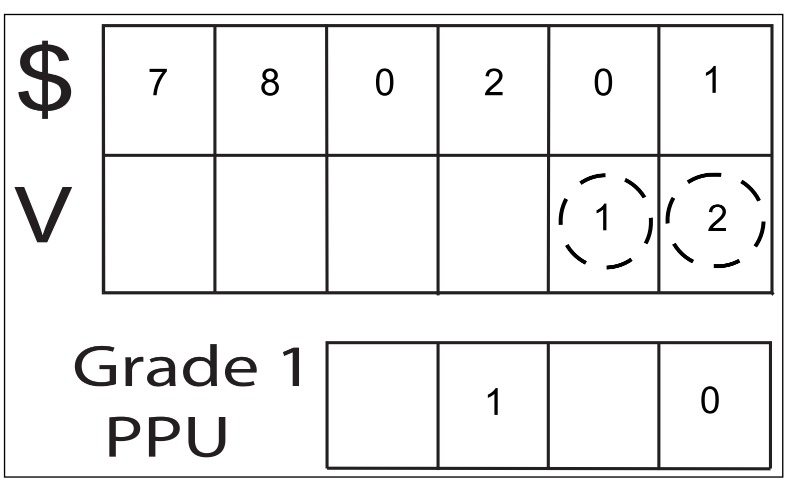


IMPORTANT INFORMATION

The procedures in step 14 on page 5-112 pertain to dispensers that have **software version 1.5.40 or later.**
The procedures in step 15 on page 5-112 pertain to dispensers that have **software version earlier than 1.5.40.**

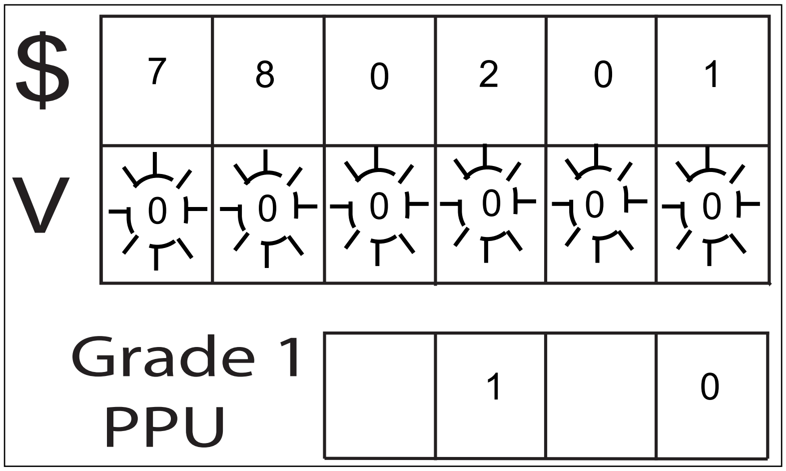
10 **If the dispenser has software version 1.5.40 or later**: In this example, 12 is the number of cubic inches read as over from the prover can. Enter 12 on the keypad to show up in the display (see Figure 5-87).

Figure 5-87: Entering on the Keypad



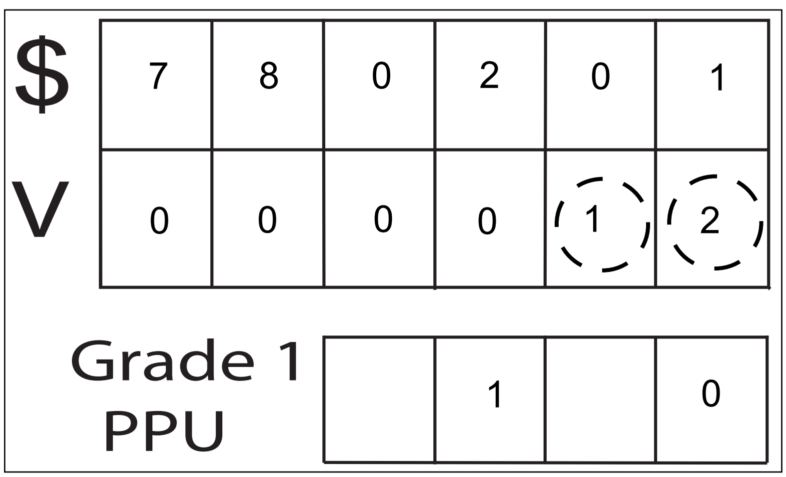
11 **If the dispenser software version is before 1.5.40**: In this example, all zeros are flashing in the volume field.

Figure 5-88: Zero Flashing in the Volume Field



12 Enter the offset number (number of cubic inches or centimeters away from zero).

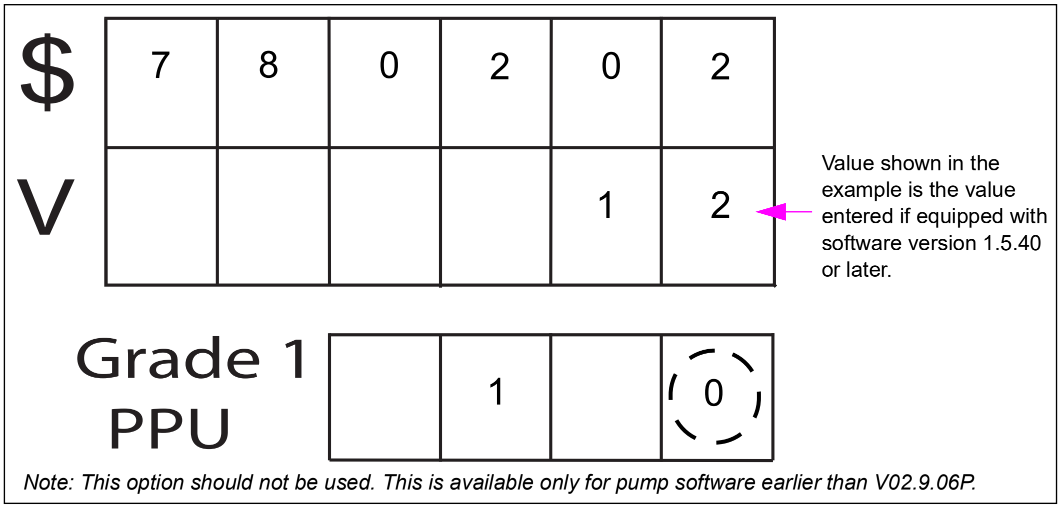
Figure 5-89: Entering the Offset Number



13 Press **ENTER**. After entering the overflow or underflow amount, the selection field will move to the underflow/overflow designation field as shown in the following example. Select the correct offset using the following table:

If Reading on Prover Can is an	Indicated by a Reading	Enter this on Keypad
Overflow	above 0 (plus)	0
Underflow	below 0 (minus)	9

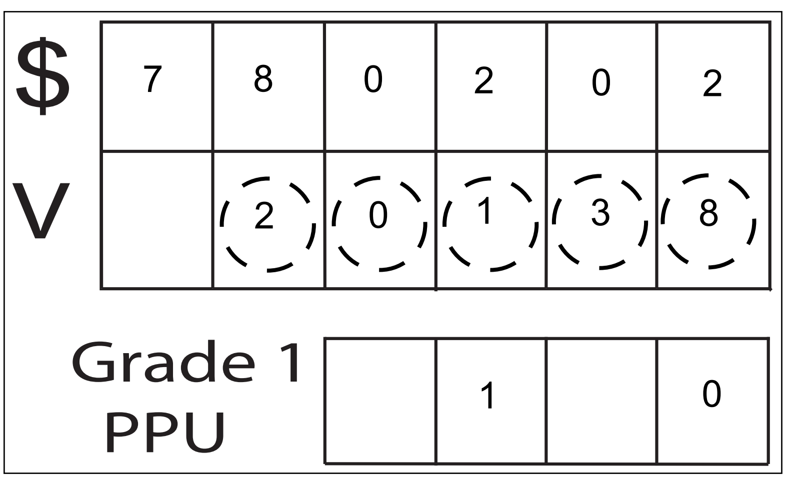
Figure 5-90: Underflow/Overflow Designation Field



14 Press **ENTER**. On pressing **ENTER**, the volume that must be in the can is shown on the display. Compare this reading to the actual volume in the prover can.

Note: If figures do not match, then it means that the number was entered incorrectly or wrong offset code was used (0.004 gallon = 1 cubic inch).

Figure 5-91: Can Volume Shown in the Display



15 Accepting/rejecting the indicated amount:

- To reject the displayed amount, press **CLEAR**. The selection field will then move back to the Volume field and allow re-entry of the overflow/underflow amount.
- To accept the displayed amount, press **ENTER** again. The selection field then moves back to the side selection in the main money display.

16 The dispenser now calculates the calibration factor based on the internal pulse count of the dispenser test volume and overflow/underflow amount. The meter is now calibrated.

F1 allows additional programming or F2 may be used to exit programming. F1 and F2 will operate only if the security switch has been set to the secure position (“OFF”). Otherwise, calibration mode cannot be exited.

CC80: Set Maximum Flow Rate

This option allows user to select maximum flow rate allowed per grade. If flow rate value entered is larger than hydraulics maximum capable flow rate, maximum hydraulic flow rate will flow. Default value flow rate is 10 Gallons per Minute (GPM) per grade for U.S. and 40 Liters per Minute (LPM) per grade for Europe.

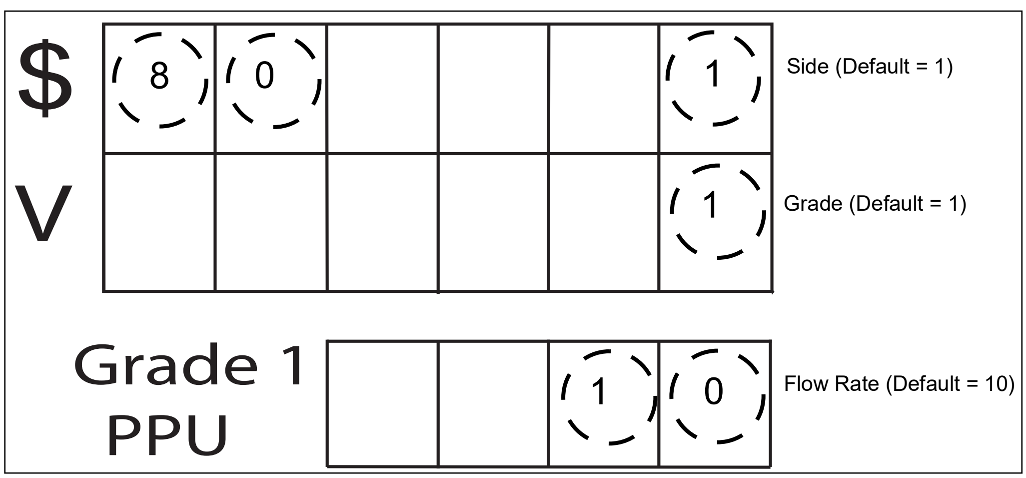
This function works properly only with proportional flow control valves. It does not work properly with digital valves.

The following table shows available options for this FC:

Selection	Option
1 or 2	Side
1-6	Grade
01-98	Flow: units per minute/grade

Layout and meaning of each digit position for this programming feature are shown in Figure 5-92.

Figure 5-92: Layout and Digit Position for the Programming Feature



1 Press **8**, **0**, and then press **ENTER** on the manager keypad.

2 Select side (1 or 2) and press **ENTER**.

3 Select Grade (1-6) and press **ENTER.**

4 Select Flow Rate [01 to 98 (Default = 10)] and press **ENTER**.

CC82: Setting Prover Can Size

Note: For E550 calibration procedures for Encore 500/S/700 S and Eclipse, refer to MDE-4281 Calibration Quick Reference Card Encore 300/500/550, and Eclipse Units.

CC82 is used with pump software version V1.7.58 and later.

Prover can size is set to default to a preset number based on the type of volume units entered. For these defaults, refer to the following table. CC82 is used if an unusual prover can size is required. When changing the VUT, enter the programming mode in the usual manner, except that the W&M calibration switch must be turned "ON" before entering CC82 and switched "OFF" before exiting this programming mode to make the changes effective.

If Volume Unit Is Set to	Prove Can Size Defaults to
U.S. gallons or Imperial gallons	5 gallons
Liters	20 liters

Figure 5-93: Setting Prover Can Size

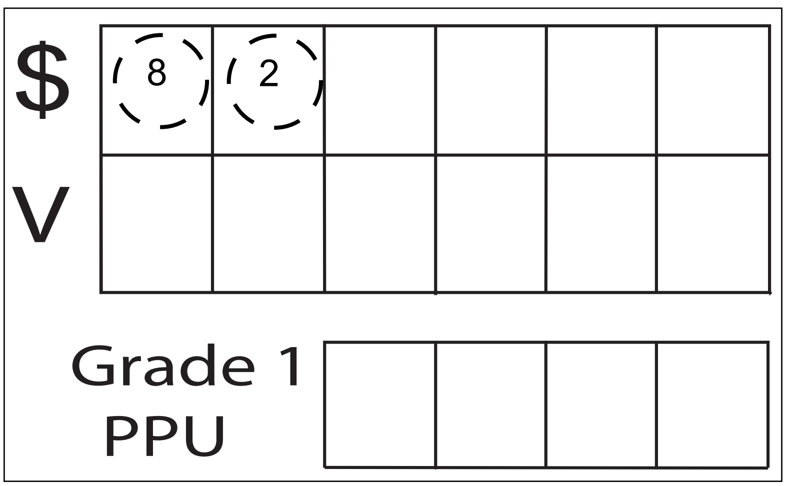
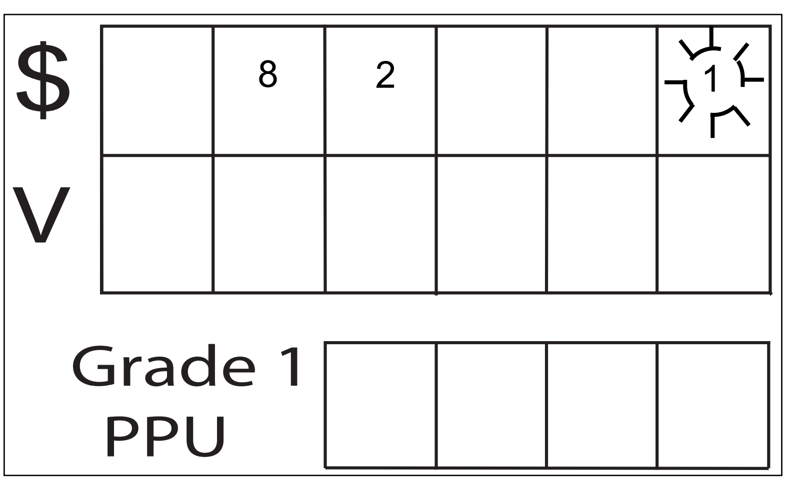


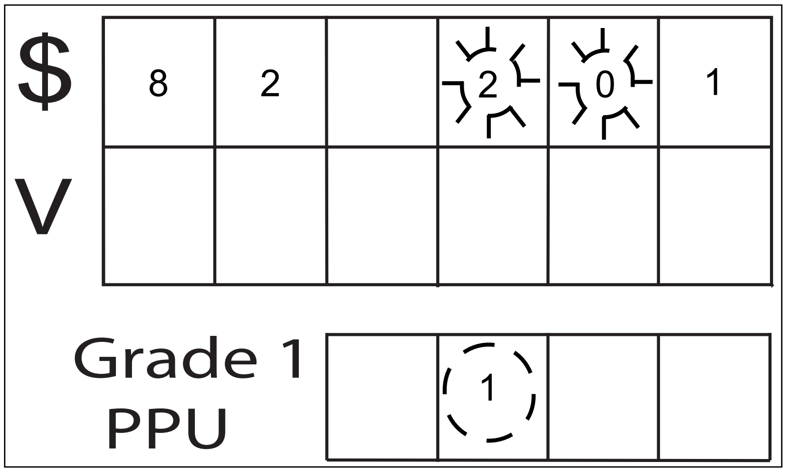
Figure 5-94 shows CC82 and unit side 1.

Figure 5-94: CC82 and Unit Side 1



1 Enter the prover can size. The default value is “0”. It is required to enter a volume from 1 to 999 to proceed to the next step. In this example, the prover can size is 20 U.S. gallons. Current volume unit is selected in CC71 and reported in the Unit code field, shown as “1” in Figure 5-95.

Figure 5-95: Inserting Prover Can Size



The following table shows the volume units and their codes:

Prover Can Volume Units	Unit Codes
No Units Programmed (Default)	0
U.S. gallons	1
Liters	2
Imperial gallons	3

2 Press **ENTER** to accept the default or press **2** to select side 2. Then, insert the calibration side. Default value is “1” as shown in Figure 5-99 on page 5-119.

CC83 - Set Alternative Fuel Mode

This function allows the programmer to enable alternate fuel modes. This CC is protected by the security switch. Without the switch being open, the unit will double beep and prevent entering a new value. All changes will be logged into the Regulatory Log.

When set to natural gas, this mode has two features.

1 When the **ENTER** key is pressed, the number in the volume display is multiplied by the factor detailed in the following table based on setting in CC83. Subsequent presses of **ENTER** key will exhibit the behavior described in “Encore 500 Audit Trail Information” on page 5-137.

Type	Conversion	Factor
GGE	Pounds (lbs.)	5.66
GGE	Kilograms (kg)	2.567
GLE	Pounds (lbs.)	1.693
GLE	Kilograms (kg)	0.768

2 The valve feedback signal is ignored to allow for third-party valve control hardware. For simulator testing, this means that the valve type defaults to DIGITAL VALVE and simulators with proportional valve will not work properly with this feature enabled.

When set to DEF (UREA), this mode has the following four features:

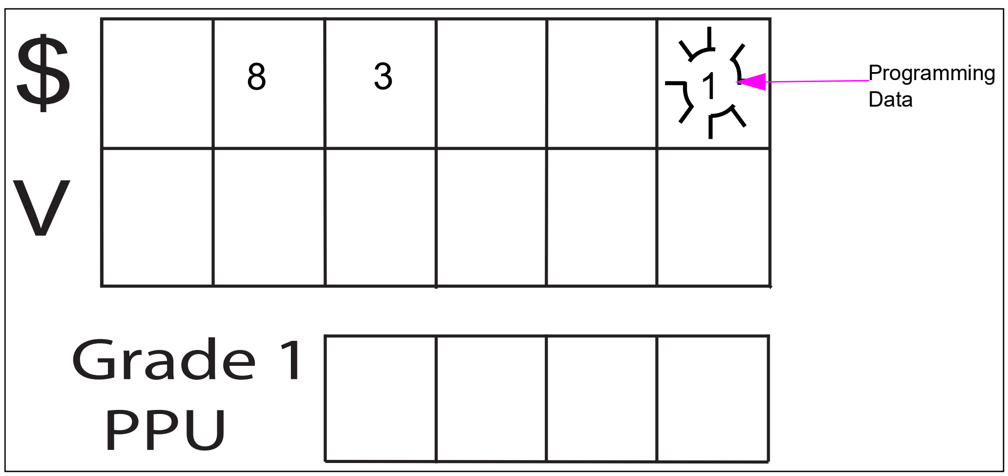
1 The proper input on the VaporVac port is monitored for “frozen” status which indicates that the hydraulic system is currently unable to operate properly.

2 Enables the display of EC31 (Pump Frozen), which also prevents the dispenser from responding to two-wire polls even if in two-wire mode. EC31 is only cleared when the frozen status is removed.

3 Assigns the slowdown value for DEF to 0.020 but does not restrict the slowdown setting for diesel when set to MPD2 or MPD3 pump types where diesel+DEF pump setup is used.

4 Does not require a fuel density setting for DEF if ATC is in use. However, CC75 can be set to DEF to force ATC and Vapor Recovery system to ignore the DEF grade.

Figure 5-96: Setting Alternate Fuel Mode



To program, proceed as follows:

1 Press **8**, **3**, and then press **ENTER** to access this CC.

2 Enter programming data in the table:

Alternate Fuel Mode	Setting
Standard Fuel (Default)	1
Third-party Fuels	2
DEF	3
Diesel + DEF	4
Third-party CNG/LNG with GGE to Pounds	5
Third-party CNG/LNG with GGE to Kilograms	6
Third-party CNG/LNG with GLE to Pounds	7
Third-party CNG/LNG with GLE to Pounds	8

3 Press **ENTER** to accept programming.

Level Four CCs

If Using Pump Software V1.7.58 or Later

In this version of software, the calibration mode is automatically entered when the W&M switch is toggled. Earlier versions of software allowed you to toggle the programming switch before entering the programming mode. Now, you must be in the programming mode before toggling the W&M switch. For additional information, refer to "Important Information About Using V1.7.58 and Later for Encore 500/S/700 S and Eclipse Dispensing Pump Software" on page 5-7.

If the Pump Uses Software Version Earlier Than V1.7.58

You must toggle the W&M switch before entering the programming mode.

Level 4 provides configuration level commands that are performed by the factory to initialize dispenser and configure software to match dispenser hardware. Under certain conditions, an ASC may have to perform these commands to repair or convert a dispenser.

Note: Improper dispenser configuration will result in display of applicable configuration ECs when pump is placed in normal operation.

Programming Level	Security PIN Code	CC Range
Level 4	0128	84-99

- Press **ENTER** after making selection in all levels.
- Press **F1** to perform additional programming.
- Press **F2** to exit programming.

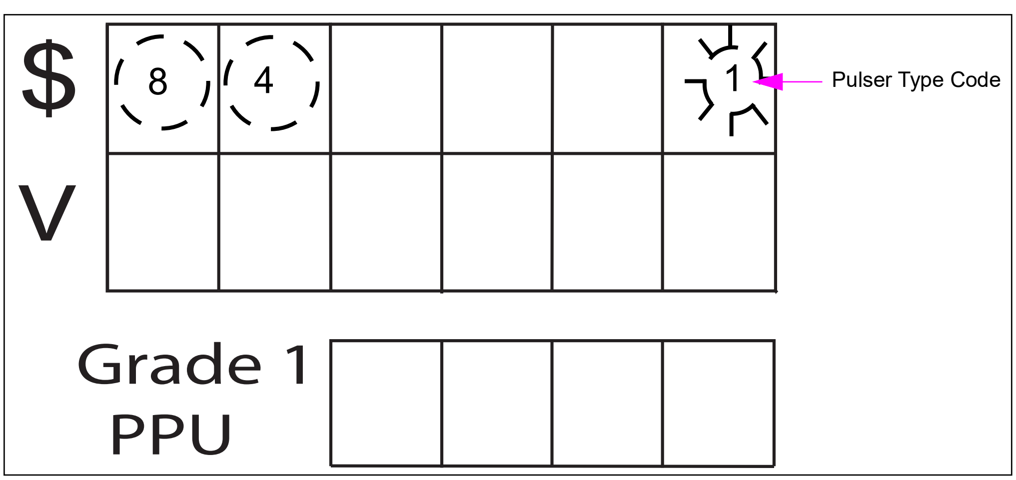
CC84: Set Pulser Type

This function allows the programmer to set the pulser type for the dispenser to work properly with either Standard (0), Standard Encrypted (1), or UHF Encrypted (2) Pulsers. The default for the dispenser is Standard Pulser (0). This setting applies to all normal pulser devices in UHF, Standard Flow, and Ecometer based devices. Encrypted pulser is a special pulser type. Once set to encrypted pulser, the software will detect pulser serial numbers and store them for pulser validation. Once set to encrypted pulser, this parameter cannot be set back to standard pulser. Each time the switch is opened and a 1 or 2 is entered again, the pulser IDs are rediscovered.

Encrypted Pulser is only supported in pump software V02.9.42 or later.

The layout of this CC is as shown in Figure 5-97.

Figure 5-97: Setting Pulser Type



To program, proceed as follows:

1 Press **8**, **4**, and press **Enter** to access the CC84 programming.

2 Enter the pulser type from the following table:

Option	Pulser Type
0 (DEFAULT)	Standard Pulser
1	Encrypted Pulser
2	UHF Encrypted

3 Press **Enter** to confirm the programming change.
Note: If the unit is configured for single-sided operation, even though encrypted pulsers may be installed in the positions for the unused side, the display will show 0 for those pulsers. For a dispenser that does not use all 8 pulsers, 0 will be displayed for the pulsers that are unused whether encrypted pulsers are installed or not.

Uploading the Latest Software

To upload the latest software version, proceed as follows:

1 Use the Gilbarco single click update to obtain the current production version (V02.9.42P is the minimum supported version).
Note: Refer to Technician Resource Page (TRPs) for any software updates.

2 Ensure that the calibration and security switch for the PCN is unsealed and turned on before uploading software.

3 Using the Laptop Tool, upload and install the correct software version.

4 When PCN software installation is complete, double check all configuration and programming options if the previous software version was earlier than 01.8.30L or 02.8.30P. For a list of changes for this software version, refer to *MDE-3860 Programming Quick Reference Guide* and current TRPs.

5 Configure the dispenser for encrypted pulsers by entering Level 4 programming and setting CC84 to 1 for Encrypted Pulser and 2 for UHF Encrypted.

Figure 5-98: Encore Encrypted Pulser Programming Screen

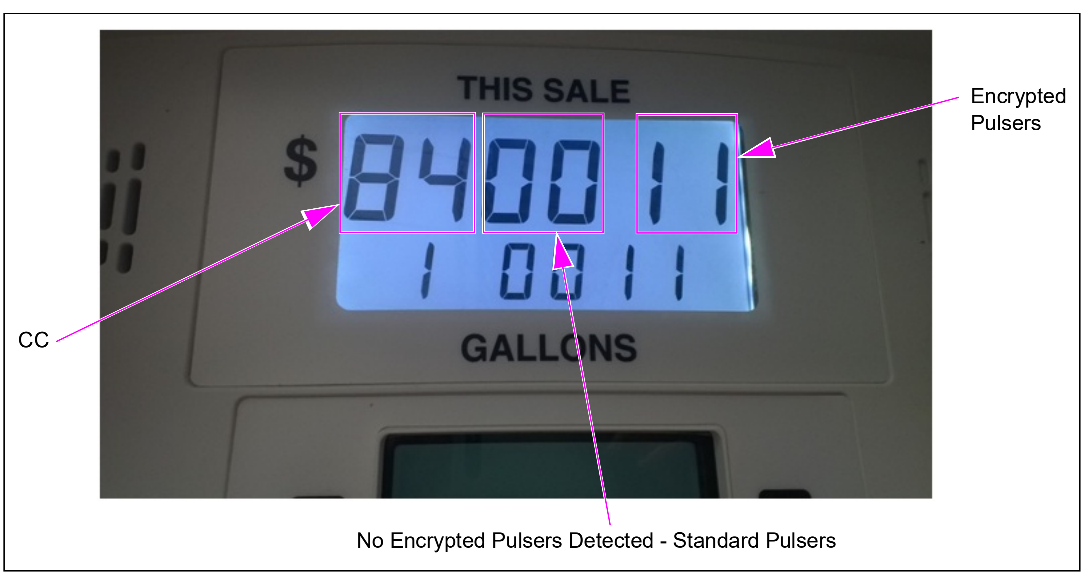


Figure 5-98 on page 5-118 is an example showing an MPD unit (NA1, NC1, NG4, or NG5 model code) or 3+0 Blender type (NN1, NN5, NJ0, or NJ1 model code).

0011 and 0011 shown in Figure 5-98 on page 5-118 indicate positions that are recognized by the PCN as having encrypted pulsers installed. View from side 1 (side A) will show a map with positions of pulsers as they are installed in the hydraulics.
Note: Configuration is the same regardless of unit type or model.

6 Exit the programming mode by pressing **F2** on the manager keypad.

7 After the dispenser reboots, EC8025 will be displayed for each grade. Enter level 4 programming again and set CC 84 again to detect pulsers and to verify installation.
Note: This allows the unique ID for each encrypted pulser to be detected and verified in each of the current physical pulser positions.

8 Exit the programming mode by pressing **F2** on the manager keypad.

9 Ensure that there are no ECs displayed on any grade for the unit. If any EC is displayed, identify the error and troubleshoot appropriately. For the latest ECs and definitions, refer to *MDE-3860 Programming Quick Reference Guide*.

10 Dispense product for each grade on both sides of the unit to verify that encrypted pulsers are installed and working properly.

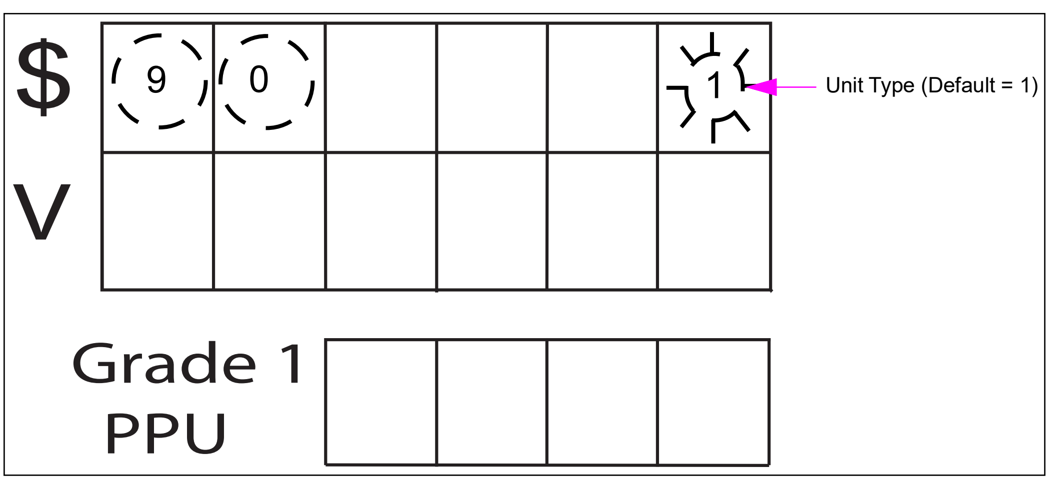
11 Check calibration and calibrate grades as necessary with the configured prover can size. For the configured calibration can size, refer to CC82. For calibration procedure, refer to *MDE-4281 Calibration Quick Reference Card Encore 300/500/550, and Eclipse Units*.

12 After verifying calibration, place a new seal on the PCN security switch.

CC90: Set Unit Type

Function allows programmer to set unit type for dispenser.

Figure 5-99: Setting Unit Type



You must perform a “cold coldstart” before changing selection in CC90 or the unit type may not be properly set, regardless of what is indicated during programming.

Unit type assignment is based on a numeric code as shown in the following table:

Unit Type Assignment Table

Unit Code	Unit Type
1 (Default)	1 Grade, 1 Hose MPD
2	2 Grade, 2 Hose MPD
3	3 Grade, 3 Hose MPD
4	4 Grade, 4 Hose MPD
5	2 Grade, 1 Hose MPD
6	3 Grade, 1 Hose MPD
7	3 Grade, 1 Hose + 1 MPD
8	3 Grade, 3 Hose Blender
9	3 Grade, 3 Hose Blender + 1
12	2 Grade, 1 Hose Blender
13	3 Grade, 1 Hose Blender
14	4 Grade, 1 Hose Blender
15	5 Grade, 1 Hose Blender
16	2 Grade, 1 Hose Blender + 1
17	3 Grade, 1 Hose Blender + 1
18	4 Grade, 1 Hose Blender + 1
19	5 Grade, 1 Hose Blender + 1
20	2 Grade, 1 Hose Blender + 2 Grade 1 Hose Blender
21	3 Grade, 1 Hose Blender + 2 Grade 1 Hose Blender
22	2 Grade, 1 Hose Blender + 1 + 1
23	3 Grade, 1 Hose Blender + 1 + 1
24	2 Grade, 1 Hose Blender + 2 Grade 2 Hose Blender
25	3 Grade, 1 Hose Blender + 2 Grade 2 Hose Blender
26	Reserved
64 (NF0)	3 Grade + 2 Grade
65 (NF1)	3 Grade + 1 Grade + 1 Grade
66 (NF2/4)	3 Grade + 1 Grade + 1
67 (NF6)	3 Grade + 1 Grade + 1 Grade + 1
68 (NF7)	4 Grade + 1 Grade + 1
69* (NF8)	3 Grade + 1 + 1 Grade, which reserves “+1” for diesel.

Note: Unit Code 64-69 Universal Multi-Hose Blender.
** Denotes 2 of 4 product blending versus 2 of 3 product blending.*

1 Press **9**, **0**, and then press **ENTER** on the manager keypad.

2 Select Unit Type (1-21) and press **ENTER**.

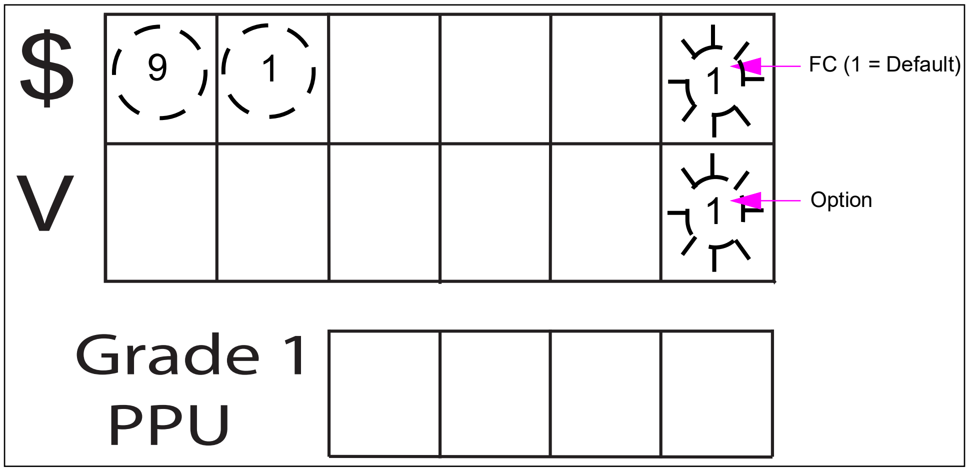
CC91: Select Installed Options

Note: Self-configuring.

Allows the factory, and under certain circumstances, a field programmer, to install optional equipment in the dispenser. FCs under this command are used to enable/disable dispenser options. By default, all options are not installed. Entry into CC91 will default to FC1. The dispenser will generate an event log for improperly configured options.

Layout and meaning of each digit position for this programming feature are shown in Figure 5-100.

Figure 5-100: Selecting the Installed Options



1 Press **9**, **1**, and then press **ENTER** on the manager keypad.

2 Select FC (1-21) and press **ENTER**.

FC and Option Selection Table

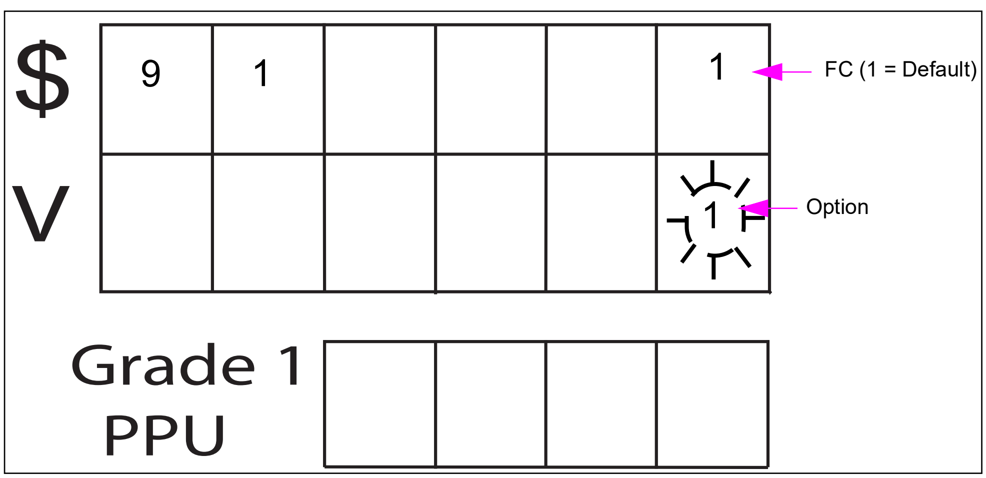
FC	Dispenser Option Description	Option Selections
1 - Default	Preset Type	1 = No Keypad (Default)
		2 = 5 Button Preset
		3 = Incremental Preset
		4 = Programmable Pump Preset
2	Push-to-start	1 = Disabled (Default)
		2 = Enabled
3	Vapor Recovery *Note: VAPORIX options are allowed only for pump software later than V02.8.60P.*	1 = Disabled (Default)
		2 = VaporVac
		3 = VMC/Vaporix Without Shutdown
		4 = VMC/Vaporix With Shutdown
5	Push-to-stop Button	1 = Disabled (Default)
		2 = Enabled
6	ATC	1 = Disabled (Default)
		2 = Enabled
7	Grade Select Function *Note: Available only for pump software V02.8.86P or later.*	1 = Grade Select Before Pump Handle (Default)
		2 = Grade Select Allowed After Lamp Test
		3 = One Grade Select Allowed
8	Miscellaneous Protection Method *Note: Available only for pump software V02.8.60P or later.*	1 = None (Default)
		2 = Mexico
		3 = Clear Last Transaction With Switch
		4 = OIML

FC	Dispenser Option Description	Option Selections
11	Totalizer Function	1 = Not Installed 2 = Totalizer per Meter/Side 3 = Totalizer per Inlet
12	Nozzle Activated	1 = Disabled (Default) 2 = Enabled
13	Meter Type *Notes: 1) Not available in pump software 01.8.30 or later.* *2) Only available in 01.7.85 or earlier.*	1 = C++ Meter (Default) 2 = LC Meter
26	Operation Mode Protection *Note: Only available with V02.9.06P software or later.*	1 = Disabled (Default) 2 = Security Switch Required for Changing Operation Mode
30	Pulser Error Reporting *Note: Only available in pump software 02.9.77 or later.*	1 = Persistent Major Error (Default) 2 = Major Error 3 = Medium Error
31	Unauthorized Flow Error Reporting *Note: Only available in pump software 02.9.77 or later.*	1 = Minor Error (Default) 2 = Medium Error 3 = Medium Error with Two-wire Offline
32	South Korea Roundup Method This FC can only be used in conjunction with the money or volume roundup feature in CC22. *Note: Only available in pump software 02.9.77 or later.*	1 = 0.100 Liter/Gallon Minimum (Default) 2 = 0.075 Liter/Gallon Minimum 3 = 0.050 Liter/Gallon
33	Price Change Method OIML Price Change Method performs a 5 second Delay at next transaction start to ensure price is Displayed for a minimum of 5 seconds before Transaction fuel flow starts. *Note: Only available in pump software 03.0.01 or later.*	1 = USA Price Change Immediate Method (Default) 2 = OIML Price Change Immediate Method
34	Door Switch Mode *Note: Only available in pump software 03.2.12 or later.*	1 = Disabled (Default) 2 = Enabled 3 = Clear Last Transaction Switch
37	Door Switch Reporting *Notes: 1) Only available in pump software 03.2.12 or later.* *2) Only available when Door Switch Method Enabled.*	1 = Logging/Information Only (Default) 2 = Stop Current Transaction Only 3 = Disable Side 4 = Disable Side (Persistent) 5 = Disable Pump 6 = Disable Pump (Persistent)
38	Door and Hydraulics Panel Alarms *Notes: 1) Only available in pump software 03.2.12 or later.* *2) Only available when Door and/or Hydraulics Panel Switch Methods are enabled and FC 37 and 42 are NOT set for logging/information only. To silence alarm (but retain the error) enter the level 4 PIN Code.*	1 = Disabled (Default) 2 = Enabled

FC	Dispenser Option Description	Option Selections
39	DEF/ARLA/Adblue Cold Weather Warning *Notes: 1) Only available in pump software 03.2.20 or later.* *2) Only available for DEF/ARLA/ Adblue dispensers.*	1 = Disabled (Default) 2 = Enabled
40	Log Language *Note: Only available in pump software 03.2.20 or later.*	1 = English (Default) 2 = Spanish 3 = French
41	Hydraulics Panel Switch Monitoring *Note: Only available in pump software 03.3.10 or later.*	1 = Disabled (Default) 2 = Enabled
42	Hydraulics Panel Switch Reporting *Notes: 1) Only available in pump software 03.3.10 or later.* *2) Only available when Hydraulics Panel Switch Monitoring Enabled.*	1 = Logging/Information Only (Default) 2 = Stop Current Transaction Only 3 = Disable Side 4 = Disable Side (Persistent) 5 = Disable Pump 6 = Disable Pump (Persistent)
45	Special Option for Door Security *Notes: 1) Only available in pump software 03.3.21 or later.* *2) Only allowed for Persistent Door Switch Reporting Options.*	1 = Disabled (Default) 2 = Enabled

Note: FC4 is available only in pump software 01.7.85 or earlier.

Figure 5-101: Selecting the Option Code



3 Select Option Code and press **ENTER**.

FC1: Preset Keypad Type

FC1 allows the programmer to pick the installed preset type. If no prior preset type has been programmed, the DEFAULT display will show no preset. Otherwise, the display will show the current preset type. This FC directly affects the function of CC22.

FC2: Push-to-start
FC2 allows the programmer to enable the push to start option. Although this option can be enabled for all unit types, the push-to-start button is most often used as the “second action” on a multi-hose dispenser. If used the dispenser must have the push-to-start keypad or the auxiliary keypad installed for proper operation.

FC3: Vapor Recovery
FC3 allows the programmer to enable the vapor recovery system installed.

FC5: Push-to-stop
FC5 allows the programmer to enable or disable the pump to stop feature of the dispenser. This FC allows use of CC56 for the pump to stop function. If used the dispenser must have the push-to-stop keypad or the auxiliary keypad installed for proper operation.

FC6: ATC
FC5 allows the programmer to enable or disable the Encore ATC feature. If used, the dispenser must have the ATC hardware set and temperature probes installed for proper operation.

FC7: Grade Select Function
FC7 allows the programmer to select methods for how grade select buttons operate. This feature has no function if no grade select buttons or installed on the dispenser or the dispenser is configured for auto-on.

FC8: Miscellaneous Protection Method
FC8 allows the programmer to select methods for special pump security operation required by European and Mexican standards.

(All listed options available after V03.0.32.)
1 = Standard (NCWM/MC Category 2)*
2 = Mexico Protection (NOM-185)
3 = Reserved
4 = OIML Protection (OIML R117)
5 = NCWM/MC Category 3
(Option 5 is allowed only if CC 91.47 = 3)

Note: First, set CC91 FC47 PC Serial Communication Protocol to RTP (Option 3), and then set CC91 FC8 Protection Method to Category 3 (Option 5). RTP must be set before Category 3 can be activated.

FC11: Totalizer Function
FC11 allows to programmer to program and select the configuration and method of use for electro-mechanical totalizers. Regardless of function, electronic totals stored in the PCN are not affected. If International Organization of Legal Metrology (OIML) security is used in the Miscellaneous Protection Method, totalizers are required to be installed for proper pump operation.

FC12: Nozzle Activated
FC12 allows the programmer to enable the nozzle activated pump handle setting. A nozzle activated pump handle eliminates the need for a physical pump handle. When a customer removes the nozzle, the pump handle will automatically be set. This option is need for internal timeouts actions.

FC13: Meter Type

FC13 is only in used in pump software 01.7.85L or earlier. This feature allows the programmer to select the meter type installed for use on the dispenser.

FC26: Operation Mode Protection

FC26 allows the programmer to set protection for the operation mode setting. When enabled, this will require the security switch to be unsealed an opened to change the operation mode from Two-wire to Standalone.

FC30: Pulser Error Reporting

FC30 allows the programmer to set the function of the standard pulser error, EC20. When set to 1 (default), an EC20 on any grade will show EC20 in the PPU of the entire side of the dispenser. This error will be persistent and will refresh after each reboot until F1, pin code and F2 is entered at the manager keypad to clear the error. When set to 2, a standard Major error is used. This will EC20 in every PPU of the affected side until reboot. A reboot clears the error until it is redetected. When set to 3, a standard Medium error is used that will be cleared at the end of transaction when the nozzle is replaced.

FC31: Unauthorized Flow Reporting

FC31 allows the programmer to set the function of the authorized flow error 5049. When set to 1 (default), an unauthorized flow detection will go unreported to the dispenser and will only be logged for information in the event log. When set to 2, an unauthorized flow error will be reported as a normal medium error. When set to 3, the detections will act like option 2 except that two-wire will go offline for 10 minutes.

FC32: South Korea Roundup Method

FC32 sets the minimum threshold for the Volume Rounding Digit or Money Rounding Digit feature of preset keypads for operation. This setting sets the threshold under the configured rounding digit that is considered for rounding to the next whole digit. For example, if the default setting is used for.100 of volume and the current dispenser volume is 1.901, pressing the roundup key on the keypad will round to 21.000 rather than 20.000.

FC33: Price Change Method

FC33 allows two similar but slightly different methods for price change processing. The default of 1 performs of an immediate price change when a price change occurs. When set to 2, this performs an immediate price change, but a 5 second delay occurs after lamp test to ensure the customer has seen prices as required by OIML. This will occur on the first transaction for each grade after a price change occurs.

FC34: Door Switch Mode

FC34 is used to enable or disable upper door security.
When set to 1 (default), disabled.
When set to 2, enabled.

FC37: Door Switch Reporting

FC37 is used to set the reporting type for upper door security. *It can be set only when CC91 FC34 is set to enabled. There are multiple settings for this function code, each of which drives a particular behavior when an "upper door open" event is detected.*

Setting	Behavior
1 (default)	Pump event is logged when an "upper door open" event is detected.
2	• Pump event is logged. • Error code 5721 is displayed on the side where the event is detected, any transaction in progress, on that side, is halted. • Error code is displayed until the door is closed. • If a transaction is halted, error code is displayed until the door is closed and the nozzle is returned to the nozzle boot. • For alarm and timeout behavior refer to CC91 FC38 and CC53 FC11.
Setting	**Behavior**
3	• Pump event is logged. • Error code 5721 is displayed on the side where the event is detected and the side is disabled, halting any transaction in progress on that side. • Error code is displayed, the side remains disabled until the door is closed and the unit is reset. • If a transaction is halted, error code is displayed, the side remains disabled until the door is closed, the nozzle is returned to the nozzle boot and the unit is reset. • For alarm and timeout behavior refer to CC91 FC38, CC53 FC10, and CC53 FC11.
4	• Pump event is logged. • Error code 5721 is displayed on the side where the event is detected and the side is disabled, halting any transaction in progress on that side. • Error code is displayed, the side remains disabled until the door is closed and the level 4 PIN code is entered from the manager keypad. • If a transaction is halted, error code is displayed, side remains disabled until the door is closed, the nozzle is returned to the nozzle boot and the level 4 PIN code is entered from the manager keypad. • For alarm and timeout behavior refer to CC91 FC38 and CC53 FC11.
5	• Pump event is logged. • Error code 5721 is displayed on both sides and any transaction in progress, on either side, is halted. • Error code is displayed on both sides, the dispenser remains disabled until the door is closed and the unit is reset. • If a transaction is halted, error code is displayed and the dispenser remains disabled until the door is closed, the nozzle is returned to the nozzle boot and the unit is reset. • For alarm and timeout behavior refer to CC91 FC38, CC53 FC10, and CC53 FC11. *Note: This option should not be used on a Single-Sided dispenser.*
6	• Pump event is logged. • Error code 5721 is displayed on both sides and any transaction in progress, on either side, is halted. • Error code is displayed on both sides, the dispenser remains disabled until the door is closed and the level 4 PIN code is entered from the manager keypad. • If a transaction is halted, error code is displayed, the dispenser remains disabled until the door is closed, the nozzle is returned to the nozzle boot and the level 4 PIN code is entered from the manager keypad. • For alarm and timeout behavior refer to CC91 FC38 and CC53 FC11. *Note: This option should not be used on a Single-Sided dispenser.*

FC38: Door and Hydraulics Panel Alarms

FC38 is used to enable or disable door security alarms. *It can only be set when CC91 FC34 or FC41 is set to 2 (enabled) and CC91 FC37 or FC42 is set to a value higher than 1.*

When set to 1 (default), disabled.

When set to 2, enabled. Once enabled, the alarm will sound whenever an "upper door open" or "hydraulics panel open" event is detected. The alarm will continue to sound until the error code being displayed is cleared or until the configured timeout is reached.

FC39: DEF/ARLA/Adblue Cold Weather Warning

FC39 is used to enable or disable a cold weather warning.
When set to 1 (default), disabled.
When set to 2, enabled.

FC40: Log Language

FC40 is used to set the log language.
When set to 1 (default), English.
When set to 2, Spanish.
When set to 3, French.

FC41: Hydraulics Panel Switch Monitoring

FC41 is used to enable or disable the hydraulics panel/side sheathing security.
When set to 1 (default), disabled.
When set to 2, enabled.

FC42: Hydraulics Panel Switch Reporting

FC42 is used to set the reporting type for hydraulics panel/side sheathing security. *It can only be set when CC91 FC41 is set to enabled. There are multiple settings for this function code each of which drive similar but different behavior when a "hydraulics panel open" event is detected*.

Setting	Behavior
1 (default)	Pump event is logged when an "hydraulics panel open" event is detected.
2	• Pump event is logged. • Error code 5723 is displayed on the side where the event is detected, any transaction in progress, on that side, is halted. • Error code is displayed until the door is closed. • If a transaction is halted, error code is displayed until the door is closed and the nozzle is returned to the nozzle boot. • For alarm and timeout behavior refer to CC91 FC38 and CC53 FC11.
3	• Pump event is logged. • Error code 5723 is displayed on the side where the event is detected and the side is disabled, halting any transaction in progress on that side. • Error code is displayed, the side remains disabled until the door is closed and the unit is reset. • If a transaction is halted, error code is displayed, the side remains disabled until the door is closed, the nozzle is returned to the nozzle boot and the unit is reset. • For alarm and timeout behavior refer to CC91 FC38, CC53 FC10, and CC53 FC11.
4	• Pump event is logged. • Error code 5723 is displayed on the side where the event is detected and the side is disabled, halting any transaction in progress on that side. • Error code is displayed, the side remains disabled until the door is closed and the level 4 PIN code is entered from the manager keypad. • If a transaction is halted, error code is displayed, side remains disabled until the door is closed, the nozzle is returned to the nozzle boot and the level 4 PIN code is entered from the manager keypad. • For alarm and timeout behavior refer to CC91 FC38 and CC53 FC11.
5	• Pump event is logged. • Error code 5723 is displayed on both sides and any transaction in progress, on either side, is halted. • Error code is displayed on both sides, the dispenser remains disabled until the door is closed and the unit is reset. • If a transaction is halted, error code is displayed and the dispenser remains disabled until the door is closed, the nozzle is returned to the nozzle boot and the unit is reset. • For alarm and timeout behavior refer to CC91 FC38, CC53 FC10, and CC53 FC11. *Note: This option should not be used on a Single-Sided dispenser.*

Setting	Behavior
6	• Pump event is logged. • Error code 5723 is displayed on both sides and any transaction in progress, on either side, is halted. • Error code is displayed on both sides, the dispenser remains disabled until the door is closed and the level 4 PIN code is entered from the manager keypad. • If a transaction is halted, error code is displayed, the dispenser remains disabled until the door is closed, the nozzle is returned to the nozzle boot and the level 4 PIN code is entered from the manager keypad. • For alarm and timeout behavior refer to CC91 FC38 and CC53 FC11. *Note: This option should not be used on a Single-Sided dispenser.*

FC45: Special Options for Door Security

FC45 is used to enable to disable door security special options.
When set to 1 (default), disabled.
When set to 2, enabled.

FC47: PC Serial Communication Protocol

(Only available for V04.1.17 or later.)
1 = ZMODEM Protocol*
2 = ZMODEM Protocol with alarm
3 = Real Time Protocol

Note: To activate W&M Category 3, set CC91 FC47 PC Serial Communication Protocol to RTP (Option 3) first, and then set CC91 FC8 Protection Method to Category 3 (Option 5). RTP must be set before Category 3 can be activated.

Laptop Tool does not communicate in Real Time Protocol. To ensure seamless communication and avoid any issues where other PCNs appear offline in the i360, it is crucial that this setting matches the corresponding settings in OMNIA.

FC49: Nozzle Guard Time

(Only available for V04.0.62 or later.)
5* - 60 seconds
This new function code has been added to provide our customers with the ability to customize the nozzle guard time. Prior to PCN software version 04.0.62, the nozzle guard was hardcoded to 5 seconds. This new functionality offers greater flexibility to meet specific customer needs.

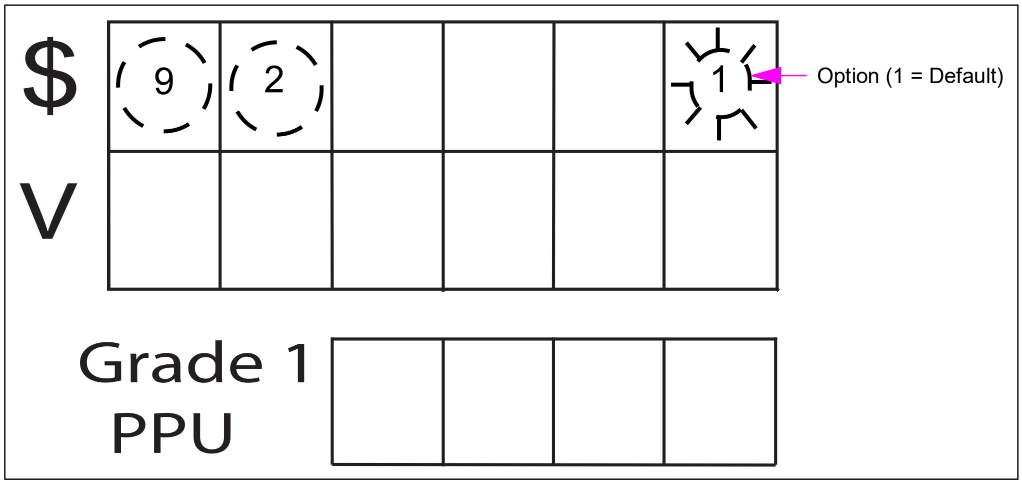
FC50: Special handling of STP in NF series

(Only available for V06.x.xx or later.)
1 = Disabled
2 = Enabled*
This new function code has been added to provide our customers with the ability to control blend errors that may occur due to line pressure drops or no flow in ethanol inlets/pipes. If 43xx errors appear without a probable cause, this function can be disabled by setting it to 1. This option can be enabled again once the root cause of the blend error is determined and corrected.

CC92: Set Side Exists

This option allows setting up for single-sided operation. Default is for both sided operation. Layout and meaning of each digit position for this programming feature are shown in Figure 5-102.

Figure 5-102: Setting up for Single-sided Operation



1 Press **9**, **2**, and then press **ENTER** on the manager keypad.

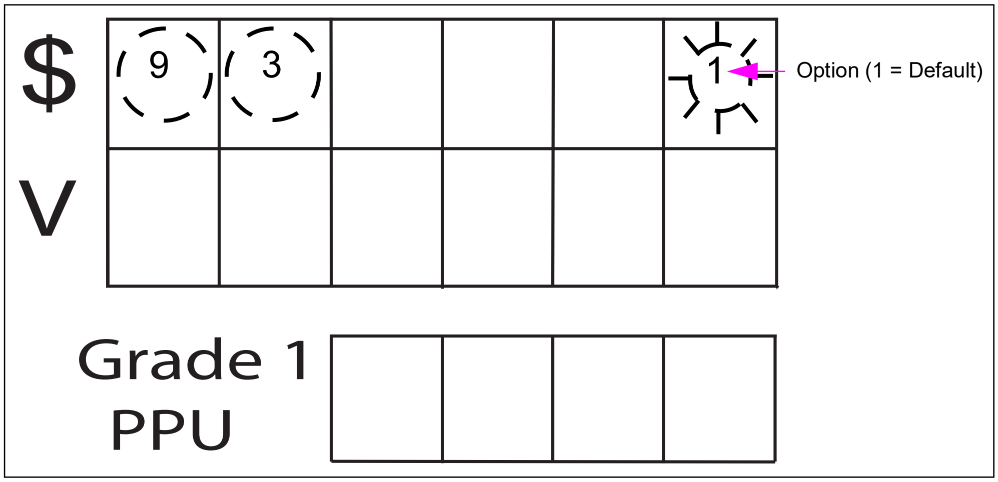
Option	Sides Active
1 (Default)	Both Sides
2	Side 1 Only
3	Side 2 Only

2 Select FC (1-3) and press **ENTER**.

CC93: Set Cash/Credit at Dispenser

CC93 allows programmer to enable Cash/Credit for each side of the dispenser. Layout and meaning of each digit position for this programming feature are shown in Figure 5-103.

Figure 5-103: Setting Cash/Credit at Dispenser



1 Press **9**, **3**, and then press **ENTER** on the manager keypad.

2 Select side (1 or 2) and press **ENTER**.

Option	Sides Active
1 (Default)	No Cash/Credit
2	Side 1 Only
3	Side 2 Only
4	Both Sides

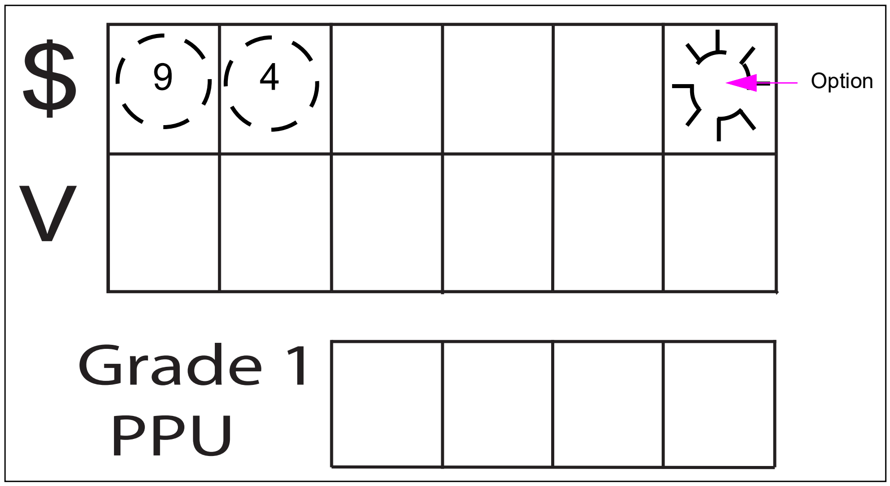
3 Select Option Code (1-4) and press **ENTER**.

CC94: Set Zero PPU Option

When this option is enabled, the dispenser is allowed to dispense fuel with a zero PPU. Default is “DISABLED”, meaning that the dispenser will not deliver fuel if any PPU is set to zero.

This CC is commonly used when a dispenser is not being used for retail sales. Layout and meaning of each digit position for this programming feature are shown in Figure 5-104.

Figure 5-104: Setting Zero PPU Option



1 Press **9**, **4**, and then press **ENTER** on the manager keypad.

Selection	Option
1 (Default)	Disabled
2	Enabled

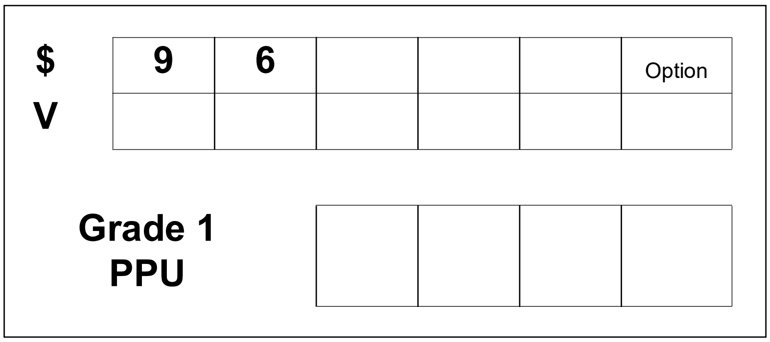
2 Select Option (1 or 2) and press **ENTER**.

IMPORTANT INFORMATION
CC95 is not available in pump software V02.8.60P or later.

CC96: Set Select Feature

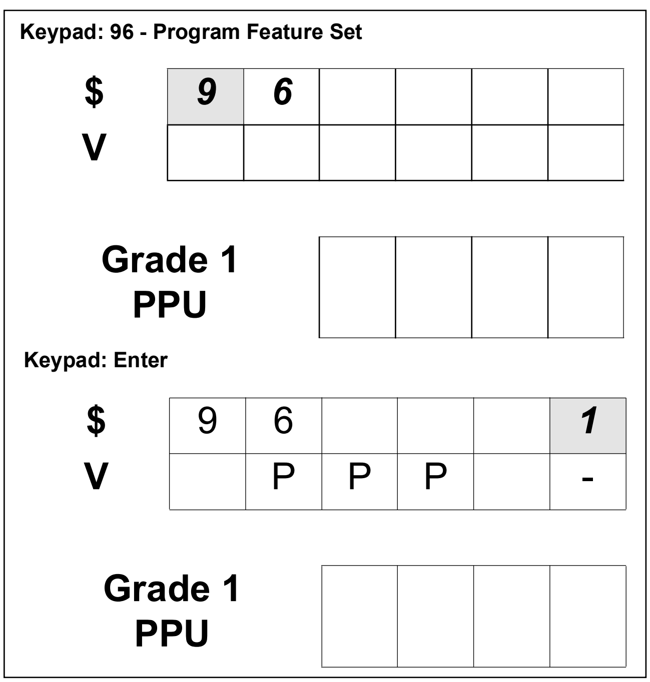
This CC allows the programmer to select the appropriate feature set for the dispenser. The only functionality currently dependent on this setting is the operation of the blinking LEDs (as opposed to blinking PPUs) for ‘S’ models. This CC is protected by the security switch. Without the switch being open, the unit will double beep and prevent entering a new value. All changes will be logged into the regulatory log.

Figure 5-105: Setting Features



1 Select Feature Set CC.

Figure 5-106: Selecting Feature Set CC



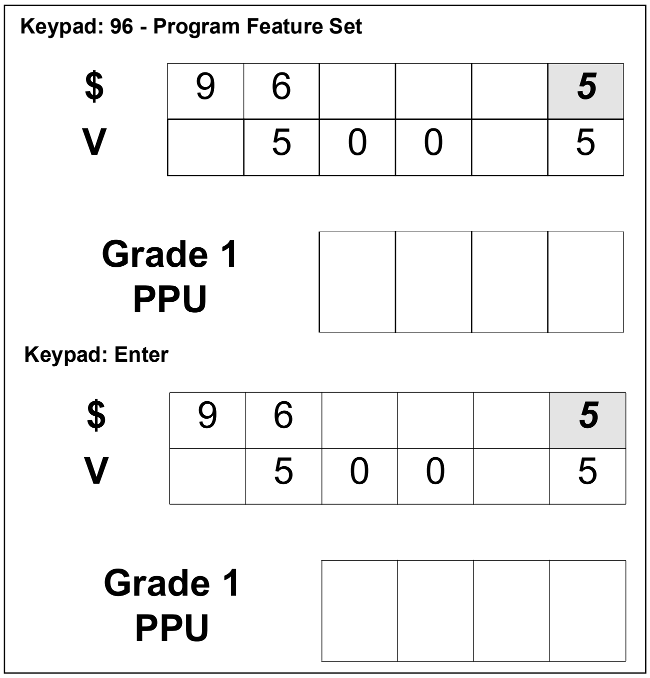
2 Select the desired feature set.

The following table reports the side code meaning:

Feature Set	Code	Display
Encore 500	1	“ 500 -”
Encore 500 S	5	“ 500 5”
Encore 700 S	7	“ 700 5”

Note: - E500 is the default value for new units.

Figure 5-107: Keypad: 5 - Select Encore 500 S Feature Set



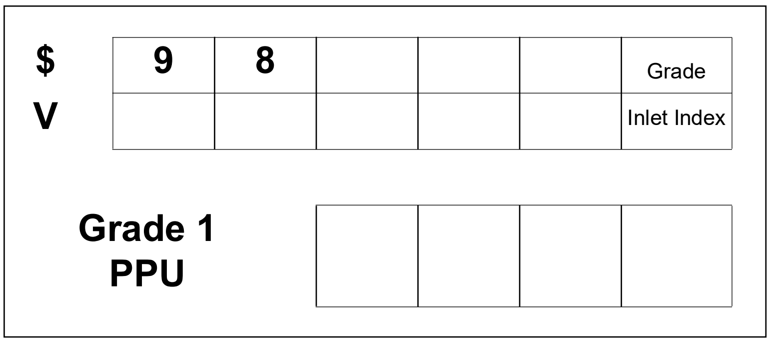
The F1 key will allow additional programming or the F2 key may be used to exit programming.

CC98: Inlet Assignment Per Grade (Multi-Hose Universal Blender)

Note: This CC was introduced in conjunction with the introduction of NF Series Dispensers (Multi-Hose Universal Blender). The CC98 can be set only after a Multi-Hose Universal Blender unit type has been set via CC90.

CC98 allows the user to assign a required pair of product inlets to each blended grade. There is no valid default setting for inlet assignments. Regardless of the Multi-Hose Universal Blender unit type, each blended grade must have an inlet pair assigned. Not having an inlet pair assigned will result in Error Code 4342.

Figure 5-108: Inlet Assignment Per Grade



1 Press **9**, **8**, and then press **ENTER** to access this CC.

2 Select Grade (1-6).

3 Press **ENTER**.

4 Select Inlet Index (1-6).

5 Press **ENTER**.
Note: No Default - MUST be programmed.

Inlet Index	Inlet Pair Assignment
1	W,X
2	W,Y
3	X,Y
4	W,Z
5	X,Z
6	Y,Z

Additional Manager Functions for Encore 500/S/700 S

Unit Totals Retrieval Convention

Totals Examples

For both $ TOTAL and VOL TOTAL keys, press **ENTER** to toggle between grade and side selection.

Non-resettable Money and Volume Totals

The dispenser maintains a set of non-resettable totals for each dispensed fuel grade. These totals reflect total money and volume since dispenser installation time. The following are keypad and display examples for viewing dispenser non-resettable money and volume totals.

Money and Volume Totals

Use the $ TOTAL key to retrieve money totals for each fuel grade. This and the VOL TOTAL key do not require a Security PIN Code. Use the CLEAR key to exit money totals mode. The “1” shown in the leftmost $ display indicates that this total is a non-resettable money total. The VOL TOTAL displays the volume for the grade selected. Volume for both sides of the unit can be viewed. Use the CLEAR key to exit the volume total mode. This convention comes from the Advantage series product line and is maintained here for ASC and site manager familiarity.

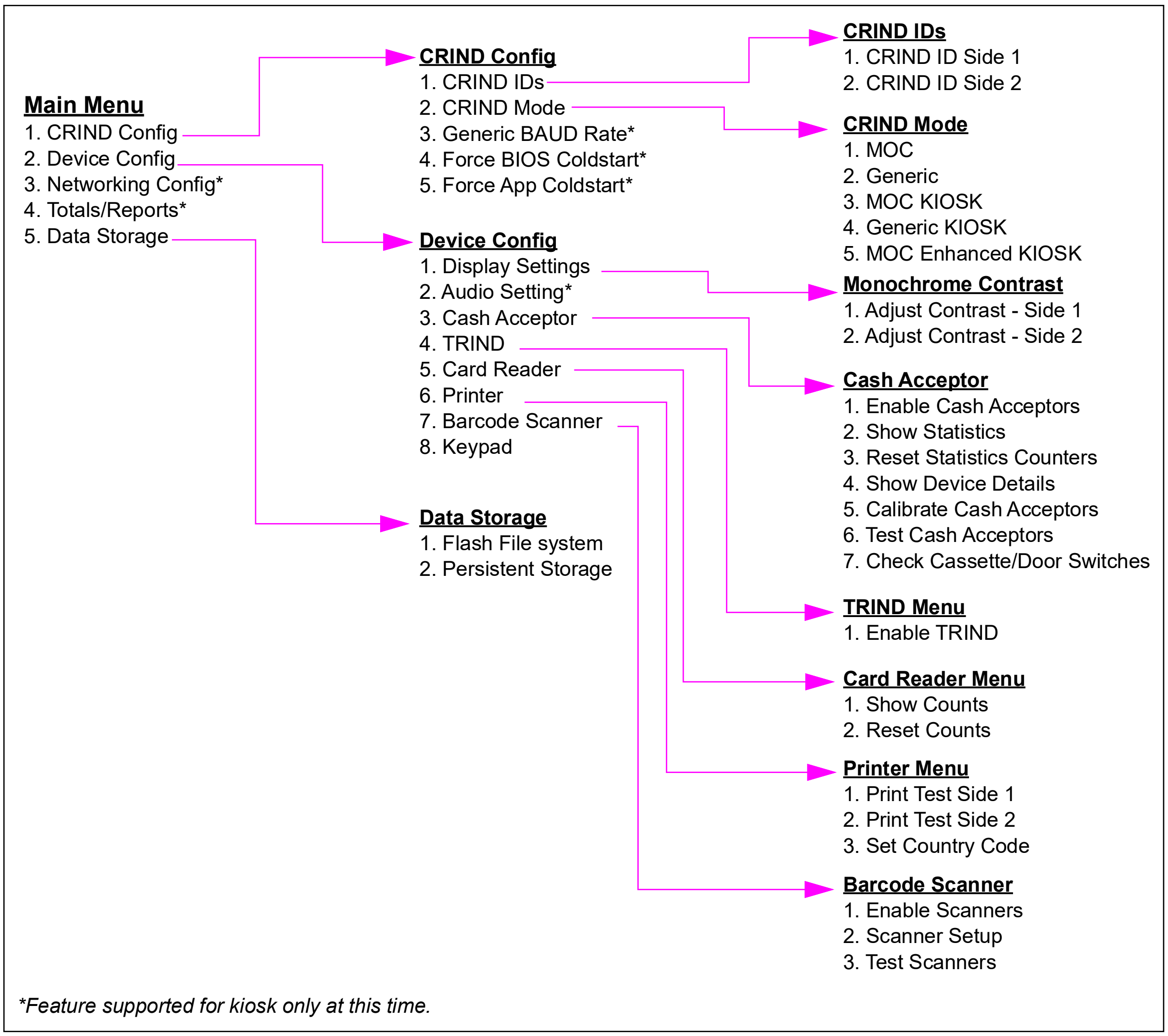
Programming and Diagnostics of CRIND and TRIND at Pump/Dispenser

These instructions apply to the monochrome CRIND only. For later CRINDs Generic color, M3 (SPOT), M5, (FlexPay II), and M7 (FlexPay IV) refer to their respective service manuals.

Menu Tree

The options that are available from the Main Menu are shown in Figure 5-109.

Figure 5-109: Options Available on the Main Menu



Encore 500 Audit Trail Information

The **Enter** Key is used to display blend ratios and the number of times the blend ratios and volume units have been adjusted. While in this mode, a flashing "P" will be displayed in the leftmost digit of the money display. The **Clear** key is used to exit blend info mode. The unit will also exit blend info mode after 1 minute has elapsed. This key does not require a security code.

W&M Data Review

All Encore 500 based systems provide a function that provides a simple method for officials to review common items for auditing a dispenser. Pressing **Enter** on the manager keypad of the dispenser with all fuelling points idle will yield the following information each time **Enter** is pressed. Below are tables of items shown on the Money, Volume and PPU displays. Different information will be shown for MPD versus blender types:

W&M Data Review for Non-blenders

Non-blender unit types include any units types 1-7 for MPD and Single Hose MPD, including Ultra-Hi Flow; DEF; 1+1 Ultra-Hi Flow and DEF; and 2+1 Ultra-Hi Flow and DEF.

For Unit Type 1-7 (MPD Types with Standard Fuels or DEF)								
Number of times **Enter** is pressed	1	2	3	4	5	6	7	8
Money Display	P	Programmed Feature Set (CC96)	P1	P2	P3	P4	P5	P6
Volume Display		Programmed Unit Type (CC90)	Blank	Software Version Number	Pump Boot Version Number	Side 1 Door Node Software Version Number	Side 2 Door Node Software Version Number	ATC Node Version (If ATC, enabled otherwise No change)
PPU Display(s)		Number of times meter has been cancelled	Pulses per volume unit for meter	MIP Version Number	Blank	Door Node Host Version	Door Node Host Version	Blank

W&M Data Review for Blenders

Blender unit types are any unit type 8-9 and 12-25 that represent Single Hose Blenders, Multi-Hose Blenders, 3+2 Dual Blenders and any of the above containing +1 single pure grade or +1 +1 with two extra pure grades.

For Unit Type 8-9, 12-25 (Blender Types with Standard Fuels or DEF)									
Number of times **Enter** is pressed	1	2	3	4	5	6	7	8	9
Money Display	P and Number of times Blend Rations of Changed	Programmed Feature Set (CC96)	P1	P2	P3	P4	P5	P6	P7
Volume Display	Number of times Volume Type has changed	Programmed Unit Type (CC90)	Blank	Blank	Software Version	Pump Boot Version Number	Side 1 Door Node Software Version Number	Side 2 Door Node Software Version Number	ATC Node Version (If ATC Enabled, Otherwise No Change)
PPU Display(s)	Blend Ratios	Days since the meter has been calibrated	Number of times meter has been calibrated	Pulses per Volume Unit for Meter	MIP Version Number	Blank	Door Node Host Version	Doot Node Host Version	No Change

W&M Data Review for Third-party Fuels

Third-party fuels are set via the Alternate Fuels programming detailed in CC83. This is a set of standard Gilbarco electronics that are sold to specifically work with a third-party set of Hydraulics for Compressed Natural Gas (CNG), Liquefied Petroleum Gas (LPG), Liquefied Natural Gas (LNG), Hydrogen and other fuel types.

Third-party CNG and other weight or mass delivered fuels are shown to the customer in a format called a Gasoline Gallon Equivalent (GGE) or a Gasoline Liter Equivalent (GLE) approved by the U.S. and Canadian measurement authorities.

The first screen of the W&M Data Review will be a review of the previous transaction to show the pounds or kilograms equivalent of the previous transaction.

In software 02.9.06 and before, the review was always in a GGE conversion converting the previous volume by multiplying 5.660 to convert to pounds.

In software after 02.9.06, Alternate Fuels programming can allow settings to convert the GGE into pounds or kilograms as well as the GLE to pounds or kilograms by the following factors:

- GGE to pounds - 5.660 pounds
- GGE to kilograms - 2.567 kilograms
- GLE to pounds - 1.693 pounds
- GLE to kilograms - 0.768 kilograms

For Unit MPD Types									
Number of times **Enter** is pressed	1	2	3	4	5	6	7	8	9
Money Display	Last Transaction Money	P	Programmed Feature Set (CC96)	P1	P2	P3	P4	P5	P6
Volume Display	Pound or Mass of Last Transaction	Number of times Volume Type has changed	Programmed Unit Type (CC90)	Blank	Software Version Number	Pump Boot Version Number	Side 1 Door Node Software Version Number	Side 2 Door Node Software Version Number	ATC Node Version (If ATC Enabled, otherwise no change
PPU Display(s)	Current PPU	Days since the meter has been calibrated	Number of times meter has been calibrated	Pulses per volume unit for meter	MIP Version Number	Blank	Door Node Host Version	Door Node Host Version	No change

Audit Trail Screen for Pump Software Version

Verification of installed software is via the W&M Review Screen P3 by use of the internal software Audit Trail. This is performed by using the manager keypad and pressing the Enter key 4 times until the “P2” is shown for MPD units or 5 times until the “P3” is shown on Blender units. This allows the auditor to display the currently installed pump software version in the second line of the main display. P02906, as approved previously, is shown as example in Figure 5-110.

Figure 5-110: Currently Installed Pump Software Version

W&M Screen Information Formats

Blend information is displayed in the following format:

- Money Display: XXXX - Number of times blend ratios have been adjusted.
  Example - 0018 would indicate that the blend ratios have been changed 18 times.
- Volume Display: XXXX - Number of times volume units have been adjusted.
  Example - 0019 would indicate that the volume unit has been changed a total of 19 times.
- PPU Grade Display: XXX - Each PPU Grade will display the percentage of the "lowest octane" of the pure product portion of the blend.

Note: "lowest product" depends on site configurations and fuel types blended. Example: PPU 1 = 100, PPU 2 = 83, PPU 3 = 75. This would indicate that grade one was composed of 100% of the low octane pure product, grade two was composed of 83% of the low octane pure product and grade three was composed of 75% of the low octane pure product.

Additionally for Blender or MPD types, the feature set is shown in the Money display during the boot process (while the software version is displayed in the Volume display). In both cases, the feature set is displayed as described in the following table:

Feature Set	Displayed as
Unprogrammed	PPP -
Encore 500	500 -
Encore 500 S	500 5

Activating the Encore 500/S/700 S CRIND Device

⚠ CAUTION

When any work activity needs to be performed outside the scope of "normal" operation (SW download, terminal activation, etc), you must disconnect all communication cables (CAT5/6) to the POS to prevent possible interference and related issues when performing the required task(s).

To activate the CRIND device, proceed as follows. For details on enabling the CRIND device, refer to *MDE-2562 MPD-3 and The Advantage Series Single-Line CRIND, CRIND with Monochrome LCD, Infoscreen, External CRIND, C-PAM CRIND Service Manual*.

1. Restore power to the fueling units.
2. Initiate CRIND Basic Input/Output System (BIOS) diagnostics using Diagnostic Card (Q12534-170).
3. Select **1. Main Menu** in the Diagnostic Startup menu window.
4. Select **1. CRIND Config** in Main Menu window.
5. Select **1. CRIND IDs** in the CRIND Config window.
6. Select **1. CRIND ID Side 1** in the CRIND IDs window.
7. Enter a value between 1 to 32 that corresponds to the address of the fueling unit in the CRIND ID side 1 window. Press **ENTER** to have the value committed to memory and revert the CRIND device to the CRIND IDs window.

8 Select **2. CRIND ID Side 2** in the CRIND IDs window.

9 Enter a value between 1 to 32 that corresponds to address of the fueling unit in the CRIND ID side 2 window. Press **ENTER** to have the value committed to memory and revert the CRIND device to the CRIND IDs window.

10 Press **CANCEL** to return to the CRIND Config window.

11 Select **2. CRIND Mode** in the CRIND Config window.

12 In the CRIND Mode window, if setup is for G-SITE, select **1. MOC**. If setup is not for G-SITE (that is, third-party POS), select **2. Generic**. Press **ENTER** to have the selected value committed to memory and revert the CRIND device to the CRIND Config window.

CRIND Configuration

Press **1** from the Main Menu. The CRIND Config menu opens.

Figure 5-111: CRIND Config Menu

CRIND Config
1. CRIND IDs 2. CRIND Mode 3. Generic BAUD Rate 4. Force BIOS Coldstart 5. Force App Coldstart
Cancel = Exit

Note: On most screens, you can navigate to the previous screens by pressing ***Cancel****.*

CRIND IDs

This function allows you to choose the side of the dispenser for which IDs must be programmed.

1 Press **1** from the CRIND Config menu. The CRIND IDs menu opens.

Figure 5-112: CRIND IDs Menu

```
CRIND IDs

1. CRIND ID Side 1
2. CRIND ID Side 2

Cancel = Exit
```

2 Press **1** to enter the CRIND ID Side 1 screen. Press **2** to enter the CRIND ID Side 2 screen.

The CRIND ID Side 1/Side 2 screen opens. The heading of the screen will depend on whether you selected side 1 or side 2.

Figure 5-113: CRIND ID Side 1/Side 2 Screen

```
CRIND ID Side 1/Side 2

CRIND ID                          1
(Range: 1 - 32, 99 = disable)

Number Keys = Direct Entry
Clear = Erase Entry
Enter = Save Value and Exit
Cancel = Discard Change and Exit
```

The current CRIND communication address for Side 1/Side 2.

3 Press numeric keys to enter one or two-digit CRIND ID. Legal values for this field are 1-32 and 99. Setting the value at 99 configures the side not to answer polls on the CRIND communication loop. This is useful for field troubleshooting and diagnostic purposes.

4 Press **Enter** to save the edited CRIND ID and exit. The CRIND ID menu opens.
~ OR ~
Press **Cancel** to discard the changes and exit. CRIND IDs menu opens.

CRIND Mode

1 Press **2** from the CRIND Config menu. The CRIND Mode menu opens.

Figure 5-114: CRIND Mode Menu

CRIND Mode		
CRIND Mode	MOC	Shows current setting. For new installations and with existing CRIND nodes that have undergone purge or BIOS coldstart, default will be **Not Configured**.
1. MOC		
2. Generic		
3. MOC Kiosk		
4. Generic Kiosk		
5. MOC Enhanced Kiosk		
Enter = Save Value and Exit		
Cancel = Discard Change and Exit		

2 Press numeric key to change the current CRIND mode setting:
- Pressing key 1 causes the unit to operate in Major Oil Company (MOC) CRIND The Advantage Series - compatibility mode.
- Pressing key 2 causes the unit to operate in Generic CRIND The Advantage Series - compatibility mode.
- Pressing key 3 causes the unit to operate in MOC Kiosk mode and will automatically set the other side's CRIND ID to 99.
- Pressing key 4 causes the unit to operate in Generic Kiosk mode and will automatically set the other side's CRIND ID to 99.

3 Press **Enter** to save the changes and exit. The CRIND Config menu opens.
~ OR ~
Press **Cancel** to discard the changes and exit. The CRIND Config menu opens.

Generic BAUD Rate

This option is currently disabled. Press **3** from the CRIND Config menu to display the Unsupported Feature screen.

Force BIOS Coldstart

This option is currently disabled. Press **4** from the CRIND Config menu to display the Unsupported Feature screen.

Force App Coldstart

This option is currently disabled. Press **5** from the CRIND Config menu to display the Unsupported Feature screen.

Device Configuration

Press **2** from the Main Menu. The Device Config menu opens.

Figure 5-115: Device Config Menu

Device Config
1. Display Settings
2. Audio Setting
3. Cash Acceptor
4. TRIND
5. Card Reader
6. Printer
7. Barcode Scanner
8. Keypad

Display Settings

This function allows you to select a display side for contrast adjustment. If the contrast setting for one side has been set to a value that renders the display temporarily unusable, you can adjust contrast from the opposite side of the unit.

1 Press **1** from the Device Config menu. The Monochrome Contrast menu opens.

Figure 5-116: Monochrome Contrast Menu

Monochrome Contrast
1. Adjust Contrast - Side 1
2. Adjust Contrast - Side 2
Cancel = Exit

2 To display the settings for side 1, press **1**. To display the settings for side 2, press **2**. A screen displays the contrast settings for the side that was selected.

Figure 5-117: Contrast - Side1/Side 2 Screen

Contrast - Side1/Side 2	
Contrast	35
(Range: 1-99)	
Number Keys = Direct Entry	
Clear = Erase Entry	Up>
Enter = Apply Entry	
Cancel = Exit	Down>

3 Use numeric keys to enter a one or two-digit contrast setting. Press the **Up** or **Down** function keys (if applicable) to increase or decrease the contrast setting.

4 Press **Enter** to apply the changes made to the contrast setting and view the display with the new setting immediately.

5 Press **Cancel** to exit the screen and return to the Monochrome Contrast menu. If the contrast setting is currently blank, the display will continue to show the last valid contrast setting that was entered.

Audio Setting

This option is currently available for kiosk only. Press **2** from the Device Config menu to access this feature.

Cash Acceptor

The Cash Acceptor menu allows the user to configure and test operation of the cash acceptors.

Press **3** from the Device Config menu. The Cash Acceptor menu opens.

Figure 5-118: Cash Acceptor Menu

Cash Acceptor
Cash Acceptor Enabled
1. Disable Cash Acceptors
2. Show Statistics
3. Reset Statistics Counters
4. Show Device Details
5. Calibrate Cash Acceptors
6. Test Cash Acceptors
7. Check Cassette/Door Switches

Indicates status of cash acceptor. After cash acceptor support is enabled for both sides of the unit, this status reflects both sides of the unit.

Enable/Disable Cash Acceptors

1 Press **1** from the Cash Acceptor main menu to toggle between enabling and disabling the cash acceptors. After the cash acceptors are enabled, the cash acceptor software starts to function, such as the Show Statistics option is chosen which requires the software to be running.

Screen shown in Figure 5-119 will display the first time you attempt to access a cash acceptor function in diagnostics mode.

Figure 5-119: Cash Acceptor Enabled Screen

Cash Acceptor
Cash Acceptor Enabled
Cash Acceptor Software Ready
Enter = Continue

2 Press **Enter** to continue to the screen option you selected.

Show Statistics

This function allows you to view the cash acceptor events statistics for each side of the unit for troubleshooting purposes. Cash acceptor statistics are collected for all cash acceptor activity, regardless of whether the cash acceptor events occur in the course of normal transactions or during a diagnostics session.

1 Press **2** from the Cash Acceptor main menu. Figure 5-120 opens.

Figure 5-120: Cash Acceptor Screen

Cash Acceptor	
Statistics for Side 1	
Insertions	000005
Rejects	000000
Escrows	000000
Jams	000000
Enter = Next Screen	
Cancel = Exit	

2 Press **Enter** to toggle between side 1 and side 2 counter screens. The first screen displays the statistics for side 1 and the second screen displays the statistics for side 2.

3 Press **Cancel** to exit and return to the Cash Acceptor main menu.

Reset Statistics Counters

This function allows you to reset cash acceptor event statistics for each side of the unit. This function must be used after correcting a cash acceptor problem so that future statistics will reflect the repaired cash acceptor state. Statistics will also be cleared each time persistent storage is cleared/purged through the Persistent Storage menu.

1 Press **3** from the Cash Acceptor main menu. Figure 5-121 opens.

Figure 5-121: Cash Acceptor Screen

Cash Acceptor
Cash Acceptor Enabled

1. Reset Statistics - Side 1
2. Reset Statistics - Side 2

Cancel = Exit

2 To reset statistics for side 1, press **1**. To reset statistics for side 2, press **2**. Screen shown in Figure 5-122 opens to ensure that the statistics have been reset.

Figure 5-122: Cash Acceptor Screen

Cash Acceptor
Cash Acceptor Enabled

Statistics Reset

3 The Cash Acceptor Reset Statistics screen displays after a few seconds. Press **Cancel** to exit and return to the Cash Acceptor main menu.

Show Device Details

This screen allows you to query the firmware version(s) installed on each of the cash acceptor units in the system.

1 Press **4** from the Cash Acceptor main menu. Figure 5-123 opens.

Figure 5-123: Cash Acceptor Screen

Cash Acceptor	
Cash Acceptor Enabled	
Side 1 Details	
Model: 80	Firmware Version: 36
Side 2 Details	
Model: 80	Firmware Version: 36
Cancel = Exit	

2 Press **Cancel** to exit and return to the Cash Acceptor main menu.

Calibrate Cash Acceptors

This function allows you to execute a calibration procedure on either side of the unit.

Note: It is possible to calibrate the cash acceptor from the single side that started the diagnostics session.

1 Press **5** from the Cash Acceptor main menu. Figure 5-124 opens.

Figure 5-124: Cash Acceptor Screen

Cash Acceptor
Cash Acceptor Enabled
1. Calibrate - Side 1
2. Calibrate - Side 2
Cancel = Exit

2 To calibrate side 1, press **1**. To calibrate side 2, press **2**. Figure 5-125 opens for the side being calibrated.
Note: Do not calibrate using currency.

Figure 5-125: Cash Acceptor Warning Screen

<u>**Cash Acceptor**</u>
Cash Acceptor Enabled
WARNING: You must insert a proper
calibration slip on the next screen
to calibrate the cash acceptor.
If you do NOT have a calibration slip,
exit NOW using the Cancel key.
Enter = Continue
Cancel = Exit

If you select any interactive cash acceptor operation for the opposite side (the side that is not being calibrated), Figure 5-126 opens.

Figure 5-126: Cash Acceptor Warning Screen

<u>**Cash Acceptor**</u>
Cash Acceptor Enabled

WARNING: You will disrupt cash
acceptor sales on the other side of
this dispenser until it is restarted

Enter = Continue
Cancel = Exit

3 Press **Enter** to continue with the calibration and proceed to the next screen.

~ OR ~

Press **Cancel** to stop the calibration process and return to the Cash Acceptor main menu.

4 If you press **Enter**, Figure 5-127 opens.

Figure 5-127: Cash Acceptor Screen

<u>**Cash Acceptor**</u>
Cash Acceptor Enabled

Insert Calibration slip

The two green "arrow" lights on the cash acceptor bezel will blink rapidly, indicating that the device is in calibration mode and ready to accept a calibration card.

A reference slip of calibration paper, about the size of U.S. paper currency is required to perform this operation.

5 Insert the calibration slip. Figure 5-128 opens.

Figure 5-128: Cash Acceptor Screen

Cash Acceptor
Cash Acceptor Enabled

Calibration in Progress. Please Wait.

The cash acceptor calibrates itself to the content it received from the calibration slip.

Figure 5-129 opens after the cash acceptor has completed calibration. The operator's result is displayed.

Figure 5-129: Cash Acceptor Screen

Cash Acceptor
Cash Acceptor Enabled

Calibration <result>

Enter = Continue

Results of calibration display here:
- **Succeeded** - if calibration completed properly
- **Failed** - if calibration did not complete properly

6 Press **Enter** to return to the Cash Acceptor Calibration menu.

Test Cash Acceptors

This function allows you to execute specific diagnostic tests on the cash acceptor. The following functions can be performed:

- Inserting a bill and reading/escrowing the bill if not rejected by the cash acceptor.
- Returning a bill from the escrow position.
- Returning a bill from the escrow position and exiting Diagnostics Mode.
- Stacking a bill from the escrow position (if a Test Cassette has been installed).

Note: A second Test Cassette is required to stack bills during diagnostics mode. The CRIND BIOS must see a set of cassette-removed/replaced events to allow bill stack testing. Otherwise, the "stack bill" option will not be provided from the Cash Acceptor Interactive Diagnostics menu.

1 Press **6** from the Cash Acceptor main menu. Figure 5-130 opens.

Figure 5-130: Cash Acceptor Screen

Cash Acceptor
Cash Acceptor Enabled
1. Run Diagnostics - Side 1
2. Run Diagnostics - Side 2
Cancel = Exit

2 To start diagnostic tests for side 1, press **1**. To start diagnostic tests for side 2, press **2**.

A message, "Please Wait..." is displayed and then the Figure 5-131 opens.

Figure 5-131: Cash Acceptor Screen

Cash Acceptor
Cash Acceptor Enabled
To allow bill stack testing, you must remove the station's cassette and attach test cassette.
Enter = Continue
Cancel = Exit

Notes:1) If the CRIND BIOS detects a pair of Cassette Removed/Cassette Replaced events as having occurred some time earlier in the current diagnostics session, this screen will not open.

2) If you select any interactive cash acceptor operation for the opposite side (the side that is not being calibrated), then Figure 5-133 on page 5-151 opens.

Figure 5-132: Cash Acceptor Warning Screen

Cash Acceptor
Cash Acceptor Enabled
WARNING: You will disrupt cash acceptor sales on the other side of this dispenser until it is restarted
Enter = Continue
Cancel = Exit

3 If you are troubleshooting a potentially defective cassette, remove the station cassette, and replace it with another similar cassette for testing purposes.

4 Press **Enter** to display the Cash Acceptor Insert Bill screen.

Figure 5-133: Cash Acceptor Insert Bill Screen

Cash Acceptor
Cash Acceptor Enabled

Insert Bill

Cancel = Exit

5 Insert a bill into the cash acceptor (on the side selected in the Cash Acceptor Diagnostics menu) to initiate diagnostics tests. Figure 5-134 opens.

Figure 5-134: Cash Acceptor Screen

Cash Acceptor
Cash Acceptor Enabled

Bill Escrowed: $X

Clear = Refund Bill
Enter = Stack Bill
Cancel = Refund Bill (if present) and Exit

Status Line displays the current state of the cash acceptor being tested. Possible values for this line:

- **Bill Escrowed $X** - bill is currently held in escrow position, awaiting command: bill has been read.
- **Bill Escrowed Bookmark/Unknown** - unknown or invalid bill type is read, or device is rejecting a bill that is currently left in the bezel, awaiting removal.

6 Press one of the following keys:

- **Clear** - instructs cash acceptor to return the bill in escrow.
- **Enter** - instructs cash acceptor to stack the escrowed bill in the test cassette.
  Note: This option is only available if the CRIND BIOS previously detected a cassette swap.
- **Cancel** - instructs cash acceptor to return the bill in escrow. You are asked to replace the station's cassette and then must press **Enter**. The Cash Acceptor Run Diagnostics screen opens.

7 When ready to exit screen, press **Cancel**. Figure 5-135 opens.

Figure 5-135: Cash Acceptor Screen

Cash Acceptor
Cash Acceptor Enabled

Please replace the station’s cassette.

Enter = Continue

You must press **Cancel** to exit this screen.

8 Remove the test cassette and return the station’s cassette. This operation must be performed by the manager so that the vault can be locked when the station cassette is restored.

9 Press **Cancel** to return to the Cash Acceptor Diagnostics menu.

Check Cassette/Door Switches

This function allows you to select which side of the unit must be used for switch testing. It is possible to check switches on either side, regardless of which side started the diagnostics session. Use this function to test the operation of the three switches that govern cash acceptor operation:

- **Cassette switch** - changes state whenever cash acceptor cassette is removed or replaced.
- **Vault switch** - changes state whenever the cash acceptor vault is opened or closed.
- **Door switch** - changes state whenever the main Encore CIM door or Eclipse oven door is opened or closed.

1 Press **7** from the Cash Acceptor main menu. Figure 5-136 opens.

Figure 5-136: Cash Acceptor screen

Cash Acceptor
Cash Acceptor Enabled

1. Check Switches - Side 1
2. Check Switches - Side 2

Cancel = Exit

2 To start the switch testing procedure for side 1, press **1**. To start the switch testing procedure for side 2, press **2**. Figure 5-137 opens.

Figure 5-137: Cash Acceptor Screen

Cash Acceptor
Cash Acceptor Enabled

Cassette Switch: Closed
Vault Switch: Open
Door Switch: Open

Cancel = Exit

If you select any interactive cash acceptor operation for the opposite side (the side that is not being tested), then Figure 5-138 opens.

Figure 5-138: Cash Acceptor Warning Screen

Cash Acceptor
Cash Acceptor Enabled

WARNING: You will disrupt cash
acceptor sales on the other side of
this dispenser until it is restarted

Enter = Continue
Cancel = Exit

- Press **Enter** to continue with the previously selected cash acceptor task.
- Press **Cancel** to revert to the Cash Acceptor main menu.

3 Press **Cancel** to exit and return to the Cash Acceptor Test Switches menu.

TRIND Device

The TRIND Menu option allows the CRIND BIOS support for TRIND device hardware to be enabled or disabled.

1 Press **4** from the Device Config menu. The TRIND Menu opens.

Figure 5-139: TRIND Menu

TRIND Menu
TRIND Disabled ← Current Status of TRIND Device

1. Enable TRIND

Enter = Save Value and Exit
Cancel = Discard Change and Exit

2 Press **1** to toggle between disabling and enabling the TRIND device.

3 Press **Enter** to save the current setting and return to the Device Config menu.

~ OR ~

Press **Cancel** to discard the current setting and return to the Device Config menu.

Card Reader

This menu allows a technician to review and manage the card reader statistics (counts) that have been collected since the last time the card reader statistics have been cleared. Card reader counts are collected both during the course of normal transactions as well as during the diagnostics session. Card reader percentages can be derived from these numbers to analyze whether this card reader is working acceptably, or requires cleaning or repair. Compare results to other units and similar stations.

Note: A 100% reading will never occur as not all cards will be inserted properly, or it may be a wrong or defective card.

Press **5** from the Device Config menu. The Card Reader Menu opens.

Figure 5-140: Card Reader Menu

Card Reader Menu

1. Show Card Counts
2. Reset Card Counts

Cancel = Exit

Show Card Counts

When this screen is displayed, you will not be able to exit diagnostics mode by using the Gilbarco CRIND diagnostics card. This feature allows you to use the diagnostics card on this screen to interactively test the card reader without unintentionally exiting during the configuration interface session.

1 Press **1** from the Card Reader Menu. The Card Reader Counts screen displays for side 1.

Figure 5-141: Card Reader Counts Screen

Card Reader Side 1	Track 1	Track 2	Track 3
Good Reads	13	13	0
Bad Reads	0	0	0
Total Reads			0
Total Reads Without Data			0

Enter = Next Screen
Cancel = Exit

Note: ISO magstripe cards support up to three "tracks" (sets) of data storage. The data follows a certain format and contains a checksum to guarantee against data corruption on the track.

The screen reports the current totals for each of the supported card reader statistics:

- **Track 1 Good Reads -** count of card insertions with good track 1 data
- **Track 1 Bad Reads -** count of card insertions with bad track 1 data
- **Track 2 Good Reads -** count of card insertions with good track 2 data
- **Track 2 Bad Reads -** count of card insertions with bad track 2 data
- **Track 3 Good Reads -** count of card insertions with good track 3 data
- **Track 3 Bad Reads -** count of card insertions with bad track 3 data
- **Total Reads -** count of all card insertions with at least some valid data on at least one track
- **Total Reads Without Data -** count of card insertions with no valid data present.

2 Press **Enter** to toggle between side 1 and side 2 card reader counts screens.

3 Press **Cancel** to exit and return to the Card Reader Menu.

Reset Card Counts

This function allows the technician to reset card reader counts independently for each side of the unit.

1 Press **2** from the Card Reader menu. Figure 5-142 opens.

Figure 5-142: Card Reader Screen

Card Reader

1. Reset Counts - Side 1
2. Reset Counts - Side 2

Cancel = Exit

2 To reset the card reader counts for side 1, press **1**. To reset the card reader counts for side 2, press **2**. Figure 5-143 opens.

Figure 5-143: Card Reader Screen

Card Reader

Card counts reset

This screen opens for several seconds, then the Card Reader Reset menu displays.

3 Press **Cancel** to return to the Card Reader Menu.

Printer

The Printer Menu allows you to perform certain configuration and diagnostics tasks on the printers.

Press **6** from the Device Config Menu. The Printer Menu screen opens.

Figure 5-144: Printer Menu Screen

Printer Menu
1. Print Test Side 1 2. Print Test Side 2 3. Set Country Code
Cancel = Exit

Print Test Sides

Press **1** or **2** from the main Printer Menu to cause a test receipt to print for that particular side and display the Figure 5-145.

Figure 5-145: Printer Menu Screen

Printer Menu
Sending Printer Test Job Please Wait...

After five seconds, the diagnostics session will return to the main Printer Menu and a sample receipt will be printed on the side 1 or side 2 printer.

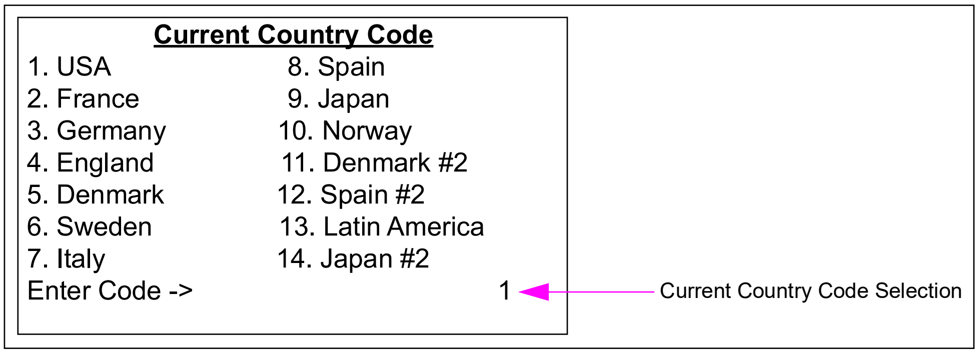
Note: Starting with CRIND Node software V02.1.60, the test receipt will include a sample UPC-A barcode printed horizontally, at the bottom of the test receipt. If barcode scanner hardware is present, then barcode must be scannable from the CRIND diagnostics session. For a description of how the barcode must be scanned, refer to the Barcode Scanner Test menu.

Set Country Code

This function changes the printer's character set to better conform with those used in the individual countries listed at this screen.

1 Press **3** from the main Printer Menu. The Current Country Code screen opens.

Figure 5-146: Current Country Code Screen



2 Press numeric keys to enter a one or two-digit country code. Values range from 01-14. The default setting is 1 USA.

3 Press **Enter** to commit the code entered into persistent storage on the CRIND node. If you enter a number incorrectly and wish to delete it, press **Clear**.

Note: If the setting you enter is out of range, "Value Out of Range" will be displayed in this field for a few seconds and then the field will return to the previous setting.

~ OR ~

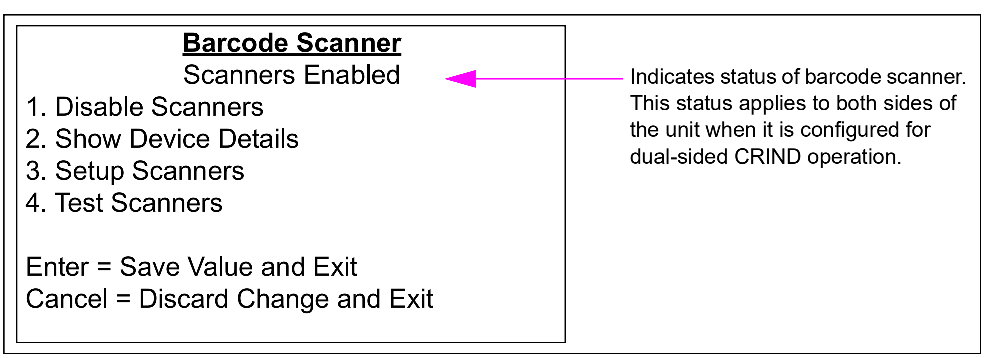
Press **Cancel** to exit and return to the main Printer Menu.

Barcode Scanner

The Barcode Scanner menu allows you to configure and test the barcode scanner operations.

1 Press **7** from the Device Config menu. The Barcode Scanner screen opens.

Figure 5-147: Barcode Scanner Screen



2 Use numeric keys to select the option from the menu.

3 Press **Enter** to save the changes and return to the Device Config menu.

4 Press **Cancel** to discard any changes and return to the Device Config menu.

Enable/Disable Scanners

Press **1** from the Barcode Scanner menu to toggle between disabling and enabling the scanner.

Show Device Details

1 Press **2** from the Barcode Scanner menu to display firmware information about side 1 and side 2 scanners.

Figure 5-148: Barcode Scanner Menu

Barcode Scanner
Scanner Enabled
Side 1 Details
Firmware Version: NBRWCAAH

Side 2 Details
Firmware Version: NBRWCAAD

Cancel = Exit

2 Press **Cancel** to return to the Barcode Scanner menu.

Setup Scanner

Use this function to adjust the barcode scanner settings.

Press **3** from the Barcode Scanner menu. If this is the first time entering the barcode setup screen, then Figure 5-149 opens.

Figure 5-149: Barcode Scanner Warning Screen

Barcode Scanner
Scanner Enabled
WARNING: The following actions could
disrupt scanner use by customers on
the other side of this dispenser.
Ensure that the other side is not in
use before continuing.
Enter = Continue
Cancel = Exit

- Press **Enter** to continue to the Barcode Scanner Setup menu.
- Press **Cancel** to exit to the Barcode Scanner menu.

If this is not the first time entering the barcode setup screen, the Barcode Setup menu opens (see Figure 5-150).

Figure 5-150: Barcode Setup Menu

> **Barcode Scanner**
> Scanner Enabled
>
> 1. Reset to Factory Defaults
> 2. View/Change POS Scanner Settings
> 3. Send POS Settings to Scanner
>
> Cancel = Exit

Reset to Factory Defaults

This function resets the scanner to its default settings. Any required POS-specific settings must be reapplied. Resetting scanners affects both sides automatically.

1 Press **1** from the Barcode Scanner Setup menu. The Barcode Scanner warning screen opens to remind the technician that all scanner settings will be reset if you proceed.

Figure 5-151: Barcode Scanner Warning Screen

> **Barcode Scanner**
> Scanner Enabled
> WARNING: This action will reset
> scanner setup to factory defaults.
> All POS specific changes must be
> reapplied.
>
> Enter = Continue
> Cancel = Exit

2 Press **Enter** to advance the session to begin the process of resetting the scanners to their default settings. All scanner settings will be reset.

~ OR ~

Press **Cancel** to return to the Barcode Scanner Setup menu without resetting the scanners to their default settings.

If you press **Enter**, Figure 5-152 opens for side 1 and then for side 2.

Figure 5-152: Barcode Scanner Screen

Barcode Scanner Scanner Enabled Resetting scanners to Factory defaults Programming Side: X Please Wait...

The scanners are currently being sent configuration commands that will reset them to their default operating parameters.

3 When the reset operations are complete, then the Figure 5-153 opens.

Figure 5-153: Barcode Scanner Screen

Barcode Scanner Scanner Enabled Scanners reset to factory defaults. Enter = Continue

4 Press **Enter** to return to the Barcode Scanner Setup menu.

View/Change POS Scanner Settings

This function allows you to enable or disable codes, and each of the supported barcode formats. Configuration changes will not be automatically sent to the scanner. The application software (POS) establishes the formats that are supported. After the configuration is complete, all settings must be sent to the scanner at the same time using the “Send POS Settings to Scanner” function.

1 Press **2** from the Barcode Scanner Setup menu. Figure 5-154 and Figure 5-155 opens.

Figure 5-154: Barcode Scanner Screen

Barcode Scanner	
Scanner Enabled	
1. AIM Code	Disabled
2. Laser Scanning Mode	Raster
0 = Next Screen	Page 1/4
Enter = Save Values and Exit	
Cancel = Discard Changes and Exit	

Figure 5-155: Barcode Scanner Screen

Barcode Scanner	
Scanner Enabled	
1. AIM Code	Disabled
2. UPCA	Enabled
3. UPCE	Enabled
0 = Next Screen	Page 1/3
Enter = Save Values and Exit	
Cancel = Discard Changes and Exit	

2 To configure the settings, proceed as follows:

- Press **1** to enable/disable the ability for an Automatic Identification Manufacturers (AIM) Code to be prepended to the data reported by the device each time a barcode is scanned.
- Press **2** to enable/disable the ability for a UPCA barcode format to be detected and reported by the scanner toggling between raster and cyclonic laser scanning modes.
- Press **3** to enable/disable the ability for a UPCE barcode format to be detected and reported by the scanner.

3 Press **0** to proceed to the next screen. Figure 5-156, Figure 5-157, and Figure 5-158 opens.

Figure 5-156: Barcode Scanner Screen

Barcode Scanner	
Scanner Enabled	
1. UPCA	Enabled
2. UPCE	Enabled
3. EAN8	Enabled
0 = Next Screen	Page 2/4
Enter = Save Values and Exit	
Cancel = Discard Changes and Exit	

Figure 5-157: Barcode Scanner Screen

Barcode Scanner	
Scanner Enabled	
1. UPCA	Enabled
2. UPCE	Enabled
3. EAN8	Enabled
0 = Next Screen	Page 2/4
Enter = Save Values and Exit	
Cancel = Discard Changes and Exit	

Figure 5-158: Barcode Scanner Screen

Barcode Scanner	
Scanner Enabled	
1. EAN8	Enabled
2. EAN13	Enabled
3. Code 39	Enabled
0 = Next Screen	Page 2/3
Enter = Save Values and Exit	
Cancel = Discard Changes and Exit	

4 To configure the settings, proceed as follows:

- Press **1** to enable/disable the ability for an EAN8 UPCA barcode format to be detected and reported by the scanner.
- Press **2** to enable/disable the ability for an EAN13 UPCE barcode format to be detected and reported by the scanner.
- Press **3** to enable/disable the ability for a Code 39 EAN8 barcode format to be detected and reported by the scanner.

5 Press **0** to proceed to the next screen. Figure 5-159 and Figure 5-160 opens.

Figure 5-159: Barcode Scanner Screen

Barcode Scanner	
Scanner Enabled	
1. EAN 13	Enabled
2. Code 39	Enabled
3. Code 128	Enabled
0 = Next Screen	Page 3/4
Enter = Save Values and Exit	
Cancel = Discard Changes and Exit	

Figure 5-160: Barcode Scanner Screen

Barcode Scanner	
Scanner Enabled	
1. Code 128	Enabled
2. UCC EAN 128	Enabled
3. Interleaved 2 of 5	Enabled
0 = Next Screen	Page 3/3
Enter = Save Values and Exit	
Cancel = Discard Changes and Exit	

6 To configure the settings, proceed as follows:

- Press **1** to enable/disable the ability for a Code 128 EAN 13 barcode format to be detected and reported by the scanner.
- Press **2** to enable/disable the ability for a UCC EAN 128 Code 39 barcode format to be detected and reported by the scanner.
- Press **3** to enable/disable the ability for a Interleaved 2 of 5 Code 128 barcode format to be detected and reported by the scanner.

7 Press **0** to proceed to the next screen. Figure 5-161 opens.

Figure 5-161: Barcode Scanner Screen

Barcode Scanner	
Scanner Enabled	
1. UCC EAN 128	Enabled
2. Interleaved 2 of 5	Enabled
0 = Next Screen	Page 4/4
Enter = Save Values and Exit	
Cancel = Discard Changes and Exit	

8 Use the following instructions to configure the settings:
- Press **1** to enable/disable the ability for a UCC EAN 128 barcode format to be detected and reported by the scanner.
- Press **2** to enable/disable the ability for an Interleaved 2 of 5 barcode formats to be detected and reported by the scanner.

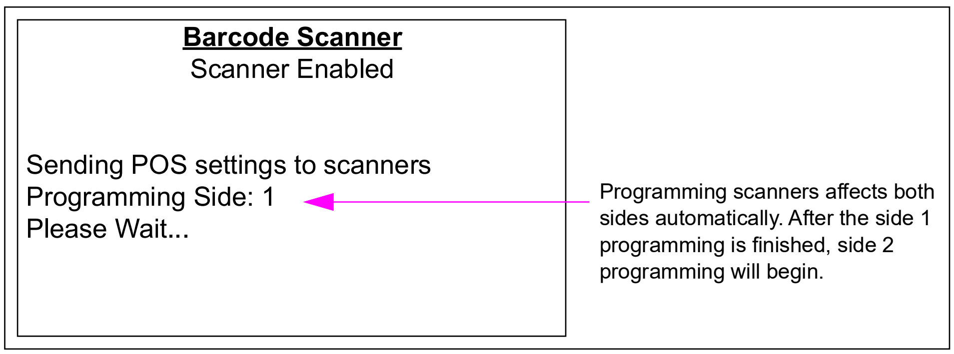
9 Press **Enter** to save the changes and exit. Press **Cancel** to exit without saving the changes.

Send POS Settings to Scanner

Use this function to send the changes you made to the scanner settings to the scanner device.

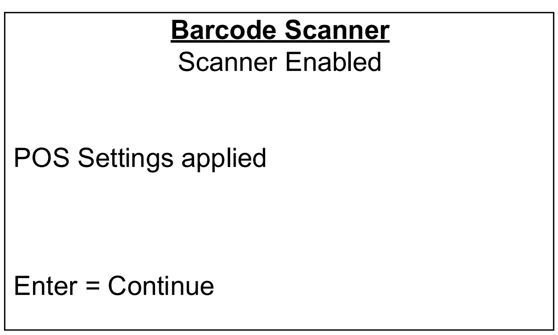
1 Press **3** from the Barcode Scanner Setup menu. Figure 5-162 opens.

Figure 5-162: Barcode Scanner Screen



When programming is complete, then Figure 5-163 opens to notify the technician that all settings from the four-page Barcode Scanner Settings menu have been applied.

Figure 5-163: Barcode Scanner Screen



2 Press **Enter** to return to the Barcode Scanner Setup menu.

Test Scanners

This function enables the technician to run interactive scanner tests on either side of the unit.

1 Press **4** from the Barcode Scanner Setup menu. Figure 5-164 opens.

Figure 5-164: Barcode Scanner Screen

Barcode Scanner
Scanner Enabled

1. Test Scanner - Side 1
2. Test Scanner - Side 2

Cancel = Exit

2 Press **1** or **2** to display the Barcode Scanner interactive test screen. This screen enables the barcode scanner for the selected side and allows the technician to present barcodes to be read.

Figure 5-165: Barcode Scanner Screen

Barcode Scanner
Scanner Enabled

Present test barcode in front of
scanner

This test will timeout in 60 seconds

Cancel = Abort

If the diagnostics session is currently running on the opposite side and this is the first time scanners have been tested in this session, then the warning screen opens (see Figure 5-166). This is a one-time warning message displayed to alert the technician that interactive barcode testing will disrupt the transactions on the opposite side.

Figure 5-166: Barcode Scanner Screen

Barcode Scanner
Scanner Enabled
WARNING: You will prevent barcode
scanner use during sales on the
other side of dispenser until it
is restarted.

Enter = Continue
Cancel = Exit

- Press **Enter** to proceed to the next screen.
- Press **Cancel** to exit and return to the Barcode Scanner Test menu.

3 Present a readable barcode in a format currently enabled on the scanner. Figure 5-167 opens.

Figure 5-167: Barcode Scanner Screen

Barcode Scanner
Scanner Enabled

Scan Data: 012345678905
Code Type: UPC/EAN

Enter = Continue
Cancel = Exit

Scan data and code type (format in which data was encoded) is read from barcode and displayed on screen.

4 Press **Enter** to return to the Barcode Scanner interactive test screen.

~ OR ~

Press **Cancel** to return to the initial Barcode Scanner Test menu.

Barcode Scanner Test Failure Screen

The Barcode Scanner screen opens when the barcode scanner interactive test operation times out after waiting 60 seconds for valid barcode data. The screen lets the technician know that the scanner laser must be off and the scan session has been stopped.

Figure 5-168: Barcode Scanner Screen

Barcode Scanner
Scanner Enabled

Test scan failed

Enter = Continue
Cancel = Exit

- Press **Enter** to return to the Barcode Scanner Test menu.
- Press **Cancel** to return to the Barcode Scanner main menu.

Keypad

The Keypad Menu displays a unique code in real-time for each key that is pressed.

1 Press **8** from the Device Config menu. The Keypad Menu opens.

Figure 5-169: Keypad Menu

Keypad Menu

Key Pressed

Cancel = Exit

2 Press a key on the keypad to display the value of that key.

Figure 5-170: Keypad Menu

Keypad Menu

Key Pressed 1

Cancel = Exit

For example, “1” displays when **1** is pressed on the keypad.

The Cancel key differs from other keys in that it displays “Cancel - exiting” and then the screen returns to the Device Config menu.

The F4 key differs from other keys in that it displays “F4” and then the screen returns to the Device Config menu.

Programming DEF Dispensers

CCs

CC54

Programming of slowdown still applies to the diesel portion of the DEF+1 unit. There is no slowdown programming capability for the DEF portion of the unit. A fixed 0.2 gallons is used. The DEF of the DEF+1 is automatically programmed to 0.2 gallons.

CC82

Calibration prover cans are typically set to 50 gallons for Ultra-Hi units. When calibrating a DEF meter, this must be set to 5 gallons (20 liters). If you do not reset the prover can size, then the calibration will be inaccurate. Ensure that you return the calibration prover can to 50 gallons for Ultra-Hi units after calibrating the DEF meter. Anytime you calibrate the DEF, you must temporarily revert the can size to 5 gallons (20 liters).

During purging, if the Coriolis meter does not have a reasonably high calibration factor installed, the DEF hose will start and stop flow rate repeatedly. This is because 10 GPM flow control valve of the dispenser turns the valve on and off to control flow rate, even if the real flow rate is well below 10 GPM. To prevent the issue, program the flow control to a number similar to the Ultra-Hi (CC80).

CC83

Programming and calibration of the DEF dispenser is similar to the standard Encore S dispensers. A new CC83 must be programmed for the DEF dispenser. The CC83 is programmed for alternate fuel types where option 1 is not programmed, option 2 is natural gas, option 3 is DEF only, and option 4 is DEF+1. Option 3 must be programmed for DEF units.

CC91: FC12

For NA4 DEF units, CC91, FC12 must be set to Option 1 (not installed).

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6 – Electronic and Electrical Components

About This Section

Topics in This Section

Section Overview

This section provides information on the electronics and electrical components contained within the pumps and dispensers. This information may consist of an illustration of the component, functional description and wherever applicable, data on connections, test points, and settings.

How to Use This Section

For a listing of topics contained in this section and its location, refer to "Topics in This Section". Under each of the main topics is a table that lists individual components. Use this table to find information on a specific component. Additionally, the index and table of contents can be used to locate information.

Abbreviations and Acronyms

Term	Description
CRIND	Card Reader in Dispenser
LON	Local Operating Network
MIP	Microprocessor Interface Program
MOC	Major Oil Company
PCA	Printed Circuit Assembly
PCB	Printed Circuit Board
PDN	Pump Door Node
POS	Point of Sale
USB	Universal Serial Bus

Terms Used in This Section

Term	Definition
LON	Local Operating Network - This is a hardware and software system devised by the Echelon company for coordinating distributed control systems, from something as small as a copying machine, to large building utility systems such as Elevators, Fire Alarms, and so on. All reference to LON in this manual refers to systems inside the dispenser electronics.
PCA	Printed Circuit Assembly - This is a dedicated component designed for specific function with limited or non-existent intelligence or processing power. For comparison, refer to "PCAs and Nodes" on page 6-5. This term may be used interchangeably with PCB.
PCB	Printed Circuit Board - This is a dedicated component designed for specific function with limited or non-existent intelligence or processing power. For comparison, refer to "PCAs and Nodes" on page 6-5. This term can be used interchangeably with PCA.
MIP	Microprocessor Interface Program - Allows all the nodes to communicate together.
Neuron	A low level processor found on all nodes. Its primary function is handling communication between higher level processors. The CRIND device and pump nodes contain this kind of higher level processors.
Nodes	Nodes contain intelligence (a processor or 'neuron' and connect to one or more peripheral boards). Nodes communicate with each other over the LON. Refer to Note.
RX	Receiving data
TX	Transmitting data
WatchDog	Monitors and resets processor in the event of a system hang-up.

Note: In and out LON cable connection on nodes are not critical as there is no specific in or out. Where a side's identity is critical, the node involved will have an identifying jumper, so that the system can determine if it is related to Side 1 or Side 2.

About Service Procedures

⚠ WARNING

You are performing inspections and maintenance in a potentially dangerous environment of flammable fuels/vapors and high voltage.

Fire, explosion, or electrical shock can result in severe injury or death if you do not follow safe procedures. Read and follow all safety precautions.

⚠ WARNING

In the event of inclement weather, including snow, ice, or flooding that makes driving conditions dangerous, please avoid servicing units. Always use available door stops to secure upper doors against unwanted/unexpected movement, especially during high winds. If necessary, reschedule service to avoid damage to the equipment. Weather may change unexpectedly, be aware of local weather conditions. During service, if conditions develop making service unsafe, close the unit(s) and proceed to a safe location.

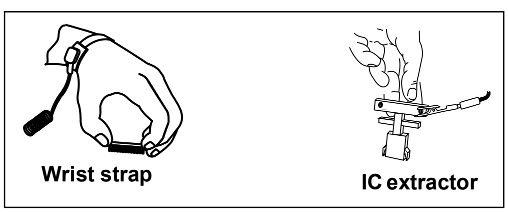
Working on Electronic and Electrical Components

Explosive vapors or liquids may be present in and around fuel dispensing equipment that can be ignited by electrical arcs or static electricity discharges. A potential shock hazard exists when working with this equipment. Additionally, static electricity discharges can damage sensitive electronic components. Follow all precautions and recommendations.

Preparing for Service

- Barricade area that you will be working in.
- Remove all power to unit when servicing electronic components. Remove power from involved devices, STPs, and so on. Multiple disconnects may be required.
- Use extreme caution during any observation or testing procedure that requires power to be applied to the unit.
- Follow Electrostatic Discharge (ESD) procedures to prevent damage to electronic components or ignition of flammable fuels or vapors.

Preventing Electrostatic Discharge



Printed Circuit Assemblies (PCAs) and Integrated Circuits (ICs) are sensitive to electrostatic discharge caused by static electricity. Electrostatic discharge can damage electronic parts.

When removing PCAs or handling sensitive parts:

- Touch an unpainted metal surface to discharge any static electricity buildup.
- Use a wrist strap connected to a grounded metal frame or chassis.
- Place removed PCAs or ICs on a grounded antistatic mat.
- Use an IC extractor tool to remove ICs.
- Place PCAs you plan to return for Credit or repair in antistatic bags.

Hazardous Materials

Some materials present inside electronic enclosures may present a health hazard if not handled correctly. Ensure that you clean your hands after handling equipment. Do not place any equipment in mouth.

WARNING

This area contains a chemical known to the State of California to cause cancer.
This area contains a chemical known to the State of California to cause birth defects or other reproductive harm.

Laser Light Energy

Scanners emit laser light that can damage eyesight when viewed directly.

WARNING

Laser light is used in this product and may present an eye hazard during service.
Direct viewing of laser light may damage eyesight.
Avoid long term viewing of direct laser light.

Replaceable Batteries

Some Printed Circuit Boards (PCBs) contain replaceable batteries. Ensure that you insert batteries correctly.

WARNING

There is a danger of explosion if battery is replaced incorrectly. Replace with only the same or equivalent recommended type. Dispose of used batteries according to the battery manufacturer's instructions.

Replacing Fuses

CAUTION

Using an incorrectly sized fuse may cause equipment damage. To ensure good equipment protection and maintain safe operation, always use the correct replacement fuse.

Helpful Service Information

For PCAs, PCBs, and nodes, wherever applicable:

- Cable connectors are labeled with the destination of the cable.
- Test Points are clearly labeled and recessed to facilitate use of meter probes.
- Test point label usually indicates acceptable range, that is 4.8 - 5.2 VDC.
- Light Emitting Diode (LED) Status Code - refer to LED Status description that is included with other board information, wherever applicable.

PCAs and Nodes

Topics in This Sub-section

About Nodes

Encore 500 and Eclipse nodes communicate over a LON within the pump or dispenser. The nodes receive power and data from the LON bus (discrete cable). All nodes, except for the printer node have downloadable software. The nodes appear to be connected serially but are actually a parallel circuit. LON terminators must be located at the end of the LON run for the system to work correctly.

If an M00317 LON printer is used, the termination is on a printer connector, forming the end of the LON bus. If a USB printer is used, the termination is built into the display node.

About PCAs

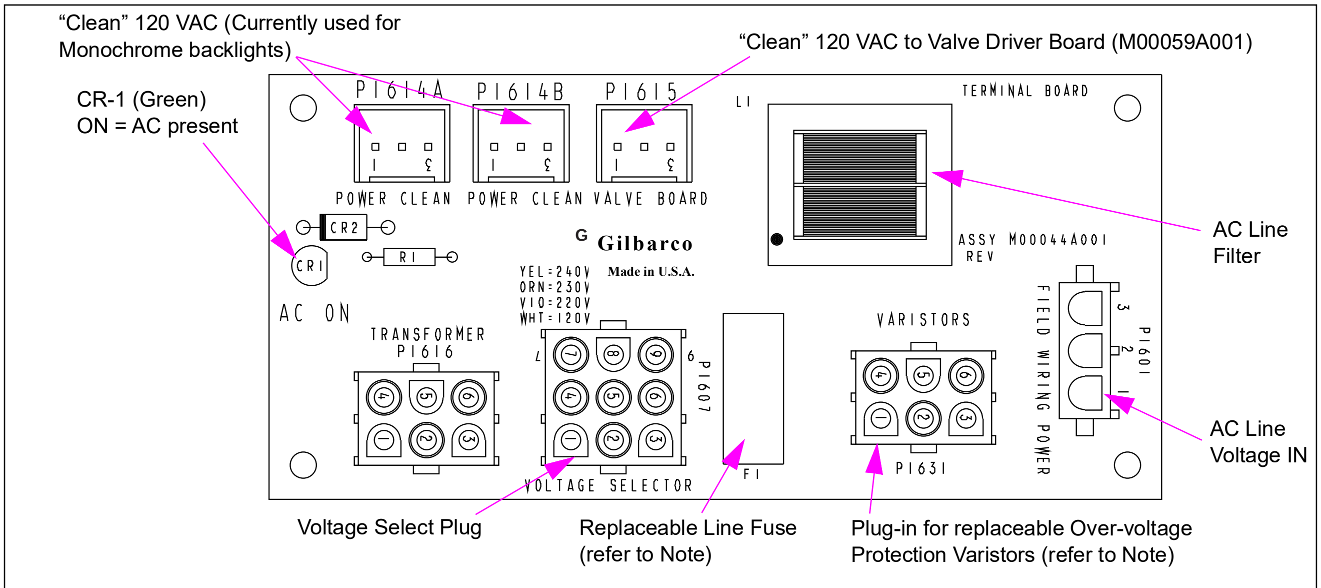
PCAs and PCBs control the hydraulics, monitor fuel delivery, compute, display, and store transaction information, interface with the POS device, and support payment options. The term PCA and PCB may be used interchangeably. Encore 500 and Eclipse employ a LON and Nodes to provide various functional information that drive operations throughout the unit. Generally, nodes contain a processor.

The M18666A001 pump board, M18830A001 display board, and M18983A001 ATC board communicate over a non-LON protocol. These boards are not interchangeable with LON-based boards.

Terminal Block PCA (M00044A001) for Encore 500/Eclipse

This board serves as the unit's field wiring terminal. Field connection received at this board includes AC power (hot, neutral, and ground). This board was obsoleted by M02274 Power Supply.

Figure 6-1: Terminal Block Board (M00044A001)



Note: Replaceable fuse is Type GDC-4 A. It is a Underwriters Laboratories (UL) recognized 4 A, Slow Blow, 5 X 20 mm. If replaced, the designated replacement must be used.

Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P1601 (3-pin)	M00153A1001	Field Wiring Power	Field wiring on terminal block
P1607	None - Plug In	Voltage Program Plug: • M00488A115 (115 VAC) • M00488A220 (220 VAC) • M00488A230 (230 VAC) • M00488A240 (240 VAC)	9-pin
P1614A (3-pin)	M00630A003 (Encore) M00630A004 (Eclipse)	Side 1 Backlight, Ballast and Heater through J1614C to P1614C on M01103A001 cable	3-pin series

Connector#	Through Cable	To Board	At Connector#
P1614B (3-pin)	M00630A003 (Encore) M00630A004 (Eclipse)	Side 2 Backlight, Ballast and Heater through J1614C to P1614C on M01103A001 Cable	3-pin series
P1615 (3-pin)	M00625A001 to J1206/ P1206 connection to M01006A001 cable	1. Transformer Card Reader Heater Sub-Assembly (M00293A001) 2. Steering Valve Driver PCA (M00059A00X)	1. N/A 2. P1206 (3-pin)
P1616 (6-pin)	Multiple Colored Wires	Primary on Power Supply	
P1631 (6-pin)	None - Plug In	Varistor Assembly: • M00487A115 (115 V application) • M00487A230 (115 V application) • M00487A231 (115 V application)	6-pin Plug In

Note: If electronics are damaged due to power surge (for example, lightning), the varistor must be replaced.

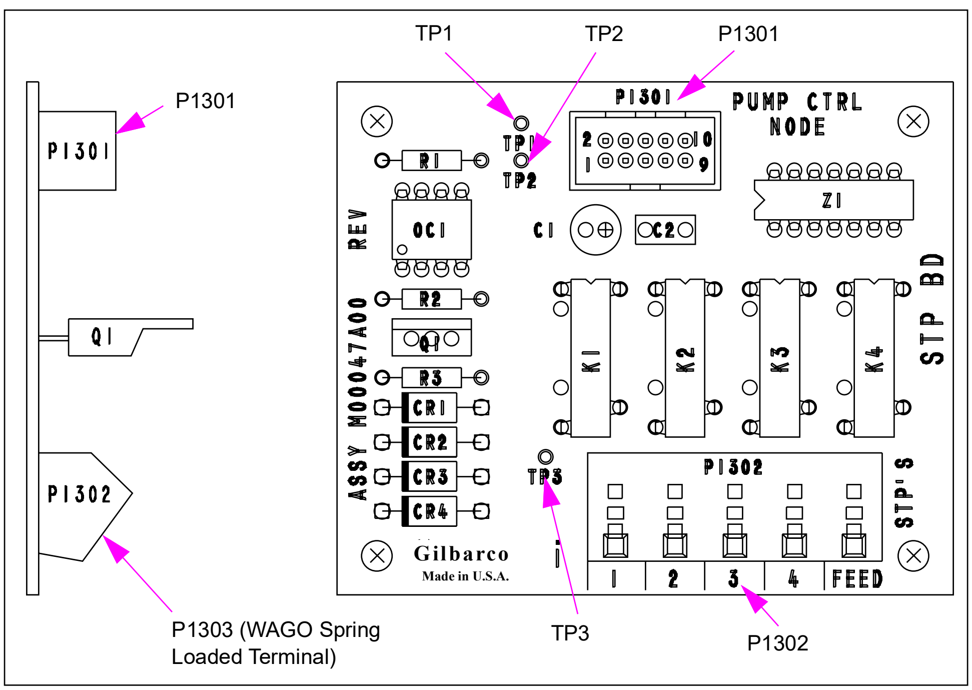
STP Driver PCA (M00047A001)

STP driver board receives line voltage from Terminal Block PCA [M00044A001 (Field Wiring)]. It receives control signal from M00056A001 or M01922A001 Pump Nodes, Totalizer Node and activates STP output signal relays.

Service Tip

Use caution during service when positioning the metal shield over this and other components. If the shield is not positioned correctly, it can short on board components to ground and permanently damage the board.

Figure 6-2: STP Driver Board (M00047A001)



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P1301 (10-pin)	M00635A001	Pump Node (M01922A001 or M0056A001)	P1101 (10-pin)
P1302	1. STP Control Out (Field) 2. Feed, In	1. Field Wiring. Refer to: • *FE-340 Field Wiring Diagram for Encore* • *FE-341 Field Wiring DiagramEclipse Series Dispensers* 2. Field Wiring	1. 4 Outputs 2. Terminal Block or Field Wiring

Test Points

Test Point	Range
TP 1	+5 VDC
TP 2	DC GND
TP 3	STP Feed Voltage

Pump Control Node-1 (M00056A001)

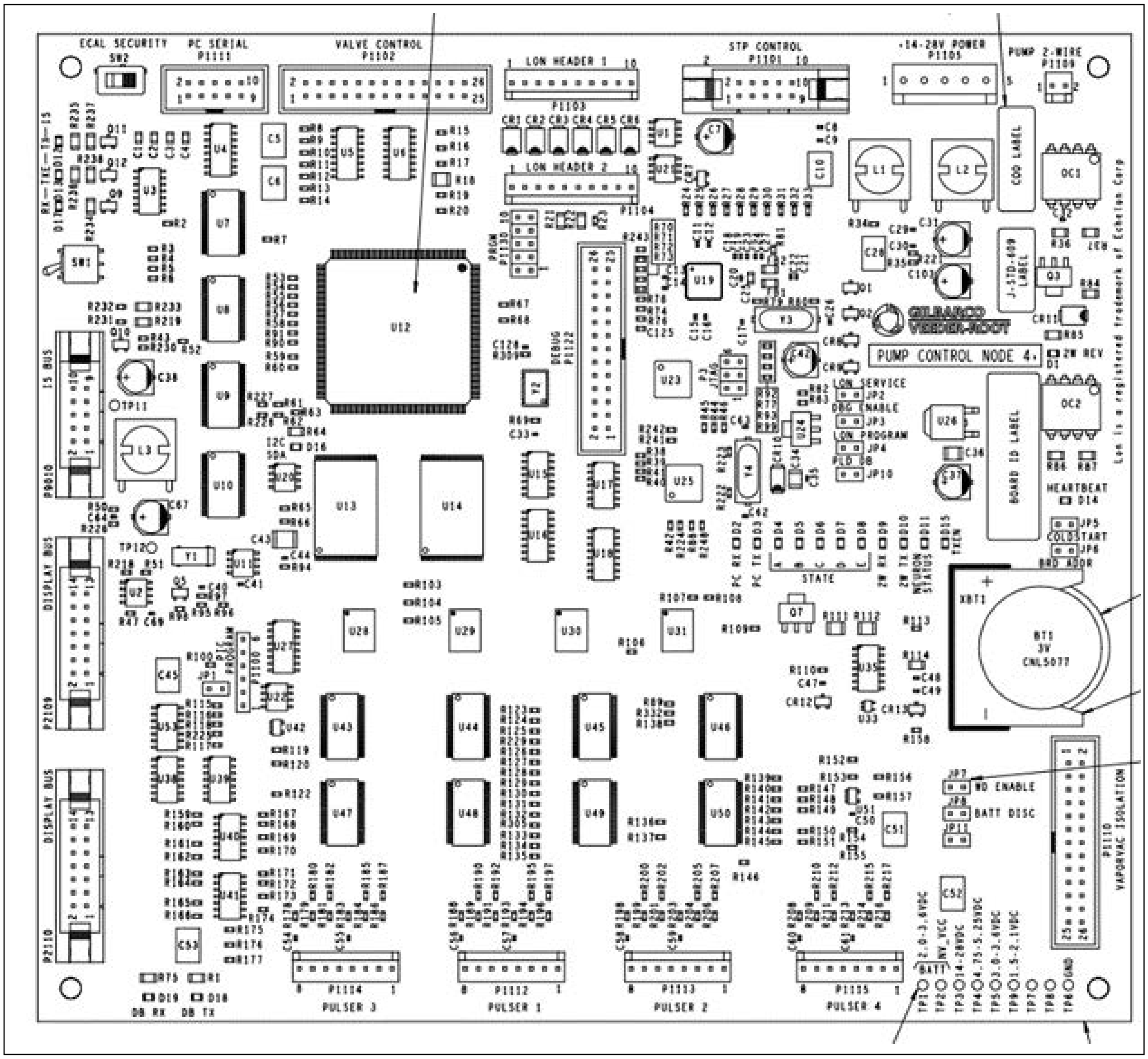
The main purpose of the first generation PCN is to provide control for fueling. This board drives the LON, stores the operating system, configuration (programming), and one "complete" set of totals and operational logs. The PCN connects directly to the LON, valve driver board, pulsers, the STP driver board, the POS two-wire, PC serial port, VaporVac and also connects to the W&M ECAL switch. For non-e-CRIND units, the PCN provides the main processing power for system. The Lithium battery on this board maintains Level 1 and Level 2 programming.

The M18666A001 is highly compatible, but uses a non-LON communication method and can only communicate to M18830 display boards.

Service Tips

- The coldstart jumpers only reset the processor of Level 1 and Level 2 programming. To completely reset the board (cold/coldstart), it is required to load V1.3.74 software and then load the latest version of software. Cold/coldstart clears all programming, the clock, totals, calibration, and resets the processor.
- Software version V1.5.40 or later provides redundant storage of programming information. Two door nodes and one pump node are required to maintain redundancy storage. Never replace two of these boards at the same time or redundant storage may be lost.
- For M18666A001, the redundant storage is only maintained in the pump board and the A-side M18830 display board. There is a new procedure for replacing the M18666A001 pump board in order to retain data.
- The LON circuit must have a terminator at the last node (Does not apply to M18666A001).
- All non-M18666A001pump control nodes are interchangeable, and are software compatible.
- Some nodes list TP3 voltage as 14-22 VDC but 14-28 VDC is also acceptable.
- The pump nodes contain two software types: Pump Microprocessor Interface Program (MIP) and pump node software (Does not apply to M18666A001).
- The pump two-wire transmit (TX) LED may flash under certain circumstances when the pump node is not transmitting.

Figure 6-3: Board Assembly with Connectors labeled

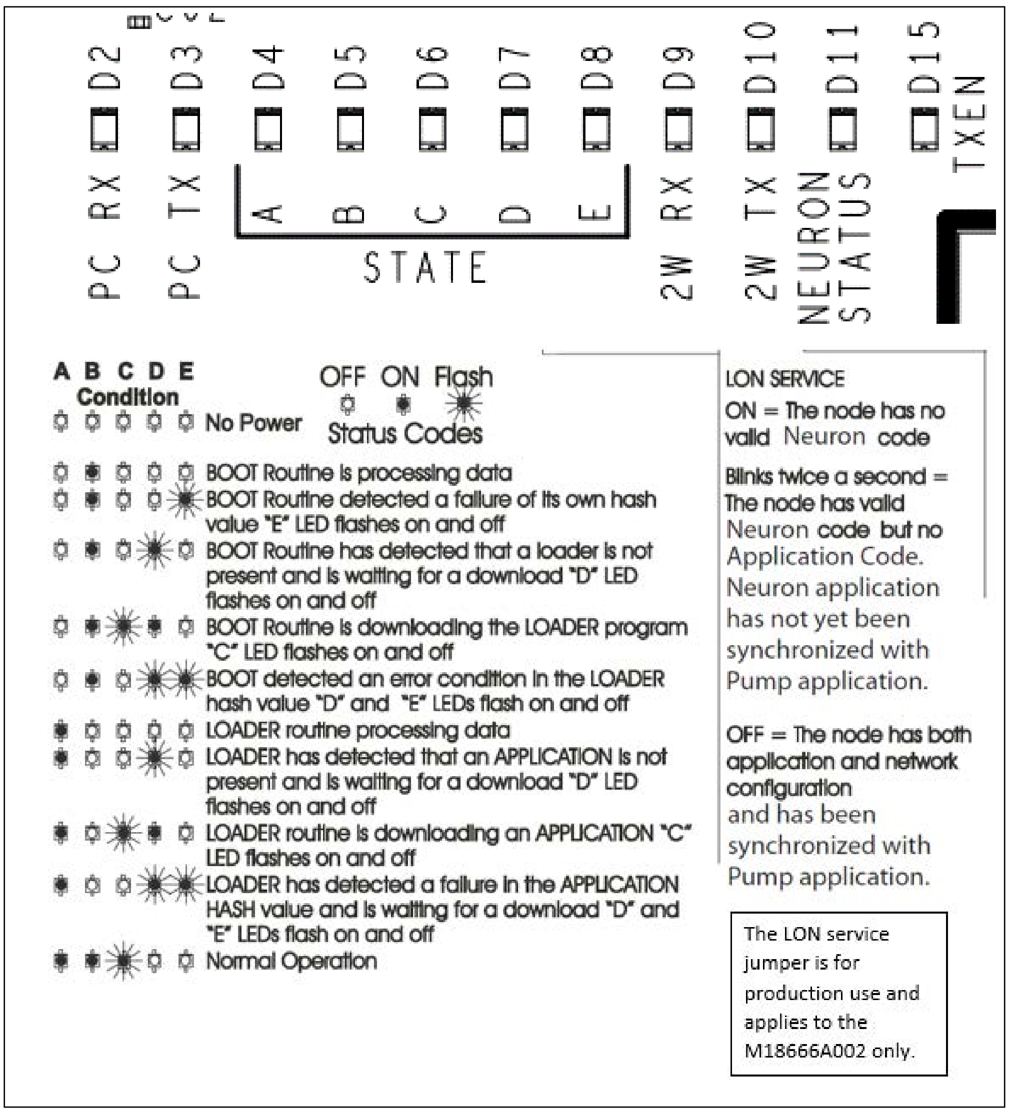


M18666 Connector list

Connector Number	Description	Connects To
SW2	Calibration switch	Secured by calibration slider
P1101	STP Control	M07121 Power distribution board
P1102	Valve Control	
P1103	LON Header	To Door Nodes (Only populated in M18666A002 variant)
P1104		
P1105	Power Input	M07121 Power distribution board
P1109	Pump 2-wire	To forecourt controller via CCP or Omnia boards
P1110	Vapor Vac	Vapor Vac assembly

Connector Number	Description	Connects To
P1111	PC Serial	For technician use. Also connects to CCP or Omnia boards.
P1112	Pulser 1	Pulsers
P1113	Pulser 2	
P1114	Pulser 3	
P1115	Pulser 4	
P1122	DO NOT USE	
P2109	Display Bus	Alternate to the LON bus. Connects to M18830 display boards via a ribbon cable.
P2110		

Figure 6-4: LEDs & function

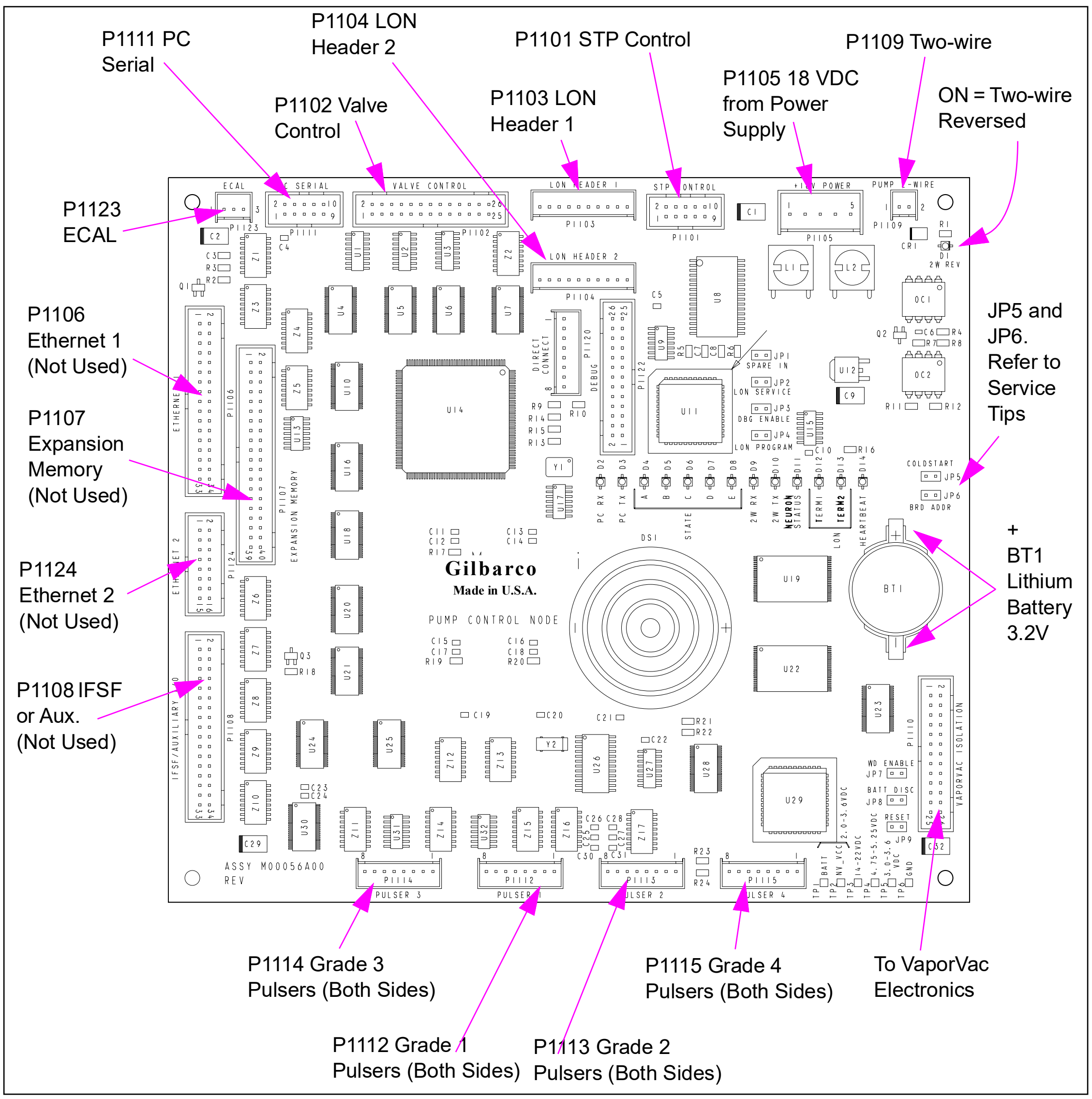


M18666 Test points and Jumpers

Jumper Number	Label	Function
JP 2	LON SERVICE	MUST BE OPEN FOR NORMAL USE
JP 4	LON PROGRAM	
JP 3	DBG Enable	
JP 10	PLD DB	
JP 5	COLDSTART	
JP 6	BRD ADDR	Unused
JP 7	WD ENABLE	MUST BE INSTALLED
JP 8	BATT DISC	Unused
JP 11	None	Pump board reset

Measure TPx to TP6 (DC ground)	Function	Range
TP 1	Battery voltage	3.4V to 2.8V DC
TP 3	Input power	18VDC to 28VDC

Figure 6-5: Pump Control Node-1 PCA

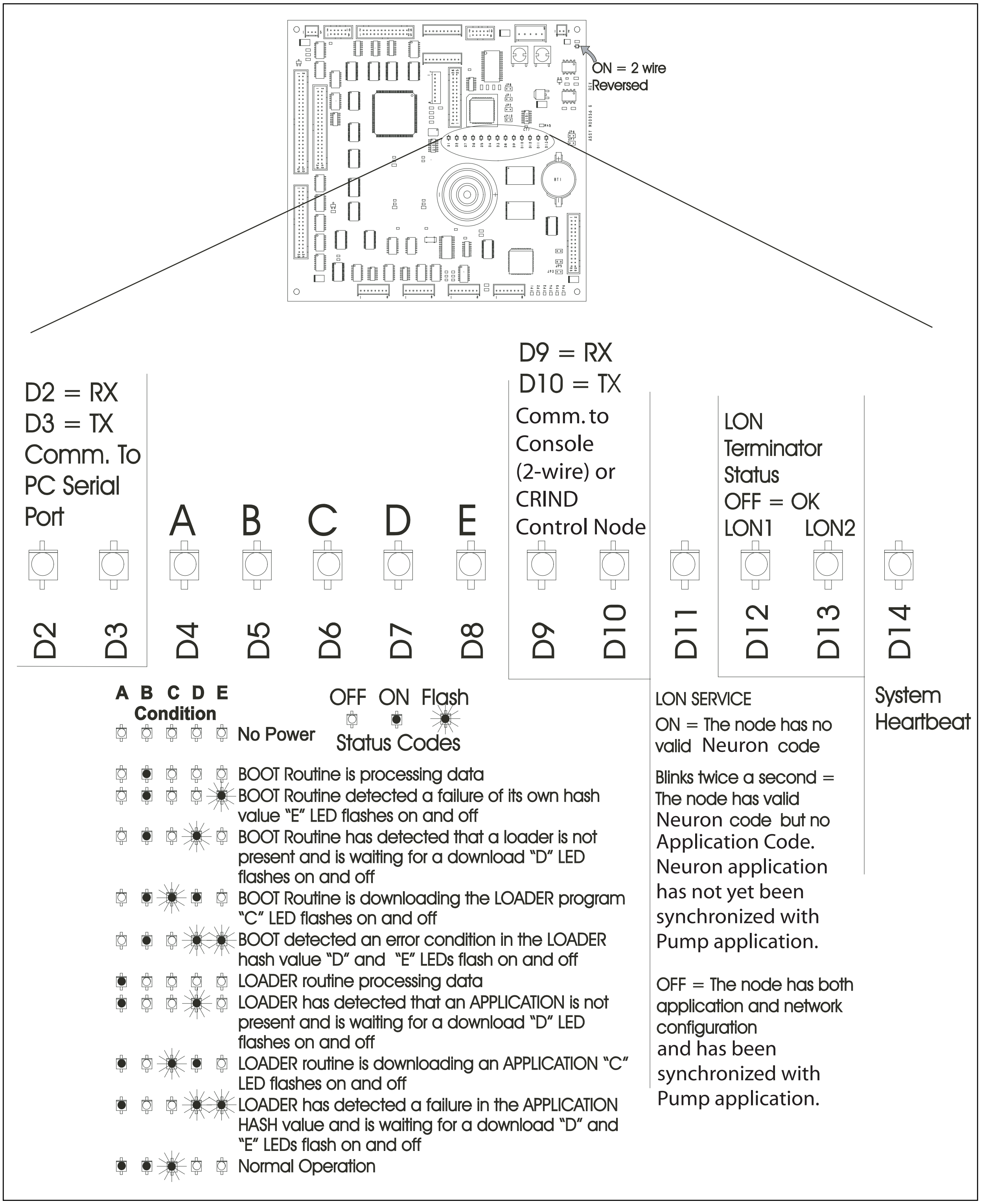


Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P1112-P1115 (Quantity 4; 8-pin)	• M00453A001 (Encore, quad) • M01026A001 (Eclipse) • T18350-G6 (Encore, requires V1.6.00 or later software)	Pulser(s)	
P1101 (10-pin) - STP Control	M00635A001	STP Board M00047	P1301
P1102 (26-pin) - Valve Control	M00549A001	M00059A00X Steering Valve Driver PCA	P1201
P1103 - LON Header 1	M02235 (LON Header 1: 10-pin)	LON system, Side 1	
P1104 - LON Header 2	M02235 (LON Header 2: 10-pin)	LON system, Side 2	
P1105 (5-pin) - 18 VDC from Power Supply	M00635A001	18V Power Supply PCA (M00050A001)	P1402
P1106 (34-pin) Ethernet 1	(not used)		
P1107 (Expansion Memory)	(not used)		
P1108 (34-pin) IFSF or Auxiliary	(not used)		
P1109 (Two-wire)	M00491A001	Pump Two-wire: • To Terminal Block in Power Supply (Standalone or generic CRIND) ~ OR ~ • M00089A001 CCN (MOC CRIND) ~ OR ~ • MOC special CCN M00089G001 (refer to M00089A001 CCN)	
P1110 (26-pin)	M00549A002 (Encore) M01036A001 (Eclipse)	VaporVac Isolation PCB (M00080)	P1405
P1111 (10-pin) - PC serial (signals to Valve Driver PCA)	M00719A001	Serial Programming Port	
P1120 Direct Connect	(not used)		
P1122 Debug	(not used)		
P1123 (3-pin) ECAL	M00150A001 (Encore) M00150A002 (Eclipse)	ECAL Security Switch	(3)
P1124 (16-pin) Ethernet 2	(not used)		

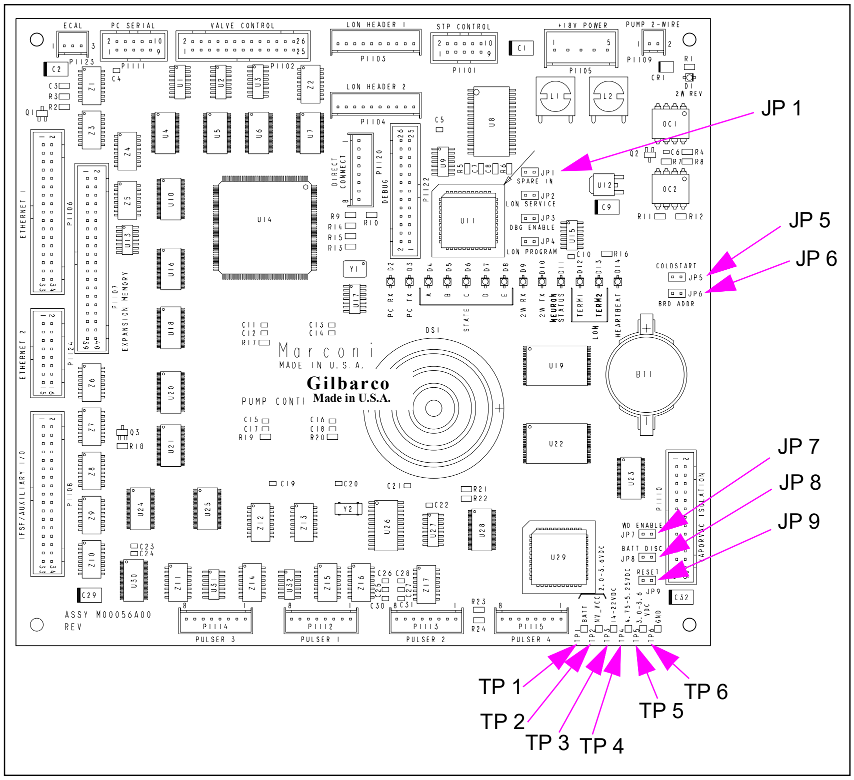
Status LEDs

Figure 6-6: PCN LCD Nomenclature



Test Points and Jumpers

Figure 6-7: PCN Jumper



Test Points

Test Point	Range	
TP 1	Battery	2.0 to 3.6 VDC
TP 2	NV_VCC	
TP 3	14 to 22.8 VDC	
TP 4	4.75 to 5.25 VDC	
TP 5	3.0 to 3.6 VDC	
TP 6	GND	

Jumpers

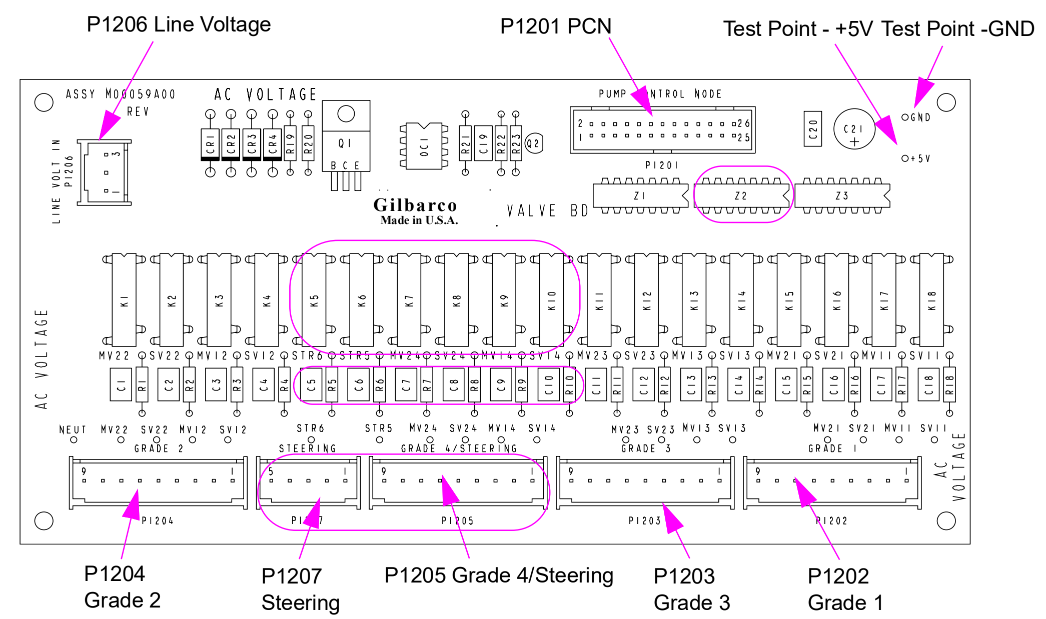
Jumper	Description
JP 1	NA
JP 5	Coldstart [if available (refer to "Service Tips" on page 6-8)]
JP 6	Board Address (IN) suppresses minor nuisance ECs, but still recorded in Event Logs.
JP 7	WatchDog Enable (always IN)
JP 8	Disconnect Battery (if IN: NVRAM is NOT battery backed - preserves power)
JP 9	Resets Processor - short with power on (if available)

Valve Driver PCA (M00059A00X)

M00059A001 Board is standard for Encore 500 and Eclipse units with digital valves. M00059A002 is a fully loaded version for 4-Grade and steering valve equipped units (multi-hose blenders). The board drives all valves and senses current through the valves for automatic configuration and EC generation. It receives its signal from the pump node and filtered AC power from Terminal Block PCA (M00044A001). It is not used with Proportional Flow Control Valve (PFCV).

Note: NJ5 and NJ6 contain two Valve Drive PCAs (M00059A002) .

Figure 6-8: Steering Valve Driver Board (M0059A00X)



Note: M00059A002 version shown. Circled components are omitted on -A001 Board.

Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P1201 (26-pin) PCN	M00549A001	Pump Node (M01922A001 or M00056A001)	P1102 (26-pin)
P1202 (9-pin) Grade 1	Dedicated Cables: • M00454A002 (Encore) • M00724A001 (Eclipse)	Slowdown Valve and Main Valve 11 + 21	
P1203 (9-pin) Grade 3		Slowdown Valve and Main Valve 13 + 23	
P1204 (9-pin) Grade 2		Slowdown Valve and Main Valve 12 + 22	
P1205 (9-pin) Grade 4/Steering		Slowdown Valve and Main Valve 14 + 24	
P1206 (3-pin)	M00625A001 (Inline connection J1206 to P1206 on cable M01006A001 on Eclipse only)	Terminal Block PCA (M00044A001)	P1615 (3)
P1207 (5-pin) Steering 6-Hose Fixed Blender	M01339A001	Steering Valve coils (M00223B002)	J701

Test Points

For location of test points (all relays K1-K18 are installed), see Figure 6-9. The A001 board does not provide G4 and steering valve control. As a result, not all test points shown in Figure 6-9 are applicable.

Figure 6-9: Test Points on A002 Version

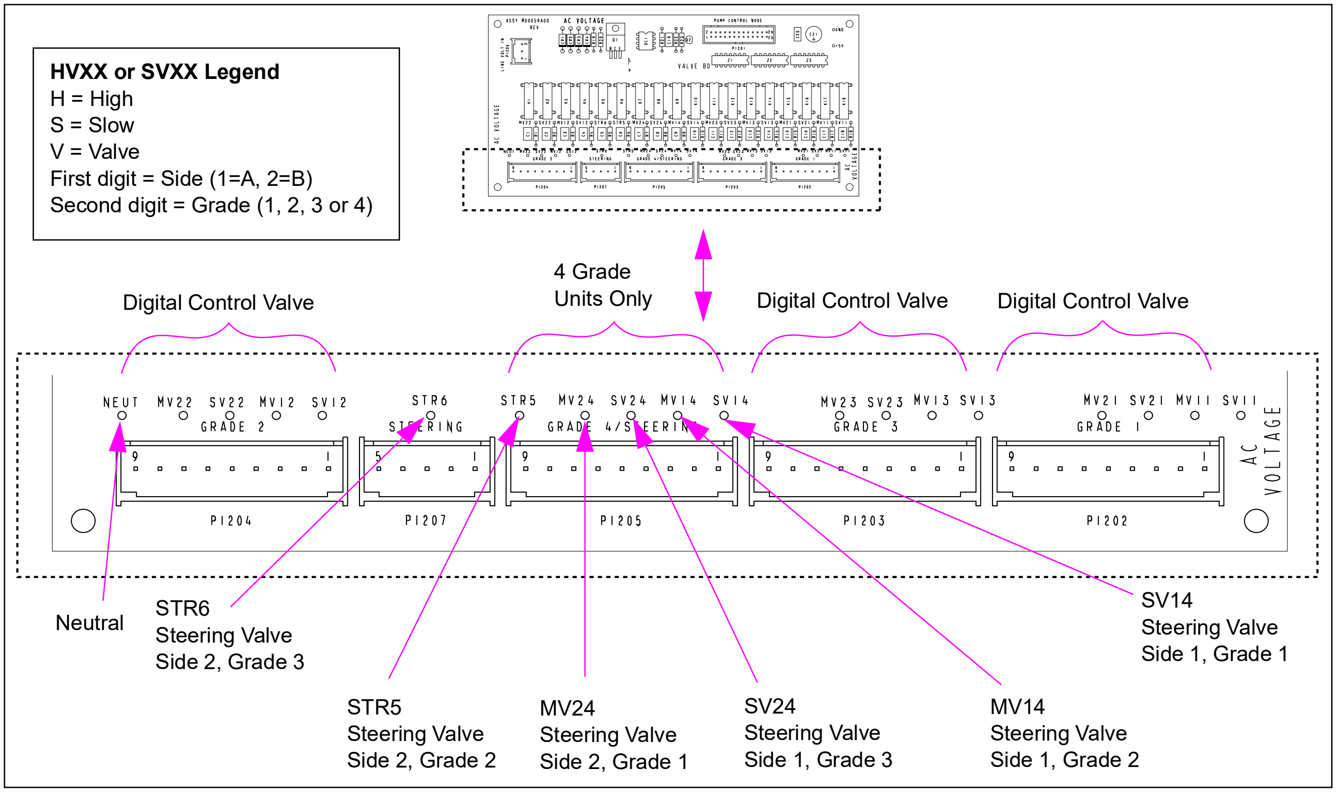


Figure 6-10: NJ5/NJ6 Block Diagram

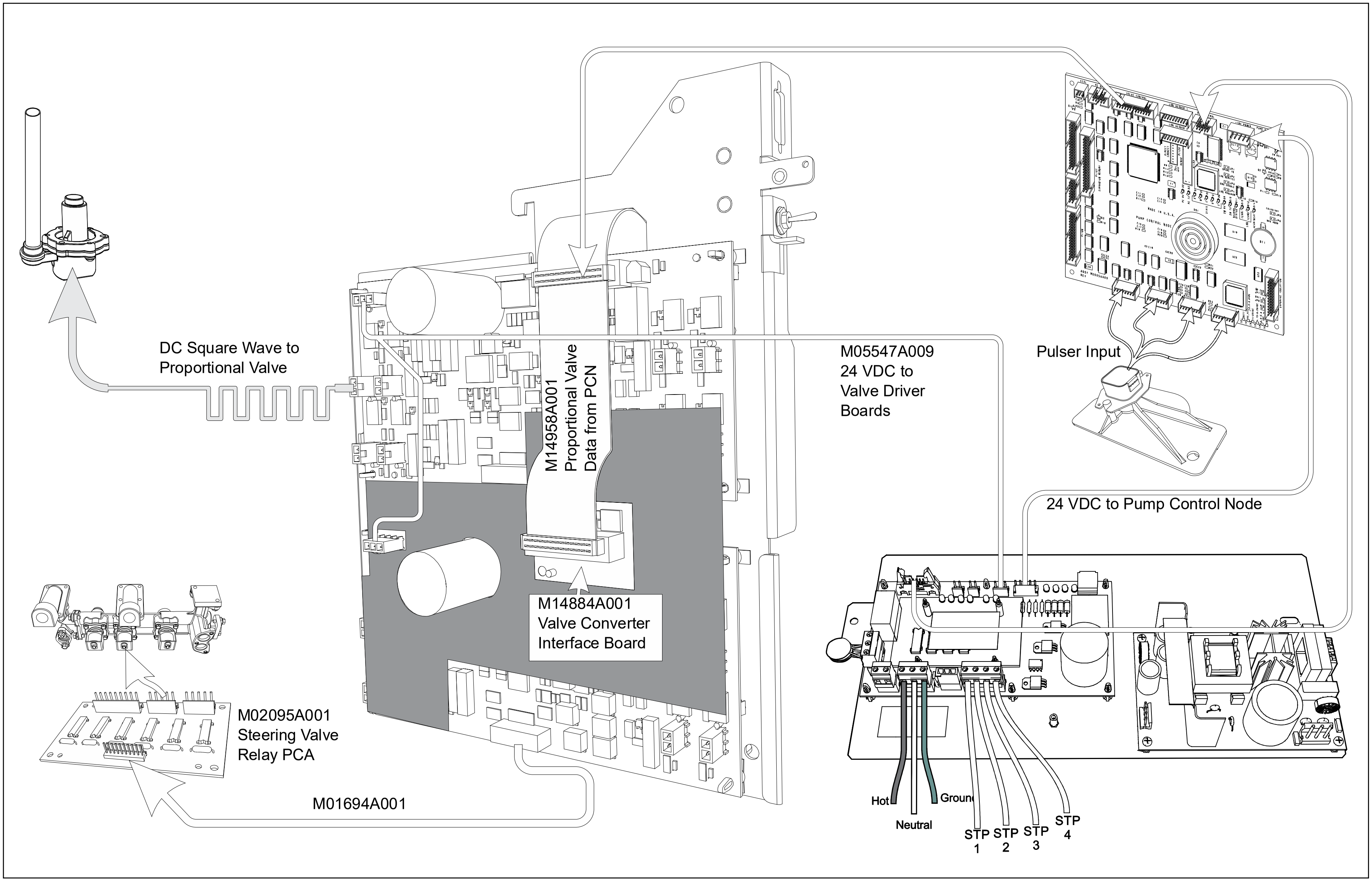
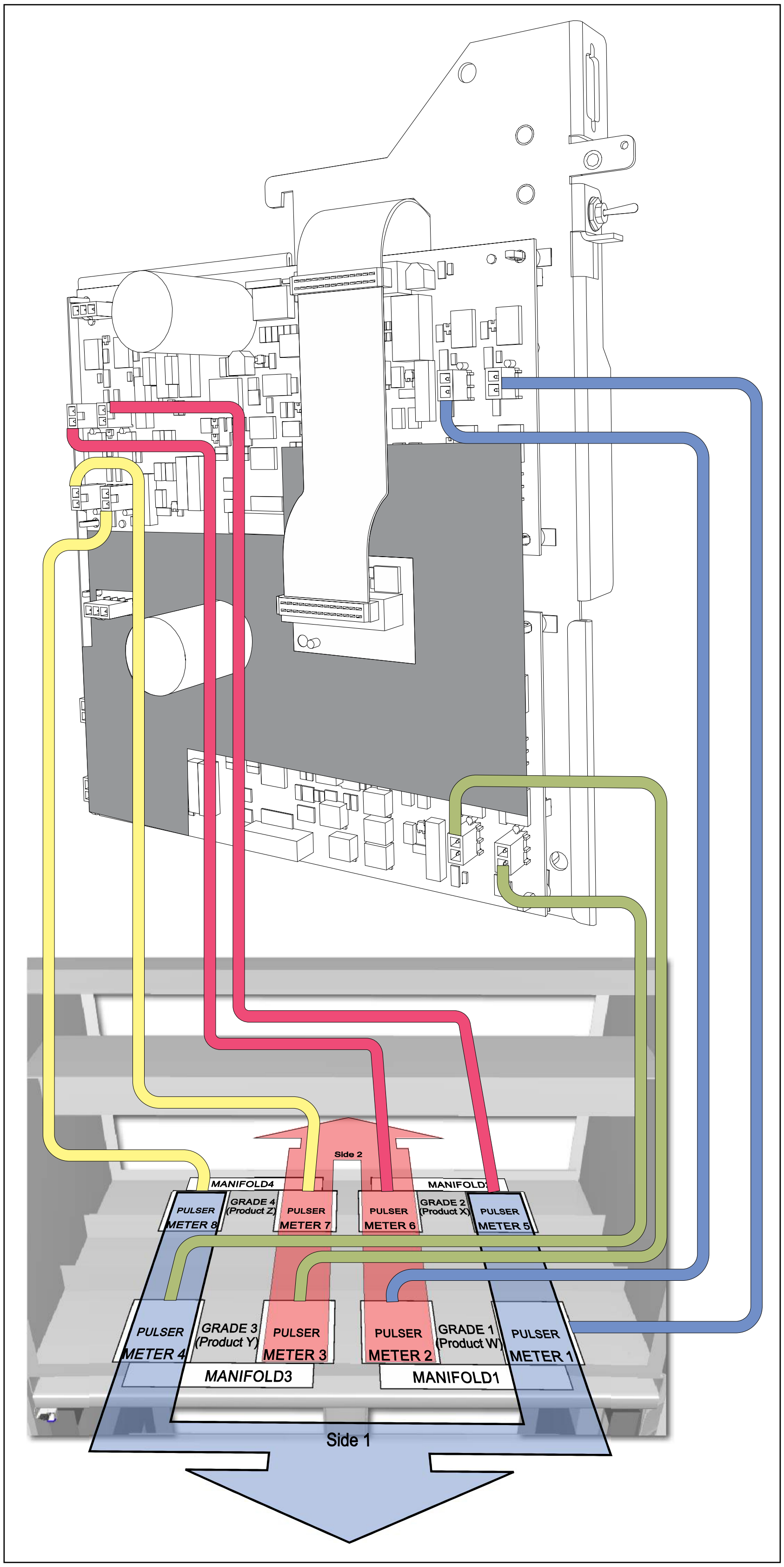


Figure 6-11: NJ5/NJ6 Valve Mapping Diagram



Door Node-1 (M00062A001)

The first generation Door Node (M00062A001) is for use with Encore 500 and Eclipse units with steering valves only. This node provides an interface for inputs and outputs from the customer, and acts as the interface to many of the devices that are found on the door. The door node also provides sales totals to the customer. There are two generations of door nodes which are interchangeable. The first generation node shown in Figure 6-12 on page 6-21 has more connectors and some additional components not seen on the second generation node. There is no loss in features in the second generation node.

The door node is part of the LON system and many devices normally found on the door will connect to the node and pass on the information through the LON. Devices directly or indirectly connected to the door node are:

- Push-to-start, call and pump stop buttons
- Customer (CRIND) keypad
- Manager keypad (side 1 only)
- Indirectly: Handle switches and grade select switches through PPU board
- Soft Keys: Monochrome and color e-CRIND
- Beeper (speaker)
- Door sensor (security)

Retaining Data While Replacing Pump Boards

As a means of retaining calibration and configuration data, this data is stored redundantly in nonvolatile memory. Redundant storage of this data allows for automatic, reliable, and quick retrieval in the event of a PCN or Door Node failure or replacement thereby minimizing service time.

LON based PCN (M12702AXXX, M01922AXXX)

This applies to all PCNs prior to PCN4+. The primary storage location for this data is on the PCN. However, it is also stored on both Door Nodes (PDNA, PDNB). Whenever calibration or configuration data is changed it is stored in all three locations when the PCN is booting up.

In the event either the PCN or one of the Door Nodes is replaced, while the PCN is booting up, it checks to see which two nonvolatile data locations are the same and then updates the third. If no two are the same the nonvolatile data is taken from the PCN. In order for this process to work, it is important that only one of the three nodes (PCN, PDNA, or PDNB) be replaced at a time.

Display Bus-based PCN (M18666Axxx connected to M18830 display boards)

Note: This ONLY applies to PCN4+.

With the Display Bus protocol, calibration and configuration data is also stored redundantly. However, it is only stored in two locations. The PCN is the primary storage location and PDNA is the secondary location.

In the event the PCN is replaced, it is important to follow the procedure above in order to minimize the risk of losing calibration and configuration data. If the above procedure is not followed, the PCN nonvolatile data will not be overwritten by the PDNA nonvolatile data.

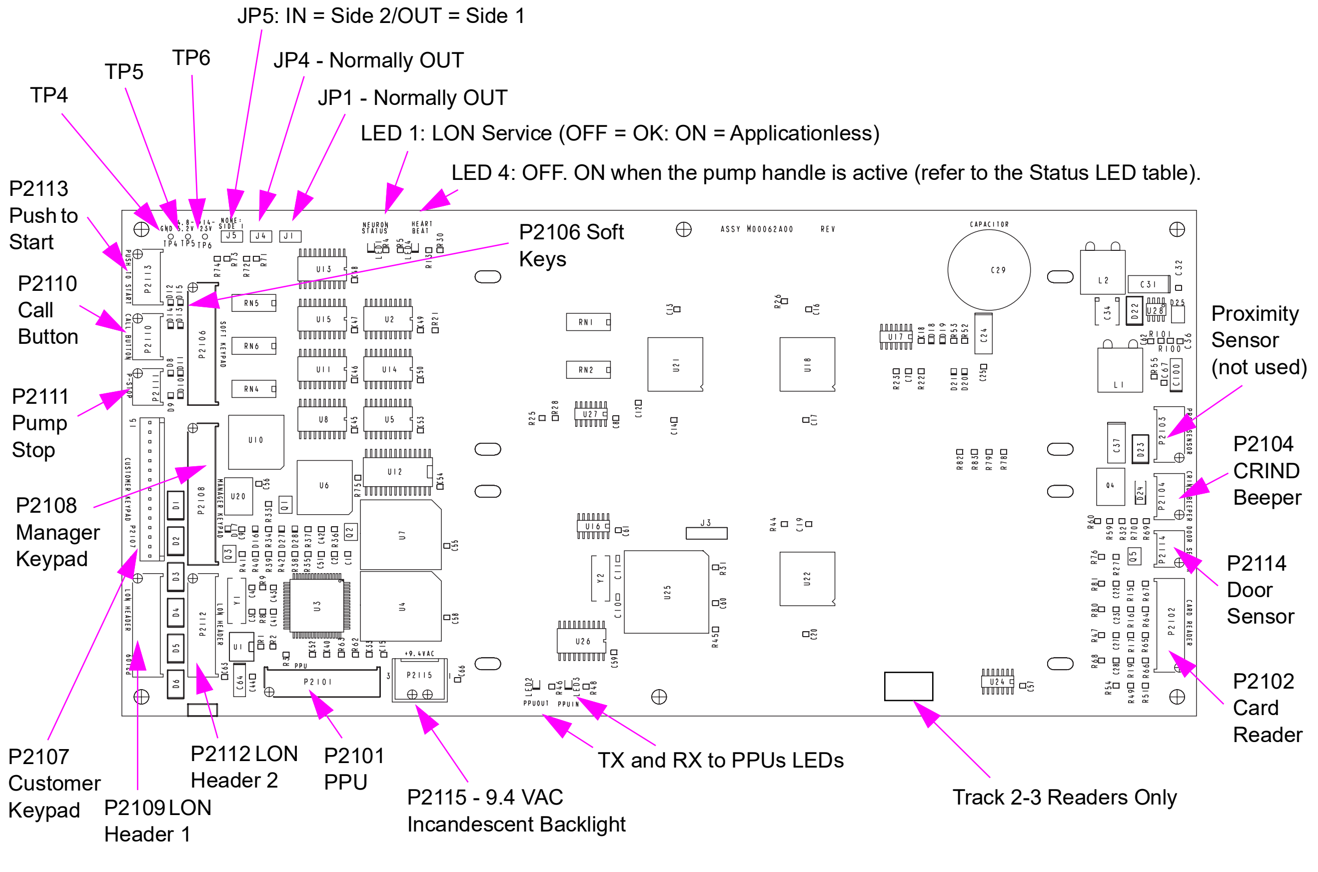
IMPORTANT INFORMATION
The below process must be followed in the exact sequence shown in the steps below. If this process is not followed, there is a risk of losing calibration.

Proceed with the following steps for replacing a M18666Axxx Pump Control Board connected to M18830 display board:

1. If applicable, record the Pump Totals which can be re-entered later in the process.
2. Retrieve the “WhoAreYou” file for later reference, if needed.
3. Isolate the two wires from the POS.
4. Disconnect the power from the dispenser.
5. Unseal the security/W&M switch.
6. Disconnect all the cables from the Pump Control Board and remove it from the mounting bracket.
7. Remove the battery from the new Pump Control Board (PCN4+) and set it aside (it will be re-installed later).
8. Wait at least 30 seconds, then replace the battery.
9. Attach the new Pump Control Board to the mounting bracket.
10. Re-connect all the cables disconnected previously.
11. Turn ON/OPEN the security/W&M switch by sliding it forward.
12. Restore power to the dispenser.
13. Wait for the Pump Control Board to boot up fully (zeros showing on the main display along with error 5056 code flashing on the Grade 1 PPU).
14. Remove then restore power on the dispenser.
15. Wait for the Pump Control Board to boot up fully.
16. Turn OFF/CLOSE the security/W&M switch by sliding it backward.
17. Enter Pump Totals, if applicable.
18. Refer to “WhoAreYou” file to verify configuration settings, if necessary.
19. Re-connect the two wires.
20. Test the dispenser for proper operation and communication.

21 Seal the security/W&M switch or if necessary, notify your local W&M office that the seal has been broken and needs to be repaired.

Figure 6-12: Door Node-1 (M00062A001)



Jumper Notes:
JP1 and **JP4** - for future options.

Status LEDs

LED	Status Indication
1	LON (Neuron) Status - ON or blinking indicates LON or Node problem
2	TX or PPU-OUT indicates communication to PPU boards
3	RX or PPU-IN indicates communication from PPU boards
4	OFF. ON when the pump handle is active

Test Points

Test Point	Range
TP4	GND
TP5	4.75 V to 5.25 VDC
TP6	+14 V to +28 VDC

Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P2101 (11-pin)	• M00509A001 or M00509A002 for Encore • M00509A003 for Eclipse	M02652A00X PPU Board (one per grade)	P2201
P2102 (9-pin)	• M00501A001 (Encore) • M00501A002 (Eclipse)	Card Reader	
P2103 (5-pin)	(not used)		
P2104 (4-pin)	• M00636A001 (Encore) • M00495A001/ M00636A002 (Eclipse)	Beeper (refer to Encore Series Cable Block Diagram or the Eclipse Series Cable Block Diagram).	
P2106 (12-pin)	M01200A001	Soft Keys Keypad (refer to Encore Cable Block Diagram M00284)	
P2107 (15-pin)	M01202A001 (Encore M01202A002 (Eclipse)	Customer Keypad Options (CRIND or PPP): • M01109 Misc Options Keypad • M00141 CRIND Keypad	Refer to *MDE-3804 Wiring and Configuration Manual*
P2108 (14-pin)	M00515A001 (Encore) M00515A003 (Eclipse)	Manager Keypad [M00147 (Side 1 only)]	
P2109 (10-pin)	M00489/M02235	LON Bus 1	
P2110 (4-pin) [Call Button]	M01201A001 (Encore) M01215A001 (Encore)	Miscellaneous Options (M01109B001)	12-pin
P2111 (3-pin) [Push-to-Stop]			
P2113 (5-pin) [Push-to-Start]			
P2112 (10-pin)	M00489/M02235	LON Bus 2	
P2114 (4-pin)	N/A	Door sense	
P2115 (3-pin)	M00614A004 (Encore) M00614A002 (Eclipse)	Power Supply 9.4 VAC (PS2 to PS3 on bulkhead connector)	PS2

Connector PIN Functions and Voltages

P2101

Pin #	Description	Input/Output Voltage
1	+9.4 VAC	9.4 VAC +/-10%
2	+9.4 VACRTN	
3	GND	0 VDC
4	GND	0 VDC
5	DLT +5 VDC	+4.7 V +/-10% (with the power on)
6	PPUCLK	0 to 5 V
7	PPUFBDATA	0 to 5 V
8	PPUSND_REC	0 to 5 V
9	PPUBP	0 to 5 V
10	PPUSDATA	0 to 5 V
11	PPUSYNC	0 to 5 V

P2102

Pin #	Description	Input/Output Voltage
1	VCC	+5 V +/-5%
2	CRRDT1	0 to 5 V
3	CRRCL1	0 to 5 V
4	CRRDT2	0 to 5 V
5	CRRCL2	0 to 5 V
6	CRCLD	0 to 5 V
7	GND	0 V
8	CRSTAT	Open to 0 V
9	GND	0 V

P2103

Pin #	Description	Input/Output Voltage
1	VCC	+5 V +/-5%
2	GND	0 V
3	PRSENIN	0 to 5 V
4	PRSENSTAT	Open to 0 V
5	+18 VUNREG	+17.5 V to +28 V

P2104

Pin #	Description	Input/Output Voltage
1	VCC	+5 V +/-5%
2	SPKRTN	Open Collector to 0 V
3	SPKR_STAT	Open to 0 V
4	GND	0 V

P2109

Pin #	Description	Input/Output Voltage
1	+18 VUNREG	+17.5 V to +28 V
2	GND	0 V
3	+18 VUNREG	+17.5 V to +28 V
4	GND	0 V
5	485B	-7 V to +12 V Max.
6	485A	-7 V to +12 V Max.
7	GND	0 V
8	+18 VUNREG	+17.5 V to +28 V
9	EXTRESET	0 V to +12 V
10	TERM RES FEEDTHRU	

P2112

Pin #	Description	Input/Output Voltage
1	+18 VUNREG	+17.5 V to +28 V
2	GND	0 V
3	+18 VUNREG	+17.5 V to +28 V
4	GND	0 V
5	485B	-7 V to +12 V Max.

Pin #	Description	Input/Output Voltage
6	485A	-7 V to +12 V Max.
7	GND	0 V
8	+18 VUNREG	+17.5 V to +28 V
9	EXTRESET	0 V to +12 V
10	TERM RES FEEDTHRU	

P2115

Pin #	Description	Input/Output Voltage
1	+9.4 VAC	9.4 VAC +/-10%
2	+9.4 VACRTN	
3	No connect	

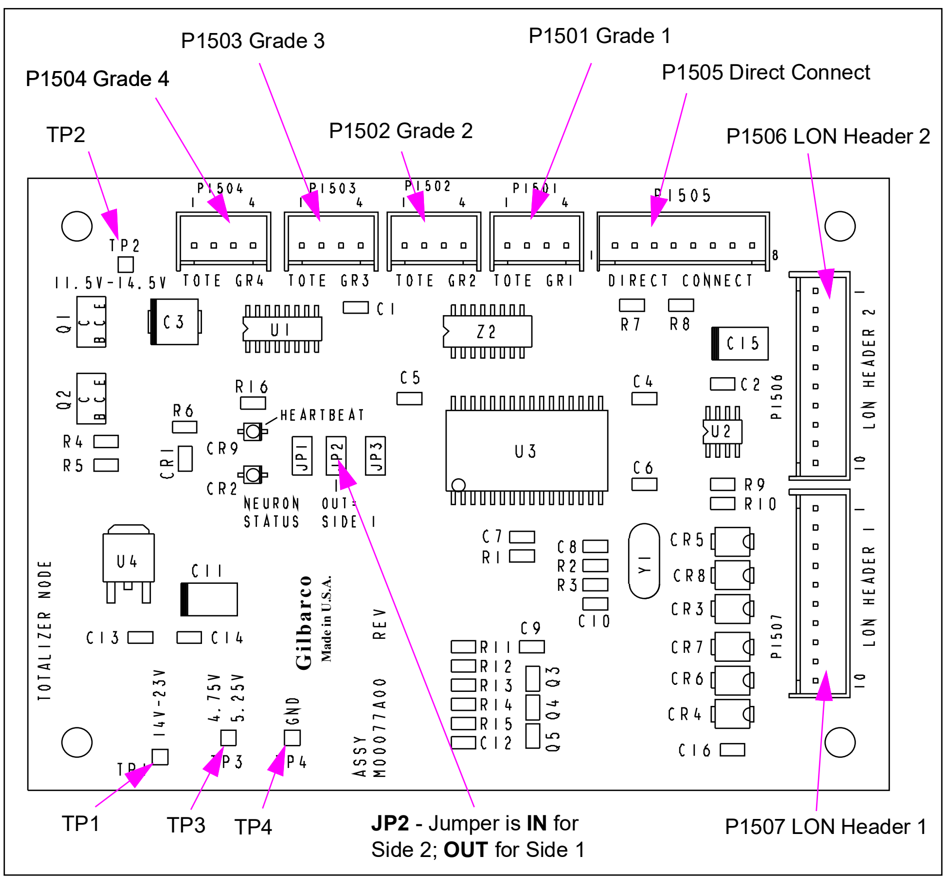
Totalizer Node (M00077A001)

Totalizer node is a dedicated peripheral control node on the LON Bus for Encore 500 and Eclipse units. This node is used to control a maximum of four electromechanical totalizers that may be found on one or both sides of the unit. Totals are non-changeable.

Totalizer operation can be programmed in selected installed options: CC91 at FC11. Totalizer can be configured as:

- Per inlet per unit (only option available for the Eclipse model)
- Per product per side

Figure 6-13: Totalizer Node PCA



Board Connections

Connector#	Through Cable	To Board
P1501 (4-pin)		Electronic Totalizer Grade 1
P1502 (4-pin)		Electronic Totalizer Grade 2
P1503 (4-pin)		Electronic Totalizer Grade 3
P1504 (4-pin)		Electronic Totalizer Grade 4
P1505 (8-pin)		N/A (not for field use)
P1506 (10-pin)	M00489	LON Header 2
P1507 (10-pin)		LON Header 1

Test Points

Test Point	Range
TP 1	17.5 to 28 V to GND
TP 2	11.5 V to 14.5 V to GND
TP 3	4.75 V to 5.25 V to GND
TP 4	GND

Status LEDs

LED	Status Indication
CR 2	LON (Neuron) Status - ON or blinking indicates problem
CR 9*	HEARTBEAT - must blink at 1 per second

**Currently blinks for every five seconds. In future, all heartbeats must be consistent with each other.*

Connector PIN Functions and Voltages

P1501

Pin #	Description	Input/Output Voltage
1	TOTE_RETURN1	0 V to open collector
2	TOTE_DETECT1	0 to +5 V
3	+5 V	+5 V +/-5%
4	TOTE_FEED1	+13 V +/-10%

P1502

Pin #	Description	Input/Output Voltage
1	TOTE_RETURN2	0 V to open collector
2	TOTE_DETECT2	0 to +5 V
3	+5 V	+5 V +/-5%
4	TOTE_FEED2	+13 V +/-10%

P1503

Pin #	Description	Input/Output Voltage
1	TOTE_RETURN3	0 V to open collector
2	TOTE_DETECT3	0 to +5 V
3	+5 V	+5 V +/-5%
4	TOTE_FEED3	+13 V +/-10%

P1504

Pin #	Description	Input/Output Voltage
1	TOTE_RETURN4	0 V to open collector
2	TOTE_DETECT4	0 to +5 V
3	+5 V	+5 V +/-5%
4	TOTE_FEED4	+13 V +/-10%

P1505

Pin #	Description	Input/Output Voltage
1	CP0	Selectable Hysteresis
2	CP1	Selectable Hysteresis
3	CP2	0 to +5 V
4	CP3	0 to +5 V
5	RX_DISABLE	0 to +5 V
6	GND	0 V
7	No connect	
8	BOARD_RESET	0 to 0.7 V

P1506

Pin #	Description	Input/Output Voltage
1	+18 VUNREG	+17.5 V to +28 V
2	GND	0 V
3	+18 VUNREG	+17.5 V to +28 V
4	GND	0 V
5	485B	-7 V to +12 V Max.
6	485A	-7 V to +12 V Max.
7	GND	0 V
8	+18 VUNREG	+17.5 V to +28 V
9	EXTRESET	0 V to +12 V
10	TERM RES FEEDTHRU	

P1507

Pin #	Description	Input/Output Voltage
1	+18 VUNREG	+17.5 V to +28 V
2	GND	0 V
3	+18 VUNREG	+17.5 V to +28 V
4	GND	0 V
5	485B	-7 V to +12 V Max.
6	485A	-7 V to +12 V Max.
7	GND	0 V
8	+18 VUNREG	+17.5 V to +28 V
9	EXTRESET	0 V to +12 V
10	TERM RES FEEDTHRU	

CRIND Control Node-1 (M00089A001)

First Generation CRIND Control Node [CCN (M00089A001)] found in Encore 500 and Eclipse units communicates through the LON. MOC applications access this node by communicating through two-wire. It controls the card reader, monochrome display, printer, barcode scanner, cash acceptor, TRIND, beeper, and associated keypads. The lithium battery on this node maintains CRIND application software and CCN programming. M00089A001 is replaced by CCN-2 [M01753A001 (refer “CRIND Control Node-2 (M01753A001)” on page 6-46)]. CCN-1 is interchangeable with CCN-2. However, certain features, such as USB support will be lost. Additionally, the CCN-2 does not use a monochrome interface board.

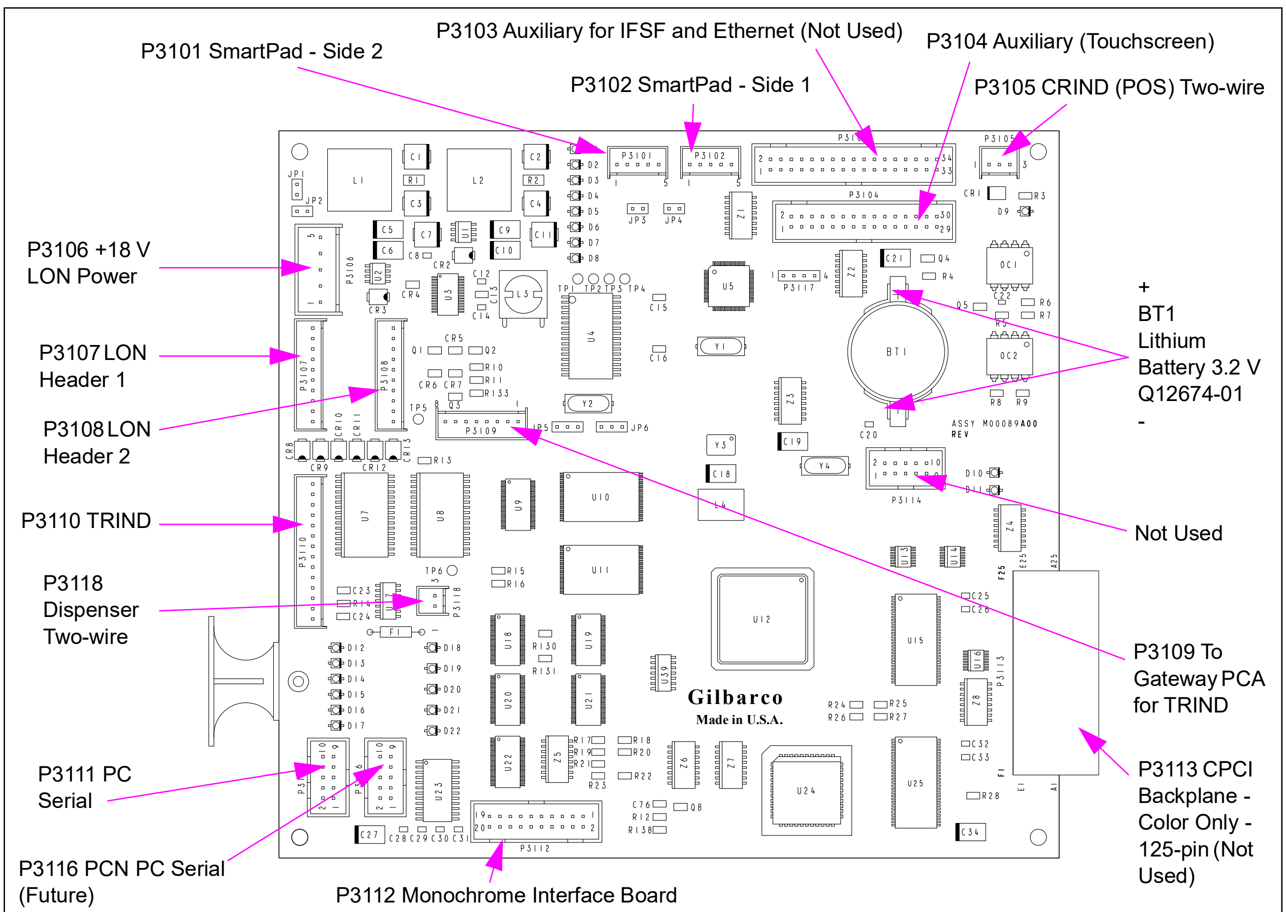
WARNING

There is a danger of explosion if battery is replaced incorrectly. Replace with only the same or equivalent recommended type. Dispose of the used batteries according to the battery manufacturer’s instructions.

Service Tips

- The coldstart jumper does work for this board. A true coldstart can be done using the diagnostic card and accessing the Clean Persistent Memory function.
- This board must be used with a monochrome interface board.
- CRIND MIP software must be loaded through the pump node.
- The CRIND node uses three types of software: CRIND MIP, CRIND Node, and CRIND application from POS software.

Figure 6-14: CRIND Control Node-1 (M00089A001)

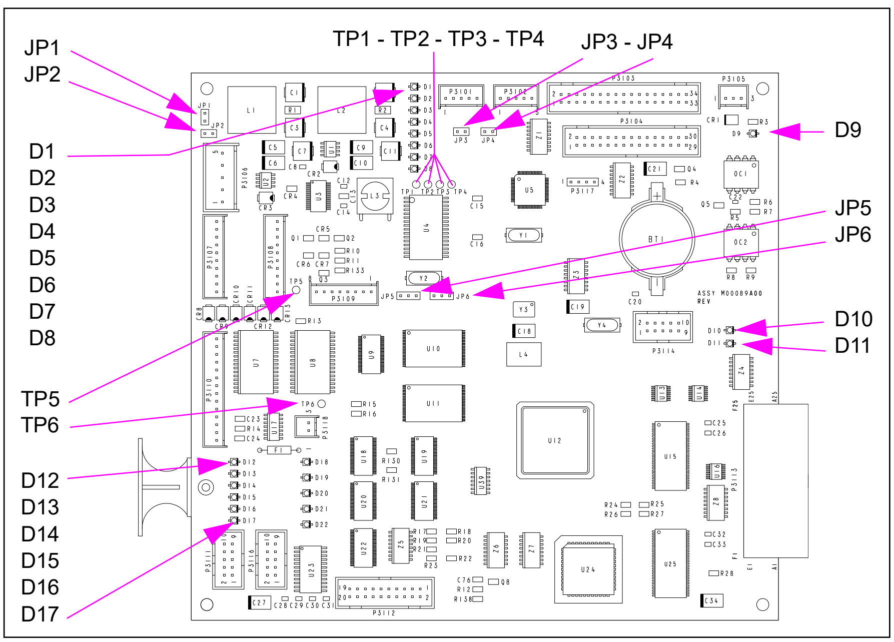


Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P3101 (5-pin)	M00497A002	SmartPad™ - Side 2	12-pin
P3102 (5-pin)	M00497A002	SmartPad - Side 1	12-pin
P3103 (34-pin)	M00716A001	CRIND Auxiliary Board M00095A001	P3403 (34-pin)
P3104 (30-pin)	M00716A002		P3404 (30-pin)
P3105 (3-pin)	M00635A001	Terminal Block 2 in Power Supply (CRIND Retrofits Only)	
P3107 (10-pin)	M00489A001	LON Header 1	10-pin
P3108 (10-pin)		LON Header 2	10-pin
P3109 (14-pin)		TI Gateway PCA	J3110
P3110 (14-pin)	M00515A001	TRIND Gateway PCA	P250
P3111 (10-pin)	M00719A001 (Encore) M00719A002 (Eclipse)	Serial Programming Port (when available)	
P3112 (20-pin)	M00545A001	CRIND Auxiliary I/O Board (M00095A001)	P3301
P3113 (125-pin)	M00101A001 CPCI Backplane Assembly (In Design)		
P3114 (10-pin)	not used		N/A
P3116	Serial Port on M00077A001 Totalizer Node (Future)		
P3118 (2-pin)	M00491A001	Pump Node (M00056A001) Pump Node (M01922A001)	P1109

Status LEDs, Jumpers, and Test Points

Figure 6-15: Status LEDs, CCN Jumpers, and Test Points



Jumpers

Jumper	Purpose/Status
JP1	Reset CCN (only)
JP2	Coldstart (not yet available)
JP3 JP4	N/A (no field use)
JP5	Selects type of communication between PCN (M00077A001 Totalizer Node) and CCN. • If JJ is installed on Pins 1-2, PCN and CCN communicate through two-wire interface. • If JJ is installed on Pins 2-3, PCN and CCN communicate through PC serial interface (not available at this time).
JP6	Not populated; also part of PCN to CCN serial interface. • If JJ is installed on Pins 1-2, pump is set for two-wire or PC serial without a Clear to send handshaking capability. • If JJ is installed on Pins 2-3, PC serial communications is set to use the Clear to send handshaking signal.

Test Points

Test Point	Range	Where Used
TP1	3.15 to 3.45 V	Memory Voltage
TP2	4.75 to 5.25 V	Logic Voltage
TP3	3.0 to 3.6 V (Battery)	Battery
TP4	GND	Common for DC Voltage
TP5	17.5 to 28 V	LON Voltage
TP6	2.0 to 3.6 V	RAM Voltage

LEDs

LED	COLOR	Status Indication
RED		
D1		Heartbeat - must blink at 1 per second
YELLOW		
D2		TX SmartPad Side 1
D3		RX SmartPad Side 1
D4		TX SmartPad Side 2
D5		RX SmartPad Side 2
RED		
D6		LON 1 Terminator
D7		LON 2 Terminator
D8		LON (Neuron) Service - ON or blinking indicates problem
D9		POS Two-wire - reversed if ON
YELLOW		
D10 D11		POS Two-wire
D12		TX TRIND
D13		RX TRIND
D14		TX PC Serial
D15		RX PC Serial
D16		TX Pump Two-wire
D17		RX Pump Two-wire

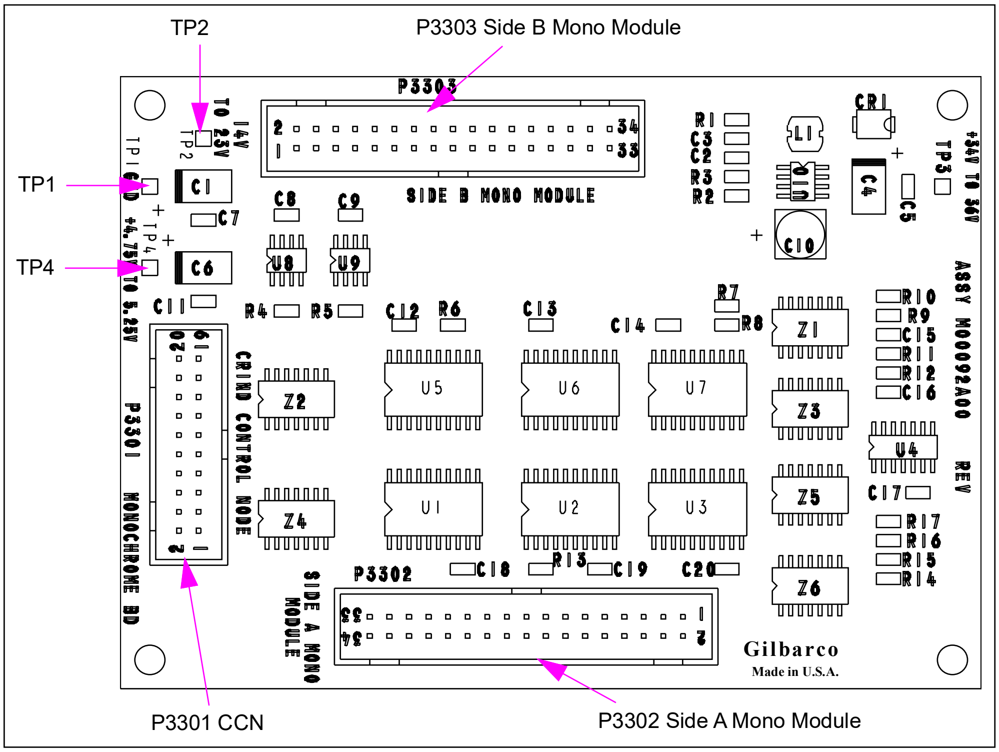
Monochrome Interface PCA (M00092A001)

Monochrome Interface Board (M00092A001) is used in Encore 500 and Eclipse units equipped with first generation CCN (M00089A001). This interface displays signals from M00089A001 to side 1 and side 2 monochrome displays. This board is not required with CCN-2.

Service Tip

A fault in this board may affect one or both monochrome displays.

Figure 6-16: Monochrome Interface Board (M00092A001)



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P3301 (20-pin)	M00545A001	M00089A001 CCN	P3112 (20-pin)
P3302 (34-pin)	M00553A001 (Encore) M00553A002 (Eclipse)	Monochrome Module Side 1	J1 (34-pin)
P3303 (34-pin)	M00553A001 (Encore) M00553A002 (Eclipse)	Monochrome Module Side 2	J1 (34-pin)

Test Points

Test Point	Range
TP1	GND
TP2	14 V to 23 V
TP3	34 V to 36 V
TP4	4.75 V to 5.25 V

LON to Serial Node [M00122A001 (LON Gateway)]

The LON to serial node converts LON data to RS-232 data on Encore 500 and Eclipse series units. RS-232 serial data is a protocol that allows the LON to interface with serial devices such as scanners and cash acceptors. This board is equipped with door indicator connections for security. This board contains a replaceable battery that allows for monitoring of the door indicator connections, even when power is disrupted.

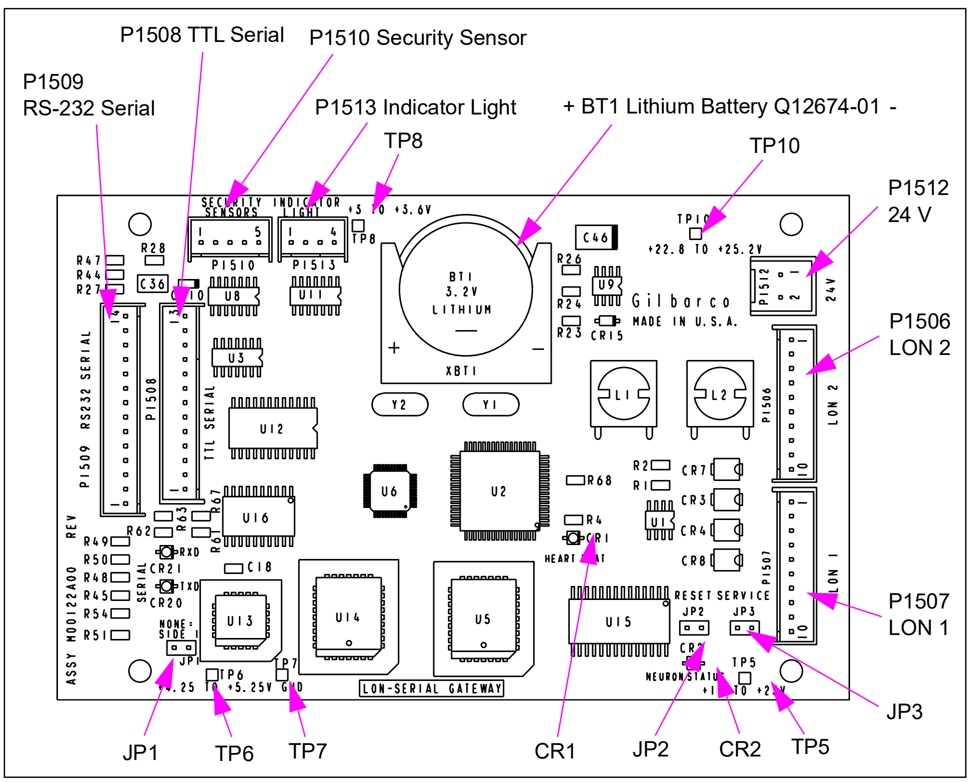
WARNING

There is a danger of explosion if battery is replaced incorrectly. Replace with only the same or equivalent recommended type. Dispose of the used batteries according to the battery manufacturer's instructions.

Service Tips

- If the neuron status LED is on or blinking, then there is a problem with the LON.
- Heartbeat LED must be blinking at a steady rate.
- Transmit and receive LEDs indicate communication on the RS 232 port.

Figure 6-17: LON to Serial Node



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P1506 (10-pin)	M00489	LON Header 2	
P1507 (10-pin)		LON Header 1	
P1508 (13-pin)	N/A	Cash Acceptor Interface	13-pin
P1509 (14-pin)	N/A	RS-232 Interface	14-pin
P1510 (5-pin)	N/A	Not available	
P1512 (2-pin)	N/A	Power Supply Board 24 VDC [M00053A002 (Barcode only)]	• P3503 A (Side 1) • P3503B (Side 2)
P1513 (4-pin)	N/A	Indicator (Barcode only)	

Test Points

Test Point	Range
TP5	17.5 V to 28 V
TP6	4.75 V to 5.25 V
TP7	GND
TP8	3.0 to 3.6 V (Battery)
TP9	11.4 to 12.6 V
TP10	22.8 to 25.2 V

Status LEDs

LED	Status Indication
CR 1	HEARTBEAT - must blink at 1 per second
CR 2	LON (Neuron) Status - ON or blinking indicates problem
CR 20	Serial TXD (ON when transmitting)
CR 21	Serial RXD (ON when receiving)

Jumpers

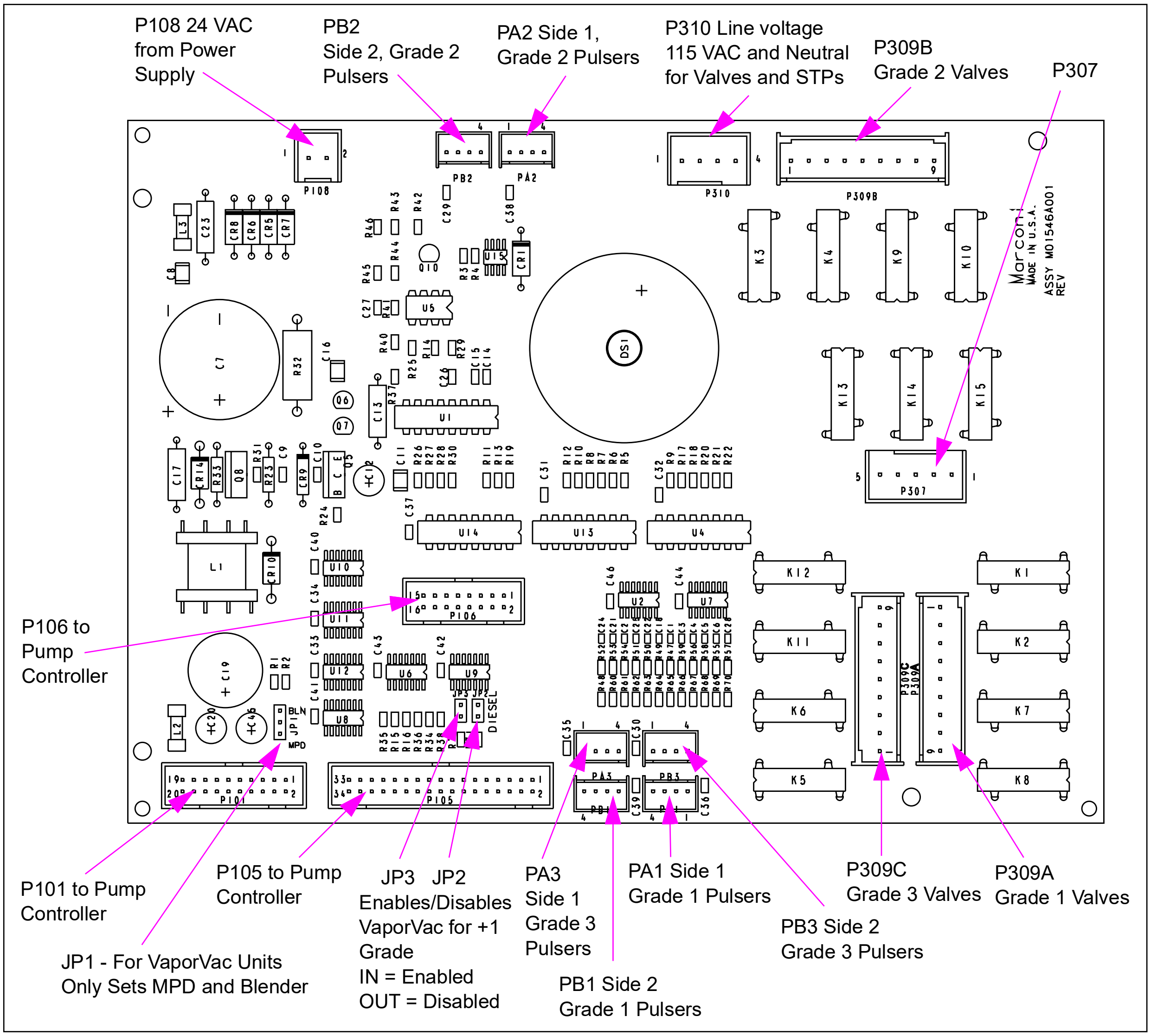
Jumper	Purpose/Status
JP1	Jumper for Side 1 (Side 1 = NONE)
JP2	Reset
JP3	Not for field use

Hydraulic Interface PCA (M01546A001)

This hydraulic interface board is used on Encore 300 series units with a digital valve. It interfaces with various devices such as pulsers, valves, VaporVac, and STP relays to the pump or dispenser electronics. This board provides a regulated 14.5 VDC for system operation, contains solid state relays that control solenoid valves, STP relays, and a pulser signal interface. This board also contains a buzzer and an interface to the VaporVac vapor recovery system.

The regulated 14.5 VDC has an under-voltage lockout circuit that prevents regulated power supply circuits from starting until the unregulated DC input voltage reaches about 23 VDC. This circuit also turns off regulated voltages when the unregulated DC input voltage drops to about 18 VDC.

Figure 6-18: Hydraulic Interface PCA (M01546A001)



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
PA1 (4-pin)	R20462-G4	Pulser Assembly T18350-G5 - A1	P1 (4-pin)
PA2 (4-pin)		Pulser Assembly T18350-G5 - A2	
PA3 (4-pin)		Pulser Assembly T18350-G5 - A3	
PB1 (4-pin)		Pulser Assembly T18350-G5 - B1	
PB2 (4-pin)		Pulser Assembly T18350-G5 - B2	
PB3 (4-pin)		Pulser Assembly T18350-G5 - B3	
P101 (20-pin)	R17995-G7	Totalizer Interface PCA (M01621A001)	P201 (20-pin)
P105 (34-pin)	R19667-G1		P205 (34-pin)
P108 (2-pin)	Two-wire/24 VAC	Main Power Supply (M01608A00)	Two-wire
P307 (5-pin)	M01601A001	From Field Wiring	STP1/2/3
P309A (9-pin)	M00454A001	Valves A1/B1	
P309B (9-pin)		Valves A2/B2 (N/A for Dual)	
P309C (9-pin)		Valves A3/B3 (N/A for Dual and Quad)	
P310 (4-pin)	115 V Valve Feed	Main Power Supply (M01608A00)	4-wire

Test Points - AC (Voltage Requirements)

Connector	Pin Number	Input	Range	Current
AC Voltage Requirements				
P108	1	24 VAC Feed	24 VAC +/- 5%	0.750 A
P108	2	24 VAC Return		
P101	16	5 V	5 V +/- 5%	0.25 A
P310	1	115 VAC	115 VAC +/- 10%	0.75 A
P310	2	AC Neutral	0 VAC	
P310	4	STP Feed	98 to 240 VAC	0.75 A
DC Voltage Requirements				
P101	6	14.5 VDC	14.5 +/- 5%	0.5 A
P101	12	DC GND		

Jumper Settings

Used only for units with VaporVac.

Jumper	Function
JP1 - MPD	Enables A2 and B2 VaporVac
JP1 - BLENDER	Disables A2 and B2 VaporVac
JP2	IN Enables/disables for VaporVac +1 Grade, Side 1
JP3	IN Enables/disables for VaporVac +1 Grade, Side 2

Note: For units with VaporVac only.

P105 Connector PIN Functions and Voltages

Pin #	Description	I/O	Max Lo	Min Hi
1	Slowdown Valve A1	I	1.00 VDC	3.50 VDC
2	Main Valve B3	I	1.00 VDC	3.50 VDC
3	Slowdown Valve A2	I	1.00 VDC	3.50 VDC
4	Main Valve B2	I	1.00 VDC	3.50 VDC
5	Slowdown Valve A3	I	1.00 VDC	3.50 VDC
6	Main Valve B1	I	1.00 VDC	3.50 VDC
7	Slowdown Valve B3	I	1.00 VDC	3.50 VDC
8	Main Valve A2	I	1.00 VDC	3.50 VDC
9	Slowdown Valve B2	I	1.00 VDC	3.50 VDC
10	Main Valve A3	I	1.00 VDC	3.50 VDC
11	Slowdown Valve B1	I	1.00 VDC	3.50 VDC
12	Pump Handle B2	I	1.00 VDC	3.50 VDC
13	Pump Handle B3	I	1.00 VDC	3.50 VDC
14	Pulser Data A3	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
15	Pump Handle B1	I	1.00 VDC	3.50 VDC
16	Pulser Data A3	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
17	Pulser Data A2	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
18	Pump Handle A3	I	1.00 VDC	3.50 VDC
19	Pulser Data A2	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
20	Pump Handle A2	I	1.00 VDC	3.50 VDC
21	Pulser _Data A1	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
22	Pump Handle A1	I	1.00 VDC	3.50 VDC
23	Pulser Data A1	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
24	Pulser Data B2	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
25	Pulser Data B2	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
26	Beeper A	I	1.00 VDC	3.50 VDC
27	Pulser Data B1	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
28	Pulser Data B3	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
29	Pulser Data B1	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
30	Pulser Data B3	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
31	STP 1	I	1.00 VDC	3.50 VDC
32	Main Valve A1	I	1.00 VDC	3.50 VDC
33	STP 2	I	1.00 VDC	3.50 VDC
34	STP3	I	1.00 VDC	3.50 VDC

P106 Connector PIN Functions and Voltages

Pin #	Description	I/O	Max Lo	Min Hi
1	Pulse B	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
2	Pulse B1	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
3	Connect	I	Feed through to pin 4	
4	Connect	O	Feed through to pin 3	
5	GND	O	0 VDC	0 VDC
6	GND	O	GND	
7	Pulse A1	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
8	Pulse A	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
9	VVac A4	O	GND	
10	VVac B3	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
11	VVac B2	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
12	VVac B1	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
13	VVac A2	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
14	VVac A3	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA
15	VVac A4	O	GND	
16	VVac A1	O	0.10 VDC @ 20 µA	4.50 VDC @ 20 µA

PA1 Connector PIN Functions and Voltages

Pin #	Description	I/O	Max Lo	Min Hi
1	5 VDC	O	4.75 VDC	
2	GND	O	0 VDC	0 VDC
3	Pulser Data A1	I	1.00 VDC	3.50 VDC
4	Pulser Data A1	I	1.00 VDC	3.50 VDC

PB1 Connector PIN Functions and Voltages

Pin #	Description	I/O	Max Lo	Min Hi
1	5 VDC	O	4.75 VDC	
2	GND	O	0 VDC	0 VDC
3	Pulser Data B1	I	1.00 VDC	3.50 VDC
4	Pulser Data B1	I	1.00 VDC	3.50 VDC

PA2 Connector PIN Functions and Voltages

Pin #	Description	I/O	Max Lo	Min Hi
1	5 VDC	O	4.75 VDC	
2	GND	O	0 VDC	0 VDC
3	Pulser Data A2	I	1.00 VDC	3.50 VDC
4	Pulser Data A2	I	1.00 VDC	3.50 VDC

PB2 Connector PIN Functions and Voltages

Pin #	Description	I/O	Max Lo	Min Hi
1	5 VDC	O	4.75 VDC	
2	GND	O	0 VDC	0 VDC
3	Pulser Data B2	I	1.00 VDC	3.50 VDC
4	Pulser Data B2	I	1.00 VDC	3.50 VDC

PA3 Connector PIN Functions and Voltages

Pin #	Description	I/O	Max Lo	Min Hi
1	5 VDC	O	4.75 VDC	
2	GND	O	0 VDC	0 VDC
3	Pulser Data A3	I	1.00 VDC	3.50 VDC
4	Pulser Data A3	I	1.00 VDC	3.50 VDC

PB3 Connector PIN Functions and Voltages

Pin #	Description	I/O	Max Lo	Min Hi
1	5 VDC	O	4.75 VDC	
2	GND	O	0 VDC	0 VDC
3	Pulser Data B3	I	1.00 VDC	3.50 VDC
4	Pulser Data B3	I	1.00 VDC	3.50 VDC

P309A Connector PIN Functions and Voltages

Pin #	Description	I/O	Max Lo	Min Hi
1	Slowdown Valve A1	O	0 VAC	115 VAC
2	Main Valve A1	O	0 VAC	115 VAC
3	Slowdown Valve B1	O	0 VAC	115 VAC
4	Main Valve B1	O	0 VAC	115 VAC
6	AC Neutral	O	0 VAC	
7	AC Neutral	O	0 VAC	
8	AC Neutral	O	0 VAC	
9	AC Neutral	O	0 VAC	

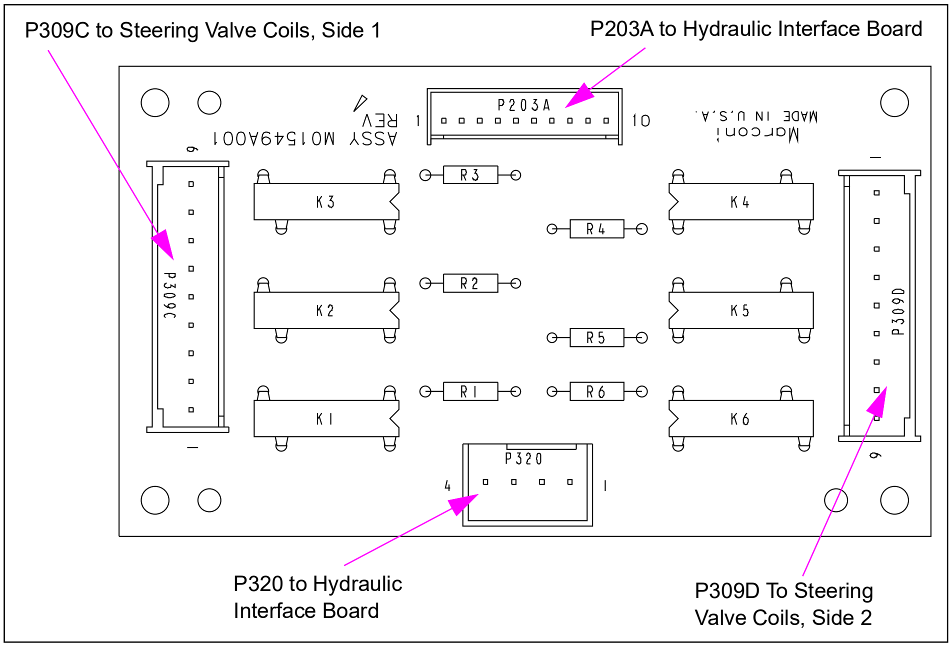
P309B Connector PIN Functions and Voltages

Pin #	Description	I/O	Max Lo	Min Hi
1	Slowdown Valve A2	O	0 VAC	115 VAC
2	Main Valve A2	O	0 VAC	115 VAC
3	Slowdown Valve B2	O	0 VAC	115 VAC
4	Main Valve B2	O	0 VAC	115 VAC
6	AC Neutral	O	0 VAC	
7	AC Neutral	O	0 VAC	
8	AC Neutral	O	0 VAC	
9	AC Neutral	O	0 VAC	

Steering Valve Relay PCA (M01549A001)

This board is used only with 6-hose blenders on Encore 300 units. It accepts 0 to 14 VDC levels that control solid state relays. These relays turn the steering valves on and off. Units manufactured after June 2002 will use M02097A001. M01549 and M02097 are electronically the same, but are not interchangeable because of connector differences.

Figure 6-19: Steering Valve Relay Board (M01549A001)



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P203A (10-pin)	M01694A001	Hydraulic Interface PCA (M01546A001)	P206 (10-pin)
P309C (9-pin)	M01685A001	Fixed Blender Manifold M00223B002 - Side 1 coils	J701 (9-pin)
P309D (9-pin)		Fixed Blender Manifold M00223A001 - Side 2 coils	
P320 (4-pin)	R20593-G1	Hydraulic Interface PCA (M01546A001)	

Test Points

Test Point	Range	Current (Max)
P203A-7 P203A-8	14 VDC,+/- 0.50 VDC	70 mA
P320-1 with respect to P320-2	115 VAC, +/- 5%	600 mA

P203A Input Requirements

Pin #	Signal Name	On	Off
1	SVA1		
2	SVA2		
3	SVA3		
4	SVB1	10 mA of current flow	0.5 mA or less of current flow
5	SVB2		
6	SVB3		
7, 8	+14 V		
9,10	GND		

P309C Output Requirements

Pin #	Signal Name	On	Off
1	Steering Valve A1	Current flow (refer to Note 1)	No current flow (refer to Note 2)
2			
3	Neutral		
4	Steering Valve A2	Current flow (refer to Note 1)	No current flow (refer to Note 2)
5			
6	Neutral		
7	Steering Valve A3	Current flow (refer to Note 1)	No current flow (refer to Note 2)
8			
9	Neutral		

Notes:
1) Requires a 300 mA load to the appropriate neutral connection. There must be current flow through the load and no more than a 1.4 V (peak) drop between P320-1 and the appropriate output.
2) There must be 100 μA or less current flow through the load.

P309D Output Requirements

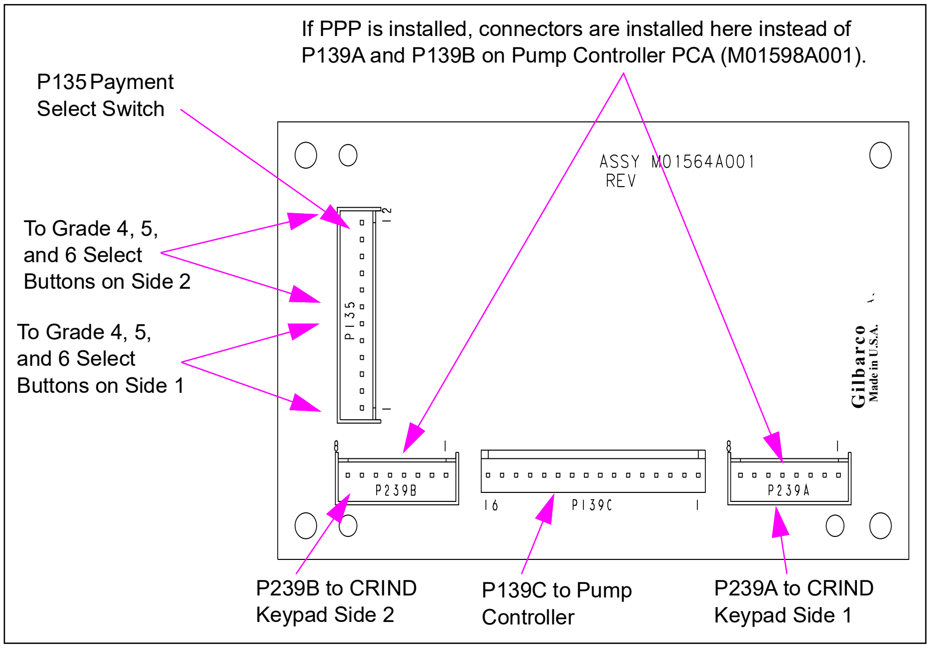
Pin #	Signal Name	On	Off
1	Steering Valve B1	Current flow (refer to Note 1)	No current flow (refer to Note 2)
2			
3	Neutral		
4	Steering Valve B2	Current flow (refer to Note 1)	No current flow (refer to Note 2)
5			
6	Neutral		
7	Steering Valve B3	Current flow (refer to Note 1)	No current flow (refer to Note 2)
8			
9	Neutral		

Notes:
1) Requires a 300 mA load to the appropriate neutral connection. There must be current flow through the load and no more than a 1.4 V (peak) drop between P320-1 and the appropriate output.
2) There must be 100 μA or less current flow through the load.

Grade Select PCA (M01564A001)

Grade select PCB is used on Encore 300 units for grade select and pump preset option for units with 4, 5, or 6 grade select options.

Figure 6-20: Grade Select Board (M01564A001)



Board Connections and Cables

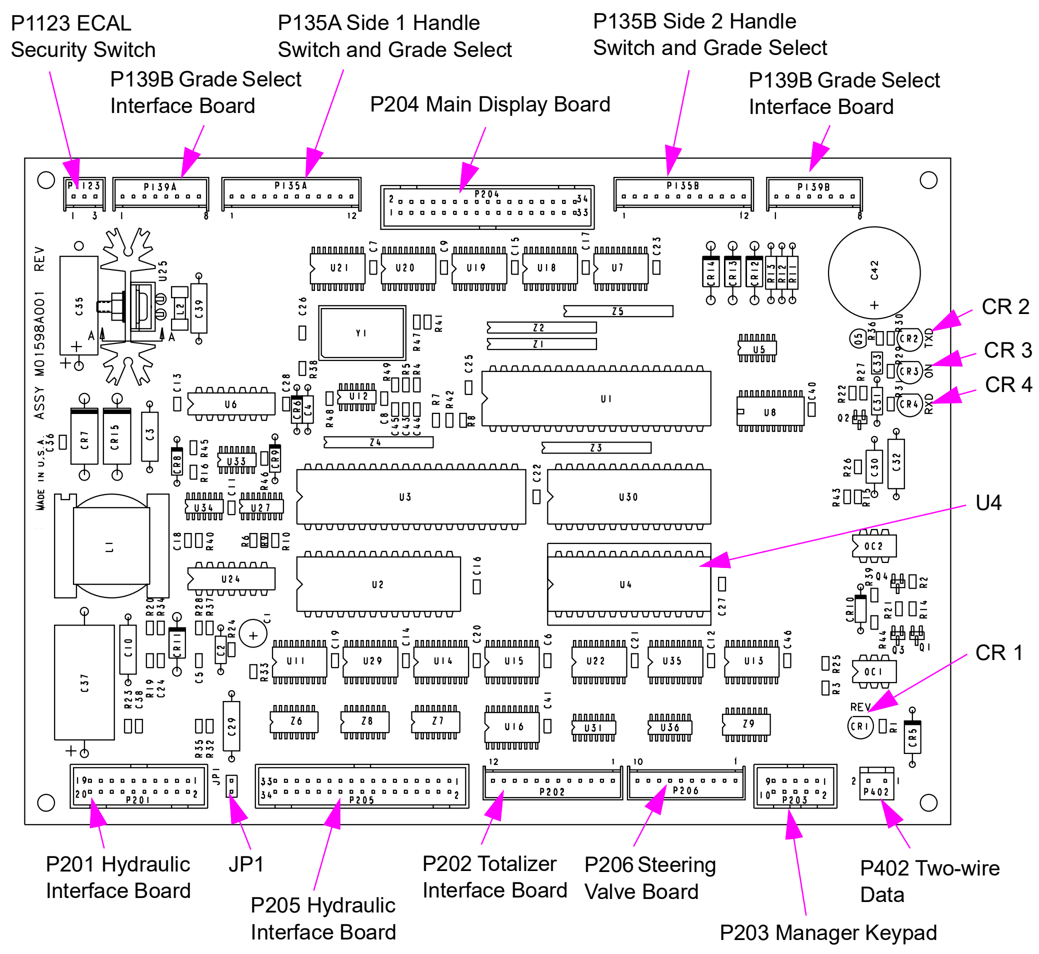
Connector#	Through Cable	To Board	At Connector#
P135 (12-pin)	M01695A001	Payment Select Membrane Switches (T19370)	P136 (4-pin)
P139C (16-pin)	M01693A001	Pump Controller PCA	P139A/B (8-pin)
P239A (8-pin) P239B (8-pin)	M01686A001	CRIND Keypad (M00141B002)	P902 Side 1 or 2 (12-pin)

Pump Controller PCA (M01598A001)

Pump controller board on Encore 300 units processes data pulses from pulser assemblies, level changes from pump handles, and two-wire serial data. Also, provides required outputs to control the transaction and price posting displays, solenoid valves, STPs, motors and converts 14 VDC to 5 VDC. The function of this board is similar to the Advantage Pump Controller Board (T20011-G1). However, the firmware on this board is not interchangeable.

IMPORTANT INFORMATION
This board runs at about 50% faster clock speed than an Optimized Advantage pump controller board.

Figure 6-21: Pump Controller Board (M01598A001)



Note: For descriptions of callouts on this board, refer to the tables on *page 6-42* *and page* *page 6-43*.

LED Indicators

LED	Indicates
CR-1	On = Two-wire loop reversed
CR-2	Blinking = transmitting data
CR-3	On = CPU is on
CR-4	Blinking = receiving data from console

Memory

IC	Description
U4	Firmware location

Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P135A/B (12-pin)	M01650A001 (harness)	CRIND Z180 Logic Board (T17764-G3/4)	P264 (6-pin)
		Totalizer Interface PCA (M01621A001)	P1507 (12-pin)
		Single PPU Board (M01522A001)	P2201 (2-pin)
		Dual PPU PCA (M01525A001)	P2201 (2-pin)
P201 (20-pin)	R17995-G7	Hydraulic Interface PCA (M01546A001)	P101 (20-pin)
P202 (12-pin)	M01650A001 (harness: see above)		
P203 (10-pin)			
P204 (34-pin)			
P206 (10-pin)	Not used		
P402 (2-pin)	M01664A001	Two-wire from Field Wiring	
P1123 (3-pin)	M00150A001 ECAL Security Switch		

P201 Connector PIN Functions and Voltages

Pin#	Description	I/O	Max Low	Max High
1	SYS RESET	O	0.10 VDC	4.50 VDC
2	VVac Error	I	0.80 VDC	2.00 VDC
3	TXB	O	0.40 VDC	4.75 VDC
4	RXDB	I	0.80 VDC	2.00 VDC
5	14.5 V	I	Refer to "P201 Power Test Points" on page 6-43.	
6	14.5 V	I		
7	14.5 V	I		
8	DC GND	I		
9	DC GND	I		
10	DC GND	I		
11	DC GND	I		
12	DC GND	I		
13	DC GND	I		
14	DC GND	I		
15	+5 VDC	O	Refer to "P201 Power Test Points" on page 6-43.	
16	+5 VDC	O		
17	DC GND	I		

Pin#	Description	I/O	Max Low	Max High
18	DC GND	I		
19	+5 VDC	O	Refer to "P201 Power Test Points"	
20	+5 VDC	O		

P202 Connector PIN Functions and Voltages

Pin#	Description	I/O	Max Low	Max High
1	Connect A	I	0 V	4.50 VDC @ 20 µA
2	STP-1	O	1.30 VDC @ 156 mA	
3	STP-2	O	1.30 VDC @ 156 mA	
4	STP-3	O	1.30 VDC @ 156 mA	
5	14.5 V	O	Refer to "P201 Power Test Points".	
6	GND	O		
7	Connect B	I	0 V	4.50 VDC @ 20 µA
8	SPARE A	O	1.30 VDC @ 156 mA	
9	SPARE B	O	1.30 VDC @ 156 mA	
10	SPARE C	O	1.30 VDC @ 156 mA	
11	14.5 V	O	Refer to "P201 Power Test Points".	
12	GND	O	0 V	0 V

Jumper Settings

Jumper	Position	Function
JP1	IN	Push-to-Start activated
	OUT	Lever on activated

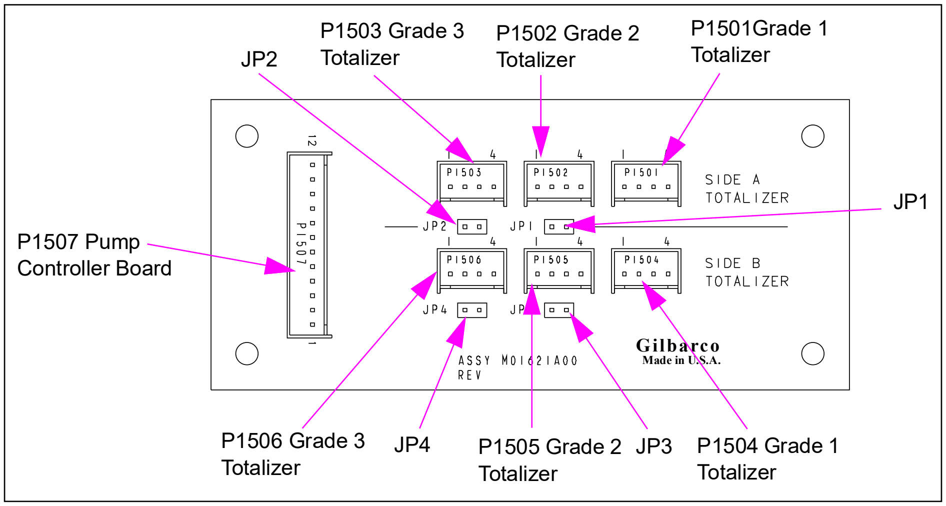
P201 Power Test Points

Connector Pin#	I/O	Signal Name	Tolerance	Maximum Current
P201-6	I	14.5 VDC	14.5 VDC +/- 5%	800 mA
P201-12	I	GND/Signal Return	0 VDC	
P201-16	O	5 VDC	5 VDC +/- 5%	125 mA

Totalizer Interface PCA (M01621A001)

This board is the interface for the electromechanical totalizers.

Figure 6-22: Totalizer Interface Board (M01621A001)



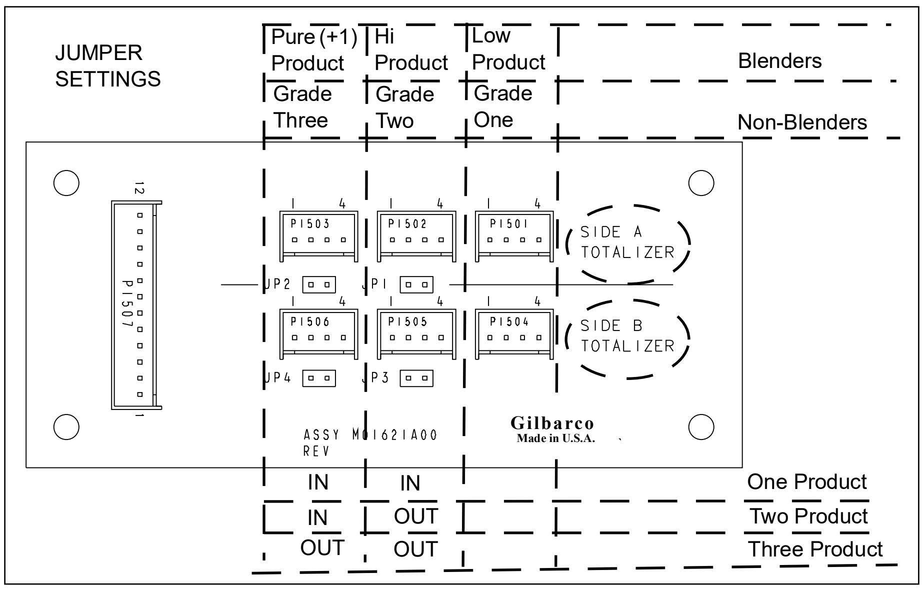
Board Connections and Cables

Connector#	Through Cable	To Board
P1501 (4-pin)	Totalizer cable	Electronic Totalizer Side 1 Grade 1
P1502 (4-pin)		Electronic Totalizer Side 1 Grade 2
P1503 (4-pin)		Electronic Totalizer Side 1 Grade 3
P1504 (4-pin)		Electronic Totalizer Side 2 Grade 1
P1505 (4-pin)		Electronic Totalizer Side 2 Grade 2
P1506 (4-pin)		Electronic Totalizer Side 2 Grade 3
P1507 (12-pin)	M01650A001 (harness)	Refer to "Totalizer Interface PCA (M01621A001)".

Jumper Settings

Jumper	Position	Function
J1	Side A	One Grade or Single Product
J2	Side A	One or Two Grades or Two Product
J3	Side B	One Grade or Single Product
J4	Side B	One or Two Grades or Two Product

Figure 6-23: Jumper Setting Diagram for M01621A001



CRIND Control Node-2 (M01753A001)

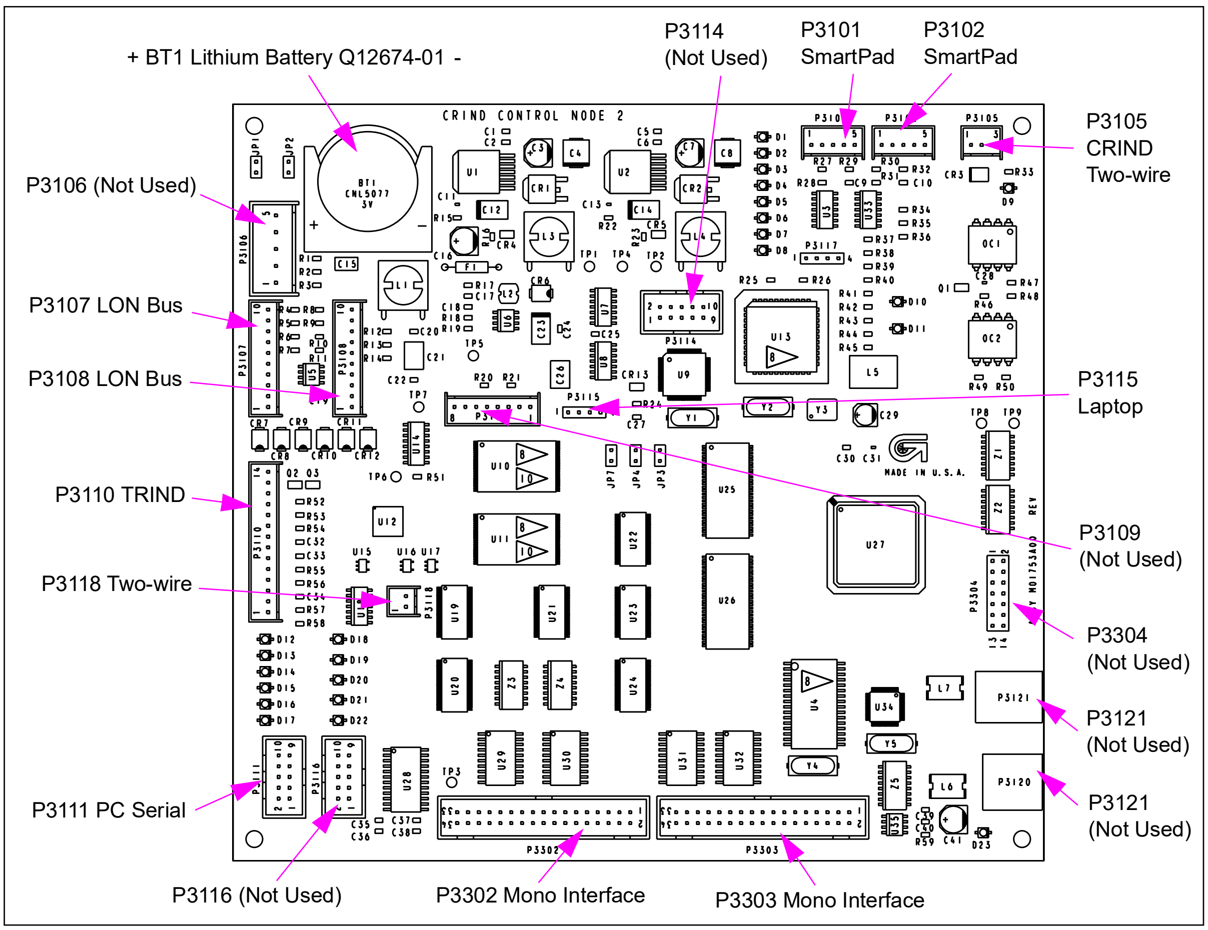
The CCN is the primary CRIND device option processor on e-CRIND-equipped Encore 500 and Eclipse units. This node receives information directly or indirectly from, and/or controls the card reader, monochrome display, printer, barcode scanner, cash acceptor, beeper, and CRIND related keypads. It also interfaces with TRIND devices and supports USB communications.

The M01753A001 CCN (now replaced by M04108A001) is backward compatible with M00089A00X CCN. It requires CNN BIOS version 2.1.80 or later. The M00089A00X is forward compatible with the M01753A001 CCN with the exception of enhanced durability improvements. M04108AXXX is replaced by M12828A001.

Service Tips

- Coldstarts are best done using diagnostic card and “Clear Persistent Memory” function.
- If the M01763A001 CCN is replaced with a M00089A00X CCN, then a Monochrome Interface PCA (M00092A001) and an Interface Cable (M00545A001) will be required.

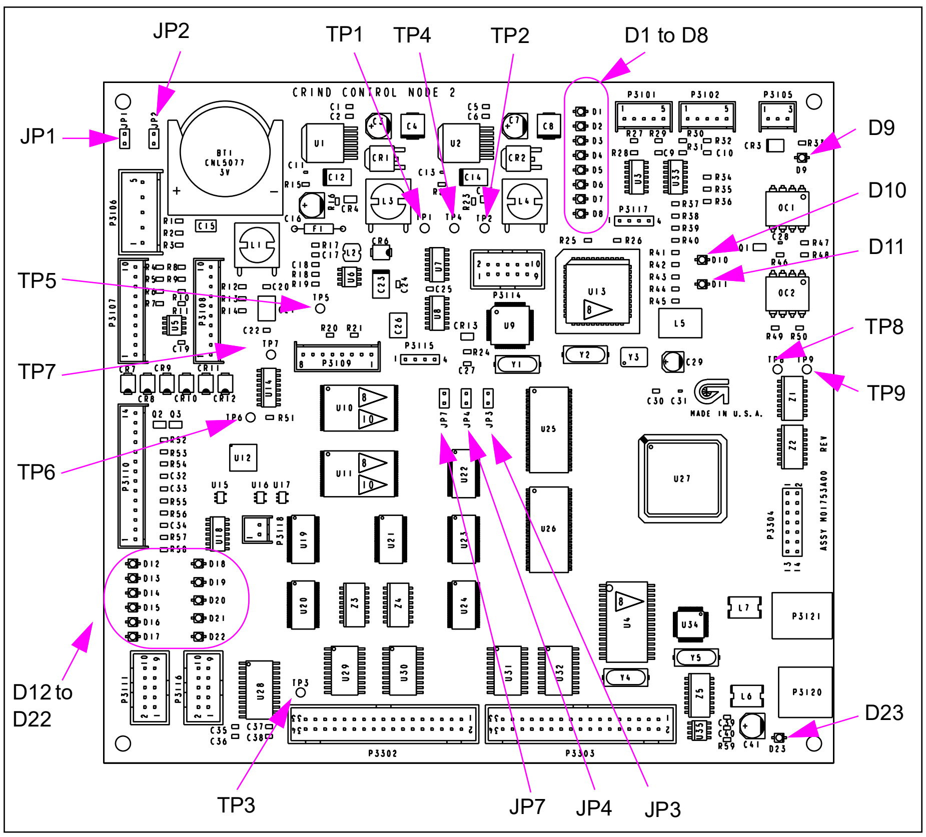
Figure 6-24: CRIND Control Node-2 Connectors (M01753A001)



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P3101 (5-pin)	M00497	SmartPad T20247-01	
P3102 (5-pin)			
P3105 (3-pin)	M02159A002	CRIND Two-wire	P412
P3107 (10-pin)		LON Bus	
P3108 (10-pin)			
P3110 (14-pin)		TRIND	
P3111 (10-pin)	M00719A001	Serial programming port	
P3115 (4-pin)	M02995A001	to Laptop	
P3117 (4-pin)			
P3118 (2-pin)		PCN	P1109 (2-pin)
P3302 (34-pin)	M00553A001	Mono module	J1 (34-pin)
P3203 (34-pin)			

Figure 6-25: CRIND Control Node-2 LEDs, Jumpers, and Test Points



Status LEDs

LED	Name	Status Indication
D1	HEART BEAT	Flashes steadily at rate of “on” for 1 second, “off” for 1 second during normal operation. Flashing begins 15-20 seconds after power-up.
D2	SMARTPAD A TX	Blinking when sending data to Side 1 SmartPad or Krone pinpad through P3101.
D3	SMARTPAD A RX	Blinking when receiving data from Side 1 SmartPad or Krone pinpad through P3101.
D4	SMARTPAD B TX	Blinking when sending data to Side 2 SmartPad or Krone pinpad through P3102.
D5	SMARTPAD B RX	Blinking when receiving data from Side 1 SmartPad or Krone pinpad through P3102.
D6	LON TERM1	• Off solid is normal condition • On solid when LON cable connected to P3107 is not terminated
D7	LON TERM2	• Off solid is normal condition • On solid when LON cable connected to P3108 is not terminated
D8	NEURON STAT	• Off solid is normal condition • On solid if Neuron is application-less (node must be returned) • Blinking at a rate of “on” for 1 second, “off” for 1 second if Neuron in not configured (node must be returned) • Short flashes every 1.5 seconds indicates PCN software is not running. This is normal during PCN software upgrade and during the first 15-20 seconds of startup.
D9	POS 2W REV	• Off solid is normal condition • ON solid if two-wire loop is reversed
D10	POS TX	Blinking when sending data to the POS through P3105
D11	POS RX	• Off when current is present but communications are idle • On solid if no current is present • Blinking when receiving data from POS through P3110
D12	TRIND TX	Blinking when sending data to TRIND system through P3110
D13	TRIND RX	Blinking when receiving data from TRIND system through P3110
D14	PC SER TX	Blinking when sending data to service laptop through P3111
D15	PC SER RX	Blinking when receiving data from service laptop through P3111
D16	PCN 2W TX	Blinking when sending data to the pump through P3118
D17	PCN 2W RX	• Off when current is present, but communications are idle • On solid if no current present • Blinking when receiving data from pump through P3118
D18	STATE A	Currently not used
D19	STATE B	Currently not used
D20	STATE C	Currently not used
D21	STATE D	Currently not used
D22	STATE E	• Off solid is normal condition • On solid if CRIND MIP version is older than V02.0.00, of if Neuron is not communicating with the CPU. Usually means MIP must be upgraded, but sometime infers board failure.
D23	USB GOOD	On solid if USB link is established

Test Points

Test Point	Value	Where Used
TP1	3.15 to 3.45 VDC	Memory voltage
TP2	4.75 to 5.25 VDC	Logic voltage
TP3	35 VDC	Display voltage
TP4	Common	For DC voltage
TP5	14 to 28 VDC	LON voltage
TP6	2 to 3.6 VDC	RAM voltage
TP7	2 to 3.6 VDC	Battery voltage

Jumpers

Jumper	Setting	Function
JP1	Out	Reset (refer to Note 1)
JP2	Out	Coldstart
JP3	Out	Service (refer to Note 2)
JP4	Out	LON Programming (refer to Note 1)
JP7	Out	Battery disconnect

Notes: 1. Used in manufacturing.
2. Used in board repair.

Door Node-2 (M01785A001)

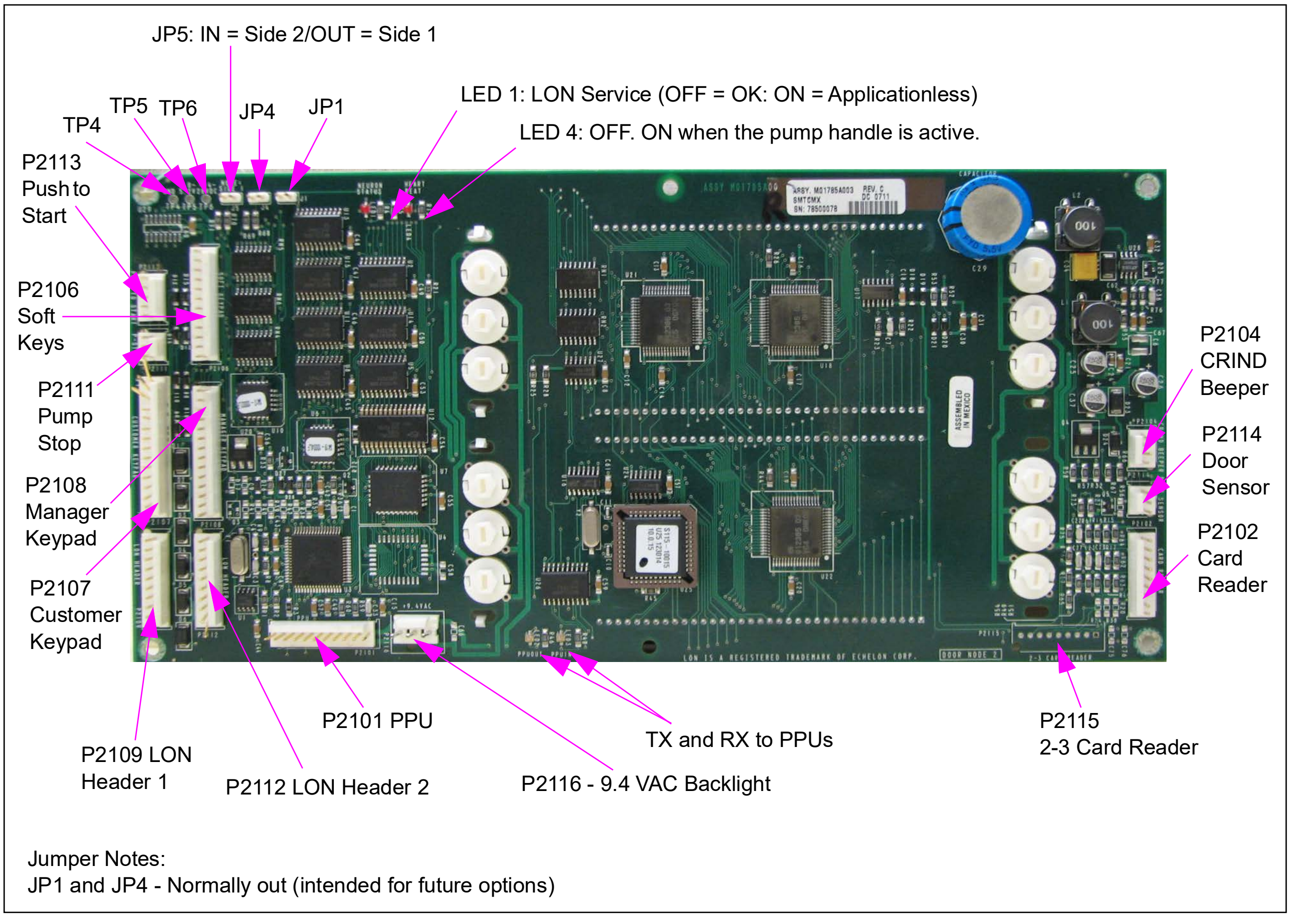
Door Node-2 is used on Encore 500 and Eclipse series units. This node displays transaction information to the customer and serves as on I/O gathering point by interfacing peripherals to the LON bus. These peripherals include card reader, customer keypad, and CRIND beeper. The door node collects pump handle and grade data, maintains a backup of the pumps data, and communicates to the PPU display boards for that side.

The Door Node-2 (M01785) is backward compatible with Door Node (M00062A00X) requires door node LON version 1.0.37 or later. The Door Node (M00062A00X) is forward compatible with M01785A001 with the exception that certain enhanced reliability components will not be available.

Service Tip

Do not disrupt the download process when loading door node MIP software or the node will require repair at the factory.

Figure 6-26: Door Node-2 for Encore 500 and Eclipse



Status LEDs

LED	Status Indication
1	LON (Neuron) Status - ON or blinking indicates LON problem
2	TX or PPUOUT indicates communication to PPU boards
3	RX or PPUIN indicates communication from PPU boards
4	OFF. ON when the pump handle is active.

Test Points

Test Point	Range
TP4	GND
TP5	4.75 V to 5.25 VDC
TP6	+14 V to +23 VDC

Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P2101 (11-pin)	• M00509A001 or M00509A002 for Encore • M00509A003 for Eclipse	Totalizer Node (M00077A001), one per grade	P2201
P2102 (9-pin)	• M00501A001 (Encore) • M00501A002 (Eclipse)	• Card Reader (7-pin) • Card Reader Heater R20379 (2-pin)	
P2104 (4-pin)	• M00636A001 (Encore) • M00495A001/M00636A002 (Eclipse)	Speaker (refer to Encore Series Cable Block Diagram or Eclipse Series Cable Block Diagram).	
P2106 (12-pin)	M01200A001	Soft Keys Keypad (refer to Encore Cable Block Diagram M00284)	
P2107 (15-pin)	M01202A001 (Encore M01202A002 (Eclipse)	Customer Keypad Options (CRIND or PPP): • Misc Options Keypad (M01109) • CRIND Keypad (M00141)	Refer to Encore Series Cable Block Diagram or Eclipse Series Cable Block Diagram.
P2108 (14-pin)	M00515A001 (Encore) M00515A003 (Eclipse)	Manager Keypad [M00147 (Side 1 only)]	
P2109 (10-pin)	M00489	LON Bus 1	
P2110 (4-pin) [Call Button] P2111 (3-pin) [Push-to-Stop] P2113 (5-pin) [Push-to-Start]	M01201A001 (Encore) M01215A001 (Encore)	Miscellaneous Options Keypad (M01109B001)	12-pin
P2112 (10-pin)	M00489	LON Bus 2	
P2114 (4-pin)	N/A	Door sense	
P2115 (9-pin)	M00614A004 (Encore) M00614A002 (Eclipse)	Power Supply 9.4 VAC (PS2 to PS3 on bulkhead connector)	PS2
P2116 (3-pin)	M00614A005 to M00614A004	Main Power Supply	J2115

Connector PIN Functions and Voltages

P2101

Pin #	Description	Input/Output Voltage
1	+9.4 VAC	9.4 VAC +/-10%
2	+9.4 VACRTN	
3	GND	0 VDC
4	GND	0 VDC
5	DLT+5 VDC	+4.7 V +/-10%
6	PPUCLK	0 to 5 V
7	PPUFBDATA	0 to 5 V
8	PPUSND_REC	0 to 5 V
9	PPUBP	0 to 5 V
10	PPUSDATA	0 to 5 V
11	PPUSYNC	0 to 5 V

P2102

Pin #	Description	Input/Output Voltage
1	VCC	+5 V +/-5%
2	CRRDT1	0 to 5 V
3	CRRCL1	0 to 5 V
4	CRRDT2	0 to 5 V
5	CRRCL2	0 to 5 V
6	CRCLD	0 to 5 V
7	GND	0 V
8	CRSTAT	Open to 0 V
9	GND	0 V

P2103

Pin #	Description	Input/Output Voltage
1	VCC	+5 V +/-5%
2	GND	0 V
3	PRSENIN	0 to 5 V
4	PRSENSTAT	Open to 0 V
5	+18 VUNREG	+14 V to +23 V

P2104

Pin #	Description	Input/Output Voltage
1	VCC	+5 V +/-5%
2	SPKRTN	Open Collector to 0 V
3	SPKR_STAT	Open to 0 V
4	GND	0 V

P2109

Pin #	Description	Input/Output Voltage
1	+18 VUNREG	+14 V to +23 V
2	GND	0 V
3	+18 VUNREG	+14 V to +23 V
4	GND	0 V
5	485B	-7 V to +12 V Max.
6	485A	-7 V to +12 V Max.
7	GND	0 V
8	+18 VUNREG	+14 V to +23 V
9	EXTRESET	0 V to +12 V
10	TERM RES FEEDTHRU	

P2112

Pin #	Description	Input/Output Voltage
1	+18 VUNREG	+14 V to +23 V
2	GND	0 V
3	+18 VUNREG	+14 V to +23 V
4	GND	0 V
5	485B	-7 V to +12 V Max.
6	485A	-7 V to +12 V Max.
7	GND	0 V
8	+18 VUNREG	+14 V to +23 V
9	EXTRESET	0 V to +12 V
10	TERM RES FEEDTHRU	

P2115

Pin #	Description	Input/Output Voltage
1	+9.4 VAC	9.4 VAC +/-10%
2	+9.4 VACRTN	
3	No connect	

Door Node-3 (M04326A001/A002)

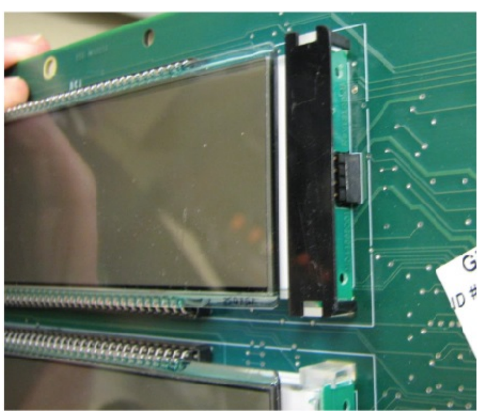
Door Node-3 is used on Encore and Eclipse units. This node displays transaction information to the customer and serves as an I/O gathering point by interfacing peripherals to the LON bus. These peripherals include the card reader, customer keypad, push-to-start, soft keypad, and CRIND beeper. The door node collects pump handle and grade data, maintains a backup of the pump data, and communicates to the PPU display boards for each side.

The Door Node-3 is backward compatible with Door Node (M00062A00X) and requires door node LON version 1.0.37 and later. The new Door Node-3 now has the LON terminator built into it, therefore only needing one LON connector. There is a 2-pin connector between the door node and the monochrome display, which provides the 24 VDC needed for the backlight power.

The following are the modified components in the new Door Node-3:

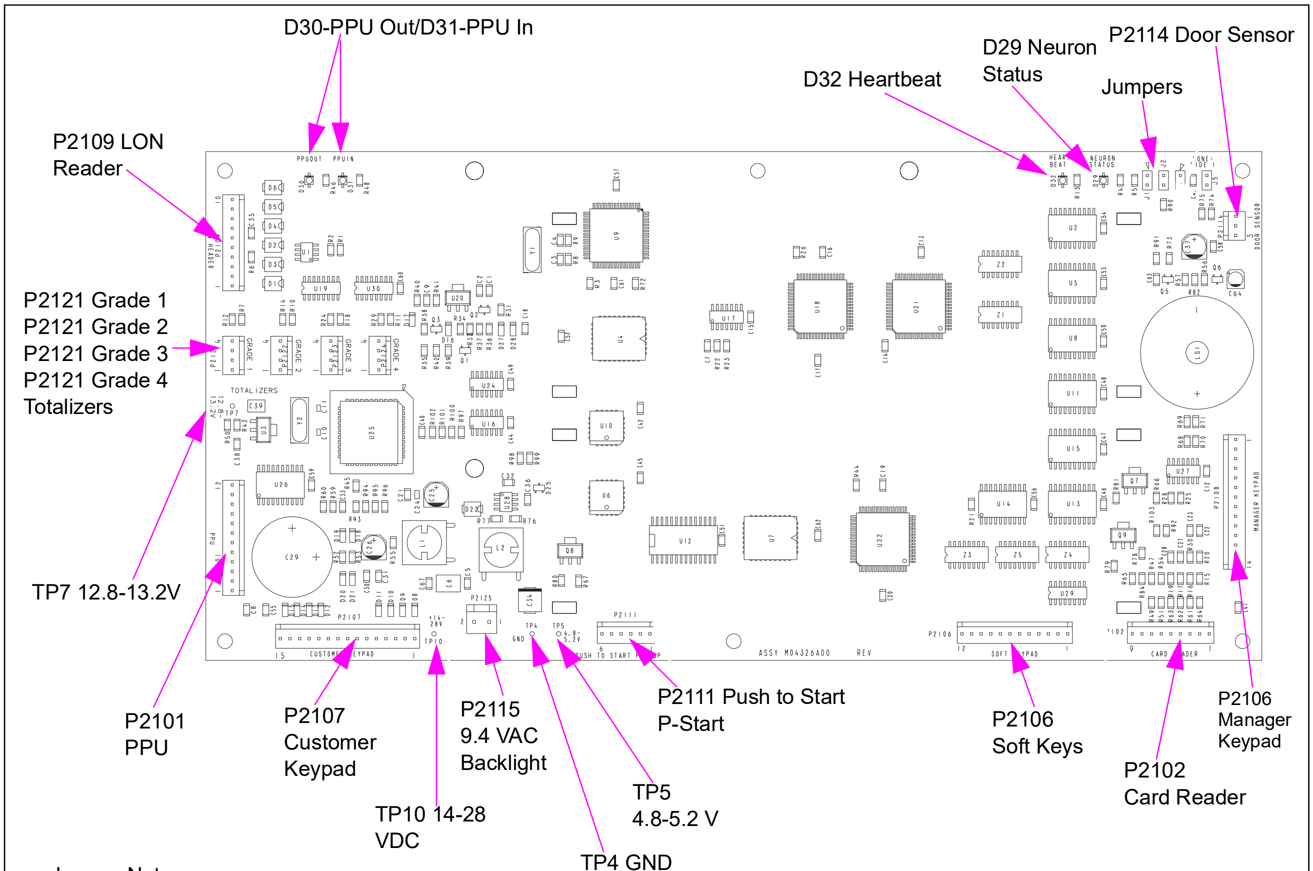
- The pump stop (P2111) and start button connectors (P2113) have been combined into one connector, labeled P2111.
- The beeper is now built into the board, but the connector for the external printer beeper still exists. The board senses the presence of the external speaker, and does not use the internal beeper if the external one is connected.
- P2121, P2122, P2123, P2124 are the four connectors that have been added to drive the totalizers when the door node software is written to support them.
- The 11-pin cable to the PPUs has been replaced with the 12-pin cable. This is to prevent mixing and matching of PPU and door node types.
- The backlight is provided by LED boards that are field replaceable. To replace the LED board, remove the backlight shield (see Figure 6-27) and then remove the LED assembly. There are three sets of four LEDs in series. All four center LEDs are in series. The top and bottom LEDs on the left side of the money and volume are in series, and the top and bottom LEDs on the right side of the money and volume are in series. They have been put in series to minimize the power consumption, and they are distributed in this way to minimize the visual impact of one string not working.

Figure 6-27: Backlight Shield



IMPORTANT INFORMATION
The minimum software version for Door Node-3 (M04326A001) is 01.0.47. The minimum software version for Door Node-3 (M04326A002) is 01.0.70.

Figure 6-28: Door Node-3 for Encore



Jumper Notes:
JP1 and JP4 - Normally Out (intended for future options)

Note: M04326 was used in E500 and Eclipse and is now replaced by M13170.

Jumper Table

Note: The setting for JP1 and JP4 is normally "Out" (intended for future options).

Jumper	Purpose	Function	
		Installed	Uninstalled
JP1	LON Service	LON Service	Normal Operation
JP2	Beeper Volume Control	Louder Beeper	Normal Beeper
JP4	Display Test	Display runs test mode at startup	Normal Operation
JP5	Side Control	Side 2 (Side B)	Side 1 (Side A)

Status LEDs

LED	Purpose	Meaning
D29	Neuron Status	On-program locked up, door node bad, or application is missing
D30	PPU In	• On - No legacy PPU communication response • Flashing/Solid - Normal operation to legacy PPU
D31	PPU Out	• Off - U25 problem or U25 not present • Flashing/Solid - Normal operation to legacy PPU
D32	Heartbeat	**Pump Door Node (PDN) - V01.0.24 and before - On - Hardware/Software issue** • Off - Hardware/Software issue • Flashing - Normal operation **PDN- V01.0.24 and after** • On - All pump handles not reporting from PPU • Off - All pump handles reporting from PPU

Test Points

Test Point	Range
TP4	GND
TP5	4.8-5.2 VDC
TP7	12.8-13.2 VDC
TP10	14-28 VDC

Door Node-4 (M05835A004)

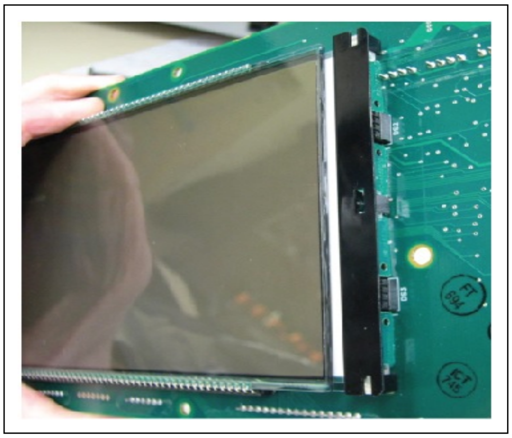
The door node/main display introduces a combined money/volume LCD display, with larger size money digits. Some of the connectors along with the bottom of the board are flipped compared to the old door node boards for ease of connection.

The Door Node-4 (M05835A004) is strictly designed to be used on the Encore S units. It **cannot** be used on the standard Encore 500 units due to their single large display segment. This node displays transaction information to the customer and serves as an I/O gathering point by interfacing peripherals to the LON bus. These peripherals include the card reader, customer keypad, push-to-start, soft keypad, and CRIND beeper. The door node collects pump handle and grade data, maintains a backup of the pump data, and communicates to the PPU display boards for each side.

The following are the modified components in the new Door Node-4:

- The pump stop (P2111) and start button connectors (P2113) have been combined into one connector, labeled P2111.
- The beeper is now built into the board, but the connector for the external printer beeper still exists. The board senses the presence of the external speaker, and does not use the internal beeper if the external one is connected.
- The 11-pin ribbon cable to the PPUs has been replaced with the 12-pin cable. This has been done to prevent mixing and matching of PPU and door node types.
- The backlight is provided by LED boards that are field replaceable. To replace the LED board, remove the backlight shield (see Figure 6-29 on page 6-57) and then remove the LED assembly. There are three sets of four LEDs in series. All four center LEDs are in series. The top and bottom LEDs on the left side of the money and volume are in series, and the top and bottom LEDs on the right side of the money and volume are in series. They were put in series to minimize the power consumption, and they were distributed this way to minimize the visual impact of one string not working.

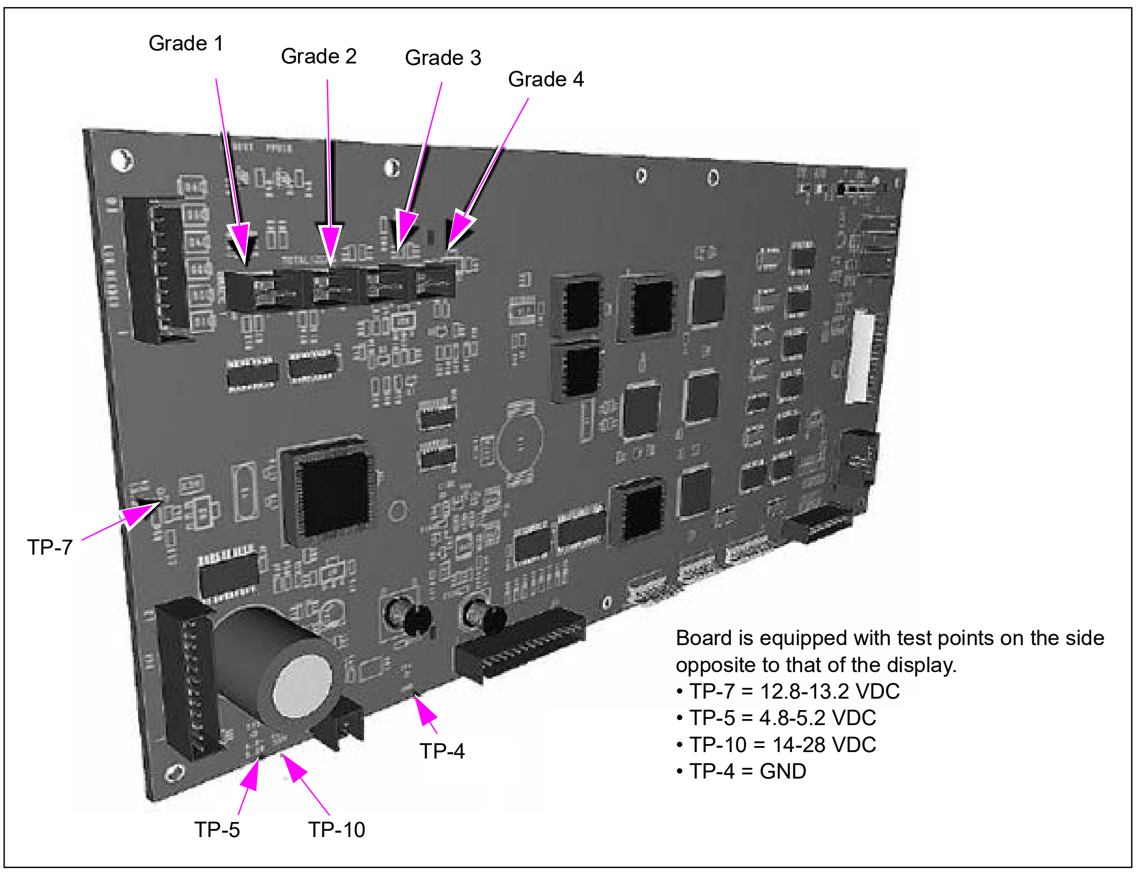
Figure 6-29: Backlight Shield



IMPORTANT INFORMATION
M05835A001-A003 (boards no longer available) uses software up to V01.0.47. M05835A004 uses software V01.0.70 and later.

Totalizer cables plug directly into Door Node-4 (same as Door Node-3).

Figure 6-30: Totalizer Connectors

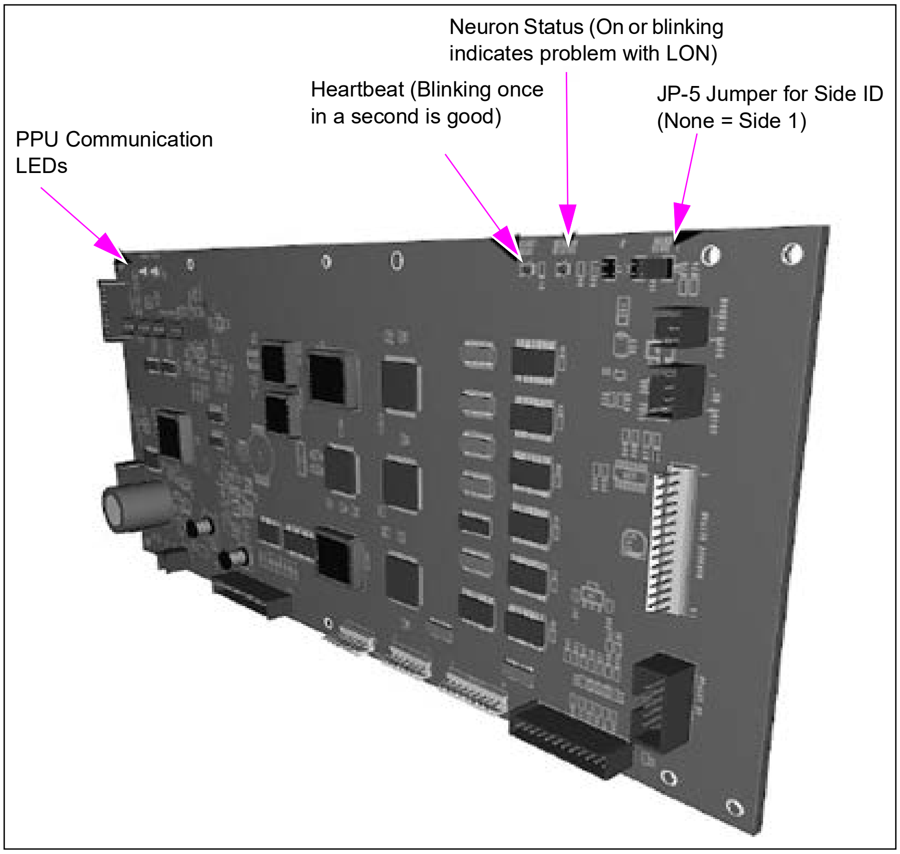


Jumper Table

Note: The setting for JP1 and JP4 is normally "Out" (intended for future options).

Jumper	Purpose	Function	
		Installed	**Uninstalled**
JP1	LON Service	LON Service	Normal Operation
JP2	Beeper Volume Control	Louder Beeper	Normal Beeper
JP4	Display Test	Display runs test mode at startup	Normal Operation
JP5	Side Control	Side 2 (Side B)	Side 1 (Side A)

Figure 6-31: Status LEDs



Note: The Door Node-4 has the same basic functions as the Door Node-3. Refer to Sandpiper 2b Launch Package for additional information.

Status LEDs

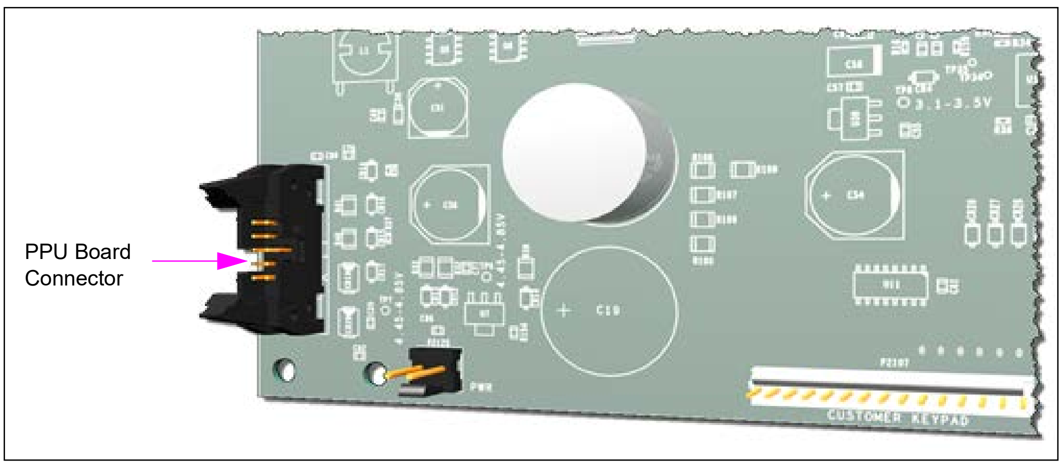
LED	Purpose	Meaning
D29	Neuron Status	On - program locked up, bad door node, or missing application
D30	PPU In	• On - No legacy PPU communication response • Flashing/Solid - Normal operation to legacy PPU
D31	PPU Out	• Off - U25 problem or U25 not present • Flashing/Solid - Normal operation to legacy PPU
D32	Heartbeat	**PDN - V01.0.24 and before - On - Hardware/Software Issue** • Off - Hardware/Software issue • Flashing - Normal operation **PDN - V01.0.24 and later** • On - All pump handles not reporting from PPU • Off - All pump handles reporting from PPU

Door Node-5 (M12803A001/A002/A003/A004)

Door Node-5 (M12803A001/A002)

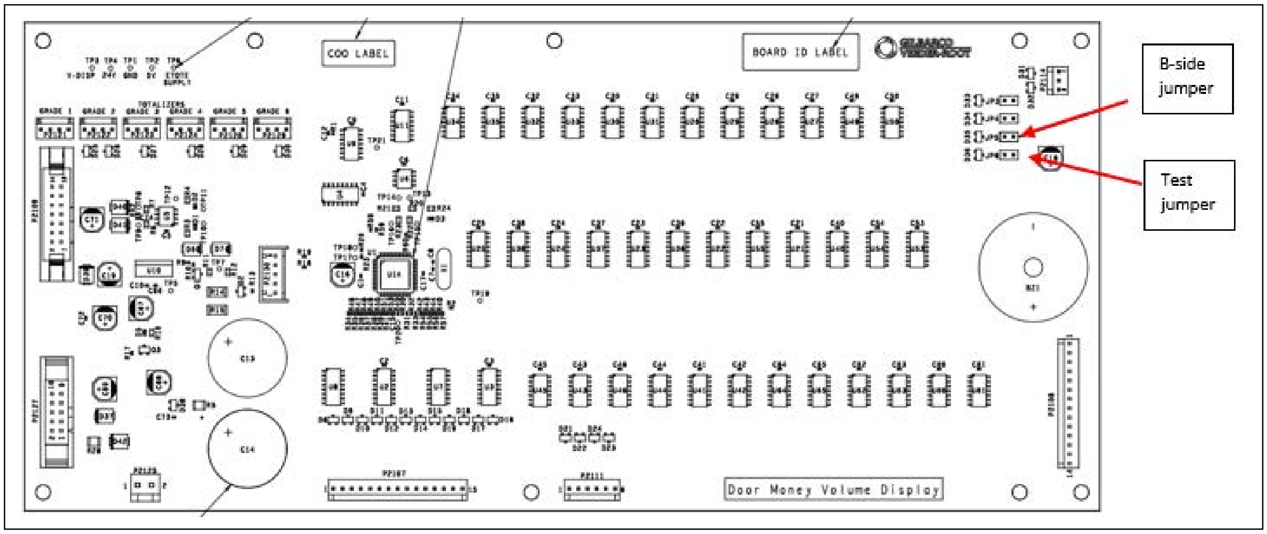
Door Node-5 can be used in Encore 500/500 S/500 S E-CIM and Encore 700 S dispensers. The major difference in this board over the last one is the newly added PPU board connector. This door node works with the new PPU board only.

Figure 6-32: Door Node-5 (M12803A001/A002)



M18830 Board Assembly

Figure 6-33: M18830 Board Assembly



Connector list

Connector	Function	Connects to
P2109	Data & power	M18666 pump board.
P2127	Data & power to PPUs	PPU displays.
P2125	+24V out	
P2107	Preset keypad inputs	Preset keypad membrane switch via ribbon cable.
P2111	Pump start/stop inputs	Pump start & stop switches via cables.
P2108	Manager keypad inputs	Manager keypad membrane switch via ribbon cable.
P2114	Door switch input	Connects to door switch(es) via cable.

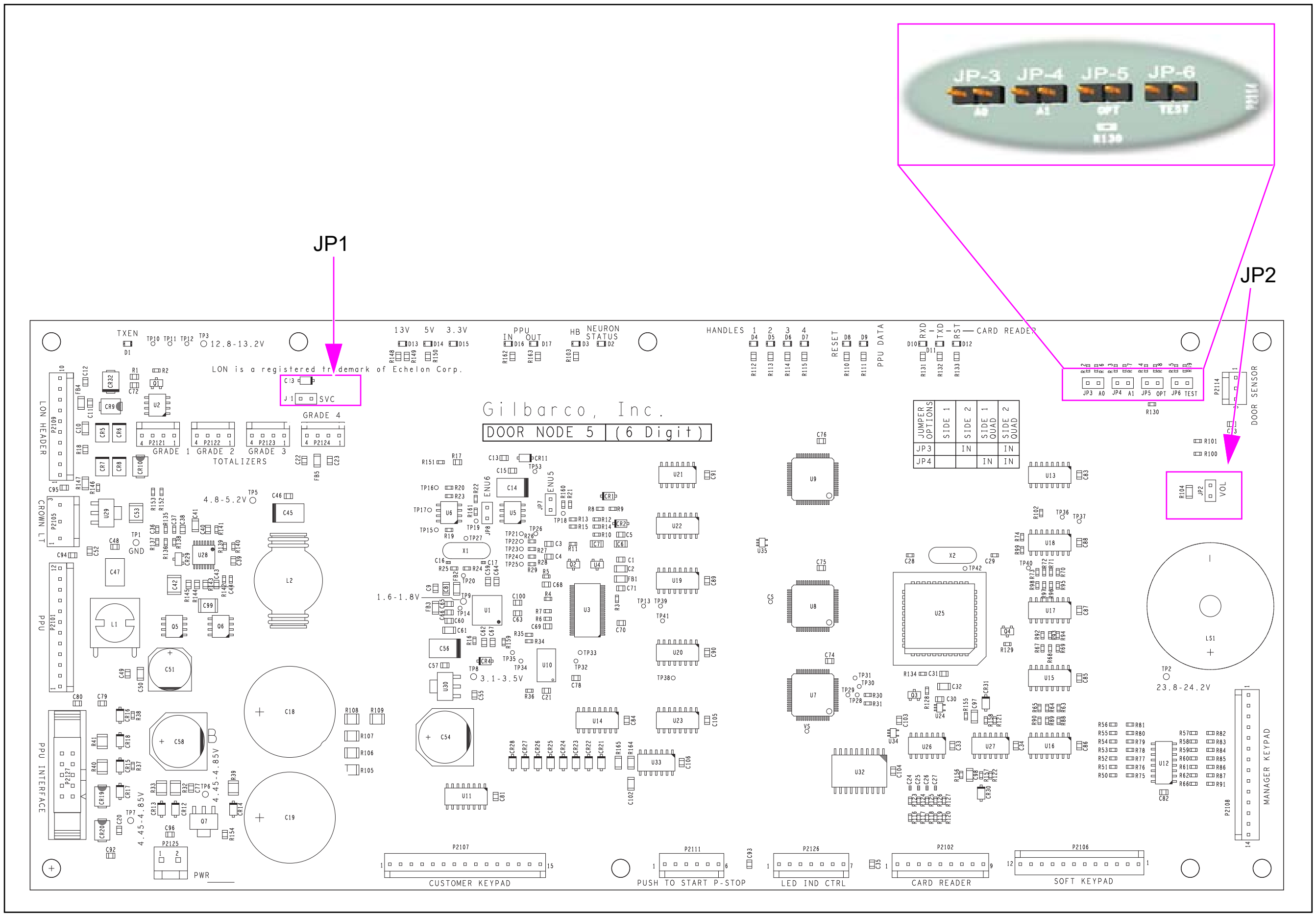
M18830 jumpers

Jumper	Description	Use
JP5	B-side Jumper	Must be installed on the B-side display.
JP6	Test jumper	This can be used to test the display and keypads. If this is installed on power-up, the display will do a repeating digit-test. If a keypad is pressed or door switch closed, the corresponding key-code is displayed.

M18830 LEDs

LED	Function	Connects to
D1	RX	Data received from pump board
D2	TX	Data to pump board
D3	Heartbeat	Display board alive (blinking)

Figure 6-34: Door Node-5 Address - (1)



Jumper Table

Note: The setting for JP1 and JP4 is normally "Out" (intended for future options).

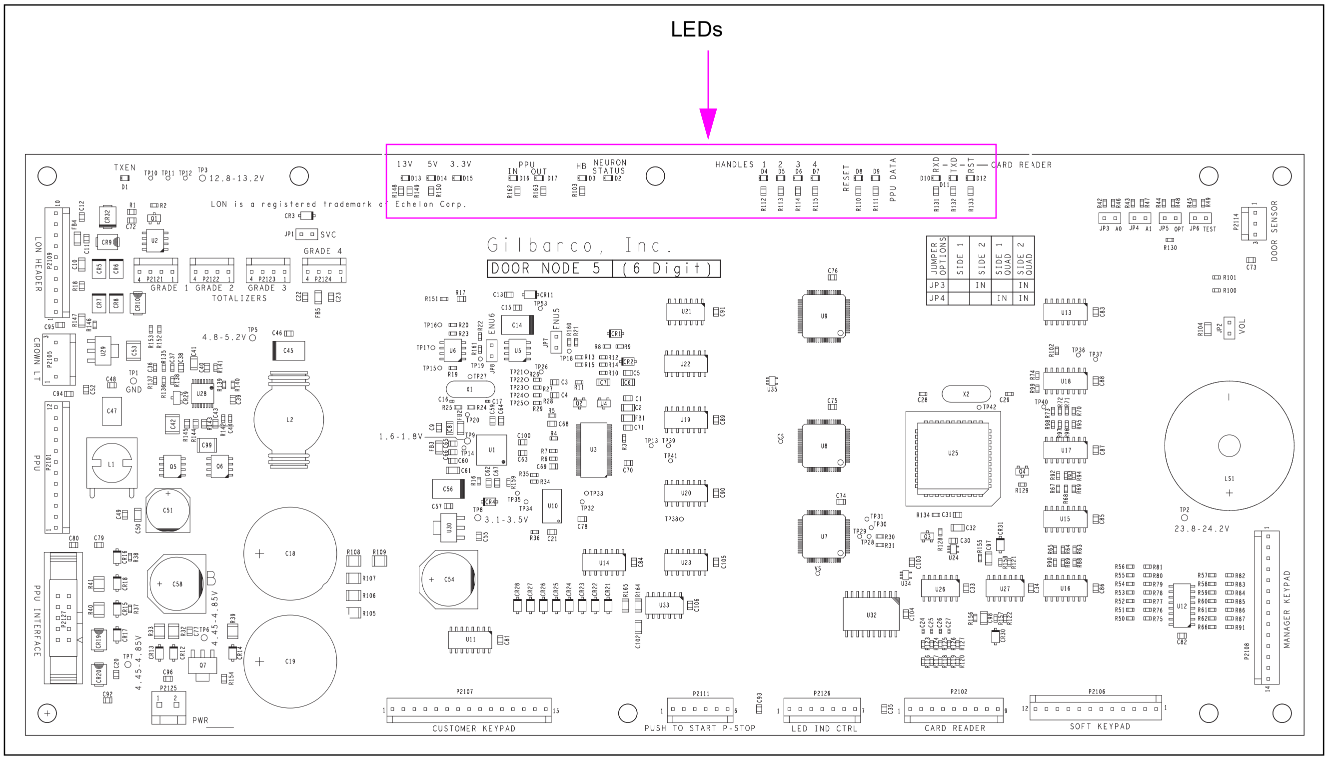
		Function	
Jumper	**Purpose**	**Installed**	**Uninstalled**
JP1	LON Service	LON Service	Normal Operation
JP2	Beeper Volume Control	Louder Beeper	Normal Beeper
JP3 (A0)	Side Control	Address - See silkscreen table on PC board	
JP4 (A1)	Side Control	Address - See silkscreen table on PC board	
JP5	PPU Option	PPU 5 is treated as PPU 6	PPU 5 is treated as PPU 5
JP6	Display Test	Display runs test mode at startup	Normal operation

If jumpering a PPU display for grades 5 or 6, a jumper must also be installed on JP-5 on the door node.

LEDs

New LEDs have been added to help during troubleshooting. The extra LEDs indicate PPU data and voltages.

Figure 6-35: Door Node-5 Address - (2)



The LEDs highlighted here include LEDs for the card reader, PPU data, reset, and handle switch.

The following table lists Door Node-5 LEDs along with their description and purpose:

LED	Description/Color	Purpose
D1	LON traffic	Off - No LON traffic On - LON BUS problem or LON node problem on BUS Flashing - Normal operation
D2	Neuron Status (Red)	On - Program locked up, bad Door Node, or lost application Flashing - Door Node in normal operation
D3	Heartbeat (Red)	On - Program locked up, bad Door Node, or lost application Flashing - Door Node in normal operation
D4	Pump Handle 1 (Red) (PPU 1)	On - Pump handle reporting from grade 1 Off - Pump handle not reporting from grade 1
D5	Pump Handle 2 (Red) (PPU 2)	On - Pump handle reporting from grade 2 Off - Pump handle not reporting from grade 2
D6	Pump Handle 3 (Red) (PPU 3)	On - Pump handle reporting from grade 3 Off - Pump handle not reporting from grade 3
D7	Pump Handle 4 (Red) (PPU 4)	On - Pump handle reporting from grade 4 Off - Pump handle not reporting from grade 4
D8	Reset Circuit (Red)	On - Power-up, power down, or board problem Off - Normal operation, or board has no power

LED	Description/Color	Purpose
D9	PPU Data (Yellow)	Solid/Fast Flashing - Normal communication to SP-III PPUs Slow Flashing - Communication problem to SP-III PPU
D10	Card Reader RX (Yellow)	On - Possible Door Node issue Flashing - Normal operation Off - No Neuron to U25 communication
D11	Card Reader TX (Red)	On - Possible Door Node issue Flashing - Normal operation Off - No Neuron to U25 communication
D12	Card Reader Reset (Red)	On - Power-up, power down, or board problem Off - Normal operation, board no power, no U25 present
D13	13 V (Green)	On - Voltage active Off - Voltage not active
D14	5 V (Green)	On - Voltage active Off - Voltage not active
D15	3.3 V (Green)	On - Voltage active Off - Voltage not active
D16	PPU In (Yellow)	Off - No legacy E500 PPU communication response Flashing/Solid - Normal operation with legacy PPU
D17	PPU Out (Yellow)	Off - No legacy E500 PPU communication response Flashing/Solid - Normal operation with legacy PPU

Test Points

Test Point	Range
TP1	GND
TP2	23.8-24.2 VDC
TP3	12.8-13.2 VDC
TP5	4.8-5.2 VDC
TP6	4.45-8.5 VDC
TP7	4.45-8.5 VDC
TP8	3.1-3.5 VDC

Software Changes

The software used on the new door node is a different version. The new version is V02.0.51. The new software is only compatible with this new door node. If you try to load the new software onto previous door nodes, the board(s) will be damaged, and will need to be replaced. Likewise, if you try to load software V01.0.47/V01.0.70 onto this door node, the board(s) will be damaged, and you will have to replace the door node.

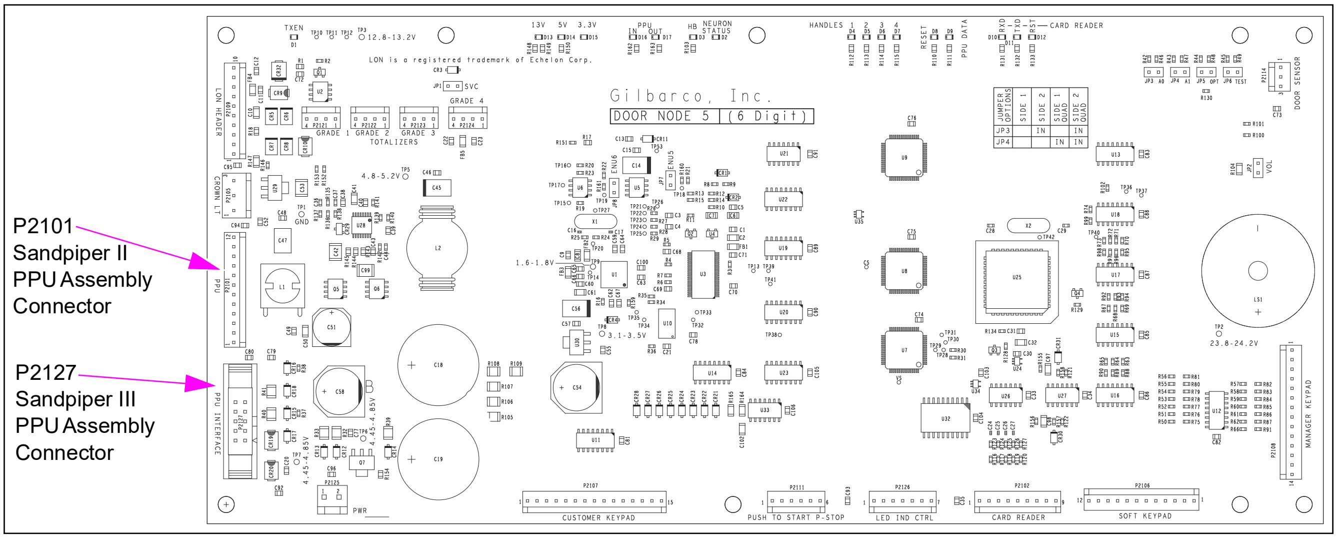
Hardware Changes

The hardware is not interchangeable. You cannot replace any previous door nodes with the new door node, and you cannot remove the current PPU assembly and install the new one in its place. The new door node and new PPU assembly must be installed together as a set, or it will not work.

Door Node-5 (M12803A003/A004)

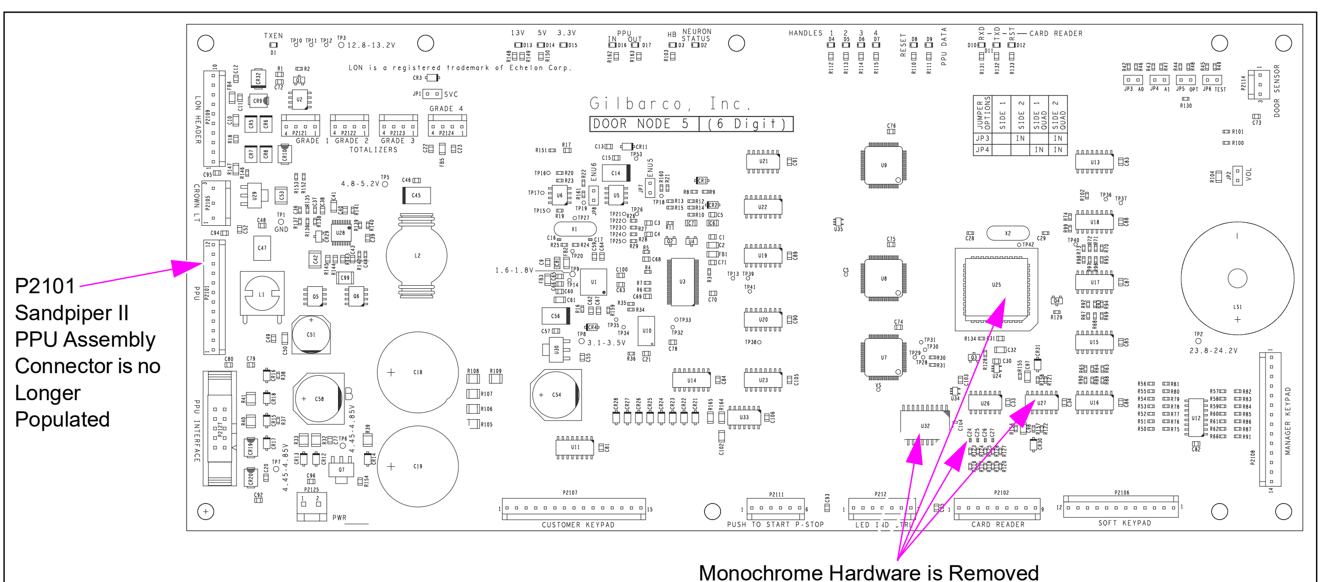
The M12803A003 Door Node-5 will replace the previous door node. If the door node needs to be replaced, order the M12803A003 Door Node-5 and it will work with any PPU assembly in the dispenser. The cable and PPU assembly do not have to be replaced. The M12803A003 Door Node-5 contains P2101 and P2127 connectors (see Figure 6-36), use whichever is needed. The LEDs and test points will remain the same, with no changes being made. For specific information on the M12803A003 Door Node-5, refer to TRP # 2384.

Figure 6-36: M12803A003 Door Node-5



The M12803A004 Door Node does not support the monochrome CRIND. It does support FlexPay™ II (E700 S) or EMV electronics. The electronics on the door node that supports the monochrome CRIND features have been removed (see Figure 6-37 on page 6-64). Therefore, the P2101 connector that supports the Sandpiper II PPU assembly is no longer populated. The LEDs and test points that supported the Sandpiper II functionality have been removed.

Figure 6-37: M12803A004 Door Node-5



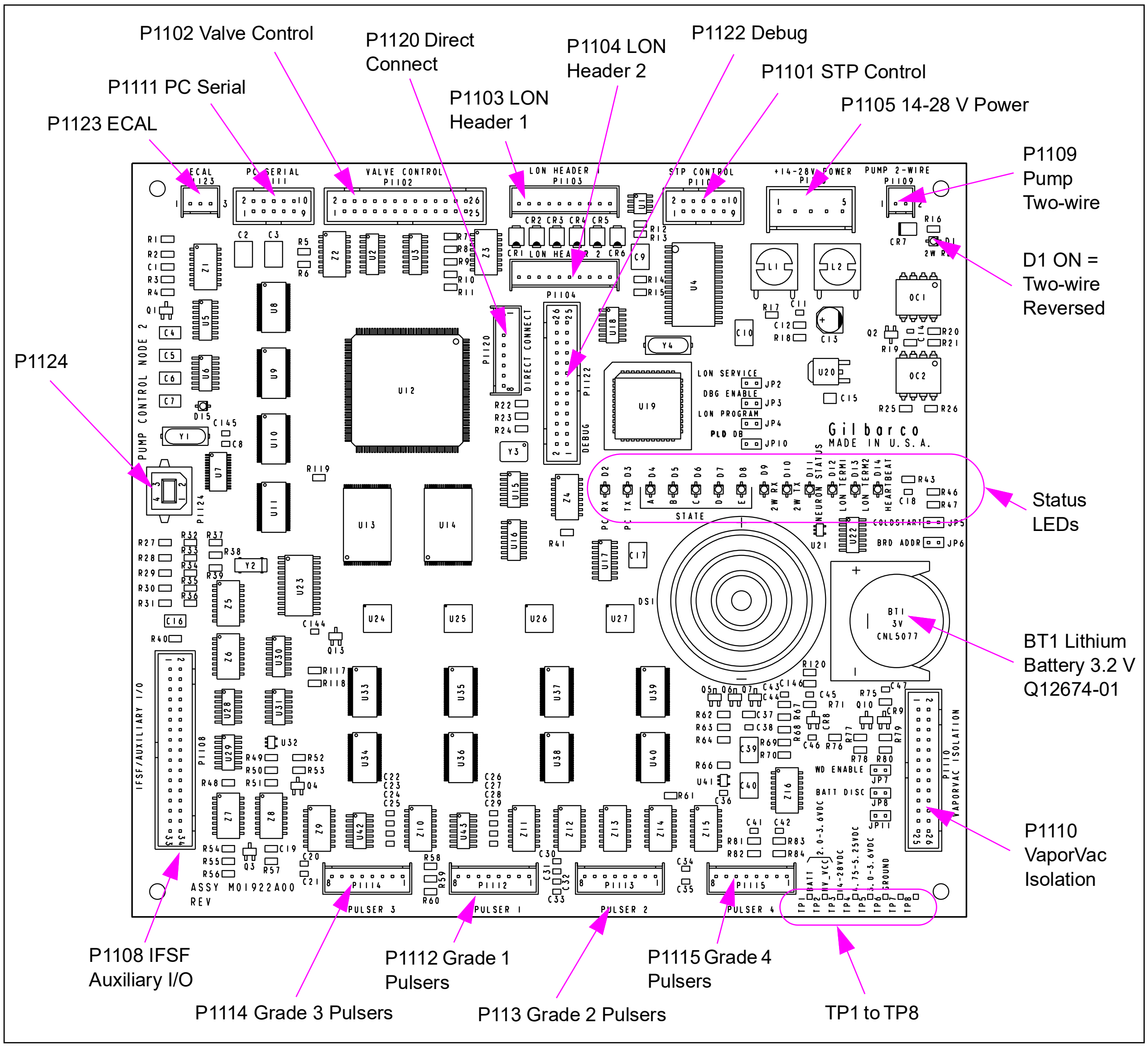
Note: P2101 Sandpiper II PPU assembly connector is not used in M12803A004 Door Node-5.

Pump Control Node-2 (M01922A001)

The PCN is used on Encore 500 and Eclipse series units to control fueling operations. It performs all totals calculations, processes handle switch information and pulser input, and stores W&M data. The PCN interfaces directly with the VaporVac controller, STP control board, and valve driver board. The PCN interfaces with the door node, CCN, and ATC node through the LON. It uses the ATC node to supply information to adjust the metered volume to a predetermined standard temperature.

The M01922A001 PCN is backward compatible with M0056A00X PCN. It requires PCN software version 1.6.00 or later. The M0056A00X node is forward compatible with M01922A001, but certain enhancements such as USB support and durability improvements will be lost. For non-e-CRIND units, the PCN provides the main processing power for system. The Lithium battery on this board maintains level 1 and level 2 programming.
Note: M01922A00X is now replaced by M12702A00X.

Figure 6-38: Pump Control Node-2 PCA



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P112-P115 (Quantity 4; 8-pin)	• M00453A001 (Encore) • M01026A001 (Eclipse)	Quad Pulser(s) M00786B001	
P1101 (10-pin)	M00635A001	STP Board m00047	P1301
P1102 (26-pin)	M00549A001	M00059A00X Steering Valve Driver PCA	P1201
P1103	M00489A001 (LON Header 1: 10-pin)	LON system, Side 1	
P1104	M00489A001 (LON Header 2: 10-pin)	LON system, Side 2	
P1105 (5-pin)	M00635A001	18V Power Supply PCA (M00050A001)	P1402
P1106 (Ethernet 1/34-pin)	M00712A001	Pump Ethernet Board (M00071A001) (not yet available)	(34)
P1108 (34-pin)	M00713A001	IFSF Interface Board	J2 (34)
P1109 (Two-wire)	M00491A001	Pump Two-wire: • To Terminal Block in Power Supply ~ OR ~ • M00089A001 CCN ~ OR ~ • MOC special CCN M00089G001 (refer to M00089A001 CCN)	
P1110 (26-pin)	M00549A002 (Encore) M01036A001 (Eclipse)	M00080 VaporVac Isolation PCB	P1405
P1111 (10-pin)	M00719A001	Serial Programming Port	
P1123 (3-pin)	M00150A001 (Encore) M00150A002 (Eclipse)	ECAL Security Switch	(3)
P1124 (Ethernet 2/16-pin)			

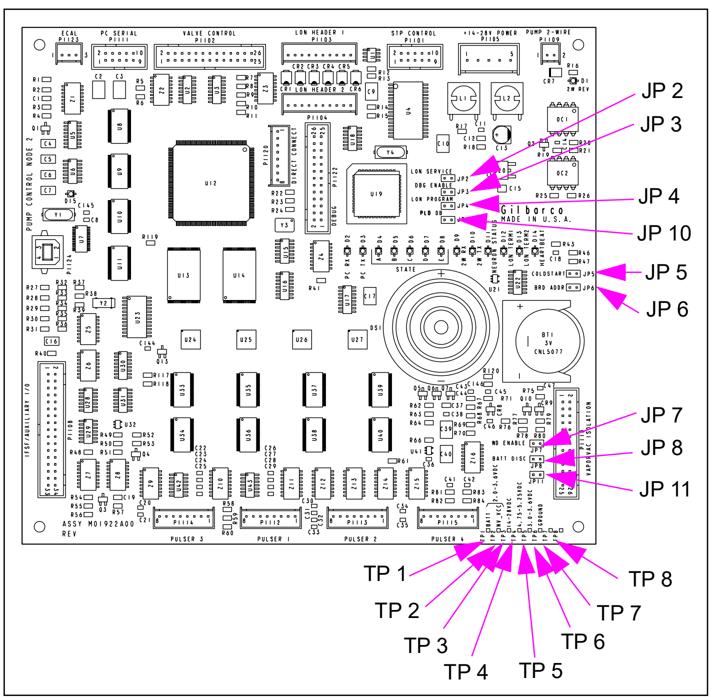
Status LEDs

LED	Name	Status Indication
D1	2W REV	On solid if two-wire loop is reversed
D2	PC RX	Blinking when receiving data from PC
D3	PC TX	Blinking when Transmitting data to PC
D4	STATE A	On solid when application is running properly
D5	STATE B	On solid when application is running properly
D6	STATE C	Blinking if PCN is unable to communicate with door nodes
D7	STATE D	On solid when handle on Side 1 is raised
D8	STATE E	On solid when handle on Side 2 is raised
D9	2W RX	Blinking when receiving data from POS or CRIND
D10	NEU 2W TX	• Off when current is present • On solid if no current is present • Blinking when transmitting to POS or CRIND

LED	Name	Status Indication
D11	NEU STAT	• Off solid is normal condition • On solid if Neuron is applicationless (node must be returned) • Blinking at a rate of “on” for 1 second, “off” for 1 second if Neuron in not configured (node must be returned) • Short flashes every 1.5 seconds indicates PCN software is not running. This is normal during PCN software upgrade, and during the first 15-20 seconds of startup.
D12	LON TERM1	• Off solid is normal condition • On solid when LON cable connected to P1103 is not terminated
D13	LON TERM2	• Off solid is normal condition • On solid when LON cable connected to P1104 is not terminated
D14	HEART BEAT	Flashes steadily at rate of “on” one half second, “off” one half second during normal operation. Flashing begins 15-20 seconds after power-up.

Test Points and Jumpers

Figure 6-39: Test Points and Jumper Locations



Test Points

Test Point	Value	Where Used
TP1	2 to 3.6 VDC	Battery Voltage
TP2	2 to 3.6 VDC	Memory Voltage
TP3	14 to 28 VDC	Lon Voltage
TP4	4.75 to 5.25 VDC	Logic Voltage
TP5	3 to 3.6 VDC	Logic Voltage
TP6	Common	For DC Voltage

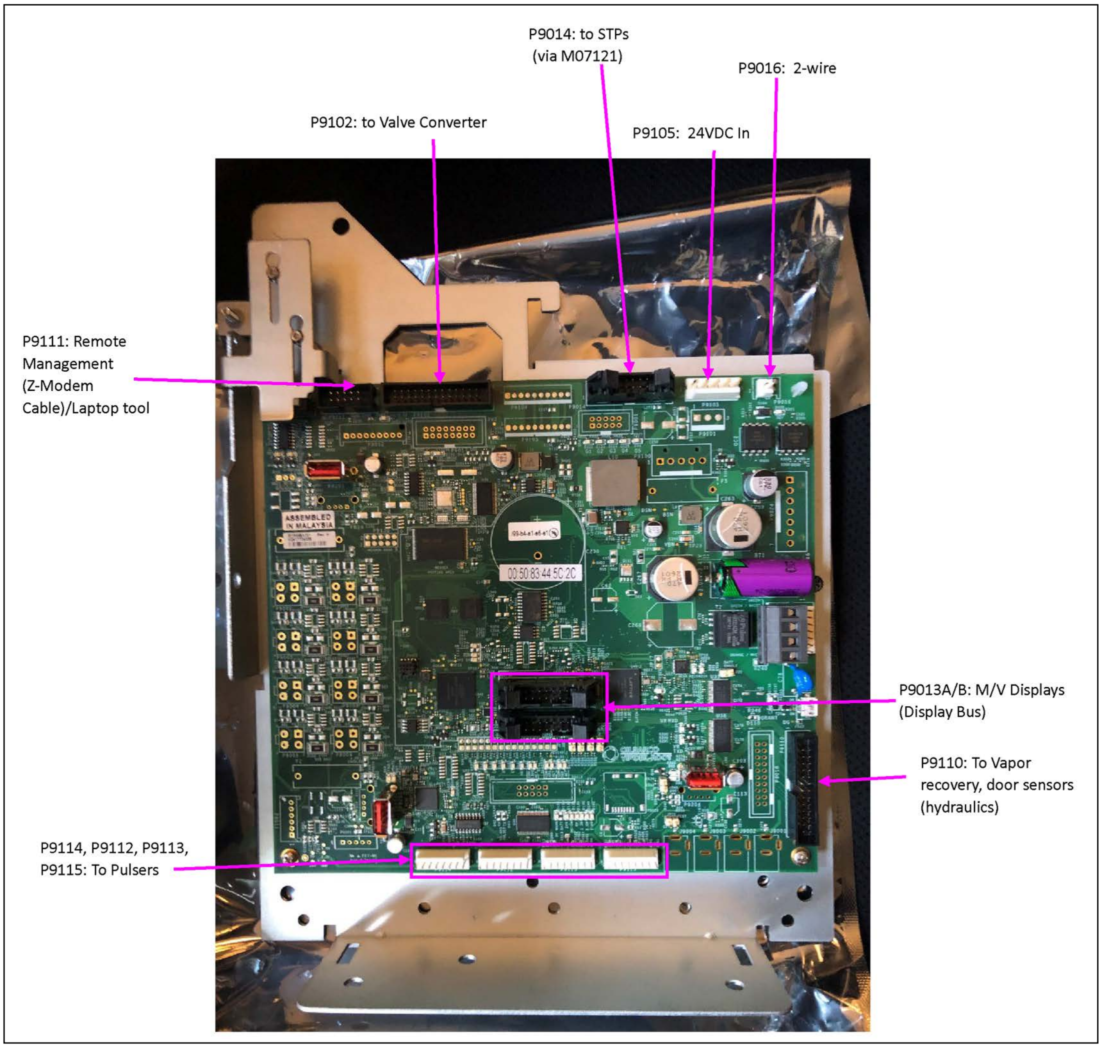
Jumpers

Jumper	Setting	Function
JP2	Out	LON service (refer to Note 2)
JP3	Out	Debug enabled (refer to Note 2)
JP4	Out	LON programming (refer to Note 1)
JP5	Out	Coldstart
JP6	In	Board address
JP7	In	WatchDog enabled
JP8	Out	Battery disconnect
JP10	Out	N/A
JP11	Out	N/A

Notes:
1. Used in manufacturing.
2. Used in board repair.

PCN5 Cable Connections and Jumpers

Figure 6-40: PCN5 Cable Connections



Jumper on PCN5

Jumper	Purpose
JP5 (COLD START)	Installed as a part of coldstart procedure, should be removed otherwise.
JP19 (TEST)	Not used
JP21 (BATT ENABLE)	Required in order for RTC to maintain time.
JP22 (A)	Not used
JP23 (B)	Not used
JP24 (C)	Not used
JP25 (D)	Installed as a part of NVData backup procedure, should be removed otherwise.

LEDs on PCN5

These LEDs are controlled by the boot loader.

LED	Purpose Boot Loader
D14 (RUN)	• ON - Normal Operation • OFF - No Board Power
D15 (SPI EN)	• ON - Normal Operation • OFF - No Board Power
D83 (PORT 1)	• OFF - Normal Operation • ON - Port Connected to Device
D85 (PORT 3)	• OFF - Normal Operation • ON - Port Connected to Device
D87 (PORT 5)	• OFF - Normal Operation • ON - Port Connected to Device
D90 (FAULT)[1] (PORT 1)	• OFF - Normal Operation • ON - Port Disconnected from Device
D92 (FAULT)[1] (PORT 3)	• OFF - Normal Operation • ON - Port Disconnected from Device
D97 (FAULT)[1] (PORT 5)	• OFF - Normal Operation • ON - Port Disconnected from Device
D159 (HW FAIL)	• OFF - Normal Operation • ON - Board is Locked out

Note: [1] After the application has loaded via USB, this LED turns ON for all ports and remains ON until power is removed.

These LEDs are controlled by the application.

LED	Purpose Pump Application
D13	OFF – FPGA Programming Done ON – FPGA Programming In Progress
D89 (SUSPEND)	OFF – Normal Operation ON – USB Hub in Suspend Mode
D102 (RXD)	Two-Wire Rx
D103 (TXD)	Two-Wire Tx
D105 (REV)	OFF – Two-Wire connected correctly ON - Two-Wire connected backwards
D108 (VR TXD)	Display Buss Tx

LED	Purpose Pump Application
D109 (VR RXD)	Display Buss Rx
D112 (TXD)	PC Serial Port Tx
D113 (RXD)	PC Serial Port Rx
D122 (G1)	OFF – STP1 OFF ON – STP1 ON
D123 (G2)	OFF – STP2 OFF ON – STP2 ON
D124 (G3)	OFF – STP3 OFF ON – STP3 ON
D125 (G4)	OFF – STP4 OFF ON – STP4 ON
D126 (VALVE ENABLE)	OFF – Valve Signals Disabled (Software/Hardware Problem) ON – Valve Signals Enabled
D127 (LED A)	OFF – Not Used
D129 (LED B)	OFF – Not Used
D131 (LED C)	OFF – Flash Checksum running (México/OIML) ON – Flash Check Completes (México/OIML)
D133 (LED D)	OFF – Side 1 Handle Down ON – Side 1 Handle Up
D135 (LED E)	OFF – Side 2 Handle Down ON – Side 2 Handle Up
D128 (PF WARN)	OFF – Normal Operation ON – 24V Going Away
D132 (24V)	OFF – Not Used
D134 (DC IN)	OFF – DC IN not Present ON – DC IN Present
D136 (5V)	OFF – 5V not Present ON – 5V Present
D138 (3.3V)	OFF – 3.3V not Present ON – 3.3V Present
D140 (1.8V)	OFF – 1.8V not Present ON – 1.8V Present
D142 (1.0V)	OFF – 1.0V not Present ON – 1.0V Present
D137 (HB)	FLASHING – Normal Operation OFF – Incorrect Operation (Software/Hardware Problem) ON – Incorrect Operation (Software/Hardware Problem)
D139 (I360 CON)	OFF – Not Used
D141 (I360 ACT)	OFF – Not Used
D143 (ATG CON)	OFF – Not Used
D144 (ATG ACT)	OFF – Not Used
D145 (IOT CON)	OFF – Not Used
D146 (IOT ACT)	OFF – Not Used
D158 SYSRESET	OFF – Not Used

LED Boot-Up Status/Activity PCN5

When power is applied it takes approximately 40 seconds for PCN5 to boot up fully. The table below describes the status and activity of the LEDs during this process when everything is correct.

LED	Status/Activity
D13	Turns ON for approximately 5 seconds (FPGA being programmed) then turns OFF.
D14 (RUN)	Both Turn and remain ON.
D15 (SPI EN)	
D134 (DC IN)	Status LEDs
D136 (5V)	Turn and remain ON.
D138 (3.3V)	
D140 (1.8V)	
D142 (1.0V)	
D126 (VALVE ENABLE)	Turns ON after approximately 5 seconds and remains ON *(FPGA has been programmed)*.
D127 (LED A)	Roughly 30 Seconds after power is applied, This group of LEDs turns ON (The pump application has started). They remain on for approximately 5 seconds after which LED A, LED B, and LED E Turn OFF. A few seconds later LED C and LED D turn OFF.
D129 (LED B)	
D131 (LED C)	

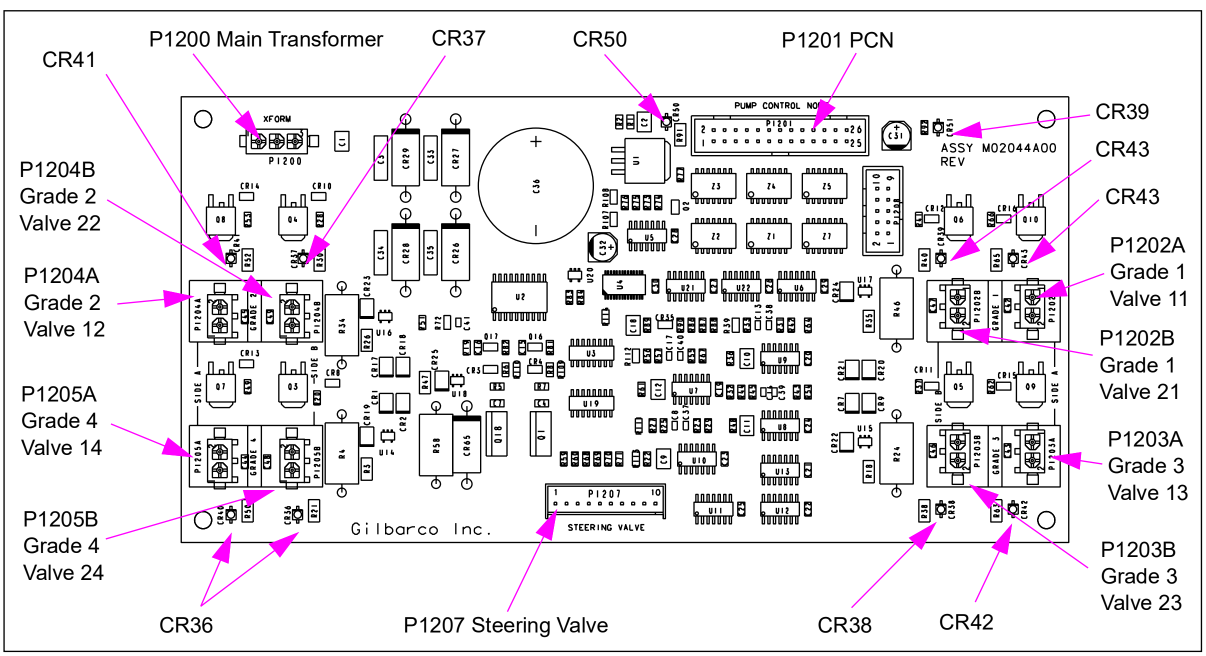
Valve Converter PCA (M02044A001)

This board is in Encore 500 and Eclipse series units with PFCV. This board is the equivalent to Encore 500 and Eclipse units with the digital valve driver board.

Service Tip

Voltage to coils typically varies from 10 to 18 VDC during operation depending on the operating conditions. Coil resistance is 40 to 45 ohms.

Figure 6-41: Valve Converter PCA (M02044A001)



Board Connections and Cables

Connector	Description	Through Cable	To Board	At Connector#
P1200	Main Transformer			
P1201 (26-pin)	PCN	M04549A001	Valve Controller	P1102
P1202A	Grade 1	M02958A001	Valve 11	Cable is part of valve assembly
P1202B			Valve 21	
P1203A	Grade 3		Valve 13	
P1203B			Valve 23	
P1204A	Grade 2		Valve 12	
P1204B			Valve 22	
P1205A	Grade 4		Valve 14	
P1205B			Valve 24	
P1207 (10-pin)	Steering Valve	M01694A001	M02097A001	P203A (10-pin)

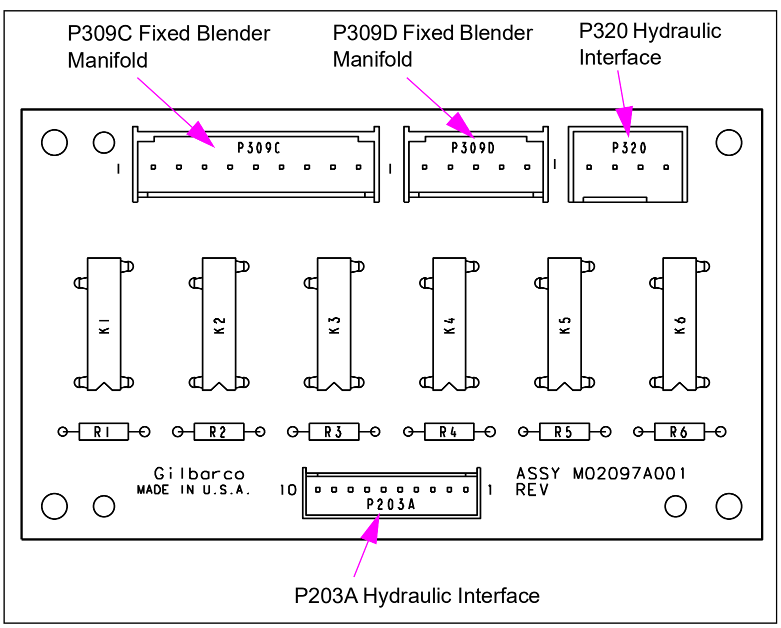
Status LEDs

LED	Status Indication
CR36 to CR43	On indicates current flow to the corresponding valve
CR50	On indicates +24 VDC valve voltage is present
CR51	On indicates +5 VDC is present

Steering Valve Relay PCA (M02097A001)

This board is on Encore 300 series units and used only with 6-hose blenders. It accepts 0 to 14 VDC levels that control solid state relays. These relays turn the steering valves on and off. M01549 and M02097 are not interchangeable.

Figure 6-42: Steering Valve Relay Board (M02097A001)



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P203A (10-pin)	M01694A002	Hydraulic Interface (M02335A001)	P306
P309C (9-pin) P309D (5-pin)	M01339A002	Fixed Blender Manifold, sides A and B (M01339A002)	P701A/P701B
P320 (4-pin)	R20593-G1	Hydraulic Interface (M02335A001)	P310

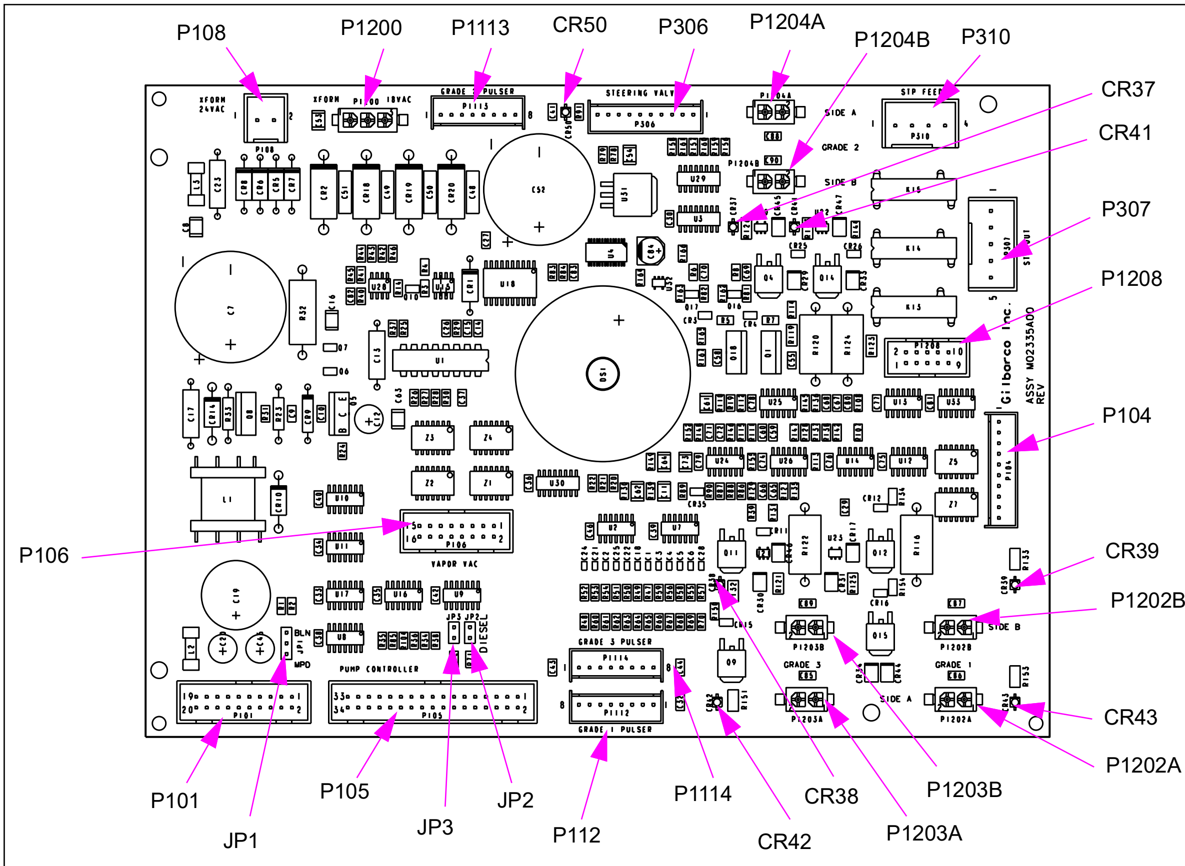
Valve Converter PCA (M02335A001)

This board is used with Encore 300 units equipped with proportional valves. Other parts to convert existing digital valve units to proportional valve units. The board is the PFCV equivalent to the digital valve hydraulic interface board. The function of this board is similar to the Advantage Hydraulic Interface PCA (T20076-G3).

Service Tip

Voltage to coils typically varies from 10 to 18 VDC during operation depending on operating conditions. Coil resistance is 40 to 45 ohms.

Figure 6-43: Valve Converter Board (M02335A001)



Note: For descriptions of callouts on this board, refer to the following tables.

Status LEDs

LED	Status Indication
CR 37	On indicates current to P1204A
CR 38 CR 39	On indicates current flow to the corresponding valve
CR 41	On indicates current to P1204B
CR 42 CR 43	On indicates current flow to the corresponding valve

Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P101 (20-pin)	R17995-G7	VaporVac/ATC option	
	M01667A001	ATC option	
P104 (10-pin)	M02971A001	Pump Controller PCA (M01598A001)	J106
P105 (34-pin)	R19667-G1	Pump Controller PCA (M01598A001)	J205
P106 (16-pin)	M01666A001	Vapor Recovery Controller (T19401-G1)	
P108 (2-pin)		Main Power Supply (M042274)	J108
P306			
P307 (5-pin)	Field wiring	J-box	
P310 (4-pin)	M01687A		J310 or P604
P1112 (8-pin)	M02651A001	Pulser -T18350-G8	P1
P1113 (8-pin)	M02651A001	Pulser -T18350-G8	P1
P1114 (8-pin)	M02651A001	Pulser -T18350-G8	P1
P1200 (3-pin)		Main Power Supply (M042274)	J108
P1202A (2-pin)	M02958A001	Valve B1	J1202A
P1202B (2-pin)	M02958A001	Valve A1	J1202B
P1203A (2-pin)	M02958A001	Valve A3 (refer to Note 1)	J1203A
P1203B (2-pin)	M02958A001	Valve B3 (refer to Note 1)	J1203B
P1204A (2-pin)	M02958A001	Valve A2 (refer to Note 2)	J1204A
P1204B (2-pin)	M02958A001	Valve B2 (refer to Note 2)	J1204B
P1208			

Notes:1. Omit for Dual and Quad.
2. Omit for Dual.

Jumpers

Jumper	Purpose/Status
JP1	For VaporVac only (blender or MPD)
JP2	Side 1, Enables or disables VaporVac for the +1 Product. In = enabled, out = disabled
JP3	Side 2, Enables or disables VaporVac for the +1 Product. In = enabled, out = disabled

CRIND Z180 Logic Board (T17764-G3/T17764-G4)

T17764 board (replaced by M03651A00X) is used in Encore series units. Each dispenser can have up to two CRIND logic boards (one per side). Side 1 communicates directly with the dispenser controller board (through the hydraulic interface board) through P262 connector, except on generic systems. Side 2 communicates to the dispenser through the side 1 logic board through P264 connector.

T17764 is a Z180 based microprocessor with input/output (I/O) ports. I/O ports connect to the card reader, keypad, display, speaker, printer controller board, and regulator board. The logic board controls CRIND two-wire communications. Jumpers set the CRIND polling address and other options. The CRIND logic board uses +5 VDC from the regulator board. The logic board transfers +68 VDC from the regulator board to the CRIND display board. The logic board has a built-in Lithium battery rated at 3.5 V and provides VRAM during loss of power. This battery is not field replaceable.

CRIND BIOS on the logic board is in ROM at U7. This firmware contains software that defines pump control, device control (I/O ports), power-up tests, communications, and diagnostics.

The site controller downloads customer-specific applications such as display prompts and ISO tables to logic board RAM through CRIND regulator board.

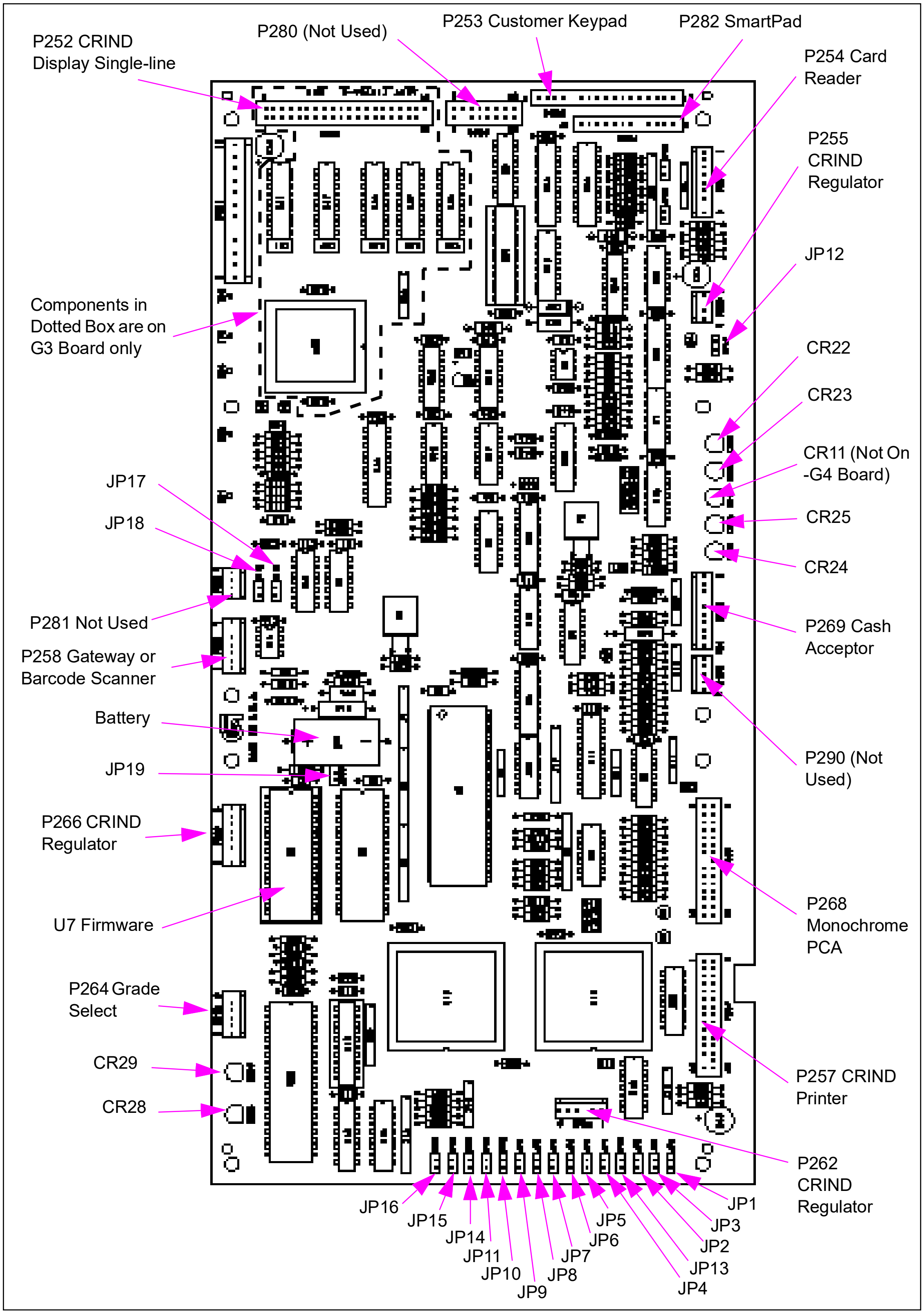
The -G4 is for InfoScreen/Monochrome only. It is a simplified version of the -G3, without the VF display interface components.

Service Tips

MOC CRINDs do not require communication with the dispenser to download. If CRIND cannot communicate to the dispenser, the CRIND will stop at pump startup after the download is complete. If MOC CRIND is not communicating with the dispenser, check the LEDs on the hydraulic interface board. During loss of communication with the dispenser, the Pump In and Pump Out LEDs on the CRIND logic boards may still flash, due to communication between the side 1 and side 2 logic boards.

T17764-G3/T17764-G4 Boards

Figure 6-44: Z180 Logic Board - G3/G4



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P251 (12-pin)	M01668A002 harness	CRIND Regulator PCA (T20306-G1)	P552 (12-pin)
P253 (17-pin)	M01555A001	Softkeys/Customer Keypad	12-pin
P254 (9-pin)		Card Reader	7-pin
P255 (4-pin)	M01668A002 harness	CRIND Regulator PCA (T20306-G1)	J255 (4-pin)
P257 (30-pin)	M01668A001 (Side 1) M01668A002 (Side 2)	Printer Controller	CN2 (30-pin)
P258 (7-pin)	M01804A001	Side 2 CRIND Logic Board TRIND Gateway Board	P258 (7-pin) J250 (14-pin)
P262 (6-pin)	M01650A001 harness	CRIND Regulator PCA [T20306-G1 (Side 1 only)]	Cable Harness
P264 (6-pin)	M01650A001 harness	Encore 300 Electronics Single PPU Board (M01522A001) Totalizer Interface PCA (M01621A001)	Cable Harness
P266 (8-pin)	M01668A001 (Side 1) M01668A002 (Side 2)	CRIND Regulator PCA (T20306-G1)	P569 (8-pin, Side 1) P570 (8-pin, B Side)
P268 (30-pin)	M01650A001	Monochrome CPU PCA (T19501-G2)	P2013 (10-pin)

Note: Connectors not shown are not used at this time.

LED Indicators

LED	Status/Indicates
CR11	On = Indicates display circuit failure, replace logic board Off = Normal operation
CR22	Blinking = Receiving data from CPIB/pump
CR23	Blinking = Sending data to CPIB/pump
CR24	Fast flash = Receiving data from cash acceptor
CR25	Fast flash = Sending data to cash acceptor
CR28*	Blinking = Receiving data from CRIND InfoScreen or Monochrome
CR29*	Blinking = Transmitting data to CRIND InfoScreen or Monochrome

Test Points

Test Points	Voltages
TP1	VRAM +3 to +3.5 VDC Power Off; 5 VDC Power On
TP2	+5 V Unregulated
TP3	+5 V
TP4	+67 to +71 V (marked 59 V)
TP5	GND

Pin to Pin Tables

P251 (Regulator)	
Test Point	**Voltage**
Pin 1	+5 V
Pin 2	+5 V
Pin 3	GND
Pin 4	GND
Pin 5	Key
Pin 6	TXD G-SITE
Pin 7	RXD G-SITE
Pin 8	N/C
Pin 9	+5 V Unregulated
Pin 10	Cut
Pin 11	+58 V Return
Pin 12	+58 V

P254 (Card Reader)	
Test Point	**Voltage**
Pin 1	+5 V
Pin 6	Card inserted: +5 VDC No card inserted: 0 VDC
Pin 7	GND

P255 (Speaker Signals)	
Test Point	**Voltage**
Pin 1	Speaker +
Pin 2	Key
Pin 3	N/C
Pin 4	Speaker –

P257 (Thermal Printer Signals)	
Test Point	**Voltage**
Pin 24	GND
Pin 25	+5 V

P258 (Thermal Printer Signals)	
Test Point	**Voltage**
Pin 2	GND
Pin 4	+5 V

P262 (Modular Two-wire A)	
Test Point	**Voltage**
Pin 1	Pump data in
Pin 2	Pump data out
Pin 3	Key
Pin 4	N/C
Pin 5	GND
Pin 6	Side A select

P268 (Spare I/O)	
Test Point	**Voltage**
Pin 6	Cash acceptor door sensor
Pin 14	GND
Pin 17	+5 VDC
Pin 18	GND
Pin 19	RXD (InfoScreen/Monochrome)
Pin 20	TXD (InfoScreen/Monochrome)
Pin 22	GND
Pin 29	InfoScreen identifier
Pin 30	Monochrome identifier

P269 (Cash Acceptor)	
Test Point	**Voltage**
Pin 1	+5 V
Pin 2	Data in
Pin 3	GND
Pin 5	Data out
Pin 6	LRC removed signal
Pin 9	+5 V
Pin 10	GND

P282 (SmartPad)	
Test Point	**Voltage**
Pin 1	+5 V
Pin 2	GND
Pin 3	Button Press
Pin 4	SmartPad RXD
Pin 6	SmartPad TXD

P290 (Voice Output Module)	
Test Point	**Voltage**
Pin 1	GND
Pin 2	VOM Data
Pin 3	+5 V
Pin 4	VOM Clock

Jumper Settings

Jumper	Setting/ Function	Setting/Function
JP1	In	In = WatchDog enable Out = WatchDog disable
JP2	Out	Card reader, Inverts Active Low Clock Signals
JP3	Out	Card reader, Inverts Active Low Data Signals
JP4-JP8	In/Out	CRIND Address* - refer to "CRIND Logic Board Address Settings" on page 6-81.
JP9	Out	Diagnostics, IN - Enters Diagnostic Mode At Power-Up
JP10	Out	Hydraulic Test, IN - Enters Hydraulic Test At Power-Up (MOC only). Refer to "FC5: Set Baud Rate" on page 5-62.
JP11	Out In - G-CAT4	Coldstart, IN - Performs Coldstart At Power-up Always IN for G-CAT 4
JP12	In Out	IN - Single-line CRIND or CRIND with InfoScreen OUT - CRIND with Monochrome LCD (OUT lowers volume)
JP13	In-Side 1 Out-Side 2	Master/Satellite Configuration (MOC only), IN - Sets CRIND As Side 1 (Master) JP13 has no affect for G-CAT4.
JP14	Out-MOC In-Generic	Selects Generic or MOC, IN - Sets CRIND for generic JP14 is not used with G-CAT4.
JP15	Out	Download selection menu
JP16	Out	Spare (MOC only)
JP17*	Out*	Spare
JP18*	Out*	Spare
JP19*	IN*	BIOS selector, IN - Generic BIOS only
JP20*	Out*	Spare
JP21*	Out*	Spare

CRIND Logic Board Address Settings

MOC and Generic CRIND and G-CAT 4 Addresses (Table 1)

Address	JP8	JP7	JP6	JP5	JP4
1	IN	OUT	OUT	OUT	OUT
2	OUT	IN	OUT	OUT	OUT
3	IN	IN	OUT	OUT	OUT
4	OUT	OUT	IN	OUT	OUT
5	IN	OUT	IN	OUT	OUT
6	OUT	IN	IN	OUT	OUT
7	IN	IN	IN	OUT	OUT
8	OUT	OUT	OUT	IN	OUT
9	IN	OUT	OUT	IN	OUT
10	OUT	IN	OUT	IN	OUT
11	IN	IN	OUT	IN	OUT
12	OUT	OUT	IN	IN	OUT
13	IN	OUT	IN	IN	OUT
14	OUT	IN	IN	IN	OUT
15	IN	IN	IN	IN	OUT
16	OUT	OUT	OUT	OUT	IN
17	IN	OUT	OUT	OUT	IN
18	OUT	IN	OUT	OUT	IN
19	IN	IN	OUT	OUT	IN
20	OUT	OUT	IN	OUT	IN
21	IN	OUT	IN	OUT	IN
22	OUT	IN	IN	OUT	IN
23	IN	IN	IN	OUT	IN
24	OUT	OUT	OUT	IN	IN
25	IN	OUT	OUT	IN	IN
26	OUT	IN	OUT	IN	IN
27	IN	IN	OUT	IN	IN
28	OUT	OUT	IN	IN	IN
29	IN	OUT	IN	IN	IN
30	OUT	IN	IN	IN	IN
31	IN	IN	IN	IN	IN
32	OUT	OUT	OUT	OUT	OUT

Generic CRIND (Table 2)

Address	JP8	JP7	JP6	JP5	JP4
1	OUT	OUT	OUT	OUT	OUT
2	IN	OUT	OUT	OUT	OUT
3	OUT	IN	OUT	OUT	OUT
4	IN	IN	OUT	OUT	OUT
5	OUT	OUT	IN	OUT	OUT
6	IN	OUT	IN	OUT	OUT
7	OUT	IN	IN	OUT	OUT
8	IN	IN	IN	OUT	OUT
9	OUT	OUT	OUT	IN	OUT
10	IN	OUT	OUT	IN	OUT
11	OUT	IN	OUT	IN	OUT
12	IN	IN	OUT	IN	OUT
13	OUT	OUT	IN	IN	OUT
14	IN	OUT	IN	IN	OUT
15	OUT	IN	IN	IN	OUT
16	IN	IN	IN	IN	OUT
17	OUT	OUT	OUT	OUT	IN
18	IN	OUT	OUT	OUT	IN
19	OUT	IN	OUT	OUT	IN
20	IN	IN	OUT	OUT	IN
21	OUT	OUT	IN	OUT	IN
22	IN	OUT	IN	OUT	IN
23	OUT	IN	IN	OUT	IN
24	IN	IN	IN	OUT	IN
25	OUT	OUT	OUT	IN	IN
26	IN	OUT	OUT	IN	IN
27	OUT	IN	OUT	IN	IN
28	IN	IN	OUT	IN	IN
29	OUT	OUT	IN	IN	IN
30	IN	OUT	IN	IN	IN
31	OUT	IN	IN	IN	IN
32	IN	IN	IN	IN	IN

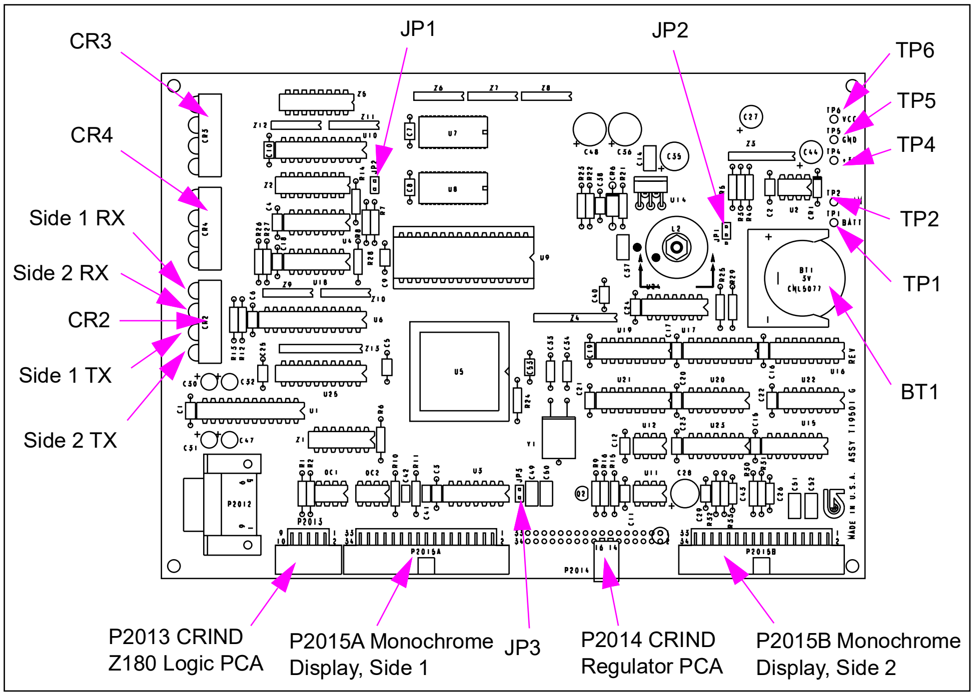
Monochrome CPU PCA (T19501-G2)

This board is used on CRIND devices with monochrome LCDs in Encore 300 series units. It stores all graphics and fonts to be displayed. Graphics and fonts are downloaded from the G-SITE system after a coldstart is performed. It then sends the graphics and fonts images to the LCD display.

Service Tip

This board functions similar to -G1 used in some early and recent Advantage Series units but has a different mounting hole pattern.

Figure 6-45: Monochrome CPU Board (T19501)



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P2013 (10-pin)	M01650A001 harness	CRIND Z180 Logic Board (T17764-G3/4)	J268A to Side 1 (30-pin) J268B to Side 2 (30-pin)
P2014 (2-pin)	M01663A001	CRIND Regulator PCA (T20306-G1)	SK2 (6-pin)
P2015A (34-pin) P2015B (34-pin)	M01668A001 (Side 1) M01668A0002 (Side 2)	Monochrome Graphic Display (Q13908-01)	J1 (34-pin)

Jumper Settings

Jumper	Position	Function
JP1	A ON B ON OFF	Resets Monochrome CPU Disables WatchDog Enables WatchDog
JP2	OFF	PPPRX
JP3	OFF	PPPTX

LED Indicators

LED	Indicates
CR2-1	Receiving data, A side
CR2-2	Receiving data, B side
CR2-3	Transmitting data, A side
CR2-4	Transmitting data, B side
CR3 CR4	EC indicators

Connectors

P2013 interfaces the monochrome. Pins 1, 2, 4, and 5 connect to P268A of CRIND logic board. Pins 3 and 6 connect to P268B.

P2013 (Monochrome Interface)

Pin	Function/Voltage
Pin 1	CRIND Power, disables monochrome communication circuits if VDC is not present.
Pin 2	CRIND A side RX
Pin 3	CRIND B side RX
Pin 4	CRIND GND
Pin 5	CRIND A side TX
Pin 6	CRIND B side TX

P2014

Pin	Function/Voltage
Pin 14	GND
Pin 16	+5 VDC supplied by the dispenser’s regulator board

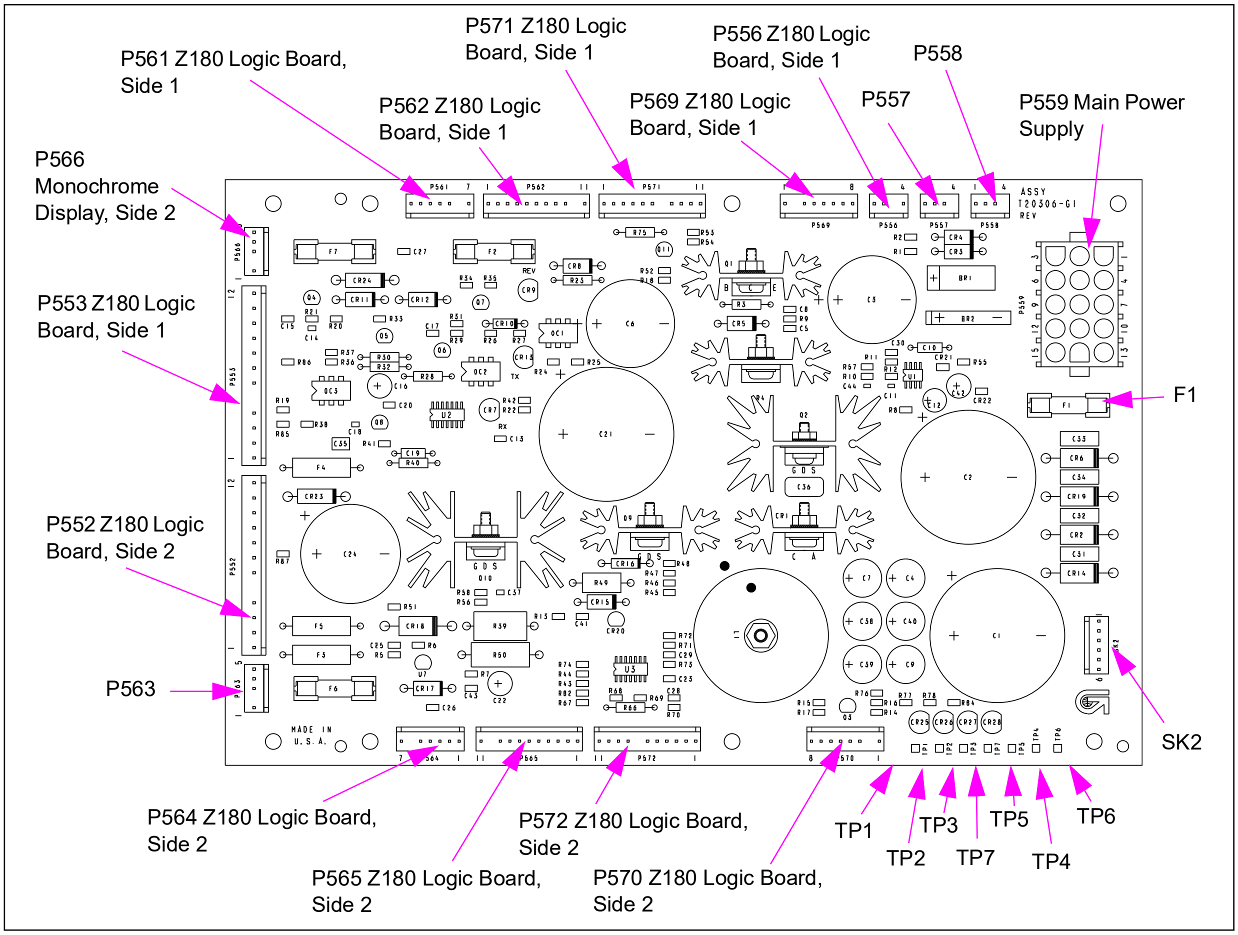
Test Points

Test Point	Voltage
TP1 (BATT)	2.8 VDC or later, with power ON or OFF. Power OFF to the monochrome CPU requires dispenser power to be removed and dispenser battery to be disabled.
TP2 (VRAM)	4.5 VDC with power ON 2.2 to 3 VDC with power OFF
TP4	+35 V
TP5 (GND)	GND
TP6 (VCC)	+5 VDC

CRIND Regulator PCA (T20306-G1)

The CRIND regulator supplies regulated voltages for CRIND devices, and receives and transmits communication signals from and to the POS system.

Figure 6-46: CRIND Regulator Board (T20306-G1)



Status LEDs

LED	Color	Function
CR7	Red	Two-wire - Receive Communication
CR9	Red	Two-wire - Reversed
CR13	Red	Two-wire - Transmit Communication
CR25	Red	5 V Present
CR26	Red	59 V Present
CR27	Red	5 V Present
CR28	Red	24 V Present

Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P552 (12-pin)	M01668A002 harness	CRIND Z180 Logic Board Side 2 (T17764-G3/4)	P251 (12-pin)
P553 (12-pin)	M01668A001 harness	CRIND Z180 Logic Board Side 1 (T17764-G3/4)	
P556 (4-pin)	M01650A001 harness	CRIND Z180 Logic Board Side 1 (T17764-G3/4) M01664A001 harness to field wiring	J262 (6-pin) P412 (4-pin)
P557 (4-pin)			
P558 (4-pin)			
P559 (15-pin)	M01662A001	Main Power Supply (M01608A00)	J606 (6-pin)
P561 (7-pin)	M01668A001 harness	CRIND Z180 Logic Board Side 1 (T17764-G3/4)	harness
P562 (11-pin)			
P563 (5-pin)		Monochrome Display [Q13908-01 (Side 1)]	J5 (2-pin)
P564 (7-pin)	M01668A002 harness	CRIND Z180 Logic Board Side 2 (T17764-G3/4)	harness
P565 (11-pin)			
P566 (5-pin)		Monochrome Display [Q13908-01 (Side 2)]	J5 (2-pin)
P569 (8-pin)	M01668A001 harness	CRIND Z180 Logic Board Side 1 (T17764-G3/4)	P266 (8-pin)
P570 (8-pin)	M01668A002 harness	CRIND Z180 Logic Board Side 2 (T17764-G3/4)	
P571 (11-pin)	M01668A001 harness	CRIND Z180 Logic Board Side 1 (T17764-G3/4)	harness
P572 (11-pin)	M01668A002 harness	CRIND Z180 Logic Board Side 2 (T17764-G3/4)	
SK2 (6-pin)	M01663A001	Monochrome CPU PCA (T19501-G2)	P2014 (2-pin)

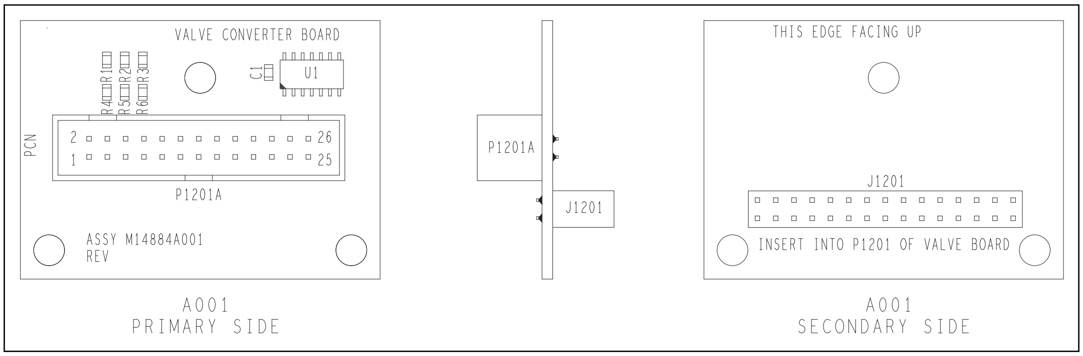
Test Points

Test Point	Voltage
TP1	+5 V (BATT)
TP2	+59 V (Single CRIND Display Voltage)
TP3	+5 V
TP4	+5 V (Unregulated)
TP5	2.5 V
TP6	DC GND
TP7	+24 V

Valve Converter Interface PCA (M14884A001)

The valve converter interface PCA inverts signals from the PCN to the second valve converter PCA so grade 3 can be blended.

Figure 6-47: Valve Converter Interface PCA Board (M14884A001)



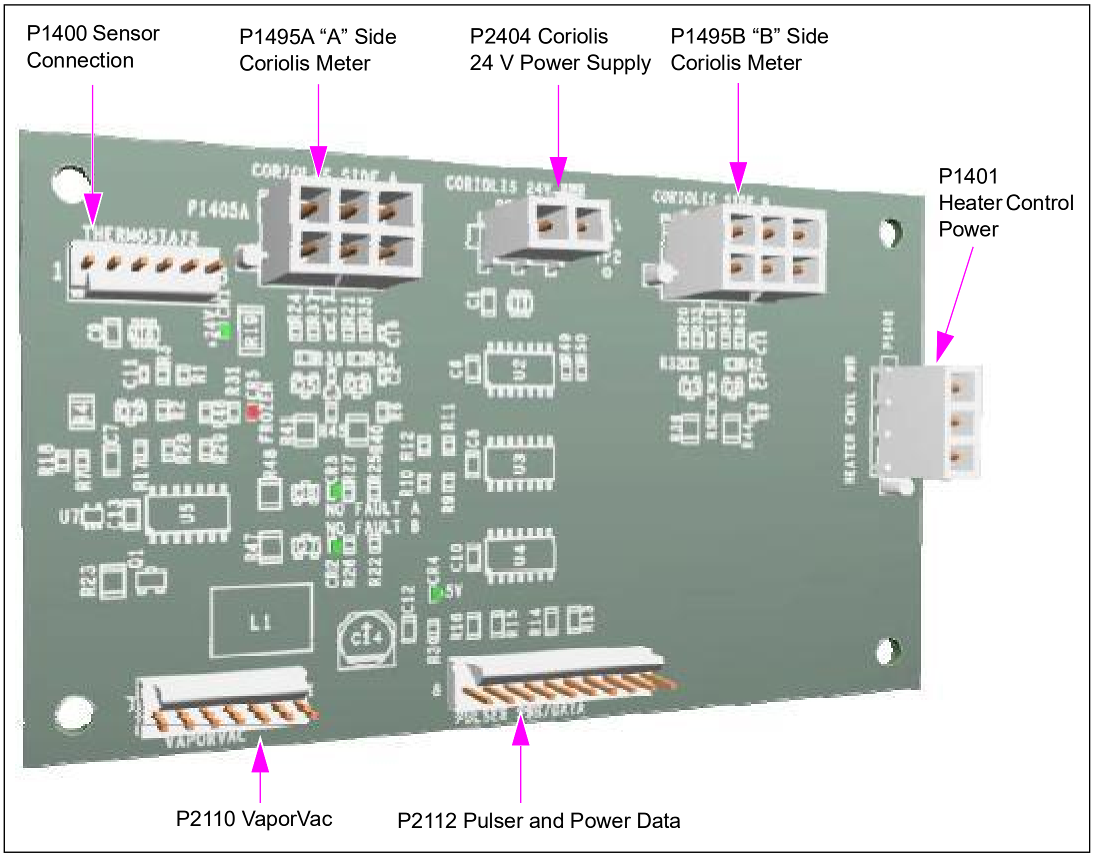
Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
J1201, J1201A	M05547A009	Valve Converter Interface PCA (M14884A001)	J1201

Coriolis Meter Interface Board

The Coriolis meter interface board interfaces the Coriolis meter and pump node, as well as to the Freeze Sensor. The meter receives a 24 V input from the dispenser electronics and supplies 5 VDC logic power to the temperature control board. The meter can handle up to two Coriolis meter inputs.

Figure 6-48: Coriolis Meter Interface Board

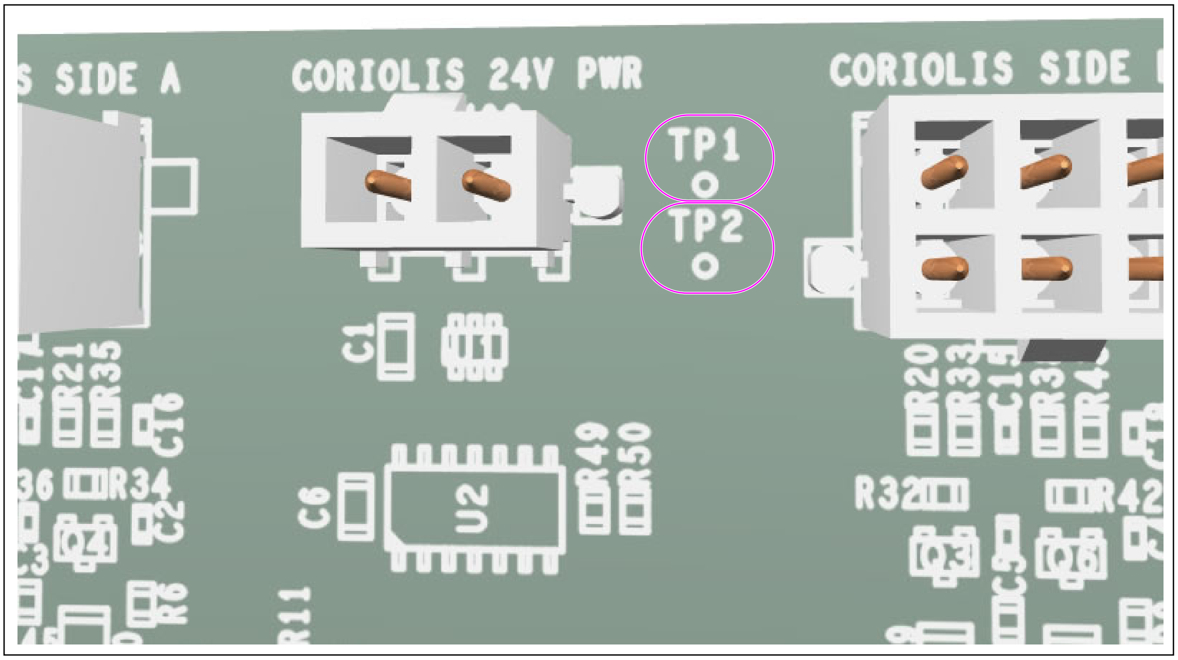


The Coriolis meter contains electronics within the meter that monitor the function of the meter and create pulses in proportion to the volume of flow through the meter. A Coriolis meter interface board provides power to the meter as well as accepts pulses and diagnostic data from the meter that are processed and directed to the standard PCN. Additional functions of the meter board are to provide power to the heater control board and to accept input from the pump freeze sensor.

Service Tips

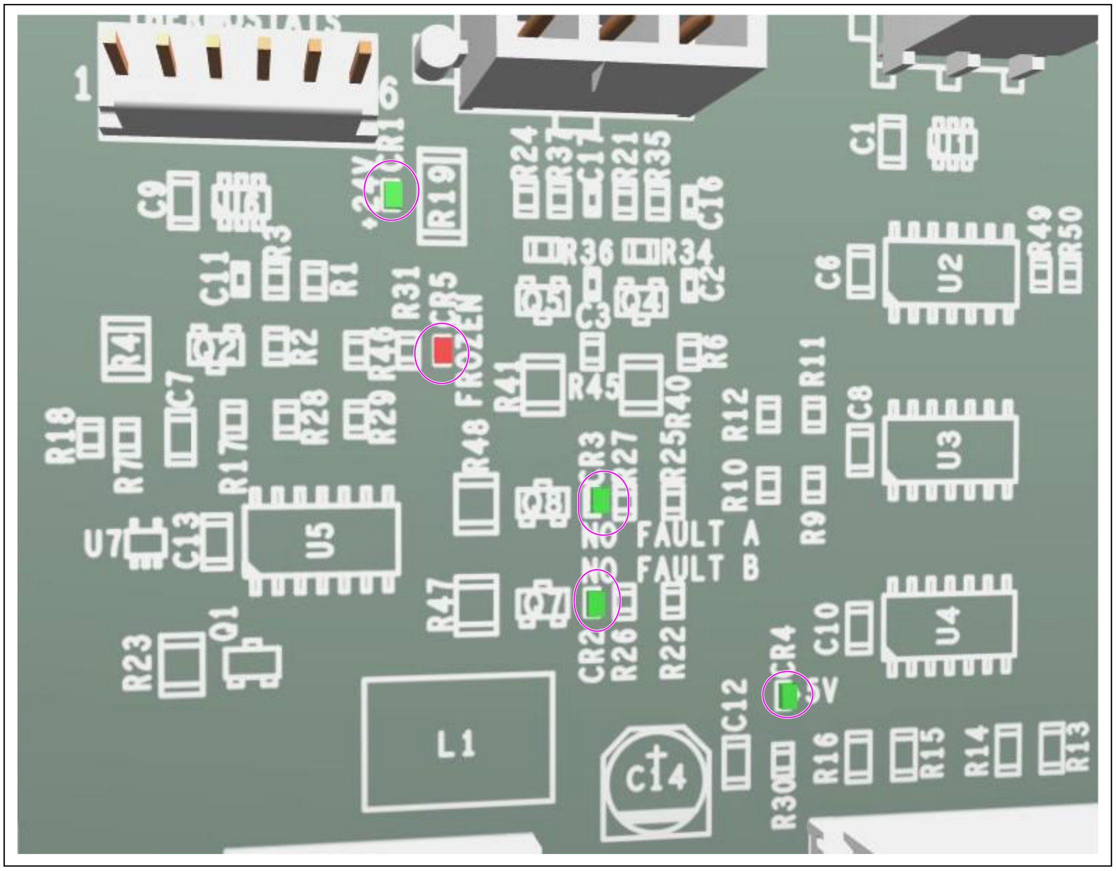
- TP1 and TP2 on the board can be used to verify if the 24 VDC power input is correct (see Figure 6-49).

Figure 6-49: TP1 and TP2 on the Coriolis Meter Interface Board



- Although the board supplies logic power to the temperature control board, it does not monitor the board's performance or state. If the temperature control board is inoperative, check the Coriolis meter interface board and cables between the boards for transference of 5 VDC.
- Pulser-related ECs can be caused by this board in addition to the cables to and from the board, as well as the Coriolis meter or occasionally the PCN.
- Side A and side B Coriolis meter inputs have the same jack, so ensure that you connect the correct meter to the jack on the board. Single-sided dispensers use side A (or side B).
- Several diagnostic LEDs exist on the board. They provide valuable troubleshooting information. Check the LEDs before replacing a board as the fault may not be with the board.

Figure 6-50: Diagnostic LEDs on the Coriolis Meter Interface Board



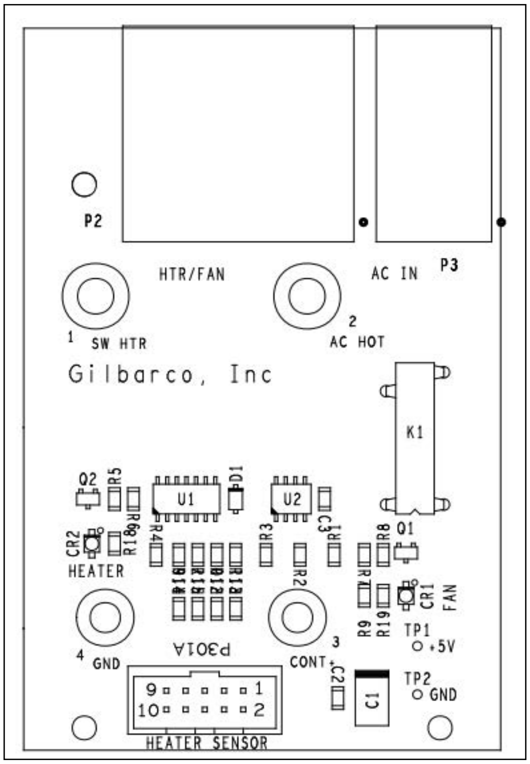
The functions of Coriolis Interface Printed Circuit Assembly (PCA) are as follows:

- Provides +24 VDC power to the Coriolis meter.
- Provides +5 VDC power to the Heater Interface PCA (M14100A001).
- Provides an interface to the pump freeze sensor and sends signals to the PCN if the unit is frozen.
- Provides Coriolis pulser data to the PCN.
- Provides the following diagnostic data:
    - If CR1 LED is on, it indicates that +24 VDC power supply is enabled.
    - If CR2 LED is on, it indicates that the Coriolis meter for side B is enabled.
    - If CR3 LED is on, it indicates that the Coriolis meter for side A is enabled.
    - If CR4 LED is on, it indicates that +5 VDC power supply is enabled.
    - If CR5 LED is on, it indicates that the pump is frozen.

Temperature Control Board

The E2 Temperature Control PCA(M19963A001) is used in full cabinet heater. DEF does not use this board. It uses the DEF Temperature Control PCA (M14100A001).

Figure 6-51: Temperature Control Board



Functions of heater control interface PCA are as follows:
- Provides 120 VAC to fan and heater.
- Monitors the heater sensor and cabinet temperature to control heater and fan.
- Provides the following diagnostic data:
  - If CR1 LED is on, it indicates that the fan is on.
  - If CR2 LED is on, it indicates that the heater is on.

For additional information, refer to "DEF Heating System" on page 8-29.

Power Supplies

The following table lists the contents in this sub-section:

For VaporVac power supplies, refer to "VaporVac Electrical Components" on page 6-148.

About Power Supplies

The function of the power supply is to provide clean voltage at the correct values required by the electronics.

Replacing Fuses

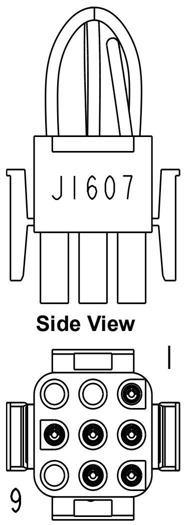
CAUTION

Using an incorrectly sized fuse may cause equipment damage. To ensure good equipment protection and maintain safe operation, always use the correct replacement fuse.

Voltage Select Plug (M00488AXXX)

The Encore 300/500 and Eclipse main power supplies use a voltage select plug to determine power supply AC voltage input. Voltage select plugs are not included and must be ordered separately. Voltage select plugs may also be reused from other power supplies or terminal block PCAs, as long as they are configured correctly. Use the following table to determine the correct voltage select plug.

Figure 6-52: Voltage Select Plug (M00488AXXX)



Voltage/Plug Designation

For This Voltage	Use This Plug
115	M00488A115
220	M00488A220
230	M00488A230

Jumper Configuration

This Plug	With This Voltage Configuration	This Terminal	Connects to This Terminal
M00488A115	115	1	3
		5	6
		8	2
M00488A220	220	1	7
		5	2
M00488A230	230	1	4
		5	2

18 V Power Supply PCA (M00050A001)

18 V Power Supply PCA supplies 18 VDC power to LON system on Encore 500 and Eclipse series units, and has LEDs for “Power” and “Fault”. It is used in M00458A002 (now obsolete) power supply assembly. In addition to rectifying the transformer secondary, it has a connection for a large filter capacitor, and has over and under-voltage detection. It also provides an advance power fail detection, has over-current sensing, and auto power-down timing. The 18 V Power Supply PCA is capable of shutting down electronics. This board along with the M00458A002 Assembly has been replaced by M02274A001. These boards are not interchangeable.

Troubleshooting Power Supply

The following procedure must be used for diagnosing 18 V Power Supply Problems. M00050 18 V Power Supply Board Revision “G” and later incorporate a very quick protective circuit to prevent the transformer fuse from blowing. With “G” boards, cycling power will allow the system to supply power again providing that an overcurrent condition does not occur.

1. With power on, measure AC voltage from the transformer to the M00050 Board. Voltage must read 12 to 25 VAC if the system is good.

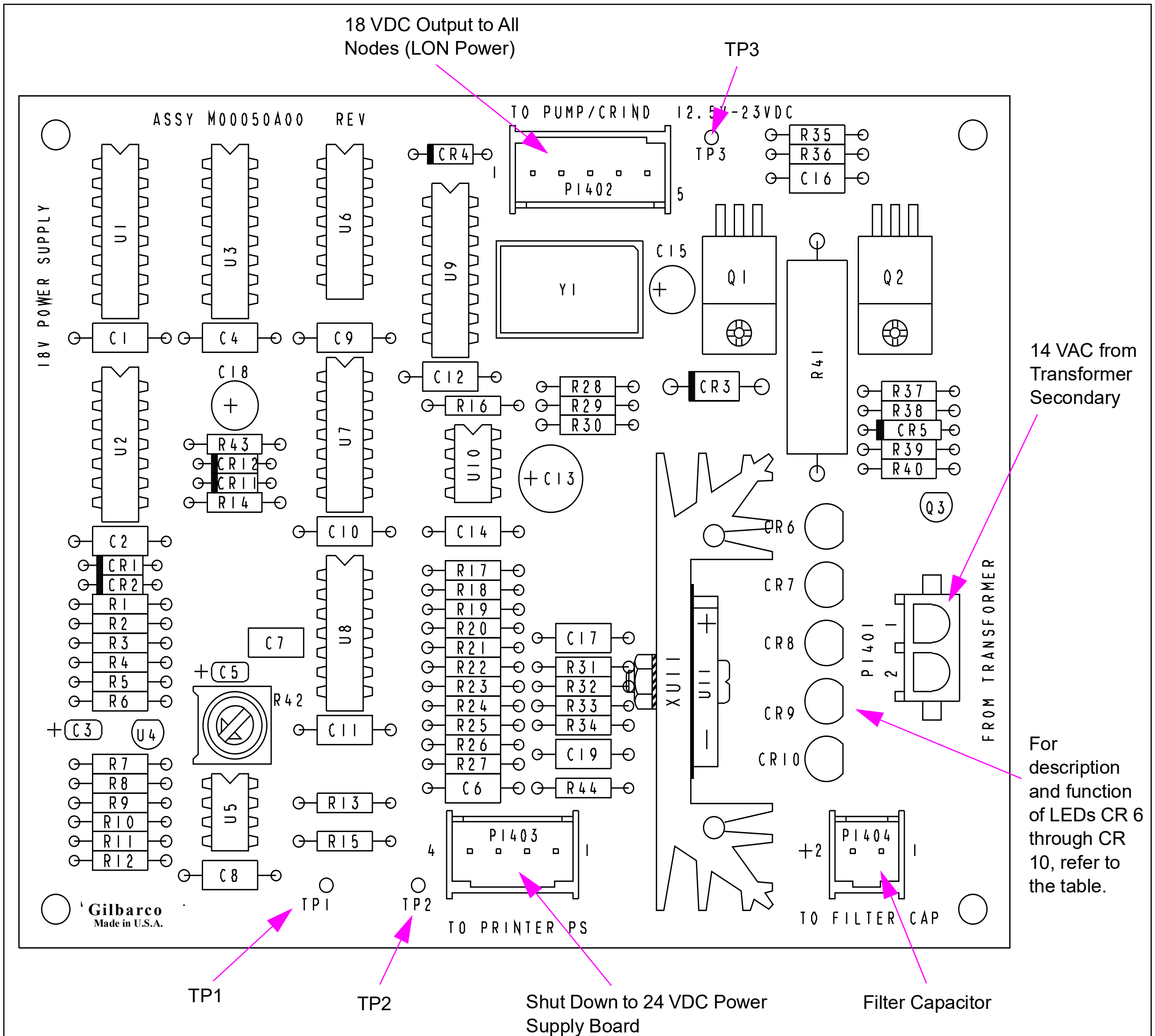
2. If the system has a fault (very low voltage), you have to check the transformer and perform the following steps.
   - With power off, you can remove the transformer connector to the M00050 Supply Board and check resistance across the wires from the transformer. Resistance must be very low, 5 ohms or less, if good.
   - If very high, the transformer is bad (open). Do not replace any transformer until you find the reason for its failure, or the new transformer may also fail.

3. With power off, at the M00050 Board, remove the 5 pin cable to the pump node.
   - Measure ohms across pins 1 and 5. If ok, you will see 800 ohms or more, depending on the number of options in the system. If bad, resistance will be very low and there is likely a fault with one of the nodes/boards using 18 V.
   - To find the bad node or board, you have to temporarily disconnect the node/board from the 18 V supply and determine if the ohm reading returns to 800 or more.
   - Although unlikely to occur, if no bad node or board is found, a cable problem can exist. Use a test cable to temporarily replace each cable in the chain, measuring ohms, until the defective cable is found.

4. If you find any faulty node, board or cable, replace it, reconnect all cables properly and then check for a proper ohm reading again across pins 1 and 5.

5. Replace the transformer if it was found defective.

Figure 6-53: 18V Power Supply PCA



Note: DC voltage measurements across these test points will indicate 1/2 the actual current (1.5 VDC = 3 A). 4 A is maximum. Readings under 1 A are unreliable.

Status LEDs

LED	Color	Function
CR 6	Green	Power IN
CR 7	Green	Power OUT
CR 8	Red	Power supply shut down
CR 9	Red	Fault
CR 10	Red	Overvoltage

Test Points

Test Point	Range
TP 1	System current draw. Refer to Note.
TP 2	DC GND
TP 3	12.5 V to 28 VDC

Note: Calculate system current draw as follows:
Current = 2X indicated voltage. Circuit cannot indicate (provide reading) lower than 0.5 V

Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P1401 (2-pin)	White Wire	14 V Secondary on Power Supply	N/A
P1402 (5-pin)	M00635A001	+18 V to M00077A001 Totalizer Node	P1105
P1403 (4-pin)	M00486A001	Power Supply Board 24 VDC (M00053A002) (Printer Option)	P3501
P1404 (2-pin)	M00484A001	Filter Capacitor	(2-pin)

Connector PIN Functions and Voltages

P1401

Pin #	Description	Voltage Range
1	Power In	12-24 VAC as measured to P1401-2
2	Power In	12-24 VAC as measured to P1401-1

P1402

Pin #	Description	Voltage Range
1	System Ground Reference (to System DC Ground)	0 V
2	System Ground Reference (to System DC Ground)	0 V
3	Power (Power) Fail Output	0.4 V = Power Fail 4 V = Power Good (Measured through external 10K pull-up)
4	System DC Power	22 V DC nominal, 17.5 VDC to 28 VDC operational
5	System DC Power	22 V DC nominal, 17.5 VDC to 28 VDC operational

P1403

Pin #	Description	Voltage Range
1	Supply Shutdown	0.4 V = Shutdown 4.0 V = Active
2	System Ground Reference	0 V
3, 4	Not Used	

P1404

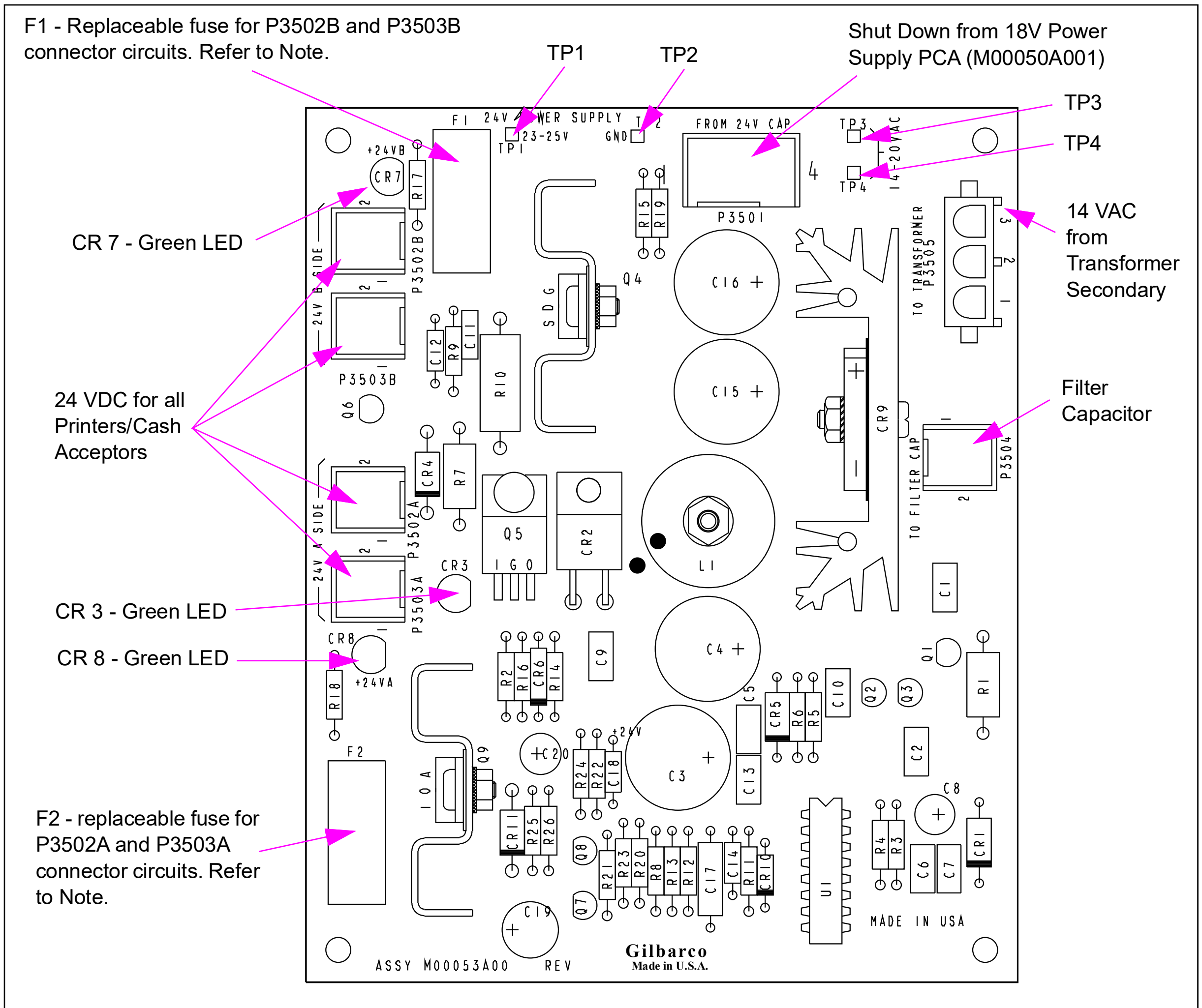
Pin #	Description	Voltage Range
1	System Ground Reference	0 V
2	18 V Filter Capacitor	18 VDC nominal, 15 VDC to 28 VDC operational

Note: Pins 3 and 4 of P1403 are presently not used. In future, +24 V board (M00053) may be powered through these pins, instead of through a dedicated transformer secondary.

24V Power Supply PCA (M00050A002)

On Encore 500 and Eclipse series units, there are two 24 VDC power circuits on this board capable of supplying power to printers, barcode scanners, and/or cash acceptors. LEDs are present on the board to indicate output for each side of this board. Board will shut down with a power fault from the 18V Power Supply PCA (M00050A001). This board has been obsoleted by the Power Supply (M02274A001). These parts are not interchangeable.

Figure 6-54: Power Supply Board (M00053A002)



Note: Replaceable Fuses are Gilbarco Part # Q12682-07; 2.5 A Slow Blow.

Status LEDs

LED	Status Indication
CR3 GREEN	ON = good 24 VDC available. Refer to Note.
CR7 GREEN	ON = good 24 VDC output to P3502B and P3503B Connectors for side 2 (Printers and/or cash acceptors). Refer to Note.
CR8 GREEN	ON = good 24 VDC output to connectors P3502A and P3503A for side 1 (Printers and/or cash acceptors). Refer to Note.

Note: If CR3 is ON and CR7 and CR8 are both OFF, one of the following two conditions exist:
- *Power Supply shutdown [check CR8 on 18V Power Supply PCA (M00050A001)]*
  ~OR~
- *Both replaceable fuses are blown (unlikely)*

Test Points

Test Point	Range
TP1	22.5 V to 25.5 VDC
TP2	GND
Reading across TP3 and TP4	14 V to 25 VAC

Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P3501 (4-pin)	M00486A001	18V Power Supply PCA (M00050A001)	P1403 (4-pin)
P3502A (2-pin)	1. M00517A001 (Encore) or M00517A006 (Eclipse) 2. M00822B002	1. Printer Controller Side 1 2. Monochrome Fan	2-pin
P3502B (2-pin)	1. M00517A001 (Encore) or M00517A006 (Eclipse) 2. M00822B002	1. Printer Controller Side 2 2. Monochrome Fan	2-pin
P3503A (2-pin)*	M00721A001	Side 1 M00122A001 LON to Serial Node (LON Gateway)	P1512 (2-pin)
P3503B (2-pin)*	M00721A002	Side 2 M00122A001 LON to Serial Node (LON Gateway)	P1512 (2-pin)
P3504 (2-pin)	M00484A001	Filter Capacitor	2-pin
P3505 (3-pin)	Blue wire	14 V secondary on Power Supply	

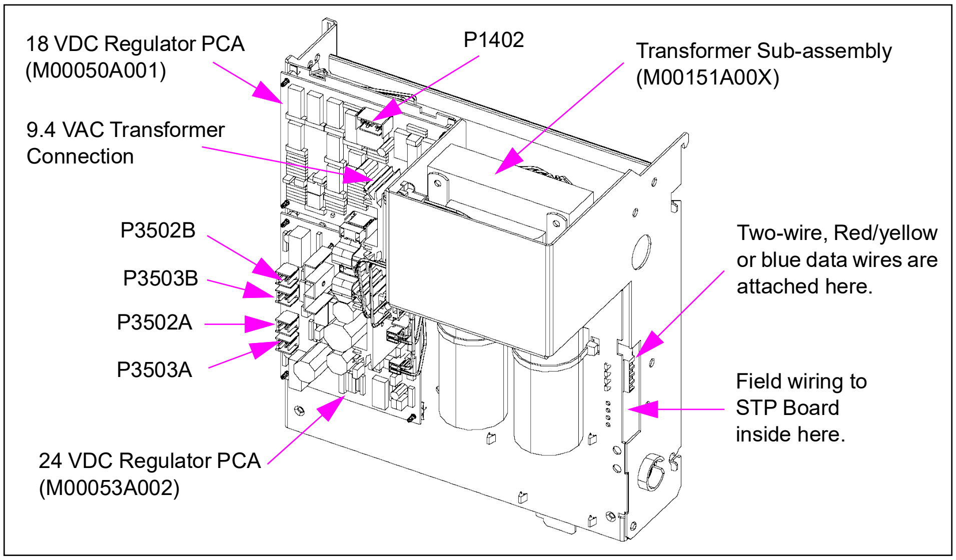
**Optional*

Encore 500 and Eclipse Power Supply (M00458A002)

Power Supply Assembly (M00458A002)

This power supply is no longer available and has been replaced by M02274. It is shown here for reference only. For replacement instructions, refer to *MDE-4133 Power Supply Kit M00016K003 Installation Manual*.

Figure 6-55: Encore Power Supply (M00458A00X)



Connections and Cables (Power Supply Connections)

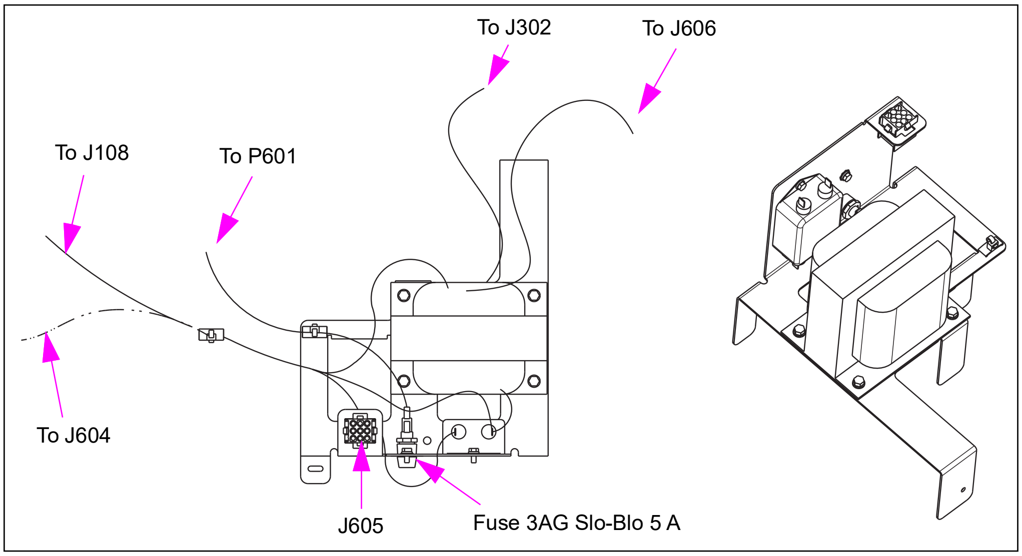
Connector#	Through Cable	To Board	At Connector#
P1402	M00635A003	PCN (M01922)	P1105
P3502A P3502B	M00715A005	Printer (M00317A001)	CN1
P3503A P3503B	Not used in this configuration		

Encore 300 and Ultra-Hi Power Supply (M01608A001)

Main Power Supply Assembly (M01608A001)

This power supply is obsolete and is replaced by M02274.

Figure 6-56: Power Supply for Encore 300



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P601	M01591A001	A2, A10, A13 field connection wires.	J401 (3-pin)
J604		Valve and STP feed	P604
J108		24 VAC Feed	P108
J302	M1650A001 to M01515A001	9.4 VAC Backlight voltage	J2115
J606	M01662A001	CRIND Regulator Board (T20306-G1)	P559

Universal Power Supply (M02274A001) for Encore 300/500 and Eclipse

The main assembly provides power to Encore 300/500 and Eclipse. This assembly was included in production units beginning December 5, 2002. Encore 500 and Eclipse units have an additional board attached to the main assembly to provide 18 V and 24 V. An STP board is also included in units that are dispensers.

Service Tip

This power supply can substitute for older power supplies using a M00016K003 Kit.

Figure 6-57: Example of Encore 300 Configuration

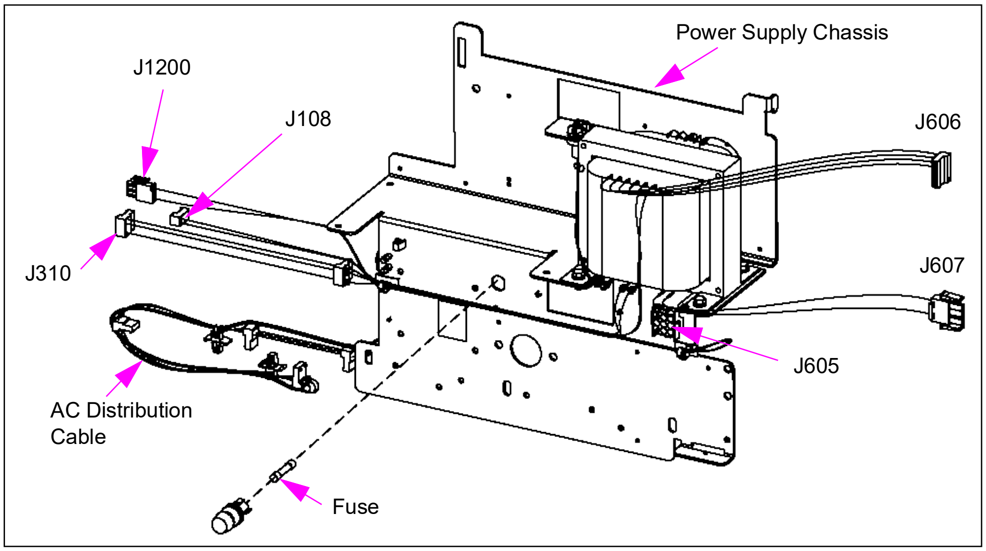


Figure 6-58: Example of Encore 500 and Eclipse Configuration

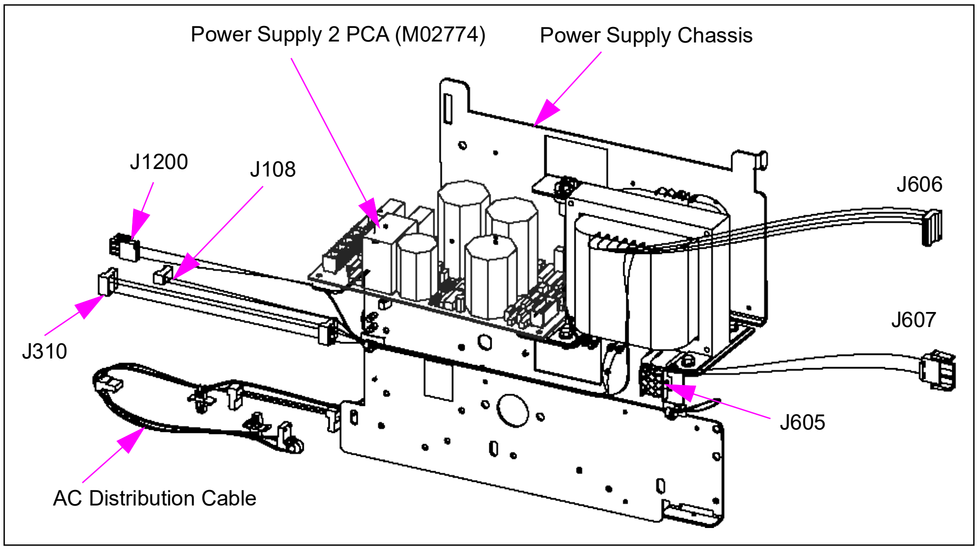
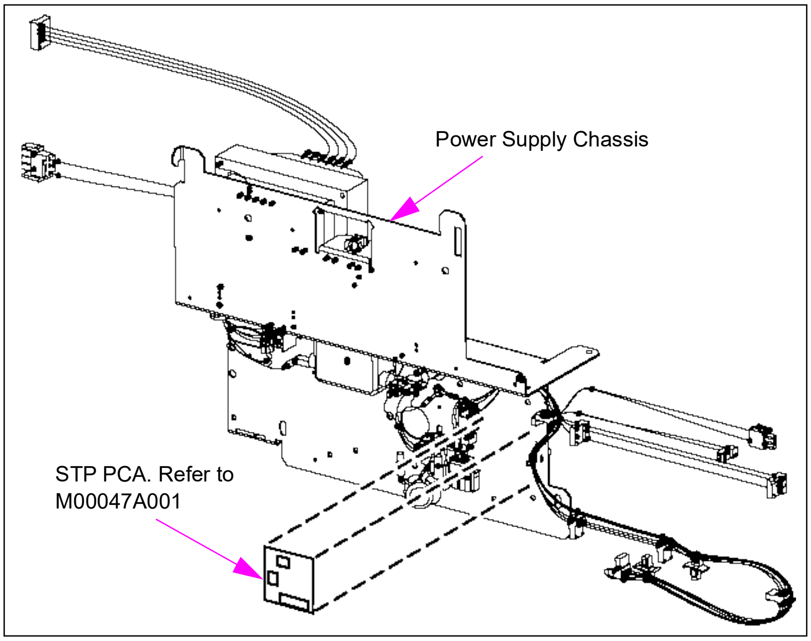


Figure 6-59: Universal Power Supply (Rear) - STP Board Location



Board Connections and Cables

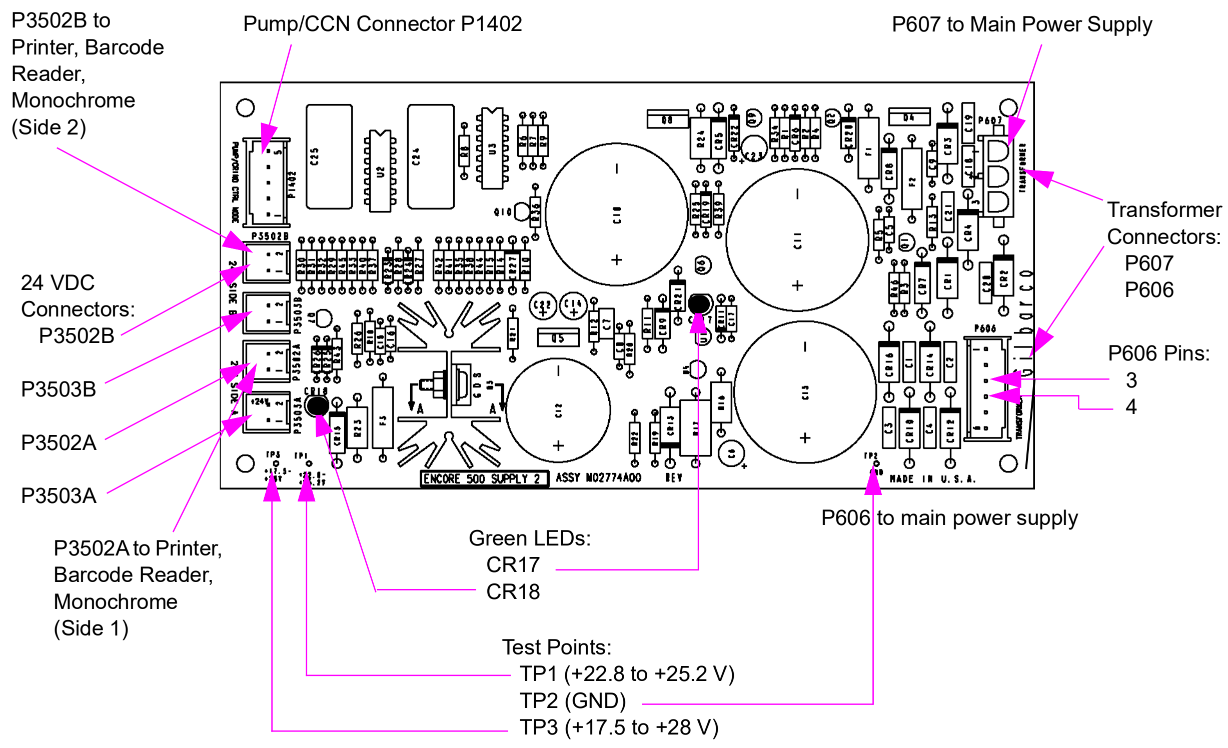
Model	Connector#	Through Cable	To Board	At Connector#
Encore 500	J108	Not applicable to this configuration		
	J302	M00614A00X	Main Display (M01785A001)	J2115
	J310	M02954A001	STP Board (M00047A001)	Terminal strip
	J604	Not applicable to this configuration		
	J605	N/A	Voltage plug (M00488AXXX)	J1607
	J606	Through attached cable	Power Supply 2 (M02774A001)	P606
	J607			P607
	J1200	Through attached cable	Valve Converter (M02044A00X)	P1200
	J1206	Not applicable to this configuration		
Encore 300	J302	M01650A001	Main Display (M01515A001)	J2115
	J605	N/A	Voltage plug (M00488AXXX)	J1607
	J108	Through attached cable	Valve Converter (M02335A00X)	P108
	J310	Through attached cable	STP Board (M00047A001)	Terminal strip
	J606	Not applicable to this configuration		P606
	J607			P607
	J1200	Through attached cable	Valve Converter (M02335A00X)	P1200
	J1206	Not applicable to this configuration		

Fuse

Description	Part Number
Fuse, 3AG Slo-Blo, 5 A	Q10131-04

Universal Power Supply-2 PCA (M02774A001) for Encore 500 and Eclipse

Figure 6-60: Universal Power Supply PCA (M02774A001)



Notes:1) P606: pins 1 and 2 are for 8 VAC; pins 3 and 4 are for 34.5 VAC.
2) P607: pins 1 and 3 are for 18 VAC.

Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P606 (6-pin)	From Transformer	Main Power Supply (M02946A001)	J606
P607 (3-pin)			J607
P1402 (5-pin)	M00635	M01922	P1105
P3502A/B (2-pin)	M00517A/B	Printer, Barcode Reader, Monochrome	CN1

Status LEDs

LED	Status Indication
CR17	18 VDC
CR18	+24 VDC

Test Points

Test Point	Range
TP1	24 VDC
TP2	GND DC
TP3	18 VDC

Troubleshooting

Node Resistance Measurement Preparation

Node	What to Do Before Measuring Resistance Between Pins 1 and 5
Pump	Temporarily disconnect the LON cables to pump node.
CRIND Device	Temporarily disconnect Monochrome Interface Board.
Last Node in Chain	Temporarily disconnect its LON cable and temporarily move the LON terminator to the node immediately upstream.
Others	Temporarily bypass them with a LON jumper cable.

LON Power Problem

A LON power problem can appear as pump power cycling off and on every 2 or 3 seconds, or as the pump logic boards having no power. Refer to the following table:

If	Then
Pump power cycles off and on every 2 to 3 seconds	1 Shut-off power to the dispenser. 2 Disconnect 5-pin P1402 connector to the pump or CRIND node. 3 Reapply power to dispenser. If the power cycling stops, the pump or CRIND node may be defective. If the cable goes to a car wash kiosk, pin 3 on the cable might be omitted. *Note: The cable to a car wash kiosk must have a 002 suffix; all other CRIND cables must have a 001 suffix.*
Pump logic boards have no power - CR17 is not lit	1 Disconnect 3-pin P607 connector from M02774 Board. 2 Measure AC voltage between pins 1 and 3 on the cable connector. If voltage across pins 1 and 3 are in 17 VAC and 35 VAC range, the M02774 Board may be defective. If voltage across pins 1 and 3 are not in 17 VAC and 35 VAC range, the transformer on the M02274 power supply assembly may be defective. 3 Disconnect 5-pin P1402 connector from M02774 Board. 4 Measure resistance across pins 1 and 5 on the cable connector. If 800 ohms or more, reconnect the cable to P1402, and power up the dispenser. This procedure is over. If under 800 ohms, proceed to next step. 5 To find bad node or board, temporarily disconnect each node/board one at a time and measure the resistance across pins 1 and 5 on the cable. If 800 ohms or more with board disconnected, replace the disconnected node/board. If under 800 ohms and each node/board has been temporarily disconnected, proceed to next step. *Note: Components using 18 V include all nodes and the monochrome board. The pump node receives 18 V from a direct plug-in while all other nodes receive 18 V over the LON cable.* If no bad board or node is found, a cable problem can exist. Use a test cable to temporarily replace each cable in the chain, measuring ohms (800 or more), until defective cable is found. If no defective cable is found, escalate the problem through normal channels. If defective cable is found, proceed to next step. 6 Replace the defective cable. 7 Reconnect all nodes/boards and cables. 8 Power up the dispenser.

24 VDC Problem

If one or more 24 VDC devices in the dispenser fails to operate, examine CR18 on M02774 and perform the action in the following table:

Note: Examples of 24 VDC devices are the CRIND printer, monochrome display backlight, and LED board for the barcode scanner.

If	Then
CR18 is lit	If you still have problems, trace out the connections from the M02774 Board to the affected device. If you find a bad connection, fix the problem. If you are still not able to locate the source of the problem, verify the voltages with respect to ground (TP2) at test points TP1 and TP3. *Note: The 24 VDC power connection is made on any of P3502A, P3502B, P3503A, and/or P3503B*
CR18 is not lit	Either the M02774 Board is bad or transformer (part of the M02274 Power Supply Assembly) feeding the 24 VDC section of the M02774 Board is bad. Measure AC voltage across pins 3 and 4 on P606 connector. The voltage must be in 30 VAC to 40 VAC range. If not in that range, suspect the transformer. If in that range, suspect the M02774 Board.

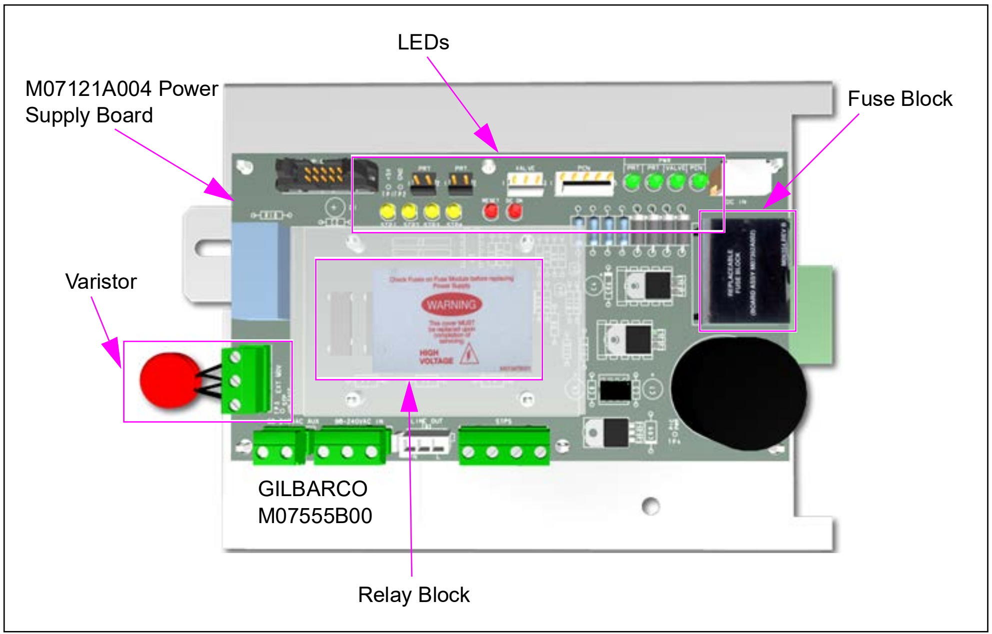
Power Supply Assembly (M07555A001) (Replaced by M07555A004)

The power supply uses the line voltage of 110-230 VAC and modifies it to supply power to the dispenser system. It converts the line voltage to 24 VDC and sends it to the M07121 Power Supply Board. The power supply board produces most of the remaining voltages needed by the system. The power supply does provide limited power conditioning and protection as well. A set of varistors provide power filtering and will even burn out if a severe surge occurs.

The following are additional capabilities:

- Four relays are used to switch the high current line voltage to the STP relay box, activating the STPs.
- The power supply has a bank of LED indicators to assist in troubleshooting.
- A replaceable fuse block to help get the power supply up and running in case the one in use is defective and needs to be replaced.

Figure 6-61: M07121A004 Power Supply Assembly

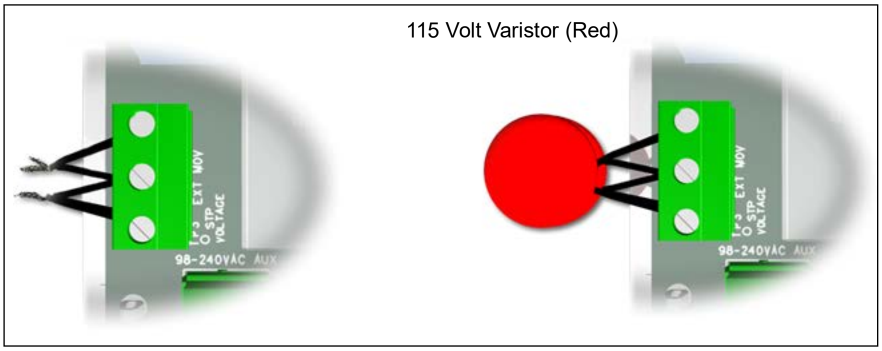


Varistors

Varistors provide limited filtering and protection to the line voltage.

In case of an extreme surge or lightning strike, they may even be destroyed. Even if they are destroyed, the power supply may still be functional. The power supply should be replaced, however, the power supply and the dispenser electronics are no longer protected because the power supply could have been damaged by the surge.

Figure 6-62: Varistors



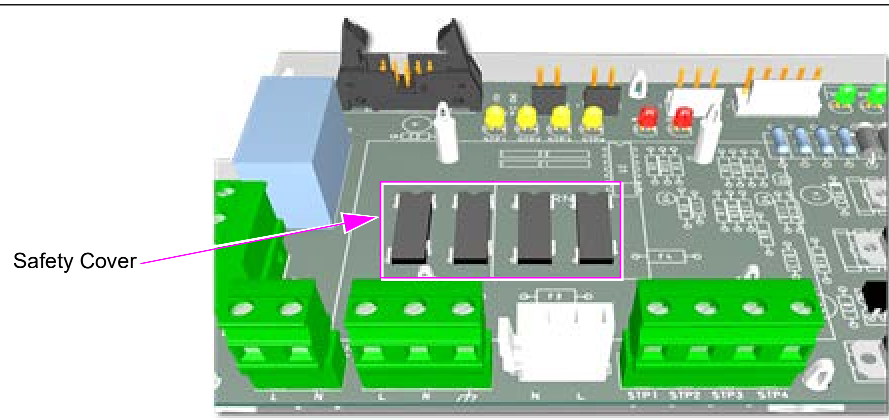
STP Relays

The four STP relays K1-K4 to switch the high current line voltage signals to activate the STP relays for the four grades.

With this power supply, the relays are not replaceable. If one of the relays is defective, the entire board must be replaced.

If the LED for the STP in question illuminates, but there is no 110 VAC on the corresponding lead, this usually indicates shorted or grounded wiring. Ensure to verify both before replacing the power supply.

Figure 6-63: STP Relays



Note: In Figure 6-63 the safety cover has been removed. This is for demonstration purpose only. The safety cover should never be removed.

Fuse Block (M07362A002)

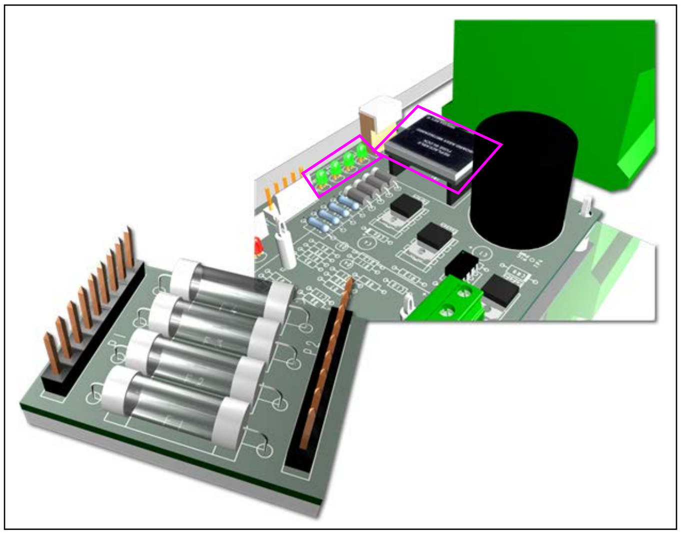
Fuse block contains the following four fuses to protect the 24 VDC outputs:

- Two Printer Outputs
- Valve Driver Board
- PCN

The LEDs indicate each 24 V output.

Note: The fuse block must be replaced as an entire assembly. Fuses cannot be replaced individually.

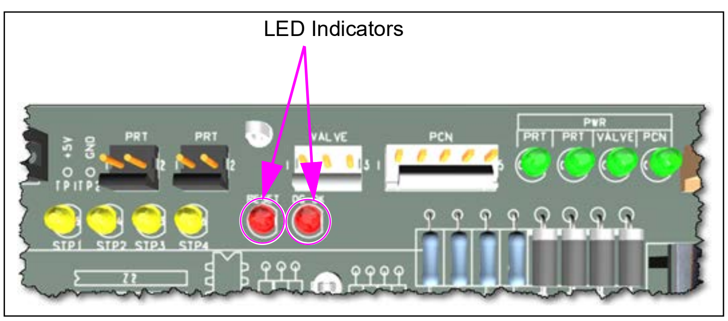
Figure 6-64: Fuse Block



LED Indicators

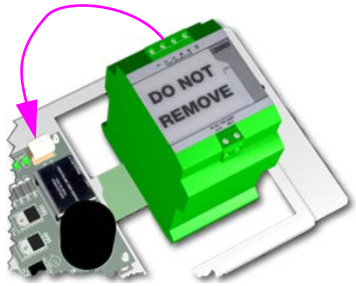
The reset LED should light up as the dispenser is powered down. This is part of the “power fail warning”, and if it is not lighting up when power is removed, memory and software lockups can occur.

Figure 6-65: LED Indicators



The DC OK LED actually is a status indicator of the +24 V. As long as the +24 V is above some minimum threshold, the LED is off. When the +24 V sags below the minimum threshold, the LED turns on until the supply voltage goes away completely. When the DC OK LED stays on solid, it is an indication that there is an issue with the +24 V source or a problem in the comparator circuit.

Figure 6-66: M04161B001 Transformer



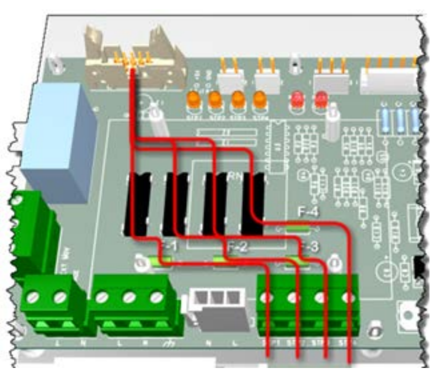
M04161B001 Transformer is shown replaced by M10996A001 Power Supply Board on the M07555A004.

CR 10 - PCN Output
CR 11 - Valve Board Output
CR12 - A Side (Left) Printer Port Output
CR17 - B Side (Right) Printer Port Output

STP Indicators

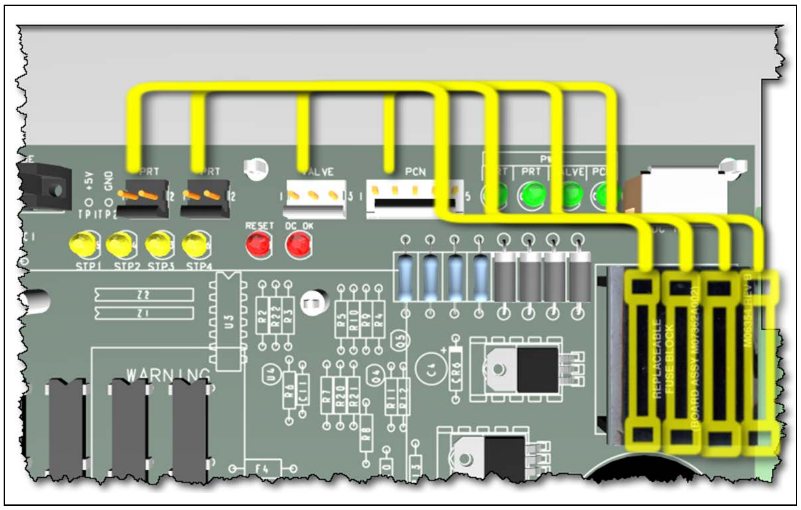
The LEDs for the STP circuits indicate that the signal from the pump control node is available for that circuit. It does not indicate line voltage output to the STP relay. If the indicator for STP1 is on, this means the pump control node signal is getting to the power supply not that the line voltage is being sent to the STP relay. This must be tested using a VOM from the STP connector to AC neutral. The LEDs are arranged in the same order as the relay and connector lead they service. (see Figure 6-67).

Figure 6-67: STP Indicators



CR15 - STP1
CR14 - STP2
CR13 - STP3
CR16 - STP4
CR19 - DC OK
CR20 - Reset

Figure 6-68: 24 VDC Indicators and Fuses



The power LEDs are an indication that the 24 VDC to both of the printer connectors, the valve driver board connector, and the pump control node connector is good. The LEDs are lined up in order with the connector it services and the fuse that protects that circuit.

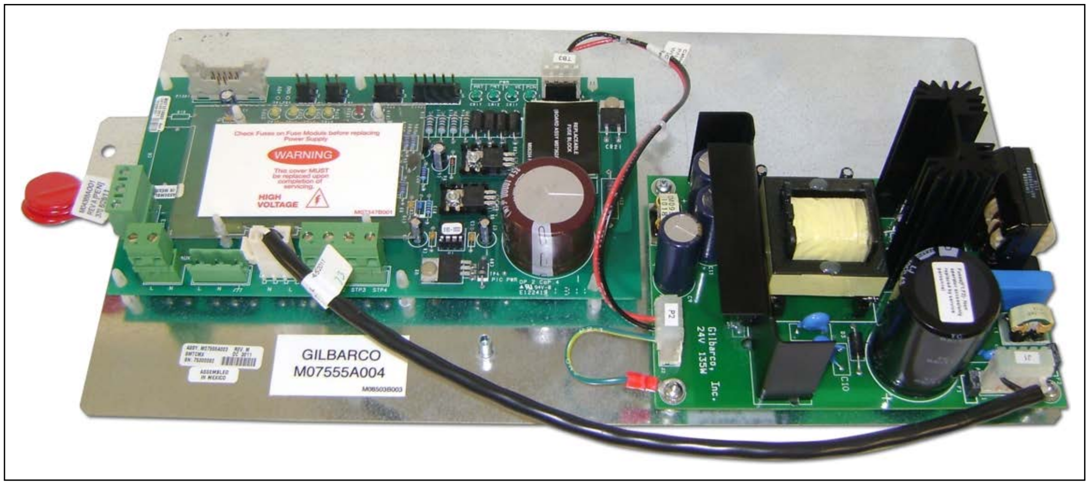
The following table lists the spare parts details:

Part Name	Part Number
AC Distribution Connector, 2 Position	M04379B002
AC Input Connector, 3 Position	M04379B003
115 VAC Output Cable, 3 Position	M07648A003
24 VDC Input Cable, 3 Position	M07648A004
STP Connector, 4 Position	M04379B004
Replaceable Fuse Module	M07362A002
Replaceable Red Varistor (115 V)	M04388A001
Replaceable Blue Varistor (230 V)	M04388A002
Power Supply Board	M07121A004

M07555A004 Power Supply (Released for Encore 500 Dispenser)

Note: This power supply is not used in Encore satellite dispensers.

Figure 6-69: M07555A004 Power Supply



The new M07555A004 Power Supply is now released to manufacturing and is a direct replacement for the M07555A001 Power Supply. It is backward compatible with previous 24 VDC power supplies.

The LEDs and voltage test points on the M07121A003 are exactly the same as the previous M07121A001 and A002 power supply board. One significant change is that this power supply handles power surge currents better, thus making it more robust, and helps to further eliminate the chance for damage to the power supply or dispenser. One other difference is there is a slight delay in the power supply coming up when power is applied so that the C5 capacitor can be fully charged.

M07555A004 Power Supply with M09922K001 Ultra-Hi Interface Board

The M07555A004 power supply ties the 24 VDC ground directly to the dispenser chassis. Therefore, the field is no longer required to add a ground wire as described in previous bulletins (TRP 1974 and 2023). When using the new power supply, you must use the M09922K001 interface board for the Ultra-Hi dispenser along with it. Failure to do so can cause the F1 fuse to open/blow on the T19301 I.S barrier board. Additionally, in previous units, tying the DC ground to chassis in an Ultra-Hi unit was not permitted, and could cause potential damage to the electronics. With the new design, this is no longer the case. The DC ground and chassis are purposely tied together.

Troubleshooting Tips

When measuring DC voltage from chassis ground to DC ground with power applied to the unit, the reading should be approximately 0 VDC. There should be no voltage potential difference between chassis ground and board ground. Even with J405 connector on the interface board is disconnected, the reading is still 0 VDC.

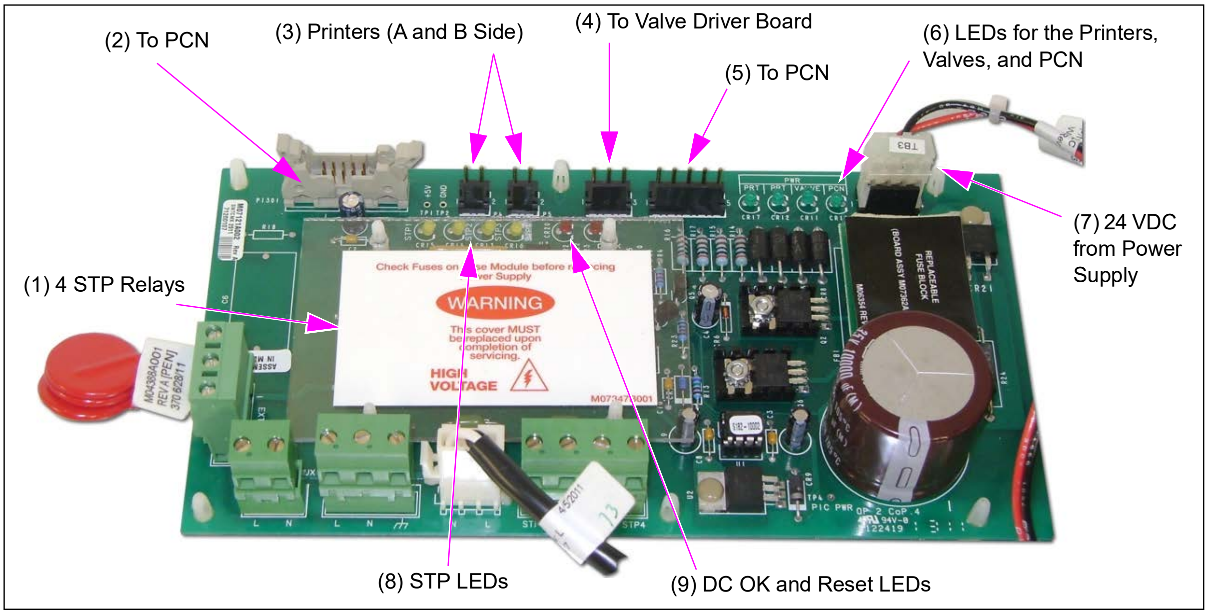
When measuring DC voltage from chassis ground to +5 V with the J405 connector disconnected, the reading should be approximately 5 V. If reading displayed is not +5 V, then you have a problem downstream between the interface board and the pulsers.

If you need to replace the IS barrier board, you should always try to find the source of the short first, otherwise you may end up replacing several boards needlessly.

⚠ WARNING

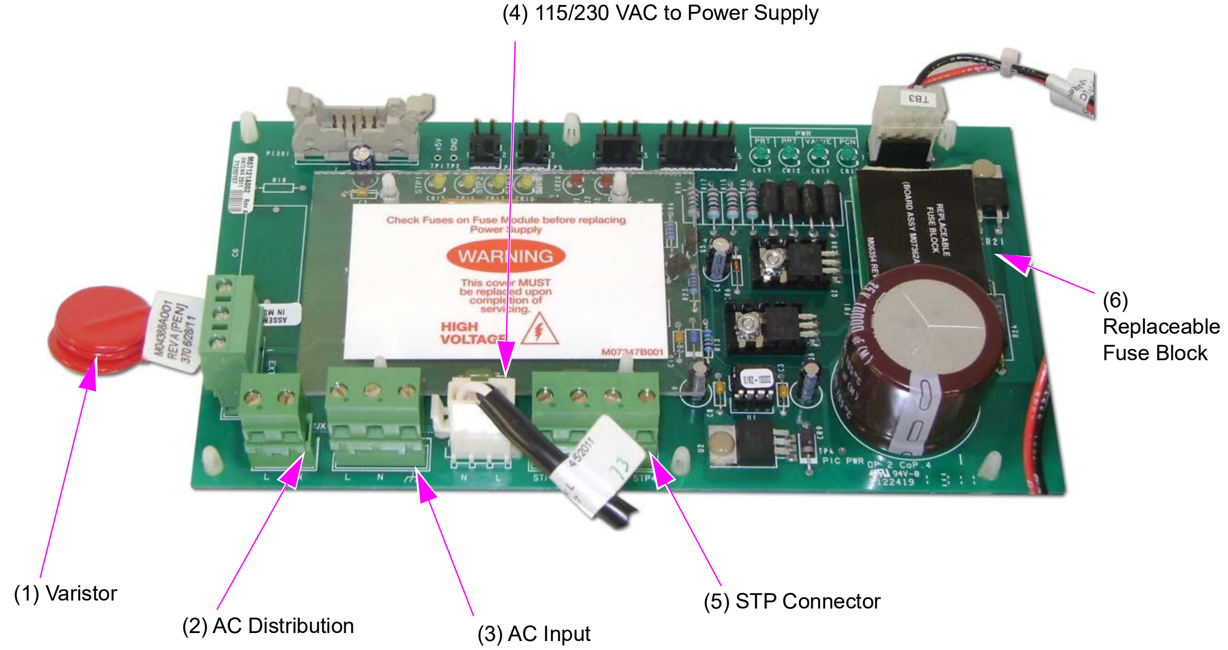
Never substitute a larger fuse in place of the F1 on the IS barrier board. This would violate UL requirements, and possibly induce a hazardous condition.

Figure 6-70: M07555A004 Power Supply Board - 1



Item	Part Name	Description
1	STP Relays	Switches low current line voltage signals to activate the STP relays for four products. *Notes: 1) Never remove the cover over the relays.* *2) The relays are not replaceable.*
2	Connector to PCN	Carries low level STP signal from the PCN to the power supply board to control the STPs.
3	Printers (A and B Side)	24 VDC feed and return
4	Connector to Valve Driver Board	24 VDC feed
5	Connector to PCN	24 VDC feed to PCN and power fail line.
6	LEDs for the Printers, Valves, and PCN	LEDs indicate 24 V available for the printers (CR12 and 17), valves (CR11), and PCN (CR10).
7	24 VDC from Power Supply	24 VDC from power supply.
8	STP relay control input signaling LEDs for each STP (CR13, CR14, CR15, and CR16)	The relay is bad if the AC control turns on, but no AC goes to the STP output connector.
9	DC OK LED and Reset LEDs	• DC OK (CR19 on green) indicates the 24 VDC from the transformer to the power supply board is present. The DC OK LED actually is a status indicator of the +24 V. As long as the +24 V is above some minimum threshold, the LED is off. When the +24 V sags below the minimum threshold, the LED turns on until the supply voltage goes away completely. If the DC OK LED stays on solid, it is an indication that there is an issue with the +24 V source or a problem in the comparator circuit. • Reset LED (CR20 on red) indicates the dispenser has been powered down.

Figure 6-71: M07555A004 Power Supply Board - 2



Item	Part Name	Description
1	Varistor	Replaceable Varistor (M04388A001) provides limited filtering and protection for the line voltage.
2	AC Distribution	AC Distribution Connector/2-position (M04379B002) hot and neutral only, the earth connection is made to the chassis through a wire nut.
3	AC Input	AC Input Connector/3-position (M04379B003).
4	Connector to Valve Driver Board	115 VAC to the Power Supply/3-position (M07648A003).
5	1415 VAC to Power Supply	STP Connector/4-position (M04379B004).
6	Replaceable Fuse Block	Replaceable fuse module (M07362A002). • Used on the M07121A002 and M07121A003 power supply. • Contains four fuses that protect the 24 VDC outputs. • Two printer outputs, 1 for valve output and 1 for the pump control node. • LEDs to show each 24 VDC output (on green when STP on). • If a printer stops working and the LED on the printer is out, and after following proper troubleshooting steps everything checks ok, always check the appropriate fuse on the power supply board. Do not replace the entire power supply for a blown fuse!

The following table provides part details for ordering the parts:

Part	Part #
Power Supply	M07555A004
Power Supply Board	M07555A003
Power Supply Assembly	M10996A001

Power Supply Assembly (M04104A001)

The power supply uses the line voltage of 110-120 VAC and modifies it to supply power to the dispenser system. It converts the line voltage to 24 VDC and sends it to the power supply board. The power supply board produces most of the remaining voltages needed by the system. The power supply does provide limited power conditioning and protection as well. A set of varistors provide power filtering and will even burn out if a severe surge occurs.

The following are the additional capabilities:

- Four relays are used to switch the low current line voltage to the STP relay box, activating the STPs.
- A replaceable fuse block to help get the power supply up and running incase the one that is in use is defective and needs replacement.

Figure 6-72: M04104A001 Power Supply Assembly Components - 1

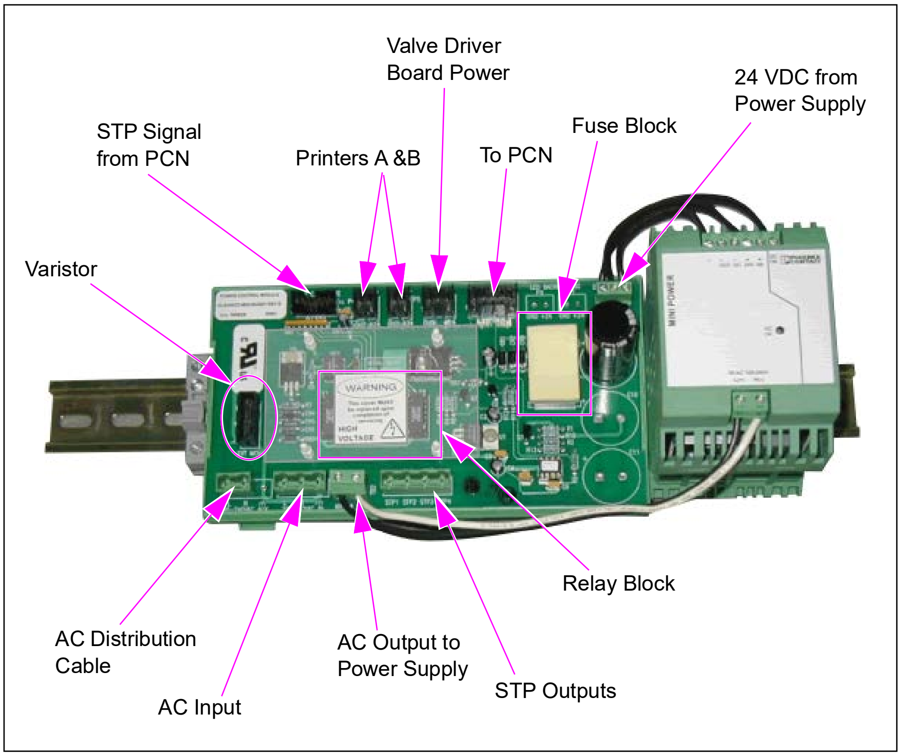
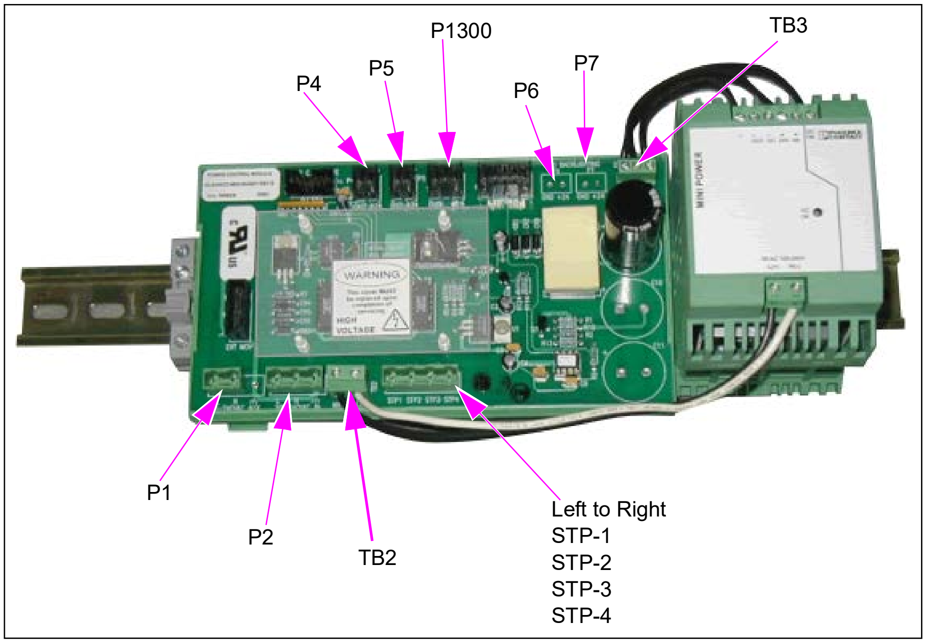


Figure 6-73: M04104A001 Power Supply Assembly Components - 2



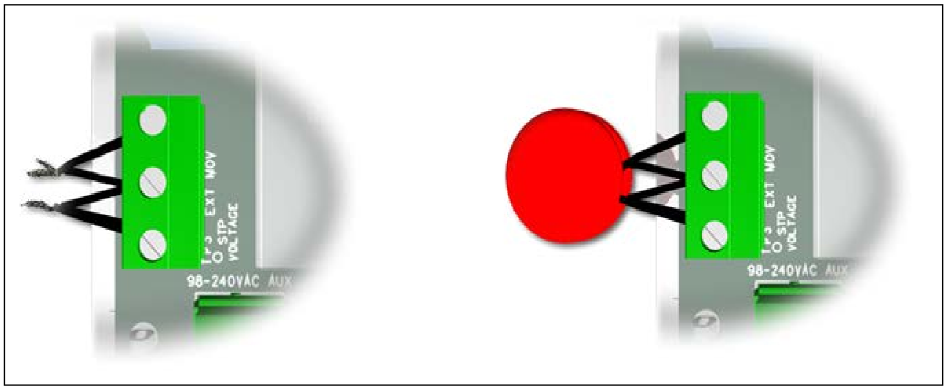
Position/Description	Voltage
P1	120 VAC/For AC Distribution Cable
P2	120 VAC/Main Supply to Dispenser
P4	24 VDC/GND/Printer
P5	24 VDC/GND/Printer
P6	24 VDC/GND/LED Backlight
P7	24 VDC/GND/LED Backlight
TB2	120 VAC/To Transformer
TB3	24 VDC/From Transformer to Power Supply Board
P1300	24 VDC/GND/Valve
P1402	Pump Control Node (PCN)

Varistors

Varistors provide limited filtering and protection to the line voltage.

In case of an extreme surge or lightning strike, they may even be destroyed. Even if they are destroyed, the power supply may still be functional. The power supply should be replaced, however, the power supply and the dispenser electronics are no longer protected because the power supply could have been damaged by the surge.

Figure 6-74: Varistors



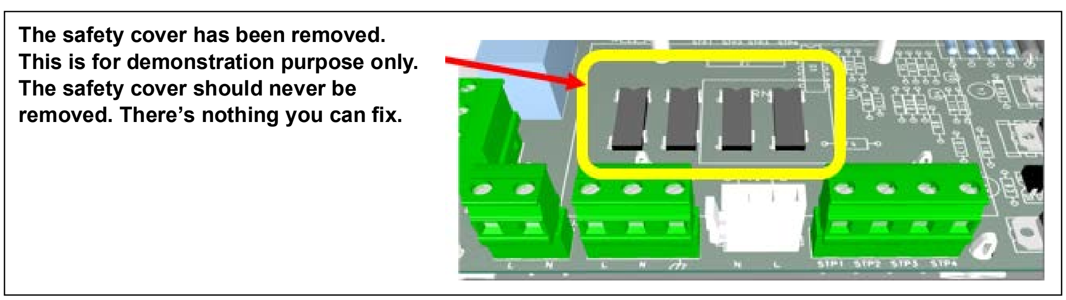
STP Relays

The four STP relays K1-K4 to switch the high current line voltage signals to activate the STP relays for the four grades.

With this power supply, the relays are not replaceable. If one of the relays is defective, the entire board must be replaced.

Ensure to verify both before replacing the power supply.

Figure 6-75: STP Relays



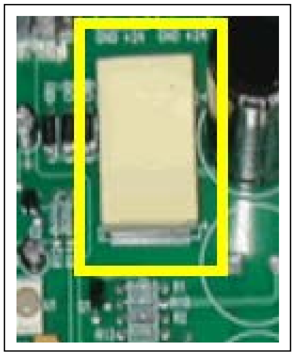
Fuse Block (M04163B001)

The fuse block contains the following three fuses to protect the 24 VDC outputs:

- Two Printer Outputs
- Valve Driver Board
- PCN

The fuse block must be replaced as an entire assembly. Fuses cannot be replaced individually.

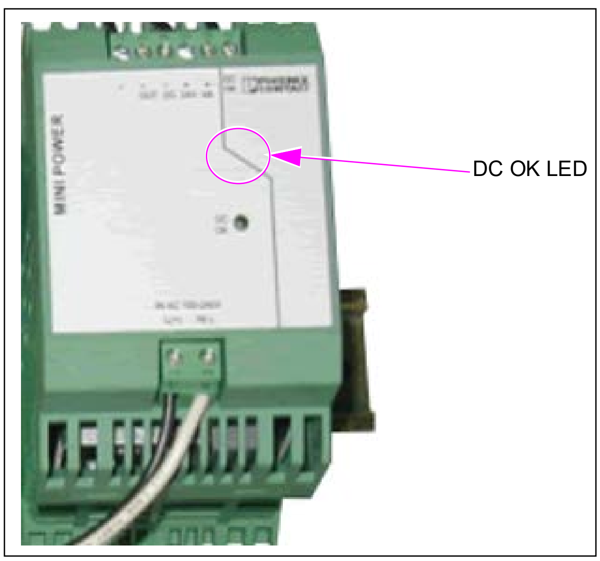
Figure 6-76: Fuse Block (M04163B001)



LEDs

There is one (green) LED indicator. It is located on the power supply. It is labeled as "DC OK". It signals that the 24 VDC circuit to the power supply board is present. The DC OK LED actually is a status indicator of the +24 V. As long as the +24 V is above some minimum threshold, the LED is off. When the +24 V sags below the minimum threshold, the LED turns on until the supply voltage goes away completely. When the DC OK LED stays on solid, it is an indication that there is an issue with the +24 V source or a problem in the comparator circuit.

Figure 6-77: DC OK LED



Displays, Main/PPU/CRIND

The components in this sub-section are grouped first by type, model, and then description.

About Displays

The main display provides the customer with information about total cost and total volume of the fuel dispensed.

PPU displays provide PPU information to the customer.

The following table shows which PPUs also have Reed Switch (Q11075-31) attached:

PPU Assembly Number	Encore Model	PPU Display Type	Reed Switch
M02652A009	Ultra-Hi	Single	None
M02652A008	300	Dual	None
M02652A007	300	Dual	Yes
M02652A006	500	Dual	None
M02652A005	500	Dual	Yes
M02652A004	300	Single	None
M02652A003	300	Single	Yes
M02652A002	500	Single	None
M02652A001	500	Single	Yes

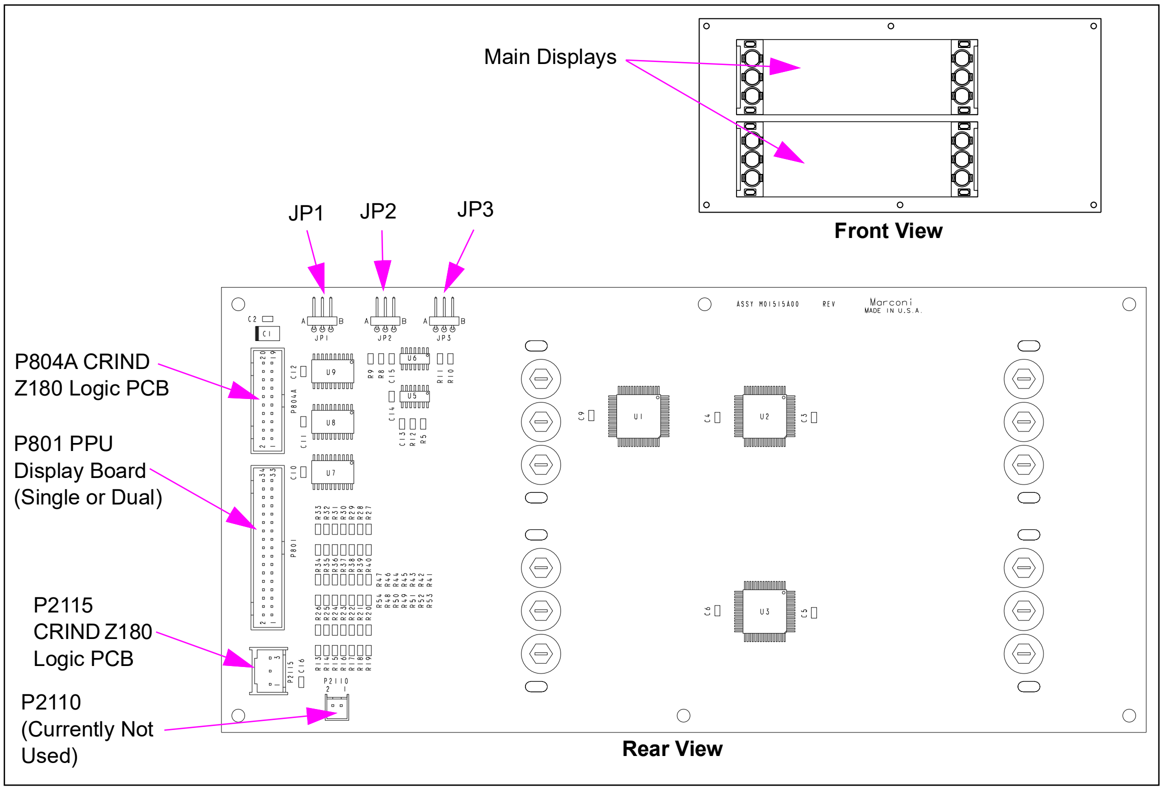
Main Display for Encore 300 (M01515A001)

This board is used in Encore 300 units to display the pump transaction information.

Service Tip

If one of the backlight bulbs burns out and the unit has been in service for quite some time, it is recommended that all backlight bulbs be replaced, as others are also probably near the end of their service life.

Figure 6-78: Main Display for Encore 300



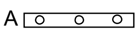
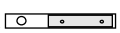
Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P801 (34-pin)	M01650A001	CRIND Z180 Logic Board (T17764-G3/4)	P264 (6-pin)
		Totalizer Interface PCA (M01621A001)	P1507 (12-pin)
		Single PPU Board (M01522A001)	P2201 (2-pin)
		Dual PPU PCA (M01525A001)	
P804A (20-pin)	M01585A001	Single PPU Board (M01522A001)	P121 (20-pin)
		Dual PPU PCA (M01525A001)	
P2110 (2-pin)	Currently not used		
P2115 (3-pin)	M01650A001	Same as P801 above.	J2115 (3-pin)

Test Points

Connector	Pin #	Range	Current
P801	18	5 VDC, +/- 5%	70 mA
P801	20	0 VDC	
P2115	1 to 3	9.4 VAC, +/1 10%	2.2 A

JP1 Jumper Settings

Side 1	A  B
Side 2	A  B
Master Reset	A  B

Connector PIN Functions and Voltage

P801 PIN Functions and Voltages

Pin #	Description	Voltage Range
1	MAIN-DATA-A3	0 to 5 V
2	MAIN-DATA-A2	0 to 5 V
3	MAIN-DATA-A1	0 to 5 V
4	MAIN-DATA-AO	0 to 5 V
5	MAIN-DATA-B3	0 to 5 V
6	MAIN-DATA-B2	0 to 5 V
7	MAIN-DATA-B1	0 to 5 V
8	MAIN-DATA-BO	0 to 5 V
9	Not used	
10	Not used	
11	COM-CONT	0 to 5 V
12	Not used	
13	CREDIT-ST	0 to 5 V
14	CASH-ST	0 to 5 V
15	DSEL1	0 to 5 V
16	DSEL0	0 to 5 V
17	ST3	0 to 5 V
18	5 V	5 V

Pin #	Description	Voltage Range
19	ST2	0 to 5 V
20	GND	0 V
21	ST1	0 to 5 V
22	Not used	
23	STO	0 to 5 V
24	Not used	
25	CASH-DP	0 to 5 V
26	MSTR-RAM-CLR	0 V
27	VOLUME-DP	0 to 5 V
28	Not used	
29	PPU-DP	0 to 5 V
30	Not used	
31	MAIN-ENABLE	0 to 5 V
32	Not used	
33	Not used	
34	Not used	

P804A PIN Functions and Voltages

Pin #	Description	Voltage Range
1	DO-DATA	0 to 5 V
2	D1-DATA	0 to 5 V
3	DGTSEL1	0 to 5 V
4	D2-DATA	0 to 5 V
5	DGTSELO	0 to 5 V
6	D3-DATA	0 to 5 V
7	ST-CRDT	0 to 5 V
8	ADD-A2	0 to 5 V
9	ADD-A0	0 to 5 V
10	ADD-A1	0 to 5 V
11	ADD-A3	0 to 5 V
12	5 V	5 V
13	ST-CASH	0 to 5 V
14	OPCOM3	0 to 5 V
15	GND	0 V
16	OPCOM4	0 to 5 V
17	OPDP4	0 to 5 V
18	OPDP3	0 to 5 V
19	OPDP2	0 to 5 V
20	OPBP	0 to 5 V

P2110 PIN Functions and Voltages

Pin #	Description	Voltage Range
1	9.4 VAC	9.4 VAC +/-10%
2	9.4 VACRTN	

P2115 PIN Functions and Voltages

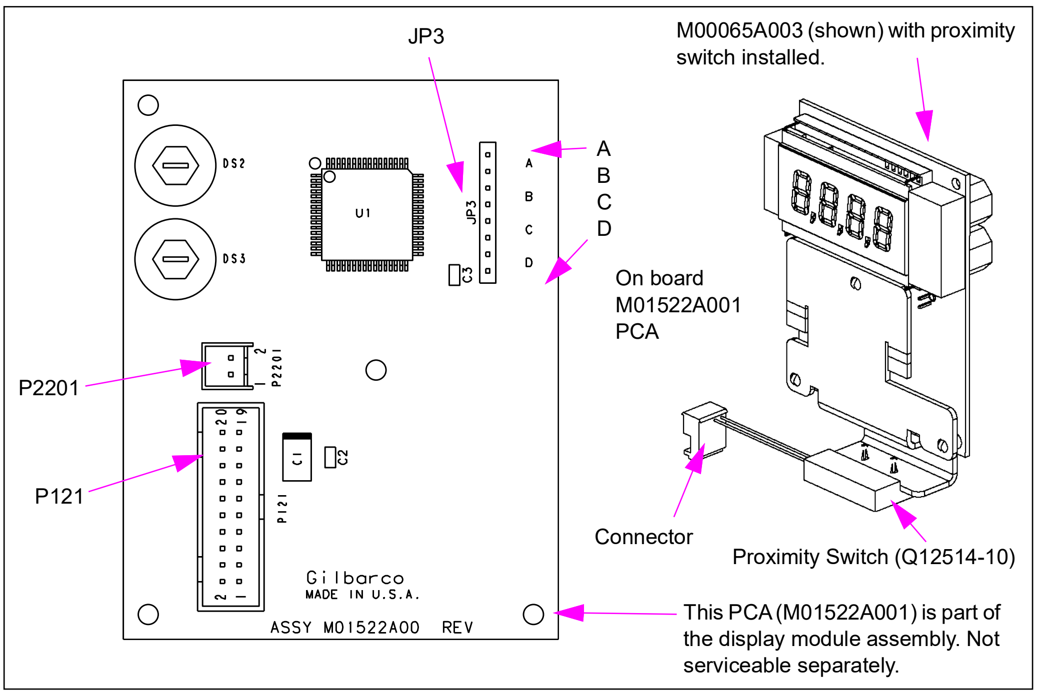
Pin #	Description	Voltage Range
1	9.4 VAC	9.4 VAC +/-10%
2	Not used	
3	9.4 VACRTN	9.4 VAC +/-10%

Single Display Module for Encore 300 (M02652A003 and M02652A004 PPU)

The single PPU display provides price-per-unit information to the customer on the Encore 300 and also reports grade selection and pump handle status. A proximity switch acts as a grade select switch. Module M02652A004 does not have the proximity switch.

Note: M02652A003 and M02652A004 are replaced by M02652A012 and M02652A013 respectively.

Figure 6-79: PPU Module with Single Display for Encore 300



Board Connections and Cables

Note: Board and connections per grade.

Connector#	Through Cable	To Board	At Connector#
P121 (20-pin)	M01585A001	Main Display PCA (M01515A001)	P804A (20-pin)
P2201 (2-pin)	M1586A001 or M01618A001	Pump Controller PCA (M01598A001)	P204 (34-pin)

JP3 - Jumper Settings

Unit Type	Side	Grade	Jumper Setting
Standard MPD, Single-hose MPD and Six-hose Blenders	1 (formerly A)	1	A
		2	B
		3	A B
	2 (formerly B)	1	A C
		2	B C
		3	A B C
Selectable Blenders	1 (formerly A)	1	A
		2	B
		3	A B
		4	C
		5	A C
		6	B C
	2 (formerly B)	1	A B C
		2	D
		3	A D
		4	B D
		5	A B D
		6	C D

PPU Dual Display Module (M02652A007 and M02652A008) for Encore 300

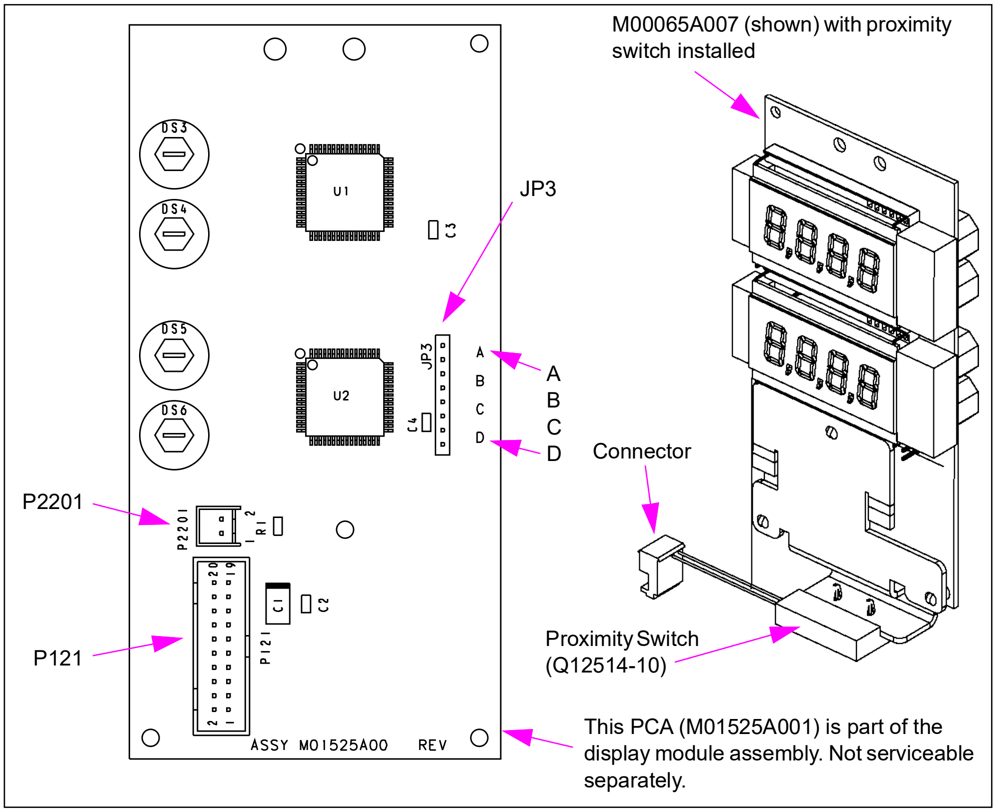
The Dual PPU Display Module (M02652A007 and M02652A008) is capable of supplying dual level price-per-unit information to the customer. Grade selection and status is reported back to the CRIND device. Jumpers are used to set the addresses. A proximity switch that reports status of the pump handle is part of the M02652A007 Module. M02652A008 Module does not have the proximity switch.

Note: M02652A007 and M02652A008 are replaced by M02652A013.

Service Tip

If one of the two backlight bulbs burns out, it is a good idea to replace both bulbs on that board.

Figure 6-80: PPU Module with Dual Display for Encore 300



Board Connections and Cables

Note: Board and connections per grade.

Connector#	Through Cable	To Board	At Connector#
P121 (20-pin)	M01585A001	Main Display PCA (M01515A001)	P804A (20-pin)
P2201 (2-pin)	M01586A001 or M01618A001	Pump Controller PCA (M01598A001)	P204 (34-pin)

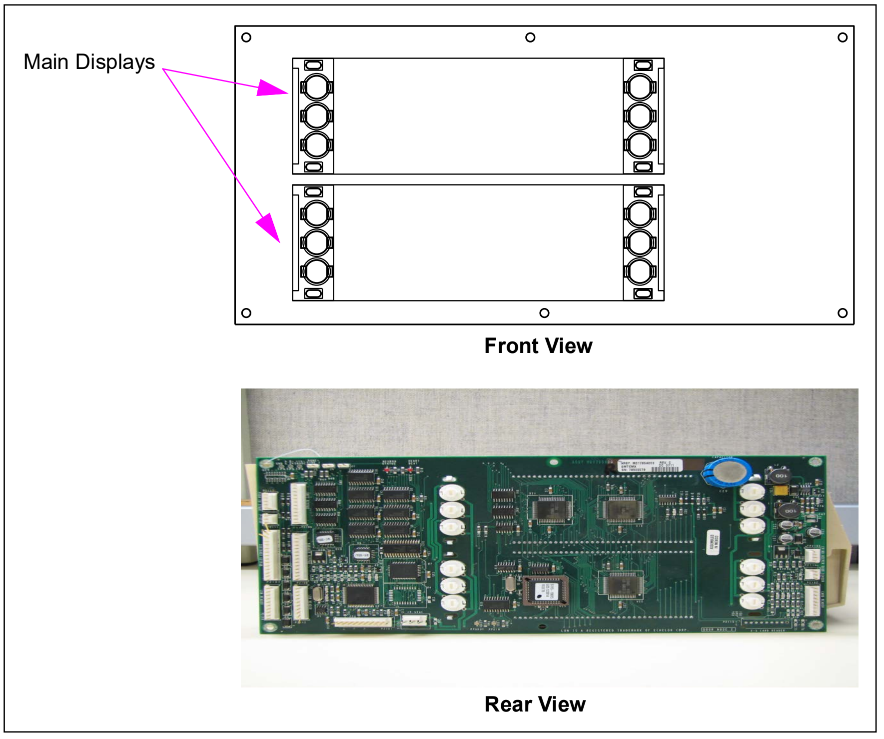
JP3 - Jumper Settings

Jumper settings are the same for single and dual PPU displays. For settings, refer to "JP3 - Jumper Settings" on page 6-121.

Main Display (M01785A001) for Encore 500 and Eclipse

The main display is part of the Door Node-2 PCA on Encore 500 and Eclipse series units. For information on connectors, status LEDs, and test points, refer to "Door Node-2 (M01785A001)" on page 6-50.

Figure 6-81: Main Display/Door Node-2 for Encore 500 and Eclipse



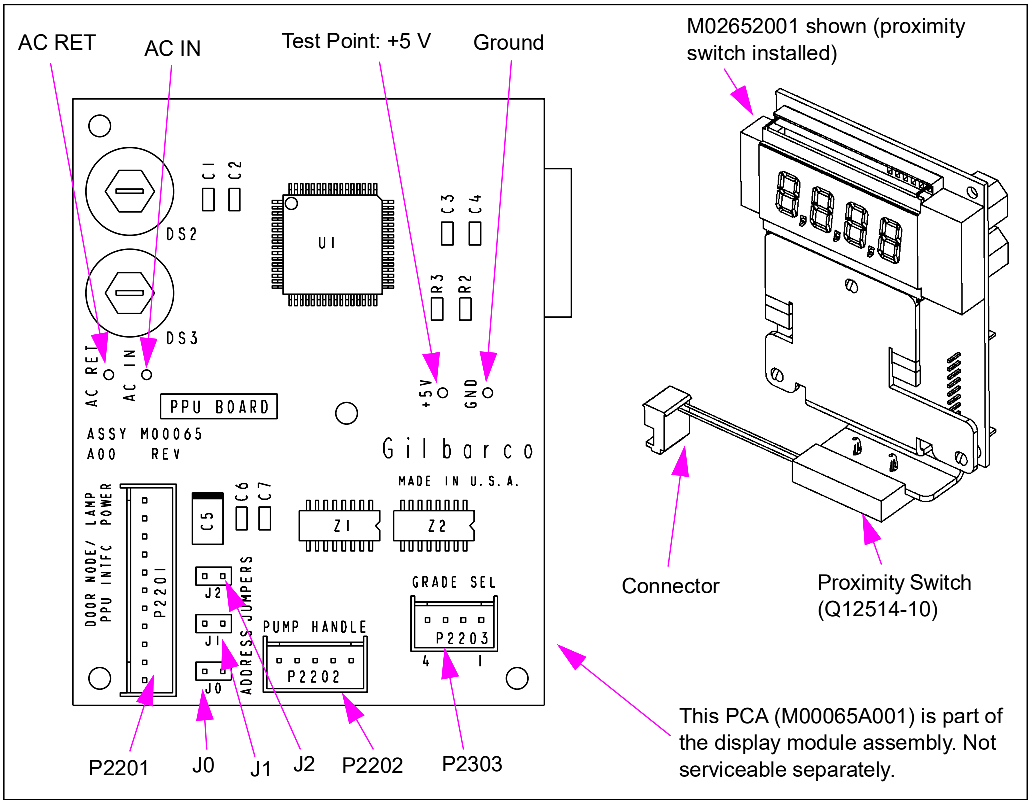
PPU-Single Display Module (M02652A001 and M02652A002) for Encore 500 and Eclipse

The single PPU display provides price-per-unit information to the customer and also reports grade selection and pump handle status information back to the pump node. A proximity switch that reports status of the pump handle is part of the M02652A001 Module. M02652A002 Module does not have the proximity switch. The PPU displayed and the STPs activated will be determined by setting the jumpers.

Service Tip

It is recommended that you replace bulbs in pairs.

Figure 6-82: PPU Module with Single Display for Encore 500 and Eclipse



PPU Address Jumpers

Addresses can be set for up to 6 grades.

Jumper	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5	Grade 6
J0	OUT	IN	OUT	IN	OUT	IN
J1	OUT	OUT	IN	IN	OUT	OUT
J2	OUT	OUT	OUT	OUT	IN	IN

Board Connections and Cables

Note: Board and connections per grade.

Connector#	Through Cable	To Board	At Connector#
P2201 (11-pin)	• M00509A001 or M00509A002 (Encore) • M00509A003 (Eclipse)	• CRIND Auxiliary I/O Board (M00095A001) • Door Node-2 (M01785A001) • Door Node-1 (M000062A001)	P2101
P2202 (5-pin)	M01152A001 (or in line with cable M00497A001 through J2202/P2202. Refer to Note)	Pump Handle	
P2203 (4-pin)	M00628A001	Grade Select	

Note: M00497A001 is used only when single-hose MPD hose is on left grade 1 or grade select button is on far right.

Connector PIN Functions and Voltages

P2201

Pin #	Description	Input/Output Voltage
1	AC_IN	9.4 VAC
2	AC_RET	
3	GND	0 V
4	GND	0 V
5	PWR (+5 V)	+5 V +/-5%
6	CLK	0 to 5 V
7	FEEDBACK	0 to 5 V
8	SND_REC	0 to 5 V
9	PBP	0 to 5 V
10	SER_DATA	0 to 5 V
11	RESET	0 to 5 V

P2202

Pin #	Description	Input/Output Voltage
1	PUMP_HANDLE	0 to 5 V
2	+5 V	+5 V +/-5%
3	+5 V	+5 V +/-5%
4	PUMP_HANDLE_LB	0 to 5 V
5	GND	0 V

P2203

Pin #	Description	Input/Output Voltage
1	GRD_SEL	0 to 5V
2	+5 V	+5 V +/-5%
3	GRD_SEL_LB	0 to 5 V
4	+5 V	+5 V +/-5%

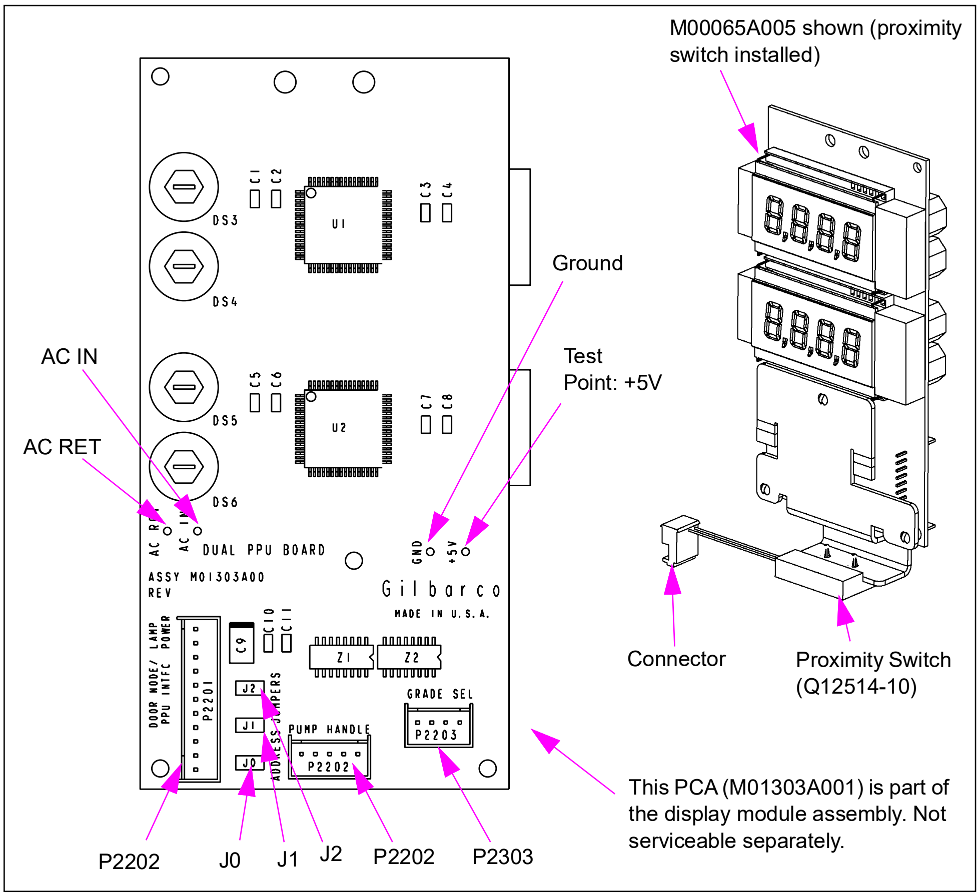
PPU Dual Display Module (M02652A005 and M02652A006) for Encore 500 and Eclipse

The Dual PPU Display Module (M02652A005 and M02652A006) is capable of supplying dual level price-per-unit information to the customer. Grade selection and status is reported back to CRIND Auxiliary I/O Board (M00095A001). Jumpers are used to set the addresses. A proximity switch that reports status of the pump handle is part of the M02652A005 Module. M02652A006 Module does not have the proximity switch.

Service Tip

If one of the two backlight bulbs burns out, it is a good idea to replace both bulbs on that board.

Figure 6-83: PPU Module with Dual Display for Encore 500



PPU Address Jumpers

Addresses can be set for up to 6 grades.

Jumper	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5	Grade 6
J0	OUT	IN	OUT	IN	OUT	IN
J1	OUT	OUT	IN	IN	OUT	OUT
J2	OUT	OUT	OUT	OUT	IN	IN

Board Connections and Cables

Note: Board and connections per grade.

Connector#	Through Cable	To Board	At Connector#
P2201 (11-pin)	• M00509A001 or M00509A002 (Encore) • M00509A003 (Eclipse)	CRIND Auxiliary I/O Board (M00095A001)	P2101
P2202 (5-pin)	M01152A001 (or in line with cable M00497A001 through J2202/P2202. Refer to Note)	Pump Handle	
P2203 (4-pin)	M00628A001	Grade Select	

Note: M00497A001 is used only when single-hose MPD hose is on left grade 1 or grade select button is on far right.

Connector PIN Functions and Voltages

P2201

Pin #	Description	Input/Output Voltage
1	AC_IN	9.4 VAC
2	AC_RET	
3	GND	0 V
4	GND	0 V
5	PWR (+5 V)	+5 V +/-5%
6	CLK	0 to 5 V
7	FEEDBACK	0 to 5 V
8	SND_REC	0 to 5 V
9	PBP	0 to 5 V
10	SER_DATA	0 to 5 V
11	RESET	0 to 5 V

P2202

Pin #	Description	Input/Output Voltage
1	PUMP_HANDLE	0 to 5 V
2	+5 V	+5 V +/-5%
3	+5 V	+5 V +/-5%
4	PUMP_HANDLE_LB	0 to 5 V
5	GND	0 V

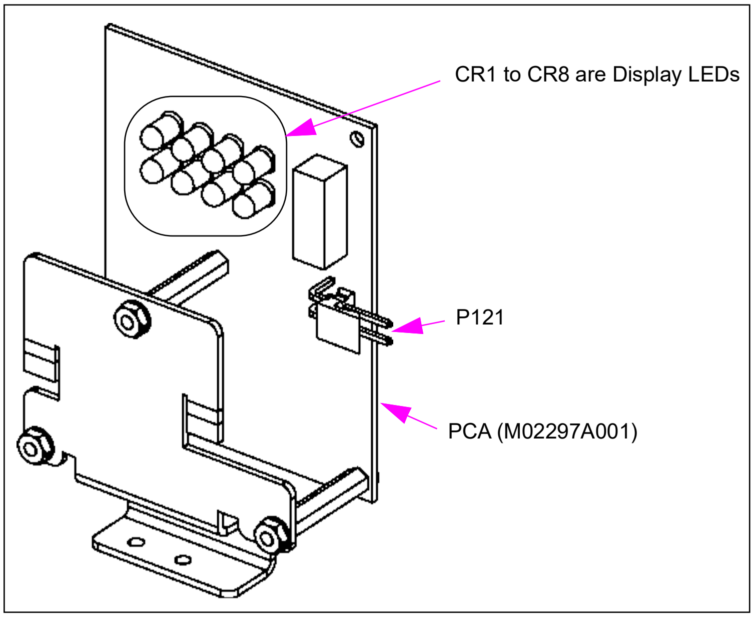
P2203

Pin #	Description	Input/Output Voltage
1	GRD_SEL	0 to 5 V
2	+5 V	+5 V +/-5%
3	GRD_SEL_LB	0 to 5 V
4	+5 V	+5 V +/-5%

Satellite Indicator Display Module (M02659A009) for Encore 300 and Encore 500

The satellite indicator display module is mounted in Encore Ultra-Hi Master or master side of a satellite/master combo unit on Encore 300 Ultra-Hi and Encore 500 Ultra-Hi units. This display indicates that the nozzle boot handle has been lifted on the satellite unit that is connected by product piping and wiring with that master.

Figure 6-84: Satellite Indicator Display (M02652A009)



PIP3+ Board
M15763A001

Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P121 (2-pin)	M02371A001	Transformer Sub-assembly (M02370A001)	JSEC

Large Touch Display Assembly (M18621A001) for Encore 700

The 15.6-inch LTD assembly components include the 15.6-inch LCD, PIP3+ Board, Touch Controller board, and the Touch Sensor Cable Shield.

Figure 6-85: Large Touch Display (LTD)

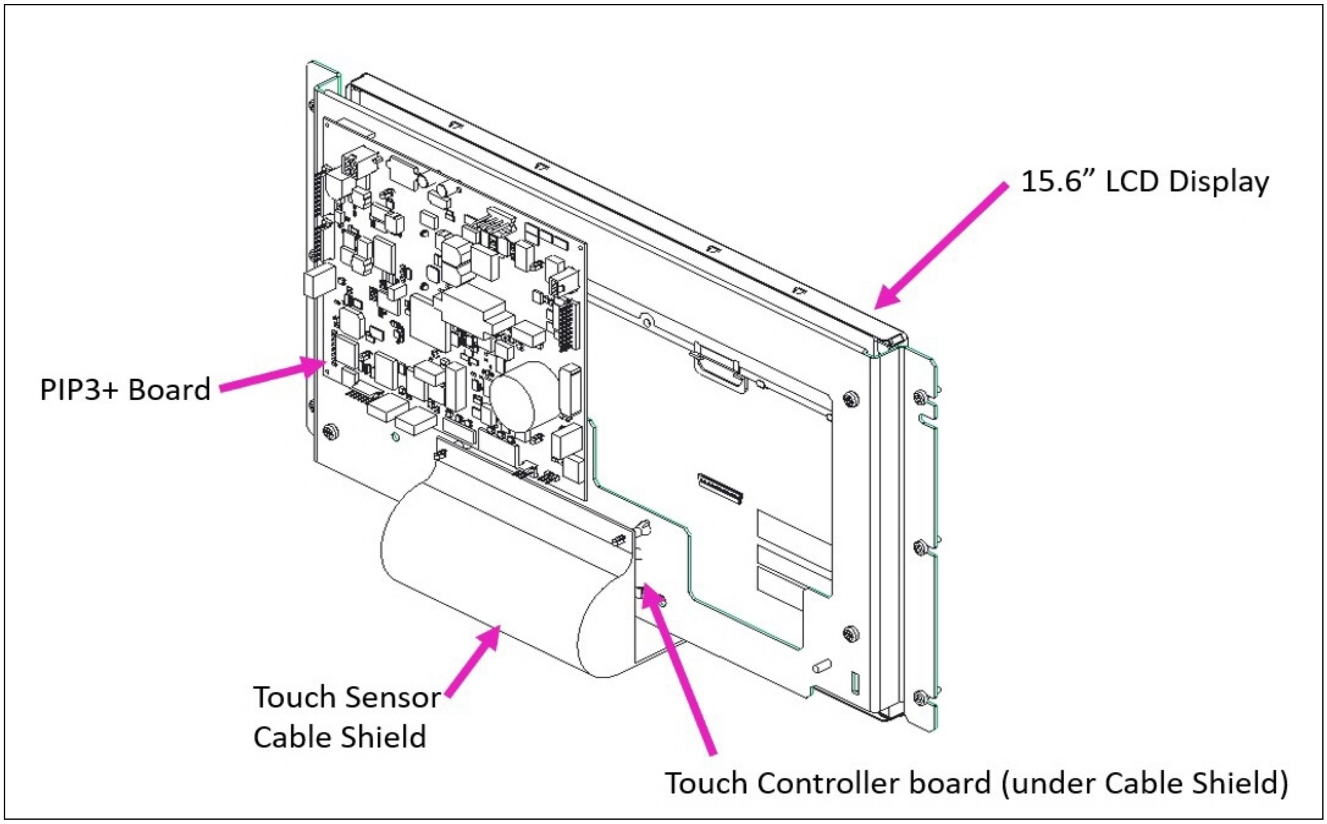
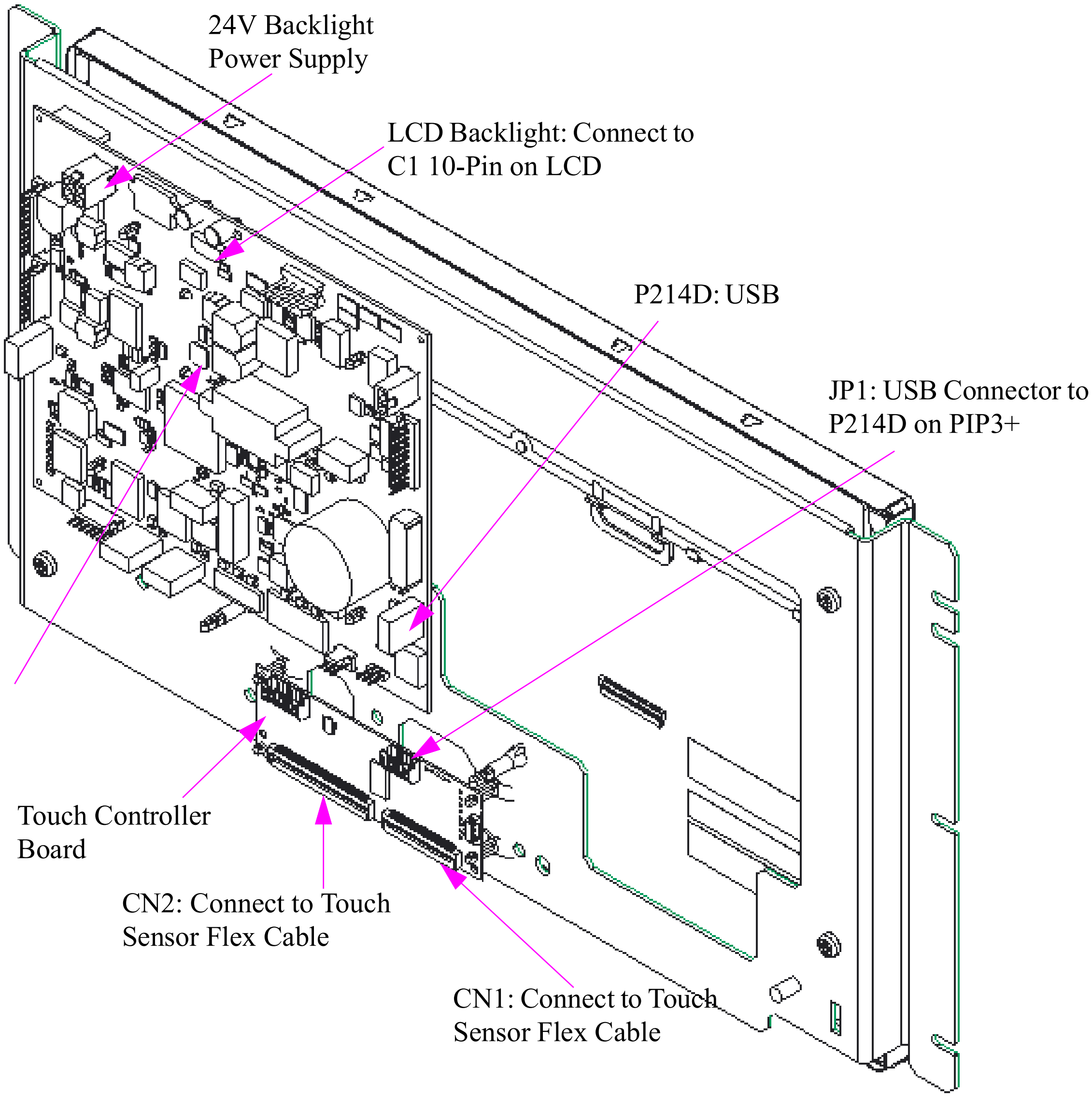


Figure 6-86: 15.6" LTD Display Assembly Cable Connections (M18621A001)



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#	Description
N/A	M03689A006	PIP3+	P220	24V Backlight Power
J2	M18934A001	PIP3+	P214D	Controller Board USB
J2	M15993A001	PIP3+	P223	15.6" Display Backlight
J1	M15994A001	PIP3+	P206	15.6" LVDS Video Data
N/A	M18399A001 Ribbon Cable	Touch Controller Board	CN1	PCAP Ribbon Cable
N/A	M18399A001 Ribbon Cable	Touch Controller Board	CN2	PCAP Ribbon Cable

Figure 6-87: 15.6” LCD Connections (M15899A001)

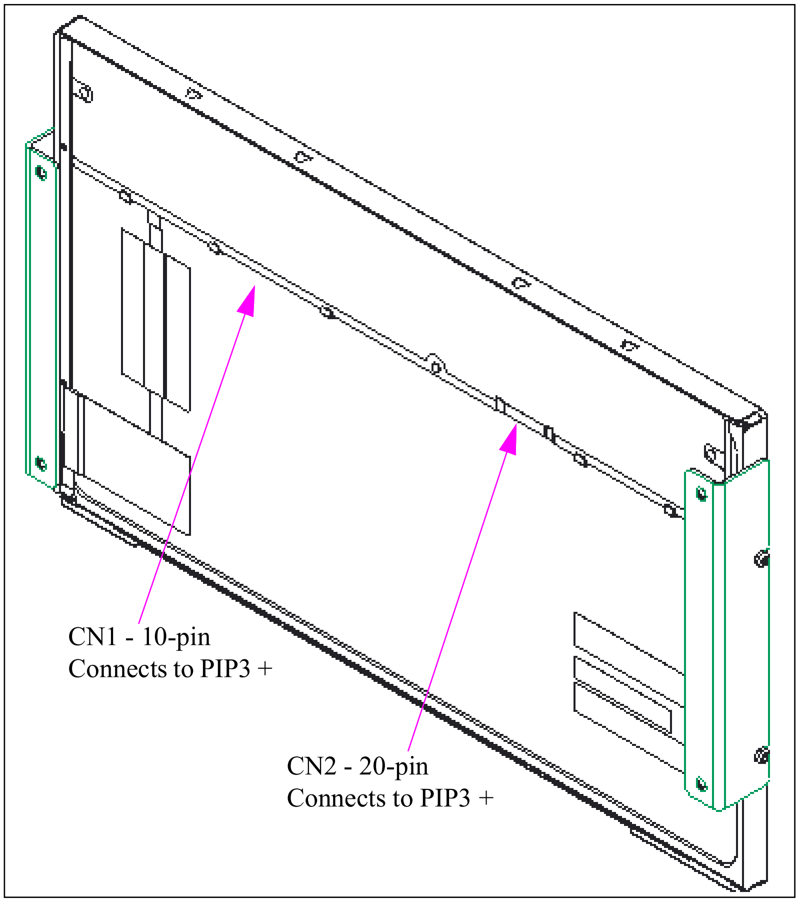
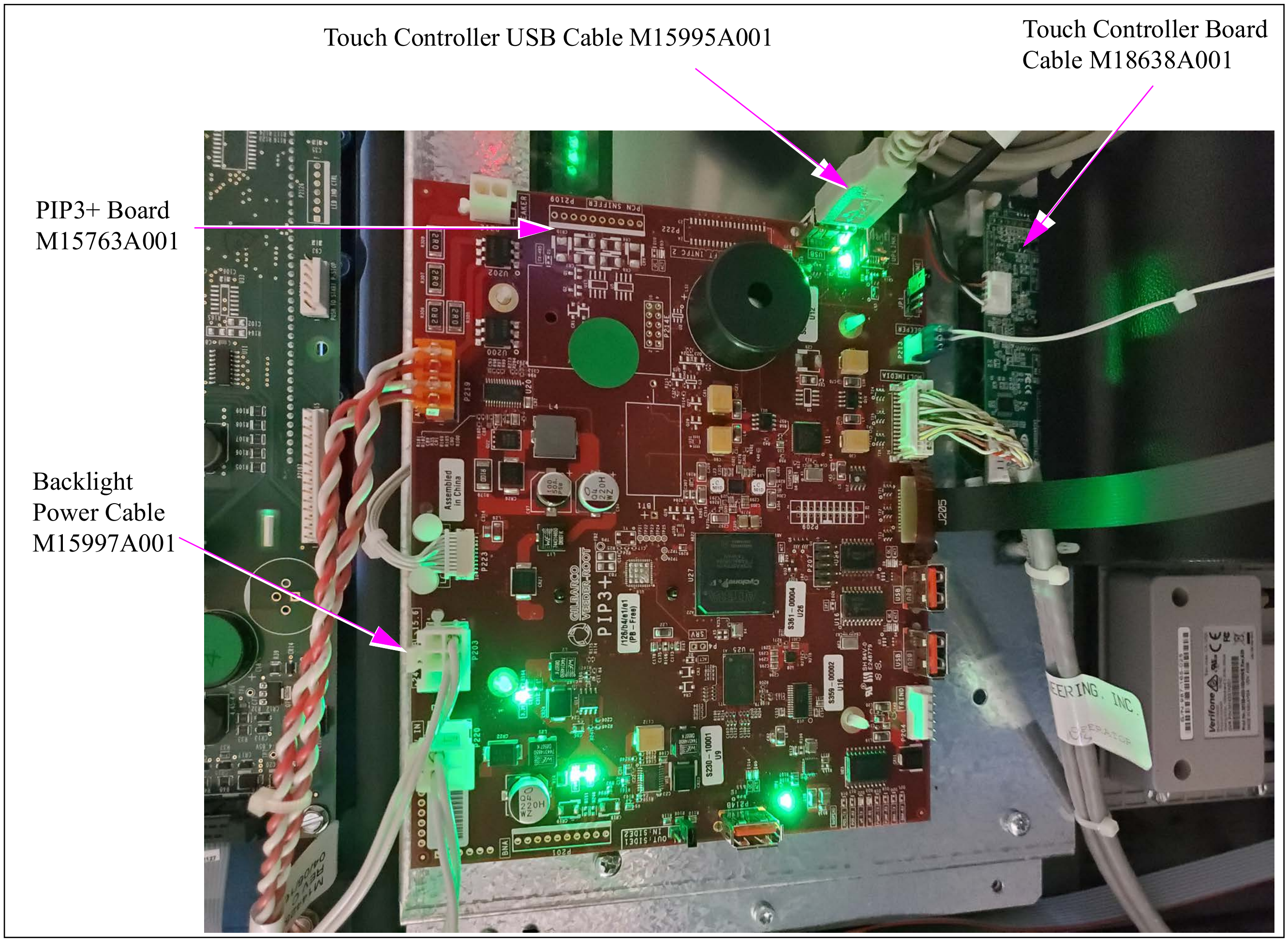


Figure 6-88: PIP3+ Board



CRIND Devices

For M3 (SPOT), M5 (FlexPay II), and M7 (FlexPay IV), refer to their respective service manuals.

The following table lists the content in this sub-section:

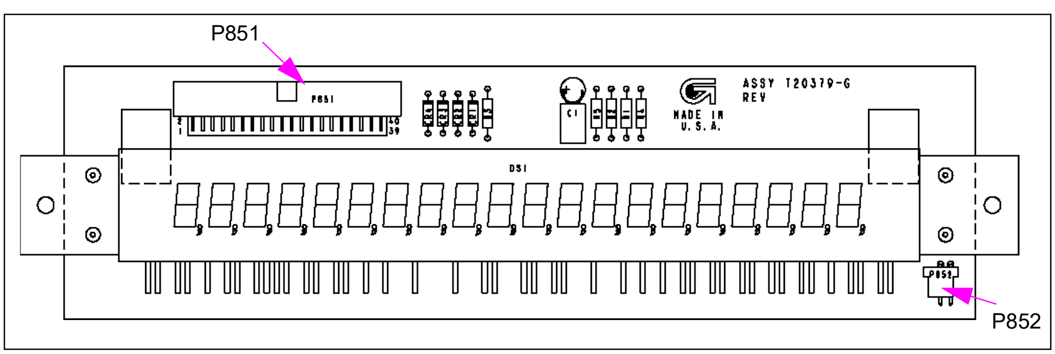
About CRIND Devices and Options

The CRIND device is a customer-activated POS device located at or near the dispenser. The CRIND device allows customers to activate dispensers and pay for fuel and merchandise without help from the station attendants. The CRIND device can be set up to accept Credit or Debit Cards. The CRIND device communicates with the site controller, which in turn communicates with Credit, and/or Debit networks. The CRIND device display prompts the customer through the transaction and depending on type, may display promotional information. The single-line CRIND displays prompts and messages on a 14 segment, 20 character display module.

Options to the CRIND include cash acceptor, that allows the customer to pay for fuel by inserting cash into the acceptance slot and the Barcode Reader option that allows customers to automatically authorize customer price discounts on CRIND device-equipped units using a barcode tag. The CRIND device printer prints customer receipts and in some cases, promotional information.

Single-line CRIND Display (T20379)

Figure 6-89: Single-line CRIND Display (T20379)



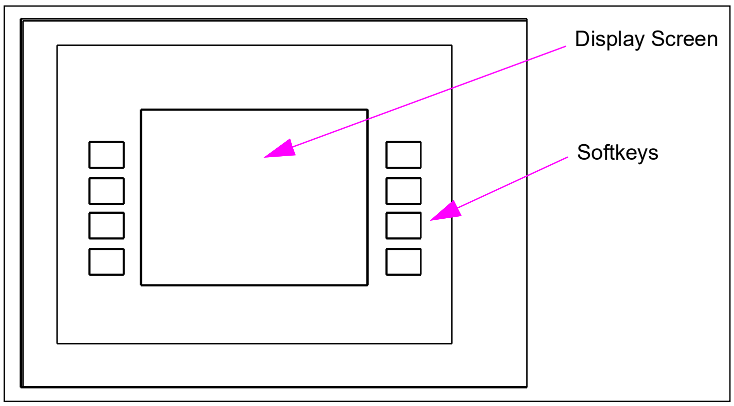
Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P851 (40-pin)	M01668A004	CRIND Regulator PCA M02705 (previously T20306)	P252
P852 (3-pin)		CRIND Logic Z180 PCA (T17764)	P558

Monochrome Display - Encore 300/500

The monochrome display screen displays multiline prompts, advertising messages, and static graphics on a monochrome LCD screen. The outside view of the display is the same for Encore 300/500 and Eclipse.

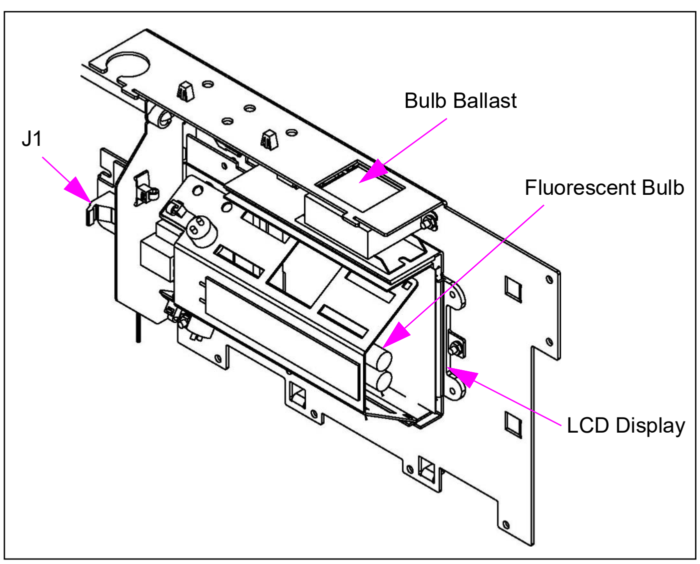
Figure 6-90: Monochrome Glass with Softkeys (M00696A001)



M01084AXXX

Monochrome displays are used in Encore 300/500 and Eclipse series units. The Monochrome display assemblies built before July 14, 2001 used replaceable fluorescent backlight bulbs to illuminate the display. M01653A001 uses LEDs that are not replaceable. Contrast adjustments are made using the CRIND diagnostic card.

Figure 6-91: Monochrome Display Assembly (M01084AXXX)



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
J1 (34-pin)	M01668A002	Monochrome PCA (T1950-G2)	P2015A for Side 1, P2015B for Side 2.
J5 (2-pin)		CRIND Regulator PCA (T20304)	P563 for Side 1, P566 for Side 2.

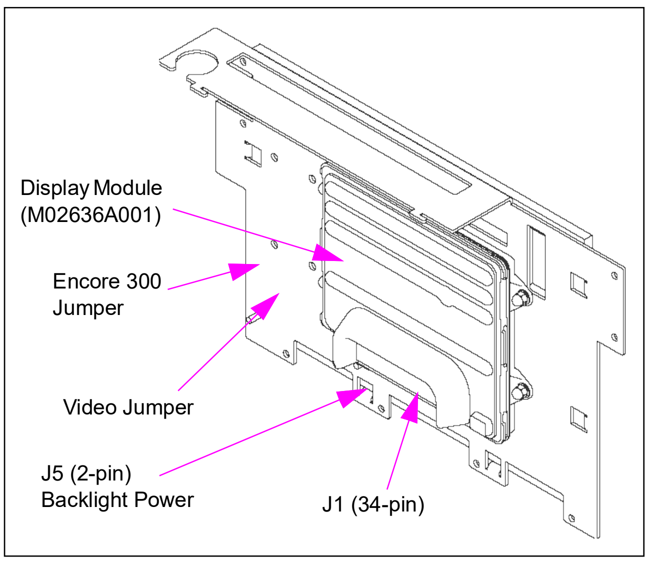
Monochrome Display (M01653A001)

The monochrome display board displays multiline prompts, advertising messages, and static graphics on a monochrome LCD screen in Encore 300/500 and Eclipse series units. Earlier version used replaceable fluorescent backlight bulbs to illuminate the display. The LEDs in this assembly are not replaceable. Contrast adjustments are made using the CRIND diagnostic card.

Service Tips

- Rain shield inside the door is required for assembly.
- Do not remove the plastic shield over the display.
- Order Display Module (M02636A001) for service.

Figure 6-92: Monochrome Display Assembly (M01653A001)



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
J1 (34-pin)	M00553A001	CCN (M01753A001)	P1302 for Side 1, P1303 for Side 2
J5 (2-pin)	R02778-G1 to M00517A005	24 V Power Supply PCA (M02774A001)	P3502A

Jumpers

Jumper	Setting	Function
J3	A	Normal image
J3	B	Image rotated 180° (inverted)

Connector PIN Functions and Voltages

J1 (34-pin) Data Interface (pin assignment not available)

J5 (2-pin)

Pin #	Description	Input/Output Voltage
1	Backlight Power	+24 V Regulated
2	Ground, Backlight Power Supply	

CRIND Device Card Reader

Production of the single head card reader was discontinued in January 2002. Use M02139K00X Kit to replace. Refer to *PT-1936 Encore Series Pumps and Dispensers Illustrated Parts Manual*.

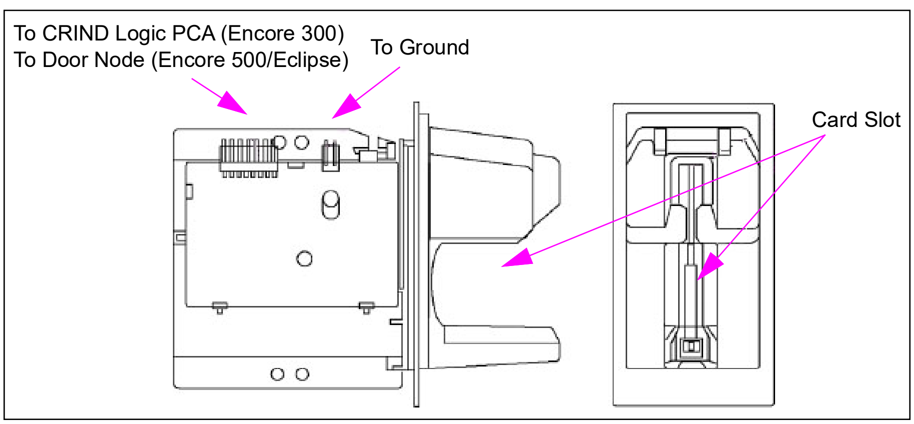
The dual head card reader option provides two magnetic heads located at the top of the card reader and directly opposite each other. This provides the capability for reading the card from either side, as long as the magnetic strip is facing up.

Encore 300 models that contain a CRIND logic board will require an upgrade to the CRIND BIOS when switching from single to dual head card reader. For details of the version of CRIND BIOS required to support the dual head card reader, contact the Gilbarco Call Center at 1-800-800-7498.

Service Tips

- Card readers must be cleaned periodically (weekly or as required) using a Cleaning Card (G11482). A dirty reader can result in reading problems. Always attempt to clean reader before deciding to replace it.
- CRIND diagnostics can be used to test card reader performance and determine statistical data useful in card reader service.

Figure 6-93: Dual Head CRIND Card Reader



Board Connections and Cables

Model	Connector#	Through Cable	To Board	At Connector#
Encore 300	7-pin	M01555A001	Door Node (M00062A001)	P254 (9-pin)
	CR (6-pin)	M02134A001	GND	Ring Terminal
Encore 500, Eclipse	7-pin	M00501A001	Door Node	P2102 (9-pin)
			GND	Ring Terminal

Cleaning Card Reader Instructions

Use a Gilbarco Cleaning Card (Q11482) to clean the card reader frequently (Gilbarco recommends cleaning the card reader weekly with a new cleaning card). Also, use a cleaning card to clean the card reader if cards are not being read in normal operation. Reorder cleaning cards from your Gilbarco distributor.

Cash Acceptor

The Cash Acceptor option for CRIND rs accepts bills and extends pay-at-the-pump convenience to cash customers. Each bill inserted and accepted by the CRIND device is recorded in the CRIND log file. The CRIND log file prints the bills accepted at the top of the receipt. Cash acceptor assembly part number M01271A002 is used in Encore 300 and M01271A004 is used in Encore 500 and Eclipse. In rare cases, a customer may dispute the amount of money they inserted into the cash acceptor. Use the G-SITE system records to help settle the dispute or to get cash acceptor information beneficial to site management.

CAUTION

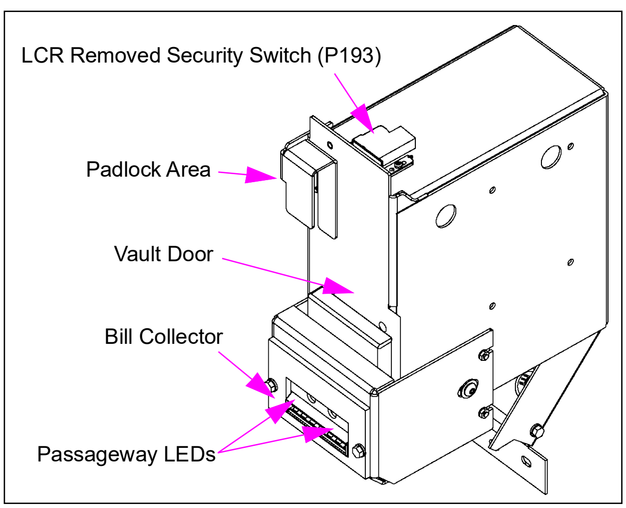
Do not open the electronics cabinet to change paper, to remove cash acceptor cassettes, or to perform any other tasks while it is raining. The moisture from the rain can damage the unit.

Service Tips

- Technicians must always request the station manager to empty the cash acceptor cassette before servicing the mechanism.
- CRIND diagnostics can be used to test cash acceptor performance and to determine statistical data.

Cash Acceptor - Encore 300

Figure 6-94: Cash Acceptor Assembly (M01271A002)

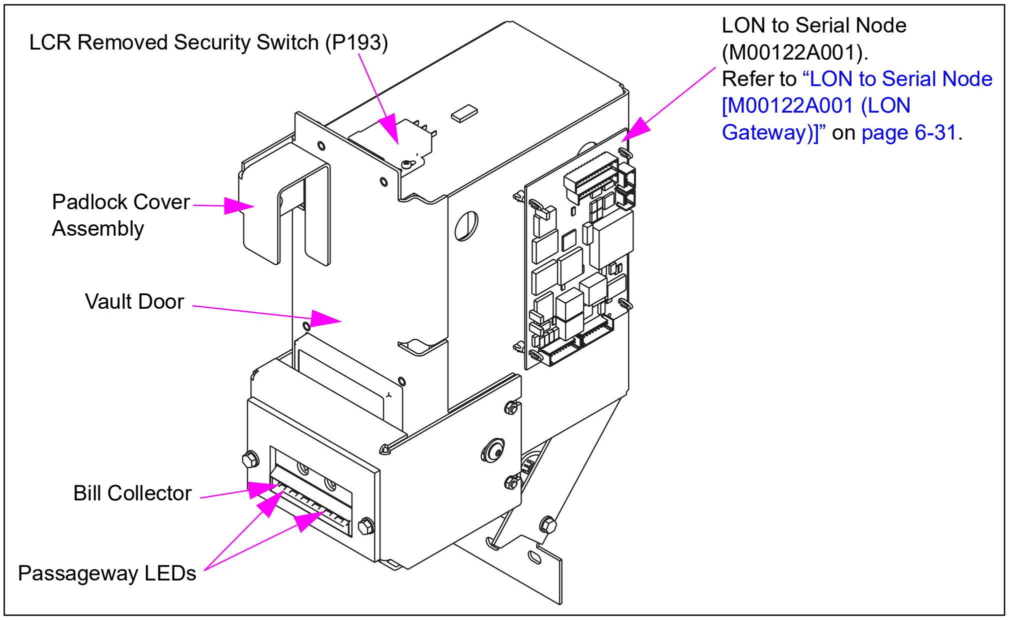


Connections and Cables (Encore 300)

Model	Connector#	Through Cable	To Board	At Connector#
Encore 300	P193 (2-pin)	M02211A001	CRIND Logic PCA (T17764-G3/G4)	P269 (10-pin)
	P1 (30-pin)	M02154A001 to M02211A001		
	J115 (2-pin)	M02160A002 to M01650A001	Monochrome PCA (T19501-G2)	P192 (2-pin)
	PWR (3-pin)	M02367A004	Power Supply (M02274A001)	Cash A/B (3-pin)
	Line Filter (2-pin)	M02154A002	Cable M02211A001	P1991 (4-pin)

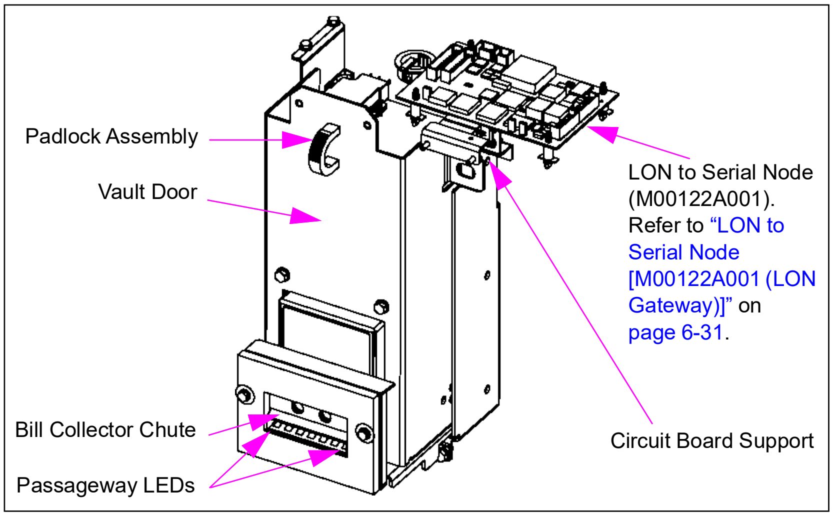
Cash Acceptor - Encore 500

Figure 6-95: Cash Acceptor Assembly (M01271A001)



Cash Acceptor - Eclipse

Figure 6-96: Cash Acceptor Assembly (M01403A001)



Connections and Cables (Encore 500 and Eclipse)

Model	Connector#	Through Cable	To Board	At Connector#
Encore 500, Eclipse	J1 (30-pin)	M02154A001 to M02153A001	Serial to LON Node (M00122A001)	P1508 (13-pin)
	J115 (2-pin)	M02160A001 to M02152A001		P1510 (5-pin)
	J193 (2-pin)	M02152A001		P193 (2-pin)
	Line Filter	M02154A001 to M02153A001		P1508 (13-pin)
	PWR (3-pin)	M02367A002	Power Supply Terminal Block	Cash A/B (3-pin)

Cash Acceptor Troubleshooting

Use the following table to troubleshoot cash acceptor problems. If you cannot resolve the problem, contact your local service provider.

Problem	Possible Cause	Corrective Action
Cash acceptor does not operate or stack bills when power is applied.	• Unit is not getting power (no LEDS lit inside passageway at door). • Unit is in calibration mode.	• Check AC power to cash acceptor. • Ensure that all cash acceptor switches are off.
Cash Acceptor does not give Credit or register accepted bills.	• The cash acceptor is in "test/pulse" mode. • The cash acceptor is defective.	• Take cash acceptor out of test mode. • Replace the cash acceptor.
Cash acceptor will not pull in a bill (motor does not start).	• The cassette is not installed. • The cassette is installed incorrectly. • The cash acceptor is defective. • The CRIND receipt printer is low on paper or out of paper.	• Ensure that the cassette Is properly installed. • Ensure that all locks are installed. • "REMOVE"/"INSTALL" latch must be turned completely to the "INSTALL" position and not beyond on the LRC. • Ensure that the CRIND receipt printer has paper.

Problem	Possible Cause	Corrective Action
Cash acceptor has excessive number of bill jams.	• The cassette is installed incorrectly. • There are bill pieces in the Cash Acceptor.	• Ensure that the cassette Is properly installed. • Ensure that all locks are installed. • “REMOVE”/”INSTALL” Latch must be turned completely to the “INSTALL” position and not beyond on the LRC. • Remove any bill pieces in the cash acceptor. • If none of the above work, try a different cassette (spare or from another unit). • If this is a recurring problem
Cash acceptor rejects all bills.	• The cash acceptor had a failure of one or more components. • The cash acceptor is not properly programmed.	• Verify if all cash acceptor switches are correctly set. • Cash acceptor requires calibration.
“General Fault” displays on G-SITE screen.	The cash acceptor is too unstable to continue operation.	Power-fail the dispenser.

CRIND Printer Module

The CRIND printer and printer node in Encore 500 and Eclipse series units is an assembly module with the node portion being connected to the LON bus. The printer node provides power to the thermal printer contained inside the unit and is designed to move information from the LON to the printer and control the printer. There is one jumper on the printer node to determine side 1 or side 2. The Encore 500 and Eclipse printer module assembly are interchangeable by relocating the printer node enclosure. The Encore 300 has a printer interface board attached to the printer interface mechanism similar to Encore 500 and Eclipse. The Encore 300 printer is not interchangeable with the Encore 500 and Eclipse printer.

Service Tips

- Printer and circuit boards are only replaceable as a unit.
- Paper size and type are critical for proper printer operation.
- Paper that does not conform may shorten the printer life and cause service problems such as jams and poor print quality. For recommended suppliers, refer to the following table.
- Use Printer Cleaning Card (Q13400), when print quality is poor.
- Intermittent cable connections can cause printer problems. Ensure that the cable connections are secure and tight.
- The anti-static brush bristles under the roller access cover must touch the paper lightly to prevent paper jams.
- Paper jams may occur if the roller access cover is not closed completely.
- Printer chute kits are available for printers manufactured before March 2003 that have jamming problems.

Figure 6-97: CRIND Printer Assembly Module - Encore 300

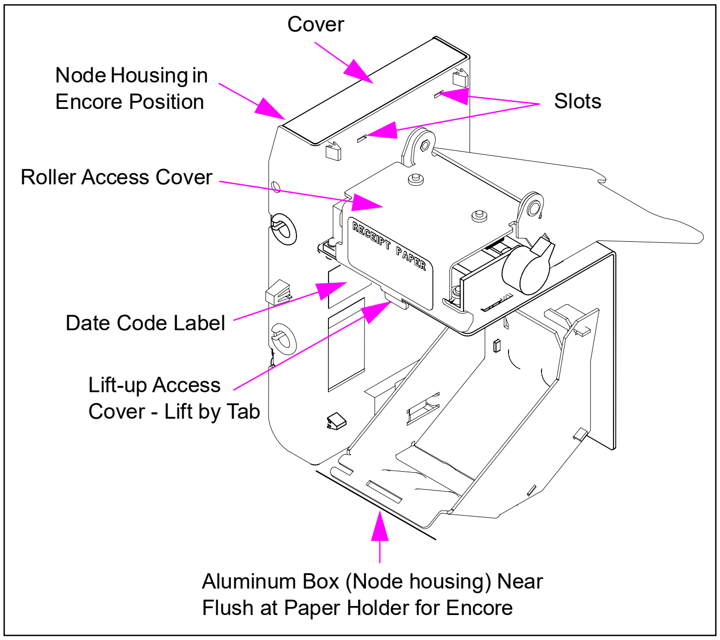


Figure 6-98: CRIND Printer Assembly Module - Eclipse

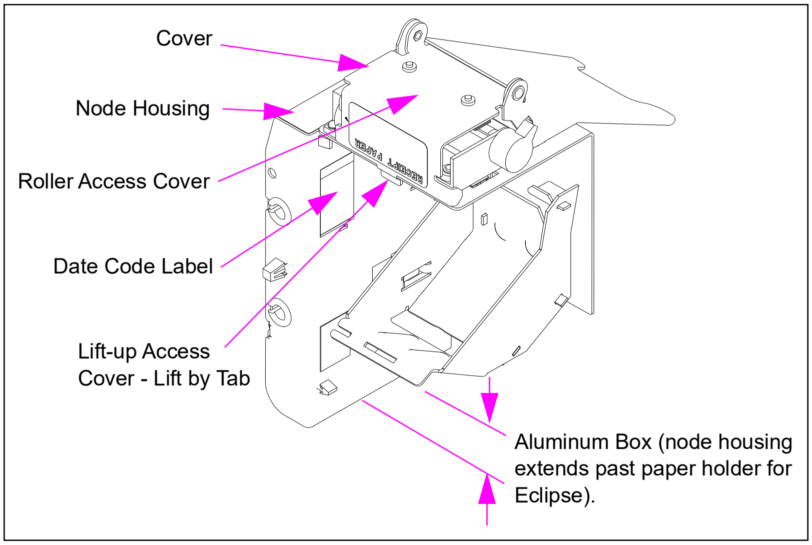
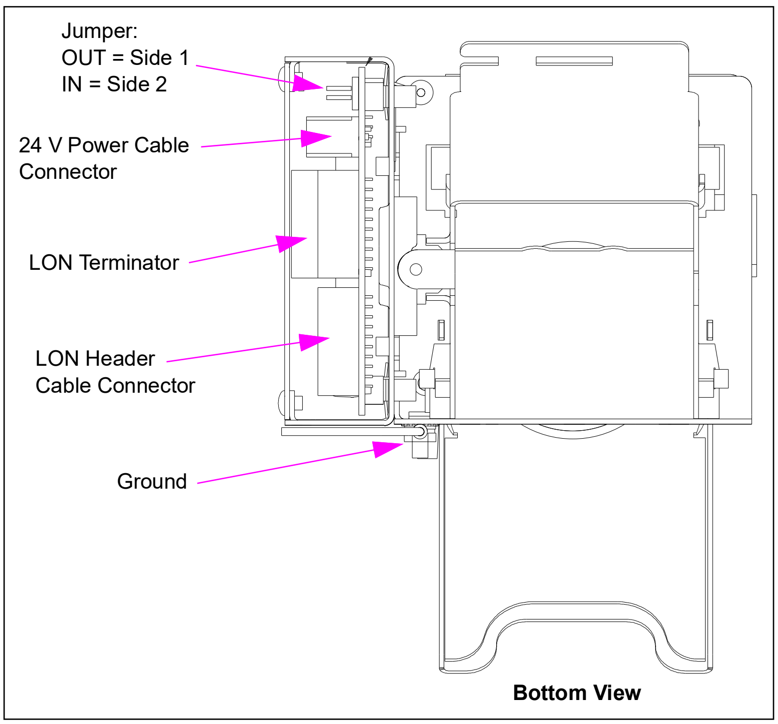


Figure 6-99: Printer LON Board Connections for Encore 500 and Eclipse



Jumper Settings

Jumper	For This Side	Jumper Setting Is
JP-1	1	OUT
	2	IN

Replacing CRIND Receipt Printer Paper

It is important to keep the CRIND receipt printer supplied with paper. If the CRIND printer is low on paper, or out of paper, the cash acceptor will not accept bills.

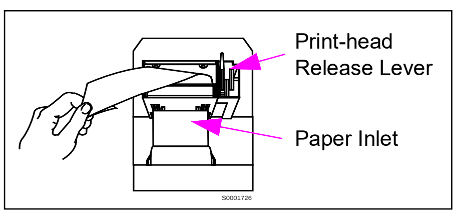
Note: You may encounter different printers. Follow the paper change instructions on the printer.

Replacing the CRIND Receipt Printer Paper

To replace the CRIND receipt printer paper, proceed as follows:

1. Remove the paper roll core.
2. Raise roller access cover and actuate print-head release lever.
3. Remove the paper from cutter and chute.
4. Feed new paper into paper inlet until paper appears between roller bars.
5. Align and feed paper through printer rollers until paper protrudes slightly past end of chute.
6. Lower the print-head release lever.
7. Fully close the roller access cover.
8. Ensure the proper installation by printing test receipts.
   Note: Certain models will automatically print a test receipt. Otherwise, use the diagnostic card to print a test receipt.

Figure 6-100: CRIND Device Receipt Printer



Recommended Suppliers

Supplier	Part Number
Fujitsu®	FTP-202PE011-64M
NCR	SLW690
Wallace	1146

Cleaning Printer Instructions

Use Printer Cleaning Cards (Q13400) periodically to clean the printer (Gilbarco recommends cleaning the printer every three months). Cleaning the printer will eliminate most printing quality problems, such as light or faint printing.

To clean the printer, proceed as follows:

1 Raise the roller access cover and activate the print-head release lever.

2 Remove the paper from the printer.

3 Insert a Printer Cleaning Card (Q13400) into the printer inlet and rollers.

4 Close the print-head release lever.

5 Manually advance the cleaning card through the roller bars by moving the gears on the side.

6 Replace the paper roll and reinstall receipt paper as described in "Replacing CRIND Receipt Printer Paper" on page 6-143.

Barcode Scanner

When interfaced with a POS system, the barcode scanner option allows customers to automatically authorize customer price discounts on CRIND device-equipped units using a barcode tag.

Encore 300 uses existing CRIND Z-180 Logic Board (T17764-XX). For Encore 500 and Eclipse units, the scanner is controlled by a LON/serial node per unit side. The Encore 500 and Eclipse barcode scanners use CRIND BIOS software version 2.0.30 or later to operate properly. The Encore 300 scanners require CRIND BIOS software version 60.7.20 or later for generic dispensers, and version 25.2.40 or later for MOC.

⚠ WARNING

Laser light used in this product may present an eye hazard during service.
Direct viewing of laser light may damage eyesight.
Avoid long term viewing of direct laser light.

Figure 6-101: Barcode Scanner Assembly (M00969A001)

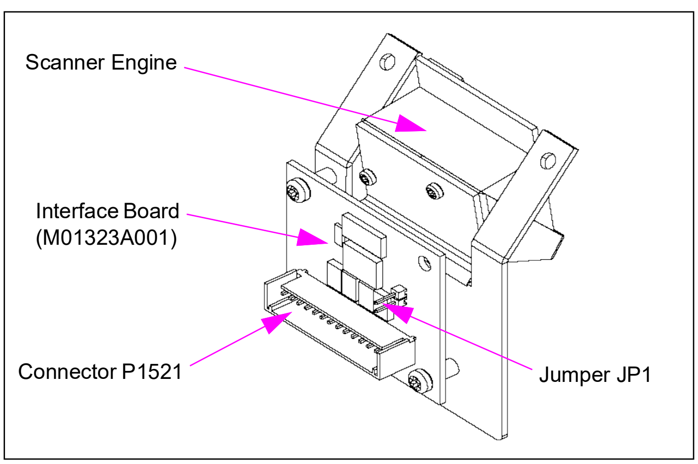


Figure 6-102: Advantage/Encore 300 Block Diagram

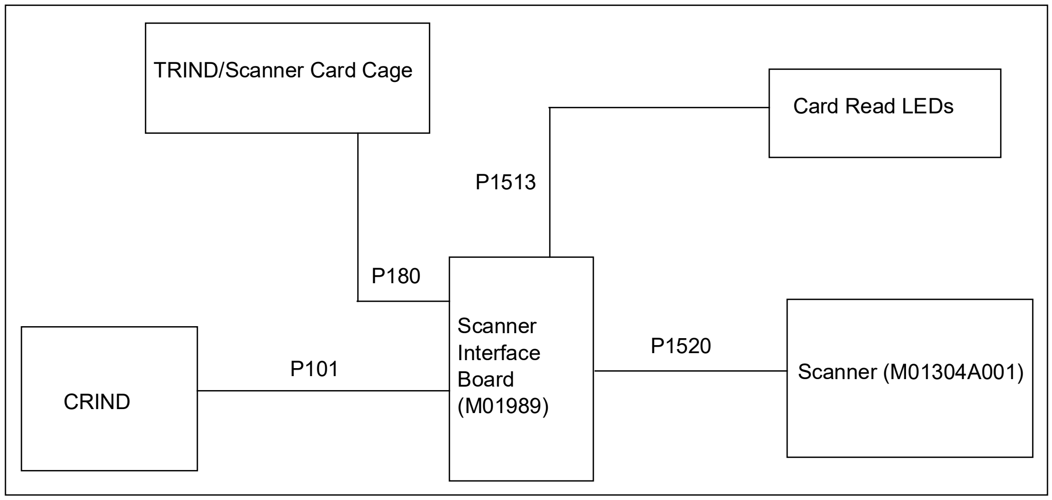
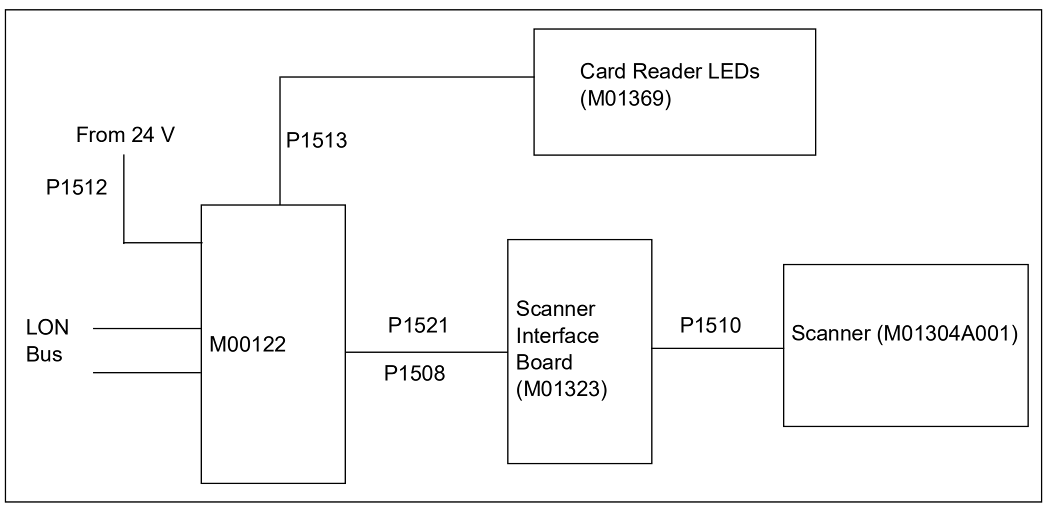


Figure 6-103: Encore 500/500 S and Eclipse Block diagram



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P1521 (14-pin)	P01378A001	Serial to LON Node (M00122A001)	P1509 (14-pin)

Jumper Settings

Jumper	
JP1	Turns scan laser on for testing (normally not installed)

J-box

The following table lists the contents in this sub-section:

About J-box

For Encore units, conduits may run into the electronics cabinet without the use of a J-box. Depending on configuration, the Encore series dispenser units may ship without a J-box or field conduit. Some retrofit units will require a junction box due to limited wire length available from sub-up conduit; in this case, the installer will have already installed a J-box to splice wires. J-box, when installed, are Class 1 Division 1 explosion proof. The J-box is mounted on side 1 of unit. Eclipse dispensers contain a factory supplied J-box on side 1. The stub up for this box is located on side 2 of the unit. Junction boxes must be secured to the dispenser's frame

J-box retrofit kits are typically installed when existing station wiring is adapted to an Encore unit. For more information, refer to *MDE-4084 Encore Junction Box Retrofit Kit M01483K00X Installation Instructions*. Check conduit routing through the air gap for compliance-no unsealed holes are allowed in the air gap plates. For specific information on junction boxes, refer to *MDE-3985 Encore Installation Manual* and *MDE-3986 Eclipse Installation Manual*.

Encore Series J-box

Figure 6-104: J-box in Encore Dispenser

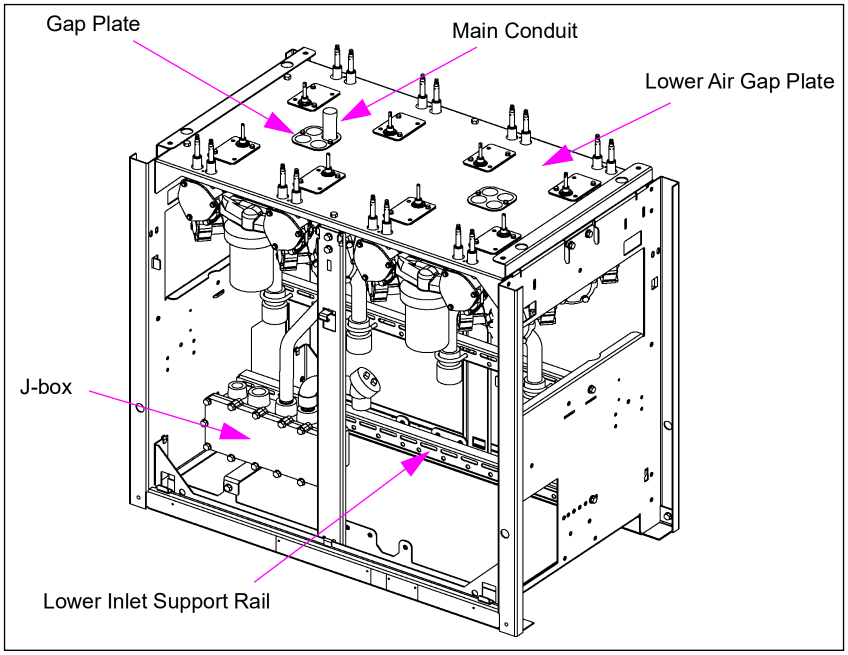
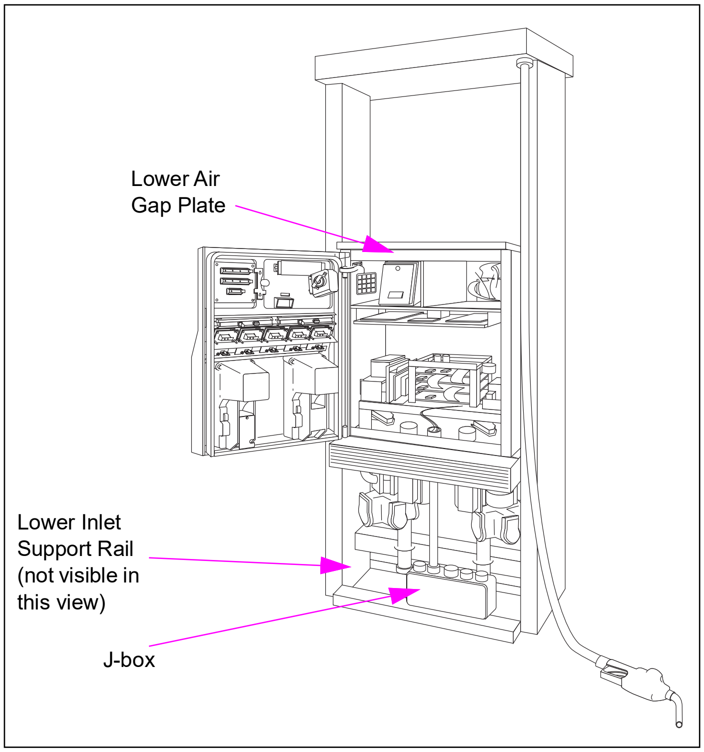
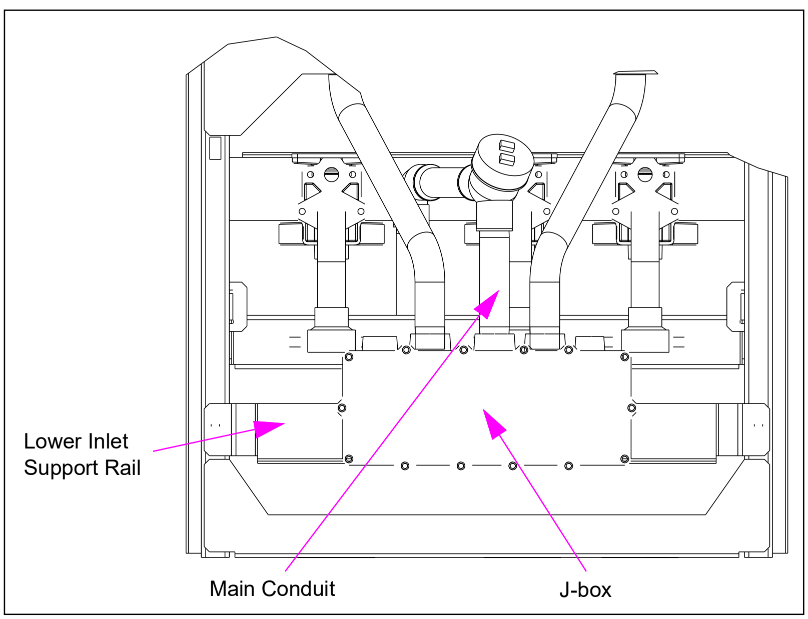


Figure 6-105: J-box in Encore Pump



Eclipse Series J-box

Figure 6-106: J-box in Eclipse Dispenser



VaporVac Electrical Components

The following table lists the contents in this sub-section:

About VaporVac Electronic Components

The fueling customer places the VaporVac nozzle in the vehicle's fill pipe. The VaporVac Controller uses the pulse signals from the meters to track real-time flow rates and determine the proper vacuum to remove vapors from the vehicle tank. Vapors are returned to the site's fuel storage tank.

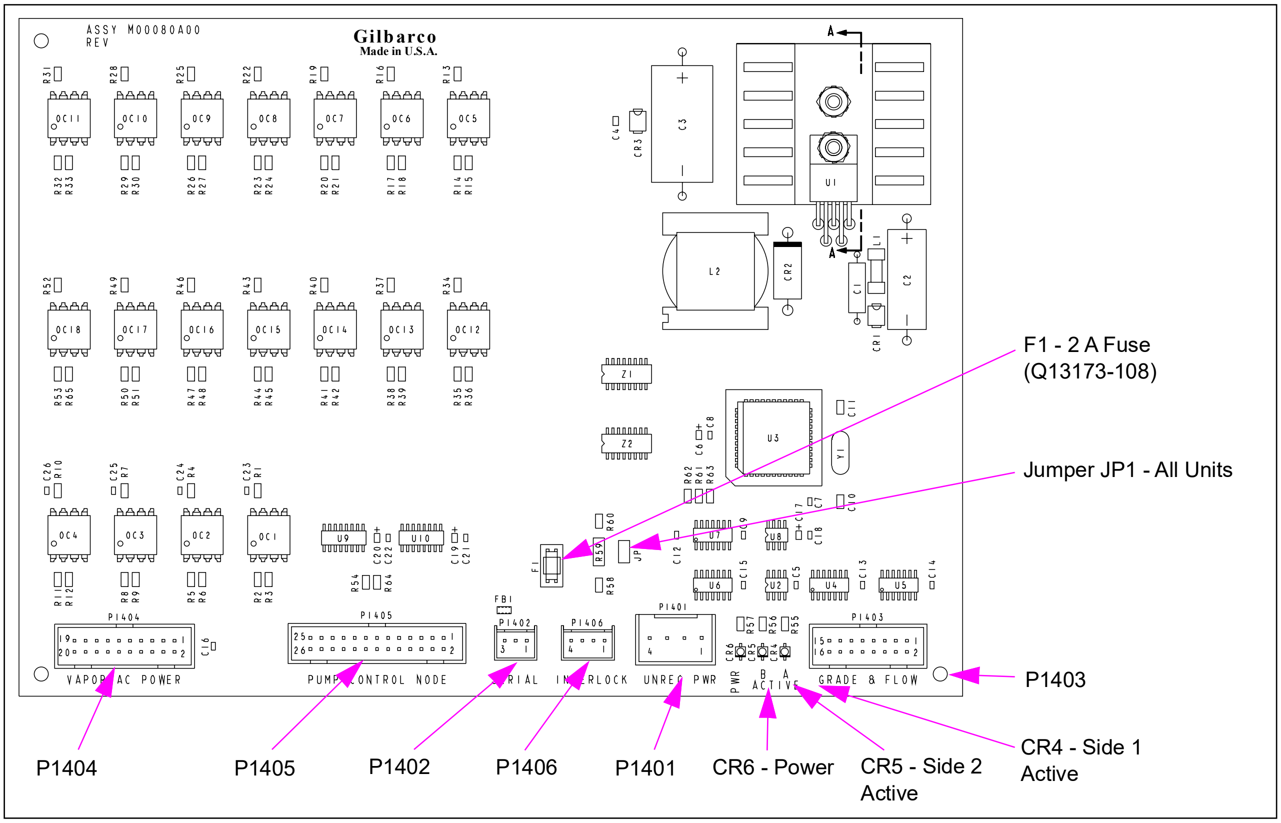
The PCAs, power supply, and card cage make up the VaporVac electronics module. The following table lists modules, PCAs, and power supply associated with each:

Model	Voltage	Module No.	PCA No.	Power Supply No.
Eclipse				
Encore 300	115 VAC/60 Hz without vapor valves	M01776A004	• T18018-G1 • T19401-G2	M01775A001
	220 VAC/50 Hz without vapor valves	M01776A006		M01775A002
Encore 500	115 VAC/60 Hz without vapor valves	M01776A003	• M00080A001 • T18018-G1 • T19401-G2	M01775A001
	220 VAC/50 Hz without vapor valves	M01776A005		M01775A002

VaporVac Isolation PCA (M00080)

The VaporVac Isolation PCB is used to interface the VaporVac system with the pump node.

Figure 6-107: VaporVac Isolation Board



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P1401 (4-pin)	R19672-G1	VaporVac Motor Drive PCA (T18018)	P1005 (4-pin)
P1403 (4-pin)	R19665-G1	VaporVac Controller PCA (T19401)	P1103 (Interface)
P1404 (20-pin)	R17995-G4		P1104 (20-pin)
P1405 (26-pin)	M00549A002 (Encore) M01036A001 (Eclipse)	Pump Control Node-1 (M00056A001)	P1110 (26-pin)
P1406 (4-pin)	R19669-G1	VaporVac Controller PCA (T19401)	P1106 (Interlock)

LED Indicators

LED	Status/Indicates
CR4	OFF = Side 1 not authorized / ON = Side 1 authorized / Blinking = Side 1 dispensing fuel
CR5	OFF = Side 2 not authorized / ON = Side 2 authorized / Blinking = Side 1 dispensing fuel
CR6	OFF = no power to PCA / ON = power to PCA

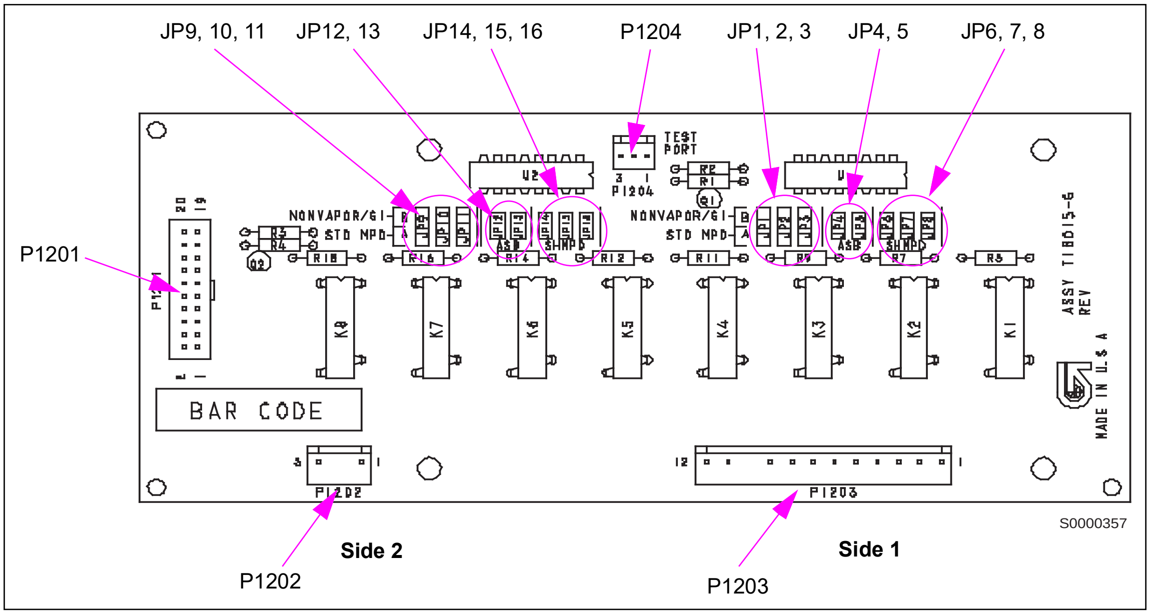
VaporVac Valve Driver PCA (T18015)

When used in Encore 300/500 and Eclipse units, this board receives 12 VDC signals from the VaporVac Controller PCA (T19401) and switches 120 VAC to the Vapor Valves (T17909).

Service Tip

Units manufactured after July 2002 do not use solenoid valves on this board.

Figure 6-108: VaporVac Valve Driver Board



Board Connections and Cables

Multi-Hose + 1 Units (Encore Series Units Only)

Unit Type		Side 1	Side 2
9: 3 Grade, 3 Hose Blender + 1	Non Diesel + 1	N/A	N/A
	Diesel + 1	JP1A, JP2A, JP3A	JP9A, JP10A, JP11A

Connector#	Through Cable	To Board	At Connector#
P1201 (20-pin)	R19666-G1	T19401 VaporVac Controller PCA	P1101 (20-pin)
P1202 (3-pin)		T18162 VaporVac Power Supply Assembly	3-pin
P1203 (12-pin)	• Encore: T17928-G2 cable to M00845A001 Cable at J/P1030, to M00830A001 Cable at J/P2030 to P1 • Eclipse: M00981A001	VaporVac Valves	• Encore: P1 on Valve(s) J11, J12, J13, J21, J22, J23 • Eclipse: J13, J21

Jumper Settings

Multi-hose Units (Encore Series Units Only)

Unit Type	Diesel On Grade (If Applicable)	Side 1	Side 2
2: 2 Grade, 2 Hose MPD	N/A	JP1A, JP2A	JP9A, JP10A
	1	JP2A	JP10A
	2	JP1A	JP9A
3: 3 Grade, 3 Hose MPD	N/A	JP1A, JP2A, JP3A	JP9A, JP10A, JP11A
	1	JP2A, JP3A	JP10A, JP11A
	2	JP1A, JP3A	JP9A, JP11A
	3	JP1A, JP2A	JP9A, JP10A
4: 4 Grade, 4 Hose MPD	N/A	N/A	N/A
	1	JP1B, JP2B, JP3B	JP9B, JP10B, JP11B
	2	JP1A, JP2B, JP3B	JP9A, JP10B, JP11B
	3	JP1A, JP2A, JP3B	JP9A, JP10A, JP11B
	4	JP1A, JP2A, JP3A	JP9A, JP10A, JP11A
8: 3 Grade, 3 Hose Blender	N/A	JP1A, JP2A, JP3A	JP9A, JP10A, JP11A
10: 3 Grade, 3 Hose 1 Meter MPD	N/A	JP1A, JP2A, JP3A	JP9A, JP10A, JP11A

Single-hose Units (Encore and Eclipse Series Units)

Unit Type	Side 1	Side 2
1: 1 Grade, 1 Hose MPD 5: 2 Grade, 1 Hose MPD 6: 3 Grade, 1 Hose MPD 11: 3 Grade, 1 Hose 1 Meter MPD 12: 2 Grade, 1 Hose Blender 13: 3 Grade, 1 Hose Blender 14: 4 Grade, 1 Hose Blender 15: 5 Grade, 1 Hose Blender	JP1A	JP9A

Single-hose + 1 Units (Encore and Eclipse Series Units)

Unit Type		Side 1	Side 2
7: 3 Grade, 1 Hose + 1 MPD 16: 2 Grade, 1 Hose Blender + 1 17: 3 Grade, 1 Hose Blender + 1 18: 4 Grade, 1 Hose Blender + 1 19: 5 Grade, 1 Hose Blender + 1	Non Diesel + 1*	JP1A, JP2A	JP9A, JP10A
	Diesel + 1	JP1A	JP9A

Encore Series units only

Voltage Test Points

Test Point	Voltage
P1202, Pin 3	120 VAC Hot
P1202, Pin 1	120 VAC Neutral

VaporVac Motor Drive PCA (T18018)

This board mounts in the electronics cabinet card cage on factory built units. The motor drive board provides high current switching to drive the motors.

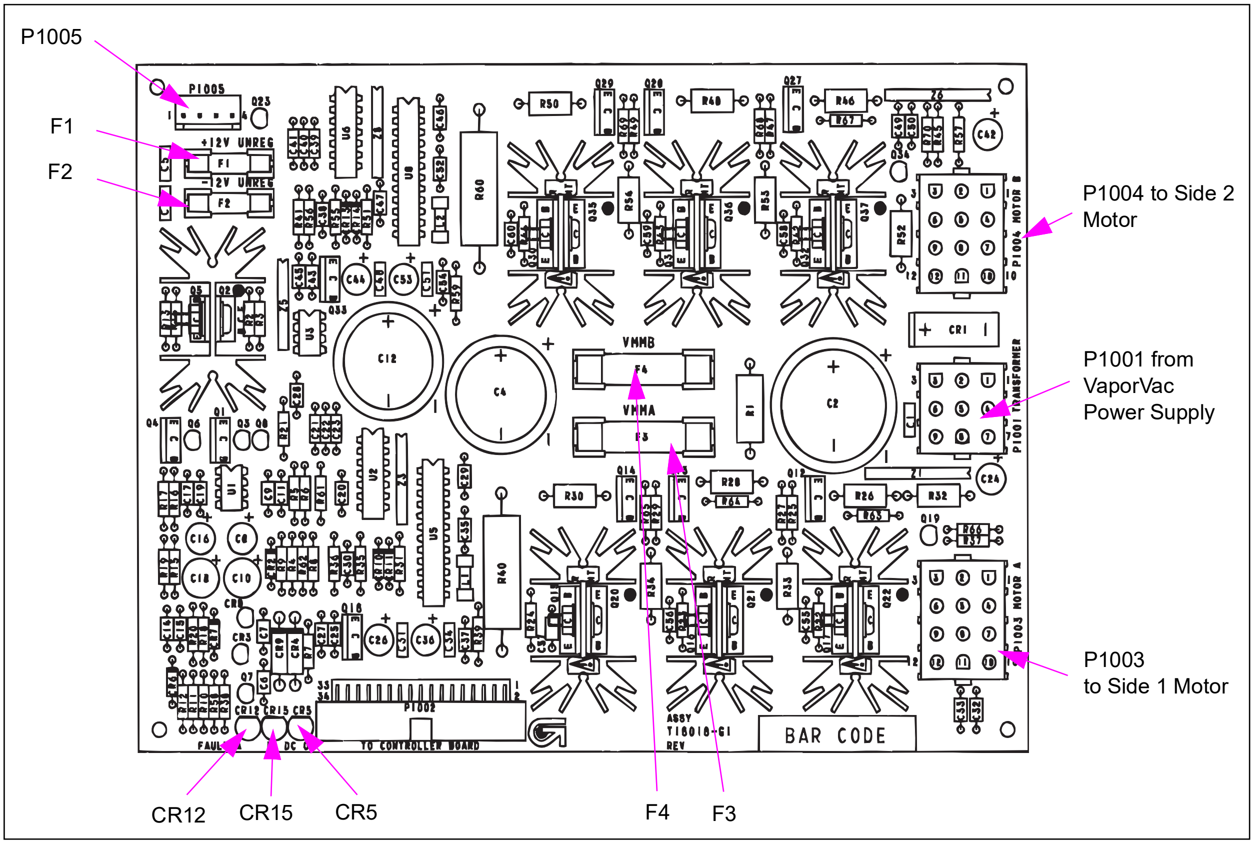
It receives the following voltages:

- +43 VDC and 25.2 VAC from the VaporVac power supply

It supplies the following voltages:

- +/-12 VDC (unregulated) to P1005
- +/-12 VDC (regulated) to the T19401 VaporVac Controller PCA
- 6.2 VDC to the T17942 VaporVac Pump Assemblies (for Hall effect sensors)
- 36-50 VDC (unregulated) to T17942 VaporVac Pump Assemblies

Figure 6-109: VaporVac Motor Drive Board (T18018)



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P1001 (9-pin)		VaporVac Power Supply Assembly (T18162)	
P1002 (34-pin)	R19667-G1	VaporVac Controller PCA (T19401)	P1102 (34-pin)
P1003 (12-pin)	• Encore: M00845A001 to M00830A001 (at J2003/P2003) to J3003 • Eclipse: M00980A001	VaporVac Pump Assemblies (Motor) Side 1 (T17942)	• Encore: J3003 • Eclipse: J11
P1004 (12-pin)	• Encore: M00845A001 to M00830A001 (at J2004/P2004) to J3004 • Eclipse: M00980A001	VaporVac Pump Assemblies (Motor) Side 2 (T17942)	• Encore: J3004 • Eclipse: J21
P1005 (4-pin)	R19672-G1	VaporVac Isolation PCA (M00080)	P1401 (4-pin)

LED Indicators

LED	Indicates
CR12 Fault A CR15 Fault B	ON = vacuum pump motors are off and fuel is not flowing. If LED is on or flickering with fuel flowing, look for problems in VaporVac pump/motor, motor drive board, or related wiring.
CR5 DC OK	ON = normal operation (+/-12 VDC for VaporVac controller board). If LED is off, look for problems in VaporVac motor drive board, controller board, or power supply.

Voltage Test Points

Test Point	Voltage
P1001, pins 3 and 6	43 VDC
P1001, pins 8 and 9	43 VDC return
P1001, pins 1 and 2	25.2 VAC
P1001, pin 4	GND
P1001, pin 7	0 VAC
P1005, pin 1	12 VDC (unregulated out)
P1005, pin 2	GND
P1005, pin 4	-12 VDC (unregulated out)

Fuses

Fuse	Rating
F1	2 A, 250 VAC Slo-Blo, (+12 VDC)
F2	2 A, 250 VAC Slo-Blo, (-12 VDC)
F3	15 A, 32 VAC, (for motor voltage A)
F4	15 A, 32 VAC, (for motor voltage B)

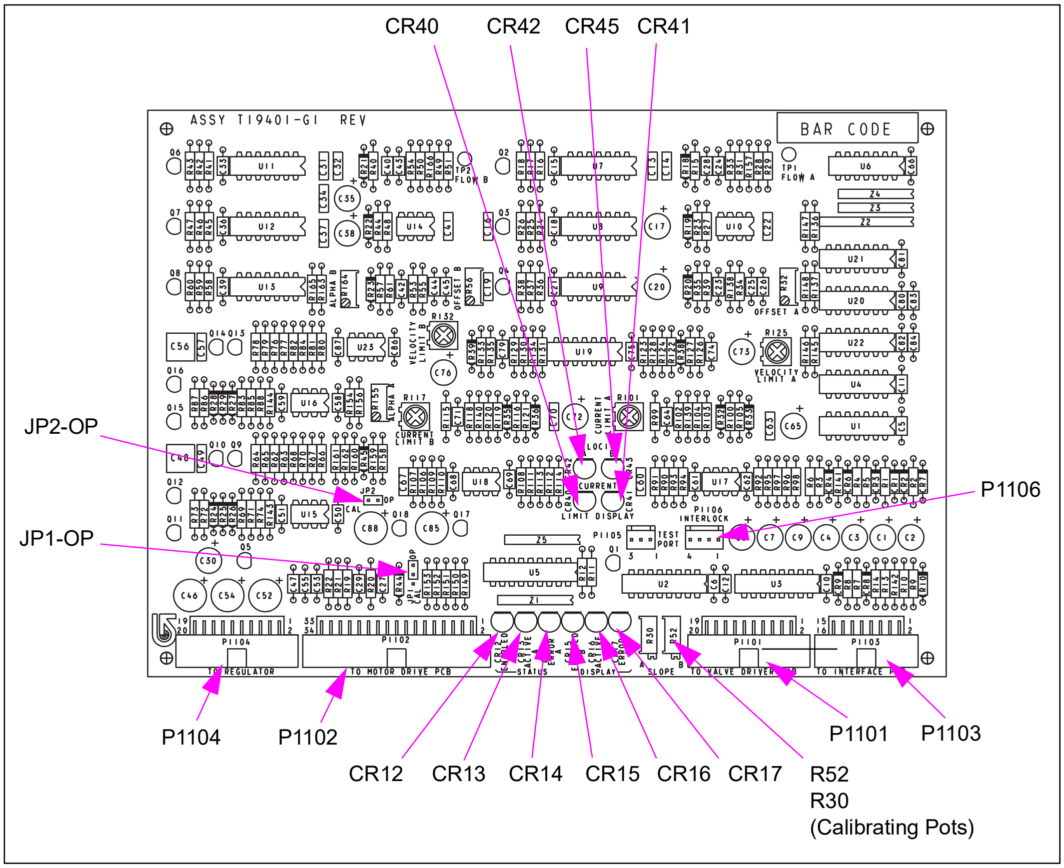
VaporVac Controller PCA (T19401)

Board controls the VaporVac system providing error checking for system status and controls.

It receives the following voltages:

- 5 VDC from the VaporVac Isolation PCA (M00080)
- +/- 12 VDC from the VaporVac Motor Drive PCA (T18018)

Figure 6-110: VaporVac Controller Board



Board Connections and Cables

Connector#	Through Cable	To Board	At Connector#
P1101 (20-pin)	R19666-G1	VaporVac Valve Driver PCA (T18015)	P1201
P1102 (34-pin)	R19667-G1	VaporVac Motor Drive PCA (T18018)	P1002
P1103 (16-pin)	R19665-G1	VaporVac Isolation PCA (M00080)	P1403 (4-pin)
P1104 (20-pin)	R17995-G4		P1404
P1106	R19699-G1		P1406

Jumper Settings

JP-1 and JP-2 are always in the ON position.

LED Indicators

LED	Indicates
CR12 ENABLED (side A) CR15 ENABLED (side B)	ON = normal operation. Monitors VaporVac Controller PCA (T19401) activation and fuel flow through pulser activity. If LED is off and pulser activates, check for problems in VaporVac controller, VaporVac Isolation PCA (M00080), or hydraulic interface boards or cabling.
CR13 ACTIVE (side A) CR16 ACTIVE (side B)	ON = normal operation. Monitors VaporVac Pump Assemblies (T17942) motor rotation. If LED is off and fuel flowing, check for problems in VaporVac pump/motor, VaporVac motor drive, or VaporVac controller boards or cabling.
CR14 ERROR (side A) CR17 ERROR (side B)	OFF = normal operation. Monitors errors in the system. If LED is on, check for problems in VaporVac pump/motor, VaporVac motor drive, or VaporVac controller boards or cabling.
CR40 (side A) CR41 (side B)	Brief flicker when motor changes speed = normal operation. Monitors excessive VaporVac motor current draw. If LED is on, check for problems in VaporVac/pump/motor, VaporVac motor drive, or VaporVac controller boards or cabling.
CR 42 (side A) CR 43 (side B)	ON = suspect unrestricted air purging. Look for problems in VaporVac pump/motor, VaporVac motor drive, or VaporVac controller boards or cabling.

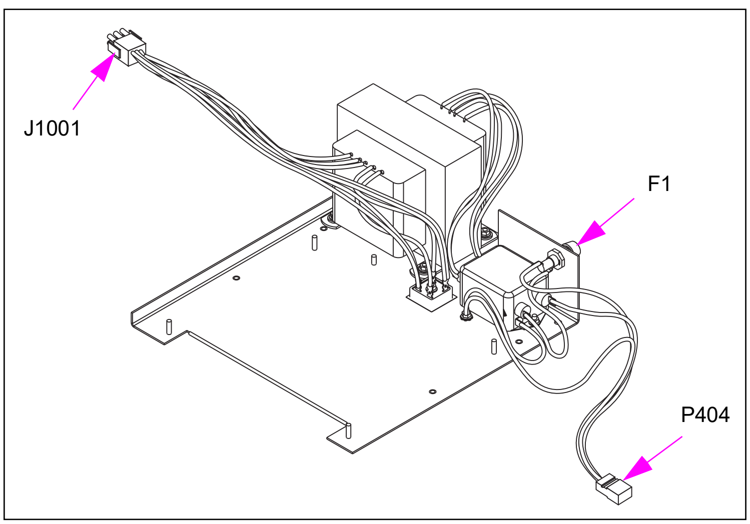
VaporVac Power Supplies

The VaporVac power supply is part of the VaporVac electronics module and provides power to the VaporVac PCAs.

VaporVac Power Supply (M01775A001 and M01775A002) for Encore

This power supply is part of the VaporVac electronics module.

Figure 6-111: VaporVac Power Supply (M01775A001)



Connections and Cables

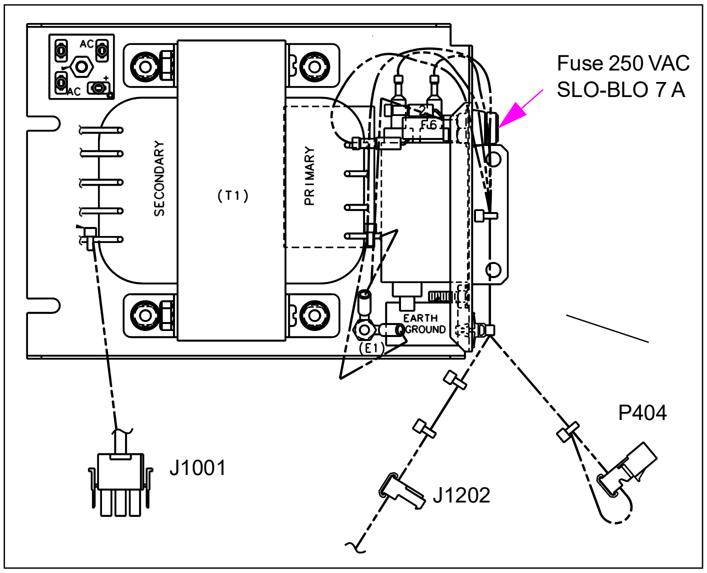
Connector#	Through Cable	To Board	At Connector#
J1001 (9-pin)	Power supply lead	VaporVac Motor Drive PCA (T18018-G1)	P1001 (9-pin)
P404 (3-pin)	M02710A002	Main power supply	J404 (3-pin)

Fuse

Part	Description
F1	Fuse 250 VAC SLO-BLO 7A

VaporVac Power Supply (T18162) for Eclipse

Figure 6-112: VaporVac Power Supply (T18162-G3)



Connections and Cables

Connector#	Through Cable	To Board	At Connector#
J1001 (9-pin)	Power supply lead	VaporVac Motor Drive PCA (T18018-G1)	P1001 (9-pin)
J1202 (3-pin)		VaporVac Valve Driver PCA (T18015)	P1202 (3-pin)
P404 (3-pin)	M02948A001	Main Power Supply	J404 (3-pin)

Fuse

Part	Description
F1	Fuse 250 VAC SLO-BLO 7 A

7 – Hydraulic/Mechanical Components

About This Section

Topics in This Section

Section Overview

This section supplies information on the hydraulic systems and fuel-handling mechanical components contained within Encore and Eclipse units. This information includes specific safety information, typical measuring procedures, information on related components and common service procedures.

How to Use This Section

For a listing of topics contained in this section with its location, refer to "Topics in This Section" on page 7-1. There are tables that list individual components at these locations. Use this list to find information on a specific component. Additionally, the index and table of contents can be used to locate information.

Abbreviations and Acronyms

Term	Description
ATC	Automatic Temperature Control
GDP	Global Dispenser Pump (Blackmer unit)
ISD	In-station Diagnostic
PCV	Pressure Control Valve Proportional Control Valve
STP	Submersible Turbine Pump
TLS	Tank Level Sensor
UST	Underground Storage Tank

Terms Used in This Section

Term	Definition
Dispenser	Fixture from which product is supplied under pressure from STP or aboveground pump.
Dispensing Device	Refers to both pump and dispenser.
Metric Fasteners	Fastener design is based on millimeter measuring system.
Pump	Unit contains motor and pumping device to draw product from storage tank (also known as a self-contained unit).
Society of Automotive Engineers (SAE) Fasteners	Fastener design is based on inch measuring system.

General Service Procedures

Read and follow all safety precautions as outlined in "Read This First" on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Working on Hydraulic Components

Dispensing devices pressurize fuels from the Underground Storage Tank (UST) by way of STPs for dispensers or by pressurizing fuel within the unit in the case of self-contained pumps. A potential hazard of fluid spray or leaks exists when working on the hydraulic system, especially if under pressure. Ensure that proper procedures are followed and required precautions taken. For additional information, refer to "About Servicing Hydraulic Components" on page 7-4.

⚠ WARNING

You are working in a potentially dangerous environment of flammable fuels/vapors and high voltage.

Fuel and its vapors may ignite, leading to serious injury or death. Fire, explosion or electrical shock can result in severe injury or death if you do not follow safe procedures.

Surfaces of Eclipse Series units are non-metallic. In any situation where power is on and gasoline and its vapors are present or potentially present (for example, calibration), do not attempt to use the manager keypad with electronics cabinet open.

When calibrating Eclipse series units, run keypad cable over door and close door, allowing keypad to be suspended on unit and accessed from outside, for all procedures done with power applied or in presence of fuel or fuel vapor.

⚠ WARNING

Some residual pressure may be present in unit after releasing pressure through the nozzle, resulting in fuel spray.

Always wear proper eye protection when servicing the hydraulic system. Take appropriate steps to contain any spraying or draining fuel.

⚠ WARNING

Static electricity may be present before or during dispensing of fuel. A spark between container and nozzle may occur.
Fire or explosion may result causing serious injury or death.

Ground the nozzle and prover can to prevent static discharge. Do not place prover cans on ungrounded surfaces such as plastic truck liners.

⚠ WARNING

In the event of inclement weather, including snow, ice, or flooding that makes driving conditions dangerous, please avoid servicing units. Always use available door stops to secure upper doors against unwanted/unexpected movement, especially during high winds. If necessary, reschedule service to avoid damage to the equipment. Weather may change unexpectedly, be aware of local weather conditions. During service, if conditions develop making service unsafe, close the unit(s) and proceed to a safe location.

About Servicing Hydraulic Components

Read and follow all safety precautions as outlined in "Read This First" on page 1-1, this section, and related sections. Follow OSHA lockout/tagout procedures.

Before Beginning

1. Ensure that the unit is not pressurized with fuel. For dispensers, close and test the shear valves. For additional details, refer to "Shear Valves" on page 7-75.
2. Isolate the unit at the Distribution Box (D-Box).
3. Turn off power to the unit. In case of dispensers, remove power from the involved STPs. Follow OSHA lockout/tagout procedure at the station's breakers.
4. Open nozzle into approved container to bleed pressure. Some residual pressure may remain.
5. Continue with required service procedures.

After Repairing

1. Restore power to the unit. In the D-Box, place the unit in normal operation.
2. Purge air and check for leaks after replacements or rebuilds. For additional details, refer to "Purging Air from the System" on page 7-7.
3. Verify proper unit operation.

Servicing Seals, O-rings, and Gaskets

Read and follow all safety precautions as outlined in "Read This First" on page 1-1, this section, and related sections. Follow OSHA lockout/tagout procedures.

When opening or disassembling hydraulic components that have been in service for more than two years, you must replace the disturbed seals. Failure to replace degenerated or old seals can result in subsequent leakage.

Note: Use of Universal Seal Kits, that contain proportional quantities of material types commonly replaced during service is recommended.

1. Inspect leaking seals for excessive swelling, hardening, softening, and other degradation.
2. Clean and inspect sealing surfaces before replacing/installing seals, O-rings, or gaskets.
   Note: Always use Gilbarco-approved O-rings and gaskets.
3. Replace the parts, as required.
4. Use a small amount of silicone grease to retain O-ring seals in position during assembly and to improve durability of dynamic seals.

Pipes and Hydraulic Plumbing

Read and follow all safety precautions as outlined in "Read This First" on page 1-1, this section, and related sections of this manual. Follow OSHA tagout and lockout procedures.

To work on piping, proceed as follows:

- Only use UL-approved pipe sealants, suitable for the fuels involved, for pipe threads, as required. Follow sealant manufacturer recommendations for use.
  Note: Avoid any sealants that will become hard over time.
- Do not tighten or disturb joints where sealant has set.
- Always clean and inspect pipe threads before applying sealant. Do not apply sealant to dirty, oily, or wet threads. Do not apply sealant to the first two threads.
- Do not use Teflon® tape (fragments will cause pump failure). Excessive tightening may fracture the component during installation or assembly or tape.
- Do not excessively torque fittings. For additional details, refer to "Torque Specifications".
- Always check for leaks after service.

Torque Specifications

Ensure that you follow these recommendations when working with fasteners on pumps and dispensers:

- Never use a metric fastener in place of an SAE (or non-metric) fastener or an SAE fastener in place of a metric fastener. This may lead to damage to the device.
- Always use same grade or higher materials for load bearing devices; for example, for lifting brackets and bolts, otherwise, there is a possibility of failure.

Note: Use of universal bolt and Fastener Kits, that contain commonly replaced bolts during service is recommended.

The following tables provide general torque guidelines for plated steel screws and bolts, pipe plugs, and tube fittings.

Screws and Bolts (Plated)

	Recommended Torque	
Thread	**in-lbs**	**Nm**
M4	10-12	1.1-1.3
M5	25-35	2.8-4.0
M6	36-40	4.1-4.5
M8	120-140	13.6-15.8
M10	150-170	16.9-19.2

Pipe and Tube Fittings

	Recommended Torque	
Thread	**in-lbs**	**Nm**
1/8-inch NPT	10-12	13.6-16.3
1/4-inch NPT	22-25	29.8-33.9
3/8-inch NPT	30-35	40.7-47.5
1/2-inch NPT	35-40	47.5-54.2
3/4-inch NPT	45-55	61.0-74.5
1-inch NPT	70-80	94.9-108.4
1-1/2 inch NPT	150-180	203-244

Flanged and Compression Tube Fittings

	Recommended Torque	
Pipe	in-lbs	ft-lbs.
1/8-inch	40 - 45	-
1/16-inch	49 - 55	-
1/4-inch	58 - 65	-
5/16-inch	72 - 80	-
3/8-inch	-	9 - 10
1/2-inch	-	19 - 21
5/8-inch	-	26 - 29
3/4-inch	-	38 - 42
7/8-inch	-	49 - 54

Self-contained Pumping Units

	Recommended Torque	
Unit	in-lbs	Nm
Pump head (tightened in star pattern)	200	23
Strainer Cover	150	17
Rear Cover	200	23

Meter Bolts

		Recommended Torque	
Bolt (In Order)		in-lbs	Kg-meters
1	Cylinder cover gasket	130 ± 10	1.50 ± 0.12
2	Piston and connecting bar	35 ± 5	0.40 ± 0.06
3	Shaft seal cap	25 ± 5	0.30 ± 0.06
4	Lower bearing housing	75 ± 10	0.90 ± 0.12
5	Packing retainer	20 ± 5	0.25 ± 0.06
6	Body cover	130 ± 10	1.50 ± 0.12

Purging Air from the System

Read and follow all safety precautions as outlined in “Read This First” on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Topics in This Sub-section

⚠ WARNING

Residual pressure and entrapped fuel may still be present and may drain or spray when removing parts and fittings.

Fire and explosion can result in severe injury or death.

- Wear eye protection.
- Remove the parts and fittings slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

About Purging System

Check with your local and state agencies for meter calibration regulations. If the regulations exceed these procedures you must follow your local and state regulations.

Before any calibration procedures can be completed, all air in the lines and dispensers must be removed. When purging air at a new site start up, an existing location with STP, or line repair the following steps must be performed.

Individual Product Line Purging Procedure

1 Ensure the shear valves in all dispensers are closed aside from the unit furthest away from the STP.

2 Purge a minimum of “500” gallons through the shear valve test port of each product in the dispenser furthest away from the STP of each trunk line,

- If multiple STP’s are used in a manifold line configuration it is required to run all STP’s for that manifold=line set at the same time. Operating STP’s in an alternating configuration should not be used during purging. Ensure that the line programming is changed to All Pumps for the purging process. Make certain that you revert the programming after purging is completed.
- The number of gallons needed for purging will vary depending on line length, line type, line diameter, and flow rate. The purge amount should not be less than 500 gallons.
- The number of fittings and other physical variances such as air traps could also adversely affect the ability to purge air out of the lines. Larger volumes of fuel may be needed to ensure all air is purged from the lines and fittings.

Individual Meter Purging Procedure

1 After the initial rush of air and full product flow is achieved from the initial product line purge procedure, purge “50” gallons through the nozzle for each straight product after the initial 500 gallon purge. This must be done through each meter of each dispenser starting with the furthest dispenser from the STP working your way back to the STP. Shear valves must be opened individually as each individual product is being provided at each dispenser while working back toward the STP.

IMPORTANT INFORMATION

For Ecometer equipped units, never exceed the nozzle slow latch setting until fuel flow is steady.

Post Service Purging Procedure

1 After service is performed on the hydraulic section of the dispenser air purging must be performed using the Individual Meter Purge Procedure. To determine how much fuel to dispense when purging through the nozzle, refer to the following table:

Actions Performed	Minimum Purge Amount
New Installation/Start-up	50 gallons (190 liters)
System with fully purged product lines	50 gallons (190 liters)
Major Services • Meter Replacement • Valve Replacement • Pumping Unit Replacement • Manifold Assembly Replacement • Product Tubing Replacement	25 gallons (85 liters)
Minor Services • Changing Filters • Cleaning/Replacement of Strainer • Hanging Hardware Replacement/ Service • Outlet Casting Replacement	10 gallons (38 liters)

Measuring Flow Rates

Read and follow all safety precautions as outlined in "Read This First" on page 1-1, this section, and related sections. Follow OSHA lockout/tagout procedures.

⚠ WARNING

Static electricity may be present before or during dispensing of fuel. A spark between container and nozzle may occur.
Fire or explosion may result causing serious injury or death. Ground nozzle to prover can to prevent static discharge. Place the prover can directly on the ground and not on any insulated surface.

⚠ WARNING

Residual pressure and entrapped fuel may still be present and may drain or spray when removing parts and fittings.

Fire and explosion can result in severe injury or death.

- Wear eye protection.
- Remove the parts and fittings slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

Dispense fuel into a container and ensure that proper equipment and procedures are being used to eliminate the hazard of fuel vapors igniting as a result of static discharge or other sources. Touch the nozzle to prover can or use a grounding clip while dispensing fuel. Return dispensed fuel to the proper storage tank. Use the appropriately sized prover can:

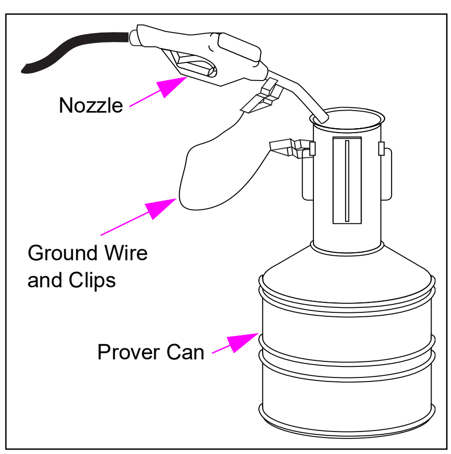
- For standard flow units, use at least a 5 gallon prover can.
- For high gallonage units, use at least a 25 gallon prover can.
- Start with displays reading 0 (zero).

To measure flow rates, proceed as follows:

1 Dispense fuel for 15 seconds (1/4 of a minute) into an approved container. Stop the fuel flow. Clean up any spills promptly.

2 To calculate gallons/liters per minute, multiply volume amount by four (for example, 5 gallons X 4 = 20 GPM).

Figure 7-1: Dispensing into Prover Can



Measuring Pressure and Vacuum

Read and follow all safety precautions as outlined in “Read This First” on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

⚠ WARNING

Residual pressure and entrapped fuel may still be present and may drain or spray when removing parts.

Fire and explosion can result in severe injury or death.

- Wear eye protection.
- Test and close the involved shear valves.
- Turn off power to unit.
- Remove the parts slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

Topics in This Sub-section

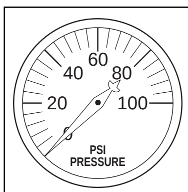
Using Pressure Gauge

Use pressure gauges and vacuum gauges to test pumps and dispensers when:

- Flow rates are too low
- Motors fail prematurely or stall
- Units take a long time to prime
- Units are noisy

Use of a combined vacuum and pressure gauge eliminates the calibrating requirement to more than one gauge.

Figure 7-2: Typical Pressure Gauge Dial



Typical Pressure Gauge Dial Face

Ensure that you:

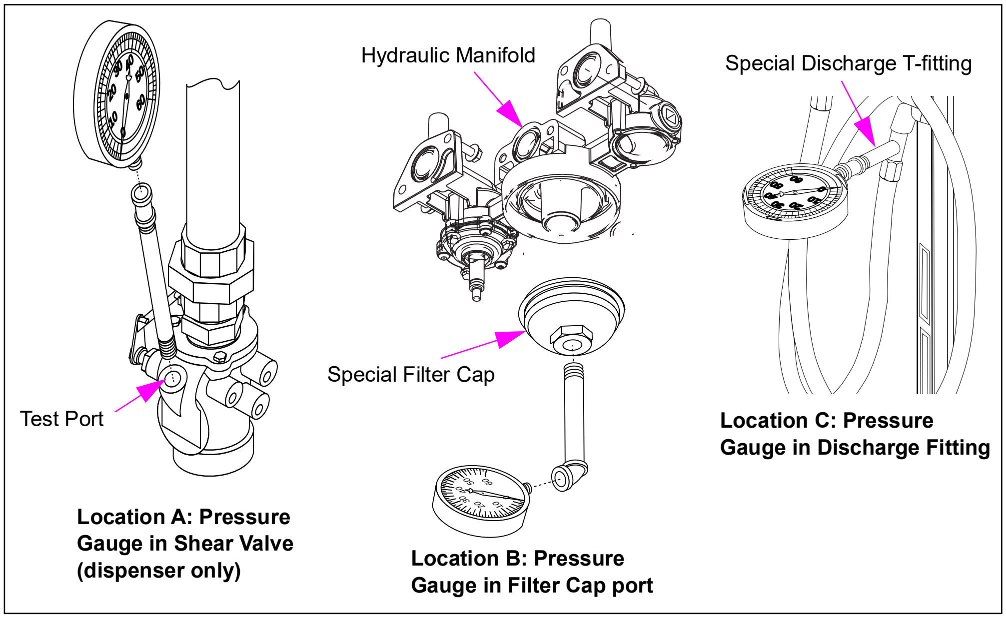
- Always use accurate calibrated gauges (using gauge snubber or dampened gauge will help to prevent gauge damage and obtain more accurate readings).
- Use gauges that have the appropriate scales for reading in pounds per square inch [scale 0-100 Pounds (of Pressure) per Square Inch (PSI)] for pressure and 0-30 Hg (inches of mercury) for vacuum.

Pressure Measuring Locations on Unit For Dispensers

Install the pressure gauge at any one of the following locations (see Figure 7-3):

To Check	Insert the Pressure Gauge in the
Inlet pressure	Shear valve test port (Location A)
Internal pressure up to first valve	Port on special modified filter cap (Location B)
Discharge pressure	Special Discharge fitting (Location C)

Figure 7-3: Typical Pressure Gauge Installations



Measuring STP Pressure for Dispensers

To measure STP pressure for dispensers, proceed as follows:

1. Turn off power to the STP.
2. Verify if the pump on STP has been deactivated.
3. Install the gauge at shear valve test port (see Figure 7-3 on page 7-11), Location A.
Note: Ensure that you use eye protection.
4. Activate the nozzle. Do not pump fuel.
5. Read PSI on gauge. Actual operating pressure depends on STP horsepower. The following approximate pressures exist when not delivering fuel (for exact performance values, consult manufacturer).
6. For a representation of typical STP performance, refer to the following table. Values for STPs with additional or fewer stages will vary from those in the following table:

	Operating Pressure			
STP Horsepower	**Gasoline**		**Diesel**	
1/3 HP	26 PSI	76 kg-cm^2	30 PSI	88 kg-cm^2
3/4 HP	30 PSI	88 kg-cm^2	34 PSI	99 kg-cm^2
1-1/2 HP	30 PSI	91 kg-cm^2	35 PSI	102 kg-cm^2
2 HP	41 PSI	120 kg-cm^2	46 PSI	134 kg-cm^2
3 HP	31 PSI	91 kg-cm^2	35 PSI	102 kg-cm^2
5 HP	38 PSI	111 kg-cm^2	43 PSI	126 kg-cm^2

For Self-contained Pumps

See Figure 7-3 on page 7-11 (Locations B and C).

- Install the pressure gauge to discharge side at a test port on special filter cap.
- Install the pressure gauge downstream at nozzle or other locations, as required.

Measuring Pressure Drop for Dispensers and Pumps

Use the pressure drop test to check specific parts (meters, shear valves, filters, and so on). An exceptionally high pressure drop for specific flow rates indicates that a part is clogged or defective.

To measure pressure drop for pumps/dispensers, proceed as follows:

1. Ensure that the pump handle is in the “off” position. For dispensers, close the shear valve. Place the nozzle in an approved container. Lift the pump handle. Squeeze the nozzle to bleed pressure.
2. Turn off power. Multiple disconnects may be required.
3. Install pressure gauges. For locations, see Figure 7-3 on page 7-11.
Note: Ensure that you use eye protection.
    - For dispensers, install at shear valve test port or filter cap port and at the discharge special T-fitting.
    - For pumps, install pressure gauge at pumping unit discharge test port and the discharge special T-fitting.

4 Restore power to unit. Open shear valve. Lift the pump handle. Activate the nozzle and pump fuel at full flow into an approved container.

5 Measure flow rate and read PSI on both gauges. Return the pump handle to “off” position.

6 Calculate pressure drop (pressure drop is the difference between PSI readings). For an example of desirable ranges, see Figure 7-4.

7 Ensure that the pump handle is in “off” position. If the unit is a dispenser, close the shear valve. Place the nozzle in an approved container. Lift the pump handle. Squeeze the nozzle and bleed pressure. Return the pump handle to “off” position.

8 Turn off power.

9 Remove pressure gauges, purge air, and restore unit to original operation.

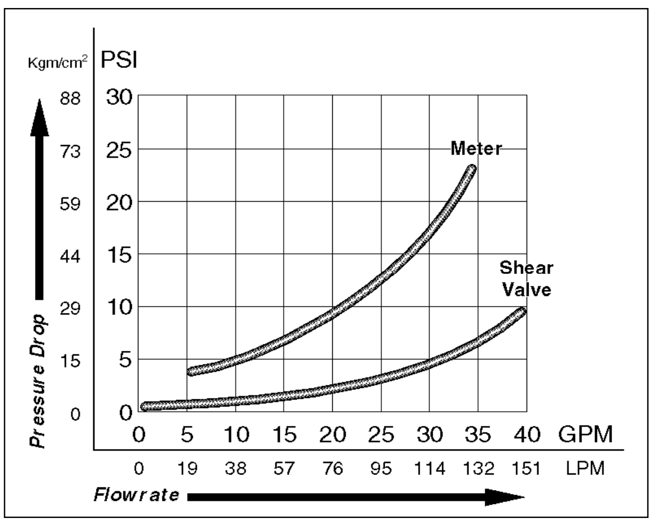
Analyzing Pressure Drop for Pump and Dispenser

Pressure drop across components increases as total flow rate increases.

- Total system pressure drop = Product Lift from bottom of STP to dispenser base
  + Pressure drop through external piping
  + Pressure drop through dispenser including hose and nozzle
- Pressure drop through external piping segments will vary depending on size and types of piping used.

If total system pressure drop at total flow rate is less than STP pressure at that flow rate, system will develop the required flow rate.

Figure 7-4: Graph for C+ Meter/V Meter and 1-1/2 Inch Shear Valve



Determining if Component Is Faulty

To determine if the component is faulty, proceed as follows:

1 Always check/replace filter and strainer before performing the test.

2 Determine the dispenser pressure and flow rate at the base of the dispenser with nozzle wide open.

Note: If the pressure at the base of the dispenser is substantially lower, the dispenser is functioning correctly.

3 Compare observed flow rates to the actual inlet pressure for the dispenser model being checked. If the flow rate is low for the pressure, it is likely that the dispenser or hanging hardware problem exists. If readings are close to the specification, it is likely that the dispenser is OK and it is a system problem.

To evaluate the probable cause of dispenser low flow rate, refer to the following table:

Low Flow Rate Symptom	Probable Causes
Observed pressure drop is more that a few PSI higher	Flow restriction
Observed and expected pressure drop is close to specification	Hanging hardware

Reaching Required Flow Rates

The following are the few methods for meeting required flow rate:

- Correct/replace the faulty component.
- Use larger STP or manifolded STP.
- Use larger inlet piping to the pump/dispenser.
- Use less restrictive hoses and nozzles.
- Reduce maximum lift of product from tank.
- Use a higher capacity pump or dispenser model.

Measuring Vacuum (Pump)

Read and follow all safety precautions as outlined in "Read This First" on page 1-1, this section, and related sections. Follow OSHA lockout/tagout procedures.

Topics in This Sub-section

⚠ WARNING

Residual pressure and entrapped fuel may still be present and may drain or spray when removing parts and fittings.

Fire and explosion can result in severe injury or death.

- Wear eye protection.
- Remove the parts and fittings slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

About Vacuum Readings

Use vacuum readings to determine if the unit has the ability to move product from the storage tank to the pump. Inability to pull a vacuum may also indicate that air is entering the system. Use vacuum gauges that read in inches of mercury (scale 0-30 Hg).

Figure 7-5: Vacuum Gauge

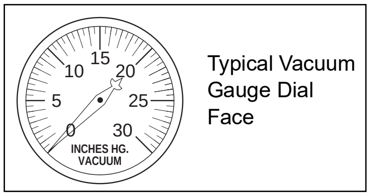


Figure 7-6: Vacuum Gauge Port on Global Dispenser Pump (GDP)

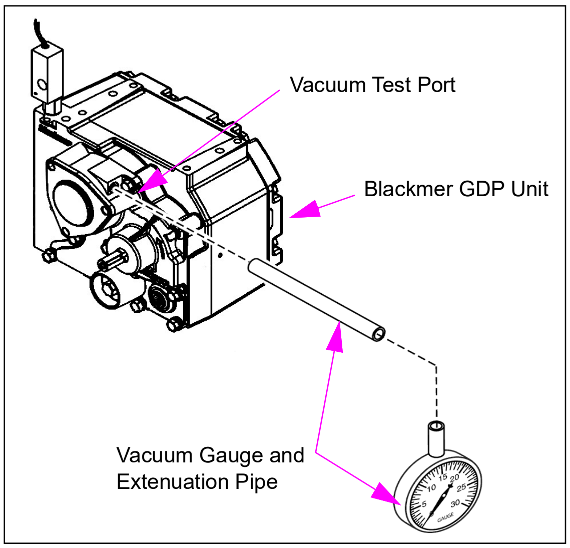
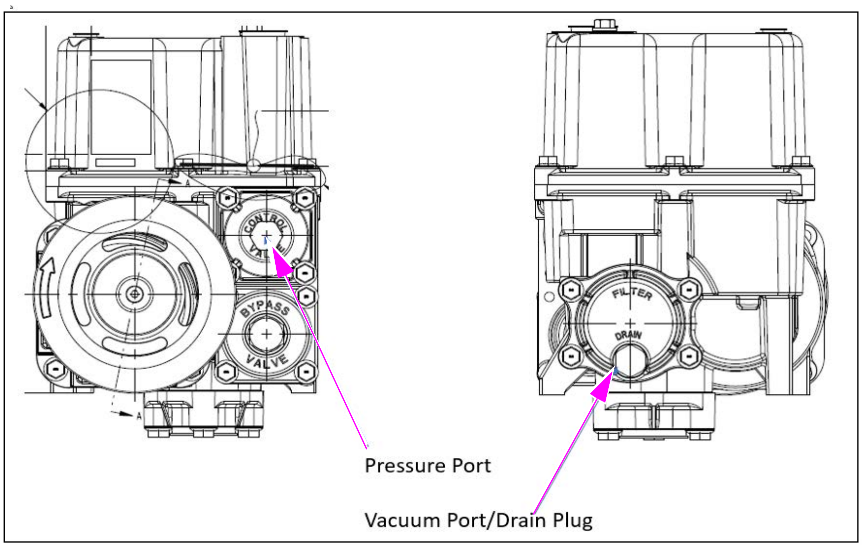


Figure 7-7: Vacuum Gauge Port on GPU+



Measuring Wet Vacuum and Pressure for Pump

This test must be performed when flow output is low or you are experiencing repetitive pump failures. Ensure that you follow all safety precautions.

1 Turn off power to the unit.

2 Check the pump to motor belt tension. For additional details, refer to "Setting Belt Tension" on page 7-66.

3 Install pressure and vacuum gauges.

4 Restore power and operate the unit with both pressure and vacuum gauges attached. Use an approved container to collect fuel. Clean up any spills promptly.

5 Record discharge pressure and vacuum pressure with nozzle open and nozzle closed.

6 Record flow rate with nozzle open.

7 Compare readings with values in the following Acceptable Range Tables.

Acceptable Range Tables

The values listed in the following tables are approximate and may vary depending on site conditions, hardware used, age of units, and so on. Flow rates for standard gallonage are based on 5/8-inch hoses, 3/4-inch swivels, and unleaded gas. Flow rates for high gallonage are based on 1-inch hoses and OPW7H nozzle.

Acceptable Ranges - Standard Measure

	Full Flow - All Nozzles Open			Bypass - Nozzles Closed	
Pump Configuration	**Inlet Vacuum (in Hg)**	**Discharge Pressure (PSI)**	**Flow Rate (GPM)**	**Inlet Vacuum (in Hg)**	**Discharge Pressure (PSI)**
One Pumping Unit per Hose:					
Standard Flow	4-10	15-30	10-13	4-7	20-35
High Flow	6-12	25-40	18-24	5-8	28-45
One Pump Unit per Two Hoses:					
Standard Flow	4-10	20-35	17-20	4-7	22-38

Acceptable Ranges - Metric Measure

	Full Flow - All Nozzles Open			Bypass - Nozzles Closed	
Pump Configuration	Inlet Vacuum (cm Hg)	Discharge Pressure (bar)	Flow Rate (LPM)	Inlet Vacuum (cm Hg)	Discharge Pressure (bar)
One Pumping Unit per Hose:					
Standard Flow	10.2-25.4	1.0-2.1	37-50	10.2-17.8	1.3-2.4
High Flow	15.2-30.5	1.7-2.8	68-90	12.7-20.3	1.9-3.1
One Pump Unit per Two Hoses:					
Standard Flow	10.2-25.4	1.3-2.4	64-76	10.2-17.8	1.5-2.6

Probable Causes of Low Flow Rate for Pumps

To determine the causes of low flow rates, refer to the following table:

Low Flow Rate Symptom	Probable Causes
The open nozzle vacuum reads high, pressure reads low, and closed nozzle pressure is normal.	• Check valve sticking, closed, or partially closed • Pumping unit strainer dirty • Supply line obstructed or restricted • Spring loaded check valve in supply line • Storage tank air vent plugged • Undersized supply line for unit • Tank too far or too deep (no further than 50-ft way and 10-ft deep) • Vapor lock
The open nozzle vacuum reads low or normal, pressure reads high, and closed nozzle pressure is normal.	• Hose crushed • Nozzle defective • Meter binding • Discharge piping obstructed • Filter dirty • Bypass valve defective or sticking • Regulator or check relief valve defective or sticking • Solenoid valve defective
The open nozzle vacuum reads low and pressure reads low; closed nozzle vacuum reads low and closed nozzle pressure reads normal.	• Belt slipping • Pump operating below specified rpm rate • Pumping unit parts worn • Sump float valve malfunctioning • Leak in supply line between tank and pump • Relief valve stuck open
Surging or hesitation occurs.	• Supply line check valve defective • Leak in supply line that traps air • High spots in supply line
Motor stalls or burns out.	• Relief valve sticking • Motor defective • Low voltage or defective wiring • Debris in pump • Broken pump elements

Calibration

Read and follow all safety precautions as outlined in “Read This First” on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Topics in This Sub-section

WARNING

Residual pressure and entrapped fuel may still be present and may drain or spray when removing parts and fittings.

Fire and explosion can result in severe injury or death.

- Wear eye protection.
- Remove the parts and fittings slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

About Calibrating Meters

Encore 300 units are shipped uncalibrated. Encore 500 and Eclipse units shipped after December 1, 2002 are shipped with initial factory calibration.

Note: All meters, including factory-calibrated meters, must still be checked for accuracy after purging air from system.

Before calibrating, you must properly purge air from the system; air in the system will affect calibration accuracy. For more details, refer to “Purging Air from the System” on page 7-7. After completing calibration, the W&M switch must be sealed per local authority. All replacement meters must be calibrated after installation and testing.

IMPORTANT INFORMATION

All meters must be checked for accuracy after purging air from system in new or existing systems. Air in the system will affect calibration accuracy.

Encore 300 NJ0 and NJ1: Six hose blends, the Grade 1 switch when selected, always calibrates the “w” product meter regardless of how the blend ratios are set for Grade 1. Grade 2 and Grade 3 switches always calibrate the “x” product regardless of how the blend ratios are set. Other blended models rely on the programmed blend ratios to determine which meter is calibrated.

For additional details on calibrating meters, refer to “Pump Programming” on page 5-1.

About Calibrating VaporVac

For additional details, refer to “Calibrating VaporVac” on page 7-84.

Note: DEF must not be contaminated. DEF is corrosive if mixed with any fuel. Use only stainless steel prover cans that are dedicated for use with the DEF for calibration.

The calibration of DEF dispensers is typical for standard dispensers. However, the following exceptions must be noted:

- DEF tends to entrap and hold air much longer than gasoline or diesel. Allow the prover to site approximate three minutes to dissipate air. To dissipate any air bubbles that stick to the interior of the can, use a rubber mallet gently hitting the prover can toward the end of waiting.
- Air entrapment will affect calibration accuracy. DEF systems can have overhead lines, pumps without air separation capability, and no means to bleed the overhead lines. In such cases, it will be difficult to remove air that in turn results in incorrect calibration.

For details on CCs related to calibration, refer to “Programming DEF Dispensers” on page 5-169.

Filters and Strainers

Read and follow all safety precautions as outlined in “Read This First” on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Topics in This Sub-section

Note: To prevent fueling problems with vehicles and to minimize dispenser down-time due to damage to valves and motor, never operate a unit without the use of a filter or strainer.

⚠ WARNING

Residual pressure and entrapped fuel may still be present and may drain or spray when removing parts and fittings.

Fire and explosion can result in severe injury or death.

- Wear eye protection.
- Remove the parts and fittings slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

About Filters

A noticeably slower flow rate very commonly indicates a dirty filter or clogged strainer. Very slow flow rates also result from water alert filters if water is in the tank.

For efficient operation and long life of pumps and dispensers, replace fuel filters and clean strainers regularly.

- After 50,000 gallons (189,000 liters) or one month for new installations
- Every 300,000 gallons (1,134,000 liters) or six months thereafter

Notes: 1) Harsh environments including old steel fuel storage tanks may require changing filters and cleaning strainers more frequently.
2) Conversions from non-alcohol to alcohol enhanced fuels will typically result in less frequent filter changes due to the cleaning effect of alcohol in the tanks and piping.

Always use Gilbarco filters (standard, water alert, and high capacity).

Filter	Description	Part Number	Where Used
Standard flow	Standard 10 micron nominal Water Alert for gasoline, diesel, and other blend fuels	R20039 M08007B010	Encore, Eclipse
	Standard 10 micron nominal for alcohol blended fuels	R19736-10 M08006B010	Encore, Eclipse
High flow	High flow 30 micron nominal Particulate for all fuels (11” long)	M02264B001	Encore Ultra-Hi

Servicing Filters

Before Removing Filters

Read and follow all safety precautions as outlined in “Read This First” on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Before removing the filters, proceed as follows:

1 Close the shear valve (dispensers only). Check operation of valve. This may require multiple shear valve closures. For pumps, turn-off all power to the pump.

2 Bleed pressure (dispensers only):
- Lift operating handle and authorize unit.
- Place nozzle in approved container.
- Close nozzle.
- Turn off operating handle.

3 Turn off the associated STP circuit breakers (dispensers only). This may require multiple STP disconnects.

4 Turn off the dispenser circuit breaker. Turn off all power to the unit.

Removing Filters

⚠ WARNING

Servicing filters without properly turning off power and ensuring that the appropriate valves are closed, may result in fuel discharge or spray. Any involved STPs with power still applied can be energized from another unit resulting in fuel pressure in the product lines for the unit you are working on. Pumping units with power applied can also be accidently energized. You may not be able to retighten filter to stop fuel flow.

Fire and explosion can result in severe injury or death.

Ensure that you:
- Wear eye protection.
- Test and close the involved shear valves.
- Remove power to pump, dispensers, and any involved STPs.
- Remove the parts slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

⚠ WARNING

Residual pressure and entrapped fuel may still be present and may spray when removing parts.

Fire and explosion can result in severe injury or death.

Ensure that you follow all safety instructions and are prepared for any residual fuel pressure.

Use an approved container to collect residual fuel. Use gasoline/fuel approved absorbent materials to clean spilled fuel. Use Gilbarco fuel collectors, if applicable.

To remove the filter, proceed as follows:

1 Remove the filter slowly by turning it counterclockwise.

2 Drain the removed filter into an approved container and dispose of properly (do not dispose of filters and fuel soaked absorbent materials in trash cans). Follow local, state, and national code requirements for disposal. Use of fuel catch cup or pan is recommended.

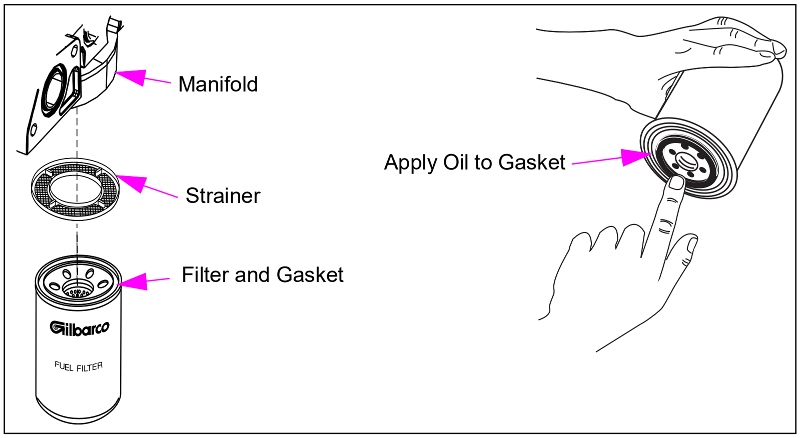
3 Remove the strainer and clean. For additional details, refer to “Servicing Strainers” on page 7-24.

Installing New Filters

To install a new filter, proceed as follows:

1 Read and follow instructions printed on the new filter.

Figure 7-8: Coating Filter Gasket



2 Install the strainer properly.

3 Coat the new filter gasket with a thin film of clean oil (see Figure 7-8).

4 Attach the filter to hydraulic manifold and turn clockwise until the gasket touches the base. Then, hand-tighten an additional 1/2 turn.

5 Restore power.

6 Open the shear valve on dispensers.

7 Lift the nozzle hook and authorize pump.

8 Check for leaks.

9 Bleed air by dispensing 10 gallons (40 liters) into an approved container for each hose involved.

10 Check for leaks again.

Service Tips

- Hand-tighten filter only. Do not use filter wrench to tighten.
- A light coating of oil on the gasket helps seal to seat properly and allows for easier release at time of removal.

About Strainers

Strainers are installed between the manifold and filter to trap larger particles leaving fine filtering to be done by the filter element. The strainer helps to prolong filter life by preventing premature filter clogging. Always reinstall or replace strainer.

Replacing Strainers for DEF Units

To replace or clean strainers for DEF units, proceed as follows:

1 For skid tank systems, shut-off the valve in the discharge line after the pump. If the system is equipped with a shear valve, it may be shut-off instead of turning off power to the pump or STP.

Note: For units with a remote tank, turn off the ball valve closest to the dispenser or if the unit is equipped with a manually closeable shear, close that shear valve.

2 Authorize the fueling position and bleed down pressure using the nozzle. If a continuous slow flow occurs, a shear valve or shut-off valve may be leaking. Correct/repair the valve before you proceed.

⚠ CAUTION

Some additional pressure may remain in the dispenser. Wear eye protection before you proceed.

3 Turn off the dispenser.

4 Wear eye protection and slowly unscrew the filter cap.

IMPORTANT INFORMATION

Use a suitable non-breakable container to collect the drained DEF. Dispose of drained fuel as per DEF guidelines mentioned in this document.

5 Remove the strainer and clean or replace.

6 Inspect the sealing washer at the strainer cap. Replace if damaged.

7 Install the new or cleaned strainer and cap.

8 Open the valve at the pump or shear valve.

9 Restore power to the dispenser.

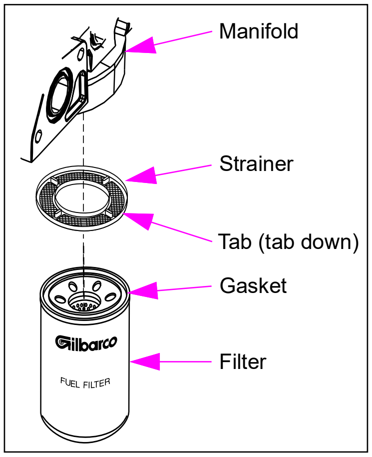
10 Dispense 5 gallons of DEF while checking for leaks.

Note: Systems with overhead lines and no shut-off or shear valve at or near the dispenser may require much greater fuel dispensing to purge air. If air is present in the system, extreme negative readings are obtained during meter calibration verification.

Servicing Strainers

Cleaning/Replacing Standard and High Flow Filter Strainers

Figure 7-9: Filter and Strainer Assembly



To clean/replace a standard and high flow filter strainer, proceed as follows:

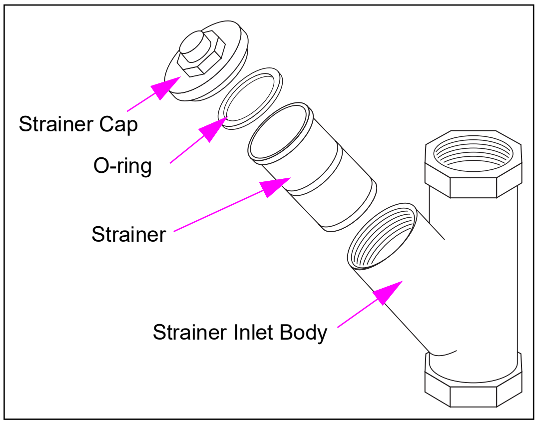
1 Remove the strainer located between filter and filter base. Remove the strainer using needle nose pliers, while being careful not to damage the strainer.

2 Clean the strainer with a soft brush and alcohol. If the strainer is permanently clogged or damaged, replace the strainer. Dispose of the rejected strainer properly.

3 Install the cleaned or new strainer back into the bore with tabs facing toward the filter cartridge.
Note: Failure to use a strainer will void the meter and valve warranty.

Cleaning/Replacing Ultra-Hi Inlet Strainer

Figure 7-10: Ultra-Hi Dispenser Strainer Assembly



To clean/replace an Ultra-Hi inlet strainer, proceed as follows:

1 Close the shear valve. Inspect operation of valve. For additional details, refer to “Shear Valves” on page 7-75.

2 Bleed pressure:
- Lift the operating handle and authorize the unit.
- Place the nozzle in an approved container.
- Open the nozzle to reduce system pressure.
- Close the nozzle.
- Turn off the operating handle.

3 Turn off the involved STP circuit breakers.

4 Turn off the dispenser circuit breakers.

5 Clean the area around strainer cap.

6 Remove the strainer cap by slowly turning it counterclockwise. Watch for residual pressure causing spray. Clean up spills promptly.

7 Remove the strainer and check for clogging. Clean with a soft brush and alcohol or replace, if required.
Note: Do not tap the strainer on a hard surface to dislodge particles.

8 Insert the strainer in housing.

9 Clean the strainer cap, seal ring, and strainer body. Inspect cap seal ring for damage (hardening or distortion). Replace as required. Coat O-ring with a thin film of oil before replacing cap. For additional details, refer to “Servicing Seals, O-rings, and Gaskets” on page 7-4.

10 Replace the cap, turning clockwise, and tighten securely.

11 Purge air after service. For more details, refer to “Purging Air from the System” on page 7-7.

12 Check for leaks after assembling and purging air from the system.

Hanging Hardware

Read and follow all safety precautions as outlined in "Read This First" on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Topics in This Sub-section

⚠ WARNING

Residual pressure and entrapped fuel may still be present and may drain or spray when removing parts and fittings.

Fire and explosion can result in severe injury or death.

- Wear eye protection.
- Remove the parts and fittings slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

About Hanging Hardware

Hanging hardware are the components that attach to the outlet casting. This includes hose, nozzles, swivels, spacer hose, and breakaways. Hanging hardware is tested for leaks and electrical continuity. Hanging hardware can be composed of other manufacturer's components. Ensure that you follow manufacturer's recommendations.

Hoses and Flow Restrictors

Inspecting Hoses

Check hoses regularly (at least once a week) for leaks and damage. Inspect and replace hose assembly, as required. The following conditions may indicate damage or an impending problem:

- Twisting and curling puts unusual stress on hose parts.
- Repeated flexing of hose in the same spot can cut, tear, or split the hose cover.
- Flattened hoses cause restricted flow and cause hose reinforcement damage.
- A soft spot is a sign of internal damage to the hose reinforcement. Reinforcement may appear through the cover.
- Loose or cracked hose couplings can cause a fuel spill.
- Hose bulges are a sign of pressurized gasoline pushing against the cover. A full rupture can occur at any time.

Hose Retrieval Mechanisms

Hose retrieval mechanisms must be operational or a trip hazard can exist. Do not use hose longer than the recommended length, without a hose retrieval system.

Replacing Hoses

Do not use soft-wall hoses. They can cause the unit to register with nozzle closed when the nozzle hook is first raised. Always use UL-approved hoses designed for the fuel being dispensed. Read and follow all safety precautions. Follow OSHA lockout/tagout procedures.

⚠ WARNING

Residual pressure and entrapped fuel may still be present and may spray when removing hoses.

Fire and explosion can result in severe injury or death.

- Wear eye protection.
- Remove the hoses and nozzles slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

To replace the hose, proceed as follows:

1. Close the shear valve on dispensers. Check for proper shear valve operation. Units may require multiple shear valve closures.

2. Turn off power to the unit and associated STPs. Multiple STP disconnects may be required. Follow OSHA lockout/tagout procedures.

3. Drain the nozzle and hose into approved container.

4. Remove the nozzle first, then hose.

5. Install swivel couplings and breakaways to unit as required.

6. Use UL-approved pipe sealant on hose coupling threads. Follow these recommendations:
   - Do not use Teflon tape. Teflon tape on these components can cause damage during assembly or installation and defeat the conductive property of the handing hardware and cause nozzles to malfunction.
   - Do not apply sealant to first two threads of hose coupling.
   - Do not use sealant on O-ring seals (a light coating of oil or silicone grease is advised).
   - For co-axial and tri-axial vapor recovery hoses, inspect and lubricate all O-ring seals with silicone grease to aid assembly.

7. Install hose assembly to pump end first, then to nozzle. Always attach hose assembly by rotating hose and coupling. Screw coupling in securely. Do not over-tighten. Do not use zinc die cast couplings on standard hoses. Use only chrome plated brass couplings. Use an open end wrench to protect chrome finish.

8. Check hose hanging hardware for continuity. For additional details, refer to "Testing Hose and Hanging Hardware Continuity" on page 7-35. Replace components, if required.

9 Install padded or contoured plastic hose clamps and retractor cables properly. In all instances, do not install hose retractor clamps downstream of the breakaway.

10 Restore power. Open shear valve for dispensers.

11 Check for leaks.

12 Bleed system of air. For more details, refer to "Purging Air from the System" on page 7-7.

13 Check for leaks again.

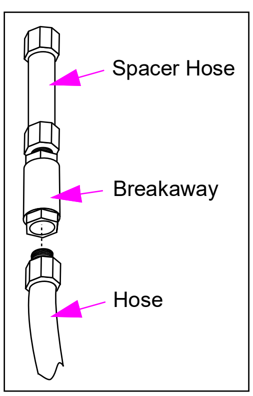
Breakaways

Emergency breakaway couplings help protect against fuel spill, fire hazard, and equipment damage. The breakaway valve stops flow during drive-offs if the nozzle or hose loop catches on the vehicle. Always use UL-listed breakaways.

- For proper operation of breakaways, always bolt the pump/dispenser securely to the island. Refer to Gilbarco installation instructions for pumps and dispensers.
- Some breakaway manufacturers require you test the pump/dispenser for pull strength before installing the device. Follow the manufacturer's instructions.
- For overhead units, install the spacer hose into the overhead outlet casting first, followed by the breakaway and the hose.
- When used, a hose retention coupling must never be installed between the breakaway and the nozzle.
- When installing breakaways, always use wrench on breakaway hex or wrench flats only.
- Always use pipe sealant for connections (except on O-ring fittings). Never use Teflon tape.
- Some breakaways are reusable. Follow breakaway's manufacturer instructions on inspection and reuse.
- Periodically and after a drive-off, check breakaway hoses and couplings for hose continuity. For additional details, refer to "Testing Hose and Hanging Hardware Continuity" on page 7-35.

Reinstalling or Replacing Breakaways

Figure 7-11: Typical Breakaway



To reinstall or replace breakaways, proceed as follows:

1. Close and test the shear valve. Units may require multiple shear valve closures. Also, refer to manufacturer's instructions.
2. Turn off power to the unit and associated STPs. Follow OSHA lockout/tagout procedures.
3. Drain the nozzle and hose into an approved container. Clean up any spills promptly. Be prepared for any residual pressure that may still exist in the hanging hardware.
   Note: Ensure that you use eye protection.
4. Check for damage to nozzle (especially the spout and threads).
5. Perform a hose continuity test. For additional details, refer to "Testing Hose and Hanging Hardware Continuity" on page 7-35.
6. Inspect the unit for structural, meter, and piping damage. Repair or replace the damaged components, as required.
   Note: Meters, piping, and connections may leak after a drive-off.
7. Replace breakaway if it is not the reusable type.
8. Check for damage to breakaway valve and seals before reassembly. Follow manufacturer's instructions.
9. Follow manufacturer's instructions for cleaning and lubricating seals and valve ends.
10. Follow manufacturer's instructions for recoupling.
11. Restore power and open shear valves.
12. Authorize the nozzle and check for leaks in hose and plumbing area (especially the meter and filter).
13. Purge system of air. For more details, refer to "Purging Air from the System" on page 7-7.
14. Check for leaks again. Ensure that you check meter and internal piping if a drive-off was involved.

Service Tips

- Do not reuse a separated breakaway if any part appears damaged.
- Some manufacturers require lubrication of breakaways every six months. Others require only periodic visual inspections. Follow their recommendations.
- Inspect breakaways for leaks at least once a week.
- If breakaway separates without a drive-off occurring, do not reuse.
- Consult manufacturer for the maximum number of times you can reset a breakaway. Some breakaways are not resettable.
- A protective sleeve supplied by manufacturer is recommended to avoid damage to dispenser and to improve appearance.

Nozzle Door Mechanism for DEF Units

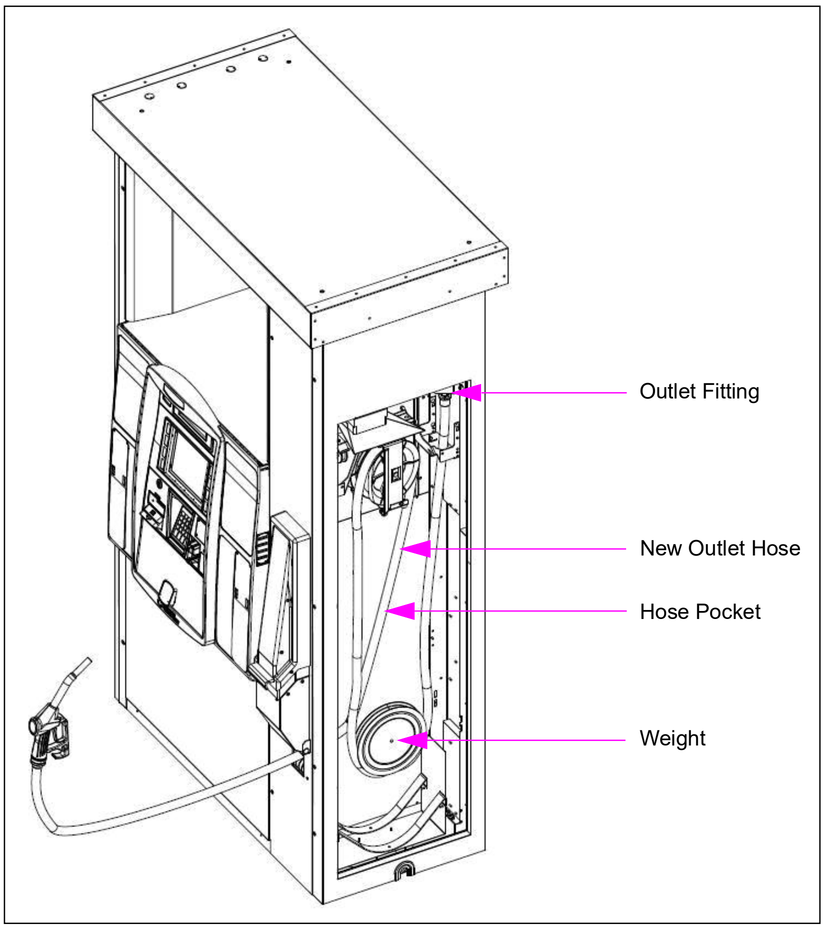
The nozzle door mechanism protects the nozzle, hose, and breakaway from freezing, by retrieving the components into the heated cabinet and covering them with a drop down or flapped door.

IMPORTANT INFORMATION
• The door mechanism must drop down completely without assistance, to cover the nozzle or other hanging hardware. Otherwise, the nozzle, hose, and breakaway may freeze. If the door does not close on its own, you must repair it. • When using the nozzle, always maintain manual control of the nozzle, as the hose retriever can pull the nozzle out of the tank, prover can, or other receptacle. • For units with flapper doors, it is very important that the curtains seal the nozzle boot chamber from the outside reasonably. If curtains are missing or damaged, they must be repaired promptly, especially during cold weather.

Outlet Hose Replacement for DEF Units

Note: Two people are required to perform this procedure.

Figure 7-12: Outlet Hose Replacement



Note: The weighted pulley is shown in Figure 7-12. The pulley in the lowered position is not visible and must be raised to be accessed.

To replace the outlet hose, proceed as follows:

1 Turn off power to the supply pump or shut-off flow to the dispenser at the closest ball valve to the dispenser or its shear valve.

2 Authorize the dispenser and relieve nozzle pressure into a stainless steel prover can.
Note: If fluid continues to flow for more than 5-10 seconds, the shut-off valve or shear valve may be leaking.

3 Turn off power to the dispenser.

⚠ CAUTION
Residual pressure may exist in the hose. Wear eye protection.

4 Remove the nozzle and breakaway. Collect any drained fluid into a stainless steel prover can. Prevent the hose from being completely drawn back into the unit (use a clamp).

5 Remove the inner sheathing, and front and rear panels.

6 Extend the hose fully until the weighted pulley is at the top-most limit of travel. Collect any drained fluid into a stainless steel prover can.

7 Remove the weighted pulley.

8 Remove the hose.

9 Install the new hose (see Figure 7-12 on page 7-30) by routing the hose through the roller passage with anti-kinking sleeve at the nozzle. Prevent the hose from being completely drawn back into the unit (use a clamp).

10 Attach the new hose to outlet fitting and tighten to 30-35 feet-lbs.
Note: Do not overtighten. Do not use a thread sealant.

11 With the hose extended, carefully place the hose and weighted pulley back in the hose pocket and retract the hose slowly to a fully lowered position.

12 Extend the hose fully out and then back in to ensure proper operation.

13 Reinstall the front panel.

14 Reattach nozzle and breakaway, and tighten to 30-35 feet-lbs.
Note: Do not overtighten. Do not use a thread sealant.

15 Extend the hose fully to ensure proper operation.

16 Restore power to the supply pump and dispenser.

17 Purge approximately 5 gallons of DEF into a stainless steel prover can.

18 Check for leaks.

19 Reinstall the rear panel and inner sheathing.

Inlet Hose Replacement for DEF Units

To replace the inlet hose, proceed as follows:

Note: Two people are required to perform this procedure.

1 Turn off power to the supply pump or shut-off flow to the dispenser at the closest ball valve to the dispenser or its shear valve.

2 Authorize the dispenser and relieve nozzle pressure into a stainless steel prover can.
Note: If fluid continues to flow for more than 5-10 seconds, the shut-off valve or shear valve may be leaking.

3 Turn off power to the dispenser.

4 Remove the inner and outer sheathing. Front and rear panels may be removed to allow greater access, but this is not required.

⚠ CAUTION

Residual pressure may exist in hose. Wear eye protection.

5 Disconnect the inlet hose end at the skid tank. Collect any drained fluid into a stainless steel prover can.

6 Disconnect the inlet hose end at the base of the meter. Collect any drained fluid into a stainless steel prover can.

7 Attach the new hose end without the anti-kinking sleeve to the old hose at skid tank end (outside the dispenser) using a 1-inch NPT X 1.5-inch long pipe nipple and finger-tighten it.
Note: Do not overtighten.

8 Wrap with tape to prevent separation.

9 Pull up the old hose out of the dispenser, while inserting the new hose at the bottom.

10 Wood or pipe of about 6 feet in length can be used inside the dispenser pocket, where the hose comes up to the meter to jockey the fittings around the 90° bend at the bottom, by pushing to one side and inserting the hose at the bottom.

11 Pull the hose until there is sufficient length to reattach the hose to the meter inlet.

12 Attach the hose to the meter inlet and tighten to 30-35 feet-lbs.
Note: Do not overtighten. Do not use a thread sealant.

13 Attach the other end to skid tank outlet and tighten to 30-35 feet-lbs.
Note: Do not overtighten. Do not use a thread sealant.

14 Restore power to the supply pump and dispenser.

15 Dispense 10 gallons of DEF to purge air.

16 Check for leaks.

17 Reinstall the external panels and sheathing.

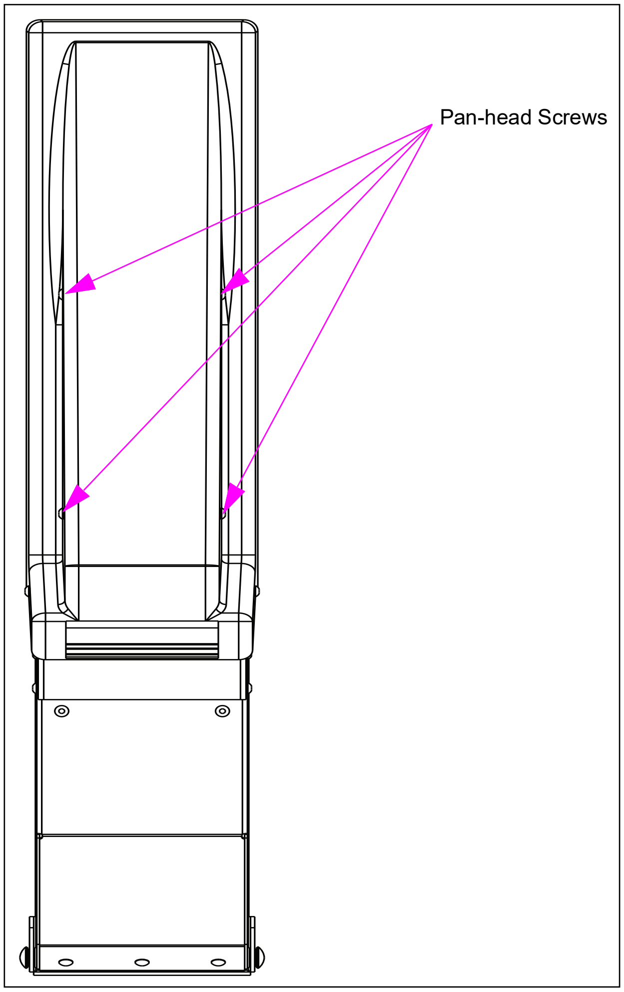
Door Pulley Cord Replacement for DEF Units

To replace the door pulley cord, proceed as follows:

1 Remove the inner sheathing.

2 Remove the fiber glass door (nozzle cover) by removing the four pan-head screws.

Figure 7-13: Fiber Glass Nozzle Cover



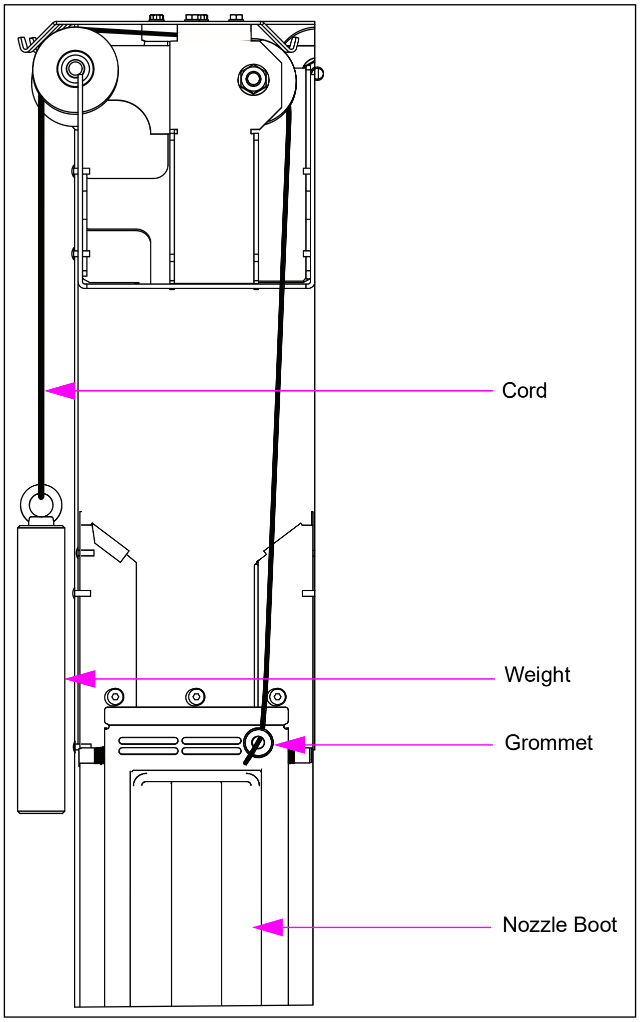
3 Pull the weight up from within the cabinet. Remove and secure the weight so that it does not fall into the cabinet.

Note: If cord is broken and weight cannot be retrieved, the old weight may be left at the bottom of the cabinet. Ensure that any old cord does not hinder the operation of the hose or cable mechanisms.

4 Untie and remove the cord and weight at the door frame.

5 Secure the new weight so that it does not fall into the cabinet.

Figure 7-14: Door Pulley



6 Thread the cord over the pulleys (one or two pulleys, depending on left-handed or right-handed unit) and through the grommet in the front panel.

CAUTION

If the grommet is missing, replace it immediately. Otherwise, the cord will fail prematurely.

7 Securely tie the cord to the door frame.

8 Carefully lower the weight into the guide tube.

9 Before installing the fiber glass door, test the door mechanism for smooth operation.

10 Reattach the fiber glass door using Loctite® Blue 242® and test again for smooth operation.

Testing Hose and Hanging Hardware Continuity

Perform an electrical continuity check when installing hoses, after a drive-off, and for periodic preventive maintenance. The STP check valve or pump regulating valve must maintain operating pressure to perform this test. The following procedure tests the continuity of the hanging hardware (that is, hose, nozzle, swivel, breakaway, and hose casting).

⚠ WARNING

Hanging hardware without the proper electrical continuity rating presents a risk of igniting fuel vapors.

Injuries or death may result if hose continuity is not maintained.

Use approved procedures to test for hanging hardware continuity. For additional information, refer to “PEI/RP400-02” in addition to equipment and component manufacturer’s instructions.

⚠ WARNING

You are working in a potentially dangerous environment of flammable fuels/vapors. Fuel and its vapors may ignite, leading to serious injury or death. Fire, explosion, or electrical shock can result in severe injury or death if you do not follow safe procedures.

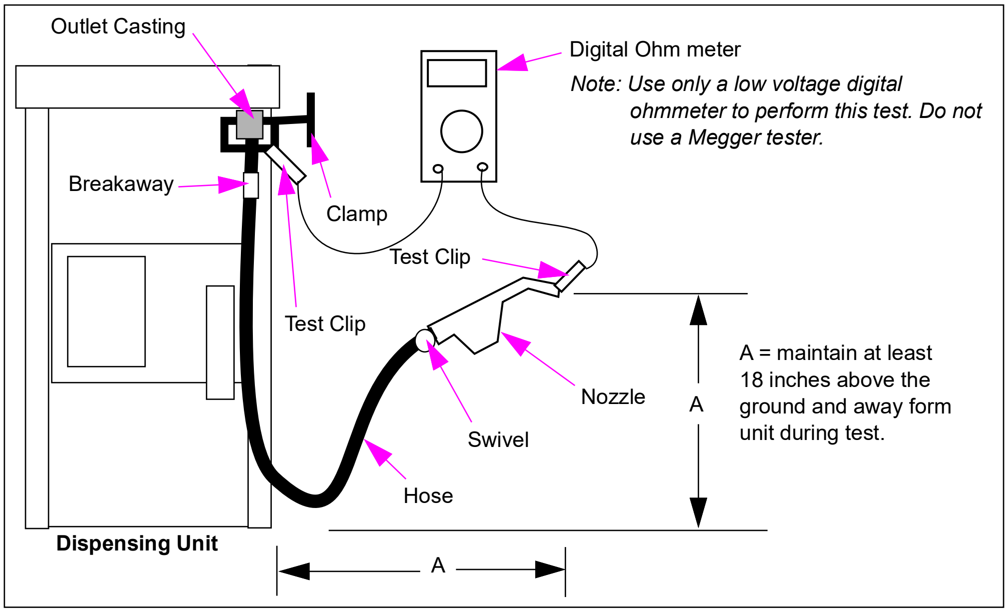
Ensure that you correct any safety hazards and barricade the area. Follow manufacturer’s instructions for the safe use, operation and maintenance of the testing equipment.

Use only a low voltage digital ohmmeter. Do not test continuity if there is spilled fuel, leaking components, or fuel vapors present in the area. Be familiar with hazardous zones requirements. Do not turn tester on until after connections are made. Turn tester off before disconnecting test clips.

To test the hose and hanging hardware continuity, proceed as follows:

1. Ensure that there is no liquid fuel or flammable vapors present. Clean up, as required.
2. Lift the nozzle hook and authorize the unit to pressurize hose.
3. Lower the nozzle hook.

Figure 7-15: Testing Hanging Hardware Continuity



4 With meter off, connect one probe to the nozzle spout, the other probe to fuel outlet fitting at outlet casting or other component grounded with the unit's frame (see Figure 7-15). Probe contact surface must be clean and free from fuel or other contamination.
- Use test leads at least 3-feet long (1 meter) equipped with insulated clips.
- Typical meter test clips are not suited for direct attachment to hanging hardware. Use appropriate clamping device to ensure good contact.
- Keep the nozzle at least 18 inches above the ground and 18 inches away from the dispenser during test.
- Wear insulating gloves if hanging hardware component is handheld during test or use an insulated device to support nozzle. For additional information, refer to "PEI/RP400-02".

5 Testing continuity: Turn on the meter to conduct test.
- Measure resistance while flexing and moving hose and nozzle into various positions that can be reasonably expected during fueling.
- Keep good electrical contact between probes and contact locations.

6 If test reading complies with UL330 which states that resistance must not exceed 70,000 ohms per foot (233,000 ohms per meter), the hanging hardware has passed the continuity test.

If test readings exceed 70,000 ohms per foot:
- Perform continuity test between J-box and hose discharge casting on pump or dispenser. This helps verify dispenser continuity and adequacy of test point locations.
- If no problem exists here, then isolate defective part by performing continuity tests between and across hose, nozzle, swivel, breakaway, and hose casting. Replace the part if resistance exceeds manufacturer's recommended maximum limit.
- Do not use components if test reading shows an open circuit, no resistance, or other defect.

Adjusting Nozzle Hook for Specific Nozzles

The following instructions are intended to adjust the position of the Gilbarco Encore dispensing unit nozzle hook to accommodate various manufacturer's manuals. Notify site personnel and secure the work area. Isolate the dispensing unit from POS or pump controller and close the shear valves. Follow all safety precautions.

Tools Required to Modify Nozzle Boot:

- Drill
- 7/32 or #22 drill bit
- 1/4-inch square-tip driver
- 7-mm metric hex nut driver or socket
- 3/8-inch nut driver or socket

⚠ WARNING

AC or battery powered drills must not be used at the dispensing unit. There exists a danger of explosion or fire due to the presence of hazardous vapors.

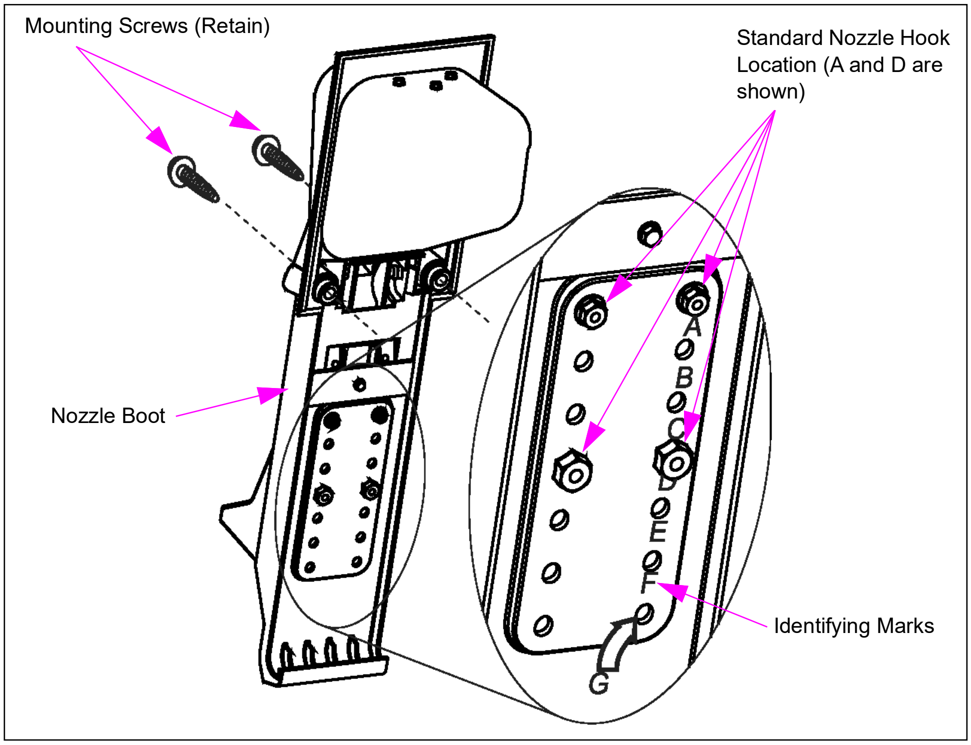
Removing Existing Nozzle Boot

To remove the existing nozzle boot, proceed as follows:

1 Remove the nozzle(s) from nozzle boot(s) and place on ground.

2 Loosen the two nozzle boot mounting screws and remove the boot from the dispenser by pulling it toward you (see Figure 7-16) using 1/4-inch square tip driver. Retain the nozzle boot mounting screws for reuse.

Figure 7-16: Nozzle Boot



3 Place the nozzle boot face down on the soft cloth to protect the nozzle boot face.

4 To identify existing nozzle hook retaining screw and nut locations, see Figure 7-16 on page 7-37. Identifying marks are located under right hand row of indented hole locations. Standard nozzle hook locations are A and D.

5 Remove the two upper hex head screws using a 7-mm nut driver or socket.

6 Remove the two nuts from lower carriage bolts using a 3/8-inch nut driver or socket.

7 Remove the nozzle hook and carriage bolts from the nozzle boot. Retain the hex head screws, carriage bolts, and nuts.

Positioning Nozzle Hook for Specific Nozzles

Determine the new nozzle hook position using the following chart as the guide to select new hole positions. To identify existing nozzle hook retaining screw and nut locations, see Figure 7-16 on page 7-37. Identifying marks are located under right hand row of indented hole locations.

Nozzle Type	Upper Hex Head Screw Location	Lower Carriage Bolt and Nut Location
Standard Factory Location All Non-Vapor	A	D
VaporVac - OPW, Husky®, Emco Wheaton	A	D
VaporVac - Catlow, Richards	B	E
Healy System	C	F
Balance - Husky Short	A	D
Balance - Husky Long, Emco Wheaton Long	E	Unmarked. Use nozzle hook carriage bolt holes as drill guide.
Balance - OPW Long	Bottom hole set (“G” in Figure 7-16 on page 7-37)	Unmarked. Use nozzle hook carriage bolt holes as drill guide.

Drilling Holes

To drill holes, proceed as follows:

1 Drill new holes using a 7/32-inch or # 22 drill bit, as required.

2 When locations “E” or “G” are used by the upper hex head screws, the lower carriage bolt and nut hole set are unmarked. Temporarily mount the nozzle hook with the upper hex head screws in location “E” or “G” (as determined by chart), then use the nozzle hook carriage bolt holes as a drilling guide for the unmarked hole set.

3 After the holes are drilled, remove the nozzle hook and clean up debris around hole set.

Assemble Nozzle Hook to Nozzle Boot

To assemble the nozzle hook to the nozzle boot, proceed as follows:

1. Locate the nozzle boot on the dispenser and insert the nozzle hook and carriage bolts into nozzle boot. Attach using hex head screws, carriage bolts, and nuts.
2. Tighten the two nuts onto the lower carriage bolts using a 3/8-inch nut driver or socket.
3. Attach the two upper hex head screws using a 7-mm nut driver or socket.
4. To identify existing nozzle hook retaining screw and nut locations, see Figure 7-16 on page 7-37. Identifying marks are located under right hand row of indented hole locations. Standard nozzle hook locations are A and D.

Dispenser Nozzle Boot Installation

To install the nozzle boot, proceed as follows:

1. Position the nozzle boot over correct positioning holes for nozzle type.
2. Attach the nozzle boot to the dispenser using the two existing screws.
3. Hold the nozzle boot upright and insert the nozzle over the nozzle hook and into boot. Wiggle the boot to verify if the nozzle does not slip out of position.
4. Power up the unit and verify proper operation.

Nozzle Boot Spring Repair (SB1483)

Encore lift-to-start (fipper) nozzle boots are being replaced to correct a broken spring condition. If the failure is just the spring, replace only the spring (M00349B003).

Manifolds

Read and follow all safety precautions as outlined in “Read This First” on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Topics in This Sub-section

⚠ WARNING

Residual pressure and entrapped fuel may still be present and may drain or spray when removing parts and fittings.

Fire and explosion can result in severe injury or death.

- Wear eye protection.
- Remove the parts and fittings slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

About Hydraulic Manifolds

The hydraulic manifold is a fluid input chamber, where fluid is distributed into one or two outlets. The manifolds included in this section are:

- Hydraulic Tree Assembly
- Steering Manifold
- Configuration Manifold

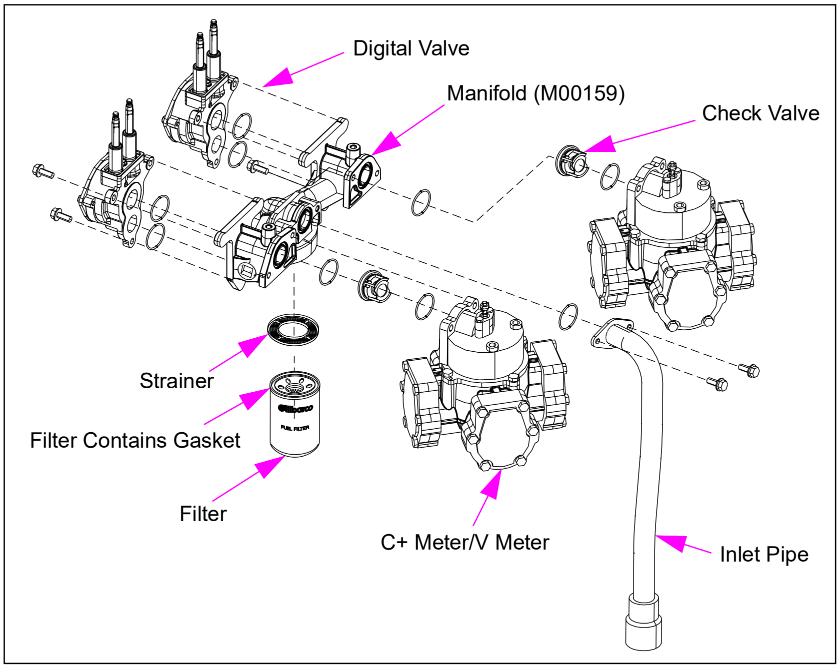
Note: Do not over-torque inlet piping into hydraulic manifold.

Hydraulic Tree Assemblies

The Encore manifold, meters, valves, and filter make up the hydraulic tree. There are two versions that are distinguished by different valves and manifolds. For units with digital valves, refer to “Manifold with Digital Valves (M00159)” on page 7-41 and for units with proportional valves, refer to “Manifold with Proportional Control Valves [PCVs (M02656)]” on page 7-41.

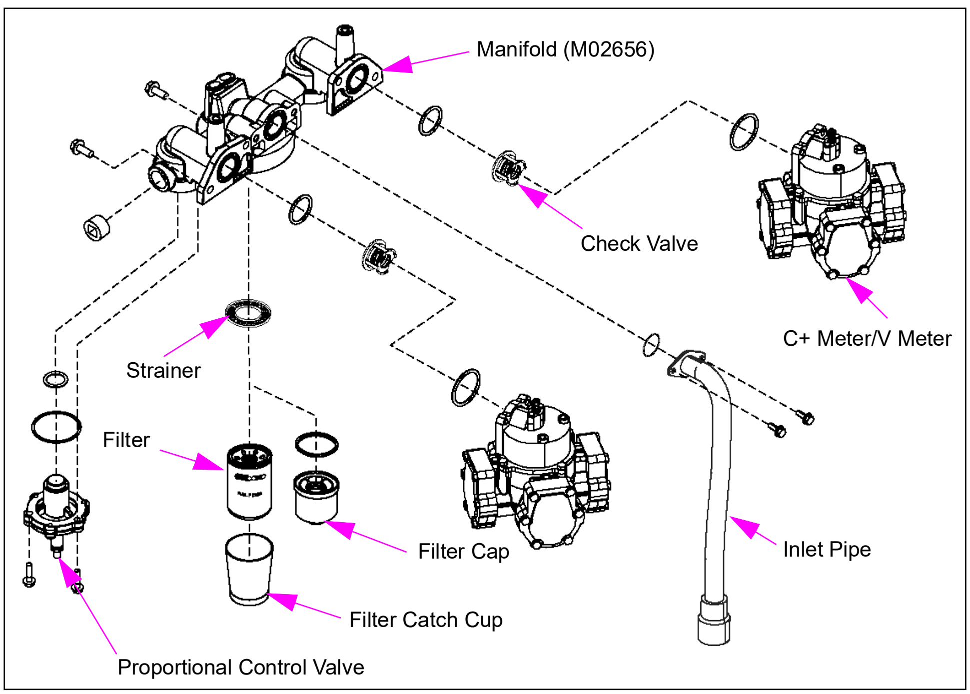
Manifold with Digital Valves (M00159)

Figure 7-17: Hydraulic Tree with Digital Valve



Manifold with Proportional Control Valves [PCVs (M02656)]

Figure 7-18: Hydraulic Tree with PCV



Service Tips

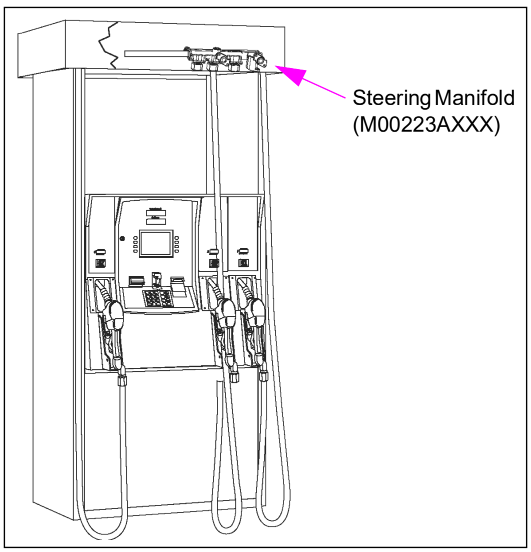
- To avoid damage to the hydraulic manifold, always keep the inlet piping support U-bolt in place.
- When replacing inlet union nipple, loosen the U-bolt only enough to allow nipple to turn with wrench. Do not pull the wrench and pipe up and toward you while turning. After completing the service, U-bolts must be tight and in place to ensure proper operation of shear valve during impact.
- When replacing filters, refer to "Filters and Strainers" on page 7-19.

Steering Manifold

The Encore series steering manifold and valves are located in the upper housing of six-hose Encore blenders (multi-hose) to direct the fuel to the proper hose. Attached to the steering manifold are three steering valves. These valves ensure that only the selected hose is activated. The valve is activated by the unit's electronics to open the appropriate chamber to direct the fuel. Fuel enters steering manifold through the Low and High product lines.

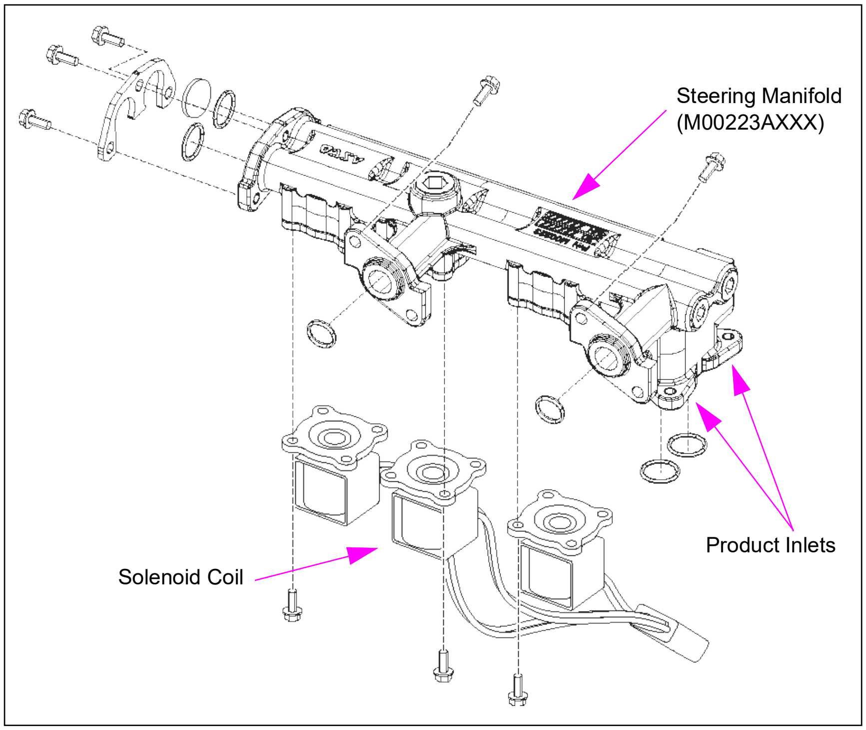
Eclipse series units are not configured as a multi-hose blender and therefore do not require a steering manifold. For more information, refer to "Solenoid Valves" on page 7-76.

Figure 7-19: Steering Valve Manifold in Multi-hose Blender



Steering Manifold (M00223AXXX)

Figure 7-20: Steering Manifold Components



Service Tips

- The main problems that you will notice with the steering valve manifold are: leaks at the connection points and valve seals, coil related problems, and no flow with certain products.
- Ensure that filters and stainers are in place. Non-use of filters and strainers can cause this and other valves to fail.
- For additional information on seal problems, refer to "Servicing Seals, O-rings, and Gaskets" on page 7-4.

Configuration Manifold

The configuration manifold channels fuel to the proper hose outlet and is incorporated into the fuel discharge assembly mounted in the upper housing of Encore series units manufactured as of March 2002. This manifold is not incorporated into Encore six-hose blenders and Eclipse series units.

This configuration manifold is of open-closed or blend design and allows unit conversions from single-hose blenders to standard units and vice-versa without major rework. For any conversion kit instructions, refer to the following description and see Figure 7-23 on page 7-45.

- Open manifold - The passages in this manifold allow fuel from all fuel discharge pipes to be channeled to one hose outlet. The open configured manifold is primarily used for single-hose MPD.

- Closed manifold -The passages in the closed manifold channel fuel from the discharge piping to be separated and directed to separate outlets. The closed manifold design is used primarily on multi-hose units.
- Blend manifold - The blend manifold allows two different grades of fuel to mix just before being discharged to the outlet casting.
  Note: The Blend manifold is used primarily on blender units.

Model	Configuration	Manufactured
Encore	All except 6 Hose Blender	Before March 2002
	All except 6 Hose Blender	March 2002 and after
	6 Hose Blender	N/A*
Eclipse	Any unit	N/A*

**Configuration manifold not applicable to these units.*

Figure 7-21: Configuration Manifold Location

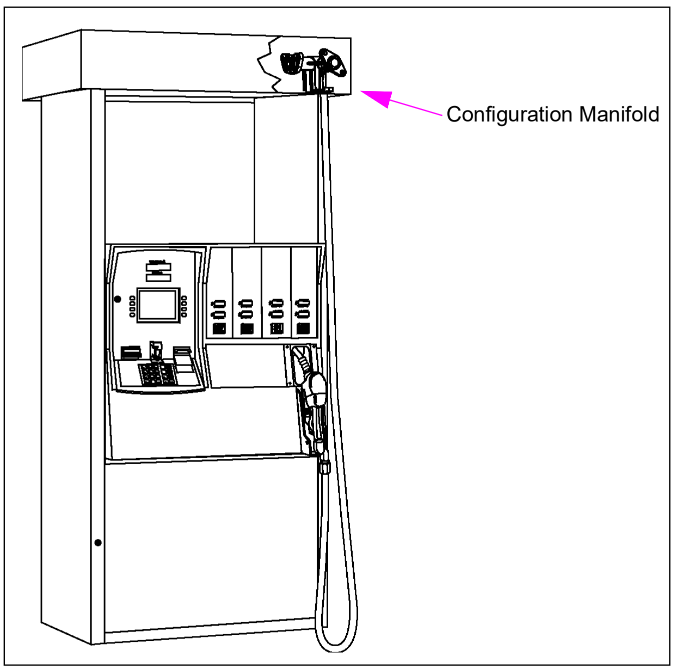
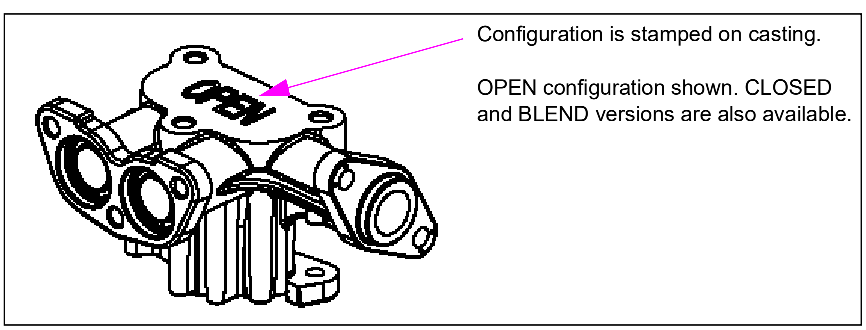


Figure 7-22: Configuration Manifold



The following cutaways show the features of each type of configuration manifold.

Figure 7-23: Open Configuration Manifold Cutaway

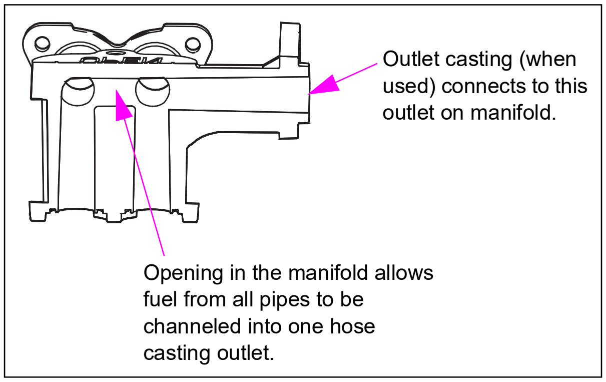


Figure 7-24: Closed Configuration Manifold Cutaway

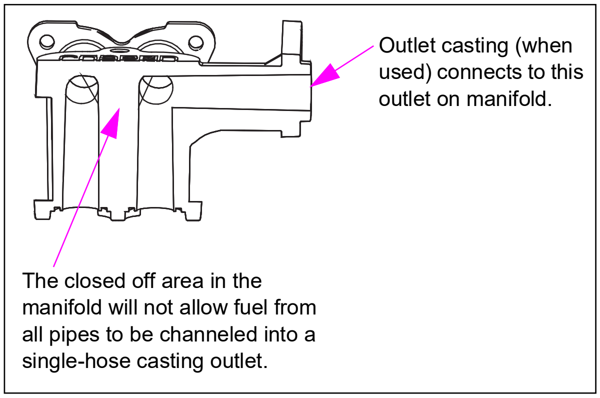
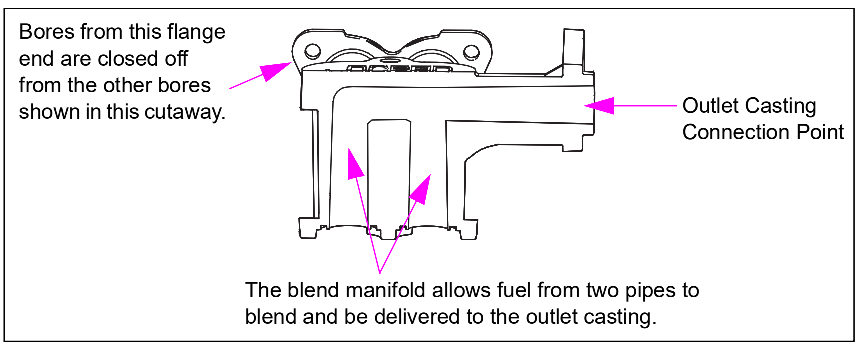


Figure 7-25: Blend Configuration Manifold Cutaway



Meters

Read and follow all safety precautions as outlined in "Read This First"on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Topics in This Sub-section

⚠ WARNING

Residual pressure and entrapped fuel may still be present and may drain or spray when removing hoses.

Fire and explosion can result in severe injury or death.

- Wear eye protection.
- Remove hoses and nozzles slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

About Meters

Gilbarco C+ meters and V meters are used on standard Encore units. For more information, refer to "C+ Meter/V Meter". For Encore Ultra-Hi units, refer to "LC Meter (Encore Ultra-Hi)" on page 7-48. Gilbarco does not recommend a major field overhaul of the meter. Suggested repairs are related to fixing external leaks. The only serviceable parts on the meter are gaskets and seals. Most repairs require removing the meter from the unit. ATC equipped C+ meters/V meters have a special cover to accommodate probe end and test wells.

Model and Date Codes

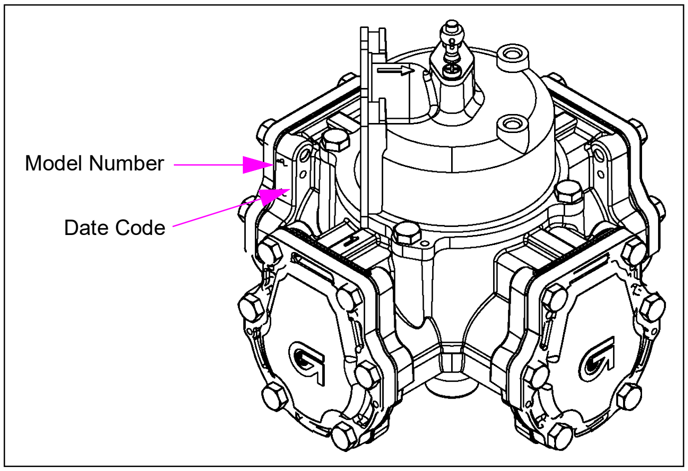
Model number is stamped on each meter (see Figure 7-26 on page 7-47) or may be listed in the serial number label on the meter. For a breakdown of this designation, refer to model configurations.

C+ Meter/V Meter

Both C+ meter and V meter are positive displacement piston meters, where fluid is pushed into a cylinder which causes the opposing piston to push a similar cylinder of fluid out. The piston connecting rod converts the piston movement into rotation of the pulser shaft. Both C+ meter and V meter have a flow path that discharges product from the center chamber on the bottom of the meter. This configuration reduces water and contaminants from building up in the meter, thus resulting in longer meter life.

The C+ meter/V meter is electronically calibrated. C+ meters/V meters for Encore 300 are not shipped calibrated from factory and precise calibration must be performed at start-up. Encore 500 and Eclipse units built after December 2002 are shipped with initial factory calibration. An error message is displayed if calibration at start-up is not completed. Meter calibration must always be verified at start-up.

Figure 7-26: C+Meter/V Meter Date and Model Code Location



Service Tips

- Purge air from the system before performing calibration. Air in the system will affect accuracy of the calibration. For additional details, refer to "Purging Air from the System" on page 7-7.
- Place the approved pan under the meter during repair to collect residual fuel.
- Replacing the top quad ring requires use of a special tool that is included with the replacement seal kit. This tool protects the new seal during installation over the drive pin cross hole in the shaft.
- Usually, it is not cost effective to rebuild a meter. If the meter exhibits problems, install a new meter.
- Remove and replace the meter using metric tools.
- Inspect all gaskets during off unit repairs. Replace as required.
- Some meter bolts are difficult to access. A 13-mm universal socket will make installation and removal easier.
- For calibration problems, refer to "Calibration" on page 7-7.
- After a drive-off, inspect the meter for leaks.
- Non-use of filters and strainers can significantly reduce meter life.
- Abnormally low pressure and flow rates can cause meter to stall.
- Meter failure may be indicated by a significant change in calibration from high to low flow rates (excessive calibration spread), or meter stall at low flow.
  Note: Other problems may also cause the same symptoms.
- Calibration problems are not always caused by a bad meter. Meter check valves, leaking nozzles, binding pulsers, defective pistons, or internally leaking vapor recovery hoses can also result in calibration problems.

Meter Seepage (SB1441)

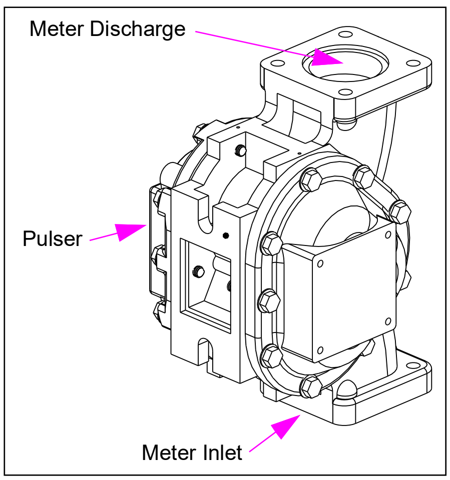
Current seals used in the C+ meter/V meter are capable of handling very low temperatures. However, some sites have noted seepage at -15 °F (-9.4 °C) or colder. If sites with C+ meters/V meters in your area have experienced extreme cold weather seepage at the shaft seals, the units may be fixed by using special seals developed for extremely cold areas. Depending on fuels, meter tolerances, and other factors, some meters will require special seals. It is generally required only to use the special seals where seepage is noted to occur when temperatures are extremely low [approximately -15 °F (-9.4 °C) or colder]. Monthly leak inspections of the dispenser hydraulics will catch any leaks that must develop.

The cold weather meter seal kit is K35222-01. The standard seal kit is K35222. The standard kit is considerably less costly than the K35222-01 Kit. Also, the standard kit is completely acceptable for even very cold sites. Hence, use of cold weather seal kits across the board is not recommended. One kit services ten meters.

LC Meter (Encore Ultra-Hi)

LC meters are used in Encore Ultra-Hi Master or Ultra-Hi Combo series units.

Figure 7-27: LC Meter and Pulser Assembly



Service Tips

- Remove and replace the meter using metric tools.
- Place the approved pan under meter during repair to collect residual fuel.
- Inspect all gaskets during off unit repairs. Replace, as required.
- Some meter bolts are difficult to access. A 13-mm universal socket will make installation and removal easier.
- For calibration problems, refer to "About Calibrating Meters" on page 7-18.
- Non-use of filters and strainers can significantly reduce meter life.
- Meter failure may be indicated by a significant change in calibration from high to low flow rates (excessive calibration spread), or meter stall at low flow.
  Note: Other problems may also cause the same symptoms.
- Calibration problems are not always caused by meters; dispenser check valves and leaking nozzles may also cause calibration problems.

Coriolis Meter

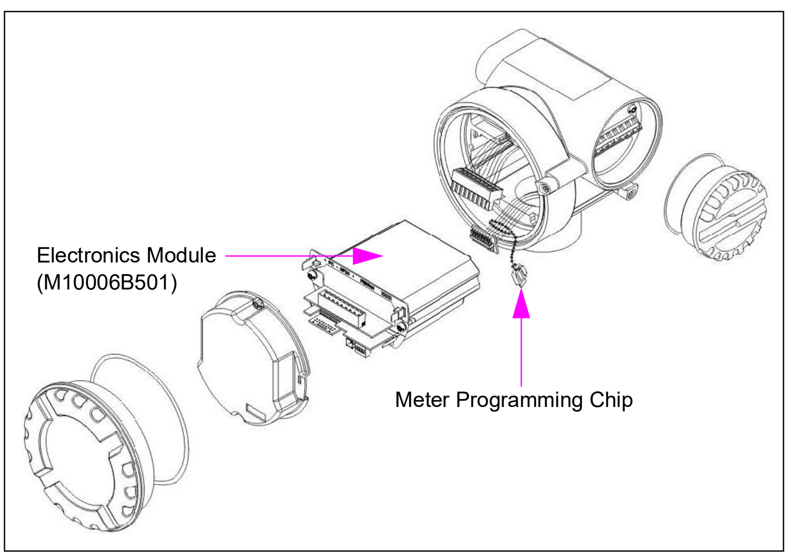
The Coriolis meter is a very precise mass flow meter that can be used to measure volume for fluids of consistent density. It is extremely durable and has no sensitive components directly exposed to fluids.

Figure 7-28: Coriolis Meter

Some important considerations for servicing the Coriolis meter include the following:

- The meter electronic parts are serviceable by replacement. Typical ECs associated with pulsers indicate symptoms of failure in the electronic parts. In case of such symptoms, replace the electronic parts.
- The expensive meter electronics and hydraulics assembly can also be replaced.
- The meter hydraulics including the transducers are not serviceable. If these components are damaged, replace the meter.
- When servicing the electronics, you must reuse the purple chip from the previous meter or the meter will not be properly adjusted. Use ESD protection when servicing the electronics.

Figure 7-29: Components of the Coriolis Meter



- As meters are very sensitive to air in the fluid, the air can cause a pulser EC or erratic calibration results at fast and slow flow. If the calibration for the meter is erratic, then check the aboveground pumping units. Ensure that the pump is not producing excessive air due to inlet leaks in the pump. If the meter freezes, then check the calibration of the meter. If there is a large difference between fast and slow flow values, then the meter is permanently damaged.
  Note: Presence of air changes the fluid density and mass flow rate.
- The meters can be permanently damaged by freezing. Usually the symptom is repeatable tests at high flow rate with a large change in calibration resulting at slow flow rate. That is, meter linearity error is high.
- When purging the air from the meter during installation, temporarily connect a standard Encore S pulser to the PCN, as the meter is sensitive to air. When dispensing DEF, spin the pulser manually until all air is purged from the system. If this is not done, an EC may be generated by the dispenser. Ensure that you turn off the dispenser before disconnecting the output between the Coriolis meter interface board and PCN when connecting or disconnecting the temporary pulser and Coriolis meter interface board.

⚠ WARNING

Do not substitute the pulser while power is being supplied to the unit. Lethal voltages exist within the dispenser and damage to the unit can occur if certain critical steps are not followed. It is recommended that only Gilbarco-certified ASCs who are trained in service and operation of Encore dispensers perform this procedure.

The meter is electronically calibrated similar to standard meters and Ecometers. There is no change to this procedure.

For additional information, refer to “Coriolis Meter Interface Board” on page 6-87.

Pulsers

Read and follow all safety precautions as outlined in “Read This First” on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Topics in This Sub-section

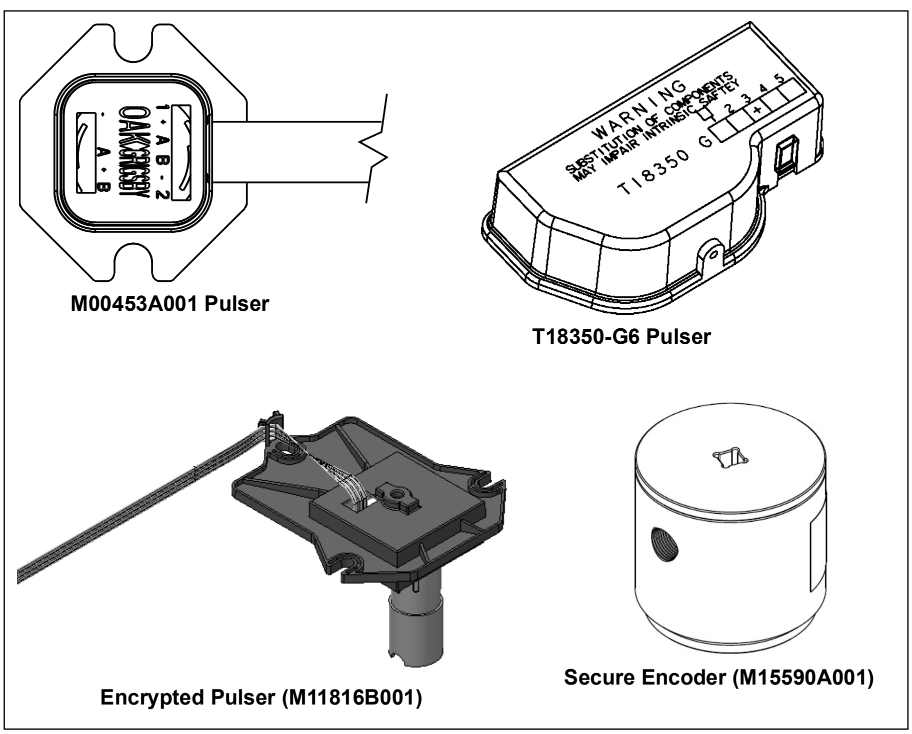
About Pulsers

Pulsers convert meter rotation to pulses per gallon (or liter) for use in some units by the controller board. You can program the unit to scale pulses for specific applications. For example, standard gallonage units use 1000 pulses per gallon. For other conversion factors, refer to “Pump Programming” on page 5-1.

Earlier, Encore 500 units used a M00453A001 Pulser, which was replaced with an Encore 300 style pulser (T18350-G6) in September 2002. The M00453A001 Pulser will continue to be available as a replacement part. The minimum version of PCN software required to support the T18350-G6 Pulser is V1.6.00. Eclipse will continue to use the M00453A001 Pulser. Ultra-Hi units use the T18350-G6 Pulser or the Secure Encoder (M15590A001).

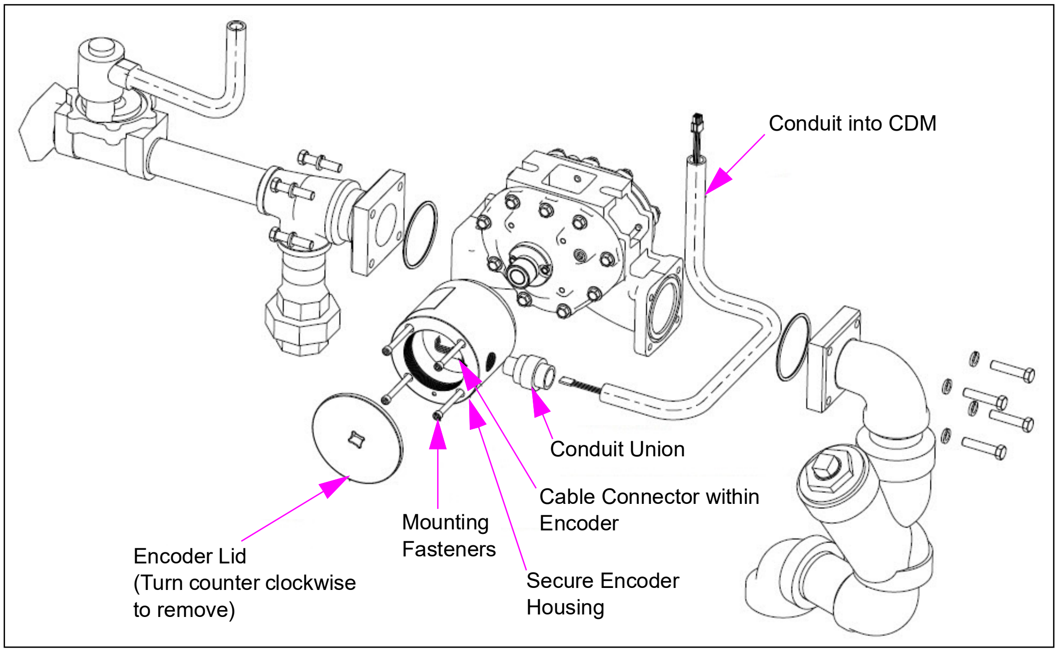
The Encrypted Pulser (M11816B001) functions like previous pulsers but includes encrypting all communications to the dispenser. Older pulsers can be replaced with the encrypted pulser.

Figure 7-30: Pulser Assemblies



Ultra-Hi Secure Encoder Assembly

Figure 7-31: Ultra-Hi Secure Encoder Assembly



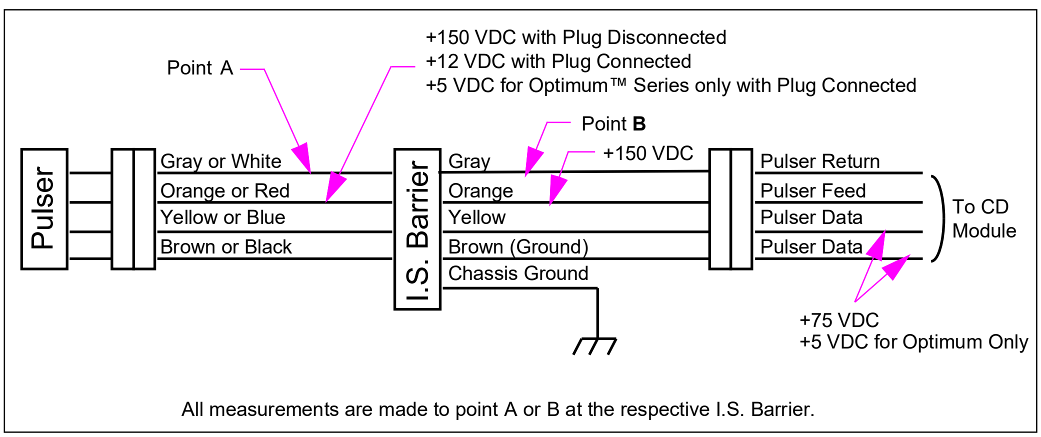
Testing Pulsers and Intrinsic Safety (I.S.) Barriers

Voltage is limited to 12 VDC in the pulser. An I.S. barrier is used to limit current.

Test the pulser by substitution:
- Unplug pulser and plug a known good one in.
- Rotate the pulser and watch the displays for count.

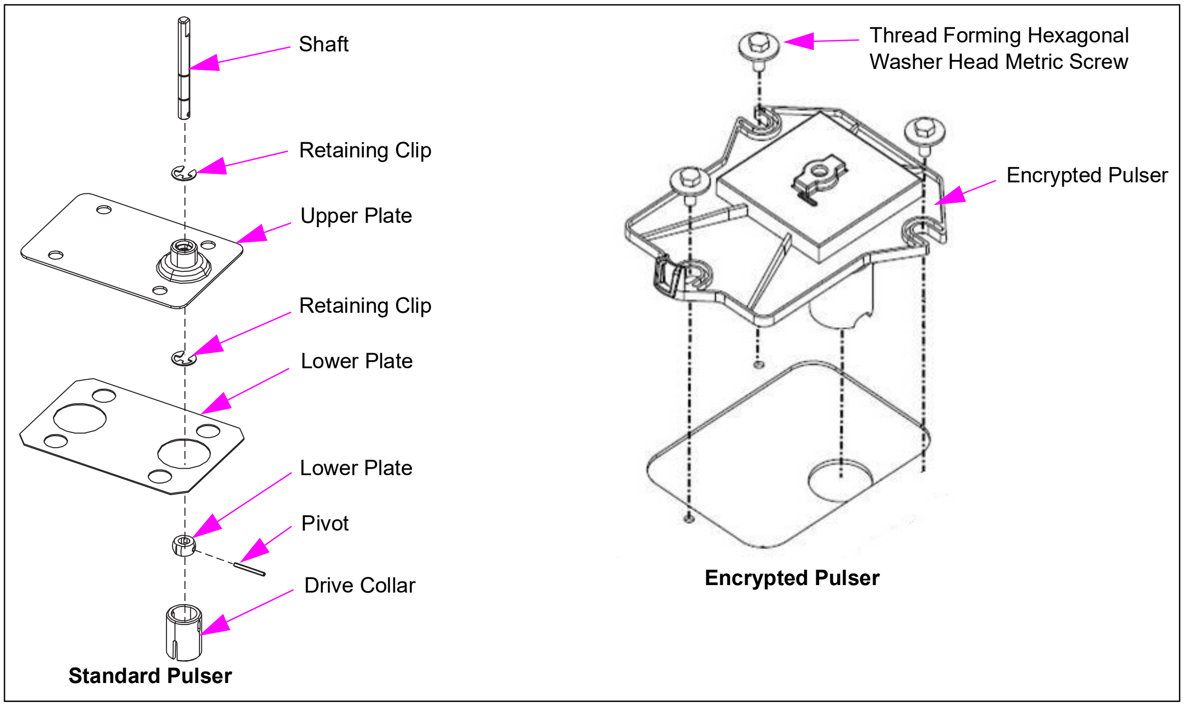
Test the I.S. barrier by substitution or disconnect it and check it with an ohmmeter.
- Measure 10,000 ohms +/- 20% from gray to gray, yellow to yellow, brown to brown, and orange to orange.
- Always read a low resistance from the green wire to the case.
- Measure between each wire. Meter must indicate an open.

Figure 7-32: Pulser and I.S. Barrier Test Points



Pulser Drive Assembly

Figure 7-33: Standard and Encrypted Pulser Drive Assembly



Special Note for Canadian Installations

An O-ring is used on the shaft in Canadian installations. If squealing is noted, lubricate with silicone grease. When replacing a meter, align the lower drive collar to meter shaft to reduce the potential for binding.

Service Tips

- During shipment, T18350-G6 Pulsers are restrained by a tie-wrap and must be removed during installation. If a tie-wrap is discovered on the pulser during service, it must be removed. A tie-wrap left in place can result in binding of the pulser shaft.
- Inconsistent communication with the M11816B001/B002 or M15590A001/A002 Secure Encoder will usually indicate a defective pulser, if the unit had been programmed and calibrated properly.
- Event Logs can be used to pinpoint a specific pulser while troubleshooting pulser errors such as 20, 5047, 5049, and 5050. Pulser locations are designated by ePulserX. To determine the location of the pulser, refer to the following table:

Pulser Name	Signifies This Side of Unit	At This Product Pulser
ePulser1	1	1
ePulser2	1	2
ePulser3	1	3
ePulser4	1	4
ePulser5	2	1
ePulser6	2	2
ePulser7	2	3
ePulser8	2	4

Troubleshooting Tips

A faulty pulser could be responsible for generating any of the following pulser errors:

EC	Error Condition	Actual/Potential Causes
20	Pulser Disconnected	Faulty or broken cable
5047	Reverse Flow	• Leaky Valve • Dispenser not configured for encrypted pulser
5050	Invalid Pulser Pattern	Dispenser not configured for encrypted pulser
8025	ID Mismatch	• Initial start-up • Different pulser installed
8040	Communication Interrupt	Faulty or broken cable
8041	Missing Data	Broken or loose data wire
8042	Missing Clock	Broken or loose clock wire
8044	Checksum Error	• Electrical noise • Faulty cable
8046	Lift-off Detection	• Pulser not engaged with meter shaft • Pulser too far away from magnet cup*
8050*	Sensor Failure/Improper Magnet Field	• Missing magnet cup • Pulser too far away from magnet cup
8051*	Lid Separation	Lid removed/not securely tightened

**Specific to Ultra-Hi Encrypted Pulser*

Note: For more information on Standard Encrypted Pulsers, refer to MDE-4950 Encrypted Pulser Retrofit Kits (M12514K00X) Installation Instructions for Encore 500/S/700 S/S E-CIM, MDE-5354 Ultra-Hi Encrypted Encoder Retrofit Kits (M15177K00X, M15178K00X, M15179K00X, M15180K00X) Installation Instructions for Encore 500/S/700 S/S E-CIM, and MDE-3860 Programming Quick Reference Guide.

Pulser Fail (SB1470)

EC20 (pulser fail) has been enhanced to further aid troubleshooting. The software has been modified to display only the pulsers affected by the EC in the PPU displays. The EC can now also be cleared by just lowering the pump handle. Previously, EC20 used to display in all PPU displays and a warmstart was required to clear the EC.

Note: EC20 and EC8046 are latching ECs. After the cause has been prepared, to clear the error, enter programming mode from the manager keypad (any level) and then exit.

Pulser Rotation Error (SB1451)

The M04012B001 Pulser currently “learns” its correct rotational direction by monitoring the first 20 pulses every time the dispenser is powered up. In case the pulser replaces an existing pulser and it turns (or be turned) in the wrong direction for the first 20 pulses or more after power-up of a unit, it will establish the rotational direction as being correct. The subsequent delivery will result in an EC5047. To avoid this, do not assemble or rotate a “live” pulser in the wrong direction just after replacement or power-up. If the pulser is inadvertently rotated incorrectly, you can simply correct the condition by powering the unit down and back up ensuring that the pulser turns in the proper direction for the first 20 pulses. Software changes are being investigated to prevent this minor potential problem (eliminate the requirement of learning rotational direction) and you will be notified of any new software release.

Note: EC5047 and EC5050 are pre-existing ECs. In case of encrypted pulser retrofitting, these errors may be dispalyed on initial start-up before the dispenser is configured for encrypted pulsers. After the dispenser is configured properly, EC5047 takes on its original meaning (reverse flow) and error 5050 becomes invalid for an encrypted pulser configuration.

Pulser Interchangeability (SB1451)

The M04012B001 Pulser is backward compatible with the old pulser and vice-versa. However, certain important EC information will not be provided if the new design is replaced with the old. Therefore, new pulsers must always be used to service new pulsers except in case of an emergency situation where it is not available. The pulser is similar in internal design to those used for Eclipse units, but is not interchangeable with Eclipse pulsers. As the pulser is of quadrature design, it can be used to reliably detect reverse flow, such as for a blender when a meter check valve is bad. EC5047 will occur if reverse flow is detected. For service, the pulser can be used with T18350-G6 Pulsers in the same unit, as long as pulsers are replaced as pairs.

The T18350-G6 Pulsers are still being used for Encore 300 units.

Pumping Device - GPU+

Read and follow all safety precautions as outlined in “Read This First” on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Topics in This Sub-section

⚠ WARNING

Residual pressure and entrapped fuel may still be present and may drain or spray when removing parts and fittings.

Fire and explosion can result in severe injury or death.

- Wear eye protection.
- Remove the parts and fittings slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

About the GPU+

Self-contained Encore pumps use the GPU+ (M09593A102) pumping unit produced by GVR. The previous pumping unit M04920B003, was not produced by GVR. For detailed service information refer to *MDE 5662 Gilbarco GPU+ Operation and Service Manual*.

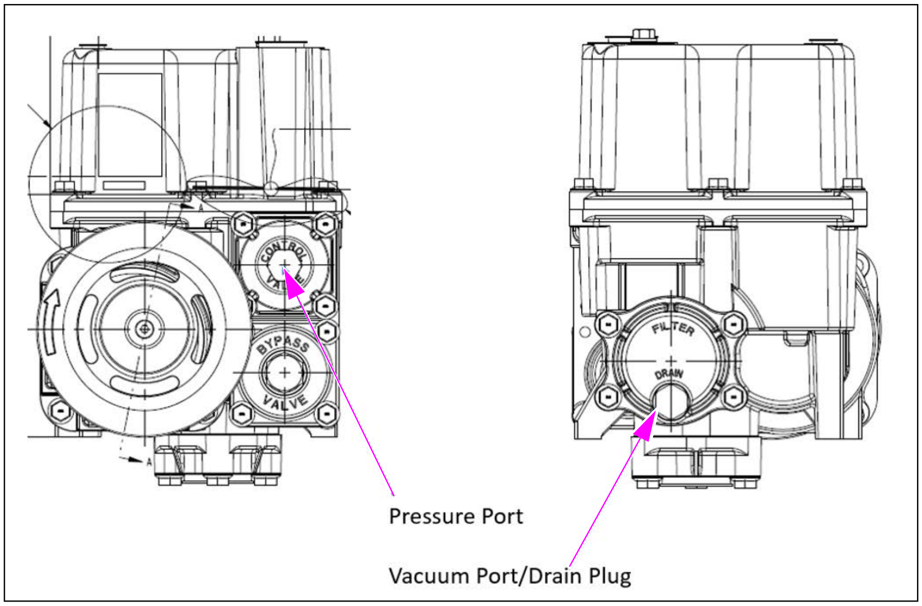
The GPU+ (M09593A102) is a direct replacement for the previous M04920B003 pumping units (vane type), except for the motor pulley to adjust speed for the GPU+ and a pump mounting bracket for use on Encore. The four grades on Encore models also require a different motor conduit (M16125B003).

Notes: 1) Pumping units do not come with Inlet Check valves and are not interchangeable between M04920B003 and M09593A102. Order check valves separately. For units sharing inlets, both pumping units are required to have internal Check Valve (M09593K207 GPU+ Inlet Check kit).
2) Inlet Check valves (other than Internal Check valves) are prone to have excessive spring preload and can be detrimental to unit performance.

The following kits are needed for upgrading to GPU+ assembly, as indicated for the dispenser:

Kit Number	Kit Description	Standard Belt (previous part)	Standard Pulley (previous part)	Noise/Power Reduction Belt (included in kit)	Noise/Power Reduction Pulley (included in kit)
M19196K001	Kit, Gear GPU Retrofit, Encore 60Hz	R06711-59 (39” belt)	R18900-34	N/A	R18900-30
M19196K002	Kit, Gear GPU Retrofit, Encore 50Hz	R06711-61 (40” belt)	R18900-41	R06711-59 (39” belt)	R18900-34

Figure 7-34: GPU+ (M09593A102) Assembly



General GPU+ Service Procedures

This service manual contains basic service procedures for the GPU+ unit. For additional GPU+ service information, refer to *MDE-5662 Gilbarco GPU+ Operation and Service Manual.*

Pumping Device - GDP

Read and follow all safety precautions as outlined in “Read This First” on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Topics in This Sub-section

⚠ WARNING

Residual pressure and entrapped fuel may still be present and may drain or spray when removing parts and fittings.

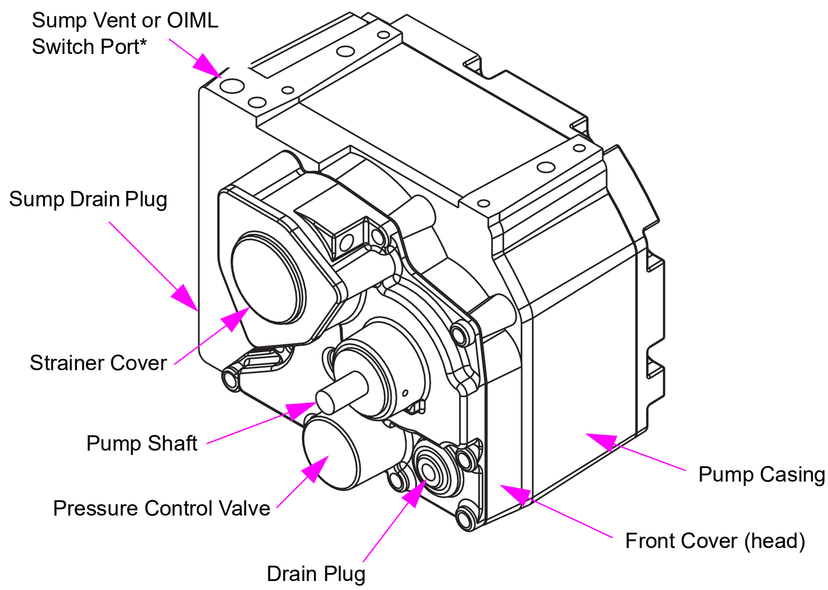
Fire and explosion can result in severe injury or death.

- Wear eye protection.
- Remove the parts and fittings slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

About the GDP

Self-contained Encore pumps use the Blackmer GDP. Service on the pump must be performed in the shop and not attempted in the field. The case is made of Aluminum and sealing surfaces may be damaged while working on an inappropriate surface. For additional information about the GDP unit, refer to *MDE-4096 Blackmer Global Dispenser Pump Operation, Maintenance and Kit Installation Manual.*

Figure 7-35: GDP Assembly (Pumping Unit GDP used until 2005)



**OIML switch is not required for U.S. domestic units.*

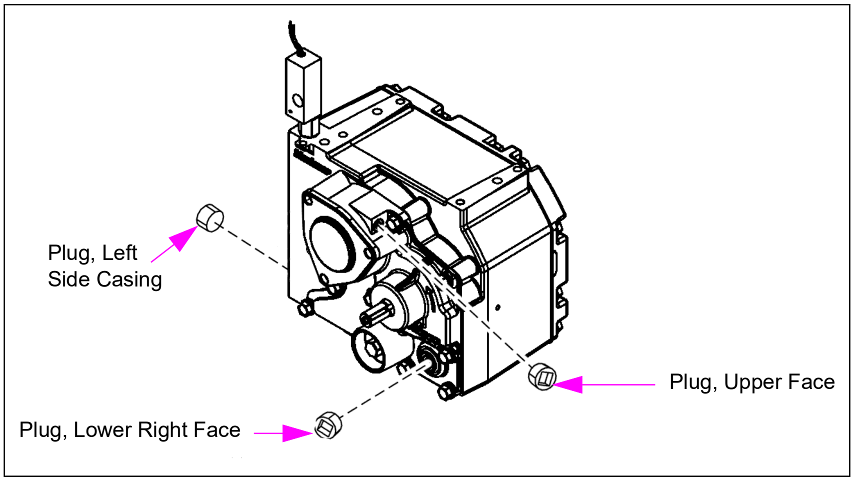
General GDP Service Procedures

This service manual contains basic service procedures for the GDP unit. For additional GDP service information, refer to *MDE-4096 Blackmer Global Dispenser Pump Operation, Maintenance and Kit Installation Manual.*

Draining the GDP

Draining the GDP unit to some extent is required when opening the unit for servicing the strainer or removing pressure valve.

Figure 7-36: Plug Locations



To drain the bulk of the fluid from the pump, proceed as follows:
Note: Residual fluid may remain in the pump.

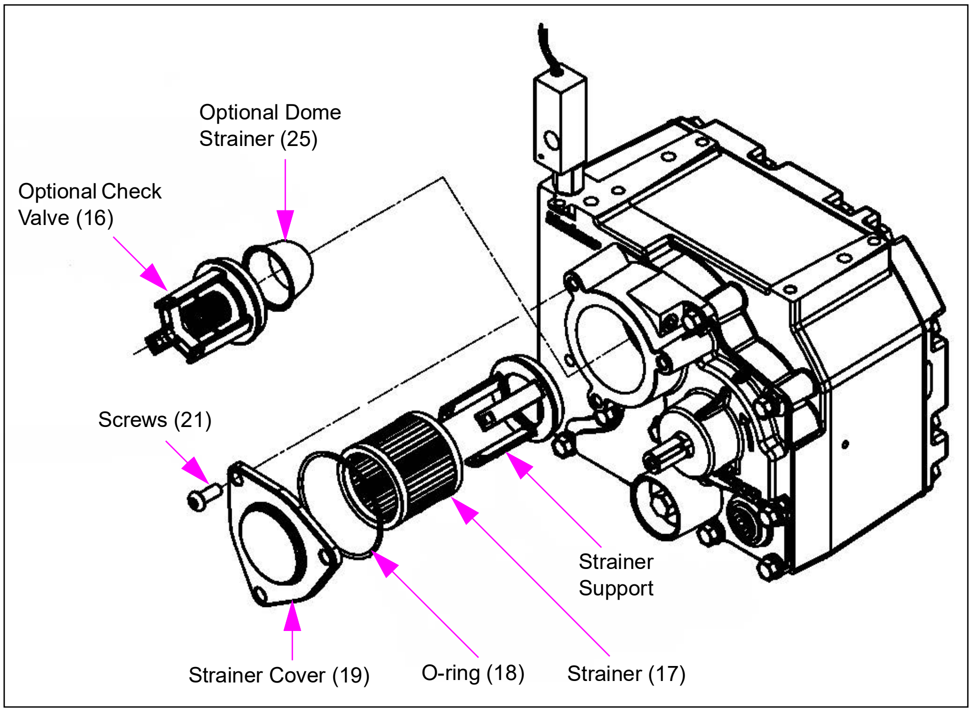
1. Put the approved tray in place to catch fuel.
2. Remove the plug on the upper face of the head, directly to the right of the strainer cover. This port may be used to monitor vacuum (see Figure 7-36).
3. Remove the plug on the lower right face of the head. Fluid will be emptied by removing this plug. Fluid must be properly contained during draining procedure.
4. Pump contains up to 0.5 liters of fluid. Do NOT use this port to monitor system pressure.
5. If sump drainage is required, the plug near the mounting hole on the left side of the pump casing may be removed.
6. Manually turn the pump pulley counterclockwise to aid in fluid removal.
7. Properly dispose of all fluid drained from pump.
8. After pump is drained, replace all plugs using a non-hardening pipe sealant suitable for type of fluids being used.
   Note: Do NOT use Teflon tape.

Strainer/Check Valve Service

Strainer/Check Valve Removal

Check the valves (if equipped) must be replaced only if the pump is experiencing frequent loss of prime, indicating that the check valve is not functioning. Check valve must be installed when the tank check valve is not available. Always use strainers with or without check valves. Strainer must be kept clean to ensure proper operation and pump life.

Figure 7-37: Strainer/Check Valve



To remove the strainer/check valve, proceed as follows:

Note: The numbers within parenthesis in the following procedural steps refer to the numbers in Figure 7-37.

1. Lower the pump fluid level by removing the plug located at the right of the strainer cover and rotating the pump shaft counterclockwise approximately ten turns.
   Note: Removing the plug allows air to enter.
2. Remove the three strainer cover screws (see Figure 7-37).
3. Remove the strainer cover. Inspect the cover O-ring (18) for damage. Discard the O-ring and replace with new O-ring, if damaged.
4. Carefully pull out the strainer (17). Keep the strainer in a horizontal position to avoid contaminating pump with strainer debris. It may be required to use a small tool to gently hook the inside of the metal strainer end cap.
5. Remove the inlet check valve assembly (16), if equipped. Inspect for cracked or damaged O-rings. Replace, if damaged.
6. Lift the dome strainer (25) out from the bore located behind the check valve assembly. Inspect the strainer for damage. Clean and replace, if required.

Strainer/Check Valve Installation

To install the strainer/check valve, proceed as follows:

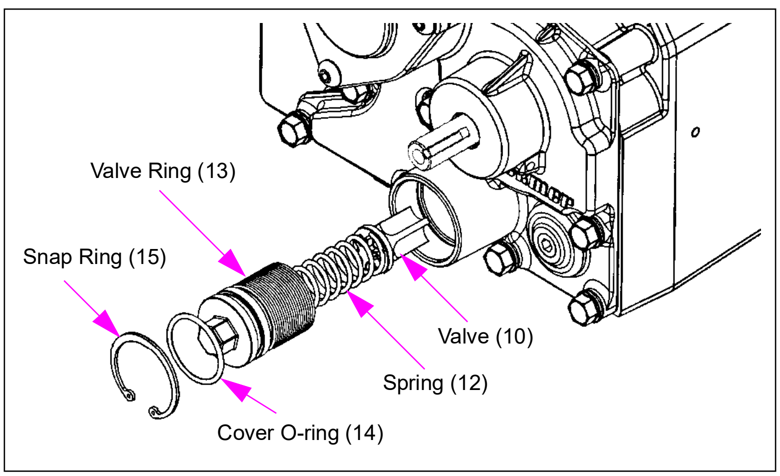
Note: The numbers within parenthesis in the following procedural steps refer to the numbers in Figure 7-37 on page 7-60.

1. If equipped with check valve option, insert the dome strainer (25) into the bore.
2. Clean debris from check valve, especially at the O-ring seats.
3. Lubricate the O-ring before installing the check valve assembly.
4. Insert the check valve assembly (16) or strainer support (30) into the corresponding pump bore as shown in Figure 7-37 on page 7-60. Press the assembly squarely over receiving cylindrical surface until O-ring sits firmly over the surface and holds the assembly in position.
5. Insert the new or cleaned strainer (17) into the bore over check valve assembly.
6. Clean debris from the strainer cover O-ring groove and install O-ring (18). If the O-ring does not stay in place, use a small amount of all purpose grease to hold the O-ring in the groove.
7. Install the strainer cover (19) with screws (21). Hand tighten, alternating between each screw to keep the cover parallel to the casing face. Torque the screws to 150 lbs-in (17 Nm). Do NOT overtighten. Overtightening may cause damage to the screw or casing.
8. Replace the plug (11) in the hole located to the right of the strainer cover.

Pressure Control Valve (PCV) Service

Pressure Control Valve Removal

Figure 7-38: Pressure Control Valve



⚠ CAUTION

Excessive wear on the pressure control valve can cause improper pump performance, including flow loss or excessive or low discharge pressure.

To remove the PCV, proceed as follows:

Note: The numbers within parenthesis in the following procedural steps refer to the numbers in Figure 7-38 on page 7-61.

⚠ CAUTION

Adjusting valve pressure beyond factory specifications to increase flow rates can result in premature motor failure or pump wear not covered by warranty.

1. Drain the pump by following the procedure outlined in "Draining the GDP" on page 7-59.
2. Remove the snap ring [15 (see Figure 7-38 on page 7-61)].
3. Turn the valve cover (13) outward or counterclockwise until completely removed using a 16-mm socket wrench.
4. Remove the spring (12) and valve (10).
5. If the valve did not come out with the spring, use needle nose pliers to remove it from the bore.

Pressure Control Valve Installation

To install PCV, proceed as follows:

Note: The numbers within parenthesis in the following procedural steps refer to the numbers in Figure 7-38 on page 7-61.

IMPORTANT INFORMATION

Modification of PCV or failure to correctly install all components of the PCV will void any regulating agency recognition applicable to this pump.

1. Inspect the valve (10) for damage or excessive wear. Replace, if required.
2. Install the PCV in reverse order of removal (see Figure 7-38 on page 7-61).
3. Apply a light coating of grease on the cover O-ring (14) to help keep the O-ring from being damaged during installation.
4. When the pump is operating at full speed, discharge pressure must NOT exceed 50 PSI (3.5 bar) with any discharge restriction (fully open to fully close discharge). If the pressure exceeds 50 PSI (3.5 bar), see Figure 7-38 on page 7-61.

PCV Adjustment

PCVs must be set to the required single-hose operation pressure. A retaining ring must be in place at all times during PCV adjustment.

> **⚠ WARNING**
>
> 
>
> Not using a retaining ring can release hazardous pressure that can cause personal injury or property damage.
>
> Always ensure that the retaining ring is in place before adjusting the PCV.
>
> 
>
> Take appropriate safety precautions. If not, personal injury or property damage may occur.

To set the PCV operation pressure, proceed as follows:

1 To increase the pressure setting, turn the PCV cover (13) inward or clockwise using a 16-mm socket wrench (see Figure 7-38 on page 7-61).

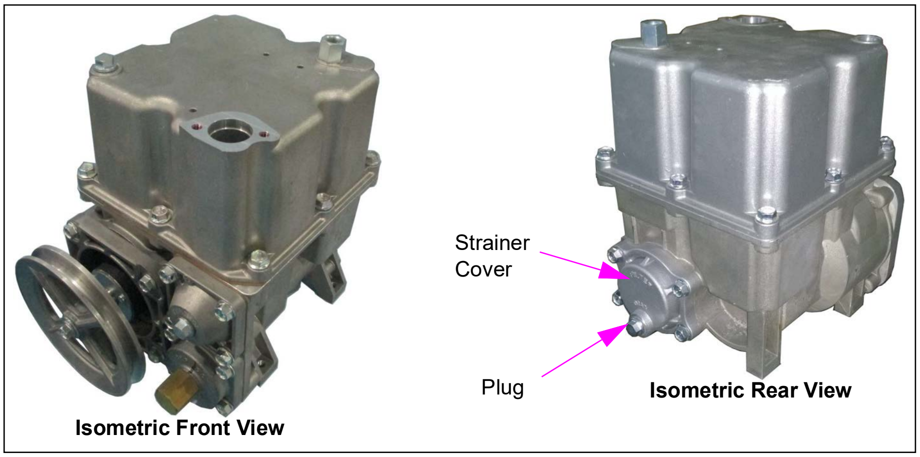
2 To decrease the pressure setting, turn the PCV cover (13) outward or counterclockwise using a 16-mm socket wrench.

Pumping Device - Gilbarco Global Pumping Unit (GPU)

Note: GPU production for Encore standard flow units began in 2005.

For service and maintenance information, refer to *MDE-4447 Gilbarco Global Pumping Unit Operation and Service Manual*.

Figure 7-39: GPU



Pump Device Motor

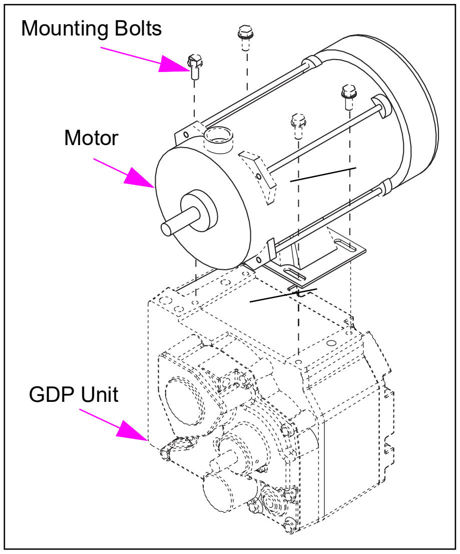
Read and follow all safety precautions as outlined in “Read This First” on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Contents in This Sub-section

About Electric Motor for Self-contained Pump Units

Read and follow all safety precautions as outlined in “Read This First” on page 1-1, this section, and related sections of this manual.

Figure 7-40: Motor for GDP pump



Thermal Overload

Most motors have a thermal overload switch which opens the circuit automatically, if the motor is drawing excessive current or overheats. When the switch cools, it activates and turns the motor back on. Motors do not require a manual reset. The 220/380 VAC motors do not have a thermal overload switch.

If overload shut-off occurs frequently:

- Check the voltage switch for proper voltage selection (220/230/240/380).
- Check for free pump and motor shaft rotation.
- Check the pump bypass valve and relief pressure setting.
- Check the belt tension.
- Check the line voltage during operation with nozzle closed and then open. Low or high voltages (greater than +/- 10%) can cause overheating or premature failure.
- Use only Gilbarco specified explosion-proof motor for replacement. National Fire Protection Association (NFPA) requires explosion-proof motors.

Belts and Pulleys

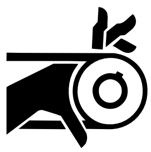
Inspecting Pulleys and Belts

Check every time that you inspect or change the belts. Replace, if the following conditions exist:

- Excessive wear in pulley grooves in pulleys or belts, excessive pulley-to-shaft clearance or excessive bearing play.
- Cracks and so on.
- Frayed belts or excessively worn belts.

Read and follow all safety precautions as outlined in “Read This First” on page 1-1, this section, and related sections of this manual. Follow OSHA tagout/lockout procedures.

⚠ WARNING



Belts and pulleys can pinch fingers and hands.

Injuries may result. Avoid placing fingers or hands between belts and pulley.

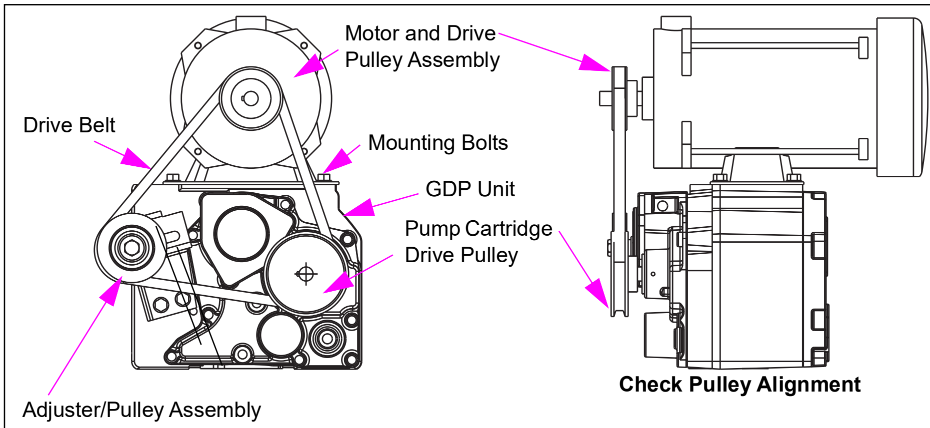
Never turn on the power to fit the belt on the pulley.

Setting Belt Tension

Tighter belt tensions can cause both pump and motor bearing wear and premature failures. Loose belts result in low flow or pulley/belt failures or noise. Use 45-55 lbs (200-240 newton) rating.

Do not run motor to install belt. Belt will be tight during assembly. This is normal. Some belts have arrows marked on them showing direction of rotation. When replacing belts, ensure that the arrow points in the direction of rotation.

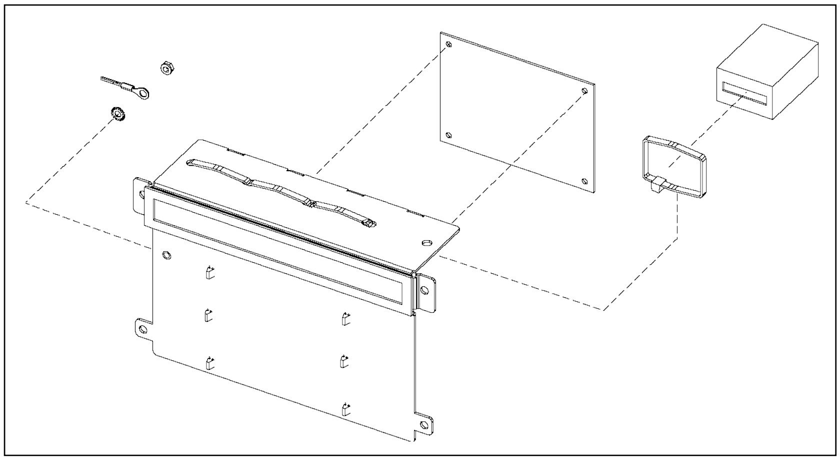
Figure 7-41: Motor and Pump Pulley Assembly



Totalizers

Read and follow all safety precautions as outlined in "Read This First" on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Figure 7-42: Electronic Totalizer Assembly



Service Tips

- Encore and Eclipse totalizers can be set to operate per product inlet, per unit, or per side.
- Programming of jumpers is required to set functionality.
- Certain problems can be indicated by ECs.

Valves

Read and follow all safety precautions as outlined in “Read This First” on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Contents in This Sub-section

⚠ WARNING

Residual pressure and entrapped fuel may still be present and may drain or spray when removing parts and fittings.

Fire and explosion can result in severe injury or death.

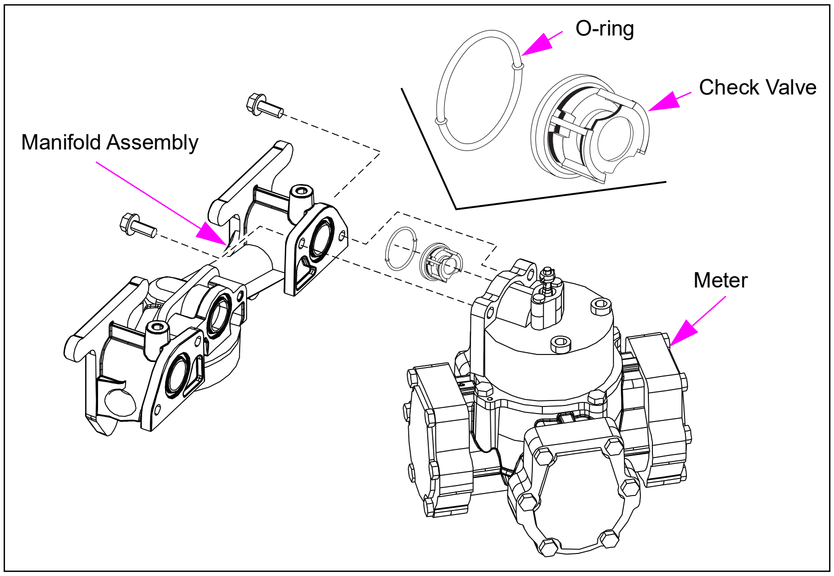
- Wear eye protection.
- Remove the parts and fittings slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

Flow control valves regulate the flow rate and blend ratio, if applicable. The Encore and Eclipse models may be outfitted with digital valves or PCVs. These valves and manifolds are not interchangeable and must be used with specific components such as air gap plates, manifolds, and electronics. The proportional flow control valve can be programmed for flow control for blenders. Non use of filters and strainers can cause premature valve failures.

Check Valves at Manifold and Meter

Meter pressure check valves help keep the meter full of fuel. They also help relieve excessive pressure due to thermal expansion that can damage parts or make nozzles difficult to open. For blending, they also prevent crossover flow from tank-to-tank. The check valves are located in the meter intake port.

Figure 7-43: Check to Meter Orientation



Some Potential Problems

- Damage to hydraulic parts can occur from thermal expansion or driving over a hose, if valve is inoperative.
- If the valve sticks open, a sale may register when you first turn the nozzle hook ON with nozzle closed. Also, check the STP check valve for problems.
- Blend ECs indicating product contamination.
- Stuck valve causing reduced flow rates
- Leaks
- Calibration problems

Service Steps

- Check valves are not rebuildable; replace any defective check valves.
- Wear eye protection for this process.

1 Close and test the shear valves (for dispensers).

2 Remove power to the unit and involved STPs. Follow OSHA lockout/tagout procedure.

3 Open the nozzle into an approved container to bleed pressure. Some residual pressure may remain.

4 Replace the check valve using Gilbarco repair kit.

5 Inspect metallic seat of valve body for damage or uneven seating surface.

6 Clean the contamination from valve body interior when servicing.

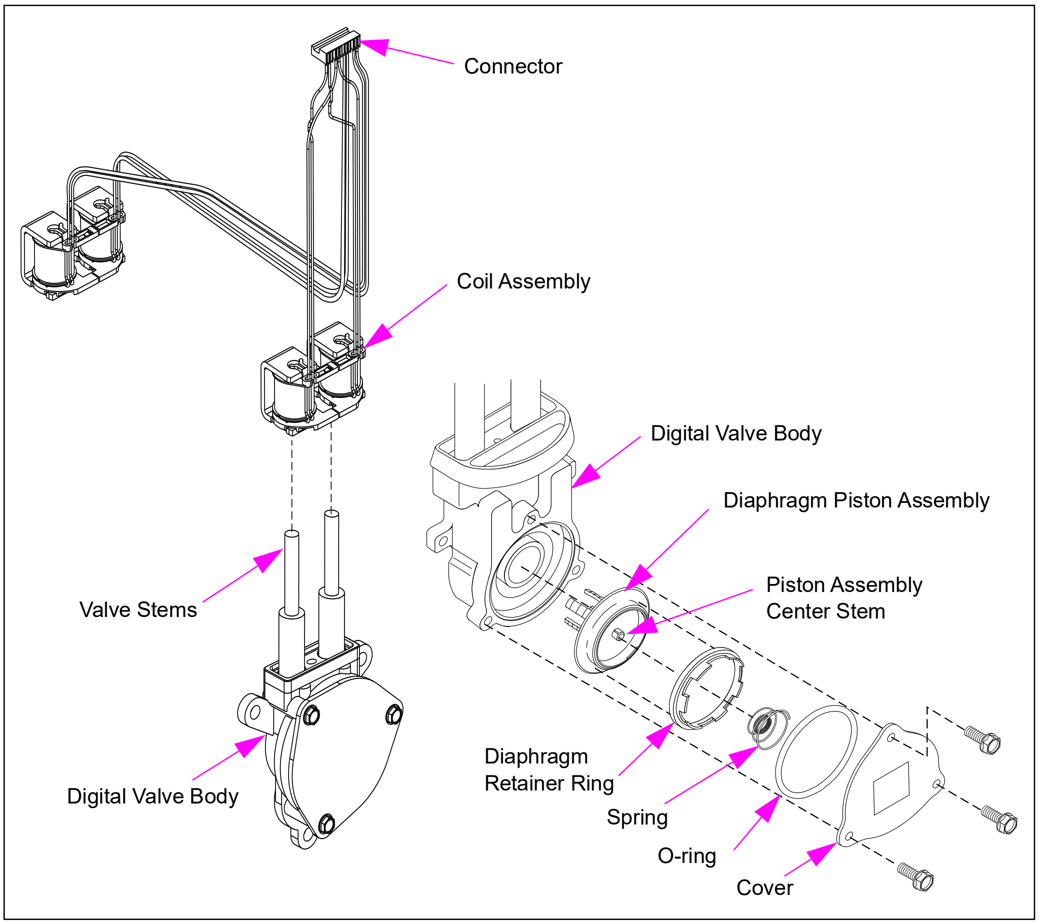
7 Torque bolts properly. Refer to "Torque Specifications" on page 7-5.

8 Follow the steps in "After Repairing" on page 7-4.

Digital Valve

The digital valve consists of a casting, a diaphragm, piston assembly, and two pilot control valves with electromagnetic actuation coils. Fuel flow is controlled by signals actuating the coils that open and close the pilot control valves. The digital valve allows precise control of fuel blends by regulating the movement in small increments of a diaphragm piston assembly. The opening/closing of the pilot control valves are continuously monitored and adjusted to maintain blend ratios. For non-blenders, the valve operates at only slow and fast flow. For more detailed information on how the digital valve works, refer to "Theory of Operation and Troubleshooting" on page 8-1.
Note: Digital valves cannot be used for programmable flow control.

Figure 7-44: Digital Valve Assembly



Some Potential Problems:

- Improperly installed (upside down) coils can result in improper valve operation.
- Retaining hardware placed incorrectly or damaged.
- Excessive buzzing
- ECs
- Leaks, internal (bypass) and external
- Coil activation handle switch if bracket is not used
- Preset overrun/underrun
- No or low flow
- Very cold weather leaks (solved by using a special valve kit)

Service Steps - Disassembling Digital Valve

Determine if the problem is with the coil or valve assembly. Only valve assembly service requires opening the fuel system. The following instructions pertain to diaphragm replacement. This is usually done to solve the problems with valve noise, internal leaks (bypass), and flow. Depending on the labor rates in your area, it may be less expensive to replace as opposed to rebuilding the valve.
Note: Other components can cause similar symptoms.

To disassemble the digital valve, proceed as follows:

1. Using approved containers, absorbent materials and practices, prepare for any residual fuel and pressure that may be in valve.
2. Close the shear valves involved and verify proper operation (dispensers only).
3. Remove power from the unit and any associated STPs at circuit breakers. Follow OSHA lockout/tagout procedures.
4. While being careful of any residual pressure and fuel spray, slowly remove three hex bolts from the diaphragm assembly cover and remove the cover. Wear eye protection. Retain the cover and bolts.
5. Use Gilbarco Kit K96621 to service the valve.
6. Remove and discard the existing O-ring.
7. Remove the spring from center of the diaphragm piston assembly. Spring is snapped onto center stem of the diaphragm piston assembly.
8. Remove the diaphragm retainer ring.
9. Retainer ring is not threaded. It is pressed in place. Removal may require use of channel lock pliers and a screwdriver to pry retainer ring loose. Use the tools carefully to avoid damage to the digital valve body.
10. Grasp the outer edge of the plastic piston cup with needle nose pliers and pull to remove the diaphragm piston assembly.
    Note: Suction within digital valve body may cause resistance during removal.

Replacing Digital Valve Parts

Follow the kit instructions to install the new parts. Reverse the disassembly procedure to reinstall the existing parts. Test for proper operation and leaks after rebuilding the valve.

Service Tips

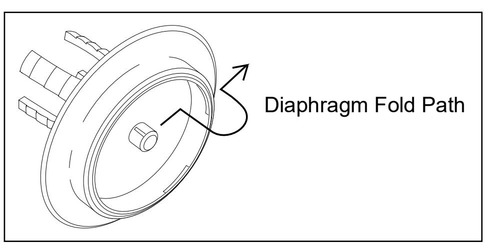
- When servicing the diaphragm assembly, ensure that the diaphragm is positioned correctly on the piston. Gently roll the diaphragm over to follow the path shown in Figure 7-45, if required. Diaphragm may have unfolded during shipping.
- Ensure that all parts are clean. Introduction of contamination can cause faulty operation or premature failure of the valve.
- Replace all parts as provided when rebuilding the valve, as some parts have design enhancements.
- Retainer ring is not threaded. It is pressed in place. Use the tools carefully to avoid damage to the digital valve body.

Digital Valve Seepage (SB1447)

Current seals used in the digital valve are capable of handling very low temperatures. However, some sites have noted seepage at -15 °F or colder. If sites with digital valves in your area have experienced extreme cold weather seepage at the valve seals, the units can be fixed by using special seals and a new valve cover developed for extremely cold areas. Depending on fuels, tolerances, and other factors, some valves will require this special kit. Generally, it is required only to use the special kit, where seepage is noted to occur when temperatures are extremely low (approximately -15 °F or colder). Monthly leak inspections of the dispenser hydraulics will detect any leaks that may develop.

The cold weather digital valve seal kit is K94444-05. The standard seal kit is K94444-03. The standard kit is considerably less costlier than the K94444-05 Kit. Also, the standard kit is completely acceptable for even cold sites. Hence, the use of cold weather seal kits across the board is not recommended. For additional information, refer to *MDE-4242 The Advantage/Encore/Eclipse Digital Valve Rebuild Kit*.

Figure 7-45: Proper Diaphragm Fold



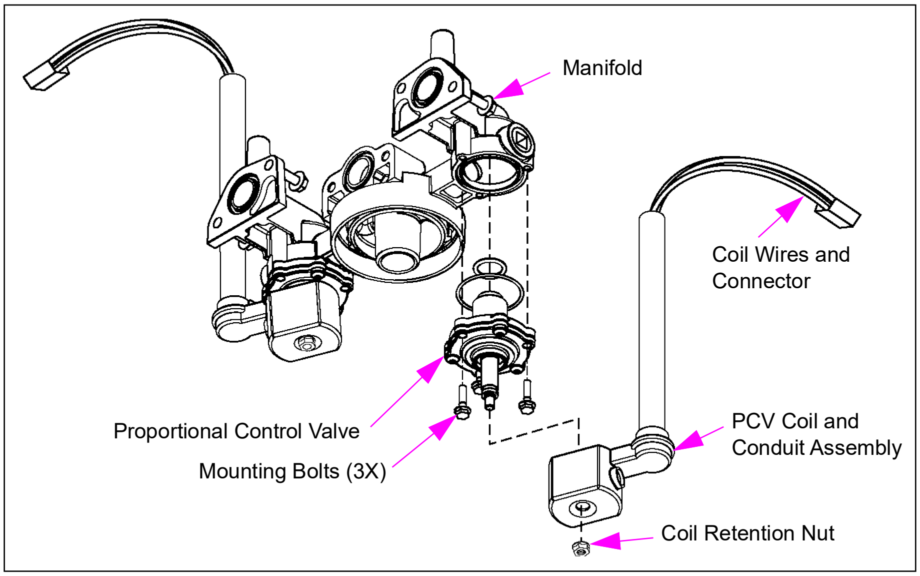
Proportional Control Valve

The Gilbarco PCV uses an analog signal to control flow. This feature provides smooth and quiet operation. Coils and conduit are a combined unit and are easily replaced. For detailed information on how the PCV works, refer to "Theory of Operation and Troubleshooting" on page 8-1.

Some Potential Problems

- Excessive noise
- ECs
- Leads external and bypass
- Preset overrun or underrun
- No flow or low flow
- Flow hesitation

Figure 7-46: Proportional Control Valve Assembly



Service Steps - Replacing Proportional Control Valve Assembly

Determine if the problem is with coil or valve assembly. Only valve assembly service requires opening the fuel system. The following instructions pertain to valve replacement. This is usually done to solve problems with valve noise, internal leaks (bypass), and flow.
Note: Other components can cause similar symptoms.

To replace the proportional control valve assembly, proceed as follows:

1. Using approved containers, absorbent materials and practices, prepare for any residual fuel and pressure that may be in valve.
2. For dispensers, close the shear valves involved and verify proper operation.
3. Remove power from the unit and any associated STPs at circuit breakers. Follow OSHA lockout/tagout procedures.

4 While being careful of any residual pressure and fuel spray, slowly remove three hex bolts from the diaphragm assembly. Do not remove the pan head bolts that require special tools. Wear eye protection. Retain the bolts.
Note: The internal parts of the PCV are not serviceable.

5 Remove and discard the existing O-rings.

6 Install new O-rings (see Figure 7-46 on page 7-73). Use Gilbarco Kit K96621-0X.

Replacing Proportional Control Valve Parts

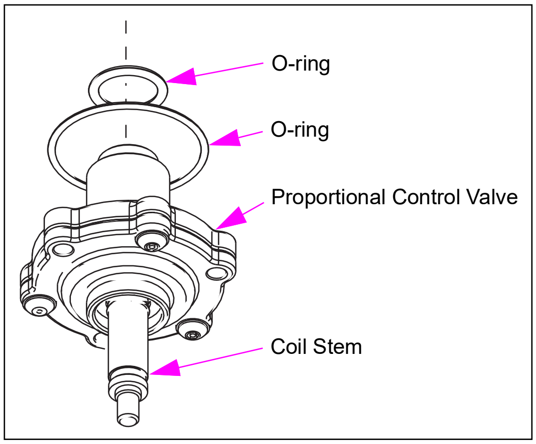
To replace the proportional control valve parts, proceed as follows:

1 Reverse the disassembly procedure to install the valve.

2 Torque three hex bolts to 100 inches pound.

3 Test for proper operation and leaks after installing the valve.

Figure 7-47: Proportional Control Valve Assembly



Service Tips

- Ensure that the coil is installed properly.
- Do not pry on coils.
- No flow or other valve problems may be caused by a bad valve coil, valve adaptor, or PCN board. Do not rebuild or replace the valve without checking these first.

Shear Valves

Shear valves have a section of the valve that is designed to break from a sufficient impact to the unit. Shear valves cut off the flow of fuel from product piping if impact or fire causes shear valve linkage to trip. Single poppet shear valves only stop fuel at the product piping. Double poppet shear valves also stop fuel from draining out of the dispenser.

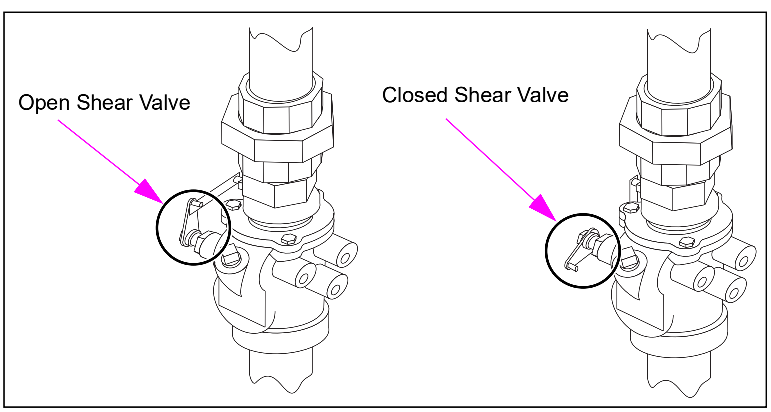
⚠ CAUTION

Improper wrench technique may damage the shear valve and lead to leaks. Excessive force applied to the shear valve during fitting and tightening may damage or weaken shear section of valve. Always use two wrenches - one wrench to hold the valve above the shear line and the other to apply force to fitting.

UL requires all dispensers to use shear valves. Follow these recommendations when installing or servicing shear valves:

- Ensure free movement of linkage.
- Lubricate valve linkage per manufacturer’s instructions.
- Follow manufacturer’s recommendations first when installing valves.
- Shear section of valves must be flush with top of islands.
- Do not mount the valve upside down.
- Anchor the shear valves rigidly. Rigid cross member and mountings are important devices for proper operation of the system.
- Inspect the valves every six months. Ensure that the valve linkage is free and clear from any obstructions (conduit, piping, and so on). Lubricate the linkage with SAE 10 W motor oil.
- Gilbarco highly recommends use of double poppet shear valve instead of single poppet when replacing valves. Use of double poppet shear valves may require some internal plumbing changes for older units. Double poppet shear valves prevent fuel from draining from a unit that has been knocked off the island.
- Do not use pliers to open or close the shear valves. Reduce pressure if the valve does not open easily. For proper tools and procedures, refer to shear valve manufacturer’s instructions.

Figure 7-48: Typical Shear Valve Linkage



Inspecting Shear Valves

⚠ WARNING

Always ensure proper operation of shear valve before performing hydraulic service that requires shear valve to be closed. In some instances, a shear valve may outwardly appear closed but can actually be open.

To inspect the shear valves, proceed as follows:

1. Authorize hose at console, as required.
2. Lift the operating handle.
3. Place the discharge nozzle in an approved container.
4. Squeeze the nozzle operating lever. If the flow continues after several seconds, the valve is defective.
5. Repair or replace the valve.
6. After testing, place a few drops of SAE 10 W motor oil or equivalent on valve body shaft. Open and close the valve by hand several times. Place the valve back in service.

Solenoid Valves

Solenoid valves use a solenoid coil to open or close a valve. Valves of this type are either fully open or closed. Servicing is relatively simple consisting mainly of replacing the coil, valve body, or board supplying voltage to the valve.

Solenoid valves are usually associated with:
- VaporVac valves
- Steering valves
- Ultra-Hi valves

Rebuilding or Replacing

Labor cost for your area may determine your service philosophy. High labor costs favor replacement. Difficult removal from the unit favors partial rebuild in the unit. Gilbarco recommends the following services for average or lower than average labor costs:

For Valves	Recommended Service
Less than two years old	Use a repair kit (replace spring, diaphragm, and O-rings)
More than two years old or high usage	Replace the valve.

To rebuild or replace solenoid valves, proceed as follows:

1. Close and test the shear valves (for dispensers).
2. Remove power to the unit. For dispensers, remove any involved STPs. Follow OSHA lockout/tagout procedure.
3. Open the nozzle into approved container to bleed pressure. Some residual pressure may remain.

4 Check the disconnected valve coils for continuity. If bad, replace the coil.

5 Replace the valves, if defective. Use UL-approved sealant as required for fuels when replacing the valves. Do not use Teflon tape (Teflon tape debris may cause the valve to malfunction). Clean the valve area. Do not introduce contamination into valve or piping. Install the valves with flow arrow pointing in the proper direction.

Service Tips

If no magnetic field or power exists, the likely problem is:

- Problem with power from the valve driver board
- A cable problem

If power exists, the problem may reside in:

- Coil
- Valve body

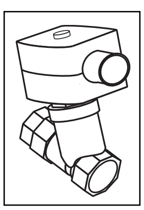
Steering Valves

Steering valves are used on multi-hose blenders. These valves are part of the steering valve manifold and are used to direct fuel flow to the proper hose. The valves ensure that only the proper hose is activated. For information on Steering Valves, refer to "Steering Manifold" on page 7-42.

Vapor Valve (VaporVac)

Vapor valves mount in the upper piping housing assembly. For additional VaporVac information, refer to "Vapor Recovery Components" on page 7-80. Vapor valves that do not operate properly can cause loss of vapor recovery system efficiency or premature nozzle cut-off (especially in Balanced Vapor Recovery systems). They are single-stage solenoid valves that isolate all hoses not in use.

Figure 7-49: Vapor Valve (T17909)



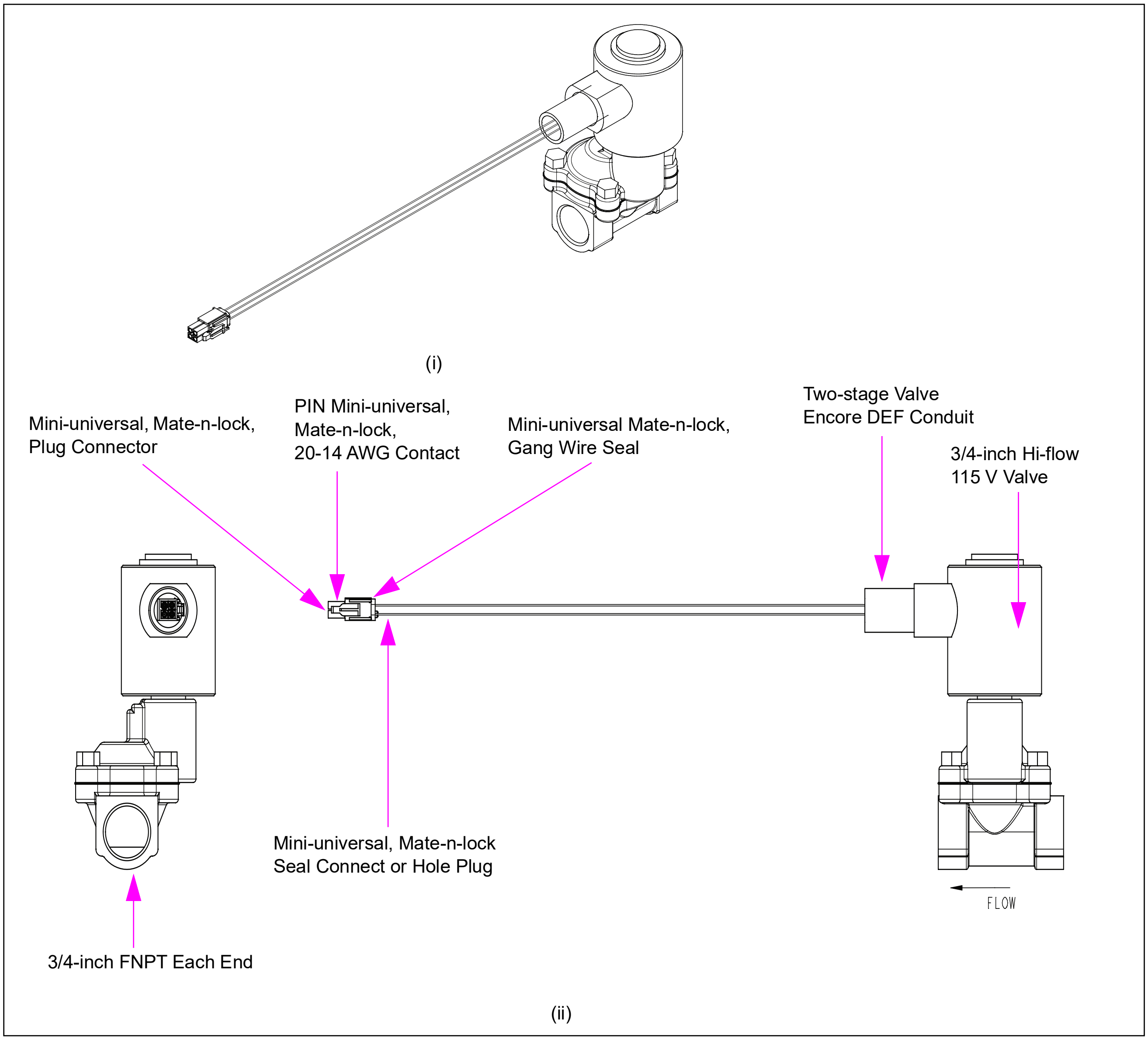
Note: New units do not use vapor valves. They require a particular type of nozzle which is different from the units containing vapor valves. Ensure that you use the correct nozzle type (refer to manufacturer's instructions) for the application, or "clean air regulations" may not be met.

Service Tip

Under normal operating conditions, vapor valves make a clicking sound when the flow begins. However, a defective valve can make the VaporVac pump/motor noisier during operation. If the vacuum pump becomes noisier only when a certain hose on a certain side is being used, suspect the valve associated with that hose.

DEF Valve

Figure 7-50: DEF Valve



The DEF valve is a two-stage proportional valve that is compatible with DEF.

⚠ CAUTION

The valve is susceptible to serious damage from freezing. The seals will partially blow out causing external leaks. The seal of the valve that is damaged by freezing, will bulge outwards from its normal assembly position. When investigating a frozen unit, always check the valve for damage/potential leaks.

IMPORTANT INFORMATION

As the valve is made of stainless steel and is expensive, it is recommended that you rebuild the valve to minimize cost. If the damage is caused due to freezing of the components, then it is recommended that you rebuild the valve using the Rebuild Kit (M02321K005), as the damage will be to the seals and diaphragm.

If the valve is defective, replace the hydraulic valve and coil. Before replacing the valve ensure that the 120 VAC power supply is supplied to the coil.

Vapor Recovery Components

Contents in This Sub-section

Read and follow all safety precautions as outlined in "Read This First" on page 1-1 and related sections of this manual. For proper test procedure, consult your local regulatory agency.

WARNING

Residual pressure and entrapped fuel may still be present and may drain or spray when removing parts and fittings.

Fire and explosion can result in severe injury or death.

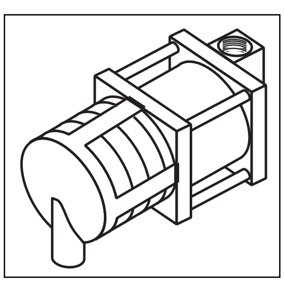
- Wear eye protection.
- Remove the parts and fittings slowly.
- Collect fuel in approved containers.
- Clean up all spills promptly.

VaporVac Assemblies

R19663-G1 and T17942

These assemblies mount in the upper piping housing of the unit on field retrofit units and in the hydraulics cabinet on factory-built VaporVac units. Retrofit units use R19663-G1 with attached connector. Factory-built units use T17942. The VaporVac motor driver board provides 36-50 VDC to operate the motor and 6.2 VDC for the sensors.

Figure 7-51: VaporVac Pump/Motor Assembly



Load Table

Unit	Load
Single-sided	3 A @ 120 VAC
Dual-sided	5 A @ 120 VAC

Wiring

Number	Wire Label (REF)	Color	Min. Wire Length	Gauge	Description
1	CHL	Green	30	22	Hall Effect sensor
2	BHL	Gray	30	22	Hall Effect sensor
3	AHL	Yellow	30	22	Hall Effect sensor
4	PHL	Orange	30	22	Hall Effect sensor
5	GHL	Black	30	22	Hall Effect sensor
6	A WYE	Blue	30	18	Power – motor
7	B WYE	White	30	18	Power – motor
8	C WYE	Brown	30	18	Power – motor

Vapor Valve

Vapor valves mount in the upper piping housing assembly. They are single-stage solenoid valves that isolate all hoses not in use. The VaporVac valve driver board supplies 120 VAC for operation. For additional information, refer to “Vapor Valve (VaporVac)” on page 7-77.

Common Vapor Recovery Testing Procedures

This section contains tests that are frequently used for vapor recovery systems.

Before performing any test on your vapor recovery system, ensure that you consult your local regulatory agency. There may be three or four different procedures that exist for each test. Only your local regulatory agency can inform you of their version, provide you with a copy of the test procedure, and provide you with the correct method of reporting test results.

Testing Air-to-Liquid (A/L) Volume with a Pulse Simulator Box

Note: This test is not for certification.

Required materials:

- A/L Test Kit (K94255)
- Five gallon or larger prover can (for changing failed nozzles)
- Standard and Phillips® screwdriver
- Allen® wrench set
- New spare nozzles to replace any that might fail
- Pulse Simulator (T19343-G1)

To test the air-to-liquid volume with a pulse simulator box, proceed as follows:

1 Place the dispenser in standalone mode.

2 Install the nozzle testing tee from the A/L Test Kit (K94255). Ensure that you use the (-05 or -07) “no flow” insert in the testing tee.

3 If air volume meter has a mechanical totalizer, record the initial value. If air volume meter has electronic output, reset the frequency counter to zero.

Note: Be aware that testing meter can measure a light breeze since there is a very low pressure drop through the meters. Protect the testing meter from wind.

4 Ensure that the nozzle hook is in the “OFF” position and the pulse simulator box is set to “Halt”.

5 Unplug the pulser from the cable assembly for the hose position to be tested.

6 Plug the pulse simulator box connector into the cable assembly.

7 Select the flow rate on the pulse simulator box that is closest to the actual dispenser/site flow rate. If the actual flow rate is not known, set the pulse simulator box to 8 GPM.

8 Raise the nozzle hook to authorize the dispenser and then turn the pulse simulator box to the “Run” position. Do not engage/squeeze the nozzle lever.

9 Switch the pulse simulator box from the “run” position to the “halt” position, when the volume display on the dispenser indicates 4.5 ± 0.3 gallons.

10 The following information obtained must be recorded on a data sheet:
- The selected flow rate setting from the pulse simulator box (6, 8, or 10 GPM).
- The actual liquid volume registered.
- The final value on the mechanical totalizer or frequency counter for the air volume meter.

11 Calculate the A/L value for the fueling position that was tested.

12 Refer to CARB Executive Order G-70-150-AD for VaporVac system. Verify if the dispenser flow rate and A/L value are within the specified acceptable limits.
- If A/L values are within the acceptable limits, the vapor recovery system is operating properly and testing must be concluded for that hose position.
- If A/L values are not within the acceptable limits, you must troubleshoot and possibly calibrate the system.

Testing Pressure Drops (Dynamic Back Pressure)

This test measures the pressure drop or resistance in the vapor recovery piping from the dispenser to the UST. Nitrogen is forced through the vapor recovery piping at various flow rates, while the back pressure or resistance is measured with pressure gauges. There are two basic versions of this test. Version 1 is called the “dry” test. Dry refers to testing the piping as is, and assumes there is no fuel in the piping to create a liquid trap. Version 2 is called the “wet” test. Wet refers to pouring fuel into the piping to create liquid traps in any low points.

Balance vapor recovery systems are tested from the nozzle to the UST. Vacuum assist systems such as VaporVac are tested from the riser at the base of the dispenser to the UST. The allowable pressure drop for balance vapor recovery systems is listed in the actual test procedure. The allowable pressure drop for vacuum assist systems is listed in the CARB Executive Orders.

Known Test Procedures	Description
CARB TP-201.4	Determination of Dynamic Pressure Performance of Vapor Recovery Systems of Dispensing Facilities
San Diego County APCD TP-91-2	Pressure Drop vs. Flow/Liquid Blockage Test Procedure For Phase II Balance System Installations
TNRCC TXP-103.1	Determination of Dynamic Pressure Performance (Dynamic Back Pressure) of Vapor Recovery Systems at Gasoline Dispensing Facilities
Bay Area AQMD ST-27	Gasoline Dispensing Facility Dynamic Back Pressure

Testing Leaks/Pressure Decay (Static Pressure)

This test evaluates the integrity of the vapor seals of the vapor recovery system components and connections, and determines if there are any leaks. The entire vapor recovery system is pressurized with nitrogen to a specified amount. The pressure is then monitored over a specified period of time. The maximum amount of pressure loss or decay is based on the ullage or vapor space in the UST.

Known Test Procedures	Description
CARB TP-201.3	Determination of Static Pressure Performance of Vapor Recovery Systems of Dispensing Facilities
San Diego County APCD TP-91-1	Pressure Decay/Leak Test Procedure For Phase I and II Vapor Recovery Installations
TNRCC TXP-102.1	Determination of Static Pressure Performance (Pressure Decay) of Vapor Recovery Systems at Gasoline Dispensing Facilities
Bay Area AQMD ST-30	Static Pressure Integrity Test For Underground Storage Tanks

Testing Performance (Vacuum Assist Systems)

Read and follow all safety precautions as outlined in "Read This First" on page 1-1, this section, and related sections of this manual.

This test verifies that the system is working in a manner similar to which it was certified by CARB. There are two basic performance tests for vacuum assist vapor recovery system. Test 1 is the A/L or V/L test. This test simultaneously measures the volume of air being collected by a vapor recovery nozzle and the volume of fuel being dispensed by that nozzle. Test 2, the vacuum test, was developed by Gilbarco. It measures vacuum generated at the nozzle by drawing air through a controlled orifice.

Known Test Procedures	Description
CARB TP-201.5	Determination (by Volume Meter) of Air to Liquid Volume Ration of Vapor Recovery Systems of Dispensing Facilities
Gilbarco *MDE-3022 Field Performance Test Kit K94123 for VaporVac*	Operating Instructions for Field Performance Test Kit (K94123) for VaporVac
Bay Area AQMD ST-39	Gasoline Dispensing Facilities, Air-Vapor to Liquid Volume Ratio
San Diego County APCD TP-95-2	Vapor to Liquid Ratio at Gasoline Dispensing Facilities Using Vacuum Assist Systems

Testing for Internal Leaks

Read and follow all safety precautions as outlined in "Read This First" on page 1-1, this section, and related sections of this manual. For proper test procedure, consult your local regulatory agency. Use this procedure to quickly test for leaks of fuel into the vapor area. Use this procedure for service, periodic tests, and regulatory agencies.

To test the tank for internal leaks, proceed as follows:

1 Raise the nozzle hook (dispenser handle) to authorize the dispensers, but do not dispense fuel.

2 Observe the gallonage readout on the main display.

3 If a slow and continual readout is noted while the nozzle is closed, a leak may exist at the inner vapor hose or seals.

Calibrating VaporVac

Before calibration, ensure that the following has occurred:

- Visually inspect for leaks, loose pipe connections, and so on.
- Use the Field Performance Test Kit (K94123) to test the system.
- Verify if all hanging hardware (for example, hose, nozzle, and breakaway) is in good working condition.

Calibrate the vapor recovery system if:

- A new vapor pump has been installed.
- The vacuum level is out of the acceptable range from testing with the field performance test kit and pulse simulator box.

To calibrate the vapor recovery system, proceed as follows:

1 Place the dispenser in standalone mode.

2 Install a new or known good nozzle on the hose position in question.

3 Install the nozzle testing tee from the Field Performance Test Kit (K94123) by following the instructions in *MDE-3022 Field Performance Test Kit K94123 for VaporVac*. Ensure that you use the (-05 or -07) "no flow" insert in the testing tee.

4 Hook up the pulse simulator box by following previous instructions in this manual.

5 Select the flow rate on the pulse simulator box that is closest to the actual dispenser/site flow rate. If the actual flow rate is not known, set the pulse simulator box to 8 GPM.

6 Raise the nozzle hook to authorize the dispenser and then turn the pulse simulator box to the "Run" position. Do not engage/squeeze the nozzle lever.

7 Measure the vacuum level after 3 gallons have registered on the volume display of the dispenser.

8 Adjust the vacuum level by turning the "slope" trim pots (R30 and R52) on the VaporVac Controller Board (T18021 or T19401).

Turn Trim Pot Screw	
Clockwise	Increases vacuum level

Turn Trim Pot Screw	
Counter-clockwise	Decreases vacuum level

9 The vacuum level must be adjusted as specified in the following table. The vacuum levels in this table are only valid when using the Orifice Plug (Q13245-01) in the Nozzle Testing Tee (T19454-G1).

GPM	Vacuum Level (Inches of Water)
6	4
8	7
10	10

10 If the vacuum level cannot be adjusted to meet the above values, the vapor pump/motor is bad and must be replaced.

11 If the vacuum level can be adjusted to meet the specified levels, one additional check must be performed.

12 Locate the "current" limit display LEDs (CR40 and CR41) on the VaporVac controller board.

13 Set the flow rate selector on the pulse simulator box to 10 GPM.

14 Reauthorize the fueling position and place the pulse simulator box in the "Run" position. Watch the LEDs as soon as the pulse simulator box has been switched to "Run".

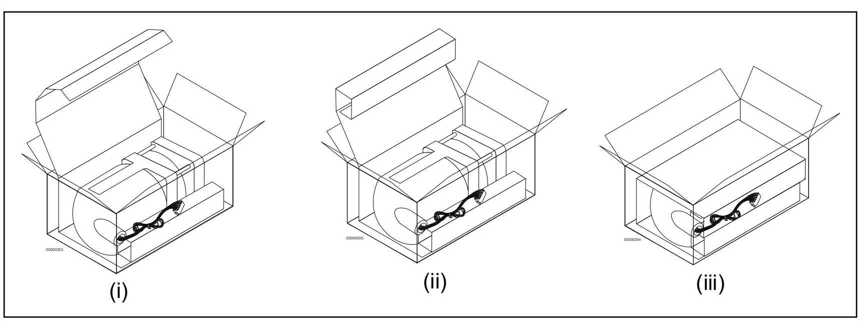
15 The LED for the side that you are monitoring, must light and turn off quickly. If the LED stays lit or flickers for more than three seconds, the vapor pump/motor is drawing too much current. The "slope" trim pot (R30 or R52) must be adjusted down (counterclockwise) until the LEDs turn on/off quickly.

Returning VaporVac Pump/Motor

Pumps with short or crushed wires cannot be rebuilt and in some cases even tested. All motors returned for warranty credit must be of rebuildable condition with wire leads at least 26 inches long.

Spare part vapor pumps are packed in a reusable shipping container. Use this box to return vapor pumps to Gilbarco. The box has an insert that keeps the wires from being crushed by the casing of the vapor pump. Put the vapor pump in this box immediately after removing it from the VaporVac system. This will keep the wires from being damaged in transit to your shop.

Figure 7-52: Packaging VaporVac Motor for Return



To return the VaporVac pump/motor, proceed as follows (see Figure 7-52 on page 7-85):

1 The VaporVac motor box has an insert that is perforated at both ends. Seat the insert into the box, with the lower end "curled".

2 Set the removed motor into the box (a). The motor nipple must be at the end of the curl.

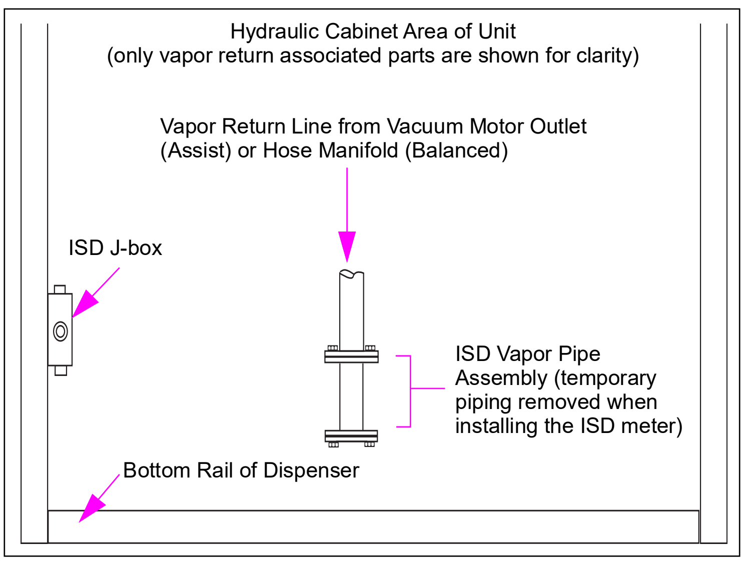
3 Curl the upper end of the insert (b). Lower the upper end down into the box.

4 Insert the wire bundle in the upper curl (c) to prevent shipping damage. Close and secure the box flaps.

In-station Diagnostic (ISD) Vapor Flow Meter

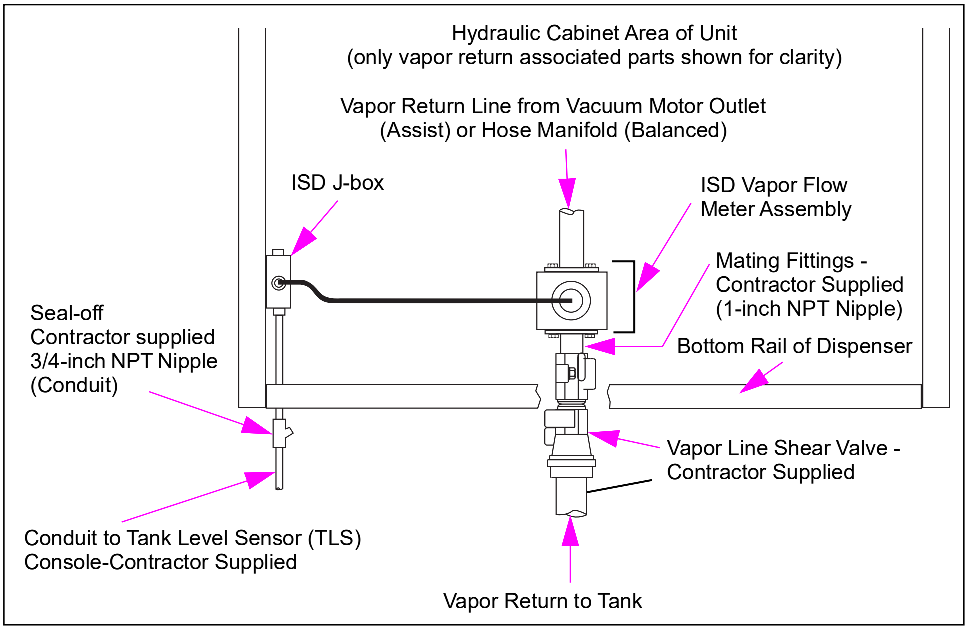
The vapor flow meter, J-box, and plumbing is part of an ISD system that measures the amount of vapor returning to the UST and determines whether the Vapor Recovery system is operating correctly. Dispenser units are available as ISD-ready as an option. ISD-ready dispensers include adaptive plumbing and a J-box (see Figure 7-53).

Figure 7-53: ISD-ready Dispenser (Facing Side B of Dispenser)



ISD-ready dispensers are integrated with the ISD system by replacing the ISD vapor pipe assembly with an ISD vapor flow meter assembly (see Figure 7-54 on page 7-87).

Figure 7-54: ISD-ready Dispenser with Vapor Flow Meter Installed (Facing Side B)



The vapor flow meter is in-line and near the vapor return shear valve in the hydraulic area of the dispensing unit. Ensure that there are no liquid flow traps. For more information on liquid traps, refer to *MDE-3985 Encore Installation Manual*.

Wiring for the ISD vapor flow meter must be intrinsically safe and run in a separate conduit. Conduit must be potted where it enters and leaves any designated hazardous zones. Potting requirements for hazardous zones must be followed so that fuel vapors do not migrate to other areas.

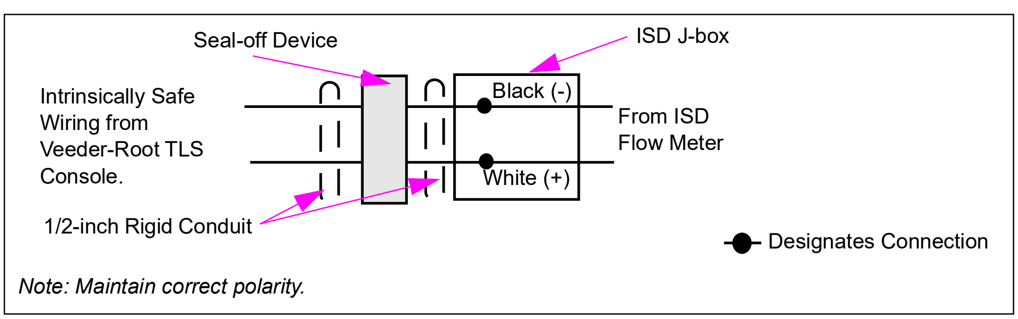
Wiring from the ISD J-box terminates at the TLS. For correct polarity connections, see Figure 7-55. For additional information, refer to *ISD Vapor Flow Meter Installation Guide* (Veeder-Root document number 577013-796).

Note: Correct polarity must be maintained to ensure proper equipment operation.

⚠ WARNING

Wiring to this J-box must be intrinsically safe power from Veeder-Root TLS-350 or Gilbarco EMC™ series tank gauge consoles, and must be installed in accordance with Veeder-Root installation document number 577013-796.

Figure 7-55: ISD Flow Meter J-box Polarity



ATC Components

Read and follow all safety precautions as outlined in "Read This First" on page 1-1, this section, and related sections of this manual. Follow OSHA lockout/tagout procedures.

Contents in This Sub-section

Purpose of the ATC Option

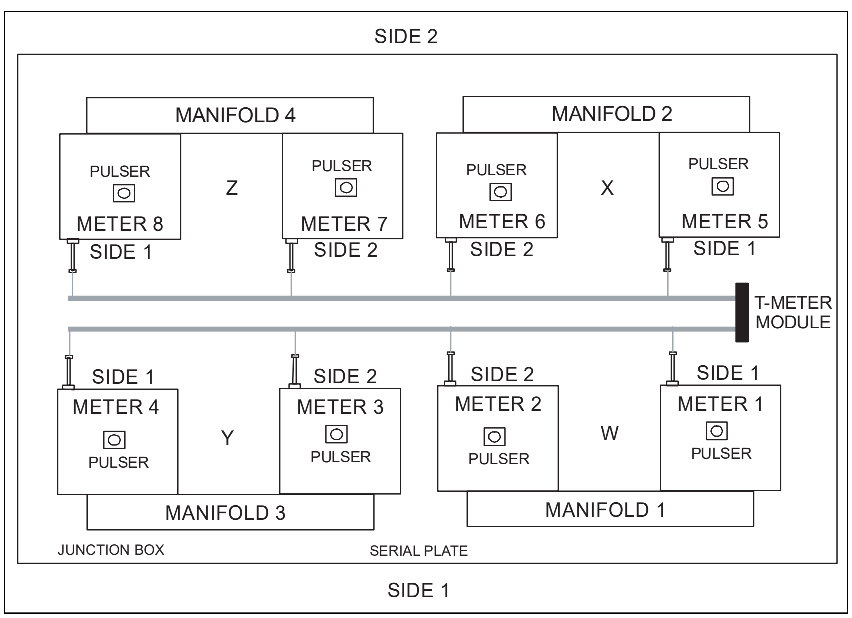
ATC is an option that measures the temperature of the product being dispensed and compensates for the changes in product volume due to extreme changes in the ambient temperature.

Note: ATC is used in Canada only.

ATC Mapping

ATC temperature probes are mapped to meters in the Encore hydraulics as shown in Figure 7-56 and the following table.

Figure 7-56: Meter Mapping Diagram



Product	Inlet	Side 1 Meter No.	Side 2 Meter No.
1	W	1	2
2	X	5	6
3	Y	4	3
4	Z	8	7
Blend Product	W and X	1 and 5	2 and 6

Real-time Data Review - ATC

When a dispenser is configured to use ATC it has a requirement to perform a running transaction to show Uncompensated Volume, Compensated Volume and Temperature for Audit Review of the system. Prior to 01.8.30 and 02.8.30, this was performed using CC27, FC7 (reference section here) to run a transaction while in dispenser programming mode. Since 01.8.30 and 02.8.30 software, when a transaction is being performed on one side of the dispenser, the user can press 7 on the manager keypad to have the Uncompensated Volume replace Money information and have Temperature replace the Price in the PPU. Pressing 7 again will return the data to normal Money/Volume/PPU display. If a handle is lowered or error occurs to end the transaction, the Money/Volume/PPU will be displayed. This transaction is also counted for all totalizers and reported to any controller over two-wire if configured.

Programming ATC

ATC programming is accomplished through dispenser programming CCs. The following instructions are intended as an overview of ATC programming. Refer to "Pump Programming" on page 5-1.

Accessing CCs

Press **F1** and then enter the 4 digit security code for the level of the CC that must be accessed. Four zeros will flash on the display. Enter the appropriate CC and then press **ENTER** on the manager keypad.

ATC Review Mode

CC27 provides read-only access to ATC operating parameters. FC1 (default) provides for audit of last transaction data. Additional ATC data will be displayed depending on the FC selected. For ATC FCs and descriptions, refer to the following table:

FC	Description	At Main Display	At Volume Display	At Grade 1 PPU Display
1 (default)	Audit Last Transaction	Gross volume	Net volume	Average temperature
2	Display Volume Correction Factor	27 2	Meter Number	Volume correction factor
3	Display Fuel Density	27 3		• 730 Gas • 840 Diesel
4	Display Temperature	27 4		Current temperature
5	Display Gross Volume Totals	27 5	Gross volume (most significant)	Gross volume (least significant)
6	Display Software Version	27 6		ATC version
7	Real-time Transaction Mode*	Gross volume	Net volume	Current temperature

**Real-time Transaction Mode is only available in software earlier than 01.8.30. For new ATC, refer to "Real-time Data Review - ATC" on page 7-89.*

Enabling ATC

ATC is enabled at Level 4 at CC91, FC4.

Selection	Description
1 (default)	Not Installed
2	Installed

ATC Specific ECs

Code	Description
5091	ATC node communication failure
5092	ATC node present but not configured for ATC
5093	ATC node software download successful
5094	ATC node software download Authentication
5095	ATC node software download CRC error
5096	ATC node software download Flash error
5097	ATC node software download Download Incomplete
5098	T-Meter not connected
5099	ATC - number of probes exceeds number of meters
5100	ATC - number or meter exceeds number of probes
5101	No Pulse train detected from T-meters (T-meter is connected)
5600	Fuel Density note not programmed
5601	ATC temperature out of range
5602	ATC node incorrect software version
5606	ATC node subscription failed
5609	ATC software not up to date

Service Tips

- ATC will not function unless the fuel density is entered.
- Encore 300 units with PFCV must be used with ATC Interface Board (T20569-T2). The G2 version can spare G1. Problems with proper display of side 2 of the unit will occur if proper version board is not used.

8 – Theory of Operation and Troubleshooting

About This Section

Topics in This Section

Section Overview

This section provides the theory of operation and troubleshooting tables to help diagnose problems with individual components and systems. Observation, EC recording, and measurement taking are used, wherever applicable, to find the symptom and a possible remedy.

How to Use This Section

To locate the appropriate topic, refer to "Topics in This Section". To familiarize yourself with how the system works, review the information in Theory of Operation on page 8-2. Cable block diagrams in "Electronic and Electrical Components" on page 6-1 and "Hydraulic/Mechanical Components" on page 7-1 may also be helpful in analyzing problems. To finalize the search, refer to the appropriate sub-section. Use the applicable table or chart to find a symptom and determine a remedy.

Terms Used in This Section

Term	Definition
PCB	Printed Circuit Board is a dedicated component designed for a specific function with limited or non-existent intelligence or processing power. For comparison, refer to "PCAs and Nodes" on page 6-5.
PCV	The Proportional Control Valve controls product blending as well as programmable maximum flow rates. This valve opens and closes by direction of electronic programming. Product flow determines the proper opening ratio of each PCV valve to maintain the required blend.

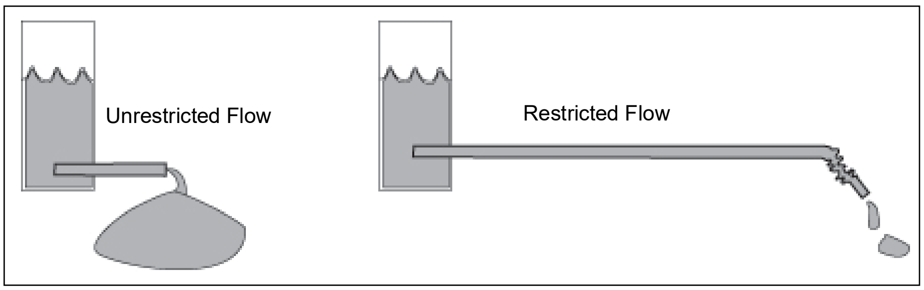
Theory of Operation

Understanding Flow Rate

Flow rate is the number of gallons dispensed per minute. In the United States, flow rate for gasoline dispensing devices is limited to 10 GPM. Flow rate may be affected by restrictions such as distance to the output, damage, and leaks in the line. These restrictions may reduce the flow rate.

For example, in Figure 8-1, the left container has minimal restrictions, therefore a greater flow rate. The right container has many restrictions, therefore a lower flow rate.

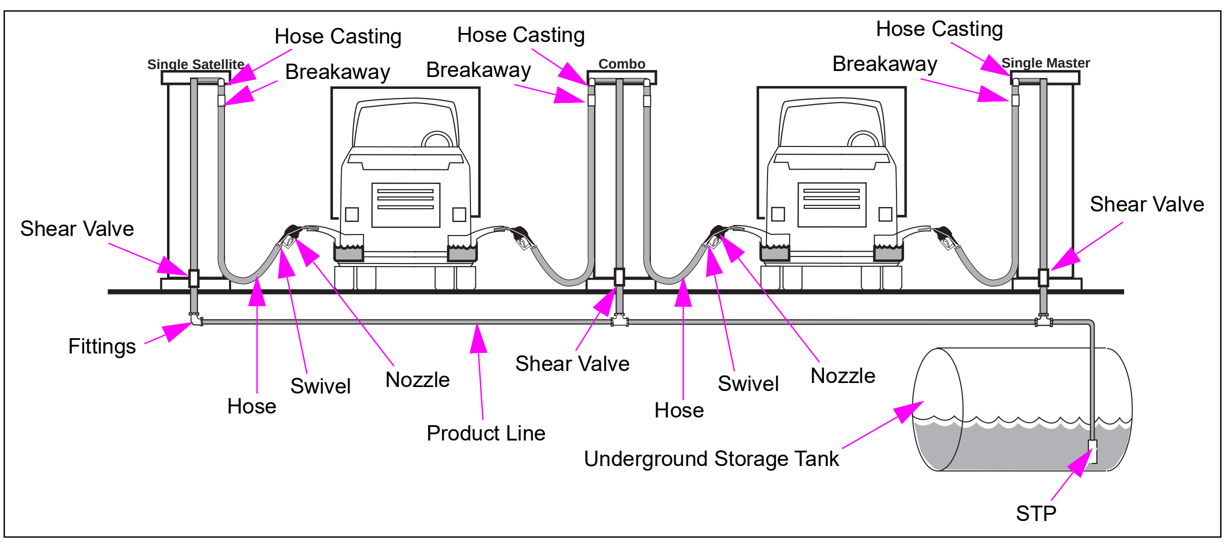
Figure 8-1: Flow Rate Examples



Flow Rate Restrictors

Each site's configuration has a number of restrictors that may affect the flow rate. Figure 8-2 shows a typical site and its possible restrictors.

Figure 8-2: Possible Flow Rate Restrictors

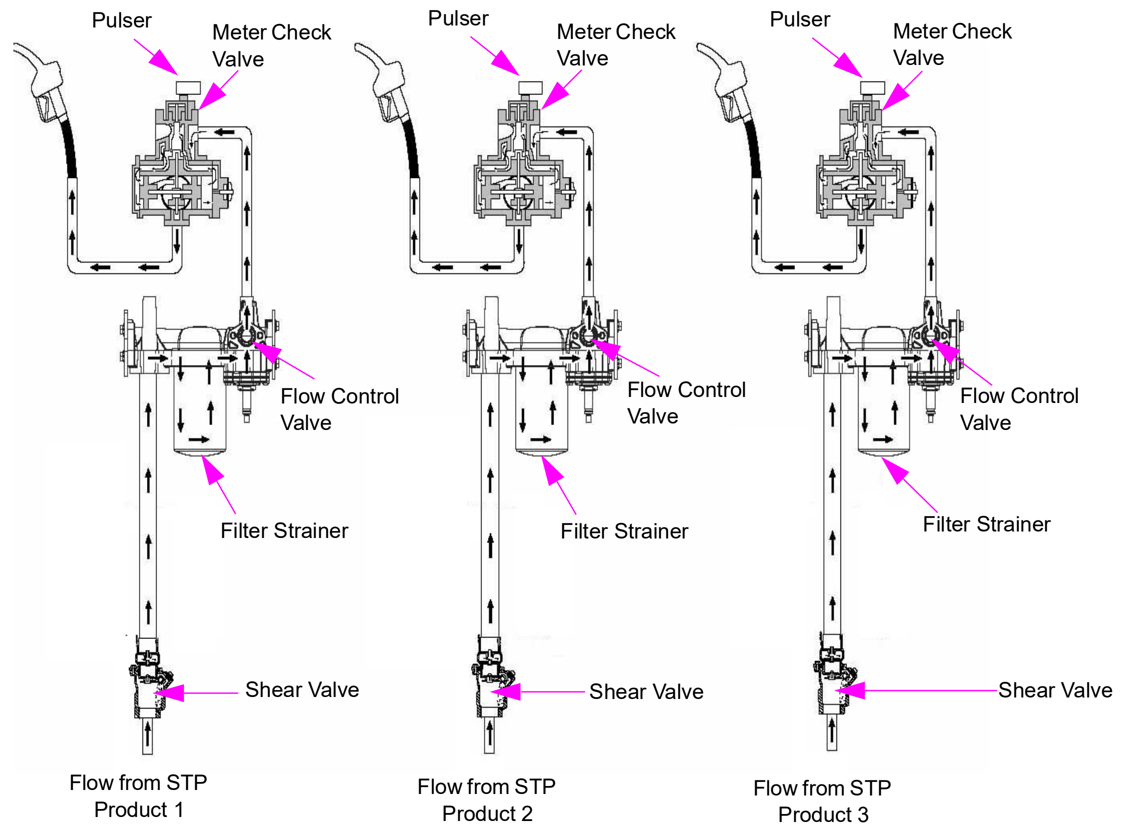


The following table identifies the possible restrictions in Figure 8-2 on page 8-2 that may affect the site's flow rate:

Possible Restrictions	Reason for Restriction
Hose casting	Size and type of hose casting
Breakaway	Type and number of breakaways
Nozzle	Type of nozzle
Swivel	Type of swivel
Hose	Size and length of hose
Shear valve	Size and type of shear valve
Dispenser model	Type of dispenser (that is, High vs. Ultra-Hi vs. Super-Hi™/master, satellite, combo, pump vs. dispenser, and so on).
Fittings	Number of fittings
Pipes	Size and length of pipe run
STP	Horsepower and output of STP
Check valves	Number of valves
Underground tank	Burial depth
Other flow considerations	Number of units dispensing per STP
Optional in line flow restrictors to limit maximum flow rate	The programmable maximum flow rates affecting valve operation available with certain models

How a Standard Dispenser Works

Figure 8-3: How a Standard Dispenser Works



Note: Shear valves are required by most regulatory agencies.

1 Fuel passes through a shear valve and enters the dispenser.

2 Fuel flows through a strainer and filter.

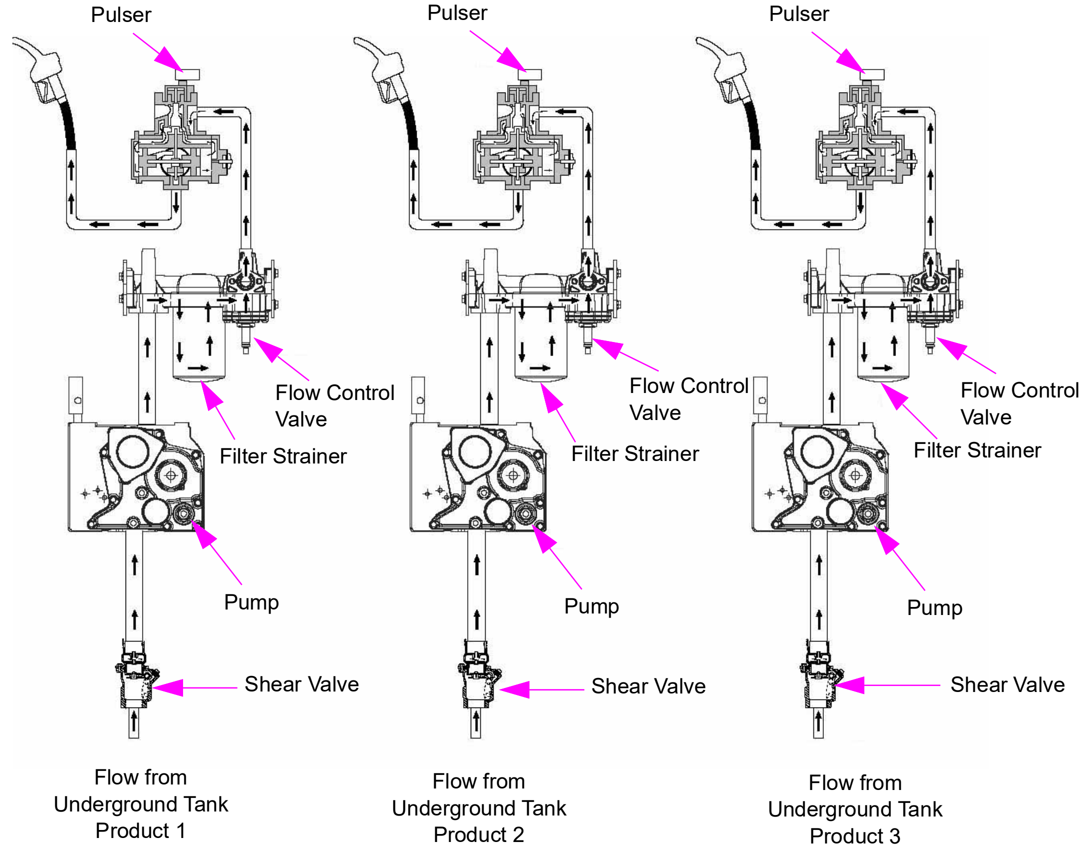
3 Filtered fuel passes through a digital or proportional flow control valve.

4 Fuel flows through a meter check valve and the meter measures fuel flow.

5 Fuel is discharged through the nozzle.
- In case of blenders, two meters supply fuel to a single-hose casting where fuel is combined before the hose.
- Directional valves are used in six-hose blenders to send the correct product for delivery to the hose.

How a Self-contained Pumping Unit Works

Figure 8-4: How a Self-contained Pumping Unit Works



Note: Shear valves are not typically used with pumps, but may be required by some regulatory agencies.

1 For aboveground tanks, a pressure regulator valve (model 52) is used at the base of the unit. Occasionally, shear valves are used for this as well as other applications.

2 Fuel enters through the pump inlet pipe, passing through a strainer or strainer/check valve located in the pumping unit.

3 Maximum pressure is regulated by the bypass valve, which recirculates fuel within the pumping unit.

4 The pump has an air separation system built within the pumping unit.

5 The sump separates air from the fuel, prevents air return to the pump when sump fuel level is low, and prevents air from passing through the nozzle and being metered when the underground tank is dry.

6 Air separated fuel flows back to the pump.

7 Air-free fuel flows out of the pump discharge line through a filter.

8 Fuel then flows through the digital or proportional flow control valve.

9 Fuel flows through a regulating (check/relief) valve. This valve keeps the meter full of fuel and helps relieve excessive pressure that can damage parts.

10 The meter measures fuel flow.

11 Fuel is discharged through the nozzle.

How a Master/Satellite - Ultra-Hi Dispenser Works

These models are for applications requiring higher flow rates. A master meters the product and controls the satellite. This configuration can deliver up to 60 GPM. These units are available in the following formats:

- Standalone unit without a satellite
- Master unit that combines a standalone unit with piping to direct flow to the satellite
- Master satellite combo combines a master and satellite combination within the same unit

Note: Actual flow rate depends on the installation and accessories used. Some local codes do not allow simultaneous operation of master and satellite.

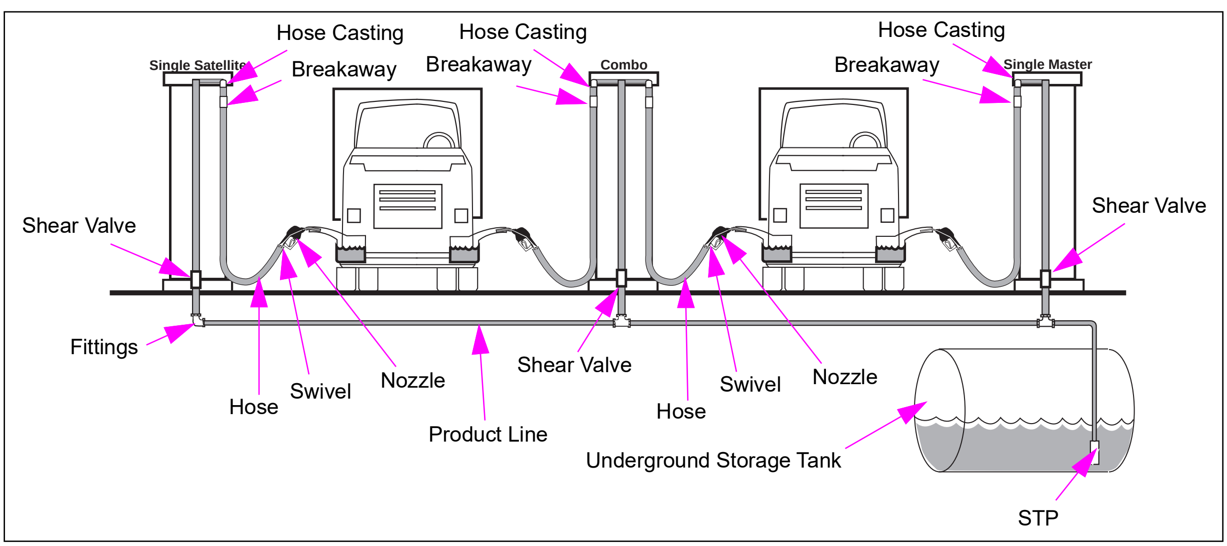
Unit Descriptions

Master - Piping in the master is teed before the master's solenoid valve, to send product to the satellite. This arrangement allows for a higher flow rate in the master and satellite units, when operating simultaneously.

IMPORTANT INFORMATION
To determine what type of operation is allowed in your area, refer to NFPA 30A, local, and state regulations.

Satellite - The satellite unit operates in conjunction with a master unit which meters the fuel. It contains minimal electronics and hydraulics- consisting of a shear valve, solenoid valves, hose, and nozzle. A combo unit is a master for a remote satellite unit and is satellite for a remote master or another combo unit.

Figure 8-5: Various Encore Ultra-Hi Configurations



Types of Operation

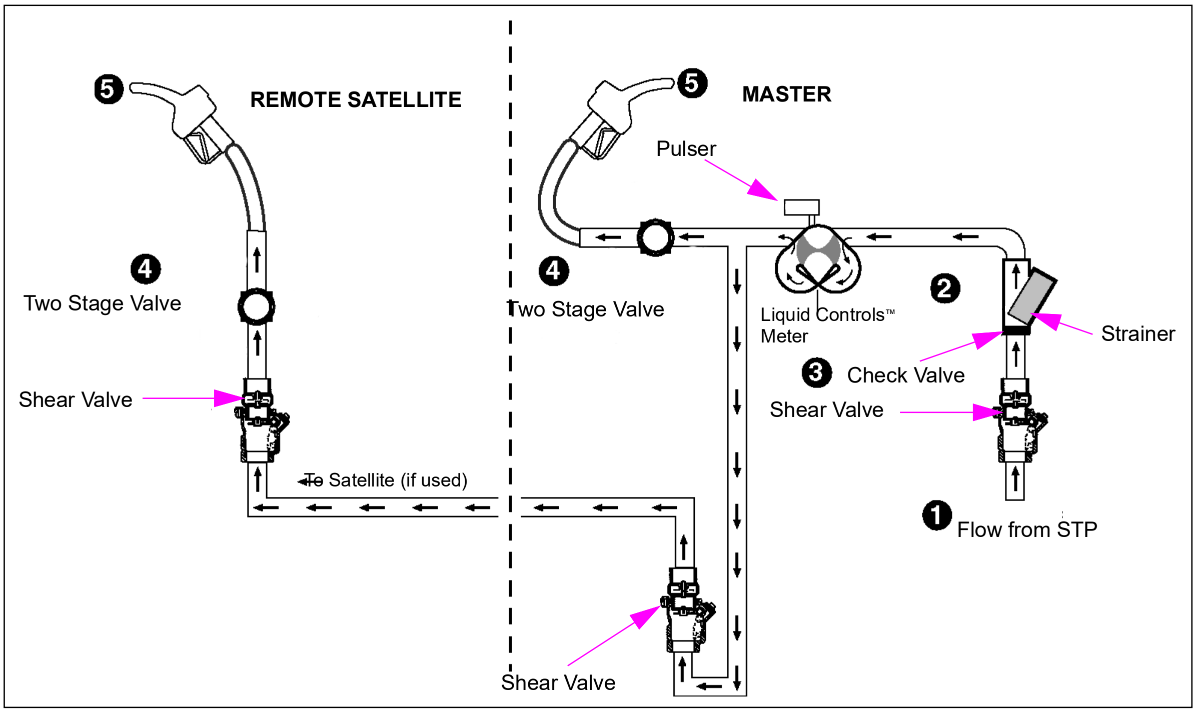
- **Simultaneous Operation** - Allows customer to fuel saddle-tank vehicle with master and satellite unit at same time.
  Note: Some regulatory agencies require a physical barrier to prevent access to another fueling vehicle, if simultaneous operation is possible.
- **Independent Operation** - Allows customer to fuel a saddle-tank vehicle only one side at a time.

Master and Remote Satellite Piping

A Master and remote satellite system is shown in Figure 8-6.

Note: The numbers in Figure 8-6 correspond to the following numbered steps.

Figure 8-6: Master/Remote Satellite Fuel Flow Schematic



Operation

1 Product passes through a shear valve and enters the dispenser.

2 Product flows through a check valve and strainer.

3 Product flows through a Liquid Controls meter.

4 Metered product passes through a single-stage shut-off valve and slowdown the valve in parallel or in more recent units, a two stage valve, and/or through piping to the satellite.

5 Product flows through the remote satellite through a shear valve, solenoid valves, and then to the nozzle.

6 Fuel is discharged through the nozzle.

How Truck Stops Are Configured

Figure 8-7, Figure 8-8, Figure 8-9 on page 8-9, and Figure 8-10 on page 8-9 show sample master/satellite/combo configurations for use at truck stops.

Figure 8-7: Truck Stop Configuration - Single Master with Remote Satellite

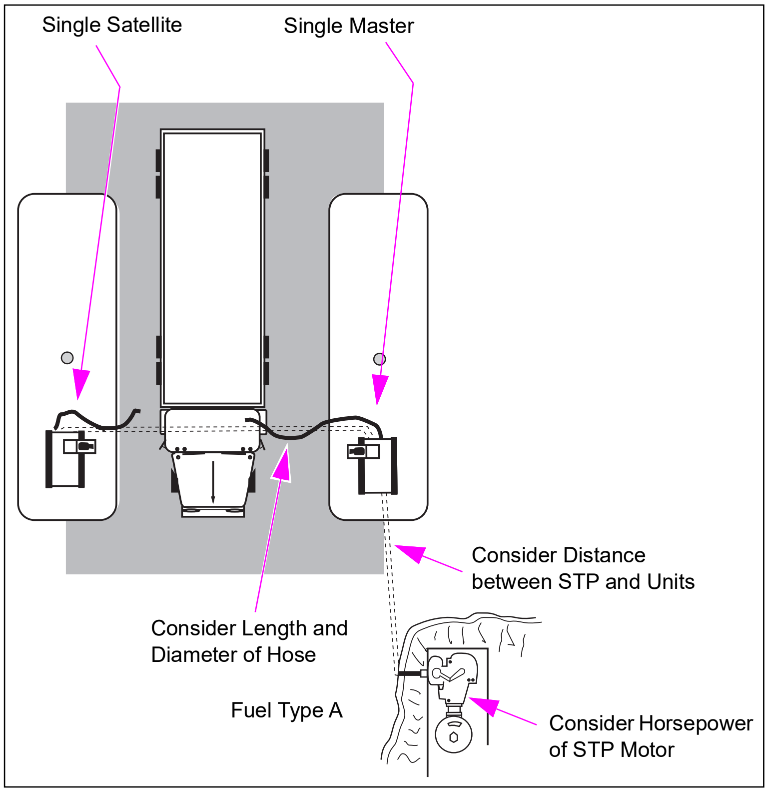


Figure 8-8: Truck Stop Configuration - Dual Master

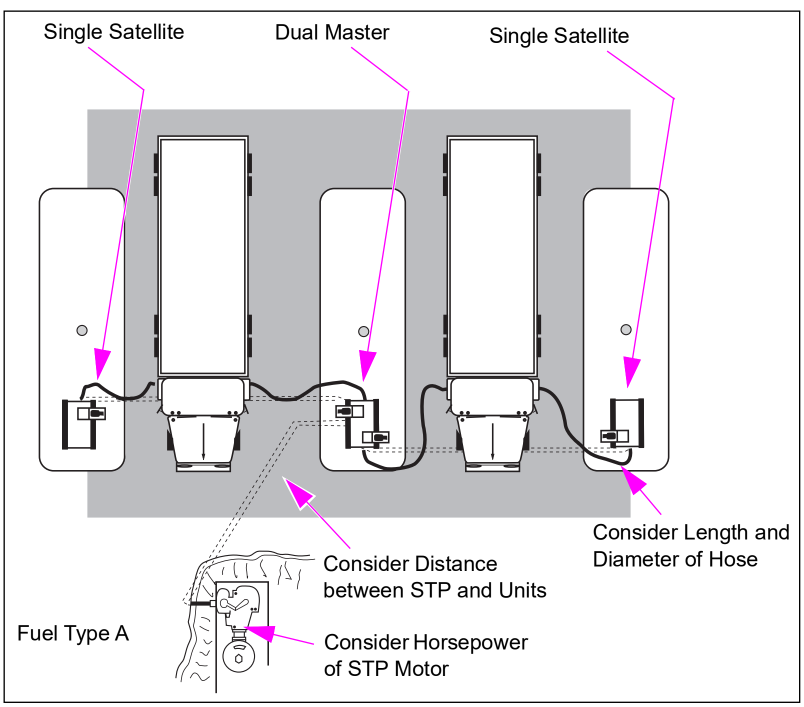


Figure 8-9: Truck Stop Configuration - Dual Satellite

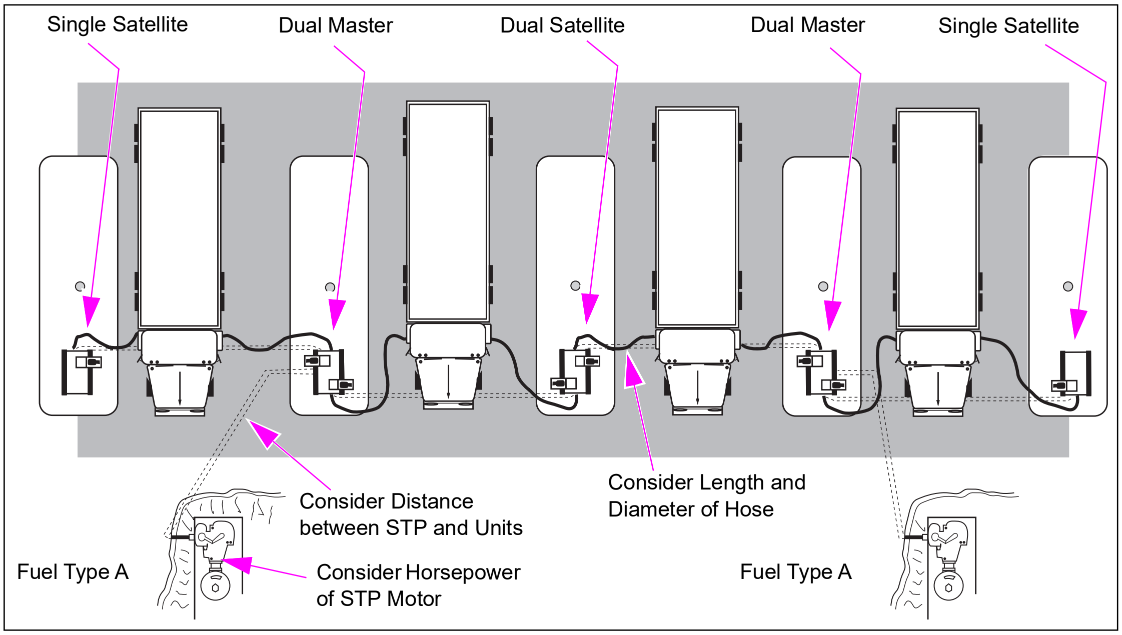
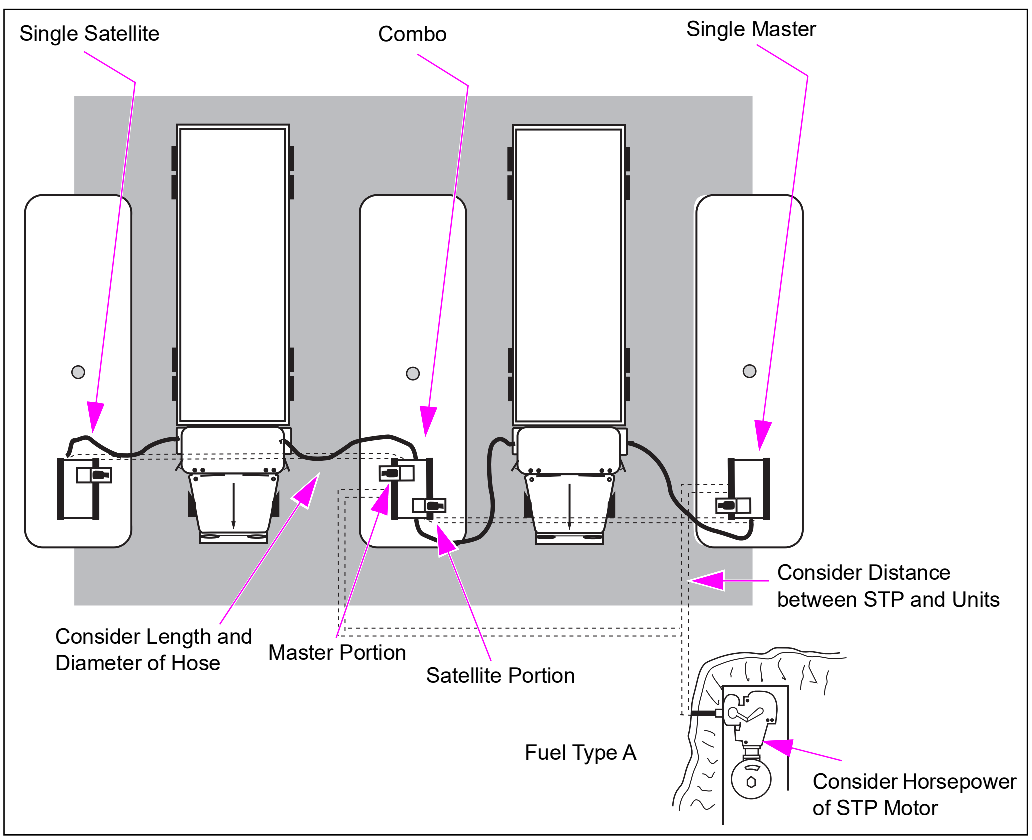


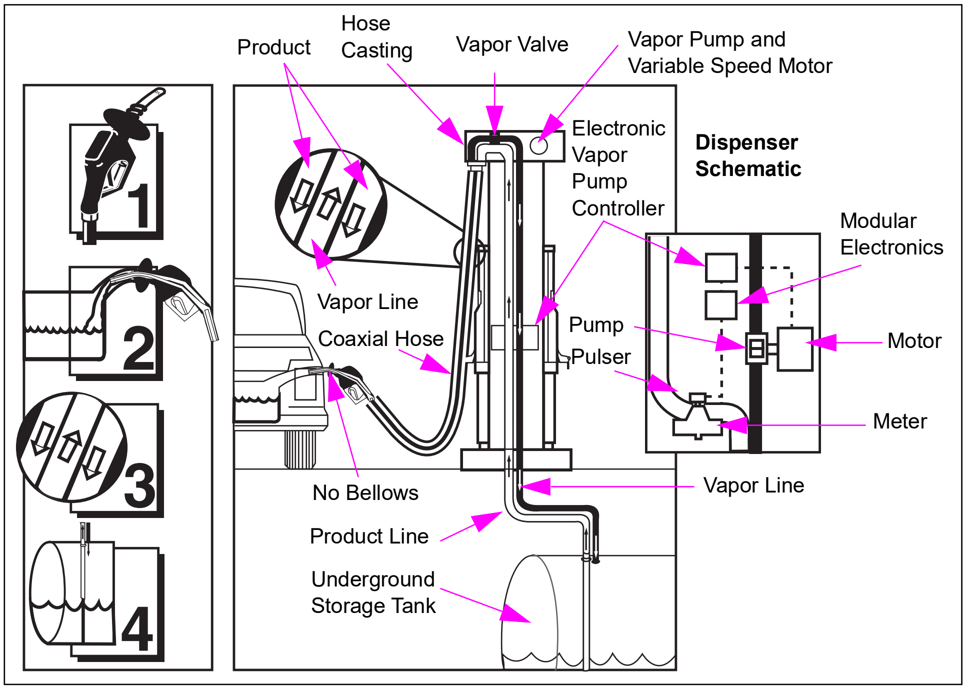
Figure 8-10: Truck Stop Configuration - Combo



How VaporVac Vacuum Assist Vapor Recovery Works

Note: The numbers in Figure 8-11 correspond to the following numbered steps.

Figure 8-11: How VaporVac Vacuum Assist Vapor Recovery Works



1 The fueling customer places the VaporVac nozzle in the vehicle's fill pipe. Vapors flow through holes in the nozzle spout.

2 As the vehicle's fuel tank fills, the VaporVac controller uses the pulser signals from the meters to track the real-time flow rate.

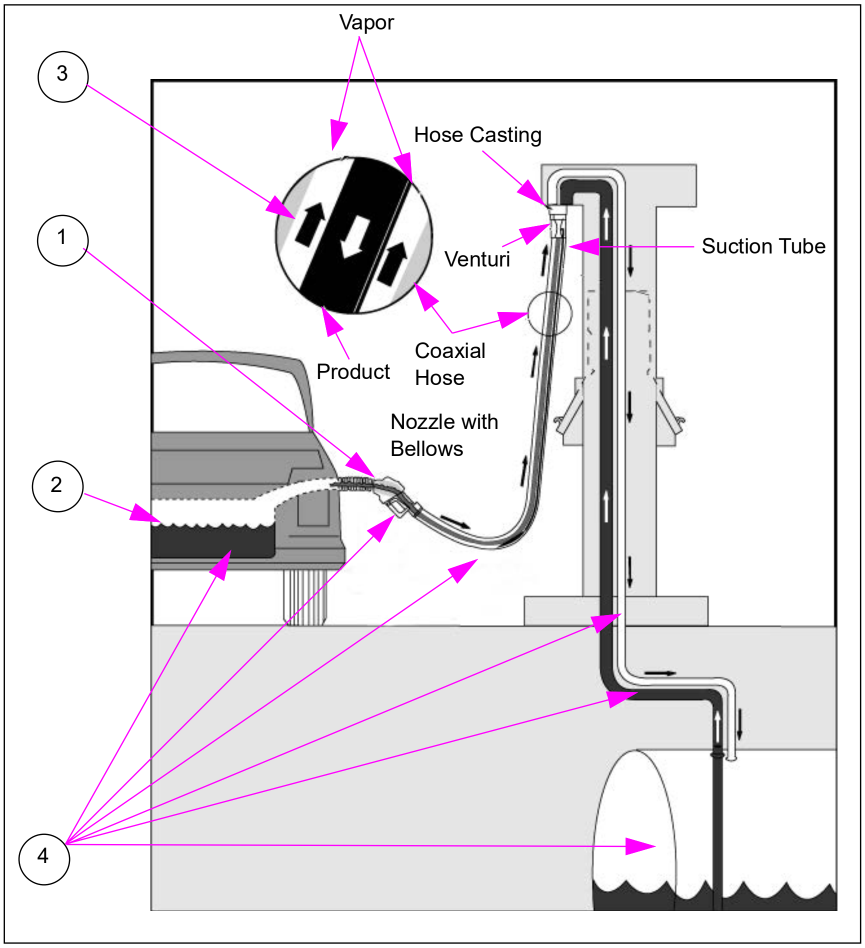
3 The variable speed VaporVac pump creates a vacuum which removes vapors from the vehicle tank at a speed proportional to the real-time flow rate.

4 The vapors return to the fuel storage tank through standard (balance system) underground vapor piping. Any additional vapor processing is not required.

How Stage Two (Balance Type) Vapor Recovery Works

Note: The numbers in Figure 8-12 correspond to the following numbered steps.

Figure 8-12: How Stage Two (Balance Type) Vapor Recovery Works



1 The fueling customer places the nozzle in the vehicle’s fill pipe and compresses the nozzle bellows to form a pressure tight seal.

2 A slight pressure develops in the car tank during fueling, which forces vapor back to the dispenser through the nozzle.

3 Balance systems use coaxial hoses to remove vapors. Fuel flows through the inner hose and vapor flows through the outer hose. A venturi pump in the hose removes liquid (through a suction tube) from the outer hose.

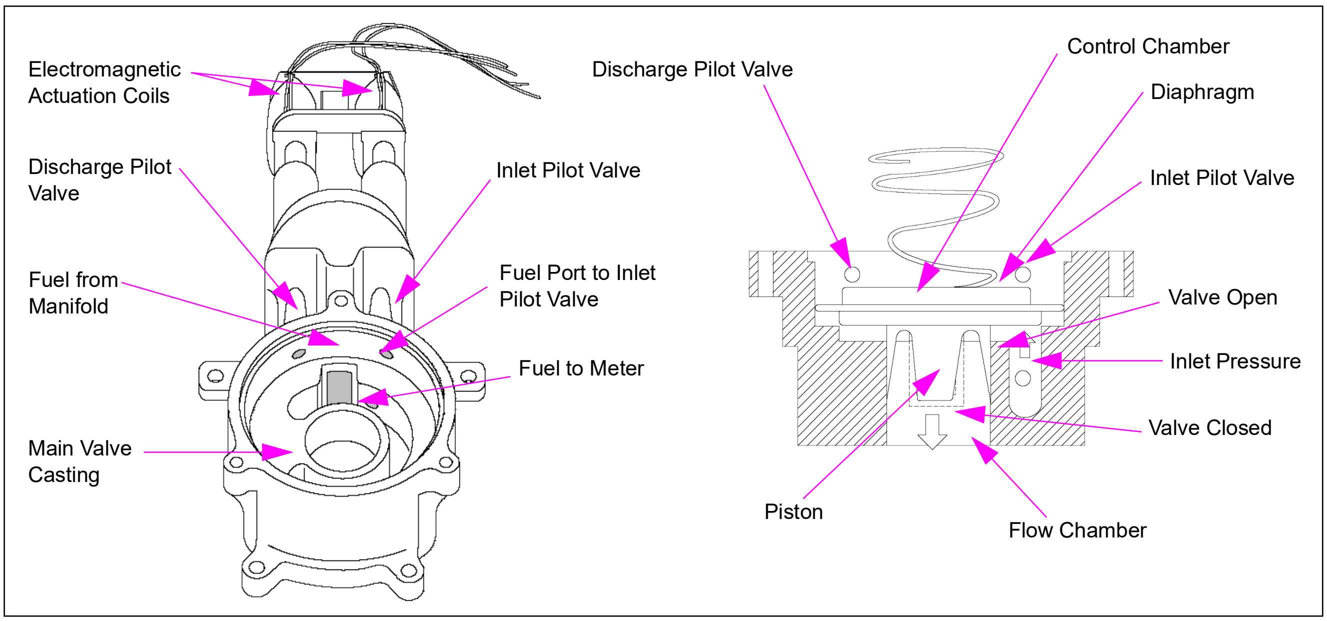
4 As system pressure balances, the vapors flow from the vehicle fuel tank through the bellows, nozzle, outer hose, dispenser vapor piping, and underground piping to the UST.

How a Digital Valve Works

The Gilbarco digital valve (see Figure 8-13) consists of an aluminum main valve casting, a diaphragm/piston assembly, and two pilot control valves with electromagnetic actuation coils. The diaphragm/piston assembly divides the main valve casting into flow, control, and inlet chambers. Fluid pressure in the control chamber is determined by inlet pressure, pilot valves, and nozzle opening state. One pilot controls the fuel flow into and out of the control chamber, thereby raising or lowering pressure in relation to the flow chamber. The other pilot valve bypasses the flow through the valve during slow flow times such as during the pulser test and just before hitting a preset target. The valve is biased through a spring to maintain a back pressure of 2.5 PSI. When the control chamber pressure is equal to or higher than flow chamber pressure, the valve closes and fuel flow stops. When the control chamber pressure is lower than the flow chamber pressure, the valve opens.

The digital valve was in production from 2000-2003 for all standard flow Encore models. All Eclipse models have the digital valve.

Figure 8-13: Digital Valve Components



The following table describes how the valve operates in various modes:

Mode	Operation
Valve at rest (unpowered)	• The inlet pilot valve is open, maintaining pressure in control chamber. • The discharge pilot valve is closed, preventing loss of pressure in control chamber. • Diaphragm/piston assembly is seated, closing main valve. • Equal pressure in the two chambers holds the diaphragm against main valve seat. • No fuel flows when the nozzle opens.
Slow flow	• Inlet pilot valve is open. • Discharge pilot valve opens (energized). • This allows fuel to flow at about 1/2 GPM (low flow) when the nozzle opens.
Full flow	• The inlet pilot valve closes (energized). • The discharge pilot valve opens (energized). • Pressure in the control chamber drops. • Diaphragm/piston assembly forced upward by pressure in flow chamber. • Fuel flows at rate set by software and nozzle opening.

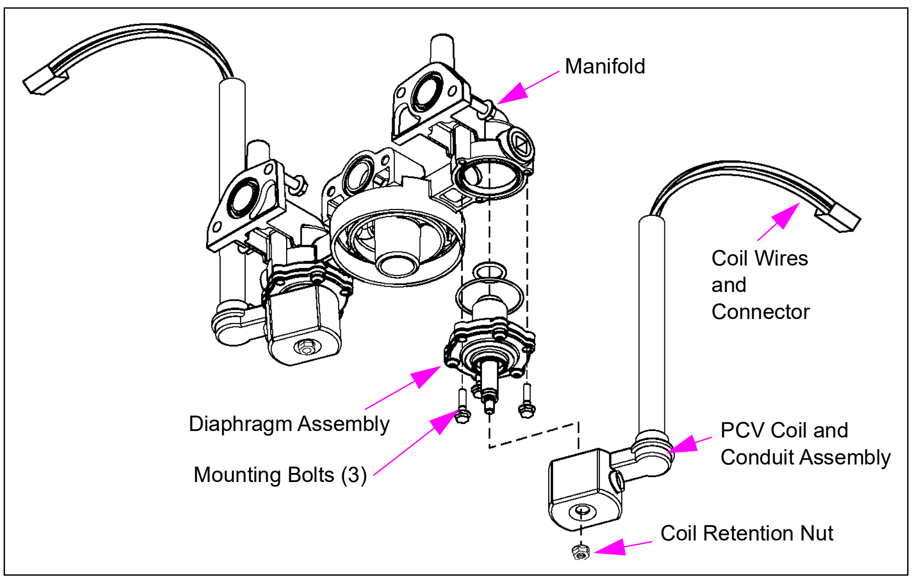
Mode	Operation
Regulate	• The inlet pilot valve closes (energized). • The discharge pilot valve is closed. The discharge pilot valve is opened variably based on software to accommodate blend ratios. • Control chamber pressure maintained at constant level. • Discharge flow for the involved valve continues or adjusts at a rate determined by software.
Power fail	• Inlet pilot valve opens. • Discharge pilot valve closes. • Diaphragm/piston assembly seats, closing main valve. • Equal pressure in two chambers holds the diaphragm against main valve seat. • No fuel flows.

How a Proportional Control Valve (PCV) Works

The PCV (see Figure 8-14) utilizes an analog signal to control the flow while providing smooth and quiet operation. Opening the pilot orifice causes pressure above the diaphragm to decrease so that the diaphragm lifts off the main valve seat. This opens the main orifice until the valve is de-energized. When controlled by the Gilbarco electronics in a closed loop system, proportional flow control is obtained by controlling the plunger position. Coils and conduit are a combined unit and the diaphragm is easily replaced. The PCV can be programmed for flow control for blenders (not for Encore 300 units).

The Proportional Control Valve went into production in 2003 for all standard flow Encore models.

Figure 8-14: Proportional Control Valve Assembly and Manifold



1 A low voltage signal is sent to the valve coil. The valve opens a slight amount to create during the pulser test.

2 After the pulser has been verified as good, a higher voltage is sent to the valve to accomplish the following:
   - Wide open if not a blender or limited by a programmable flow rate.
   - Partially open to maintain a blend ratio in conjunction with a parallel valve or if programmable flow control is present.

3 For a blender or preprogrammed maximum flow rate, the valve will be open or closed based on a revised signal from the electronics to maintain the blend ratio and/or maximum flow during delivery.

4 For preset sale, near the end of that sale, the valve signal is reduced to create slow flow and to allow the valve to close on target.

The following table describes how the valve operates in various modes:

Mode	Operation
Valve at Rest (Unpowered)	• Supply pressure is allowed above the diaphragm through the eyelet bleed hole. The plunger is pressed against the pilot orifice by a spring maintaining pressure above the diaphragm. • Slightly greater or equal pressure in the inlet chamber and above the diaphragm holds the diaphragm against main valve seat. • No fuel flows.
Slow Flow	• The solenoid is energized such that the plunger raises slightly above the pilot orifice. • Flow occurs through the eyelet bleed hole and through the pilot orifice. • Pressure above the diaphragm is still high enough to keep the diaphragm against the main seat.
Full Flow	• The solenoid is energized and the plunger raises further above the pilot orifice. Flow increases though the pilot orifice decreasing pressure above the diaphragm. • The diaphragm is forced upward by pressure in the inlet chamber. • Fuel flows through the main valve seat and main orifice. • Flow rate is set by software by raising or lowering plunger slightly.
Regulate	• The dispenser electronics adjust the solenoid signal to the plunger to maintain any programmed maximum flow rate or to maintain a blend ratio. • Discharge flow continues at rate determined by software.
Power Fail Solenoid is de-energized	• Plunger is pressed against the pilot orifice by a spring. • Pressure increases above diaphragm pushing diaphragm against main valve seat. • Equal pressure in the inlet chamber and above the diaphragm holds the diaphragm against main valve seat. • No fuel flows.

How the Electronics Work in Encore 300

1 Customer raises the pump handle.

2 Signal travels to the pump controller board through the hydraulic interface board.

3 The pump controller sends signals to the displays for reset and price display.

4 The pump controller signals the console for authorization through the hydraulic interface.

5 The pump controller signals the hydraulic interface to activate the STP for dispensers (or self-contained pump motor for pumps).

6 Attendant authorizes sale at console. Console sends preset sale data (if applicable) to the pump controller through the hydraulic interface.

7 Customer begins fueling, which initiates valve slow-flow.

8 The pump controller receives pulses through the hydraulic interface from the pulser. The meter turns with fuel flow which rotates the pulser shaft.

9 After ten pulses, the pump controller signals the hydraulic interface to initiate full-flow in the valves.

10 During delivery, the pump controller receives volume pulses and translates them into cost information.

11 This information is sent to the displays and console. For a preset delivery, the pump controller signals the hydraulic interface to slow the product flow and stop at the preset amount.

12 Customer lowers the pump handle.

13 The pump controller signals the hydraulic interface to remove power from the valves and STP or self-contained pump motor to stop product flow.

14 The pump controller loads delivery amount into memory and sends this information to the console through the hydraulic interface and two-wire circuit.

CRIND Device

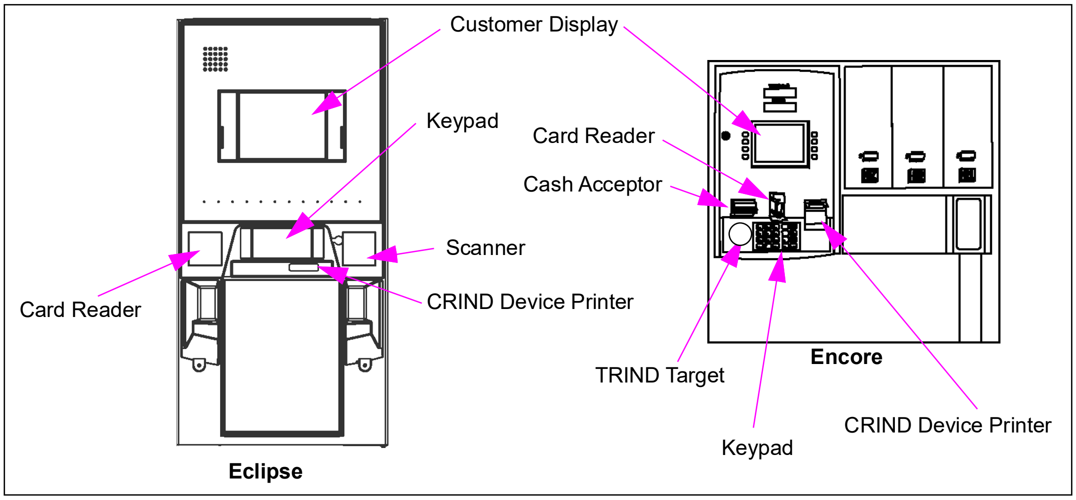
The CRIND device is a customer activated POS terminal located at the dispenser that accepts Cash, Credit Cards, and Debit Cards. It allows customers to activate the dispenser without help from station attendants. It communicates with the site controller (G-SITE or Passport™ for MOC CRIND devices or third-party controller for generic CRIND devices). The site controller communicates with Credit/Debit networks. The CRIND device contains the following assemblies:

- Keypad
- Customer display (single-line, monochrome or color)
- Receipt printer
- Magnetic stripe card reader
- Printed circuit boards
- Cash acceptor (optional)
- TRIND device (optional)
- Scanner (optional)
- Cold weather option kits

Information in this section applies to CRIND devices in all Gilbarco dispensers, unless otherwise noted.

- The single-line CRIND displays prompts and messages on a 14-segment, 20-character, single-line display on the CRIND door.
- The Monochrome LCD CRIND displays multiline prompts, advertising messages, and graphics (no video) on a monochrome LCD screen on the CRIND door.
- The color display is a CRIND option that allows CRIND devices to display multiline prompts, advertising messages, graphics, and videos on a full-color screen on the CRIND door.
- The cash acceptor option allows CRIND devices to accept cash payments, in addition to Credit and Debit Cards.
- The External CRIND (e-CRIND) adds card reader capabilities to a competitive pump. The card reader reads the magnetic stripes on standard Credit and Debit Cards.

Figure 8-15: Various CRIND Devices on Eclipse and Encore Units



How a TRIND Device Works

Overview

The TRIND device is an option available as a retrofit kit or factory installed device. The device is connected to the CRIND device through a gateway or serial interface board. In this application, the TRIND device is a software peripheral to the CRIND and associated POS Controller/Host. This means that a properly working TRIND system will look for and read tags, through the CRIND application, but only when "told" to do so by the controller.

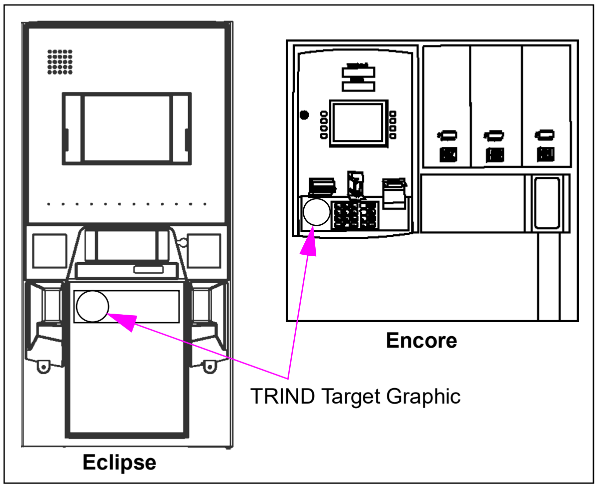
The TRIND system uses an electronic system located in the pump or register to "talk" with a miniature radio-like device (a tag). Together, these electronic devices provide "cashless" access to gasoline, food, and merchandise by charging purchases to a Credit Card, Check Card, or other account you already have. The TRIND system operates on a dedicated tag identification code. Your Credit Card or Check Card account numbers are not typically used with the tag signal system, which protects your account from unauthorized use.

TRIND utilizes Radio Frequency (RF) waves to communicate with a customer's transponder. For car tag systems, transmitters located on overhead assemblies transmit radio waves that serve as a "wake up" call. As a car mounted transponder enters the transmitter's effective read zone, the transponder is activated. The transponder then transmits a code which is received by the TRIND antennas mounted overhead or on the option doors. For hand-held tag systems, the transmitter is located behind the target graphic (see Figure 8-16 on page 8-17).

This transmitted code is communicated from the TRIND to the CRIND unit and from there to the POS. The POS system communicates with a host to obtain authorization.

Car mounted transponders are activated when they are within 6-feet of the front of the dispenser perpendicular to front of the dispenser. Hand held transponders, usually on a key chain, function when pointed at a target graphic on the option door. Two way communication is indicated when the option door target graphic light comes on, whether by car mount or hand-held transponder signal.

Figure 8-16: Encore and Eclipse TRIND Target Graphic Location



How a Cash Acceptor Works

1 Insert a currency bill (or quarters on some units). The cash acceptor accepts bills face up or down, and from either end depending on the setup.
- There will be a short delay when the system “reads” the bill.
- Listen for the stacking of the bills.
- The system will register the cash acceptor total on the pump screen and on the printout at the pump.

2 Repeat step 1 with another bill or lift the nozzle.

3 Select the fuel grade and begin pumping fuel (or select car wash and follow prompts). After the fuel grade (or car wash) is selected, the customer cannot put more money into the cash acceptor until the fueling transaction is complete.
Note: The customer cannot pay for a completed sale at the cash acceptor. Incomplete sales result in receipt that shows a refund due to the customer. This refund is tendered by the cashier. The printer must be operational and contain paper for the cash acceptor to work.

How a Scanner Works

When interfaced with a POS system, the barcode scanner option allows the customers to scan the barcoded loyalty card on CRIND device equipped units. G-SITE systems require that the AIM code be enabled for proper operation. CRIND device scanner diagnostics can be used to enable and disable the AIM code, as required. For information particular to that system, refer to Passport documentation.

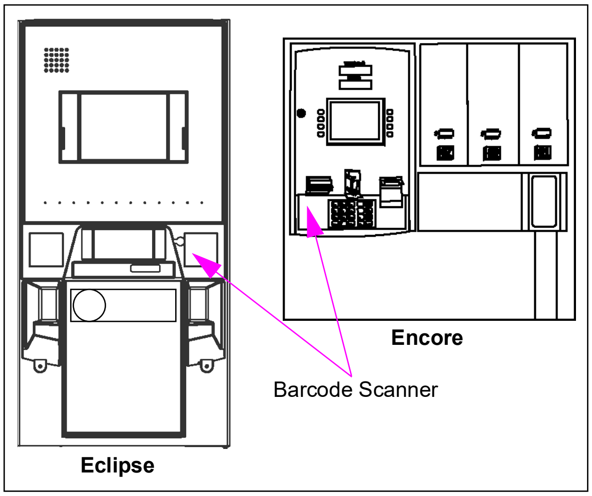
The scanner mechanism uses a low power laser directed downward that reads barcodes, that are presented directly below the scanner window. The barcodes are interpreted by the scanner components and information is sent through the CRIND to the POS for processing. The POS system responds to the information by sending special commands to the CRIND for authorizing specialized loyalty card discounts and so on.

In Encore 500 and Eclipse units, the scanner is controlled by a Local Area Network (LAN)/serial node per unit.

Service Tip

The Encore 500 and Eclipse barcode scanners require CRIND BIOS software version 200.30 or greater to operate properly.

Figure 8-17: Encore and Eclipse Barcode Scanner Location



How ATC Works

The ATC option measures the country or region involved. The ATC temperature of the dispensed product compensates for changes in product volume due to changes in fuel temperature to a standard defined temperature. The ATC option uses the following components:

- ATC Meter Cover (contains the thermal test well and ATC probe - senses fuel temperature).
- ATC Meter Manifold (manifold contains the thermal test well and ATC probe).
- T-meter Module (gathers and processes information from ATC probes).
- I.S. Barrier
  - Provides power to T-meter module.
  - Provides data path from T-meter module to ATC controller board.
- ATC Controller Board (replaces the valve driver board - controls the fuel flow valves).

In an ATC equipped unit, fuel temperature is monitored through a thermocouple probe commonly integral to the meter or optionally in close proximity. This temperature is compared against a predetermined “Standard” temperature for the country or region involved. The ATC equipment intercepts pulser signals before information is sent to the PCN or pump controller, and adjusts the pulse count to reflect the pulse count that will have occurred if the fuel was metered at the “Standard” temperature. If fuel being metered is hotter than the standard temperature, pulse count is reduced. If fuel being metered is colder than the “Standard” temperature, pulse count is increased. Adjustment changes as the fuel temperature into the meter changes.

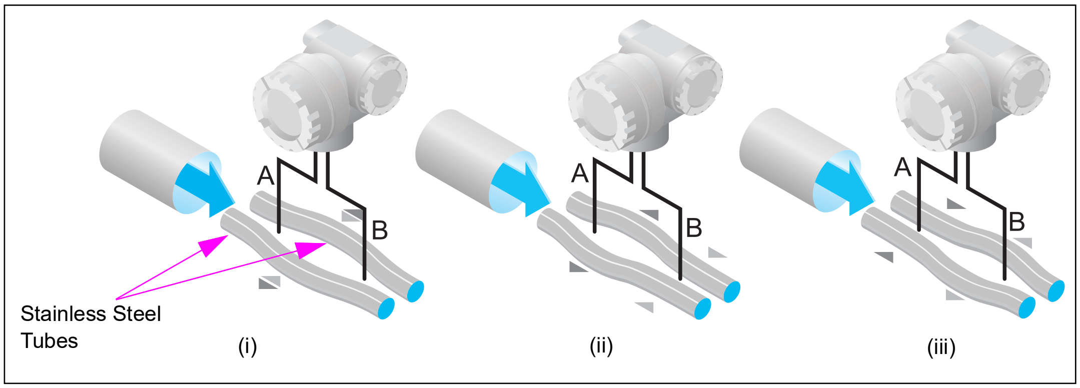
Coriolis Meter

The fluid passes through two specially shaped stainless steel tubes to which transducers are connected. The meter determines the flow rate by vibrating the tubes in a special manner with transmitting transducers.

The two parallel measuring tubes containing flowing fluid oscillate in antiphase, acting like a tuning fork. The Coriolis forces produced at the measuring tubes cause a phase shift in the tube oscillations (see Figure 8-18).

- When the fluid is at a standstill, the two tubes oscillate in phase (see Figure 8-18 (i)).
- Mass flow causes deceleration of the tube oscillation at the inlet (see Figure 8-18 (ii)) and acceleration at the outlet (see Figure 8-18 (iii)).

Figure 8-18: Coriolis Meter - Stainless Steel Tubes



The phase difference (A-B) increases with increasing mass flow. Electrodynamic sensors register the tube oscillations at the inlet and outlet. System balance is ensured by the antiphase oscillation of the two measuring tubes. The measuring principle works independently of temperature, pressure, viscosity, conductivity, and flow profile of the fluid.

Due to flow through the vibrating tubes, the displacement of the tubes varies with the mass flow rate. This change in displacement is monitored by the receiving transducers. The change in displacement of the tubes is a function of the flow rate and is based on the Coriolis Effect (from which the name of the meter is derived). The mass flow rate is converted to volume flow rate by the electronics. The meter must be factory-adjusted to work with the dispensed fluid.

Error Codes

Encore 300 ECs

20 - 44

Code	Description
20	Pulser Fail
21	Non-existent memory
22	Not used
23	Grade assignment changed
24	Conversion Factor Changed - Not programmed
25	Two-wire/Stand Alone changed
26	No conversion factor table
27	Side A Two-wire ID changed
28	Side B Two-wire ID changed
29	Pump Timeout error
30	VaporVac fail both sides or Vapor Sense
31	Totals data error
32	Pulser count error
33	VaporVac fail both or Push-to-start button activated
34	Battery charge low
35	Configuration data error
36	Unit type configuration code changed
37	PIN Code 1 changed
38	PIN Code 2 changed
39	Cash/Credit Option changed
40	Master Reset jumper in place
41	Side Exists option changed
42	PPU Option changed
43	Not used
44	Pump handle on at power-up

4301 - 4329 (Blend ECs)

Code	Description
4301	Communication Timeout
4302	Blend Controller busy
4303	Communication limit
4304	Communication limit
4305	Received Data Error
4306	Received Block Error
4307	Blends not programmed
4308	No blend table
4311	Lo Product Low Flow
4312	Hi Product Low Flow
4313	Lo Product No Flow
4314	Hi Product No Flow
4315	Hi Product Contaminated by Lo
4316	Lo Product Contaminated by Hi
4317	Lo Product Surge
4318	Hi Product Surge
4319	Undefined Surge
4320	Valve configured for but not responding
4321	Valve stuck
4322	Valve board not detected
4323	Pulser Fail
4324	Invalid Command/Software
4325	Maximum Pulse Lag exceeded
4326	Lost Communications
4327	Software Stack error
4328	BPC Slow in Removing Pulses
4329	Leaky Valve

Encore 500 and Eclipse ECs

ECs (E1 - E11) are displayed on the Main Money display.

ECs (20 - 5614) that open on the PPU display when an error occurs are useful in diagnosing problems with the pump/dispenser.

E1 - E12

Code	Description	Level
E1	Database Corrupted	Super Major
E2	Flash Corrupted. Press **F1**/**F2** to clear.	Super Major
E3	LON configuration error	Super Major
E4	One or more tasks not started	Super Major
E5	Coldstart was forced	Super Major
E6	Redundant data storage error	Super Major
E8	Data base changed, set as default the new values	Super Major
E10	Software updated, Security Switch NOT ON	-
E11	Software DSS changed, Security Switch NOT ON	-
E12	Restore Calibration from Door Node	-
E-16	FPGA programming failed	Power cycle the PCN

0 - 50

Code	Description	Level
11	Two-wire communications lost (disconnected)	Minor
20	Pulser connected and product is not mapped	Super Major
20	Pulser Disconnected/Error	Super Major
24	Volume unit type not set	Major
26	Pump not calibrated	Major
27	Two-wire ID changed side A	Undefined
28	Two-wire ID changed side B	Undefined
29	Valve not open	Medium
30	VaporVac power supply or cabling failure	Super Major
31	DEF dispenser lines frozen	-
33	Push-to-Stop activated	Medium
35	Configuration data error	Medium
37	Pin code 1 changed	Undefined
38	Pin code 2 changed	Undefined
39	Cash/Credit option changed	Undefined
41	Side exists option changed	Undefined
42	Fill/Authorization Timeout	Undefined
44	Pump handle on at power up	Medium
46	Minimum Preset	-
99	Running Database Check Failure	-
100	Running Flash Check Failure	-

4000 - 4999

Code	Description	Level
4307	Blends not programmed	Super Major
4311	Blender-Lo Octane Product Low Flow	Minor
4312	Blender-Hi Octane Product Low Flow	Minor
4313	Blender-Lo Octane Product No Flow	Minor
4314	Blender-Hi Octane Product No Flow	Minor
4315	Blender-Hi Octane Product contaminated by Lo Octane	Minor
4316	Blender-Lo Octane Product contaminated by Hi Octane	Minor
4320	Valve configured for, but not responding	Super Major
4321	Valve stuck	Super Major
4322	Valve board not responding	Super Major
4342	NF Model - CC 98 is not programmed	Super Major

5000 - 5025

Code	Description	Level
5000	Meter 0 calibration changed	Undefined
5001	Meter 1 calibration changed	Undefined
5002	Meter 2 calibration changed	Undefined
5003	Meter 3 calibration changed	Undefined
5004	Meter 4 calibration changed	Undefined
5005	Meter 5 calibration changed	Undefined
5006	Meter 6 calibration changed	Undefined
5007	Meter 7 calibration changed	Undefined
5008	Blend ratio changed for product 1	Undefined
5009	Blend ratio changed for product 2	Undefined
5010	Blend ratio changed for product 3	Undefined
5011	Blend ratio changed for product 4	Undefined
5012	Blend ratio changed for product 5	Undefined
5013	Blend ratio changed for product 6	Undefined
5014	Blend ratio changed for product 7	Undefined
5015	Blend ratio changed for product 8	Undefined
5016	Flow Control Setting changed per product 1	Undefined
5017	Flow Control Setting changed per product 2	Undefined
5018	Flow Control Setting changed per product 3	Undefined
5019	Flow Control Setting changed per product 4	Undefined
5020	Flow Control Setting changed per product 5	Undefined
5021	Flow Control Setting changed per product 6	Undefined
5022	Flow Control Setting changed per product 7	Undefined
5023	Flow Control Setting changed per product 8	Undefined
5024	ATC enabled	Undefined
5025	ATC disabled	Undefined

5026 - 5099

Code	Description	
5026	ATC base value changed	Undefined
5027	Sealable switches have been accessed	Undefined
5028	Measurement unit volume changed to liters	Undefined
5029	Measurement unit volume changed to gallons	Undefined
5030	Measurement unit volume changed to Imp. gallons	Undefined
5031	Measurement unit volume changed Hawaiian gallons	Undefined
5041	Power Fail-Power circuit register a power loss	Undefined
5047	Reverse flow detect	Medium
5048	No flow detect eMINOR	Minor
5049	Unauthorized flow detected	Medium
5050	Invalid Pulser-Pulser Pattern does not fit profiled meter	Medium
5056	Transaction started with Security Switch ON	-
5066	STP 1 configured for, but not connected	Super Major
5067	STP 2 configured for, but not connected	Super Major
5068	STP 3 configured for, but not connected	Super Major
5069	STP 4 configured for, but not connected	Super Major
5070	VaporVac option set, but board not detected	Super Major
5071	VaporVac option NOT set, but VV board is present	Minor
5072	VaporVac board detected, but option not set	Super Major
5074	VaporVac motor not responding	Medium
5079	Valve Not Configured for, but is responding	Minor
5081	Air sensor NOT connected, but option is set	Super Major
5091	ATC option set, but ATC Node not detected	Super Major
5092	ATC node present, but not configured	Minor
5098	T-Meter Not Connected	Super Major
5099	ATC - number of probes exceeds number of meter	Minor

5100 - 5121

Code	Description	
5100	ATC - number of meters exceeds number of probes	Medium
5101	ATC T-Meter pulse train not detected	Super Major
5111	Door Node Communication Failure	Super Major
5115	Door Node Download Error	Medium
5116	Door Node software download incomplete	Medium
5118	PPU Board Communication Failure	Major
5119	PPU Board Present but, not configured for	Minor
5120	PPU Board pump handle not connected	Major
5121	PPU Board Grade select button not connected	Major

5126 - 5199

Code	Description	Level
5126	Key stuck on the Manager Keypad	Minor
5131	PPP keypad option set, but not present	Super Major
5132	PPP keypad present, but not configured	Minor
5133	Key stuck on PPP keypad	Minor
5134	Call button option set, but Not Present	Super Major
5135	Call button present, but not configured	Minor
5136	Call button stuck	Major
5137	Push-to-Stop option set, but not detected	Super Major
5138	E-Stop button present, but not configured	Minor
5139	Push-to-Stop button stuck	Major
5140	Push-to-Start option set, but not detected	Super Major
5141	Push-to-Start button present, but not configured	Minor
5142	Push-to-Start button stuck	Major
5150	PPU(s) not addressed correctly	Major
5151	Door Node not addressed correctly	Major
5171	Power Fail Manager Keypad mode exited	Undefined
5174	Totalizer Node Communication Failure	Major
5175	Totalizer Node detected, but not configured	Minor
5183	Totalizer present, but no return pulse detected	Major
5198	Measurement unit changed not programmed	Undefined
5199	Measurement unit changed 500 pulses per gallon	Undefined

5235 - 5241

Code	Description	Level
5235	Database corrupted	Minor
5236	Database build process started	Minor
5239	Database init forced	Minor
5240	Database wrong read	Minor
5240	Database read failed	Minor
5240	Database wrong write	Minor
5241	Database write failed	Minor

5378 - 5724

Code	Description	Level
5378	Door Node configuration not received	Major
5413	Zero PPU set, but not allowed	Medium
5414	Display Type Mismatch	-
5416	Display Type Changed	-
5417	Door Node 5 firmware not compatible w/ PCN software	Major
5600	Fuel density not programmed	Super Major
5601	ATC temperature out of range	Medium
5602	Door Node Software updated, Security Switch NOT ON	Minor
5603	ATC Node Software updated, Security Switch NOT ON	Minor

Code	Description	Level
5604	Totalizer node incorrect software version	Minor
5605	MIP node incorrect software version	Minor
5606	ATC Node Communication Failure	Super Major
5607	Door Node Communication Failure	Super Major
5608	Totalizer node subscription failed	Super Major
5612	TW converter error	Minor
5613	Pump updated with data from door node storage	Minor
5614	Door node A updated with data from pump node	Minor
5616	Pump and door node storage match	Minor
5647	Printer node current software version	Undefined
5721	Door Opened	-
5722	Door Closed	-
5723	Hydraulics Panel Opened	-
5724	Hydraulics Panel Closed	-

6042 - 6047

Code	Description	Level
6042	VAPORIX Alarm Side A	-
6043	VAPORIX Alarm Side B	-
6044	VAPORIX option set, but board not detected	-
6047	VAPORIX board detected, but option not set	-

8025 - 8051

Code	Description	Level
8025	Secure Pulser: ID Mismatch	-
8041	Secure Pulser: Missing Data	-
8042	Secure Pulser: Missing Clock	-
8044	Secure Pulser: Invalid Data	-
8046	Secure Pulser: Lift-Off Detection	-
8050	Secure Pulser: Sensor Failure	-
8051	Secure Pulser: Lid Separation	-

EC31

EC31 indicates that the DEF may be frozen. If an EC31 is initiated when the pump node software version is 1.8.54, the dispenser will go offline and remain offline until the EC disappears. If an EC31 is initiated with later versions of the pump node software, the dispenser will go offline and remain offline until a warmstart occurs. Inform the station that the DEF may be frozen (so that they are alert), if the dispenser goes offline with the POS at very low temperatures. Also, to prevent further damage or a DEF spill, immediately contact the ASC.

5049

When operating a DEF+1 unit with no DEF in the dispenser and the nozzle locked against use, 5049 ECs occasionally opens. 5049 ECs indicate unauthorized flow. The ECs open due to the presence of air in the Coriolis meter. It is recommended that you change CC90 temporarily to option 1, so that the unit will ignore the DEF. This method can also be used to temporarily disable a DEF with fluid in the meter. For example, when a DEF unit is causing entire dispenser problems.
Note: A price must be programmed for DEF before purging.

Troubleshooting

Common Failure Modes with Solutions

This section describes some common failure issues in the field that can be solved by performing the steps included for the observed problem. This information and similar other significant and common problems is published at the TRP under:
http://www.gilbarco.com/interactive/login.cfm.

You must be a certified technician with pre-approved access through your company Gatekeeper to access this information. It is strongly suggested that you monitor the TRP frequently for similar useful information.

Electrical

Blank Monochrome Display (SB745)

If the monochrome displays are blank on the Encore 300 (the F1 Fuse (8.5 V) may also be blown on the CRIND regulator board) or for Encore 500/Eclipse, the overload LED (CR 10) on the 18 V DC power supply is on. The first thing that must be checked is the C45 capacitor on the back of the monochrome display. C45 may show signs of being burnt. Use an ohmmeter to determine if the reading across the capacitor is zero to a few ohms. If so, the capacitor is bad (approximately 4.5 K ohms or higher is a good reading). If C45 is bad, the monochrome display and F1 Fuse (Q12682-06) must be replaced on Encore 300. For Encore 500/Eclipse, the monochrome display must be replaced. Current production units use Monochrome Displays (Q13908-05) that do not have a C45 capacitor.

18V Power Supply PCA (M00050) Shutting Down (SB910)

The in rush current for an Encore 500/Eclipse at power-up may create a false over-current fault that will shutdown the dispenser. If a unit shuts down and LEDs are not properly lit on the 18 V power supply, inspect the 18V Power Supply PCA for its revision level. If you have revision F, G, or, H you may have to clip R9 off the 18 V board to correct a false over-current shutdown problem. To determine if a false over-current is involved with these revision boards, you must perform the following test. You will be checking both channels of the 18 V Power Supply Board:

- Power down the unit.
- Disconnect one LON cable from the pump node.
- Power up and check the two green LEDs on the 18 V Power Supply Board (M00050). If the two green LEDs are "ON", the other channel is good.
- If both channels and 18 V power supply board pass the test, then a false over-current fault is likely occurring and you must clip off R9 to solve the problem.

Hydraulic Components

Meter Jump (SB1470)

Hose pressurization feature (CC55) enacted: Previous software programming codes anticipated release of the feature, but it was never made functional. Hose pressurization is potentially useful for the following conditions:

- Extremely cold temperatures resulting in sale indication, when the nozzle hook is activated but the nozzle is closed (commonly called meter jump).
- Infrequent intermittent problems with check valves, significant temperature changes while dispenser is inactive, large temperature differences between tank and ambient temperature, and other unusual conditions that may affect calibration.

Three options exist for programming (for current versions of software and firmware, this feature is automatically enacted and cannot be programmed):

- Option off (default)
- Unit is inactive for 10 minutes
- Unit is inactive for 30 minutes

The option must not be used as a fix for defective leaking check valves, vapor hoses, or nozzles which can affect calibration or sale indications with the nozzle closed after the unit has dispensed fuel. For information on calibration, refer to *MDE-3860 Programming Quick Reference Guide*.

Software/Firmware

Preset Overruns, Encore 300 (SB1465)

The information in this service bulletin applies to: Encore 300 single-hose selectable blenders with proportional flow control valves manufactured after 4/26/03. V15.0.95
Encore 300 software has been released (K93734-701) to production and fixes the following issues:

- Preset over runs and EC46.
- Fixes problem with hose pressurization not working on side B.
- Changes CC24 FC1 to 10 minutes.

If you have any Encore 300 Models NL (1-3) 3 or NN (1-6) 4 experiencing preset overruns or displaying EC46, upgrade the software to V15.0.95 (K93734-701). After the software has been installed, reprogram the dispenser, as required.

In addition, V15.0.91 is required for any units having preset issues that are controlled by a Wayne® Nucleus or NCR POS.

EC Explanations

31 and 35 for Encore 300 (SB1478)

- EC31 = Totals Data Error
- EC35 = Configuration Data Error

EC31 and EC35 cannot be removed by a warmstart or a coldstart/master reset. Troubleshoot the error in the following order:

1. Record all totals.

2. Perform CC6, FC1 through the manager keypad.
   Note: A CC6 works only if EC31 or EC35 occurs.

3. Reprogram the unit and re-enter previous totals.

4. Check for loose cables.

5. Investigate other problem possibilities, if EC31 or EC35 reappears again later.

   EC31 or EC35 can be caused by various problems (noise, a bad controller board, loose cables, bad controller board battery, and so on).

   Always perform CC6 first before warmstarting, coldstarting, or replacing any parts. Refer to "CC6: Clearing Memory" on page 5-15.

DEF Heating System

The heater control board receives logic voltage from the Coriolis meter board and 120 VAC power supply for the heater and fan from line voltage. A heater sensor is located within the DEF cabinet. It monitors the temperature of the cabinet and is connected to the heater control board. There is also a second sensor in the meter interface board and both the sensors must be cold. Within the approximate range of 40 °F (4 °C) to 45 °F (7 °C) or lower, the heater control board activates the fan and heater to maintain the temperature within the cabinet. The heater circuit will activate and deactivate depending on the temperature within the cabinet. The heater interface board, sensor, heater, and fan are located within the DEF cabinet.

When the DEF freezes, it can cause extensive damage to hydraulic components. Hence, the DEF hydraulics cabinet is heated. The service technicians must be aware of the following important information regarding DEF freezing:

- The heater is turned on when the sensor detects a temperature of approximately 6 °C (43 °F) in the cabinet. It will commonly cycle on and off and may not run at all times during cold weather. To test the system, electronic spray circuit coolant can be used on the cabinet heater sensor (not the freeze sensor which is attached to the piping). At the same time, coolant must also be sprayed on U2 on the heater control PCA to test the system.

- Damage incurred due to freezing of components when there is power loss in cold weather is not covered under warranty. If power to the dispenser is lost during cold weather and temperature is below 20 °F (-6 °C), or if there are possibilities to drop below 20 °F (-6 °C), then the customer must call an ASC immediately to the site to drain fluid from the following components to avoid serious damage to the components:
  - Meter
  - Breakaway and Hose
  - Nozzle
  - Check Valve
  - Dispenser Filter

  Similar service may be required for any tank vault and intermediate plumbing components, as required by the tank system manufacturer. ASCs must respond very quickly to such customer requests.
- If the unit freezes, it is required to thaw all components and repair the damage that may have occurred. Units can be thawed using a non-flame heating device or directed exhaust from vehicle, for areas where no boards or electronic connections are present. DEF is not flammable or explosive. However, before using electrical heating devices, ensure that such devices or their power cords are not in the hazardous vicinity near diesel, gasoline, or other fueling equipment containing hazardous vapors or liquids. To determine any hazardous zone involved, refer to *MDE-3985 Encore Installation Manual*. Monitor the equipment closely for leaks when the lines are pressurized.
- Meter damage during a freeze may occur and can only be detected by performing calibration checks. For additional information, refer to “Coriolis Meter” on page 7-49.
- The insulated cabinet is critical for the proper performance of the heating system. Never remove the insulation from any area or damage it during service. Such removal can cause the system to freeze. If insulation drops or is disturbed from its original area, reglue using a quick curing outdoor compatible adhesive.

For additional information, refer to “Special Fuels/Fluids” on page 1-13.

NJ5/NJ6 Units

NJ5/NJ6 units contain two Valve Driver PCAs (M00059A002). Improper functioning of the unit could be due to failure of either of these PCA boards or due to issues(s) with any other component. Swapping boards on the unit helps to determine the faulty board.

Multi-Hose Universal Blenders (NF Series)

Multi-Hose Universal Blender NF Series dispensers expand blending to include up to all four product inlets (W, X, Y, and Z), but still limit blending to only two products at a time. For NF8 (which reserves "+1" for diesel), this may be thought of as 2 of 4 blending. For the other NF models that contain a Z grade, the Z grade is always a pure product, and will be 2 of 3 blending. With these unit types, rather than being restricted to one designated inlet position, flex-fuel products may enter through inlets W, X, Y, or Z, provided the dispenser hydraulics have the proper flex-fuel components. Although typically reserved for diesel, inlet Z may be used for gasoline or any flex-fuel product if customer requirements warrant it.

Another distinguishing feature of NF unit types is the requirement that an inlet pair be assigned to each non-diesel grade before fueling is allowed. Inlet assignments can be made only after selecting an appropriate unit type via CC90. Similar to how blend ratios are set via CC72 (select the grade, and then enter a ratio made up of 0, 5 to 95, and 100), inlet assignments are set via CC98 as described in the table on page 8-31 (select the grade, and then choose an inlet index between 1 and 6). Like CC72, an inlet assignment must be made for each blended grade.

Note: Command Code 98 is accessible only for NF unit types.

CC98	
Grade	1 - 6
Inlet Index	1 - 6

Inlet Index	Inlet Pair Assignment
1	W,X
2	W,Y
3	X,Y
4	W,Z
5	X,Z
6	Y,Z

Note: Not all inlet pairings are allowed for each NF unit type. Refer to the actual dispenser model to know which pairings are allowed. (Most models allow pairings that include W, X and Y inlets. Fewer models allow pairings that include the Z inlet). When there is a Diesel product, the Z inlet is dedicated to that product and is not allowed to be paired with another inlet.

NF unit types contain the flexibility to allow any blended grade to have a ratio of any allowable inlet pair, the flexibility to allow inlet parings to be changed at any time, and the flexibility to allow all blended grades to use the same inlet pair, if required. However, for any NF unit type, the number of grades per hose is fixed. The number of grades per hose is not a part of the flexibility of the NF series.

Blend Ratios

Once the unit type (CC90) and inlet assignments (CC98) have been made, blend ratios can be assigned. Blend ratios for NF unit types are set the same way as for other blender unit types. **However, extreme ratio settings of 1, 2, 3, 4, 96, 97, 98 and 99 are not allowed.**

Prior to setting blend ratios, care should be taken to note the inlet pairings for each grade. Confirm with the customer the correct blend ratios per the actual tank products. Gilbarco is not responsible for blend ratios specified by the customer. As blend ratios are entered, refer to these pairings and recognize that blend ratios are entered with respect to the first inlet of the inlet pair. For example, if the inlet pair for a given grade is W,Y and the blend ratio is set to 85, that grade is assigned a ratio of 85 percent of product W and 15 percent of product Y. The value of the blend ratio entry is subtracted from 100 to derive the blend ratio of the second inlet in the pair.

Grade	Inlet Index (Inlet Pair)	CC72 - Blend Ratio
3	2 (W,Y)	85

Note: Changing any of the inlet pair settings (CC98) will clear all blend ratios.

Error Codes

EC 5417

Dispenser software versions (04.0.07 and later) that support NF unit types require that Door Node 5 door nodes have a minimum software version of 02.0.59. **This only applies to Door Node 5**. On power up, if the dispenser detects a Door Node 5 software version less than 02.0.59, a 5417 error code will be displayed on the PPUs. This error code can only be cleared by updating the door node software.

EC 4342

It is imperative that inlet pairs be set through CC98. Failure to assign an inlet pair for any blended grade will result in a 4342 error code. Once an inlet pair has been set for each blended grade, this error will clear. Switching from or between NF unit types will clear inlet pair settings for all grades.

E4

This Error Code will be displayed if you are upgrading software in an existing NJ5 or NJ6 unit to an NF version. If E4 is displayed during this upgrade, the Error Code can be ignored. Continue with other programming as required.

Calibration

Given the variety of products available that can be blended to provide different fuel octanes, during normal operation it is possible that none of the blended grades has a blend ratio of 0 or 100 percent. This is important to note because before checking or setting calibration, a grade representing each product in use must have a 0 or 100 percent ratio setting. Refer to the inlet pairings (CC98) to determine the grades that should have a blend ratio set to 0 or 100.

After calibration has been checked or set, the correct ratio for each grade can be set to the appropriate value.

*Note: As part of dispenser software version 04.0.07 and later versions, when using the recalibration procedure to set a new calibration factor for a grade that does not have a pure product assignment (0 or 100 percent), "**noECAL**" will be displayed in the money and volume positions of the main display, after the W&M switch is closed to accept the new calibration.*

Examples for setting blend ratios in order to calibrate:

Grade	Inlet Index (Inlet Pair)	CC72 - Blend Ratio
1	1 (W,X)	100
2	2 (W,Y)	0
3	3 (X,Y)	100
4	1 (W,X)	--
5	2 (W,Y)	--

Grade	Inlet Index (Inlet Pair)	CC72 - Blend Ratio
1	1 (W,X)	--
2	2 (W,Y)	--
3	3 (X,Y)	100
4	1 (W,X)	100
5	2 (W,Y)	0

Grade	Inlet Index (Inlet Pair)	CC72 - Blend Ratio
1	3 (X,Y)	100
2	1 (W,X)	100
3	3 (X,Y)	0
4	2 (W,Y)	--
5	2 (W,Y)	--

‘--’ - No ratio restrictions; can be any blend ratio.

These examples illustrate the following rules related to blend ratios and calibration:

1 When a blend ratio entry is entered via CC72, this value is stored as the ratio for the first inlet position of the inlet pair. The ratio for the second inlet position of the pair is calculated as 100 minus the ratio entry.

2 In order for a calibration factor to be stored, blend ratio entries have to be 0 or 100.

3 Not every grade is required to have a 0 or 100 blend ratio setting. It is only required that grades for each inlet pair have a 0 or 100 setting.

Note: A 0 setting for the first inlet position of a pair by default is a 100 setting for the second inlet position of a pair and vice versa. (Another way to view the blend ratio settings is that each inlet position in use should have a grade setting of 100, whether entered explicitly or computed.)

Reference Only: (EC26 No Calibration Factors)
Included in dispenser software version 04.0.07 and later versions is a visual aid aimed to help with calibration in the event calibration factors are cleared or lost, resulting in EC26. There are two meters associated with each product inlet. Meter IDs for Side-1 of the dispenser are 1, 2, 3 and 4. Meter IDs for Side-2 of the dispenser are 5, 6, 7, and 8. The following table illustrates the inlet-to-meter relationship.

Inlet	Side-1 Meter ID	Side-2 Meter ID
W	1	5
X	2	6
Y	3	7
Z	4	8

When EC26 is displayed, after entering calibration mode, “9999” “9999” will be displayed across PPU grades 2 and 3 for each meter position that has a 0 calibration factor. Starting with grade 2, each “9” represents a meter position (meters 1 through 8). After a meter is calibrated, its ID shows in its respective position on the PPU displays.

This feature helps in the following cases:
- Segregating the calibrated meters from the non-calibrated meters
- Identifying which meters have been calibrated in the event of pause or interruption in the calibration process

Examples:
Meter **1** calibrated

1999	9999

Meters **1** and **3** calibrated

1939	9999

Meters **1**, **2**, **3, 5,** and **6** calibrated

1239	5699

Only meters associated with inlet positions being used require calibration. It is important to know the inlet assignments and have the proper blend ratio settings in order to eliminate confusion during calibration.

For NF unit types, apart from searching the W&M logs for calibration factors, pressing “Enter” on the manager keypad 4 times will show the calibration factor for each meter required for that unit type on the PPU displays. The PPU displays on side-1 show the calibration factors for meters associated with side-1 and the PPU displays on side-2 show the calibration factors for meters associated with side-2. To return the PPUs to displaying prices, press “CLEAR” on the manager keypad.

Note: For the NF Series, calibration factors are displayed in reference to inlet position as highlighted in the following table:

Calibration Factor for Inlet - (Meter ID Side-1/Meter ID Side-2)	Displayed on PPU Grade
W - (1/5)	1
X - (2/6)	2
Y - (3/7)	3
Z - (4/8)	4

Electronic Hardware

Just as with NJ5/NJ6 unit types, to achieve blending across more than two products with NF series, two Valve Converter Assemblies (M14613A001) are required. With these unit types, valve connections for products W and X always connect to “Grade 1” and “Grade 2” connectors, respectively, on Valve Board 1. The valve connection for product Y always connects to “Grade 3” on Valve Board 2. The valve connection for product Z in most cases connects to “Grade 4” on Valve Board 1. In rare cases where product Z is gasoline or a flex-fuel product intended for blending, the valve connection connects to “Grade 4” on Valve Board 2.

Valve Connection Chart - Product Z: Diesel - (Not Intended For Blending)

Product Valve	Valve Board 1 Connection	Valve Board 2 Connection
W	Grade 1	-
X	Grade 2	-
Y	-	Grade 3
Z	Grade 4	-

Valve Connection Chart - Product Z: Gasoline or Flex-Fuel - (Intended For Blending)

Product Valve	Valve Board 1 Connection	Valve Board 2 Connection
W	Grade 1	-
X	Grade 2	-
Y	-	Grade 3
Z	-	Grade 4

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